John W. Tukey

EXPLORATORY DATA ANALYSIS

exhibit **2** of chapter 3: reference table

Break table for two-decimal logs

A) MAIN BREAK TABLE

Break	log	Break	log	Break	log	Break	log	Break	log
9886		**1567**		**2483**		**3936**		**6237**	
1012	.00	**1603**	.20	**2541**	.40	**4027**	.60	**6383**	.80
1035	.01	**1641**	.21	**2600**	.41	**4121**	.61	**6531**	.81
1059	.02	**1679**	.22	**2661**	.42	**4217**	.62	**6683**	.82
1084	.03	**1718**	.23	**2723**	.43	**4315**	.63	**6839**	.83
1109	.04	**1758**	.24	**2786**	.44	**4416**	.64	**6998**	.84
1135	05	**1799**	25	**2851**	.45	**4519**	.65	**7161**	.85
1161	.06	**1841**	.26	**2917**	.46	**4624**	.66	**7328**	.86
1189	.07	**1884**	.27	**2985**	.47	**4732**	.67	**7499**	.87
1216	.08	**1928**	.28	**3055**	.48	**4842**	.68	**7674**	.88
1245	.09	**1972**	.29	**3126**	.49	**4955**	.69	**7852**	.89
1274	.10	**2018**	.30	**3199**	.50	**5070**	.70	**8035**	.90
1303	.11	**2065**	.31	**3273**	.51	**5188**	.71	**8222**	.91
1334	.12	**2113**	.32	**3350**	.52	**5309**	.72	**8414**	.92
1365	.13	**2163**	.33	**3428**	.53	**5433**	.73	**8610**	.93
1396	.14	**2213**	.34	**3508**	.54	**5559**	.74	**8810**	.94
1429	.15	**2265**	.35	**3589**	.55	**5689**	.75	**9016**	.95
1462	.16	**2317**	.36	**3673**	.56	**5821**	.76	**9226**	.96
1496	.17	**2371**	.37	**3758**	.57	**5957**	.77	**9441**	.97
1531	.18	**2427**	.38	**3846**	.58	**6095**	.78	**9661**	.98
1567	.19	**2483**	.39	**3936**	.59	**6237**	.79	**9886**	.99

When in doubt use an even answer; thus, 1462 gives .16 and 1496 gives .18.

B) SETTING DECIMAL POINTS

1	+0	−1	**1**
10	+1	−2	**0.1**
100	+2	−3	**0.01**
1000	+3	−4	**0.001**
10,000	+4	−5	**0.0001**
100,000	+5	−6	**0.00001**
1,000,000			**0.000001**

C) EXAMPLES

Number	**B**	**A**	**log number**
log 137.2	2	+ .14 =	2.14
log 0.03694	−2	+ .57 =	−1.43
log 0.896	−1	+ .95 =	−0.05
log 174,321	+5	+ .24 =	5.24

See text, page 63, for example of use.

311.2
T81

exhibit **7** of chapter 3: reference table

Break table for (square) roots

A) EXAMPLES

Start by dividing number into periods of two digits, decimal point falling between periods. Thus 124.2 is 1 24 2, but 1242 is 12 42. Like wise 0.00654 is 00 65 4 or 65 4.

Number	Periods	from B	from C	Number
124. 2	1 24 2	**ab.**	112	**11.2**
1242	12 42	**ab.**	35	**35.**
. 00654	00 65 4	**.0x**	80	**.080**

B) BREAK TABLES to SET DECIMAL POINT--enter between bold figures, leave with light figures

a	.x	**1**
ab.	.0x	**. 01**
abc.	.00x	**. 00 01**
abcd.	.000x	**. 00 00 01**
		. 00 00 00 01

C) MAIN BREAK TABLE--in and out as in panel B

Break	Root	Break	Root	Break	Root	Break	Root	Break	Root
98 01	100	**2 49 64**	160	**5 66**	240	**15 60**	40	**35 40**	60
1 02 01	102	**2 62 44**	164	**5 86**	246	**16 40**	41	**37 40**	62
1 06 09	104	**2 75 56**	168	**6 20**	252	**17 22**	42	**39 69**	64
1 10 25	106	**2 89 00**	172	**6 50**	258	**18 06**	43	**42 25**	66
1 14 49	108	**3 02 76**	176	**6 81**	264	**18 92**	44	**44 69**	68
1 18 81	110	**3 16 84**	180	**7 13**	270	**19 80**	45	**47 61**	70
1 23 21	112	**3 31 24**	184	**7 45**	276	**20 70**	46	**50 41**	72
1 27 69	114	**3 45 96**	188	**7 78**	282	**21 62**	47	**53 29**	74
1 32 25	116	**3 61 00**	192	**8 12**	288	**22 56**	48	**56 25**	76
1 36 89	118	**3 76 36**	196	**8 47**	294	**23 52**	49	**59 79**	78
1 41 61	120	**3 92 04**	200	**8 82**	30	**24 50**	50	**62 41**	80
1 48 84	124	**4 08 04**	204	**9 30**	31	**25 50**	51	**65 61**	82
1 58 76	128	**4 24 36**	208	**9 92**	32	**26 52**	52	**68 89**	84
1 69 00	132	**4 41 00**	212	**10 56**	33	**27 56**	53	**72 25**	86
1 79 56	136	**4 57 96**	216	**11 22**	34	**28 62**	54	**75 69**	88
1 90 44	140	**4 75 24**	220	**11 90**	35	**29 70**	55	**79 21**	90
2 01 64	144	**4 92 84**	224	**12 60**	36	**30 80**	56	**82 81**	92
2 13 16	148	**5 10 76**	228	**13 32**	37	**31 92**	57	**86 49**	94
2 25 00	152	**5 29 00**	232	**14 06**	38	**33 06**	58	**90 25**	96
2 37 16	156	**5 47 56**	236	**14 82**	39	**34 22**	59	**94 09**	98
2 49 64		**5 66 44**		**15 60**		**35 40**		**98 01**	

See text, page 75, for example of use.

exhibit **8** of chapter 3: reference table

Break table for (negative) reciprocals (using −1000/number)

A) BREAK TABLES for SETTING DECIMAL POINT

Break	Start	Start	**Break**
1000	.x	a.	**1000**
10,000	.0x	ab.	**100**
100,000	.00x	abc.	**10**
1,000,000	.000x	abcd.	**1.**
10,000,000	.0000x	abcde.	**0.1**
100,000,000			**0.01**

Examples

Number	**A**	**B**	−1000/number
124.2	a.	−80	−8.0
.04739	abcde.	−212	−212**.
1242.	.x	−80	−.80

B) MAIN BREAK TABLE--digits of negative reciprocal

Break	Value	Break	Value	Break	Value	Break	Value	Break	Value
990	−100	**1639**	−60	**2469**	−40	**4115**	−240	**617**	−160
1010	−98	**1681**	−59	**2532**	−39	**4202**	−236	**633**	−156
1030	−96	**1709**	−58	**2597**	−38	**4274**	−232	**649**	−152
1053	−94	**1739**	−57	**2667**	−37	**4347**	−228	**666**	−148
1075	−92	**1770**	−56	**2740**	−36	**4425**	−224	**685**	−144
1099	−90	**1802**	−55	**2816**	−35	**4504**	−220	**704**	−140
1124	−88	**1835**	−54	**2899**	−34	**4587**	−216	**725**	−136
1149	−86	**1869**	−53	**2985**	−33	**4672**	−212	**746**	−132
1176	−84	**1905**	−52	**3077**	−32	**4762**	−208	**769**	−128
1205	−82	**1942**	−51	**3175**	−31	**4854**	−204	**793**	−124
1235	−80	**1980**	−50	**3287**	−30	**4950**	−200	**820**	−120
1266	−78	**2020**	−49	**3367**	−294	**505**	−196	**840**	−118
1299	−76	**2062**	−48	**3448**	−288	**515**	−192	**855**	−116
1333	−74	**2105**	−47	**3509**	−282	**526**	−188	**870**	−114
1370	−72	**2151**	−46	**3584**	−276	**538**	−184	**885**	−112
1408	−70	**2198**	−45	**3663**	−270	**549**	−180	**901**	−110
1449	−68	**2247**	−44	**3745**	−264	**562**	−176	**917**	−108
1493	−66	**2299**	−43	**3831**	−258	**575**	−172	**935**	−106
1538	−64	**2353**	−42	**3922**	−252	**588**	−168	**952**	−104
1587	−62	**2410**	−41	**4016**	−246	**602**	−164	**971**	−102
1639		**2469**		**4115**		**617**		**990**	

See text, page 78, for example of use.

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Exploratory Data Analysis

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Dedicated to the memory of
CHARLIE WINSOR, biometrician, and
EDGAR ANDERSON, botanist,
data analysts both,
from whom the author learned much that
could not have been learned elsewhere

Preface

This book is based on an important principle:

It is important to understand what you CAN DO before you learn to measure how WELL you seem to have DONE it.

Learning first what you can do will help you to work more easily and effectively.

This book is about exploratory data analysis, about looking at data to see what it seems to say. It concentrates on simple arithmetic and easy-to-draw pictures. It regards whatever appearances we have recognized as partial descriptions, and tries to look beneath them for new insights. Its concern is with appearance, not with confirmation.

Examples, NOT case histories

The book does not exist to make the case that exploratory data analysis is useful. Rather it exists to expose its readers and users to a considerable variety of techniques for looking more effectively at one's data. The examples are not intended to be complete case histories. Rather they show isolated techniques in action on real data. The emphasis is on general techniques, rather than specific problems.

A basic problem about any body of data is to make it more easily and effectively handleable by minds--our minds, her mind, his mind. To this general end:

- anything that makes a simpler description possible makes the description more easily handleable.
- anything that looks below the previously described surface makes the description more effective.

So we shall always be glad (a) to simplify description and (b) to describe one layer deeper.

In particular:

- to be able to say that we looked one layer deeper, and found nothing, is a definite step forward--though not as far as to be able to say that we looked deeper and found thus-and-such.
- to be able to say that "if we change our point of view in the following way . . . things are simpler" is always a gain--though not quite as much as to be able to say "if we don't bother to change our point of view (some other) things are equally simple".

Thus, for example, we regard learning that log pressure is almost a straight line in the negative reciprocal of absolute temperature as a real gain, as compared to saying that pressure increases with temperature at an evergrowing rate. Equally, we regard being able to say that a batch of values is roughly symmetrically distributed on a log scale as much better than to say that the raw values have a very skew distribution.

In rating ease of description, after almost any reasonable change of point of view, as very important, we are essentially asserting a belief in quantitative knowledge--a belief that most of the key questions in our world sooner or later demand answers to "by how much?" rather than merely to "in which direction?".

Consistent with this view, we believe, is a clear demand that pictures based on exploration of data should *force* their messages upon us. Pictures that emphasize what we already know--"security blankets" to reassure us--are frequently not worth the space they take. Pictures that have to be gone over with a reading glass to see the main point are wasteful of time and inadequate of effect. **The greatest value of a picture** is when it *forces* us to notice **what we never expected to see.**

We shall not try to say why specific techniques are the ones to use. Besides pressures of space and time, there are specific reasons for this. Many of the techniques are less than ten years old in their present form--some will improve noticeably. And where a technique is very good, it is not at all certain that we yet know why it is.

We have tried to use consistent techniques wherever this seemed reasonable, and have not worried where it didn't. Apparent consistency speeds learning and remembering, but ought not to be allowed to outweigh noticeable differences in performance.

In summary, then, we:

- ◇ leave most interpretations of results to those who are experts in the subject-matter field involved.
- ◇ present techniques, not case histories.
- ◇ regard simple descriptions as good in themselves.
- ◇ feel free to ask for changes in point of view in order to gain such simplicity.
- ◇ demand impact from our pictures.
- ◇ regard every description (always incomplete!) as something to be lifted off and looked under (mainly by using residuals).
- ◇ regard consistency from one technique to another as desirable, not essential.

Confirmation

The principles and procedures of what we call confirmatory data analysis are both widely used and one of the great intellectual products of our century.

In their simplest form, these principles and procedures look at a sample--and at what that sample has told us about the population from which it came--and assess the precision with which our inference from sample to population is made. We can no longer get along without confirmatory data anlysis. **But we need not start with it.**

The best way to **understand what CAN be done is no longer**--if it ever was--**to ask what things could,** in the current state of our skill techniques, **be confirmed** (positively or negatively). Even more understanding is *lost* if we consider each thing we can do to data *only* in terms of some set of very restrictive assumptions under which that thing is best possible--assumptions we *know we CANNOT check in practice.*

Exploration AND confirmation

Once upon a time, statisticians only explored. Then they learned to confirm exactly--to confirm a few things exactly, each under very specific circumstances. As they emphasized exact confirmation, their techniques inevitably became less flexible. The connection of the most used techniques with past insights was weakened. Anything to which a confirmatory procedure was not explicitly attached was decried as "mere descriptive statistics", no matter how much we had learned from it.

Today, the flexibility of (approximate) confirmation by the jackknife makes it relatively easy to ask, for almost any clearly specified exploration, "How far is it confirmed?"

Today, exploratory and confirmatory can--and should--proceed side by side. This book, of course, considers only exploratory techniques, leaving confirmatory techniques to other accounts.

Relation to the preliminary edition

The preliminary edition of *Exploratory Data Analysis* appeared in three volumes, represented the results of teaching and modifying three earlier versions, and had limited circulation. Complete restructuring and revision was followed by further major changes after the use of the structure and much of the material in an American Statistical Association short course. The present volume contains:

- ◇ those techniques from the first preliminary volume that seemed to deserve careful attention.
- ◇ a selection of techniques from the second preliminary volume.
- ◇ a few techniques from the third preliminary volume.
- ◇ some techniques (especially in chapters 7, 8, and 17) that did not apppear in the preliminary edition at all.

It is to be hoped that the preliminary edition will reappear in microfiche form.

About the problems

The teacher needs to be careful about assigning problems. Not too many, please. They are likely to take longer than you think. The number supplied is to accommodate diversity of interest, not to keep everyone busy.

Besides the length of our problems, both teacher and student need to realize that many problems do not have a single "right answer". There can be many ways to approach a body of data. Not all are equally good. For some bodies of data this may be clear, but for others we may not be able to tell from a single body of data which approach is preferred. Even several bodies of data about very similar situations may not be enough to show which approach should be preferred. Accordingly, it will often be quite reasonable for different analysts to reach somewhat different analyses.

Yet more--to unlock the anlysis of a body of data, to find the good way or ways to approach it, may require a key, whose finding is a creative act. Not everyone can be expected to create the key to any one situation. And, to continue to paraphrase Barnum, no one can be expected to create a key to each situation he or she meets.

To learn about data anlysis, it is right that each of us try many things that do not work--that we tackle more problems than we make expert analyses of. We often learn less from an expertly done analysis than from one where, by not trying something, we missed--at least until we were told about it--an opportunity to learn more. Each teacher needs to recognize this in grading and commenting on problems.

Precision

The teacher who heeds these words and admits that there need be *no one correct approach* may, I regret to contemplate, still want whatever is done to be digit-perfect. (Under such a requirement, the writer should still be able to pass the course, but it is not clear whether he would get an "A".) One does, from time to time, have to produce digit-perfect, carefully checked results, but forgiving techniques that are not too distributed by unusual data are also, usually, *little disturbed by SMALL arithmetic errors*. The techniques we discuss here have been chosen to be forgiving. It is to be hoped, then, that small arithmetic errors will take little off the problem's grades, leaving severe penalties for larger errors, either of arithmetic or of concept.

Acknowledgments

It is a pleasure to acknowledge support, guidance, cooperation, and hard work. Both the Army Research Office (Durham), through a contract with Princeton University, and the Bell Telephone Laboratories have supported the writing financially. Charles P. Winsor taught the writer, during the 1940's, many things about data analysis that were not in the books. Although its formal beginning came after his death, this book owes much to S. S. Wilks, whose

leadership for statistics in Princeton made possible the gaining of the insights on which it is based.

Careful reading of earlier versions by friends and colleagues, especially David Hoaglin and Leonard Steinberg, was most helpful, as were comments by those who have taught the course at various institutions. As noted above, student reaction led to many changes. Frederick Mosteller took his editorial responsibilities very seriously; the reader owes him thanks for many improvements. The arithmetic is much more nearly correct because of the work of Ms. Agelia Mellros.

Careful and skilled typing, principally by Mrs. Mary E. Bittrich and by Mrs. Elizabeth LaJeunesse Dutka (earlier versions), with significant contributions by Mrs. Glennis Cohen and Mrs. Eileen Olshewski, has been of vital importance. The cooperative attitude and judgment of the Addison-Wesley staff, particularly that of Roger Drumm (without whose long-continued encouragement this book might not be a reality), Mary Cafarella (production editor), Marshall Henrichs (designer), and Richard Morton (illustrator), were of great help.

Princeton, New Jersey
Christmas, 1976

John W. Tukey

To the Student or Teacher

Everything illustrated or set as a problem can be done with pencil and paper. (If you have a hand-held calculator, fine.) The only tools the illustrator used (except for a few pictures that do not reflect the analysis of data) were a pen and a straightedge. Each of you could make pictures that are almost as nice if you tried moderately hard. (How to use graph paper effectively is discussed in Sections 2C and 5A.)

The first six chapters are the basic trunk from which everything grows. They belong in any version. What comes next?

It would be more conventional, and more useful for what people are used to doing, to go to chapters 10 and 11 on two-way plots, and, if time permitted, to their extensions in chapters 12 and 13.

I was, however, sufficiently impressed with the usefulness of smoothing (mainly from using it myself) to put chapter 7, and its applications to (x,y) batches in chapters 8 and 9, next. Both students-on-their-own and classes may wish to experiment by following chapters 1 to 6 with either 10, 11, etc., or 7, etc.

Chapter 14 is of central importance, but it is not easy. I would have liked to put it earlier, but could not find a good enough excuse. (It has to follow chapter 7.)

Chapter 15 deals with the important question of fractions based upon counts. It could follow either chapters 1 to 6, chapters 10 and 11 (etc.), chapter 7 (etc.), or chapter 14. Where it fits depends on the kinds of data a particular group of students are likely to be interested in.

Chapter 17, which is based on chapter 16, deals with making sense out of ordinary "distributions" of numerical values into bins. Chapters 16 and 17 could follow chapters 1 to 6 directly, but it seems unlikely that they would deserve such priority. Chapter 18 deals with a special kind of distributional situation that is neither very common nor all that rare. Many teachers and many students-on-their-own may wish to put it on the back burner.

Chapters 19 and 20 are based on Chapter 17. They help tie the techniques used in that chapter back to what may be more familiar ground. For those who have already studied such standard distributions, these chapters will surely earn their keep. For others, their priority is quite low. These chapters are not likely to help anyone analyze data better; they may help in thinking about the results of chapter 17 analyses.

In summary, then, the basic group of chapters (1 to 6) can be followed by any or all of: smoothing (chapter 7ff), two-way analyses (chapters 10 and 11 and ff), essential points about (x,y) batches (chapter 14), counted fractions (chapter 15), empirical distributions (chapters 16 and 17 and ff). After the basic group, chapters 7, 10–11, 14, and 15 are probably next most essential, unless chapter 17 (really chapter 16ff) is specially important to the particular group involved.

John Tukey

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1A. Quantitative detective work

Exploratory data analysis is detective work--numerical detective work--or counting detective work--or graphical detective work.

A detective investigating a crime needs both tools and understanding. If he has no fingerprint powder, he will fail to find fingerprints on most surfaces. If he does not understand where the criminal is likely to have put his fingers, he will not look in the right places. Equally, the analyst of data needs both tools and understanding. It is the purpose of this book to provide some of each.

Time will keep us from learning about many tools--we shall try to look at a few of the most general and powerful among the simple ones. **We do not guarantee to introduce you to the "best" tools, particularly since we are not sure that there can be unique bests.**

Understanding has different limitations. As many detective stories have made clear, one needs quite different sorts of detailed understanding to detect criminals in London's slums, in a remote Welsh village, among Parisian aristocrats, in the cattle-raising west, or in the Australian outback. We do not expect a Scotland Yard officer to do well trailing cattle thieves, or a Texas ranger to be effective in the heart of Birmingham. Equally, very different detailed understandings are needed if we are to be highly effective in dealing with data concerning earthquakes, data concerning techniques of chemical manufacturing, data concerning the sizes and profits of firms in a service industry, data concerning human hearing, data concerning suicide rates, data concerning population growth, data concerning fossil dinosaurs, data concerning the genetics of fruit flies, or data concerning the latest exploits in molecular biology. A full introduction to data analysis in any one of these fields--or in any of many others--would take much more time than we have.

The Scotland Yard detective, however, would be far from useless in the wild west or the outback. He has certain general understandings of conventional detective work that will help him anywhere.

In data analysis there are similar general understandings. We can hope to lead you to a few of them. We shall try.

The processes of criminal justice are clearly divided between the search for the evidence--in Anglo-Saxon lands the responsibility of the police and other

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★ = includes problem(s)

Comments about the Index Page

The index page for this chapter--and for others--is *nonstandard.*

Two sorts of additional entries are given between the section titles:

◇ subtopics, printed in lightface type in the index, and centered in text.

◇ vital terms defined, printed in italics in the index when short, and displayed like formulas in text.

To the right of the index page we give pages for numbered "exhibits". This neutral word is used for tables, charts, graphs, plots, and the like. They are all numbered in a single sequence within each chapter, so they will be easy to find.

Their first main appearance will be identified by using bold numbers, as **14**, for *that* appearance and not for others.

investigative forces--and the evaluation of the evidence's strength--a matter for juries and judges. In data analysis a similar distinction is helpful. Exploratory data analysis is detective in character. Confirmatory data analysis is judicial or quasi-judicial in character. Only exploratory data analysis will be our subject here.

Unless the detective finds the clues, judge or jury has nothing to consider. **Unless exploratory data analysis uncovers indications, usually quantitative ones, there is likely to be nothing for confirmatory data analysis to consider.**

Experiments and certain planned inquiries provide some exceptions and partial exceptions to this rule. They do this because one line of data analysis was planned as part of the experiment or inquiry. **Even here, however, restricting one's self to the planned analysis--failing to accompany it with exploration--loses sight of the most interesting results too frequently to be comfortable.**

As all detective stories remind us, many of the circumstances surrounding a crime are accidental or misleading. Equally, many of the indications to be discerned in bodies of data are accidental or misleading. To accept all appearances as conclusive would be destructively foolish, either in crime detection or in data analysis. **To fail to collect all appearances because some--or even most--are only accidents would, however, be gross misfeasance deserving (and often receiving) appropriate punishment.**

Exploratory data analysis can never be the whole story, but nothing else can serve as the foundation stone--as the first step.

We will be exploring numbers. We need to handle them easily and look at them effectively. Techniques for handling and looking--whether graphical, arithmetic, or intermediate--will be important. The simpler we can make these techniques, the better--so long as they work, and work well. When details make an important difference, they deserve--and will get--emphasis.

review questions

What is exploratory data analysis? How is it related to confirmatory data analysis? How is preplanned analysis related to exploratory data analysis? Should we look only at appearances we are sure are correct?

1B. Practical arithmetic

A few arithmetic details deserve our attention before we go further.

precision of arithmetic

We shall feel free to be inconsistent about keeping extra decimal places or rounding off. We shall try to use reasonable care and good judgment in view of

the total behavior of the data involved. Thus, for example, we may do either of these:

$(1/3)(19 + 19 + 20) = 19.3$ (rest of data close by)
$(1/3)(19 + 19 + 20) = 19$ (rest of data farther away)

rounding

When we do round, it will be to the nearest number, with ties broken by preferring even numbers. Thus:

$17.5 \rightarrow 18$
$18.5 \rightarrow 18$
$19.5 \rightarrow 20$
$20.5 \rightarrow 20$

As a result, we may have

$(1/2)(19 + 20) = 19.5$ (rest of data close by)
$(1/2)(19 + 20) = 20$ (rest of data farther away)
$(1/2)(20 + 21) = 20.5$ (rest of data close by)
$(1/2)(20 + 21) = 20$ (rest of data farther away)

cutting

We often prefer to cut numbers by just dropping off the extra digits. This is somewhat quicker and easier, especially if one is not used to rounding.

It is hard to say just what is the proper balance between cutting and rounding. If more detail is easily available (say on the same page or an adjacent page), then surely one may as well cut. If the data is a matter of deep personal concern, then it is almost certain that one will round, if only for conscience's sake. We shall do both. We recommend to the reader that he or she practice both.

units and decimal points

Numbers can be written down for storage--live or dead--or examination. We will almost always write them down to be looked at, either as part of the description of a computation or as part of results--final or intermediate.

When writing down a single number, keeping both raw units and decimal points often helps. To say that there are 200,000,000 people in the U.S., or that a sheet of plastic film is 0.00127 inches thick, is likely to convey about as much as we can.

Once we have a few numbers to look at, it usually helps us to move to easily understandable units that are more helpfully related to the numbers at hand. A population given as 201 millions--or 201.2--is likely to be seen more clearly than if it is given as 201,234,567. To give the thickness of a plastic

film as 1.27 thousandths of an inch, instead of 0.00127 inches, is equally likely to be helpful.

Once we have more than a few numbers to look at, decimal points are a liability. They break up the appearance of the numbers, making it harder to make comparisons, especially among more than two numbers. The cure is to shift to units that are as convenient for the numbers at hand as possible. Instead of 201.2, 127.3, 63.4, . . . millions of people, we do better to work with 2012, 1273, 634, . . . hundreds of thousands. Instead of thicknesses of 1.27, 2.52, 0.62, 3.83, . . . thousandths of an inch, we do better to work with 127, 252, 62, 384, . . . hundredths of thousandths of an inch.

To be clear about what we are doing, it is often convenient to label parts (panels) of a table with the units involved. Thus, such headings as:

A) POPULATION — in hundreds of thousands

A) POPULATION — unit = 100,000 people

A) POPULATION — unit = 10^5 people

B) FILM THICKNESS — unit = 0.00001 inch

B) FILM THICKNESS — unit = 10^{-5} inch

will occur fairly often

h for "and a half"

Even if the data is in whole numbers (integers), the numbers we calculate from it are often fractions. We try to keep our arithmetic simple. As a result, quite a few of these fractions are halves.

We would like to avoid 4.5, 13.5, and other distracting appearances of ".5". While not yet used extensively, the use of an "h", without space or decimal point, to stand for ".5" or "and a half" is very helpful. We will often use it (especially in later chapters).

When we want to count by halves, then, we may go 4, 4h, 5, 5h, 6, instead of 4, 4.5, 5, 5.5, 6, or 4.0, 4.5, 5.0, 5.5, 6.0.

"*" for place filler

Digits in numbers often serve only as place-holders. A population reported as 201234567 is not likely to be accurately known to so many figures (though the figures given may accurately represent the number of people found and counted). We often gain by having a simple notation for a place filler. Here we shall use "*", always cutting the value. Thus 20123**** stands for a number between 201230000 and 201239999 that we do not choose to give as 20123(ent) tens of thousands.

"#" for count

We shall often use "#" for "count" or "number" wherever its use seems helpful.

review questions

Do we have to be consistent about keeping extra decimal places? How does rounding differ from cutting? How do we round halves? Why is a good choice of working units important? What is the effect of decimal points when handling numbers in bulk? What is meant by "137h"? By "137*"? By "#"?

1C. Scratching down numbers

Probably the simplest task that the reader is likely not to be either equipped or accustomed to do is to write down a batch of numbers in such a way as to give a general feel of "what they are like". Here, by a

batch

of numbers we mean a set of similar values, obtained however they may have been. Simple examples might be (i) the weights of 21 students in a freshman class, (ii) the total winter snowfall at a selected place for each of the last 10 winters, (iii) the total sales this year of the 14 life-insurance agents who sold the most insurance last year, (iv) the number of electricity failures during the last decade in each of 11 high-income suburbs of New York City, (v) the numbers of mites found on each of 49 rats.

In each of these cases we are likely to desire a feel for the set of values as a whole. We may also want to copy them down, from whatever source we may have found them in, in a form that will be easy to use for almost any purpose. If one technique will serve both ends, so much the better. If two versions of the same technique are needed, we will use them.

Let us begin with the used-car advertisements in the *Sunday Standard-Times* of New Bedford, Massachusetts for 18 August 1968. Some 18 ads list three or more cars. The numbers listed are, in order: 6, 7, 7, 3, 5, 7, 3, 11, 16, 4, 17, 17, 6, 7, 9, 4, 7, 5. Just writing them down gives some idea of how many cars to an ad. Surely, however, we can do better. Exhibit **1** shows a way to write these numbers down that gives us more of a picture of them as a whole. In panel A each number is indicated by an "X". In panel B we gain a little by substituting the last digit for the X's, thus giving us something more to grasp with the flying eye, and providing a partial check on whether we put each mark in the correct line.

bold figures

In the first exhibit of the first chapter we have already begun to use bold-face figures to distinguish one kind of entry from another. What is the user of pencil and paper supposed to do? Surely, if boldface helps in communication from writer to reader, something ought to be used to help in communication from scratcher-down to looker (who may be the same person at slightly different times).

Whenever we can do it, using **two colors** (or more than two colors in more complex situations) is likely to be the clearest method. (If the four-color BIC

ballpoint is still available at a reasonable price, it provides an easy way to carry more than one color.) **Otherwise, the use of pen for bold and pencil for not bold works fairly well.**

At the last extreme, circling the numbers that would be bold is useful, though both more work and somewhat less effective.

review questions

What is a batch? What is an example of a set of numbers that is not a batch? Why do we use bold figures? What choices have we in making a similar effect with pencil or pen? What are two ways of tallying numbers?

1D. Doing better with stem-and-leaf

The technique we have just been learning works well in some cases, but it is easy to find cases where it is not good enough. If we turn to the prices of second-hand Chevrolets (including Chevelles and Impalas) in the same 18 ads, we find the following (in dollars): 250, 150, 795, 895, 695, 1699, 1499, 1099, 1693, 1166, 688, 1333, 895, 1775, 895, 1895, 795. Running the eye over this string of numbers doesn't help too much. Neither would a direct application of the technique of the last section, since going from 150 to 1895 in single-dollar steps takes 1746 lines, of which only 17 would be marked.

exhibit **1** of chapter 1: cars/advertisement

Tallying numbers of cars per advertisement in such a way as to give some feeling for the 18 numbers AS A WHOLE

A) CONVENTIONAL

3	××
4	××
5	××
6	××
7	×××××
8	
9	×
10	
11	×
12	
13	
14	
15	
16	×
17	××

B) DIGIT-REMINDER

3	33
4	44
5	55
6	66
7	77777
8	
9	9
10	
11	1
12	
13	
14	
15	
16	6
17	77

C) PROBLEMS

1a) Collect a batch of data of interest to you.

1b) Display this data as in panel A.

1c) Do the same as in panel B.

We could, of course, drop off the last two digits, and keep only the complete hundreds of dollars. Cutting the numbers in this way would give: 2, 1, 7, 8, 6, 16, 14, 10, 16, 11, 6, 13, 8, 17, 8, 18, 7, which could easily be treated as in exhibit 1. We shall set this possibility aside for the moment, planning to come back to it shortly.

Let us return to panel B of exhibit 1. Exhibit 2 sets out the data when each line corresponds to two adjacent numbers of cars per ad--and when each corresponds to three adjacent numbers. Panels C and D of exhibit 2 show what happens when we condense 10 possible values on each line.

The panel-D version is quite efficient. One has even less to write down than if one were to write the given values down as is. Thus, 1*|1677, for example, uses seven characters to represent 11, 16, 17, 17. The as-is form requires eight characters--or twelve if we count punctuation--rather than five--or seven.

The central idea of panel D is to give part of the information once for all at the start of each line and then give the rest of the information in the line, packed together as closely as makes sense. We shall call such displays

stem-and-leaf displays

Each line is a **stem**, each piece of information on a stem is a **leaf**. When, as

exhibit 2 of chapter 1: cars/advertisement

The data and notation of panel B of exhibit 1 applied when each line includes 2, 3, or 10 possibilities

(Note: Entries are final digits of observed numbers.)

A) 2 per LINE

2–3	33
4–5	4455
6–7	6677777
8–9	9
10–11	1
12–13	
14–15	
16–17	677

B) 3 per LINE

3–5	334455
6–8	6677777
9–11	91
12–14	
15–17	677

C) 10 per LINE

0–9	33445566777779
10–19	1677

D) as C, ABBREVIATED--to be called SIMPLE STEM-and-LEAF

0*	33445566777779
1*	1677

E) PROBLEMS

2a) Display the data collected for (1a) as a simple stem-and-leaf display.

2b) Collect a batch of size at least 50, doing the collection in the form of a stem-and-leaf display.

here, the label for the stem is the first part of a number, to be followed by each leaf in turn, we call the label the **starting part**.

We like to use one-digit leaves, but--as we will see shortly--we can find the use of two- or more-digit leaves a good thing.

Any stem-and-leaf display is in terms of some unit. For our used-car prices, for example, \$10 seems a convenient unit. Thus \$250 becomes 25 and appears as **2*** | 5.

What shall we do with \$1099? We could cut to 109 or round to 110. Which should we do? This was discussed in the introduction to this chapter, but left open, as it will be here.

exhibit **3** of chapter 1: Chevrolet prices

Three displays of 17 Chevrolet prices

A) The PRICES--in dollars
250, 150, 795, 895, 695, 1699, 1499, 1099, 1693, 1166, 688, 1333, 895, 1775, 895, 1895, 795.

B) The PRICES--cut to \$10 units
25, 15, 79, 89, 69, 169, 149, 109, 169, 116, 68, 133, 89, 177, 89, 189, 79.

C) UNIT = \$100[†]
CHECK-OFF

1	×
2	×
3	
4	
5	
6	××
7	××
8	×××
9	
10	×
11	×
12	
13	×
14	×
15	
16	××
17	×
18	×

D) UNIT = \$10[†]
STEM-and-LEAF
one-digit leaf

1*	5
2	5
3	
4	
5*	
6	98
7	99
8	999
9*	
10	9
11	6
12	
13*	3
14	9
15	
16	99
17*	7
18	9

E) UNIT = \$1
STEM-and-LEAF
two-digit leaf

1**	50
2	50
3	
4	
5**	
6	95, 88
7	95, 95
8	95, 95, 95
9**	
10	99
11	66
12	
13**	33
14	99
15	
16	99, 93
17**	75
18	95

[†] **Values cut, not rounded.**

P) PROBLEM

3a) Explain, in words, the entries after each stem = 16.

Panel D of exhibit **3** shows the result of cutting our Chevrolet prices to a $10 unit and giving the results in standard stem-and-leaf form, namely with:

- ◇all but the last digit remaining as the starting part, labelling the stem,
- ◇the last remaining digit entered on that line as the leaf,
- ◇enough starting parts followed by * to remind us, without unnecessary confusion, how many digits (here, one) are left for the leaf.

This display also shows the prices in $100 units à la panel A of exhibit 1, and the full prices in a different stem-and-leaf display with two-digit leaves. Clearly:

- ◇each of the three panels conveys a quite similar general picture of the 17 prices as a whole.
- ◇the amount of detailed information increases as we go from panel C to panel D to panel E.
- ◇the cost of this increase in information is negligible in going from panel C to panel D, but may not be negligible in going from panel D to panel E.

As a result, we see that we will do well to use the style of either panel D or panel E--or, often, both together--in such a situation as this.

checking

One thing we regretfully learn about work with numbers is the need for checking. Late-caught errors make for painful repetition of steps we thought finished.

Checking is inevitable; yet, if it is too extensive, we spend all our time getting the errors out of the checks. Our need is for enough checks but not too many.

In scratching down data in a stem-and-leaf, we ought to make at least one check, to ask one question: Have we made as many entries as there were values to enter? If there are but few numbers in all, we can point at them in turn and count them up without writing anything down. For the present, though, in order to remind us of the need for checking, we will carry a "(check count)" column in parentheses. (An example will appear in the next exhibit.) To keep the calculation of such check columns from being an effort, we need to learn how to count the number of leaves on a stem rapidly and easily. Consider the 19 leaves that will appear on one stem of the next exhibit, namely

0121243122301214202

The recommended procedure is the following:

- ◇place the left hand near the paper, with the forefinger pointing to the right.

◇cover blocks of three digits at a time counting "3' , "6", "9", . . . to one's self as this is done.

In this example, we would have, successively--the black figures are visible, the light ones concealed:

01212431223012142O2	"0"
01212431223012142O2	"3"
01212431223012142O2	"6"
01212431223012142O2	"9"
01212431223012142O2	"12"
01212431223012142O2	"15"
01212431223012142O2	"18"

Clearly, "18" and "1" make 19.

If we find an error in the total count, or wish to check more carefully, we need to see whether we have the right leaves on the right stems. The easy way to do this is to go through the given values again, placing a dot over each leaf that is (either originally or after adjustment) correct. This process goes rapidly and seems quite effective.

review questions

How can we enter different numbers in the same line of a tally? What is a stem-and-leaf display? What is a stem? A leaf? A starting part? How many digits in a leaf? How do we check the making of a stem-and-leaf display overall? In detail?

1E. Using the right number of stems

Sometimes splitting stem from leaf between adjacent digits does not seem satisfactory. One choice may involve so few stems that the values are too crowded for us to see them at all well, yet moving one more digit into the starting part makes so many stems that we are bothered by being too spread out. In such cases we may need to abandon the simple stem-and-leaf for a modification.

stretched stem-and-leaf

One way out is to use two stems--two lines--for each starting part. In such a display, we use one line for leaves 0, 1, 2, 3, 4 and the other for leaves 5, 6, 7, 8, 9. The starting parts are repeated, with the * shown only for those that include leaves of 0 and 1. (Dots can be used to fill in.)

Exhibit **4** shows an example using the land areas of the 82 counties of Mississippi. In this instance, using the conventional stem-and-leaf with one-digit starting parts would have not worked badly (as the reader may easily verify). However, the two-line version gives us a better picture of how county areas behave in Mississippi. Note that a few counties have been identified. This both heightens realism and stirs the imagination.

squeezed stem-and-leaf

Five stems--instead of one, two, or ten--can also be a good choice. Here we can use a scheme for tagging lines that makes it easy to avoid confusion, namely * for "zero" and "one", t for "two" and "three", f for "four" and "five", s for "six" and "seven", · for "eight" and "nine". (English digit names are surprisingly well adapted to a five-stem pattern.) Exhibit **5** shows a sample.

using mixed leaves

Sometimes we use stem-and-leaf more as a way to collect numbers than as a way to look carefully at them. When we are doing this, we may be able--or

exhibit **4** of chapter 1: county areas

The areas of the counties of Mississippi in a two-line stem-and-leaf display (stretched stem-and-leaf)

A) SAMPLE DATA--in square miles

448, 405, 729, 724, 412, 917, 592, . . .

B) Same ROUNDED--to 10's of square miles--(5's rounded to evens)

45, 40, 73, 72, 41, 92, 59, . . .

C) The DISPLAY--in 10's of square miles

			(#)
3·	8	Tate	(1)
4*	0121243121300214202		(19)
4·	597886556569		(12)
5*	142010		(6)
5·	977899958797		(12)
6*	412441		(6)
6·	898598		(6)
7*	320341203		(9)
7·	86657		(5)
8*	303		(3)
8·	8	Hinds	(1)
9*	24	Bolivar, Yazoo	(2)
			(82, √)

Doubtful roundings (originally ending in 5) are made to nearest even number.

P) PROBLEM

4a) Find another of data deserving a stretched stem-and-leaf display. Display it.

S) SOURCE

The World Almanac, 1966, page 370.

we may need--to do something special with values that trail out away from the bulk of the batch, as illustrated in exhibit **6**, including:

◇ changing the division between starting part and leaf as seems convenient. (Here, and often, at multiples of ten; sometimes at other points.)

◇ being careful to show this change **both** by a change in the number of *'s **and** by a blank line.

◇ giving the batch in two forms, one for the quick look and one for storage for future use. (Sometimes we shall omit one or the other.)

Notice carefully, for example, what has happened to the second value in the original list. In the form for looking at, 1345 has become **1*** | 3, namely

exhibit **5** of chapter 1: later example

A five-line stem-and-leaf display (squeezed stem-and-leaf)

A) FIVE-LINE VERSION

		(#)
1*	1	(1)
t	2333	(4)
f	445555	(6)
s	66677	(5)
·	88	(2)
2*	0000011	(7)
t	23	(2)
f	445	(3)
s	6	(1)
·	9	(1)
3*	1	(1)
t	3	(1)
f		
s		
·		(34√)

B) The SIMPLE VERSION

		(#)
11	8	(1)
12	0	(1)
13	488	(3)
14	08	(2)
15	1266	(4)
16	058	(3)
17	08	(2)
18	58	(2)
19		(·)
20	03688	(5)
21	38	(2)
22	1	(1)
23	5	(1)
24	05	(2)
25	8	(1)
26	3	(1)
27		·
28		·
29	5	(1)
30		·
31	2	(1)
32		·
33	0	(1)
34		·
		(34, √)

P) PROBLEM

5a) Find another batch deserving a squeezed stem-and-leaf display. Display it.

S) SOURCE

Example in later chapter, related to exhibit 6, following.

13**, something between 1300 and 1399. In the form for storage, 1345 has become **1*** | 345**, namely 1345.

If we want to save detail for later use, there is no substitute for the "for storage" form. If we want to look at the pattern, there is no substitute for the

exhibit **6** of chapter 1: ultimate powers

Ultimate power of hydroelectric plants of the Bureau of Reclamation (in megawatts)

A) SAMPLE VALUES

30, 1345, 225, 900, 120, 162, . . .

B) TWO FORMS of STEM-and-LEAF DISPLAY

(for looking at)		(#)	(for storage)	
1*	65	(2)	**1***	6, 5
2	5244	(4)	**2**	5, 2, 4, 4
3	03626	(5)	**3**	0, 3, 6, 2, 6
4	508	(3)	**4**	5, 0, 8
5*	0	(1)	**5***	0
6	0	(1)	**6**	0
7	51	(2)	**7**	5, 1
8			**8**	
9*			**9***	
1**	26350210	(8)	**1****	20, 62, 34, 50, 00, 20, 14, 08
2	285	(3)	**2**	25, 85, 50
3	7	(1)	**3**	79
4	2	(1)	**4**	24
5**			**5****	
6			**6**	
7			**7**	
8			**8**	
9**	0	(1)	**9****	00
1***	39	(2)	**1*****	345, 974
2			**2**	
3			**3**	
		(34√)		

C) NOTES

Source: The World Almanac, 1966, p. 263 (1967, p. 267).

Their source: U.S. Bureau of Reclamation.

Identification: If only one form had been used, leaving more space, some of the large and small plants would have been identified on the display, including Boysen 15, Elephant Butte 16, . . . , San Luis 424 Glenn Canyon 900, Hoover 1345, Grand Coulee 1974.

P) PROBLEM

6a) Find another set of data worthy of being shown in both of these ways together. Show it.

"for looking at" form. (In special circumstances, we may want to use compromise forms.)

A useful approach is illustrated by the righthand side of panel B in exhibit **6**. Notice the use of a comma between each pair of leaves to make clear the leaf length. If we want to look harder at such a display, we can revert to single-digit leaves, as on the left of panel B.

This type of stem-and-leaf is very effective in accumulating information about values spread from many quite small to a few quite large. Often, as in this example, the numbers tend to pile up on--and just after--each "1"--an example of what is sometimes called the "abnormal law of large numbers".

exhibit **7** of chapter 1: arbitrary example

Mixed-leaf stem-and-leaf applied to 25 values, some trailing

A) THE VALUES

5, −52, −27, −83, 8, −14, −122, −110, 112, 58, −119, 33, 18, −52, −19, 12, −82, 14, 25, −182, −40, 64, −56, 5, 13.

B) The STEM-and-LEAF for STORAGE

		(#)
1**	12,	(1)
0**	64,	(1)
5*	8	(1)
4		
3	3	(1)
2	5	(1)
1*	2, 4, 3, 8	(4)
0	5, 8, 5	(3)
−0		
−1*	4, 9	(2)
−2	7	(1)
−3		
−4	0,	(1)
−5*	2, 6, 2	(3)
−0**	83, 82	(2)
−1**	22, 10, 19, 82	(4)
		(25, √)

P) PROBLEMS

7a) Make a "for looking at" stem-and-leaf of the same data.

7b) Make a stem-and-leaf of the same data using only one size of leaves.

7c) Criticize the display mode in (7b). How could it be improved?

7d) Improve it.

When this is so, we see a triangular appearance for each "1" to "9"--as if the leaves had washed against the blank lines--an appearance which covers up most of whatever interesting things we might see. (We will learn how to deal with this before long.)

It often makes things even simpler, however, to feel free to jump to longer leaves in the middle of a digit-length, as illustrated in exhibit **7**, where one-digit leaves are used from −59 to +59, and two-digit leaves begin at −60 and +60. This enables us to use 16 lines where 31 (+11 to +0 and −0 to −18) would have been required if all starting parts had used the same number of *'s.

Notice in this exhibit:

◇the use of stem-and-leaf for both positive and negative values--and the consequent need for both a "+0" stem and a "−0" stem.

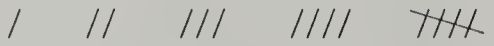
review questions

How many stems is it natural to use for one starting part? How do we choose? What can we gain? Do we need to be consistent in our starting parts within one display? What is at stake? What form of stem-and-leaf is intended for storage? What form for looking-at? How free can our choices be? How many "0" starting parts can there be?

1F. How to count by tallying

The basis of stem-and-leaf technique, entering an additional digit--or digits--to mark each value, works well for batches of limited size. Once we have much more than 20 leaves on a stem, however, we are likely to feel cramped--and our stems begin to be hard to count. We ought to be able to escape to some other way of handling such information, whenever the other way gives us enough detail.

The fast methods involve one pencil (or pen) stroke per item. One method counts by fives in this style:



This has been widely used. The writer finds it treacherous, especially when he tries to go fast. (It is too easy for him to do

for this approach to give satisfactory performance.)

The recommended scheme uses first dots, then box lines, then crossed lines to make a final character for 10. Thus:

4 is ⁘

8 is ⊡

10 is ⊠

The order in which the four dots are placed around the square is unimportant, as is the order in which the four sides of the square are filled in, and the order in which the two diagonals are crossed.

Once anyone is used to this scheme, the occurrence of errors is much less than for the other approach. The closure of the four dots completing the outline of the square is clear and definite; there is no tendency to go on to lines after three points--or to add a fifth point. The same applies to the completion of the sides of the square. (A few novices report a tendency to move on to the next character before crossing the diagonals. Their eights--☐--are, however, distinctive enough from tens--☒--to be picked up once the tallying has been completed and the result is being converted into digits.

Exhibit **8** shows a variety of simple examples.

exhibit **8** of chapter 1: arbitrary examples

Tallying by tens variously illustrated

A) TALLY — COUNT

.	1
: or . .	2
⁘ or ∴ (etc.)	3
::	4
⁙ or ⁞ (etc.)	5
⌜⌝ or ⌞⌟ (etc.)	6
⊓ or ⊏ (etc.)	7
☐	8
⧅ or ⧄	9
☒	10
☒ ☒ ⊓	27
☒ ☒ ☒ ☒ ⁖	42

B) A SIMPLE EXAMPLE

0	☒ ☒ ☐
1	☒ ☒ :
2	☒ ⧅
3	☒ ⌜⌝
4	☒ .
5	☒ ☒ ⁘
6	⧅
7	⊓
8	⁞
9	⁘
10	.

C) A TWO-WAY EXAMPLE

	Freshmen	Sophs	Juniors	Seniors
Boys	☒ ☒ ☒ ☒ ⁘	☒ ☒ :	☒ ⌜⌝	☒ .
Girls	☒ ☒ ⌜⌝	☒ ☒ ☒ ::	☒ ⁘	☒ ☒ :

P) PROBLEM

8a) Collect a batch of size at least 100 by tallying by tens, and display as in B and A.

used cars again

For exhibit **9** we return to the *Sunday Standard-Times* and its used-car ads. This display gives--in both tallied and digit form--the number of cars of each model year advertised:

- ◇in ads for one or two cars
- ◇in ads for three or more cars.

We can get a quick grasp of appearances easier and faster from the tallies-by-ten than from the same counts expressed as numbers. (It may even be worthwhile, on occasion, to turn digit-given counts into tallies-by-ten to look at them.)

The general impressions easily gained from this display include:

- ◇a tendency for small ads (one or two cars) to offer somewhat older cars.
- ◇a question as to why '55 appears to be popular.

exhibit **9** of chapter 1: auto years

Model years of used cars advertised (comparison of numbers with tallies-by-tens)

A) ADS for 1 or 2 CARS

Number	Tally
1	'54 | .
2	'55 | . .
—	'56 |
1	'57 | .
2	'58 | . .
5	'59 | ⊓
5	'60 | ⊓ :
2	'61 | : .
11	'62 | ⊠ .
12	'63 | ⊠ : .
10	'64 | ⊠
14	'65 | ⊠ : :
12	'66 | ⊠ : .
13	'67 | ⊠ : .
3	'68 | : .

B) ADS for 3 or MORE CARS

Number	Tally
—	'54 |
1	'55 | .
—	'56 |
1	'57 | .
1	'58 | .
1	'59 | .
4	'60 | : :
10	'61 | ⊠
14	'62 | ⊠ : :
17	'63 | ⊠ ⊓
25	'64 | ⊠ ⊠ ⊓ :
34	'65 | ⊠ ⊠ ⊠ : :
27	'66 | ⊠ ⊠ ⊓
12	'67 | ⊠ : .
4	'68 | : :

P) PROBLEM

9a) Collect two related batches of at least 80 values each, and display as above.

S) SOURCE

Sunday Standard-Times, New Bedford, Mass. for 18 August 1968 (page 51, Fourth Section).

Each of us can provide or invent reasons for the first of these. As for the second, reexamination of the ads shows that two of the three 55's are "classic Thunderbirds".

While tally-by-tens is an effective technique, it does not tell us as much as stem-and-leaf. It should be almost exclusively used as an overflow procedure. Indeed, overflow need not occur as often as we might think, for it is often very helpful to accumulate information in parts--parts which we are then free to combine or keep apart. Exhibit 9 shows collection in two parts--had we gone further and collected in three or four parts, it would have been reasonable--so far as size of counts go--to collect each part using stem-and-leaf. If there were further information to collect, it would be not only reasonable but desirable to do this.

review questions

What trouble does tallying by fives get into? What are the steps in tallying by tens? Can it pay to turn numerical counts into tallies by 10's? Why? Can it pay to collect tallied information in parts? What do we gain? What tends to shift our choice between stem-and-leaf and tallying-by-tens?

1G. What does it mean to "feel what the data are like"?

We have been looking at well-behaved batches of numbers. True, there was something to pick up about 1955 used cars, but generally we saw little that was striking in any of our stem-and-leaf displays.

Exhibit **10** shows the height of the highest point in each of the 50 states. Clearly the states fall into three groups, which turn out to be:

- ◇Alaska.
- ◇the Rocky Mountain States, California to Washington, and Hawaii.
- ◇all others.

The things that are frequently thrust upon us by stem-and-leaf displays are:

- ◇appearances of separation into groups.
- ◇unsymmetric trailing off, going farther in one direction than another.
- ◇appearance of unexpectedly popular (or unpopular) values.
- ◇about where the values are "centered".
- ◇about how widely the values spread.

The stem-and-leaf itself guides us to the first three. A little help with the last two may be useful, as we shall shortly see.

Notice that, where apparent "breaks" occur, it may be useful to identify values near the breaks as well as at the ends.

review questions

What do stem-and-leaf displays frequently thrust upon our attention? Would tallyings-by-ten do the same?

1H. How far have we come?

This first chapter has been devoted to (i) a discussion of what exploratory data analysis is all about, and (ii) an introduction to the simple tools that allow us to scratch down information about batches of numbers in ways that:

exhibit **10** of chapter 1: state heights

The heights of the highest points in each state

A) STEM-and-LEAF--unit 100 feet

			(#)
0*	43588	Del, Fla, La, Miss, RI	(5)
1	237886		(6)
2	484030		(6)
3	45526		(5)
4*	80149		(5)
5	34307		(5)
6	376		(3)
7	2	S. Dak	(1)
8*	8	Texas	(1)
9			
10			
11	2	Oregon	(1)
12*	768		(3)
13	81258		(5)
14	544	Calif, Colo, Wash	(3)
15			
16*			
17			
18			
19			
20*	3	Alaska	(1)
			(50, √)

P) PROBLEMS

10a) Make a similar display for the lowest point in each of the 50 states.

10b) Make the comments suggested by the results of (10a).

S) SOURCE

The World Almanac, 1966, page 269.

Their source: National Geographic Society.

◇give us a feeling of how each batch seems to behave as a whole

◇or help to collect and store the values in a form that is convenient for later use.

Sometimes we use different forms for storage and for insight.

what have we learned to do ?

Exploratory data analysis is detective work--in the purest sense--finding and revealing the clues.

Confirmatory data analysis, which we do not discuss in this book, goes further, assessing the strengths of the evidence.

The techniques for dealing with batches of numbers we have met are all closely related. All of them display the numbers before our eyes by putting down the most informative mark that we have thought of and that will be effective. (Too much detail or too much sprawl can detract from what we see.) Each technique deserves to be checked at least as far as the total count. (Overdotting makes checking of stem-and-leaf displays easy.)

When we make a stem-and-leaf display, we have a choice among one-digit leaves, two-digit leaves, etc., and mixed leaves. We make this partly for convenience, partly in view of our purposes, partly for necessity. Sometimes we make two different styles at the same time.

When we have many values to collect or display, and can give up further details, tallying-by-tens offers us real advantages.

Among the more important things that detailed displays of batches can show us are separation, asymmetry, irregularities, centering, and width.

where do we stand ?

As this chapter has begun to illustrate, one thing the data analyst has to learn is how to expose himself to what his data are willing--or even anxious--to tell him. Finding clues requires looking--in the right places and with the right magnifying glass.

Most of us know something about looking at a single number. It may not be easy to draw all the implications from the fact that there were 46,930 deaths from motor vehicle accidents in the U.S.A. in 1964, but the difficulty is not in understanding the one number.

The same is often true of comparing a few firmly based numbers, like these 1962–64 average death rates per hundred million person miles:

◇trains	0.09
◇planes	0.16
◇buses	0.17
◇cars (turnpike)	1.2
◇cars (general)	2.3

exhibit **11** of chapter 1: auto makes

Collection of model-year and name information for advertised used cars

A) The NAME CODE

Bonneville	b
Buick	B
Cadillac	A
Chevelle	h
Chevy II	2
Chevrolet	C
Chrysler	H
Classic	1
Comet	3
Corvair	4
Dodge	D
Eldorado	E
Ford	F
Futura	f
Galaxie	g
Grand Prix	G
Impala	I
Jaguar	J
Lincoln	L
Mercury	M
Mustang	m
Oldsmobile	O
Plymouth	p
Pontiac	P
Rambler	r
Simca	5
Studebaker	s
Sunbeam	S
Tempest	t
T-bird	T
Valiant	V
Volkswagon	v

B) The COLLECTION

'55	T
'56	
'57	E
'58	C
'59	I
'60	AlAv
'61	ClHvFFFHT
'62	4ClP2O3CTIIP2
'63	IIFrBFrCrLOICCbP
'64	Pg1BDD4CIIrBI3PshCAOCM
'65	5H4C4OCpDCFCCCbBICSCPOIDCCBACC2CCCr
'66	ftvh1Cv3B4hFlvACOhmPJFFCp3
'67	VPPFGmvPmCAG
'68	HpDp

C) The COLLECTION ORGANIZED

·55	T
'56	
'57	E
'58	C
'59	I
'60	AAlv
'61	CFFFHHltTv
'62	CCIIIOPPT2234
'63	bBCCCFFIIILOPrrr
'64	ABBCCCDDghIIIMOPPrs134
'65	CCCCCCCCCCCCCCC
'65	AbBBDDFHIIOOpPrS245
'66	ABCCCfFFFhhhIJmOpPtvvv1334
'67	ACFGGmmPPPvV
'68	DHpp

P) PROBLEM

11a) Find another set of data that deserves this type of display. Display it.

S) SOURCE

Ads for three or more cars, page 51, Fourth Section, Sunday Standard-Times, New Bedford, Mass. for 18 August 1968.

Again we are likely to have difficulty with the implications, but not with understanding the numbers themselves.

The simplest case where we need to learn to grasp the numbers more firmly or more rapidly comes when we face a batch of similar numbers. Hence the techniques of this chapter and those that follow.

(The end of the section called "How far have we come?" is the formal close of this chapter--and of each chapter that follows. Any further sections are--in one sense or another--optional. Some of them may introduce convenient techniques, but these techniques are not essential to our main aims or practices.)

1K. How to use stem-and-leaf to pick up additional information (optional technique)

The basic principle--stem to locate where we write it down; leaf to say more--need not be confined to the sort of examples we have so far seen. Let us return to the *Sunday Standard-Times* again, and the 18 ads for three or more cars. Let us now use the stem-and-leaf technique to collect combined information on the model year and name of the cars offered. Exhibit **11** shows the technique and the results. Some aspects of the coding scheme are worth noting:

- ◇we have used suggestive letters where available, but have not hesitated to step out to numbers when convenient letters are not available.
- ◇we have distinguished capital and small letters whenever their handwritten forms are clearly distinct, but have not tried to use both otherwise.

Panel C of this exhibit shows the result of organizing the entries alphabetically for each model year (and providing two lines for '65). This makes quite clear a tendency for certain name–year combinations to appear frequently. The most obvious example is the fifteen '65 Chevrolets, but others can be found.

pairs of numbers

In the last examples, we were really storing pairs of the form

(number, make)

Rather clearly we can go on to store pairs of the form

(1st number, 2nd number)

Exhibit **12** does this for logarithmic expressions of bank deposits (1st number) and governor's salary (2nd number) for which the identified data and source are given in exhibit 21 of chapter 7. Here the stem 10 is followed by

leaves (4, 574) and (6, 556), thus specifying the pairs

(104, 574) and (106, 556).

exhibit **12** of chapter 1

Using a stem-and-leaf display for 50 pairs of numbers (see exhibit 21 of chapter 8 for identification)

−3	(7, 505)
−2	
−1	(5, 301)
−0	(2, 398), (0, 398)
0	(6, 243), (9, 301)
1	(2, 255), (3, 255), (4, 525), (5, 366), (6, 544)
2	(3, 342), (7, 477), (7, 301)
3	(0, 398), (6, 398)
4	(2, 477), (2, 0), (3, 398), (6, 439), (9, 255)
5	(0, 653), (9, 301)
6	(0, 455), (3, 398), (4, 301), (7, 477), (9, 398)
7	(4, 398), (6, 453), (6, 477), (9, 267)
8	(0, 512), (0, 544), (1, 628), (4, 477)
9	(3, 398), (6, 439), (7,544), (8, 398)
10	(4, 574), (6, 556)
11	
12	(0, 544), (8, 602)
13	(0, 544), (0, 602), (7, 602)
14	(7, 653)
15	(0, 653)
16	(6, 644)
17	
18	
19	
20	(9, 699)

1P. Additional problems

exhibit **13** of chapter 1: data and problem

Viscosity at a temperature of 190°C and a low rate of shear

A) DATA

Run (for) each sample	Viscosity in 100,000's of poises		
	Sample I	Sample II	Sample III
1	.384	.661	3.54
2	.376	.671	3.66
3	.376	.688	3.42
4	.371	.644	4.10
5	.385	.668	4.09
6	.377	.648	3.77
7	.365	.706	4.17
8	.384	.715	3.91
9	.365	.647	4.61
10	.384	.682	3.87
11	.378	.692	
12		.729	

P) PROBLEM

13a) In 1963, McGlanery and Harban gave the values in panel A, showing how well they could measure the viscosity of liquids with a device called a capillary rheometer. Make appropriate stem-and-leaf displays for each of the three samples; comment on the appearance of each.

S) SOURCE

R. M. McGlanery and A. A. Harban 1963. Two instruments for measuring the low-shear viscosity of polymer melts. *Materials Research and Standards 3*: 1003–1007. Table 2 on page 1004.

exhibit **14** of chapter 1: data and problem

Radial velocities of stars near α Persei

A) The RADIAL VELOCITIES--in kilometers per second

+50.0	−2.4	+1.0	−0.5	+2.2	−11.8	−11.9
−36.1	+10.0	+3.7	−15.9	−4.1	−19.1	−16.2
−7.0	+7.0	−9.5	+2.0	−3.0	−22.1	−10.8
+2.2	0.0	−8.4	+5.0	+3.2	−9.1	−6.0
−2.8	−1.0	+0.5	+4.4	−4.4	−17.2	−23.9
+3.0	−0.7	+2.2	+1.6	−0.3	+12.8	−8.2
+24.7	+15.9	+18.0	+6.0	+14.5	−10.5	−13.6
+4.8	+9.0	−17.5	+5.9	−18.4	−17.2	−4.4
						+15.9
						−25.7

P) PROBLEM

14a) In 1958, Heckmann and Lübeck gave the radial velocities in panel A for stars near the star Alpha Persei. Make an appropriate stem-and-leaf. Comment on its appearance.

S) SOURCE

O. Heckmann and K. Lübeck 1958. "Das Farben-Helligkeits-Diagramm des Bewegungshaufens um Alpha Persei". *Zeitschrift für Astrophysik 45*: 243–263. Figure 2 on page 248 is based on Tables 2 and 3 on pages 247 and 248.

exhibit **15** of chapter 1: data and problem

Association constant of the first nitrate complex ion of plutonium (IV), measured in perchloric–nitric acid solutions with 2.00-M hydrogen ion concentration

A) RESULTS OF REPEATED EXPERIMENTS

1.97, 2.13, 2.02, 1.55, 2.83, 2.92, 2.40, 3.24, 3.21, 3.88, 3.85, 3.89, 2.84, 3.08, 4.17, 2.64, 3.67, 3.06, 2.70, 1.91, 2.34, 2.49, 2.76, 3.50

P) PROBLEM

15a) In 1949, Hindman gave the numbers in panel A, which refer to the stability of a complex ion combining 4-valent plutonium ions with nitrate ions. Make an appropriate stem-and-leaf diagram. Comment on its appearance.

S) SOURCE

J. C. Hindman 1949. Complex ions of plutonium. The nitrate complex ions of plutonium (IV). Pages 388–404 of *The Transuranium Elements;* edited by G. T. Seaborg, J. J. Katz, and W. M. Manning, McGraw-Hill (volume 14B of the Plutonium Project Record). Table 1 on page 390.

Easy summaries--numerical and graphical 2

chapter index on next page

We have learned to scratch down batches. Stem-and-leaf displays let us do this easily and rapidly. Even more importantly, they let us look at the general pattern of the batch.

When this pattern is distinctive, we can easily say so in words. Exhibit 10 of chapter 1, for example, shows a pattern of three quite distinct groupings. (This does not happen very often.) The patterns of exhibits 4 and 5 of that chapter, for another, stop much more abruptly toward smaller values than toward larger ones. (This happens all too often.)

We would like to be able to summarize the most frequently occurring characteristics of the pattern of a batch. It will be good to do this in terms of a few numbers that are easily understood, and to agree on what these numbers are, what they are called, how they are marked, how they are routinely written down, and how they are usefully and easily shown graphically!

To do this well, we must limit our objectives. It would be wrong to expect a standard summary to reveal the unusual--no matter how important the unusual may be when it occurs. Things like the separation of the pattern into groupings--illustrated by exhibit 10 of chapter 1--ought not to show in an easy summary of routine form. On the one hand, such a separation deserves words. On the other, trying to show such occasional happenings would complicate our procedures unduly and confuse the summaries for all the many batches where nothing of the sort happens.

We ought to depend on stem-and-leaf displays--or other visual devices for details--to show us the unexpected. Standard summaries--whether numerical or visual--should only be able to reveal the more or less expected.

Summaries can be very useful, but they are not the details. So long as the detail is not so great as to confuse us hopelessly, there will be no substitute for having the full detail where we can look at it, set out in as clear a way as we can easily manage. Summaries are necessary for vast amounts of data--and often convenient for smaller amounts. They are not supposed to--and cannot be expected to--replace the corresponding details. Often, of course, the details will add little, but it is important to prepare for the occasions when they add much.

review questions

Why would anyone want to summarize a batch? How completely can we expect to do this? How broad can our objectives be? What is the reasonable

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★ = includes problem(s)

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★ = includes problem(s)

purpose of standard summaries? What ought more detailed presentations do for us? How frequently ought they be important?

2A. Extremes and median

If we are to select a few easily-found numbers to tell something about a batch as a whole, we certainly want:

- the extremes--the highest and lowest values--which we will mark "1" (for their depth).
- a middle value.

We would like these values to be easy to find and write down, whether the total count of the batch is 8, or 82, or 208.

Once we have the batch in stem-and-leaf form, we have its values almost in order. Once we have the check count, we can count up or down a stem at a time. The easy values to find are those that are fixed by counting.

If the batch has an even number of values, there are two in the middle by count. If there are 50 in all, halfway from 1 to 50 is 25.5, so that the 25th and 26th are the middle pair.

If the count is an odd number, there is a single value in the middle by count. If there are 13 in all, halfway from 1 to 13 is 7, so the 7th is it.

The median is defined to be

$$\text{Median} = \begin{cases} \text{the single middle value} \\ \text{the mean of the two middle values} \end{cases}$$

We shall mark it "M".

For exhibit 10 of Chapter 1, we find these values

1	203	
M	46	(hundreds of feet)
1	3	

Once we have a stem-and-leaf display, the extremes are very easy. The total count is 50. The 25th and 26th values are 44 and 48, so the median is

$$46 = \tfrac{1}{2}(44 + 48)$$

counting-in

To get an easy number in the middle of the batch--assuming that the batch is in stem-and-leaf form--we count in from either end, starting by counting the extreme value "1". This procedure assigns a

rank

to each number in the batch.

We can start from either extreme. In such phrases as "he was first in his class", it seems natural to start by ranking the highest value as "1", to

rank down

Many times, by contrast, we like to have the rank increase as the value increases, to

rank up

This calls for ranking the lowest value "1".

The least rank — either up or down -- that we can give to an observation is its

depth

The depth of an extreme is always 1. Exhibit 1 shows an example.

exhibit 1 of chapter 2: Chevrolet prices

The two directions of ranking illustrated (data of exhibit 3 of chapter 1)

Upward rank	Prices	Downward rank	Depth
1	150	17	1
2	250	16	2
3	688	15	3
4	695	14	4
5	795	13	5
6	795	12	6
7	895	11	7
8	895	10	8
9	895	9	9
10	1099	8	8
11	1166	7	7
12	1333	6	6
13	1499	5	5
14	1693	4	4
15	1699	3	3
16	1775	2	2
17	1895	1	1

Remark:

upward rank + downward rank = 1 + total count

for each and every entry. This is a general property.

P) PROBLEM

1a) Find a small set of numbers that interests you, put them in order, and write them out with both sets of ranks.

Whichever way we rank a batch of 17, the median has rank 9, where $9 = \frac{1}{2}(1 + \text{total count})$

In a batch of six, say for example,

7
12
18
20
27
54

the median is the mean of the two center values, here

$$\tfrac{1}{2}(18 + 20) = 19$$

The ranks of these values are 3 and 4, no matter which end we rank from. Half-way between rank 3 and rank 4 should come rank 3h. (Recall that we use "h" for "and a half".)

Again, 3h is $\frac{1}{2}(1 + \text{total count})$. The same sort of thing happens for batches of any even count. If we do the natural thing, and use half-integer

interpolated ranks

for the midvalues half-way between the values to which we have given adjacent integer ranks, we will have the same expression

$$\text{depth or rank of median} = \frac{1 + \text{count}}{2}$$

for batches of all counts, odd or even. Exhibit 2 shows some examples.

To deal with medians, we need ranks or depths in halves and the corresponding interpolated values. The same ideas can, as we shall soon see, make it easy to describe other useful quantities. We can--and very occasionally do--use ranks or depths in quarters (possibly even more closely), but all such can be considered quite exceptional.

review questions

What is an extreme? How do we mark it? How is a median defined? Where in the batch is it? What is an interpolated rank? What is a rank? What is the difference between ranking up and ranking down? What is a depth? How do we find the value corresponding to an interpolated rank? What is an interpolated depth? What formula gives the interpolated depth of the median? Its interpolated rank? Is there any value whose interpolated rank up is the same as its interpolated rank down?

2B. Hinges and 5-number summaries

We have grasped 3 numbers to define the ends and the middle of a batch. These are useful, but are rarely enough.

The median can be found by counting half-way from one extreme to the other. If we are going to add two more numbers to form a 5-number summary, it is natural to find them by counting half-way from each extreme to the median.

exhibit **2** of chapter 2: Chevrolet prices

Examples of interpolated ranks

A) In the DATA of EXHIBIT 1

Interpolated rank	**Value** and its genesis
1h	**200** = $\frac{1}{2}$(150 + 250)
2h	**469** = $\frac{1}{2}$(250 + 688)
3h	**691.5** = $\frac{1}{2}$(688 + 695)
4h	**745** = $\frac{1}{2}$(695 + 795)
5h	**795** = $\frac{1}{2}$(795 + 795)

(and so on)

B) In the BATCH of 6

Ranks		Interpolated	
Rank	**Number**	**Number**	**Rank**
1	7		
		9.5	**1h**
2	12		
		15	**2h**
3	18		
		19	**3h**
4	20		
		23.5	**4h**
5	27		
		40.5	**5h**
6	54		

P) PROBLEMS

2a) Find the values corresponding to upward ranks of 10, 10h, 11, 11h, and 12 in exhibit 1.

2b) Find the values corresponding to downward ranks of 10, 10h, 11, 11h, and 12 in exhibit 1.

2c) Find the two values corresponding to each of depths 4, 4h, and 5 in exhibit 1.

Since we can think of finding first the median and then these new values as a folding process, it is natural to call the new values

hinges.

If we have 9 values in all, the 5th from either end will be the median, since $\frac{1}{2}(1 + 9) = 5$. Since $\frac{1}{2}(1 + 5) = 3$, the third from either end will be a hinge. If we have 13 values, the 7th will be the median--and the 4th from each end a hinge. In folded form, a particular set of 13 values appears as follows:

```
−3.2                1.5                9.8
    −1.7          1.2  1.8          6.4
        −0.4  0.3          2.4  4.3
            0.1                3.0
```

The five summary numbers are, in order, −3.2, 0.1, 1.5, 3.0, and 9.8, one at each folding point.

We usually symbolize the 5 numbers (extremes, hinges, median) that make up a

5-number summary

by a simple summary scheme like this:

```
#13
M7 |    1.5    |
H4 |  0.1  3.0 |
 1 | −3.2  9.8 |
```

where we have shown the count (marked #), the depth of the median (marked M), the depth of the hinges (marked H), and the depth of the extremes (always 1; needs no other mark) to the left of an inverted-U enclosure, and have written the corresponding values in the enclosure. We put the median in the middle, the upper values on one side, and the lower values on the other (which is which does not matter, and can change from time to time and from person to person). Such a scheme we call a

letter-value display.

(We will slowly learn about more and more detailed letter-value displays.)

Since the depth of the extremes is ONE, the depth of the median is half of the count plus ONE. When the depth of the median is a half-integer, we first discard this half and then add ONE, before halving. Thus for a batch of 20, $\frac{1}{2}(1 + 20) = 10\text{h}$, and $\frac{1}{2}(1 + 10\text{h})$ is taken as $\frac{1}{2}(1 + 10) = 5\text{h}$.

examples

Exhibit **3** shows three examples of the calculation of hinges and 5-number summaries. Panel A is in folded form. Panel B starts from a modified stem-and-leaf where (a) the leaves are divided into columns and (b) rows (stems) have been divided in such ways as to make the folded structure clearer.

exhibit **3** of chapter 2: varied data

Hinges illustrated

A) The 17 AUTO PRICES of EXHIBIT 1--in folded form

```
(1)                    (M)                    (1)
150                    895                   1895
   250              895     1099          1775
      688        895           1166     1699
         695  795                 1333 1693
            795                       1499
            (H)                       (H)
```

[17 prices, 1HMH1: 150, 795, 895, 1499, 1895 dollars]

B) The 34 ULTIMATE POWERS of EXHIBIT 6 of CHAPTER 1--folded stem-and-leaf; note irregular splitting of stems and letter display

Extremes and median	Intermediate values	Hinges
1*	5	
1·	6	
1·		
2*	2445	
3*	02	
3·		3
3·	66	
4*	058	
5*	0	
6*	0	
7*	1	
7·	5	
8*		
9*		
10	08	
11	4	
12	00	
13*	4	
1**	50	62
2	25, 50, 85	
3**	79	
0***	424	
0	900	
1***	345	
1	974	

34 ult. powers

M17h	73	
H9	33	162
1	15	1974

→

Panel C shows what will be the conventional stem-and-leaf display, one that includes cumulative counts from each end. In this example we have put the cumulative counts on the half-lines, where they really belong. In routine practice, we let the cumulative counts fall outward into the last lines included in them, as will be illustrated in Panel E. Panel D shows how we can count in, leaning on the cumulative counts. For the median, for example, we know from panel C that the cumulative count through value 54 is 38, and that rank 39 corresponds to 55, and so on. Once we have followed this pattern a few times,

exhibit **3** of chapter 2 (continued)

C) The 82 AREAS in exhibit 4 of chapter 1 with CUMULATIVE COUNTS ADDED--on the half line

		(#)		
3·	8	(1)		
			1	
4*	0000111112222223344	(19)		
			20	
4·	555566678899	(12)		
			32	
5*	001124	(6)		
			38	
5·	577778899999	(12)		√ 82 = 38 + 12 + 32
			32	
6*	112444	(6)		
			26	
6·	588899	(6)		
			20	
7*	001223334	(9)		
			11	
7·	56678	(5)		
			6	
8*	033	(3)		
			3	
8·	8	(1)		
			2	
9*	24	(2)		

Note that 82 = 38 + 12 + 32 is the count check.

D) CALCULATIONS for PANEL C

first steps			counting-in						result		
			(near 41h)		(near 21↓)		(near 21↑)				
82			**38**	54	**20**	44	**23**	68	82		
M41h			**39**	55	**21**	45	**22**	69	M41h		57
H21			**40**	57	**22**	45	**21**	69	H21	45	69
1	38	94	**41**	57	**23**	45	**20**	70	1	38	94
			42	57							

→

we will find it easy to do just by pointing at digits in the stem-and-leaf, writing down only the answers.

Though it is somewhat concealed, the tendency of the data in panel C to trail out further toward high values can be seen in the summary of panel D, whose heart is

	57	
45		69
38		94

exhibit **3** of chapter 2 (continued)

E) CUMULATIVE COUNTS in VARIOUS FORMS

As scratched down (right of display)		For record (left of stub)	
(1)	⟨1	1	**3·**
(19)	⟨20	20	**4***
(12)	⟨32	32	**4·**
(6)	⟨38	38	**5***
(12)	12	82 √ (12)	**5·**
(6)	⟨32	32	**6***
(6)	⟨26	26	**6·**
(9)	⟨20	20	**7***
(5)	⟨11	11	**7·**
(3)	⟨6	6	**8***
(1)	⟨3	3	**8·**
(2)	⟨2	2	**9***

P) PROBLEMS

Imitate Panel A for:

3a) The 17 Chevrolet prices of exhibit 3 of chapter 1.
3b) The 25 values of exhibit 7 of chapter 1.
3c) A batch of 12 to 25 values of interest to you.

Imitate Panel B for

3d) A batch of 25 to 50 values of interest to you.
3e) The 50 heights of exhibit 10 of chapter 1.

Imitate Panel C for

3f) The 58 radial velocities in exhibit 13 of chapter 1.
3g) The 21 measurements of an association constant in exhibit 14 of chapter 1.
3h) A batch of 100 or more values of interest to you.

Find the 5-number summaries for

3i) The 93 model-years of panel A of exhibit 9 of chapter 1.
3j) The 141 model-years of panel B of exhibit 9 of chapter 1.
3k) A batch of size at least 100 that interests you.

The hinges, 45 and 69, are each 12 away from the median, thus being quite symmetrical, to be sure. The extremes, however, are respectively, 7 and 25 units outside the hinges, which shows a considerably longer trail-off toward higher values.

Panel E shows two forms of cumulative counts. First, the scratch-down version--placed where we have been putting the stem counts, and distinguished by a row of ⟨'s. (Note the omission of ⟨ for the line containing the median. If the median falls between lines, we draw a horizontal line, as in exhibit 16 of chapter 3.) Then the more formal version, used in most exhibits, but often avoided in paper and pencil work. This version goes to the left of the stem-heads. The lines separating off the line with the median--and the giving of that line's individual count--are important.

further examples

Exhibit **4** shows two further examples.

The summary of panel A with its

	46
20	112
3	203

could also be interpreted as just another trail-off toward higher values. As we know, this batch is clearly separated into 3 groupings. As we have said, the sort of summary we are making should not detect this separation. Clearly, in this example, it does not.

The populations of the 50 states are shown in panel B. Note that dealing with mixed leaves is not a complication. This batch shows a more common pattern, where extremes, hinges, and median come quite close to telling the tale. The second version of a five-number summary below the stem-and-leaf--the one converted to millions--summarizes the situation quite well. We could--probably equally well--have rounded the values further to a box

	2.5
0.9	4.3
0.2	17

which would still remind us that:

- ◇50 states are considered.
- ◇the smallest state was about 200,000 population in 1960.
- ◇about half the states fell between 0.9 million and 4.3 million--with about a quarter below 0.9 and a quarter above 4.3.
- ◇about half the states were smaller and about half larger than 2.5 million.
- ◇the largest was about 17 million.

These reminders really give us a fairly good feel for this whole batch of 50 populations.

The populations of the states clearly trail off much farther above the middle than below. In this instance, this keeps us from condensing this summary still further, and is something we will need to be able to remedy.

review questions

What is a hinge? How is its depth found? What approximation is almost always made? What values make up a 5-number summary? What are two ways to write them down as a summary scheme? How are total count and depths shown in a summary scheme? What is the natural order to find these depths?

exhibit **4** of chapter 2: various

Further examples of hinges and 5-number summaries

A) The 50 HEIGHTS of EXHIBIT 10 of CHAPTER 1--with cumulative counts in the easy places

5	**0***	34588	Fla, Del, La, Miss, RI
11	**1**	236788	
17	**2**	003448	
22	**3**	24556	
√5	**4***	01489	
23	**5**	03347	
18	**6**	367	
15	**7**	2	S.Dak
14	**8***	8	Texas
·	**9**		
·	**10**		
13	**11**	2	Oregon
12	**12***	678	Idaho, Ariz, Mont
9	**13**	12588	Nev, N.Mex, Utah, Wyo, Hawaii
4	**14**	445	Wash, Colo, Calif
·	**15**		
·	**16***		
·	**17**		
·	**18**		
·	**19**		
1	**20***	3	Alaska

count check is
22 + 5 + 23 = 50 √

50 heights in hundreds of feet

M25h	46	
H13	20	112
1	3	203

50 heights in thousands of feet

M25h	4.6	
H13	2.0	11.2
1	0.3	20.3

Note: As you get more used to simple one-digit stem-and-leafs, you may want to use a single * above the vertical line in place of all the *'s and spaces just to the left of that line. (This lets you snuggle the stems up to the line.) Wholly optional.

→

What do we include in a stem-and-leaf to make counting-in easier? What kinds of behavior do the two panels of exhibit 4 illustrate?

2C. Box-and-whisker plots

We always want to look at our results. We usually can. In particular, we want to look at 5-number summaries. Exhibit **5** shows an easy way to do just this. We draw a long, thinnish box that stretches from hinge to hinge, crossing

exhibit **4** of chapter 2 (continued)

B) The 1960 POPULATIONS of the 50 STATES--unit = 10,000 people (rounded)

↓	**1***		
2	**2**	3,9	Alaska, Nevada
4	**3**	9,3	Vermont, Wyoming
5	**4**	5,	Delaware
	5*		
11	**6**	3,7,7,1,3,8	
	7	6,	
13	**8**	9,	
15	**9***	7,5	
21	**1****	30,79,74,41,77,86	
√(7)	**2**	54,76,18,18,33,38,85	
22	**3**	27,94,04,26,10,41,57,97,95	
13	**4**	95,66,32,56	Massachusetts
9	**5****	15,	
8	**6**	07,	New Jersey
7	**7**	82,	Michigan
	8		
6	**9****	71,58	Ohio, Texas
4	**1*****	572,008,678,132	Calif, Ill, NY, Pa
↑	**2**		

50 populations in tens of thousands

M25h	246	
H13	89	432
1	23	1678

50 populations in millions

M25h	2.5	
H13	0.89	4.3
1	0.23	17

P) PROBLEM

4a) Find another set of data deserving such treatment. Give the treatment and plot the display.

S) SOURCE

The World Almanac, 1966, p. 325.

Their source: U.S. Bureau of the Census

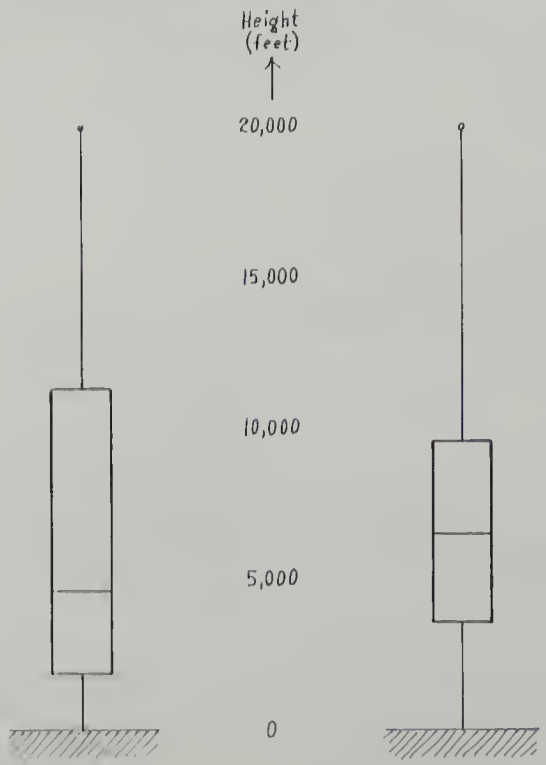
it with a bar at the median. Then we draw a "whisker" from each end of the box to the corresponding extreme.

This process shows us the five-number summaries quite clearly, so clearly as to give us a clear idea of (some of) what we may have been missing. There is, inevitably, more empty space in a box-and-whisker plot than in a listing of a 5-number summary. There is more space for identification. We can at least identify the extreme values, and might do well to identify a few more.

exhibit 5 of chapter 2: various heights

Box-and-whisker plots of 5-number summaries

A) HIGHEST POINTS in 50 STATES B) HEIGHTS of 219 VOLCANOS



P) PROBLEMS

Imitate these plots for the data of:

5a) Panel A of exhibit 3.

5b) Panel B of exhibit 3.

5c) Panel C of exhibit 3.

5d) Panel B of exhibit 4.

5e) A set of data of interest to yourself.

Exhibit **6** shows the same examples as in exhibit 5, with some identification added. Clearly, we learn more from such displays.

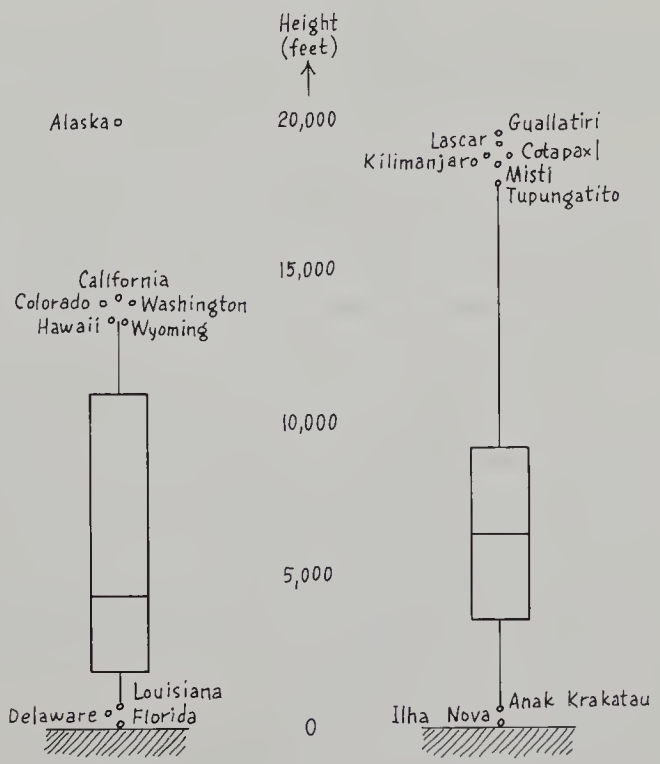
In exhibit 6 we have stopped the whiskers at the innermost identified values. This is recommended.

exhibit **6** of chapter 2: various heights

Box-and-whisker plots with end values identified

A) HEIGHTS of 50 STATES

B) HEIGHTS of 219 VOLCANOS



P) PROBLEMS

Imitate these plots for the data of:

6a) Panel A of exhibit 3.

6b) Panel B of exhibit 3.

6c) Panel C of exhibit 3.

6d) Panel B of exhibit 4.

6e) A set of data of interest to yourself.

6f) Make a similar plot for the heights of some of the highest mountains in the U.S.

6g) Make a similar plot for the heights of some of the highest mountains in the world.

basic ideas of plotting

With box-and-whisker plots we are beginning our plotting education. The first thing we need to do is separate, in our mind, what it takes to make plotting easy from what it takes to make plotting effective. The lines ruled on graph paper help to make plotting easy, but they do not make plotting effective for seeing what is going on--instead they get in the way of seeing what we ought to see. (If we are using a graph as a substitute for a table--as a way to look up values of a function--things are likely to be different. This, however, is something we almost never need to do in exploratory data analysis.)

tracing paper

If we want to see what our plots ought to tell us, there is no substitute for the use of tracing paper (or acetate). If we slip a well-printed sheet of graph paper just below the top sheet of a pad of tracing paper, we can plot on that top sheet of tracing paper almost as easily as if it were itself ruled. Then, when we have the points plotted, some boundary or reference lines drawn, and a few scale points ticked, we can take away the graph sheet and look at the points undisturbed by a grid. We often gain noticeably in insight by doing this. (And we have had to pay for a sheet of tracing paper rather than for a sheet of graph paper.) Alternatively, we can plot on a sheet of graph paper and then trace the result.

In doing either of these we gain very much by having:

- well-printed graph paper, which means (a) good quality lines, (b) every 10th line heavy, (c) every 5th line medium heavy. ("Five and dime" graph paper or quadrille-ruled paper will make our work much harder. For more detailed discussion, see section 5A, below.)
- tracing paper that erases cleanly and easily (quality equal to that of the Clearprint Paper Co. of San Francisco is worth the difference).
- a transparent plastic rule or triangle.

An alternative that:

- can be even more effective,
- is no more expensive,
- takes a little more trouble to prepare for,

replaces the tracing paper by the thin sheets of transparent plastic (acetate) made for use in overhead projectors. Two cautions are important:

1. You can only use markers specially made for the purpose. (A number of manufacturers make satisfactory products, but it may not be easy to find any of them. The writer prefers the temporary--wash-off--styles to those that cannot be erased.)
2. It is important to keep one's fingers off the plastic until the picture is completed. (A piece of thin graph paper, placed upside down, works very well as a hand shield.)

scale values

We ought to put as many scale values on the graph paper preliminary as will help us make the plot easily. On the tracing paper final, however, we ought not show more than three or four numbers along a scale. More clutters up the picture and distracts the eye from what it ought to see. (Scales for dates are sometimes an exception. It can matter whether an appearance came in 1929 or 1928, in 1776 or 1775.)

People are used to scales on the left and below. So be it — for the picture, perhaps. When one is plotting the points, however, it is much more convenient to put the horizontal scale ABOVE the plot, where you do not have to move your hand to see it. (It would be rational to plot from detailed scales above and left, and to produce a final picture with a few scale points shown below and right; but such rationality is usually not worth the possibility of occasional confusion.)

plotting without graph paper

We almost always want to look at numbers. We do not always have graph paper at hand. **There is no excuse for failing to plot and look.**

We usually have ruled paper at hand. For emergency graph paper, take out one sheet of ruled paper, turn it on its side, and place it beneath another sheet of ruled paper. If these two sheets have a light-colored backing — often provided by the rest of the pad or notebook — the vertical lines on the lower sheet are almost certain to show through well enough, combining with the horizontal lines on the top sheet to form a grid on which plotting is reasonably easy. (The first step in this sort of plotting is to mark — by ticks or unobtrusive dots — enough information on the top sheet to make it easy to get the lower sheet back to its original position after it slips.)

With this technique, one can make useful, if not decorative, plots almost anywhere.

review questions

What is a box-and-whisker plot? What do its parts show forth? What rules does it obey about showing values individually? About identifying values? What must we separate in our minds about plotting? What are the essentials of convenient, effective plotting? How can we, in an emergency, plot without graph paper?

2D. Fences, and outside values

Hinges are for our convenience. They can — and will — serve various purposes for us. Their role in 5-number summaries is only the beginning.

When we look at some batches of values, we see certain values as apparently straying out far beyond the others. In other batches straying is not so obvious, but our suspicions are alerted. It is convenient to have a rule of

thumb that picks out certain values as "outside" or "far out". To do this, we set up appropriate "fences" and use "outside" and "far out" accordingly.

A useful rule of thumb runs as follows:

◇"H-spread" = difference between values of hinges.

◇"step" = 1.5 times H-spread.

◇"inner fences" are 1 step outside hinges.

◇"outer fences" are 2 steps outside hinges (and thus 1 step outside of inner fences).

◇the value at each end closest to, but still inside, the inner fence is "adjacent".

◇valucs between an inner fence and its neighboring outer fence are "outside".

◇values beyond outer fences are "far out".

Exhibit **7** shows some examples. Notice the practice of writing the value of the H-spread just outside the box (under a horizontal "eave"), and then putting the value of the step in a "penthouse" on the second part of the display, which contains the fences and is written after the main letter display, either below it or to its right. Panels C to F show a convenient standard form, combining summary scheme, fences, and identification of outside values. We can call it a

fenced letter display

exhibit **7** of chapter 2: various examples

Examples of calculation of fences and identification of adjacent, outside, and far outside values (based on exhibits 3 and 4)

A) For panel A of exhibit 3

17 Chev. prices

M9		895		
H5	795		1499	704
1	150		1895	
		1056		
f	−261		2555	
	xxx		xxx	out
F	−1317		3611	
	xxx		xxx	far

adj: 150, 1895

Note that, here:

704 = 1499 − 795
1056 = (1.5)(704)
−261 = 795 − 1056
2555 = 1499 + 1056
−1317 = −261 − 1056
3611 = 2555 + 1056

exhibit **7** of chapter 2 (continued)

B) For panel B of exhibit 3--see exhibit 6 of chapter 1 for details

34 ultimate powers

M17h	73		
H9	33	162	129
1	15	1974	

	194	
f	−161	356
	xxx	two
F	−355	550
	xxx	three

adj: 15, 285

out: 379, 424

far: 900, 1345, 1974

Note that, here: 121h = 156 − 34h and 182 = 3(121h)/2

C) REARRANGED VERSION of (B), with IDENTIFICATION

34 ultimate powers

M17h	73		
H9	33	162	129
1	15	1974	

	194	
f	−161	356
	xxx	two
F		550
		three

adj: 15(Boysen), 285(Hungry Horse)

out: 379(Shasta), 424(San Luis)

far: 900(Glenn Canyon), 1345(Hoover)

far: 1974(Grand Coulee)

D) For panel C of exhibit 3--see panel D of exhibit 3 also

82 areas

M41h	57		
H21	45	69	24
1	38	94	

	36	
f	9	105
	xxx	xxx
F		

adj: 38(Tate), 94(Yazoo)

E) For panel A of exhibit 4--hundreds of feet

50 heights

M25h	46		
H13	20	112	92
1	3	203	200

	138	
f	−118	250
	xxx	xxx

adj: 3(Florida), 203(Alaska)

ranges

One thing that a box-and-whisker plot brings to eye and mind is the spread from one extreme to the other, called the

range

of the batch. For the various panels of exhibit 7 we have:

(for patch A) range = 1895 − 150 = 1745 dollars.

(for panel B) range = 1974 − 15 = 1959 megawatts.

(for panel C) range = 74 − 38 = 36 tens of sq. miles.

(for panel D) range = 203 − 3 = 200 hundreds of feet.

(for panel E) range = 1678 − 23 = 1655 tens of thousands.

If we were to quote only one measure of spread, we usually do better to quote the H-spread, unless we are specifically concerned with either extremes or stray values. Giving both the range and the H-spread is often useful, however. Since the extremes are in the 5-number summary, finding the range is easy. We recommend writing it down rather often. Clearly the place for it in a letter display is on the line for extremes, just to the right of the box--as in panels E and F.

trimeans

Another thing that box-and-whisker plots convey to us is an impression of location or centering that combines both median and hinges. The arithmetic that comes closest to matching this impression is probably the

trimean

defined to be

$$\frac{\text{lower hinge} + 2(\text{median}) + \text{upper hinge}}{4}$$

exhibit **7** of chapter 2 (continued)

F) For panel B of exhibit 4--10,000's of people

50 populations

```
M25h
H13   89          432   343
 1    23         1678  1655
```

adj: 23(Alaska), 782(Mich)

out: 958(Tex), 971(Ohio), 1008(Ill) 1132(Pa)

far: 1572(Cal), 1678(*NY*)

```
        4
 f   −425          946
     xxx          four
 F                1460
                   two
```

P) PROBLEMS

7a/b/c/d/e/f/g/h/i/j/k) Make a fenced letter display for the batch described in the correspondingly lettered problem of exhibit 3.

Sometimes we like to use trimeans instead of medians to give a somewhat more useful assessment of location or centering. We can use the trimean almost any place where the median is used. Readers should feel free to do so.

review questions

Do hinges have more than one purpose? Vaguely, what is a stray value? How might we identify stray values? What is an H-spread? A step? An inner fence? An outer fence? Which values are outside? Which far outside? Which adjacent? What is a standard way of combining a summary scheme with fences and identification of outside values? What is a range? How useful is it? Can it replace an H-spread? Can it supplement an H-spread?

2E. Schematic plots

Sometimes it is worthwhile to have some rules about which values deserve to appear individually in a box-and-whisker plot. The simplest convenient set of rules appear to be these:

- ◇outside, and far out values should appear separately.
- ◇whiskers should be dashed, ending with dashed crossbars at the adjacent values.
- ◇far out values should be marked impressively and identified in capital letters
- ◇outside and adjacent values should all be identified--in small letters--unless identification would confuse the picture (this rarely happens unless there are more than six at a single end). Identification of adjacent values may be less obtrusive.
- ◇abbreviated identifications are welcome whenever likely to be understood.

We expect to be consistent about the whiskers: always using dashes for **schematic plots** according to these rules, always using solid lines for free-form box-and-whisker plots that do not (necessarily) conform to these rules.

For an example where these rules are well illustrated, we can turn to the areas of the 83 counties of Michigan (the stem-and-leaf will be presented as part of exhibit 4 of chapter 3). The fenced letter-value display is:

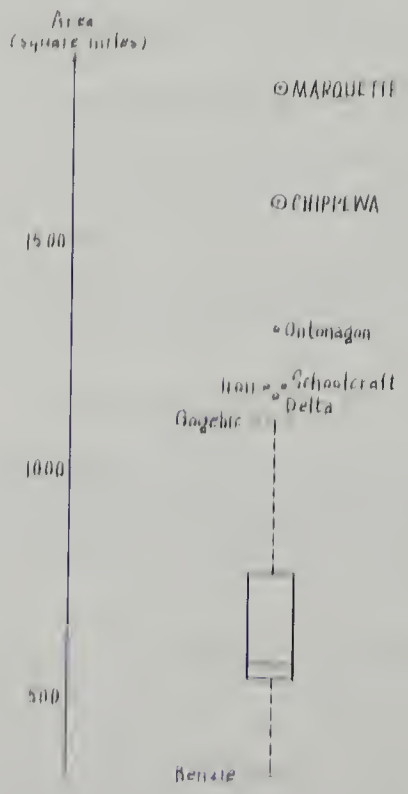
```
  83  areas
 M42        57
 H21h   54h     78h   24
   1    32      184
             36
     f   18h    114h      adj:  32(Benzie), 111(Gogebic)
         none   four      out:  118(Delta), 120(Iron)
     F          150h      out:  120(Schoolcraft), 132(Ontonagon)
                two       far:  158(CHIPPEWA), 184(MARQUETTE)
```

The corresponding schematic plot appears as in exhibit 8. Notice the treatment of a tie.

exhibit 8 of chapter 2: county areas

Michigan counties, area in mi²

A) SCHEMATIC PLOT



B) PROBLEMS

8a) Do you think the areas of counties in most states would include detached values? Why or why not?

8b) Make a suitably identified schematic plot of the county areas for the state in which you were born (or of corresponding subdivisions of the country in which you were born).

8c) What can be said about the counties in Michigan whose areas are identified in panel A? (Use map.)

8d) What display would your answer to (8c) lead you to propose?

8e) Make that display.

review questions

What is a schematic plot? What points are plotted separately? What points are identified? How do the whiskers show whether or not a box-and-whisker plot is a schematic plot? What is a fenced letter display? What rules are used in writing one down?

2F. Pros and cons; the Rayleigh example

We need now to look harder at the advantages and disadvantages of schematic plots. Let us begin with an example that shows both their weakness and their strength. During the winter of 1893–94, Lord Rayleigh was investigating the density of nitrogen from various sources. He had previously found indications of a discrepancy between the densities of nitrogen produced by removing the oxygen from air and nitrogen produced by decomposition of a chemical compound. The 1893–94 results established this difference with great definiteness, and led him to investigate further the composition of air chemically freed of oxygen. This led to the discovery of argon, a new gaseous element. Rayleigh's observations are summarized in exhibit **9**.

exhibit **9** of chapter 2: gas densities

Rayleigh's weights of a standard volume of "nitrogen"

A) DATA

	Source of "nitrogen"		
Date	Origin	Purifying agent	Weight
29 Nov. '93	NO	Hot iron	2.30143
5 Dec. '93	"	"	2.29816
6 Dec. '93	"	"	2.30182
8 Dec. '93	"	"	2.29890
12 Dec. '93	Air	"	2.31017
14 Dec. '93	"	"	2.30986
19 Dec. '93	"	"	2.31010
22 Dec. '93	"	"	2.31001
26 Dec. '93	N_2O	"	2.29889
28 Dec. '93	"	"	2.29940
9 Jan. '94	NH_4NO_2	"	2.29849
13 Jan. '94	"	"	2.29889
27 Jan. '94	Air	Ferrous hydrate	2.31024
30 Jan. '94	"	"	2.31030
1 Feb. '94	"	"	2.31028

S) SOURCE

Lord Rayleigh, 1894. "On an anomaly encountered in determinations of the density of nitrogen gas," *Proc. Roy. Soc.* (Lond.) *55*, 340–344 (also in his *Scientific Papers*, Vol. *4*, pp. 104–108).

Exhibit **10** shows dot and schematic plots for Rayleigh's 15 weights of nitrogen, treated as a single batch. Here the main issue is the violent separation into two quite isolated subgroups. This is made quite clear by the individual values of the dot plot--and almost completely covered up by the schematic plot. (Only **almost,** because the experienced viewer--finding the whiskers so short, in comparison with box length--is likely to become suspicious that he should see more detail.)

Clearly we cannot rely on schematic plots to call our attention to structure near the center of the batch; we must use some other approach when this is needed. (Either dot plots or stem-and-leaf displays will serve.)

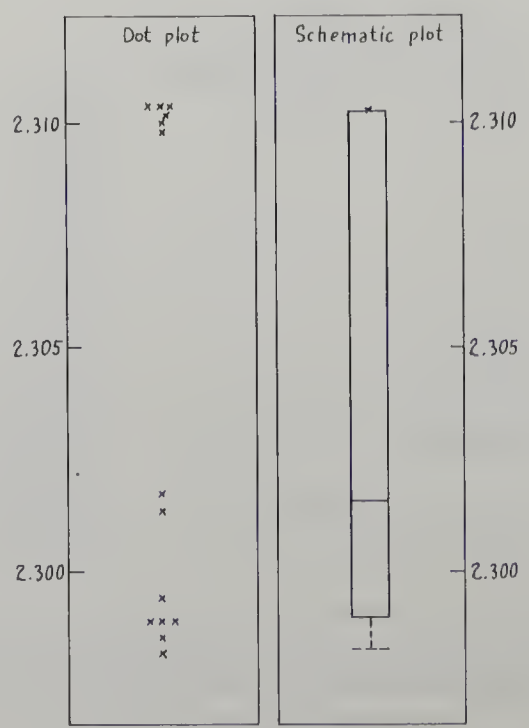
a better use

Exhibit **11** uses the schematic plots for one of the purposes for which they are best fitted: comparison of two or more batches. In it, the two batches of

exhibit **10** of chapter 2: gas densities

Rayleigh's 15 weights of "nitrogen"

A) DOT and SCHEMATIC PLOTS



Rayleigh's weights, one for "nitrogen" from air, the other for "nitrogen" from other sources, are set out and compared. Clear to every eye are:

- ◇the relatively huge difference between the batches,
- ◇a slight tendency of the "other sources" weights to straggle upward — noticeable, but far from offering any basis for explaining their difference from the air values,
- ◇a possible greater variability of the "other sources" weights, over and beyond this straggling.

Here schematic plots have done everything we could ask.

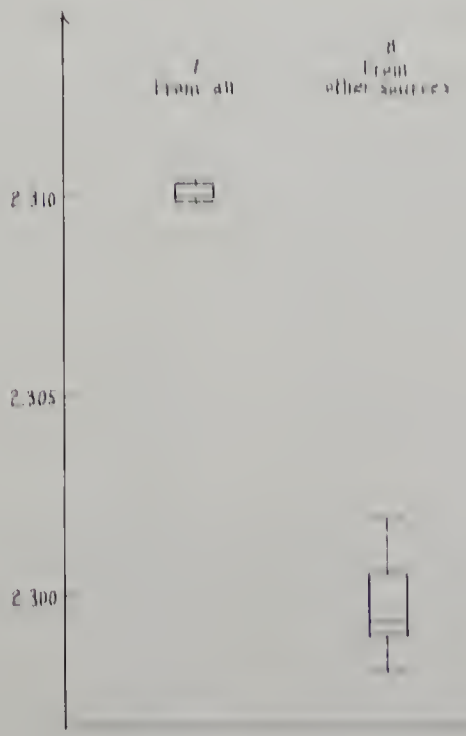
aim reiterated

The Rayleigh data lets us make an old point once more. **Aims can differ, and plots should follow aims.** We can try to show either **general behavior** or **detail.** We should almost always choose one or the other.

Many books on graphical presentation stress getting the proper impression of relative size across to the reader. This may be sound for most articles in the

exhibit **11** of chapter 2: gas densities

Rayleigh's two batches of weights of "nitrogen"



daily newspaper, where making small differences more visible by starting the vertical scale far from zero is usually thought of as a "journalistic" or "political" trick.

We need to remember that doing just this is a vital scientific tool. If we had started the vertical scale at zero and used bar graphs, Rayleigh's results--so clear in exhibit 11--would have appeared as in exhibit **12.** Whatever exhibit 12 may look like, it is neither evidence for a new chemical element nor a basis for a Nobel prize.

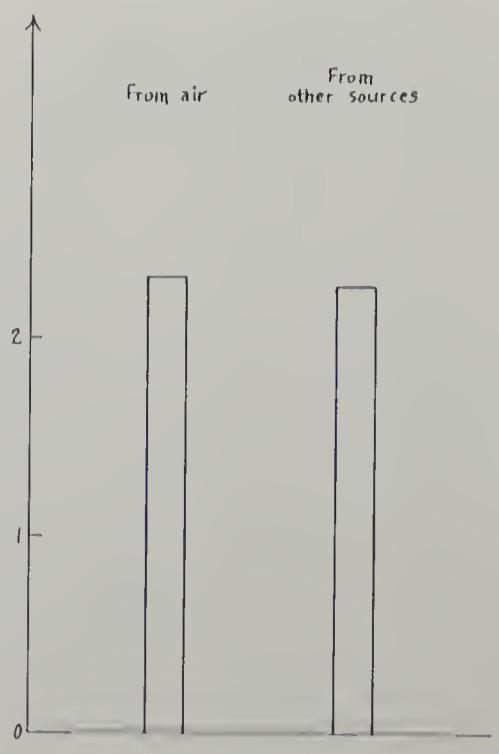
From the point of view of freight charges for compressed gases, exhibit 12's impression of "negligible difference" gives a useful impression, but it takes exhibit 11 to make clear the scientific issue. **There is often no substitute for the detective's microscope--or for the enlarging graph.**

choice of plots

We are now well equipped. Schematic plots will be the standard form, but we will feel free to use other plots instead whenever we feel that this will help.

exhibit **12** of chapter 2: gas densities

Rayleigh's weights again



Clearly, we can use schematic plots in comparing three, four, or any reasonable number of batches. (When the number of batches becomes large, we will probably want to squeeze our bars down to solid lines, and thin our dotted lines out.)

As a visual approach to most comparisons, especially when supplemented by more detailed looks at the residuals, these plots work very well, often providing very effective comparisons.

Comparing more than a few is something that stem-and-leaf cannot do at all well. A display involving more than two--or perhaps three--stem-and-leaf displays tends to leave us with no clear impression.

The key advantages of schematic plots are, then:

- ◇focusing attention where it is most likely to be useful.
- ◇reducing "busyness" to a point where many batches can be compared.

review questions

What was Lord Rayleigh up to? What did he discover? Does exhibit 9 show that he discovered anything? Does the schematic plot of exhibit 10 hint at unusualness? If so, how? Does exhibit 11 show that something was discovered? What do books on graphical presentation often insist upon? Dare we pay attention to what they say? What does exhibit 12 show? What does it fail to show?

2G. Eighths, sixteenths, etc. (almost optional here; used in later chapters)

We sometimes want to go beyond the 5-number summary. Especially in larger batches, there can be advantages to pinpointing one or more values between the hinges and the extremes. A simple and useful approach is to continue halving in terms of depths (naming the results eighths, sixteenths, etc., since the hinges and median divide the batch into approximate quarters). This leads to:

$$\text{Depth of eighths} = \frac{1 + \text{depth of hinges}}{2}$$

$$\text{Depth of sixteenths} = \frac{1 + \text{depth of eighths}}{2}$$

and so on, where, as for hinges, we drop any h before dividing by 2. It seems natural to use E for eighths and then go on to "D", "C", "B", "A", "Z", "Y", . . . as may be needed. The corresponding spreads will be used in chapter 19.

Exhibit **13** shows some examples of 7-number and 9-number summaries, adding first E's and then also D's.

When we need a name for M, H's, E's, etc., we call them

letter values

often including the extremes under this name also.

review questions

What is an eighth? A sixteenth? A thirty-second? How do we mark them? What formulas help us find them? What approximation is usually needed? What is an E-spread? What is a D-spread? What values appear in a 7-number summary? In a 9-number summary? In an 11-number summary? How do the corresponding summary schemes appear? What is an extended schematic summary? How is it extended?

exhibit **13** of chapter 2: varied examples

Examples of 7- and 9-number summaries

A) The 1960 POPULATIONS of the 50 STATES--unit = 10,000 people; stem-and-leaf at exhibit 4, panel B

(Seven-number summary)

50	populations		
M25h	246		
H13	89	432	343 (H-spread)
E7	63	782	719 (E-spread)
1	23	1678	1655 (range)

(Nine-number summary)

50	populations		
M25h	246		
H13	89	432	343 (H-spread)
E7	63	782	719 (E-spread)
D4	39	1008	969 (D-spread)
1	23	1678	1655 (range)

B) The 82 COUNTY AREAS of MISSISSIPPI--in 10's of square miles; stem-and-leaf at exhibit 3, panel C

82	areas		
M41h	57		
H21	45	69	24
E11	42	75	33
1	38	94	56

82	areas		
M41h	57		
H21	45	69	24
E11	42	75	33
D6	41	80	39
1	38	94	56

P) PROBLEMS

(13a) Imitate these schemes for the 50 heights of panel A of exhibit 4.

13b) Extend panel A to an 11-number summary.

13c) Extend panel B to first an 11-number summary and then a 13-number summary.

2H. How far have we come?

This chapter has been concerned with the simplest summaries for batches of numbers, as written down in numbers or as displayed graphically. Our central themes have been (1) summarization by picking out five numbers from among the values--and from among the means of pairs of adjacent values--that will tell us about the general behavior of the batch, and (2) plotting and identification of individual values that may be unusual. In summary, we have focused on extremes, hinges, and median, and have introduced both schematic summaries and box-and-whisker plots to show their values and relationships.

what have we learned to do ?

Ranks are defined by counting in, from either one end or the other, of a list of the values in order. (We usually do this on a stem-and-leaf display.) The lesser of the two ranks for a value is its depth. Half-integer ranks or depths (like rank 4h) refer to the mean of the values with adjacent ranks or depths (ranks 4 and 5 for rank 4h).

The median is half-way through the ordered list. Its two ranks (and its depth) are the same--half of the total count PLUS one. The depth of each hinge is half of one more than the integer part of the depth of the median. (For instance, median at 8h, integer part 8, one more 9, hinge at 4h.)

The difference between the extremes is the **range;** that between the hinges is the **H-spread.**

Extremes, hinges, and median make up a 5-number summary. When written down schematically, we give, outside an enclosure, the total count and the depths of median, hinges, and extremes, giving the values in the enclosure. We also give far out, outside, and adjacent values, often with identification. Often we add the corresponding spread to the right of the enclosure (under an eave).

Making even as simple a picture as a box-and-whisker plot deserves care and technique. Use of tracing paper (or acetate) and proper labeling are important.

The **step**--taken from hinge to first fence and from first fence to second fence--is one and one-half times the H-spread, which is the difference between the hinges. Values outside the second fences are **far out.** Values outside the first fences are **outside.** Values nearest to--but still inside--the first fences are **adjacent.**

Box-and-whisker plots show the same information more pictorially (sticking to our conventions matters). A schematic plot is a box-and-whisker plot that displays separately and identifies all outside values (all far out values with extra emphasis), whose whiskers are dashed and stop at the adjacent values.

We can include 8ths, 16ths, etc., in our schematic summaries when we wish. We use E, D, C, . . . as tags and stick to the rule

$$\text{new depth} = \tfrac{1}{2}(1 + \text{integer part of old depth}).$$

where do we stand ?

We have not looked at our results until we have displayed them effectively. In this chapter we have learned how to display summaries of batches effectively. Now we owe it to ourselves to make such displays routinely.

Except when learning the numerical part of a new technique, no problem of exploratory data analysis is "solved" without something to look at. A stem-and-leaf display is something to look at; so is a letter-value display, a schematic plot, or a more general box-and-whisker plot. Words that are both sufficiently descriptive and clearly informing can be something to look at--but it may not be possible to find words to express what a display or plot may show.

Easy re-expression 3

chapter index on next page

We have learned how to collect the values of a batch--in most cases the natural collection scheme lets us look easily and effectively at the "shape" of the numbers as a whole. There are, however, rather frequent exceptions, as exhibit 6 of chapter 1 and panel B of exhibit 4 of chapter 2 illustrate. It is time to learn how to deal with these exceptions.

Only one broad approach seems natural. If the way the numbers were gathered or collected does not make them easy to grasp, we should change them--preserving as much information as we can use--into a form where we can grasp them more easily, both personally and with our displays and our arithmetic.

Data for us usually means numbers--whether given in digits, by tallies, or on a graph. Three broad classes of numbers are important enough to be worth both names and some discussion:

◇ amounts and counts can never be less than zero and could be arbitrarily large. Heights, powers, areas, distances, numbers of deaths or of people, all fall here. The simplest indicator of whether re-expression is likely to help us look at a single batch is the ratio of the largest value to the smallest value. If this is small, close to 1, re-expression cannot change the appearance of the batch seriously. If this ratio is quite large, say 100 or more, we are almost sure to need re-expression, just to see what is going on. (More about counts when we reach section 3F.)

◇ balances, where both positive and negative values occur. Profit-and-loss, actual minus predicted, and observed-minus fitted are all natural examples. One can usually think of this sort of data as the difference of two amounts or counts. Re-expression of the balance rarely helps, but re-expression of the amounts or counts before the subtraction sometimes helps a lot.

◇ counted fractions and percentages, where the reasonable values are hemmed in on both sides. Re-expression is often very helpful, though the techniques are more special. (For the present we will avoid such cases. We will eventually come to dealing with counted fractions and percentages in chapter 15.)

◇ grades and other ordered versions--including A, B, C, . . . , E and −, +, ++, +++, ++++ as examples--often respond well to slightly more complex techniques. (We may get to them in a later volume.)

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	3F	
16★		85
	3G	
17★		86
18		90
19		91
	3H 3K 3P	
20★		94
21★		95
22★		96

review questions

What is an amount? A balance? A count? A counted fraction? What is the most urgent reason for re-expression? What suggests that re-expression will probably be needed?

3A. Logarithms = logs

What kinds of re-expression we will most frequently find helpful will depend upon the habits of our fellow men, on the ways men habitually choose to write down the numbers that come to us. Civilized "people" evolved from bears, for example, might have different habits than those evolved from apes. If so, they would be likely to find other forms of re-expression most useful.

As we have noticed, numbers that are always positive and that are not too closely alike are likely to gain from some form of re-expression. The most likely form is taking logarithms. (The next most likely form is taking square roots, to which we will come in turn.)

Logarithms--or as we shall regularly say, logs--can be made either simple or mysterious. We shall try to keep them simple.

What do you need to know about logs? In the first instance,

- how to find good enough logs fast and easily (see next section),
- that equal differences in logs correspond to equal ratios of raw values. (This means that wherever you find people using products or ratios--even in such things as price indexes--using logs--thus converting products to sums, and ratios to differences--is likely to help.)

Some readers may like a review of the few simple things they need to know about logs; this they can find in section 3K. There are different kinds of logs, but this makes little difference to us here. Of any two kinds, each must be a constant multiple of the other, so that no more care is needed than is needed in keeping feet and inches straight. We will thus routinely write merely "log"--meaning the kind of log that makes arithmetic easiest, logs to the base 10, also called common logarithms. (Should we need any other kind, we would make a special note.)

The relation of logs to various powers, particularly fractional ones, can also be of interest. (We will come to this in section 3G.)

So much for generalities. Let us take up the example of exhibit 6 of chapter 1. If we look back to that display, we see a strong tendency for the leaves to be washed into the corners where the leading digits are 1 or 2. What happens if we look at the logs of the same values? Exhibit 1 shows the result.

Notice that extremes, hinges, and median are given both in logs and in megawatts. If often helps to display both re-expressed and raw values when we are working with re-expressed numbers.

This stem-and-leaf display lets us get a rather good feel for the ultimate sizes of the Bureau of Reclamation's power plants. We see, perhaps more easily in the compressed form, a tendency to spread out to high values that is

greater than the tendency to spread to low values. The statement that only plants of 15 megawatts or more are included now begins to be related to something we can see. Clearly, the stem-and-leaf in logs is more effective than the stem-and-leaf in raw megawatts. Getting the values where we can look at them better has meant getting them much more nearly symmetrical.

exhibit 1 of chapter 3: ultimate powers

Ultimate power of 34 hydroelectric plants of the Bureau of Reclamation (logs found from powers in megawatts)

A) STEM-and-LEAF — for logs, unit = 0.01

		†	
1	11*	8	
2	12	0	
5	13	488	
7	14	08	
11	15*	2606	
14	16	508	
16	17	08	
√(2)	18	85	
·	19*		
16	20	80863	
11	21	38	
9	22	1	
8	23*	5	
7	24	60	
5	25	8	
4	26	3	San Luis
	27*		
	28		
3	29	5	Glenn Canyon
	30		
2	31*	3	Hoover
	32		
1	33	0	Grand Coulee

B) Same, SQUEEZED, — unit = 0.1

		†	
1	1*	1	Boysen
5	t	2333	
11	f	445555	
16	s	66677	
(2)	·	88	
16	2*	0000011	
9	t	23	
7	f	445	
4	s	6	San Luis
3	·	9	Glenn Canyon
2	3*	1	Hoover
1	t	3	Grand Coulee

† = does not include plants smaller than 15 megawatts (118 for panel A, 11 for panel B)

C) LETTER-VALUE DISPLAYS — logs and raw

34 pwrs (log megawatts)

M17h		1.86	
H 9	1.52	2.21	.69
1	1.18	3.30	2.12

34 pwrs (megawatts)

M17h		73	
H 9	33	162	129
1	15	1974	1959

S) SOURCES

The World Almanac, 1966, page 263.
Their source: U.S. Bureau of Reclamation.

Raw version: Exhibit 6 of chapter 1. (Also appears in exhibit 5 of chapter 1, and exhibit 3 of chapter 2.)

review questions

What is the single most needed form of re-expression? What is the minimum that we should know about logs? How many letter-value displays are likely to help in summarizing numbers re-expressed to be looked at?

3B. Quick logs

We have just seen an example where two-decimal logs were quite precise enough. Indeed, as the righthand side of exhibit 1 shows, we can do quite well in this example with one-decimal values. This is but one example of many. Though there are exceptions, in most situations where re-expression in logs really helps, two-decimal logs help as much as many-decimal ones.

Since we can find two-decimal logs easily, as we shall soon see, our sailing instructions are clear:

- ◇ plan to use two-decimal logs first always.
- ◇ in rare cases where we need more precision, go back and start again with many-decimal logs.

Overall, we will do much less work this way.

To get few-decimal logs easily, special kinds of tables are very helpful. They make looking up logs just a matter of finding our place and reading the answer. (No interpolation!)

For one-decimal logs, we can use the first two columns of the following little table, whose use is easily learned. (It gives only the fractional part of the log; you have to find the integer part as usual.)

First nonzero digits of x	First decimal of log x		Break
			8913
8913 to 1122	.0	.0	**1122**
1123 to 1412	.1	.1	**1413**
1413 to 1778	.2	.2	**1778**
1779 to 2238	.3	.3	**2239**
2239 to 2818	.4	.4	**2818**
2819 to 3548	.5	.5	**3549**
3549 to 4466	.6	.6	**4466**
4467 to 5623	.7	.7	**5624**
5624 to 7079	.8	.8	**7079**
7080 to 8912	.9	.9	**8913**

If we use the standard code--"when in doubt, go to an even answer"--we can compress this table quite a lot further. We have only to use the last two columns, entering between the breaks. (We go to an even answer when we come in exactly at the break.)

Tables that work like this are often called critical tables. We will try to be more specific and call them

break tables.

Exhibit 2 uses two break tables to make finding two-decimal logs easy

exhibit 2 of chapter 3: reference table

Break table for two-decimal logs

A) MAIN BREAK TABLE

Break	log	Break	log	Break	log	Break	log	Break	log
9866		**1567**		**2483**		**3936**		**6237**	
1012	.00	**1603**	.20	**2541**	.40	**4027**	.80	**6383**	.80
1035	.01	**1641**	.21	**2600**	.41	**4121**	.81	**6531**	.81
1059	.02	**1679**	.22	**2661**	.42	**4217**	.82	**6683**	.82
1084	.03	**1718**	.23	**2723**	.43	**4315**	.83	**6839**	.83
1109	.04	**1758**	.24	**2786**	.44	**4416**	.84	**6998**	.84
1135	.05	**1799**	.25	**2851**	.45	**4519**	.85	**7161**	.85
1161	.06	**1841**	.26	**2917**	.46	**4624**	.86	**7328**	.86
1189	.07	**1884**	.27	**2985**	.47	**4732**	.87	**7499**	.87
1216	.08	**1928**	.28	**3055**	.48	**4842**	.88	**7674**	.88
1245	.09	**1972**	.29	**3126**	.49	**4955**	.89	**7852**	.89
1274	.10	**2018**	.30	**3199**	.50	**5070**	.70	**8035**	.90
1303	.11	**2065**	.31	**3273**	.51	**5188**	.71	**8222**	.91
1334	.12	**2113**	.32	**3350**	.52	**5309**	.72	**8414**	.92
1365	.13	**2163**	.33	**3429**	.53	**5433**	.73	**8610**	.93
1396	.14	**2213**	.34	**3508**	.54	**5559**	.74	**8810**	.94
1429	.15	**2265**	.35	**3589**	.55	**5689**	.75	**9016**	.95
1462	.16	**2317**	.36	**3673**	.56	**5821**	.76	**9226**	.96
1496	.17	**2371**	.37	**3759**	.57	**5957**	.77	**9441**	.97
1531	.18	**2427**	.38	**3846**	.58	**6095**	.78	**9661**	.98
1567	.19	**2483**	.39	**3936**	.58	**6237**	.79	**9886**	.99

When in doubt use an even answer, thus, 1462 gives .16 and 1486 gives .18.

B) SETTING DECIMAL POINTS

1	10	1	**1**
10	11	2	**0.1**
100	12	3	**0.01**
1000	13	4	**0.001**
10,000	14	5	**0.0001**
100,000	15	6	**0.00001**
1,000,000			**0.000001**

C) EXAMPLES

Number	**B**		**A**		**log number**
log 137.2	2	+	14	=	2.14
log 0.03694	−2	+	57	=	−1.43
log 0.0896	−1	+	95	=	−0.05
log 174,321	15	+	24	=	5.24

Let us give an example of how this works in practice. Exhibit 3 gives the ultimate power in megawatts of the hydroelectric stations and power plants of the Corps of Engineers, first as raw values and then as (unsorted) logs. Let us trace a few examples. 14 comes between 1396 and 1429, so its two-decimal log ends in .15. Similarly, 18 comes between 1799 and 1841, so its two-decimal log ends in .26. Note the entry--15, 26,--of these logs in the first line of panel B.

The gain from going to logs is shown in the stem-and-leaf for logs in Panel C. Although a little long and gangling--something we will soon learn to cure --it does give us a reasonable picture of how these ultimate powers are spread out. (Panel A gave good storage but little help in visualization.)

exhibit **3** of chapter 3: ultimate powers

Re-expression of the ultimate power in megawatts of 62 hydroelectric stations and power plants of the Corps of Engineers

A) STEM-and-LEAF			B) RE-EXPRESSED in LOGS (unit = 0.01)		
2	1*	4,8	1**	15,26	Philpott, St. Mary's
4	2	8,6	··	45,42	J. Percy Priest, Narrows
9	3	6,0,0,4,0	··	56,48,48,53,48	
11	4	3,5	··	63,65	
13	5*	4,2	··	73,72	
17	6	0,8,8,1	··	78,83,83,79	
21	7	5,6,0,6	··	88,88,85,88	
22	8	6	··	93,	
24	9*	0,6	1··	95,98	
30	10	0,0,0,0,0,0	2**	00,00,00,00,00,00	
√(4)	11	0,2,8,0	··	04,05,07,04	
28	12	4	··	09	
27	13*	0,5,5,0	··	11,13,13,11	
23	1**	75,65,40	··	24,22,15	
20	2	50,80,04,00,70	··	40,45,31,30,43	
15	3	40,20,30	··	53,51,52	
12	4	68,00	··	67,60	
10	5**	18,40,95	··	71,73,77	
7	6	00	··	78	Dworshak
	7		··		
6	8	10,10	··	91,91	Little Goose, Lwr Monumental
	9**		··		
4	1***	728,400,743	3**	24,15,24	Cheatham, McNary, The Dalles
1	2	700	3··	43	John Day

62 powers (megawatts)

M31h	110		
h16	68	280	212
1	14	2700	2686

62 powers (log megawatts)

M31h	2.04		
H16	1.83	2.45	.62
1	1.15	3.43	2.28

→

review questions

What is a break table? How do we use exhibit 2? Did re-expression help in exhibit 3? Why?

3C. Comparisons of two batches

One notable use of stem-and-leaf displays is for comparison of two batches. Use of logs can also help here. Exhibit **4** compares the land areas of

exhibit **3** of chapter 3 (continued)

C) STEM-and-LEAF for LOGS––in 0.01

11*	5	(Philpott)
12	6	(St. Mary's)
13		
14	25888	(Narrows, J. Percy Priest, ––, ––, ––,)
15*	36	
16	35	
17	2389	
18	335888	
19*	358	
20	00000044579	
21	11335	
22	24	
23*	01	
24	035	
25	123	
26	07	
27*	1378	
28		
29	11	(Little Goose, Lower Monumental)
30		
31*	5	(McNary)
32	44	(Cheatham, The Dalles)
33		
34	3	(John Day)

P) PROBLEMS

3a) Sort out the log values into a conventional stem-and-leaf.

3b) Do you like the result? How would you improve it?

3c) Carry out the improvement.

S) SOURCE

The World Almanac, 1966, and Book of Facts, page 265.

Their source: Corps of Engineers, U.S. Army.

the 83 counties of Mississippi with those of the 82 counties of Michigan. We see that county areas in Michigan:

- ◇ tail out much more toward large areas.
- ◇ are strikingly concentrated between 545 and 595 square miles (26 out of 83).

Panel B shows another way of displaying the same information. Some will prefer the one, others the other.

Let us look to see what the effect is of going over to logs. Exhibit **5** shows the same comparison for logs. On the whole, the comparison is clearer:

- ◇ setting aside the 23 Michigan counties with logs of 1.75 or 1.76, the rest of the Michigan counties appear as a rather dinosaur-shaped pattern--running from, say, 1.6 (blunt end) to 2.2 (tail)--while the pattern of the Mississippi counties appears rather rectangular--running from near 1.6 to near 1.9.

exhibit **4** of chapter 3: county areas

The land areas of the counties of Michigan and Mississippi

A) SIDE-by-SIDE STEM-and-LEAF--unit 10 square miles

Michigan, 83 counties		**Mississippi,** 82 counties
3*	2	3*
3·	75	3· 8
4*	75	4* 0121243121301214202
4·	8596689	4· 597886556569
5*	1042043214	5* 142010
5·	75877677685776776666678666	5· 977899958797
6*	4011	6* 412441
6·	865	6· 898598
7*	120142	7* 320341203
7·	65	7· 86657
8*	3212	8* 303
8·	668	8· 8
9*	101	9* 24
9·	6	9·
10*	313	10*
10·		10·
11*	1	11*
11·	8	11·
12*	00	12*
12·		12·
13*	2	13*
13·		13·
1**	58,84	

→

exhibit **4** of chapter 3 (continued)

B) BACK-to-BACK STEM-and-LEAF––leaves sorted, as well

Michigan			**Mississippi**	
Benzie	2	**3***		
Leelanau, Arewac	75	·	8	Tate
Charlevoix	1	**4***	0001111112222223344	
	9988665	·	555566678899	
	4443221100	**5***	001124	
88877777777776666666666655		·	577778899999	
	4110	**6***	112444	
	865	·	588899	
	422110	**7***	001223334	
	65	·	56678	
	3221	**8***	033	
	866	·	8	Hinds
	110	**9***	24	Bolivar, Yazoo
	6	·		
	331	**10***		
		·		
Delta, Gogebic	8	**11***		
		·		
Iron	00	**12***		
		·		
Ontonagon	2	**13***		
		·		
		14*		
		·		
Chippewa		**15***		
	8	·		
		16*		
		·		
		17*		
		·		
Marquette	4	**18***		
	(83, check		(82, check)	

P) PROBLEMS

4a) Imitate panel A with some data of interest to you.

4b) Imitate panel B with same.

S) SOURCE

The World Almanac, 1966, pages 369–370.

Here it might seem that the gain from re-expression came mainly from improving the picture for the separate stem-and-leaf pictures. Actually, however, the spreads are much more similar for logs, and the shape difference between states is more easily described in the log display. Thus the comparison is considerably improved.

Making stem-and-leafs "easier to look at" has so far mainly involved making patterns more symmetrical--sometimes the one and only pattern, sometimes some or most of the patterns. We might as well plan to try to make stem-and-leafs as symmetrical as we can do easily. This will do us no harm if we do not take doing it too seriously, and its byproducts will often reward us.

five-figure logs may not be enough

A different sort of example comes by turning to what was known in 1927 about the atomic weights of the chemical elements. The values in exhibit **6** are the results found by different investigators in the years 1922 to 1925, which were the basis for the German 1927 standard values for the atomic weights of

exhibit **5** of chapter 3: county areas

Logs of areas of counties of Michigan and Mississippi (from exhibit 4)

A) BACK-to-BACK STEM-and-LEAF--unit = 0.01 in log

Michigan		Mississippi
41	**15***	
7	·	8
1	**16***	0001111112222223344
9988665	·	55556667889
443332221100	**17***	0011234
99866666666666655555555555	·	66666677777999
3211	**18***	111133344
88766555	·	55566666788899
4332111	**19***	9224
8665	·	67
110	**20***	
8875	·	
2	**21***	
	·	
60	**22***	
	·	

Identifications on panel B of exhibit 4.

B) PROBLEMS

5a) Imitate panel A of exhibit 4 with the log ultimate powers of exhibits 1 and 3.

5b) Imitate panel A above with same.

exhibit 6 of chapter 3: atomic weights

Atomic weights of hydrogen and bromine regarded as best available in 1927

A) RAW ATOMIC WEIGHTS

Hydrogen	Bromine
1.00779	79.916
1.00781	79.920
1.00777	79.920
1.00782	79.927
1.00775	79.912
1.00766	79.926
1.00779	79.915
1.00769	79.915
1.00783	79.918
	79.909

B) STEM-and-LEAF for RAW VALUES--
unit = 0.001, values cut

Hydrogen		Bromine	
100*	777777777	**7990***	9
		7991	62558
		7992*	0067

C) LOG ATOMIC WEIGHTS--
six decimal values

Hydrogen	Bromine
.003370	1.902634
379	656
361	656
383	694
353	612
314	688
370	628
327	628
387	645
	596

D) COMPARATIVE STEM-and-LEAF--
log values, unit = .000001

Hydrogen		Bromine	
00331*	4	**190259***	6
00332	7	**190260**	
00333		**190261**	2
00334		**190262***	88
00335*	3	**190263***	4
00336	1	**190264**	5
00337	090	**190265**	66
00338*	37	**190266**	
		190267*	
		190268	8
		190269	4

E) NOTES

1. **The comparison of panel B shows us very little.**
2. **In panel D the bulk of the two batches is about equally spread out, and we see clearly the tendency of some (i.e., one or two) individual values to stray away from most others.**

S) SOURCE

W. A. Roth and K. Scheel, 1927. Erster Ergänzungsband (1st supplement to the 5th edition of) Landolt-Bornstein's Physikalische Chemische Tabellen.

hydrogen and bromine. Clearly, any attempt to compare by stem-and-leaves with equal units fails for the raw values.

When we turn to logs, we find that five-decimal logs are not good enough, so we use six-decimal values. Panel D, which compares log atomic weights in units of the sixth decimal, shows that stem-and-leaf pictures with a common unit can do well in comparing these batches, if applied to the right expression. Moreover, they can reveal the spread and straying characteristics of the original batches.

Here, the gain from re-expression comes from matching the stem-and-leaf displays better to each other. As we will see in the next chapter, this tends to be more important than gaining symmetry.

That log expression works well for atomic weights should not surprise us. For a given sort of amount, the best measurement accuracy--at a given date --is likely to be characterized by fluctuations and errors that are, roughly, a given fraction of that amount. We usually speak of having about so many significant figures. To have about a certain number of decimals in the log is usually closer to the truth, because this is closer to making "smallest change/value" a constant fraction.

review questions

Are stem-and-leaf displays useful for comparison? Can logs help? Always? Why/why not? Was exhibit 5 better than exhibit 4? Why/why not? How many decimal places should we use in logs to start with? May we find we need more? Fewer? Why should we expect logs to do fairly well with atomic weights? Name two sorts of advantages that we may be able to gain by re-expression? Which is the more important? Which gains its value mainly from byproducts?

3D. Quick roots and quick reciprocals

quick roots

We are also likely to want to find quick square roots. Here we can still do well with a break table, although the situation is not quite as nice, since we can't reuse a square-root table until x has changed by a factor of 100--not 10, as for logs--as we see from the following simple example:

$$\sqrt{2} = 1.42$$
$$\sqrt{20} = 4.47$$
$$\sqrt{200} = 14.2$$

We could get two-figure accuracy in our roots--we shall drop the "square-" except where it is really needed--with a table of the same form as

exhibit 3, but we would not be likely to like it when we had it. Going from

$$\sqrt{100} = 10 \quad \text{to} \quad \sqrt{121} = 11$$

and from

$$\sqrt{10000} = 100 \quad \text{to} \quad \sqrt{12100} = 110$$

is a big step. Going from

$$\sqrt{96.04} = 9.8 \quad \text{to} \quad \sqrt{98.01} = 9.9$$

and from

$$\sqrt{9604} = 98 \quad \text{to} \quad \sqrt{9801} = 99$$

is a much smaller step--only about one-tenth the size, however you figure it. These would be the smallest steps in the answers from an exactly two-figure table of square roots.

We can do better by taking smaller steps near 11 and larger steps near 98. Exhibit **7** is a break table that does just this.

This table will be put to use in the next example.

quick reciprocals

From time to time, we find re-expressing numbers by reciprocals useful. When we do this, it is often convenient to preserve order, to have the more positive raw value correspond to the more positive reciprocal. This doesn't happen automatically--in fact, quite the reverse. Thus, 3 is greater than 2, while 1/3--the reciprocal of 3--is less than 1/2--the reciprocal of 2.

The way out is simple:

Use negative reciprocals

These work well, since $-1/3$ is more positive than $-1/2$. Since we are likely to start with numbers like 57, it is often convenient to work with $-1000/\text{number}$.

Exhibit **8** gives a break table for negative reciprocals. Except for the minus sign, no new feature. To look up $-1000/43.7$, we note that 437 comes between 4347 and 4425, so that the nonzero figures of $-1000/43.7$ are -228 (to this accuracy). Fixing the decimal point, using panel A, gives us -22.8 for $-1000/43.7$.

a volcano example

Examples where steam-and-leaf displays of roots clearly give us the best grasp are harder to find than examples where this happens for logs--the latter seem to turn up all over.

Exhibit **9** sets out the heights of those 219 volcanoes that the National Geographic Society and the 1966 *World Almanac* found interesting enough to list. Clearly we have more trailing off to large heights than to small ones.

exhibit **7** of chapter 3: reference table

Break table for (square) roots

A) EXAMPLES

Start by dividing number into periods of two digits, decimal point falling between periods. Thus 124.2 is 1 24 2, but 1242 is 12 42. Like wise 0.00654 is 00 65 4 or 65 4.

Number	Periods	from B	from C	Number
124. 2	1 24 2	**ab.**	112	**11.2**
1242	12 42	**ab.**	35	**35.**
. 00654	00 65 4	**.0x**	80	**.080**

B) BREAK TABLES to SET DECIMAL POINT--enter between bold figures, leave with light figures

a	.x	**1**
ab.	.0x	**. 01**
abc.	.00x	**. 00 01**
abcd.	.000x	**. 00 00 01**
		. 00 00 00 01

C) MAIN BREAK TABLE--in and out as in panel B

Break	Root	Break	Root	Break	Root	Break	Root	Break	Root
98 01	100	**2 49 64**	160	**5 66**	240	**15 60**	40	**35 40**	60
1 02 01	102	**2 62 44**	164	**5 86**	246	**16 40**	41	**37 40**	62
1 06 09	104	**2 75 56**	168	**6 20**	252	**17 22**	42	**39 69**	64
1 10 25	106	**2 89 00**	172	**6 50**	258	**18 06**	43	**42 25**	66
1 14 49	108	**3 02 76**	176	**6 81**	264	**18 92**	44	**44 69**	68
1 18 81	110	**3 16 84**	180	**7 13**	270	**19 80**	45	**47 61**	70
1 23 21	112	**3 31 24**	184	**7 45**	276	**20 70**	46	**50 41**	72
1 27 69	114	**3 45 96**	188	**7 78**	282	**21 62**	47	**53 29**	74
1 32 25	116	**3 61 00**	192	**8 12**	288	**22 56**	48	**56 25**	76
1 36 89	118	**3 76 36**	196	**8 47**	294	**23 52**	49	**59 79**	78
1 41 61	120	**3 92 04**	200	**8 82**	30	**24 50**	50	**62 41**	80
1 48 84	124	**4 08 04**	204	**9 30**	31	**25 50**	51	**65 61**	82
1 58 76	128	**4 24 36**	208	**9 92**	32	**26 52**	52	**68 89**	84
1 69 00	132	**4 41 00**	212	**10 56**	33	**27 56**	53	**72 25**	86
1 79 56	136	**4 57 96**	216	**11 22**	34	**28 62**	54	**75 69**	88
1 90 44	140	**4 75 24**	220	**11 90**	35	**29 70**	55	**79 21**	90
2 01 64	144	**4 92 84**	224	**12 60**	36	**30 80**	56	**82 81**	92
2 13 16	148	**5 10 76**	228	**13 32**	37	**31 92**	57	**86 49**	94
2 25 00	152	**5 29 00**	232	**14 06**	38	**33 06**	58	**90 25**	96
2 37 16	156	**5 47 56**	236	**14 82**	39	**34 22**	59	**94 09**	98
2 49 64		**5 66 44**		**15 60**		**35 40**		**98 01**	

exhibit **8** of chapter 3: reference table

Break table for (negative) reciprocals (using −1000/number)

A) BREAK TABLES for SETTING DECIMAL POINT

Break	Start	Start	**Break**
1000			**1000**
10,000	.x	a.	**100**
100,000	.0x	ab.	**10**
1,000,000	.00x	abc.	**1.**
10,000,000	.000x	abcd.	**0. 1**
100,000,000	.0000x	abcde.	**0. 01**

Examples

Number	**A**	**B**	−1000/number
124.2	a.	−80	−8.0
.04739	abcde.	−212	−212**.
1242.	.x	−80	−.80

B) MAIN BREAK TABLE--digits of negative reciprocal

Break	Value	Break	Value	Break	Value	Break	Value	Break	Value
990	−100	**1639**	−60	**2469**	−40	**4115**	−240	**617**	−160
1010	−98	**1681**	−59	**2532**	−39	**4202**	−236	**633**	−156
1030	−96	**1709**	−58	**2597**	−38	**4274**	−232	**649**	−152
1053	−94	**1739**	−57	**2667**	−37	**4347**	−228	**666**	−148
1075	−92	**1770**	−56	**2740**	−36	**4425**	−224	**685**	−144
1099	−90	**1802**	−55	**2816**	−35	**4504**	−220	**704**	−140
1124	−88	**1835**	−54	**2899**	−34	**4587**	−216	**725**	−136
1149	−86	**1869**	−53	**2985**	−33	**4672**	−212	**746**	−132
1176	−84	**1905**	−52	**3077**	−32	**4762**	−208	**769**	−128
1205	−82	**1942**	−51	**3175**	−31	**4854**	−204	**793**	−124
1235	−80	**1980**	−50	**3287**	−30	**4950**	−200	**820**	−120
1266	−78	**2020**	−49	**3367**	−294	**505**	−196	**840**	−118
1299	−76	**2062**	−48	**3448**	−288	**515**	−192	**855**	−116
1333	−74	**2105**	−47	**3509**	−282	**526**	−188	**870**	−114
1370	−72	**2151**	−46	**3584**	−276	**538**	−184	**885**	−112
1408	−70	**2198**	−45	**3663**	−270	**549**	−180	**901**	−110
1449	−68	**2247**	−44	**3745**	−264	**562**	−176	**917**	−108
1493	−66	**2299**	−43	**3831**	−258	**575**	−172	**935**	−106
1538	−64	**2353**	−42	**3922**	−252	**588**	−168	**952**	−104
1587	−62	**2410**	−41	**4016**	−246	**602**	−164	**971**	−102
1639		**2469**		**4115**		**617**		**990**	

exhibit **9** of chapter 3: volcano heights

Heights of 219 of the world's volcanoes

A) STEM-and-LEAF--unit =100 feet (rounded)

```
  8    0 | 99766562
 18    1 | 9761009630
 40    2* | 6998776654442221109850
 58    3 | 876655412099551426
 80    4 | 9998844331929433361107
103    5* | 97666666554422210097731
√(18)  6 | 898665441077761065
 98    7 | 98855431100652108073
 78    8 | 653322122937
 66    9* | 377655421000493
 51·  10 | 0984433165212
 38   11 | 4963201631
 28   12 | 45421164
 20   13* | 47830
 15   14 | 00
 13·  15 | 676
 10   16 | 52
  8   17* | 92
  6   18 | 5
  5   19 | 39730
```

B) VARIOUS SUMMARIES--identifications at exhibit 6 of chapter 2

219 hghts (hunds. of feet)

```
M110 |      65      |
H 55h| 37h    95h   | 58
E 28 | 24     121   | 97
   1 | 2      199   |197
```

```
          87
f |-49h      182h  |  adj:  2 and 179
  | xxx       six  |  out:  185, 190,
F |           269h |  out:  193, 193
  |           xxx  |  out:  197, 199,
```

S) SOURCE

The World Almanac, 1966, pp. 282–283.

Their source: National Geographic Society.

exhibit **10** of chapter 3: volcano heights

Logarithmic heights of volcanoes (from exhibits 9 and 2)

A) STEM-and-LEAF -- units 0.01 in log height in hundreds of feet

```
  1    3*|    0
  .     4|
  .     5|
  .     6|
  5    7*|8808
  6     8|5
  8     9|55
 12    10|4000
 13    11*|1
 18    12|83080
 28    13|8884442200
 43    14|266533220650989
 58    15*|876644319944316
 80    16|9998844331929433361107
107    17|765555554433222100766219898
√(33)  18|34322111333221998876555558865595 96
 79    19*|033221111152479988876655577
 52    20|00432211032101686554655999889
 22    21|00344215599
 11    22|0215479998
  1    23*|0
```

B) VARIOUS SUMMARIES -- some identifications at exhibit 6 of chapter 2

219 logs (in 0.01) of hghts.†

```
M110 |       181       |
H 55h|157h      198|  40h
F 28 |138       208|  70
   1 |30        230|200

        |61|
   ┌──────────┐
F  |96h   259|    adj: 99, 99, 230
   |seven xxx|    out: 70, 78, 78, 78
F  |35h      |    out: 85, 95, 95
   |one      |    far:  30 (Ilha Nova)
   └──────────┘
```

(† heights in 100's of feet)

219 logs (hghts in 100's of feet)

```
M110 |      1.81      |
H 55h|1.58      1.98|
   1 |0.30      2.30|
```

219 logs (hghts in feet)

```
M110 |      3.81      |
H 55h|3.58      3.98|
   1 |2.30      4.30|
```

Note: 70 (Anak Krakatau), 78 (Surtsey), 78 (Matthew I), 78 (Fonualei), 85 (Tavurvur), 90 (Nuafo'ou), 95 (Didicas), 99 (Taal), 99 (Tinguan), 230 (Kilimanjaro).

Exhibit **10** shows the result of taking logarithms. We now have more trailing toward low logs than we have toward high logs. Clearly, we want something in between raw values and logs. Square roots are just the thing.

Exhibit **11** shows the stem-and-leaf for the roots. Now we have a nice symmetric picture, one that fits in well with a median of nearly 6500 feet and extremes near 200 and 19,900. If one wanted to compare interesting volcanoes with other interesting mountains, or to compare the interesting volcanoes of the northern and southern hemispheres--in each case as to height--we would do well to look at square roots of heights.

looking easier

Rather than peering at the three stem-and-leafs of exhibits 9, 10, and 11, we can do better by looking at the corresponding schematic plots. Exhibit **12** shows them to us. To many of us the message is even clearer.

exhibit **11** of chapter 3: volcano heights

Square roots of heights of 219 volcanoes (from exhibits 9 and 7, latter rounded)

A) STEM-and-LEAF--units 0.1 of roots of heights in hundreds of feet (equivalently, roots of height in feet)

1	**1***	4
6	**2**	65525
13	**3**	0032262
28	**4**	410409997776655
51	**5***	14432211043099867599687
73	**6**	2000220666644666684448
107	**7**	000000066444444444222200666228888
√(34)	**8**	24222000222220888666644448644864486
78	**9***	22220000042468888866644466
52	**10**	0044222202200068664686
29	**11**	0222000226886488
13	**12**	44488
8	**13***	2266
4	**14**	0000

B) LETTER-VALUE DISPLAY

219 $\sqrt{\text{hghts}}$ (in feet)

M	110	80		
H	55h	61	98	37
E	28	49	110	61
	1	14	140	126

adj: 14 (Ilha Nova), 140 (Kilimanjaro)

out: none

far: none

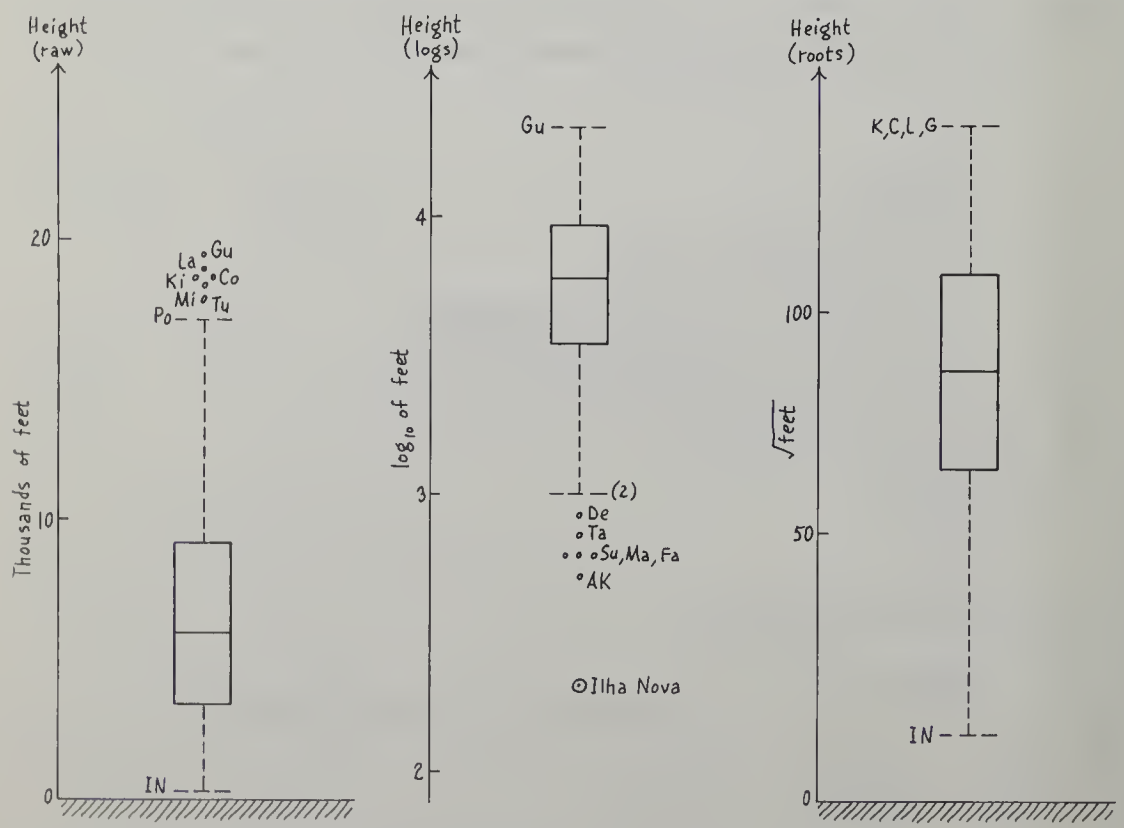
reciprocal times

It often seems natural to measure times for things to happen. How long does it take a rat--in training to learn a maze--to run through the maze (and get his reward)? How long does it take individual fish to stop swimming when exposed to a given concentration of poison? How long after the earthquake did this house fall down? A major difficulty with using times is the rat that never finishes the maze, the fish that never stops swimming, the house that never falls down.

Clearly, the rat who never finishes the maze is much more like the slow rat who takes 1000 seconds than either is like the very fast rat who runs in four seconds. Indeed, the 1000-second rat is very much like the 800-second rat, much more so than the 204-second rat is like the four-second rat.

exhibit **12** of chapter 3: volcano heights

Heights of 219 volcanoes, using three different expressions (schematic plots; for further identification see exhibit 6 of chapter 2)



The visible difficulty with "never" cases is a warning of a concealed difficulty with "very long time" cases.

We ought not to be using times in such cases; rather we should be using the reciprocal of time. For the rats we have mentioned as illustrations, and some others, we get the following sorts of results:

Time	−1000/time	1000/time
"never"	−0	0
1000 sec.	−1	1
800 sec.	−1.2	1.2
204 sec.	−49	49
10 sec.	−100	100
6 sec.	−167	167
5 sec.	−200	200
4 sec.	−250	250

If we want to continue to talk about slowness, the negative reciprocals are much better than the time. If we are willing to shift to fastness instead, the 1000/time values without the minus signs work very well indeed. (They are, of course, speeds or rates.)

Exhibit **13** shows examples from a rat-running study by Hull. Panel B shows some long times for the first pair of rats and an overall tendency for the second pair of rats to take less time. The pattern of spread is somewhat uncertain. Panel D shows the stem-and-leaf patterns for the reciprocal times. Three of the four rats now show a roughly symmetric scatter of about the same size, while the scatter for "Rat 2" is reasonably symmetrical, though somewhat more spread out. The general pattern--and the comparison--seems clearer for the reciprocal times.

If we had stuck to panel B, we would have thought that, so far as spread of results go, those for rats 0 and 1 were unusually spread out, while those for rat 3 were unusually compressed. Looking at panel D, we get a quite different impression--one that is more likely to be correct. In these terms, only rat 2 looks unusually spread out (because of his two very quick trips). **How we look at numbers can matter.**

Once in a long while, data will come to us with both "never" and "zero" as raw values. In such cases, we can usually replace "zero" by "half the smallest nonzero value that would have been written down if observed." We should, however, complain about the data collection.

review questions

What are the most common re-expressions besides logs? How do we use exhibit 7? Exhibit 8? What re-expression seemed to work best for volcano heights? Can schematic plots help us pick a re-expression? What does difficulty

with values that are either possibly or actually infinite suggest about large values? What is the simplest way to avoid such difficulties? What is the most likely re-expression for rats' running times? Would you expect this to help with horse-racing times?

exhibit **13** of chapter 3: rat running

Times and reciprocal times for rats on second day of training (from start through section 2 of a runway)

A) TIMES--in 0.1 second

Rat 0	Rat 1	Rat 2	Rat 3
76	119	108	56
127	186	39	70
261	93	65	81
137	224	29	57
74	128	59	46

B) STEM-and-LEAF displays for TIMES

0*			43	
0·	87	9	66	67865
1*	34	23	1	
1·		9		
2*		2		
2·	6			

C) RECIPROCAL TIMES--in 1000/seconds

132	84	92	180
78	54	258	144
38	108	152	124
72	45	340	176
136	78	168	216

D) STEM-and-LEAF displays for RECIPROCAL TIMES

0*	4	4		
0·	87	858	9	
1*	34	1	57	42
1·				88
2*				2
2·			6	
3*			4	
3·				

S) SOURCE

Clark L. Hull, 1934, "The rat's speed-of-locomotion gradient in the approach to food." *J. of Comparative Psychology 17:* 393–422. (Data from table 3 on page 401.)

3E. Looking quickly

Making three stem-and-leaf displays seems to be a hard way to find out what re-expression makes the batch reasonably symmetrical. Let us find an easier way.

The sort of re-expressions we ordinarily make--including roots, logs, and negative reciprocals--have the simple property of preserving order. Thus, for example, because 57 is larger than 43, $\log 57 = 1.76$ is larger than $\log 43 = 1.63$.

Preserving order necessarily preserves ranks and depths. Thus, for example, median, hinges, eighths, . . . , and extremes of log values are logs of the corresponding summaries of raw values; and the same is true for roots or negative reciprocals.

We have already noticed (in chapter 2) that the spacing of summaries throws light on the symmetry of the batch. How can we use such summaries most easily to judge symmetry? Probably by defining midsummaries, including a midhinge midH, a mideighth midE, and a midextreme mid1--often elsewhere called midrange--by

$$\text{Midhinge} = \frac{\text{lower hinge} + \text{upper hinge}}{2}$$

$$\text{Mideighth} = \frac{\text{lower eighth} + \text{upper eighth}}{2}$$

.
.
.

$$\text{Midextreme} = \frac{\text{lower extreme} + \text{upper extreme}}{2}$$

We then have only to look at the sequence

median, midhinge, midextreme

or at a longer sequence, such as

median, midhinge, mideighth, midextreme

and see if there is an indication of a trend.

To do this for a particular re-expression requires only re-expressing the corresponding summaries and then finding the midsummaries. Exhibit **14** does this for the volcano example. Whether we look at 3 midsummaries (from a 5-number summary), or 4 midsummaries (from a 7-number summary), or 5 midsummaries (from a 9-number summary), the answer is clear and strong: "use roots". (Only one stem-and-leaf--that for raw heights--was needed.) Panel D shows how midsummaries can be neatly attached to our "letter-value displays."

Several comments are in order:

◇ since the midextreme can be greatly affected by a single stray value, we are often reluctant to trust it as a base for picking out a re-expression.

◇ accordingly, starting from a 7-number or 9-number summary is often worthwhile.

◇ we often work with batches smaller than 219. When we do, we cannot expect as clear signals as we have found for this example.

◇ working with small batches, it is often worthwhile, when it is possible, to combine evidence from several batches.

exhibit **14** of chapter 3: volcano heights

Use of midsummaries to pick a re-expression for volcano heights

A) WORKING with the VALUES of a 5-NUMBER (MH1) SUMMARY

		(raw)	(root)	(log)	**Negative reciprocal**
219	**1**	199	140	230	−5
	H	95h	98	198	−10
110	**M**	65	80	181	−15
55	**H**	37h	62†	157	−26
1	**1**	2	14	30	−500
	M	65	80	181	−15
	midH	66h	80h	177h	−18
	mid1	100h	77	130	−152
	(trend)	(up)	(small)	(down)	(DOWN)

† Re-expression of 37h is 62; mean of re-expressions of 37 and 38 is (60 + 62)/2 = 61.

B) WORKING with the VALUES of a 7-NUMBER SUMMARY—added values bold

219	**1**	199	140	230	−5
	E	**121**	**110**	**208**	**−8**
	H	95h	98	198	−10
110	**M**	65	80	181	−15
55	**H**	37h	62	157	−26
28	**E**	**24**	**49**	**138**	**−42**
1	**1**	2	14	30	−500
	M	65	80	181	−15
	midH	66h	80	177h	−18
	midE	**72h**	**79**	**173**	**−25**
	mid1	100h	77	130	−252
	(trend)	(up)	(small)	(down)	(DOWN)

→

exhibit **14** of chapter 3 (continued)

C) WORKING with the VALUES of a 9-NUMBER SUMMARY––added values bold

	(raw)	(root)	(log)	Negative reciprocal
1	199	141	230	−5
D 14h	140	118	215	−7
E 28	121	110	208	−8
H 55	95h	98	198	−10
M110	65	80	181	−15
H 55	37h	62	157	−26
E 28	24	49	138	−42
D 14h	16	40	120	−62
1	2	14	30	−500
M	65	80	181	−15
midH	68h	80	177h	−15
midE	72h	79h	173	−25
midD	**78**	**79**	**167h**	**−34**
mid1	100h	77	130	−252
(trend)	(up)	(close)	(down)	(DOWN)

D) EXTENDED LETTER-VALUE DISPLAYS––of two alternative lengths

219	(mid)	raw		(spr)
M110	65	65		
H 55h	66h	37h	95h	58
1	100h	2	197	195

	(mid)	root		(spr)
	80	80		
	80	62	98	36
	77	14	140	126

	(mid)	log		(spr)
	181	181		
	177h	157	198	41
	130	30	230	200

	raw			
219	(mid)			(spr)
M110	65	65		
H 55h	66h	37h	95h	58
E 28	72h	24	121	97
D 14h	78	16	140	124
1	100h	2	197	195

	root			
	(mid)			(spr)
	80	80		
	80	62	98	36
	79h	49	110	61
	79	40	118	78
	77	14	140	126

	log			
	(mid)			(spr)
	181	181		
	177h	157	198	41
	173	138	208	70
	167h	120	215	95
	130	30	230	200

P) PROBLEMS

14a) Make a set of extended letter displays (MHE1) intermediate between the two sets of panel D.

14b) Extend the analysis of the volcano data to mid-summaries to an 11-number (MHEDC1) summary.

14c/d/e) Examine the areas of the counties of Michigan––data in exhibit 4––for a preferred re-expression using 5/7/9-number summaries.

14f/g/h) Do the same for Mississippi (same source) using 5/7/9-number summaries.

14i) Return to the log values of exhibit 1 and find midsummaries. What further re-expression is needed for symmetry?

14j) How much sense do you think this further re-expression makes?

exhibit **15** of chapter 3: warp breaks

Tippett's 54 counts of warp breaks (9 in each of 6 conditions)

A) DATA

AL 26,30,54,25,70,52,51,26,67
AM 18,21,29,17,12,18,35,30,36
AH 36,21,24,18,10,43,28,15,26
BL 27,14,29,19,29,31,41,20,44
BM 42,26,19,16,39,28,21,39,29
BH 20,21,24,17,13,15,15,16,28

B) SUMMARIES and MIDSUMMARIES, RAW--in counts

	1	E	H	M	H	E	1	M	midH	midE	mid1
AL	25	26	26	51	54	67	70	51	40	46h	47h
AM	12	17	18	21	30	35	36	21	24	26	24
AH	10	15	18	24	28	36	43	24	23	25h	26h
BL	14	19	20	29	31	41	44	29	25h	30	29
BM	16	19	21	28	39	39	42	28	30	29	29
BH	131	15	15	17	21	24	28	17	18	19h	20h
					sum of all batches			170	160h	176h	176h
					sum except AL			119	120h	130	129

(Except for the sum for all of either M or midH, increasing outwards)

C) SUMMARIES and MIDSUMMARIES, ROOTS--in 0.1's

	1	E	H	M	H	E	1	M	midH	midE	mid1
AL	50	51	51	72	74	82	84	72	62h	66h	67
AM	35	41	42	46	55	59	60	46	48h	50	47h
AH	32	39	42	49	53	60	66	49	47h	49h	49
BL	37	44	45	54	56	64	66	51h	50h	54	51
BM	40	44	46	53	62	62	64	53	54	53	52
BH	36	39	39	41	46	49	53	41	42h	44	44h
					sum of all batches			315	305h	317h	311h
					sum except AL			243	243	250h	244h

(reasonably level)

P) PROBLEMS

15a/b/c/d/e/f) Make stem-and-leafs and check the calculations of summaries for: AL/AM/AH/ /BL/BM/BH.

15g) Imitate panel C for log counts.

15h) Find a set of counts involving several batches and make a similar computation.

S) SOURCE

L. H. C. Tippett (1950). **Technological Applications of Statistics.** New York: John Wiley and Sons. London: Williams and Norgate. Page 106.

combining evidence across batches

Exhibit **15** presents counts, used by Tippett, of the number of warp breaks during a fixed amount of weaving for each of six types of warp--AL, AM, AH, BL, BM, BH--differing in kind of cotton and tightness of twist. The results for AL are a little peculiar (E and H both 26, M 51), so we look at both the sums over all six batches and sums over all five except AL. For raw counts, while there is some irregularity involving either the sum of medians or the sum of midH's when we look at all six, there is little doubt that the sum of midsummaries increases outward. For square roots of counts, it is clear that the sums are relatively constant.

If we want symmetrical spreading of our warp counts, we are thus told to use roots.

review questions

What is easy and natural to use in looking for symmetry--or lack of it--for a batch? What is a midhinge? A mideighth? A midextreme? How do we label each? Why do the sorts of summaries and midsummaries we use save us work? How clearly do midsummaries point to a particular re-expression for volcano heights? How do we fit midsummaries into letter-value displays? Should we believe midextremes? Implicitly? Not at all? How can we combine midsummaries across batches?

3F. Counted data

Most of our examples have so far involved measured data. We have slighted counted data (except for our last example, Tippett's warp counts).

(Some readers feel quite happy re-expressing heights, areas, megawatts, amounts of money, and the like, but develop a curious, almost unmanageable conscience when someone suggests re-expressing the number of times something occurred. All that listening to such a "conscience" can do is to make it more likely for them to **miss what they could have found.**)

When we have to deal, in later chapters, with special cases, such as

- ◇ rather small counts (especially when zero appears).
- ◇ how many out of a given number survived, smiled, were satisfied, etc.,
- ◇ counts arranged in special patterns,

there will be some refinements we can wisely use in re-expression.

For the present, though, so long as we deal with reasonably large counts, we may as well plan to re-express batches of counts **just as freely, and for all the same reasons,** as batches of amounts. The re-expressions most frequently useful for counts are logs and (square) roots.

In fact, it is rather hard to find a set of counts that are better analyzed as raw counts than as root counts. (Some sets of counts are all so nearly alike as to make their expression unimportant, but we may as well go at least as far as root counts with them.)

Exhibit **16** shows in logs the 1960 population of the 50 states, shown in raw counts in panel B of exhibit 4 of chapter 2. Clearly, it takes re-expression for us to look at these values. We do particularly well with the squeezed stem-and-leaf of panel B.

Exhibit **17** shows a still more extreme example--the numbers of deaths for selected causes in 1964 (omitting "all other diseases"). What appears to have led to the selection of these "causes" is a curious mixture of popular interest (acute poliomyelitis), ease of diagnosis (motor vehicle accidents), and importance (arteriosclerotic heart disease). Again, we see stem-and-leafs both for raw values and for logs. Panel B provides extra detail about the very smallest values. The condensed stem-and-leaf for logs in panel C gives us our first clear picture. We see:

- ◇ a pronounced tendency to trail off toward low values.
- ◇ a weak tendency to trail off toward high values.
- ◇ a tendency toward asymmetry, even when these extreme trailing values are eliminated, the lower values being stretched out farther.

Even if we disregard the extreme trailing values, the use of square roots deserves a trial. This is done in panel D. We find that, even in the compressed form, we still do not get too clear a picture of all these values as a batch. The tendency of the three highest values--for "malignant neoplasm of digestive organs and peritoneum", "vascular lesions affecting central nervous system", and "arteriosclerotic heart disease"--to appear disconnected from the main body of causes is now more marked.

We are thus led to think that this particular collection of causes of death includes:

- ◇ a number of unusually small (by comparison with the others) counts, mainly associated with infectious diseases.
- ◇ a general mass of medium-sized counts.
- ◇ three unusually large counts, which seem rather like "catch-alls".

By re-expressing our counts, we have learned something about even this unpromising set of data.

review questions

Should we be doubtful about re-expressing counts? Which re-expressions are most frequently useful for counts? What did we learn about causes of death? How much did re-expression help?

exhibit **16** of chapter 3: state populations

The 1960 populations of the 50 states (raw values in tens of thousands as in panel B of exhibit 4 of chapter 2)

A) LOGS--unit 0.01 **(for storage)**

1	**13***	6
2	**14**	6
4	**15**	29
5	**16**	5
6	**17***	9
12	**18**	003338
15	**19**	589
	20	
17	**21***	15
21	**22**	4557
25	**23**	4478
√ 25	**24**	04589
20	**25***	1135
16	**26**	0004679
9	**27**	18
7	**28**	9
6	**29***	89
4	**30**	05
	31	
2	**32**	02
	33*	

B) SAME, SQUEEZED--unit = 0.1 **(for viewing)**

	1*		
1	**t**	3	Alaska
4	**f**	455	
6	**s**	67	
15	**1·**	888888999	
17	**2***	11	Ariz, Nebr
√ 25	**t**	22223333	
25	**f**	444445555	
16	**s**	666666677	
7	**2·**	899	
4	**3***	00	Ill, Pa
2	**t**	22	Cal, NY

C) VARIOUS SUMMARIES of 50 STATE POPULATIONS

	in millions				in logs of 10,000's				in logs of 1's			
M25h	2.5	2.5			2.39	2.39			4.39	4.39		
H13	2.6	0.89	4.3	3.4	2.29h	1.95	2.64	.69	4.29h	3.95	4.64	.69
1	8.6	0.23	17		2.29	1.36	3.22		4.29	3.36	5.22	

P) PROBLEMS

16A) Imitate panels A and B for the 1970 populations of the 50 states. Does an apparent gap between smaller populations and larger populations continue?

16b) Imitate panels A and B for

log 1970 population − log 1960 population

Compare your results for those for the separate years.

16c) Go back to earlier censuses, and continue the comparison.

S) SOURCE

The World Almanac, 1966, page 325.

Their source: U.S. Bureau of the Census

3G. Relation among powers and logs (optional)

We have said that logs and square roots are more likely to be useful ways of re-expression than any others. We have seen--by way of the heights of interesting volcanoes--that the square root does seem to fit in halfway between raw values and logs. This is enough to take us through most situations.

As preparation for the future, though, we may wish to look at a wider variety of powers, and to learn in more detail how logs fit in among the powers.

exhibit **17** of chapter 3: deaths by cause

Deaths for 59 selected causes, 1964 (total 1,798,051, less "all other diseases 54,000")

A) SMALL COUNTS

Raw	Log	Cause
17	1.23	Polio
42	1.62	Diphtheria
93	1.97	Whooping cough (WC below)
95	1.98	Scarlet fever and strep throat (SFST below)

B) RAW VALUES--in 100's

0*	7,3,8,4,2
1*	7,8,3,1
2	6,5
3	5,8
4	9,4,6,4
5*	9,4
6	5
7	6
8	2
9*	9,9,8
1**	35,35,67,59,22,11,10
2	62,77,57,34,32,03,52,53,06
3	28,23,72,07
4	92,00,69
5**	32,74,78,69
0***	932,
1***	982,
2	
3	
4***	454,

C) LOGS--in 0.1's

·		
s	7	Polio
f		
t	3	Diph
−0*	00	WC,SFST
0*		
t	3	Abortion
f	4	Dysentery
s	6	Measles
·	89	
1*	01	
t	22	
f	4455	
s	666677	
·	8899	
2*	00000011	
t	23333	
f	444444555	
s	6667777	
·	9†	
3*		
t	3†	
f		
s	7†	
·		

† See text for identification

→

exhibit **17** of chapter 3 (continued)

D) ROOTS--of counts (cut)

0**	001111
t	22233
f	44555
s	6666777
·	889999
1*	00111
t	22
f	4455
s	666667
·	889
2*	01
t	23
f	444
s	
3*	
t	1†
f	
s	
·	
4*	
t	
f	5†
s	
·	
0**	7†

† **See text for identification**

P) PROBLEMS

17A) Collect (as in panel B) similar data for deaths in 1969.

17b) Imitate panel C.

17c) Imitate panel D.

17d) Find and display changes in logs from 1964 to 1969.

17e) Find display changes in roots from 1964 to 1969.

17f) Discuss the results of (17d) and (17e).

S) SOURCE

The World Almanac, 1966, page 299

Their source: National Center for Health Statistics

We are now trying to think--and learn--about re-expression in general, not just re-expression for making better stem-and-leaf displays. Thus we should be fairly general, which is likely to mean saying some things that may seem trivial.

Changing the numbers in a batch--or in some more complicated data structure--by multiplying or dividing all of them by the same constant usually does not affect the ease with which they can be looked at--either directly or after most standard arithmetic analyses. To make such a change is to do no more than to change feet to inches or meters, or to change kilowatts to megawatts. We usually regard this sort of change as quite trivial--both so far as easing our analysis goes (or making it more difficult) and in its effects on what the answers will be. (We do need to keep careful track of what we have done, so that we can find our way back when the analysis is completed.) Indeed, we will call such changes

trivial re-expressions.

We noticed above that the sort of re-expression we are discussing is mainly used for numbers that are all of one sign. If all the numbers we have--or might reasonably have had--are of one sign, there is likely to be a special significance for zero.

There are many purposes for which adding or subtracting the same constant to each number in a batch--or in some other data structure--is just as trivial as multiplying or dividing by a common constant. The most important exception is this:

- ◇ if there is a natural zero for the initial numbers, then, once we have added or subtracted a constant, shifting the zero, we are not likely to take a power, a root, or a log of the modified numbers--unless we are dealing with small counts, where shifting "none counted" to a small positive valuc is often followed by taking a root or a log.

Surely the simplest important change in how something is expressed in numbers is to replace the raw numbers by the same simple power of each of them. It is not usual for this power to be a square or a cube. Because of the ways people choose to make instruments and write numbers down--as we have noticed--we much more frequently change them in thc opposite direction, as by taking:

- ◇ square roots (1/2 powers).
- ◇ reciprocals (−1 powers).
- ◇ reciprocals of square roots (−1/2 powers).

(For all these but one we have already seen critical tables.)

A look at these powers

$$-1, \quad -1/2, \quad +1/2, \quad +1$$

(where the last entry refers to the raw data itself), reveals an obvious gap. What about the zero power?

We were almost all onee taught that anything raised to the zero power was 1. Our teachers were not wrong. But that does not mean that there ean be nothing to fill the gap.

It turns out that the role of a zero power is, for these purposes of re-expression, neatly filled by the logarithm.

Exhibit **18** shows how $y = (\text{eonst}) \cdot (\log x)$ fits neatly in among the simple powers. Note that negative powers need negative signs to make y increase as x inereases.

A central point is that moving up the scale--whether from x to x^2 to x^3, or from $-1/x^2$ to $-1/x$ to $\log x$--**corresponds to emphasizing differences among larger x's in comparison with differences among smaller x's.** To do the opposite, emphasizing more strongly differenees among smaller x, we need to move down the scale--as from x to $\log x$ to $-1/x$.

In re-expressing data, we ordinarily work with a narrower range of powers than those just illustrated. Not +3 to −3 but +1 to −1 is what we usually cover. Exhibit **19** shows similar information for the narrower range. All the same remarks apply. (Note that the horizontal scale for this plot was re-expressed to make the overall appearanee elear.)

quick reciprocal roots

To eomplete a set of break tables for the simple ladder requires filling in the (negative) reciproeal square root. We could give a break table for this form of re-expression. Since it seems to be very rarely useful--and sinee we ean do well enough by using two break tables in sueeession, one for roots and one for (negative) reciproeals--we shall omit it.

quick squares

The frequent peneil- and-paper user of more classical statistical proee-dures soon learns the table of squares up quite a way. We ean also do fairly well by using the break table for roots backwards. (Details left to the reader.)

review questions

What re-expressions are considered trivial? Always? Sometimes? What simple powers are rarely used for re-expression? What did (or should) our teaehers tell us about zero powers? Were they right or wrong? Is there any paradox on logs playing the role that zero powers fail to play? What happens as we move up the scale of powers? Down the scale? How ean we find semiquick reciproeal roots when needed?

exhibit **18** of chapter 3: powers and logs

The "shape" of simple functions of x

(Expressions of the form $A + B \cdot f(x)$, with A and B chosen to make the result tangent to $x - 1$ at $x = 1$.) Labels give $f(x)$. For A and B see below.

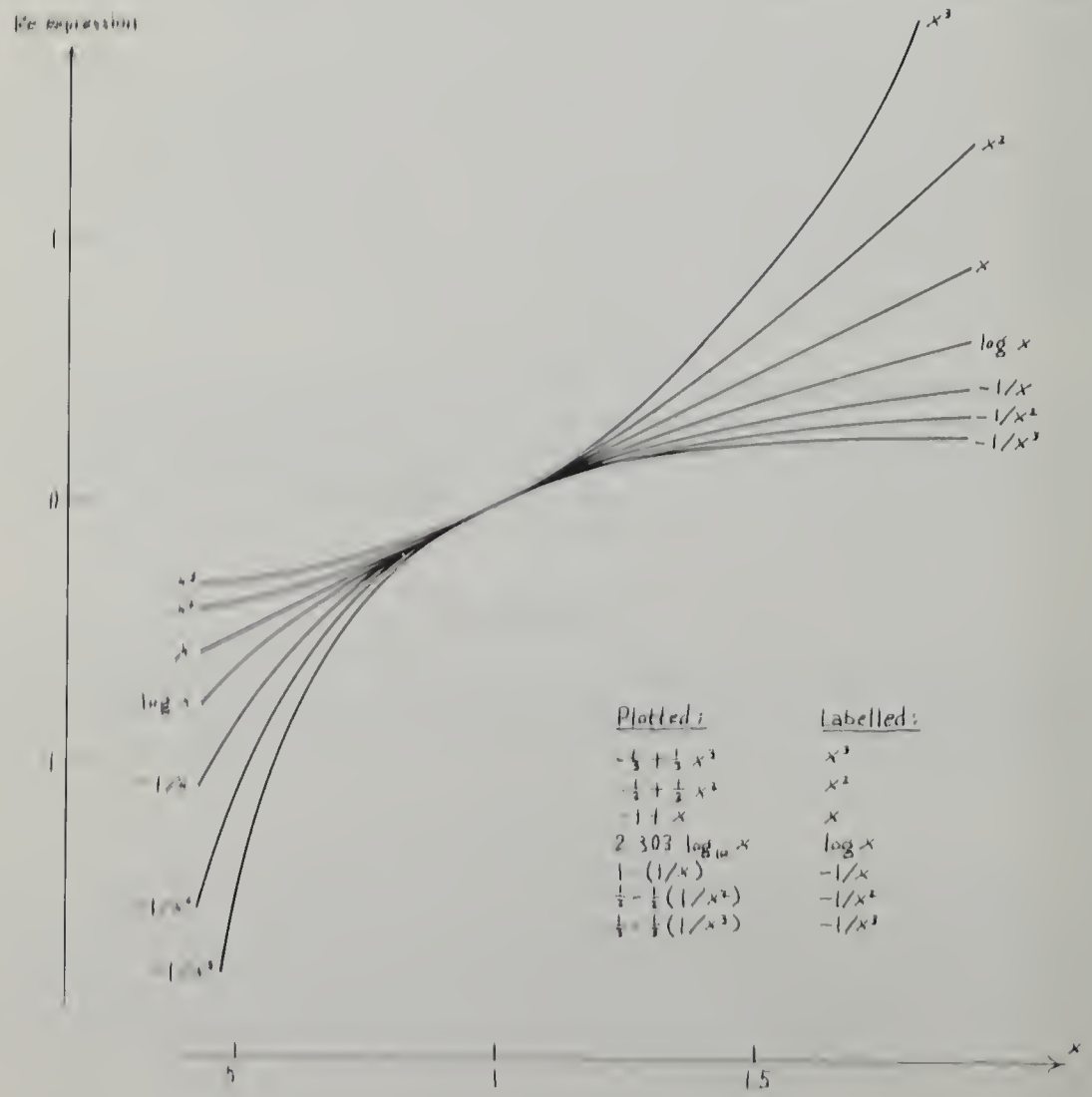
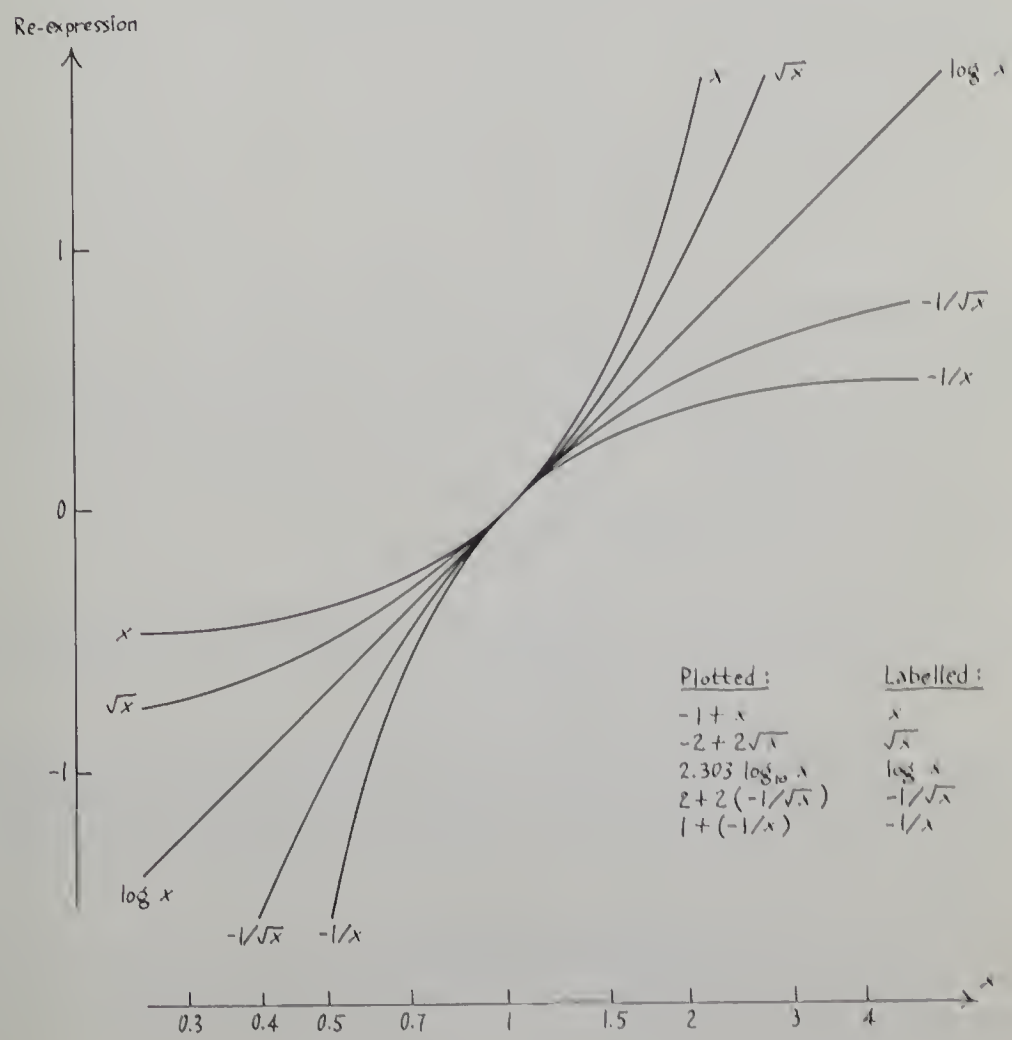


exhibit **19** of chapter 3: powers and logs

Logs and the usual powers



3H. How far have we come?

Data can be broadly divided into:

◇ amounts and counts. These are likely to need re-expression unless the ratio of largest to smallest value present is close enough to 1 for re-expression not to matter. Re-expression will usually be by logs, though other roots or powers can sometimes help.

◇ balances. Here re-expression is less frequent, and must usually be done before the balances are formed, not afterward. (Logs of anything are always balances.)

◇ counted fractions, percentages, and grades. Here re-expression is often essential, but detailed methods for doing it are going to have to wait for several chapters.

Break tables make re-expression in the commonest ways:

◇ first, logarithms,

◇ second, square roots,

◇ third, (negative) reciprocals,

◇ and even, (negative) reciprocal square roots,

both easy and quick, so long as few-digits answers will do, as is usually the case.

Counted data, unless we are dealing with quite small counts, are handled in the same way as amounts. We rarely lose by going as far as roots, and often find it pays to go to logs.

Whenever it matters, we do well to treat logs as playing the role of the zeroth power, so far as re-expression goes.

We are ready:

◇ to use break tables for few-digit re-expression, planning to occasionally come back and use more precision (efficient laziness).

◇ to "start" counts--and amounts that can be zero--by adding the same constant to each count that we are dealing with BEFORE we re-express them. (When counts are all large, starting is not likely to be worth the trouble.) If the counts include zeros, and we are to take logs, starting is essential.

◇ to calculate midsummaries, and to look toward their trends for more delicate indications of asymmetry than are given by just looking at a stem-and-leaf display of the values.

We now regard re-expression as a tool, something to let us do a better job of grasping data. In this chapter we have only begun: The grasping is with the eye and the better job is through a more symmetric appearance. As we go on, we shall learn about a variety of other ways in which re-expression can help us.

We have begun to realize that taking a firm hand with the data, before we display it or make detailed calculations, is often either the best or the only thing to do.

3K. How to think about logs (background)

We want to use logarithms--"logs"--freely as a tool. The points that follow make such use easier and are recalled for reference.

1. Logs can be looked up in tables.
2. Logs come "to" different bases; however, logarithms to different bases differ only by multiplicative constants. As a result, in data analysis, any base is almost (usually exactly) as good as any other.
3. We shall use "$\log_{10}$" for logs to the base 10 when we need to be sure (and will later use "$\log_e$" or "ln" for logs to the base e).
4. The log of 1 is 0.
5. The log of zero is not defined.
6. The log of a negative number can be defined, but is not a real number. (We shall not try to use logs of negative numbers or zeros, though sometimes we may write "L" in their place, meaning "treat as lower than any number.")
7. The log of a product is the sum of the logs of the factors, thus

$$\log uvw = \log u + \log v + \log w$$

8. The log of a ratio is the difference of the logs of numerator and denominator, thus:

$$\log(s/t) = \log s - \log t$$

9. The log to base 10 of 10 is 1 (that of 100 is 2, of 1000 is 3, of 0.1 is -1, of 0.01 is -2, of 0.001 is -3, etc.).

3P. Additional problems

See exhibits **20** to **22**.

exhibit **20** of chapter 3: data and problems

A) Tonnage shipped from ports from Maine to Key West in 1964, (B) Largest lakes in each U.S. State

A) TONNAGE--nearest 1000 tons

8*	4,9	Newport, Rockland
9*	6	Cambridge, Md.
1**	63,65,68,71,78	
2	86	
3	18,63,64,71,87	
4	33,58	
5**	66	
6	24	
7	·	
8	14,70	
9**	09,	
1***	127,207,249,358	
1···	460,771,826	
2***	127,223,349,397,699	
3	162,717	
4	158,643,992	
5***	107	
6	·	
7	161	
8	192,937	
9***	232	
1****	2206,3206,5206	
1···	7622,7841,8830	
2****	0011	Boston
3	·	
4****	3949†,7042†,8220†	
0*****	·	
1	49151	New York

†Norfolk, Philadelphia, Baltimore

B) AREAS in SQUARE MILES

0*	0‡,1‡,4,4,4,5,6,7,7,7,8,9	
1*	1,5	
2	0,5,9	
3	1,1,8	
4	7	
5*	1,5,8,9	
6	0,3	
7	2,6,8	
8	0	
9*	4	
1**	05,08,11,11,17	
1··	23,33,36,37,40	
1··	49,57,60,88,92	
2**	15,47,52	
3	60,82	
4	30,51	
5**	·	
6	09,30	
7	00	Okeechobee
1***	033‡,500‡,	
0****	9910	Erie
1	·	
2	2400	Michigan
3****	1800	Superior

‡0.31 Candlewood in Connecticut
0.66 Kolna in Hawaii
1033 Iliamna in Alaska
1500 Great Salt in Utah

P) PROBLEMS

20a) Panel A gives the amounts of tonnage shipped in 1964 from ports from Maine to Key West. What expression is indicated? Make the corresponding stem-and-leaf. Comment.

20b) Panel B gives the area of the largest lake in each state. What expression is indicated? Make the corresponding stem-and-leaf. Comment.

S) SOURCE

The World Almanac, 1966, (A) page 766, (B) page 285.

exhibit **21** of chapter 3: data and problems

(A) Areas of 4 oceans and 21 seas (including "Malayan seas"), (B) Lengths of "important" rivers of the U.S. (some multiply named)

A) AREAS--in 1000 sq. miles

1*		
2	9	Bass Strait
3		
4		
5*		
6	3†,9†	
7		
8		
9*	2†,2†	
1**	63,69,78,80	
2	22	
3	08,89	
4	76,82	
5**	90,96	
6	·	
7	·	
8	76	
9**	67	
1***	063,146,998	
2***	1	
0****	5440	Arctic
1	·	
2	8356	Indian
3	1839	Atlantic
4	·	
5****	·	
6	3802	Pacific

B) LENGTHS--in miles

1**	10,28,10,00,31,00,12
1··	37,44,40,40,48,50,64,60,64
1··	71,78,83,75,69,85,90
2**	02,30,10,17,33,10,15,37,46,50
2··	65,60,59,60,55,50,50,73,68
2··	82,90,86,80,92,91,81,80,87,76
3**	15,25,27,06,00,32,10,00,14,01
3··	30,29,36,40,38,50,52,50,50,50
3··	60,60,60,92,75,80,80,82
4**	07,09,20,24,25,31,35,44,48
4··	50,60,75
5**	20,00,05,38,40,45,50,50,60
6	00,05,18,25,30,52,87,90
7	10,30,35,60
8	40,60,62,70,90
9**	06,81
1***	018,038,171,214,306,360
1···	450,450,885‡
2***	315‡,348‡,553‡
3	710

† 63 Gulf of California
69 Engl. Channel & Irish Sea
92 Gulf of St. Lawrence
92 Persian Gulf

‡ 1450 Arkansas
1450 Colorado (of Texas)
1885 Rio Grande
2315 Missouri
2348 Mississippi
2533 Missouri-Red Rock
3710 Mississippi-Missouri-Red Rock

P) PROBLEMS

21a) Panel A gives the areas of 4 oceans and 21 seas. What expression is indicated? Make the corresponding stem-and-leaf. Comment.

21b) Panel B gives the lengths of "important" rivers of the U.S. What expression is indicated? Make the corresponding stem-and-leaf.

S) SOURCES

The World Almanac, 1966: (A) page 275, (B) pages 266–267.

exhibit **22** of chapter 3: data and problems

(A) heights of "famous" waterfalls (both highest fall and total fall may appear for one waterfall) and (B) reservoir capacity of major world dams

A) Heights--in feet

0**	40,54,65,66,68,
0··	70,70,75,90,96,98
1**	01,09,15,20,25,25
1··	30,30,30,32,40,44,
1··	50,51,65,68,86
1··	93,95,98
2**	00,07,07,13,14,18
2··	20,30,40,45,51,51
2··	56,66,70,75,68
3**	00,00,08,11,15,17
3··	20,30,30,35,44,45
3··	55,60,70,70,94,94
4**	00,00,06,27
4··	50,59,59,70
5**	00,05,08,18,25
5··	40,42,90,94,97
6	00,20,26,30,40,50,56
7	00,26,41
8	20,30,48,80,89,90
9**	74,84,84
1***	000,100,170,218,250,312
1···	312,312,325,350,385
1···	430,535,600,612
1···	640,650,696,904
2***	000,425,600,648
3	110†,212†

† **3110 Tugela (5 falls), 3212 Angel**

B) CAPACITY--in 1000's of acre feet

0*	8	Speicheri
1*	·	
2	·	
3	2	Curnera
4	1	Zeuzier
5*	4	Alpe Gera
0**	61,70,70,81,81	
1**	14,37,39,46,48	
1··	52,61,86,95	
2	19,65	
3	24	
4	87	
5**	49	
0***	600,602,746,756,930	
1***	261,325,375,405,586,709	
2***	000,030,030,092,095	
2···	106,367,446,717	
3***	024,453,468,484,648,789	
4	413,493,500,500	
5***	·	
6	100,550,600	
7	055,060	
8	000,000,512	
9***	171,402,730,890	
1****	0945,2940,4755	
1····	9715,9000,9400	
2	3600,4500,4800,7000,7160	
3	1618,2471	
4	7020	Kuibyshev
5****	·	
0*****	62000	Portage Mt
1*****	15000‡,27281‡,45115‡	

‡ **115000 Manicouagan #5. 127281 Sadu-El-Aali (High Aswan) 145115 Bratsk**

P) PROBLEMS

22a) Panel A gives the heights of "famous" waterfalls. What expression is indicated? Make the corresponding stem-and-leaf. Comment.

22b) Panel B gives the reservoir capacity of major dams. What expression is indicated? Make the corresponding stem-and-leaf. Comment.

S) SOURCE

The World Almanac, 1966: (A) page 286; (B) page 260.

Effective comparison, including well-chosen expression

4

chapter index on next page

In dealing with batches, we have already given some attention to two reasons for choosing one form of expression rather than another:

- ◇ symmetry of spreading within each batch separately.
- ◇ agreement from batch to batch in amount of spreading.

Neither is a "big deal."

Symmetry of spread is, by itself, probably a "little deal."

Fortunately for all of us, however, conflict between these reasons is infrequent. A choice good for one is, much more often than not, a choice good for the other.

Agreement of spread is rather more important; it deserves to be a "middle deal".

Indeed, in situations where more important reasons apply, what is good for the present minor reasons is, again much more often than not, good for the major reasons. There are exceptions, and when we recognize one, the major reason will have to take control, but such instances are not common. Choice of expression is rarely a balancing of conflicting reasons. Much more often it is a matter of stretching the information the data offers about alternative choices far enough to make a choice.

Most batches of data fail to tell us exactly how they should be analyzed. In making our most careful choices, we typically have to depend upon other bodies of data on similar subjects and upon experience.

Choice of expression is only one of such choices.

This does not mean that we cannot do relatively well on the basis of the data before us; often, indeed we can do tremendously better than if we used the data in a raw form, or made other choices without thought.

We will not meet "big deals" until we come to deal with more structured data--data with more "handles" for us to work with.

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review questions

What are two reasons for choice of expression? How important are they? Do they usually agree with each other? With major reasons about which we have not yet learned? Does a body of data usually indicate clearly how it should be analyzed?

4A. Alternative forms of display of summaries

Suppose we know how we want to analyze our data and what sorts of summaries we want to suggest or present. There is always still a question of presenting them. There are choices among alternatives, sometimes among more alternatives than we think of at first.

Rainfall (including snow converted to rain) at New York City offers a simple and convenient example. Exhibit **1** shows the data for the first six years of each of seven decades. Exhibit **2** shows four ways to display the data in more or less summarized form:

- ◇ as stem-and-leaf displays.
- ◇ as medians (unadorned).
- ◇ as slightly graphic medians.
- ◇ as schematic plots, modified to emphasize medians.

exhibit **1** of chapter 4: New York City rainfall

Precipitation* in New York City (to nearest inch) in years ending with 0, 1, 2, 3, 4, and 5

A) DATA

	189-	190-	191-	192-	193-	194-	195-
–0	52	42	36	49	35	45	45
–1	41	47	40	34	36	36	47
–2	39	47	38	43	39	50	46
–3	53	49	44	37	50	40	38
–4	44	42	34	38	45	52	43
–5	36	44	41	37	33	46	41
median	**42h**	**45h**	**39**	**37h**	**37h**	**45h**	**44**

* Rain plus rain equivalent of snow.

P) PROBLEM

1a) Find the data for 1960 to 1965, and use it to extend exhibit 2, below, to this additional decade.

S) SOURCE

Available from a variety of sources, including The World Almanac.

exhibit **2** of chapter 4: New York City rainfall

Four kinds of summaries of New York City precipitation compared

A) STEM-and LEAF COMPARISON––unit 1 inch

	1890–95	1900–05	1910–15	1920–25	1930–35	1940–45	1950–55
5·							
5*	23				0	02	
4·		779		9	5	56	567
4*	14	224	014	3		0	13
3·	69		68	778	569	6	8
3*			4	4	3		

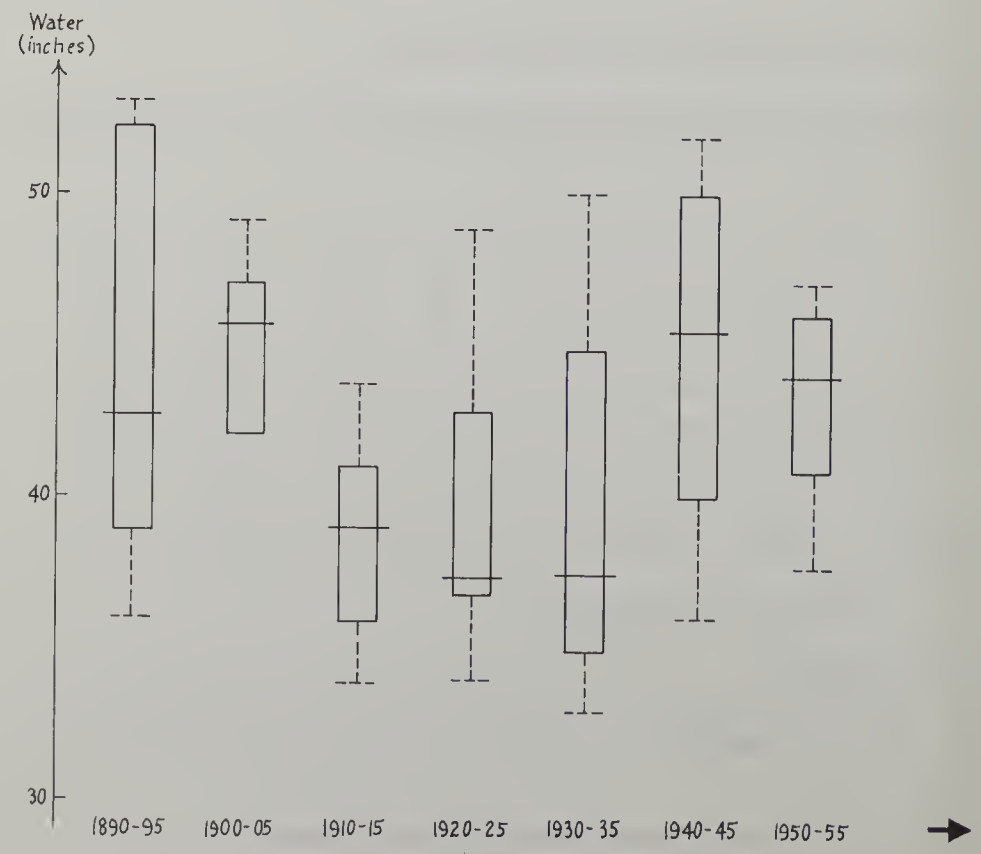
B) MEDIANS––unadorned

1890–95	1900–05	1910–15	1920–25	1930–35	1940–45	1950–55
42.5	45.5	39.0	37.5	37.5	45.5	44.0

C) PARTLY GRAPHIC MEDIANS

44:45		45h			45h	44
42:43	42h					
40:41						
38:39			39			
36:37				37h	37h	

D) SCHEMATIC PLOTS––modified to stress medians



All four displays tell the same story:

- ◇ in the 1890's, 1900's, 1940's, and 1950's, high typical annual rainfalls, near 44 inches;
- ◇ in the three intervening decades, lower typical rainfalls, near 38 inches.

There is now an understandable message--1910 to 1935 seems to have been a period of lowered rainfall. Whether or not this ought to be regarded as accidental is a confirmatory question, a question we have not tried to answer--and will not try to answer here. However, it should be regarded--since anything that happens once may happen twice--this earlier low-rainfall period might well have been a warning to prepare for the "drought" of the 1960's. Our concern is that use of any one of these displays uncovers a message in this instance--and that their use in other instances may also uncover messages.

How well does each display do? There will undoubtedly be some differences of opinion, as well there may be. If we compare the stem-and-leaf (panel A) with the schematic plot (panel D), I find little to choose. (If there were 600 values in each batch, rather than 6, we would almost surely prefer the schematic plot.) It is easier to compare the other pair of panels, in which the information about each batch is pared down to one number--the median--with each other. The expenditure of a few extra lines to make the medians partly graphic has shown us directly--with one *coup d'oeil*--most of what is going on, yet there has been no loss of detail in panel C as compared with panel B. We should be on a constant lookout for opportunities to use partly graphic displays.

Comparison of one pair of displays with the other is--and should be--harder. Sometimes we need the additional detail about the spread of the batches. Sometimes it merely confuses us slightly. We have to be guided by what we know or feel about the users, be they us or be they someone else.

review questions

What were the four kinds of display? What story did each tell? How did they pair off? How do the two displays in each pair compare? How did one pair compare with the other?

exhibit **2** of chapter 4 (continued)

P) PROBLEM

2a) Make a display like panel C, showing values for hinges (light) as well as for medians (bold), omitting hinges when they would overlap. How well do you like the result? Compare its effectiveness with panel C.

4B. Comparing several batches (continued)

When medians alone do not tell us enough, we want at least to look at schematic summaries, probably in the form of schematic plots. We want these summaries to tell us the story as clearly and as simply as they can. Both symmetry of spread within batches and, especially, balance of spread between batches, will help.

Exhibit **3** gives--in numbers--summaries of three batches from Winsor and Clarke's (1940) analysis of the catch of plankton using different kinds of nets. Three choices of expression are compared in panels B and C--raw count, root count, and log count. Whether we concentrate on symmetry--looking for midsummaries that do not drift within a batch--or on balanced spread--looking for spreads that agree from batch to batch--we come to the same position:

- ◇ raw counts are unacceptable.
- ◇ we are torn between root counts and log counts, and might even like to find a compromise.

exhibit **3** of chapter 4: Plankton hauls

Estimated number of three kinds of plankton caught (in six hauls of each of two nets)

A) ESTIMATED COUNTS

Kind I: 387, 428, 470, 497, 537, 540, 620, 760, 845, 895, 1020, 1050
Kind IV: 6060, 7600, 7900, 8260, 8600, 8900, 9250, 9830, 10200, 11000, 15500
Kind III: 189, 223, 278, 281, 288, 290, 314, 328, 328, 346, 395, 433 (hundreds)

B) RAW COUNTS--summaries (rounded) in units

	Kind I				Kind IV				Kind III			
M6h	580	580			9075	9075			302**	302**		
H3h	677	484	870	386	9048	8080	10,015	1935	308**	280**	337**	57**
1	718	387	1050		10,780	6060	15,500		311**	189**	433**	

H-spreads INcrease rapidly from left to right; mids INcrease from M toward 1.

C) ROOT COUNTS--summaries (rounded) in units

	Kind I				Kind IV				Kind III			
M6h	24	24			96	96			172	172		
H3h	26	22	29h	7h	95	90	100	10	176	168	184	16
1	26	20	32		101	78	124		172	136	208	

H-spreads INcrease slowly from left to right; mids INcrease very slightly from M towards 1. →

exhibit **3** of chapter 4 (continued)

D) LOG COUNTS––summaries in 0.01 (rounded)

	Kind I				Kind IV				Kind III			
M6h	276	276			396	396			448	448		
H3h	281	268	294	26	396	391	400	9	449	445	453	8
1	280	259	302		398	378	419		446	428	464	

H-spreads DEcrease moderately slowly from left to right; mids nearly neutral.

E) HOW MUCH RE-EXPRESSION for SIMILAR SPREAD?

Kind	**log M**	**log Hspr**
I	2.76	2.59
IV	3.96	3.29
III	4.48	3.76

Diff			**Ratio**
III − I	1.72	1.17	.7
IV − I	1.20	.70	.6
III − IV	.52	.47	.9

Ratios between .5 and 1.0; hence look at $\sqrt{\ }$ and log.

P) PROBLEMS

3a) Extend panel A to include eighths and mideighths. Any change in conclusions?

3b) Extend panel B to include EH-spreads. Any change in conclusions?

3c) Extend panel C to include log EHspr's and corresponding differences and ratios. Any change in conclusions?

3d) Invent a set of batches where logs will make spreads reasonably constant. Do the calculations of panel C. Are the ratios what you would expect?

3e) Invent a set of batches where roots will make spreads reasonably constant. Do the same.

S) SOURCE

C. P. Winsor and G. L. Clarke (1940). Journal of Marine Research (Sears Foundation) 3: 1.

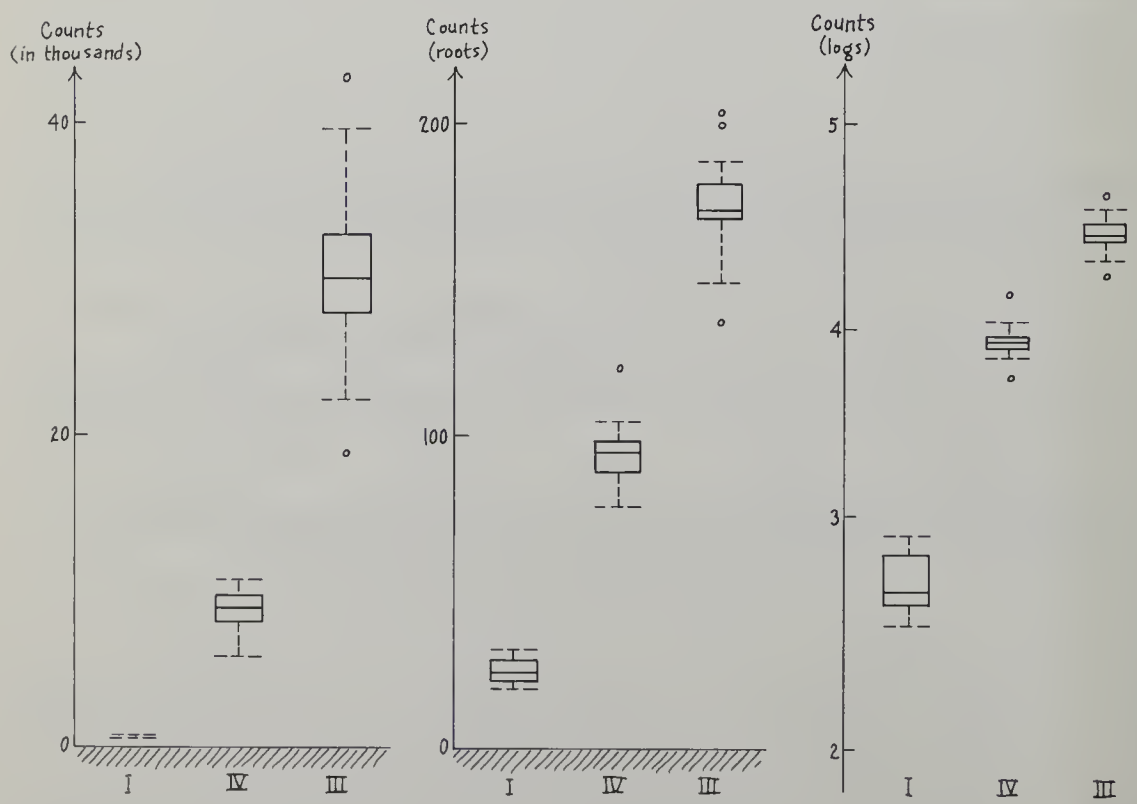
Also used by: G. W. Snedecor (1946). *Statistical Methods,* 4th ed., page 451.

Exhibit 4 shows the schematic plots graphically. The leftmost, using raw counts, is of little use as a set of schematic plots--we can see hardly more than if the medians were shown. The schematic plot for kind I is too small to see any detail. That for kind III is so large as to claim much undeserved attention. Either of the other two expressions gives about the same information, makes about the same impression. We cannot say that one is clearly better. We can say that either will clearly do.

We ought not to have to calculate through all three cases in order to find which one or two are likely to be satisfactory. Panel E of exhibit 3 introduces a way of learning a fair amount from much less work. If we look at the relation between median and H-spread--also, in suitably large batches, between median and E-spread, etc.--we can learn roughly what change of expression is likely to help. This is easier to do in terms of logs.

exhibit 4 of chapter 4: plankton hauls

Schematic plots for the data of exhibit 3, according to three choices of expression



Accordingly, panel E gives first log median and log H-spread of each batch of raw values, and then the differences in these logs as we step from one batch to another. As the ratios show, the change in log H-spread is rather more than 1/2 the change in log median and rather less than 1 times that change.

Had these ratios been close to 1/2, we would expect roots to help us. Had they been close to 1, we would expect logs to help us. Since they are in the wishy-washy in-between, we may expect either to help us, usefully but not perfectly.

review questions

What do we ask of schematic plots? Why? Where in exhibit 3 do we look to see how well what we ask is provided? Does exhibit 4 convey the same message about choice of expression for these particular data? How can we use the results of analyzing one choice of expression to guide us in what expression is likely to do well for us?

4C. A more extensive example

In 1950, Bruner, Postman, and Mosteller reported the detailed results of a simple experiment in which subjects looked at a flat picture--the Schroeder staircase--which can be easily seen in perspective in two ways. They were given different instructions as to how they were to try to manage their perspective, and the number of changes of perspective were counted in each of 10 successive minutes under each instruction. Setting aside the first two minutes, to protect somewhat against "starting-up" effects, there are 19 batches (one per subject) of 8 counts under the "alternate" instruction.

The early columns of exhibit **5** show the median M and hinges H for each of the 19 batches. To make both looking at and plotting easier, the subjects (rows in the exhibit) have been arranged in increasing order of median counts. These early columns also give log M, H-spread, and log Hspr. The logs are plotted in exhibit **6** and clearly trend upward.

If we ask how fast they trend upward, the easiest way to get a rough answer is to pick out some "typical" points near each end, and then to hold your clear plastic triangle or ruler across the picture, turning and slipping the edge to what seems to you to be a good fit. In the upper panel of exhibit 6, it seems natural to take the two points at the far lower left and the one point at the far upper right. The lower portion of panel A of exhibit 5 calculates the tilts of the two lines thus fixed as

$$\frac{134 - 48}{216 - 108} = \frac{86}{108} = .8 \quad \text{and} \quad \frac{134 - 30}{216 - 122} = \frac{104}{94} = 1.1.$$

exhibit **5** of chapter 4: Schroeder staircase

Calculations for perspective reversals (all logs in 0.01)

A) BASIC CALCULATIONS for THREE EXPRESSIONS

				raw counts		
					H spread	
#	M	log M	H	H	raw	log
2	12	108	11	14	3	48
(*) 9	16h	122	16	18	2	30
13	22	134	21	23	2	30
19	22	134	20	22	2	30
12	28	145	22	48	26	142
4	29h	147	24	32	8	90
16	33h	152	32	35	3	48
15	34	153	33	36	3	48
11	34	153	30	38	8	90
6	36	156	30	41	11	104
3	36h	156	34	38	4	60
18	36h	156	34	44	10	100
5	38h	159	37	41	4	60
14	44	164	43	48	5	70
7	45	165	42	46	4	60
8	64	181	54	67	13	111
1	74h	187	64	98	34	153
10	92	196	86	95	9	95
(*) 17	144h	216	132	154	22	134
17 – 2		108			19	86
diffs 17 – 9		94			20	104
(ratios) (17 – 2)						(0.8)
(tilts) (17 – 9)						(1.1)

→

P) PROBLEMS

5a) Plot the raw H-spread for raw counts against raw medians (also for raw counts). Is there a trend? Which points near each end of the point cloud would you choose as reasonably typical?

5b) Do the same for the raw H-spread for root counts.

5c) Do the same for the raw H-spread for log counts.

5d) Look up the source and apply a similar analysis to the last 8 minutes under the "natural" instruction.

5e) Do the same for the "hold" instruction.

5f) Find logs of median root counts, and plot against logs of median raw counts. What do you see? Should you have expected it?

5g) Find logs of median log counts and plot against logs and median row counts. What do you see? Should you have expected it?

→

exhibit **5** of chapter 4 (continued)

Panel A continued

root counts (*p)				log counts			
		H spread				H spread	
H	H	raw	log	H	H	raw	log
33	37	4	60	104	115	11	104
40	42	2	30	120	126	6	78
46	48	2	30	132	136	4	60
45	47	2	30	130	134	4	60
47	69	22	134	134	168	34	153
49	57	8	90	138	151	13	111
57	59	2	30	151	154	3	48
57	60	3	48	152	156	4	60
55	62	7	85	148	158	10	100
55	64	9	95	148	161	13	111
58	62	4	60	153	158	5	70
58	66	8	90	153	184	11	104
61	64	3	48	157	161	4	60
66	69	3	48	163	168	5	70
65	68	3	48	162	166	4	60
74	82	8	90	173	183	10	100
80	99	19	128	181	200	19	128
93	98	5	70	193	198	5	70
115	124	9	95	212	219	7	85
		5	35			−4	−19
		7	65			+1	7
			(.3)				(−.2)
			(.7)				(.1)

PROBLEMS (continued)

5h) In view of (5f) and (5g), do you think it would have mattered if the center and lower panels of exhibit 6 had used logs of median root counts and log counts, respectively, in place of logs of median row counts? Check your answer for the lower panel.

5i) Do you think such a choice will ever matter? If not, why not? If yes, give an example.

5j) Look up the Bruner, Postman, and Mosteller paper, and apply the condensed approach (see text) to the data for the "natural" instruction.

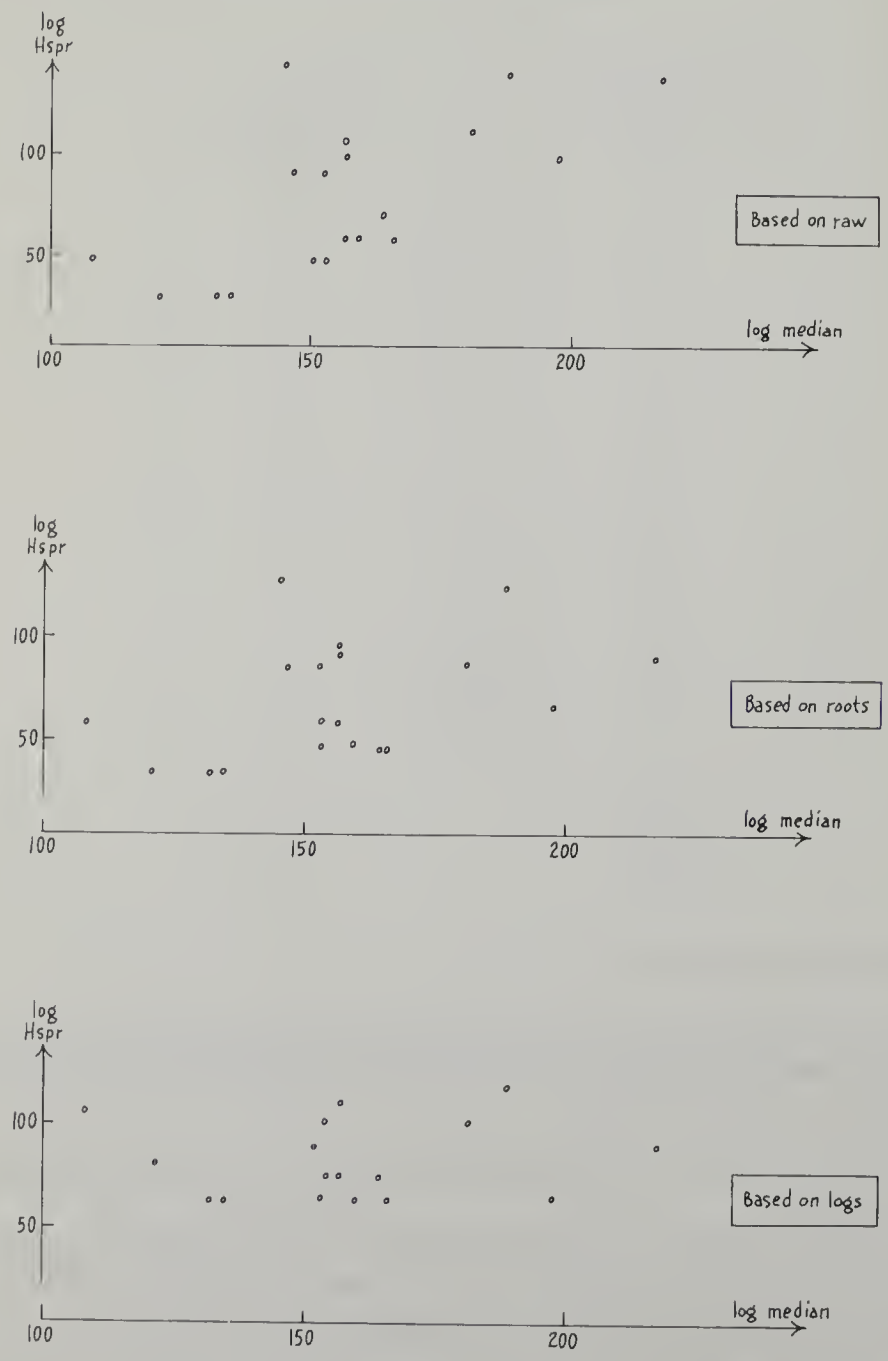
5k) Find a set of data of about the same size that interests you and carry through a similar analysis.

S) SOURCE

J. S. Bruner, L. Postman, and F. Mosteller (1950). "A note on the measurement of reversals of perspective." Psychometrika 15: 63–72. Table 1 on page 65.

exhibit **6** of chapter 4: Schroeder staircase

Plots of log H-spread against log median for the perspective reversal examples



Clearly, the slope is about 1. If the rule of thumb mentioned above works, this would lead us to expect that logs will do well as an expression that keeps the H-spread from trending.

Later columns of exhibit 5 show calculations of hinges, H-spread, and log H-spread for both expression in roots and expression in logs. The center and lower panels of exhibit 6 show the corresponding plots. The trend is clearly smaller for roots and negligible for logs.

For reasons pretty well explained by the answers to problems (5f) to (5i), we have not bothered to change the horizontal scales in the center and lower panels of exhibit 6. Instead we have used log median (raw count) throughout. (This makes the three pictures easier to compare; but this is a little thing, since our next step is to learn how to make one picture do.)

a condensed approach

If we had wanted to hold our calculation to a reasonable minimum, we could have avoided most of exhibits 5 and 6. Since we are dealing with counts, we expect to use either roots or logs. So we may go ahead as follows:

- ◇ find raw median and hinges; find log raw median.
- ◇ find root hinges; find H-spread for roots; find logs of same.
- ◇ make picture of log H-spread (for roots) vs. log M (for raws).
- ◇ select typical points; find tilts for same.
- ◇ working with the typical points only, find hinges for other expressions; find H-spreads; find logs of same.
- ◇ compare tilts of log H-spread for typical points.
- ◇ choose expression. (If we're lucky, it will be the one we started with.)

In this example, the tilt (against log median of raw counts) drops by almost 1/2 at each step of the re-expression scale (1.1 for raws, 0.7 for roots, 0.1 for logs). Dropping by about 1/2 happens very often.

Note also the unusually high position (on any of the three plots) of one subject who turns out to be #12. Turning back to the original data, we find that his ten counts, in order of occurrence were:

30, 22, 14, 22, 24, 32, 18, 48, 52, 53.

It would appear that after seven minutes this subject suddenly learned how to make his perspective reverse rapidly.

review questions

To speed our calculations what kind of points should we select from our first picture? How can we use them to find a tilt? What did the rule of thumb say about tilts? How well did it apply in this example? What expressions should we try first with counted data? Was any one of the 19 subjects used by Bruner, Postman, and Mosteller unusual? How? Are you surprised? Why/ why not?

4D. The meaning of comparison

We have said much about comparison. We have used many pictures and many numbers to make comparisons. We have not yet said enough about comparison itself. It is time we did.

Two kinds of comparisons come up in the simplest of common language: "Bill is a head taller than Jim." "George weighs twice as much as his brother Jack." Each of these statements says what you must do to one person to make him match the other. The first statement bases itself on a plus sign, and says how much we would have to add to Jim's height to match Bill's. The second bases itself on a times sign, and says what we would have to multiply Jack's weight by to match George's.

There are many ways in which addition is simpler than multiplication; doing hand arithmetic and sliding things bodily along graphs are only two. Logarithms were invented centuries ago to reduce multiplication to addition, thus making hand arithmetic much easier. We can, do, and should use them to avoid comparisons by multiplication. When multiplication would otherwise seem needed, taking logs will let us deal with comparison by addition.

When we think of comparison, then, we want to think of what needs to be added or subtracted to the data--as carefully expressed for analysis--in order to make one thing match another. If the data is originally, or commonly, expressed in another way, we will try to translate back to that expression, too.

Thus, if boys' weights are really better thought of as "twice" or "three-quarters" of one another, we will do our analysis in log weight, finding, perhaps, that we have to add 0.30 to Jack's to match George's. Having found this, we would also convert back to raw weight, and say that Jack's had to be doubled in order to match George's. Both forms of answer help.

review questions

What kinds of comparison come up in common language? How ought we think of such comparisons? How many forms of answer help?

4E. Adjustments, rough and exact

rough adjustments

If we want to put several batches under the same microscope at the same time, we need to have them centered at about the same level. If they are not

naturally somewhat similarly centered, we will have to adjust them so that they will be.

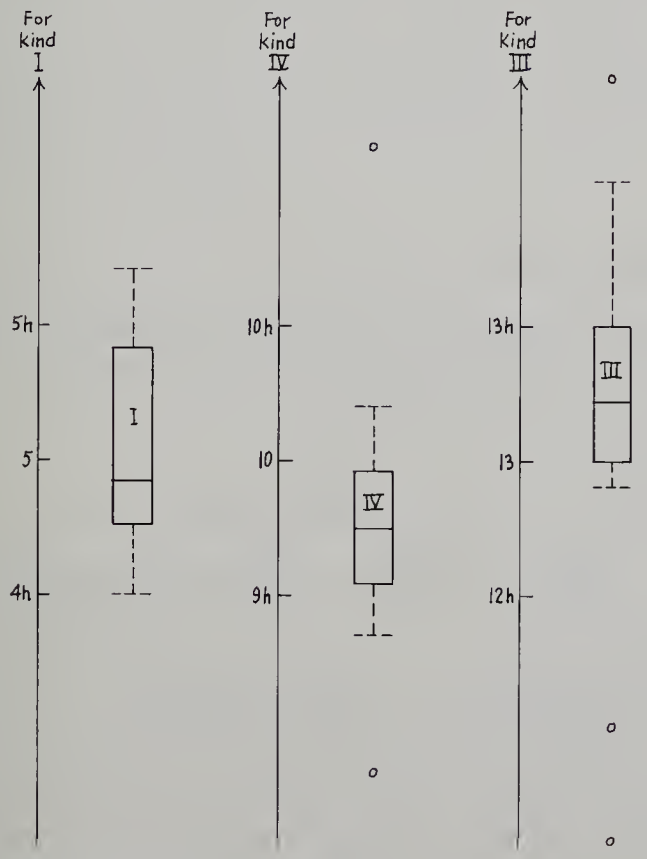
Thanks to the flexibility of the human eye, it is not important--if we are to look at a picture--that the centering be exactly the same. Rough adjustments will serve us essentially as well as precise ones.

Exhibit **7** returns to the catch-of-plankton example. For variety, we use an expression that compromises between roots and logs. (Recall that logs play the

exhibit **7** of chapter 4: plankton hauls

Rough residuals for the plankton counts plotted in terms of $\sqrt[4]{\text{count}} = \sqrt{\sqrt{\text{count}}}$ (see exhibit 1 for data)

A) PLOT of ROUGH RESIDUALS--of plankton counts



B) PROBLEMS

The fourth roots used to make the plots above were NOT obtained by two applications of exhibit 7 of chapter 3.

7a) Obtain them by two such applications, when the values will differ somewhat, and make the plots analogous to the above.

7b) Do you think the difference is important?

role of a zero power; clearly 1/4 is a natural compromise between 1/2 and 0.) Thus expression is

$$\sqrt[4]{\text{count}} = \sqrt{\sqrt{\text{count}}}$$

As exhibit 7 shows, when expressed in this way, the catches are in the vicinity of 5, 10, and 13 for Kinds I, IV and III, respectively.

In exhibit 7, then, we have shifted each schematic plot by a different amount. We have chosen round amounts, so that the three schematic plots do not exactly line up. We do have them under a common microscope, however, and can see clearly their general similarity in spread, both from hinge to hinge and from extreme to extreme.

exact adjustment

Round values and approximate adjustment save arithmetic if we are going to make a plot. If we want to present the same information numerically, we need to do subtractions to get small numbers. Since it is about as easy to subtract one number as another, it usually pays to line something up exactly. Exhibit **8** shows what happens in the catch-of-plankton example if we line up the medians exactly. Clearly the results are rather easy to compare, or even to combine.

exhibit **8** of chapter 4: plankton hauls

Results of lining medians up by adjustment (subtracting a constant) in the catch-of-plankton example expressed in log counts (all logs in .01's)

A) LETTER-VALUE DISPLAYS--for logs in 0.01

Kind I		Kind IV		Kind III	
	276		396		448
268	294	391	400	445	453
259	302	378	419	428	464

B) SAME for SHIFTED LOGS

Kind I down 276		Kind IV down 396		Kind III down 448	
	0		0		0
− 8	18	− 5	4	− 3	5
−17	26	−18	23	−20	16

P) PROBLEMS

8a) Imitate panel B for the catches of plankton expressed in roots (see exhibit 3 for original form). What seems to be going on?

8b) Do the same for the raw counts.

8c) Can you find the median of the three displays in panel B? Why/why not?

review questions

Can rough adjustments serve as well as exact ones? What light does exhibit 7 shed on this question? If we are not making pictures, should we use rough adjustments? Why?

4F. Residuals

If we wanted to know about the variations in plankton catches in rather general terms, we might like to combine all the knowledge that we can gain from the 3 batches of 12 at hand. If we define

residual = given value − summary value

we can turn each given value into a residual, for example by using the median of the corresponding batch as the summary value.

For the catch-of-plankton data, we dare not combine residuals based on raw counts, but we might try doing this for either root counts or log counts.

Exhibit **9** gives the log counts, their residuals from the median, and--in stem-and-leaf form--the result of pooling the three sets together. (A tendency to straying may be suggested by the E-spread being considerably more than twice the H-spread).

While residuals are useful as an aid to pulling together information from several batches (a rudimentary form of what might be called mustering and borrowing strength), **we will soon learn much more important uses for them:**

- ◇ as keys to the successive step-by-step improvement of our analyses.
- ◇ as keys to turning an investigator's eye toward the adequacy of our current analysis.

exhibit **9** of chapter 4: plankton hauls

Combining residuals for the catches of plankton expressed in logs (data and source, exhibit 3)

A) LOGS of COUNTS in 0.01's--two in median bold

Kind I: 259, 263, 267, 270, 273, **273, 279,** 288, 293, 295, 301, 302

Kind IV: 378, 388, 390, 392, 393, **395, 397,** 398, 399, 401, 404, 419

Kind III: 428, 435, 444, 445, 446, **446, 450,** 452, 453, 454, 460, 464

B) RESIDUALS from MEDIAN--for logs of counts in 0.01's.

Kind I: −17, −13, −9, −6, −3, −3, 3, 12, 17, 19, 25, 26

Kind IV: −18, −8, −6, −4, −3, −1, 1, 2, 3, 5, 8, 23

Kind III: −20, −13, −4, −3, −2, −2, 2, 4, 4, 6, 12, 16 →

review questions

What is a residual? Can there be more than one set of residuals for a given set of data? What are three ways in which residuals can serve us? Which are the more important?

exhibit **9** of chapter 4 (continued)

C) RESIDUALS POOLED

2	2·	56
3	2*	3
6	1·	796
8	1*	22
11	0·	586
18	0*	3123244
18	−0*	334314322
9	−0·	9686
5	−1*	33
3	−1·	78
1	−2*	0

36 pooled residuals

adj: −20(III), 23(IV)

out: 26(I)

	(mid)			(spr)
M18h	0		0	
H 9h	1	−5	7	12
E 5	2	−13	17	30
1	3	−20	26	46

	18	
f	−23	25
	xxx	one
F		43
		xxx

P) PROBLEMS

9a) Make a comparative plot of 4 batches, the 3 separate batches of residuals above and the pooled batch.

9b) Pool the residuals for the data and mode of expression of (5j) above.

9c) Same for (5k) above.

9d) Same for a new set of self-selected data.

4H. How far have we come?

In this chapter we have begun to take seriously one of the most important things we can do with data--comparison. Most of our attention has been devoted to learning how to strengthen comparisons--so that more nearly the full value of what the data offers us is put to use. We had already learned many of the techniques we need to do this--re-expression, useful display, subtraction. We now put them more carefully to work.

We are now ready:

- ◇ to make some kinds of slightly graphic summaries--displays that show forth the main issue without loss of numerical detail.
- ◇ to use plots of log spread against log level to help us judge what re-expression to try next.
- ◇ to calculate residuals by subtracting summary values from individual values. (We will learn about more general kinds of residuals in later chapters.)

Both **symmetry** of spreading-out and **similar extent** of spread make for useful comparison. While these are little and middle deals, respectively, the re-expression they point to is likely to be one that a BIG DEAL (when we have one) will ALSO prefer. We can seek such an expression by trial and error. With more data, though, it helps to use a systematic approach, plotting the log of a simple measure of spread against the log of a simple measure of level, and using the apparent tilt as a guide to both direction and amount of change.

If we want to compare in detail--rather than looking at the broadest questions--we want to adjust what is being compared so that the gross differences are taken out of the way.

Comparison, in common language, is either a matter of **difference** or a matter of **ratio**. Ratios almost always are of amounts or counts, which are never negative. Logs were invented to bring such ratios to differences. (Having values that are not fenced in, in particular are not kept from being negative, generally is likely to be helpful.)

Most importantly, we have just begun to learn that NO BODY OF DATA TELLS US ALL we need to know ABOUT ITS OWN ANALYSIS. It always takes information and insight gained from other, parallel bodies to let us analyze our body of data as well as we can. (If we don't have it, we do as well as we can.)

We are now well started on the analysis of the simplest kinds of data. We have a good chance of making an effective analysis: expressing the given values usefully, summarizing batches well enough for exploration, calculating residuals so that we can try to look deeper.

4P. Additional problems

See exhibits **10** through **17**.

exhibit **10** of chapter 4: data and problems

Percent of major party vote for Republican presidential candidates in eight southwestern states

A) DATA

	% for Nixon in 1960	% for Goldwater in 1964
Arizona	55.6%	50.5%
California	50.3%	40.8%
Colorado	54.9%	38.4%
Nevada	48.8%	36.1%
New Mexico	49.6%	40.6%
Oklahoma	59.0%	44.3%
Texas	50.5%	36.6%
Utah	54.8%	45.3%
(All U.S.)	(49.9%)	(38.7%)

P) PROBLEMS

10a) The snowfalls in New York City for 20 consecutive winters from 1918–19 to 1937–38 were (in inches) 3.5, 55.4, 18.2, 29.7, 55.2, 26.3, 27.9, 35.8, 21.9, 14.3, 13.3, 13.5, 9.7, 5.1, 24.5, 53.1, 29.0, 32.8, 11.9, 13.9. Those from 1938–39 to 1957–58 were: 31.9, 22.2, 35.0, 10.2, 27.6, 26.0, 26.7, 26.6, 33.2, 61.5, 43.0, 10.4, 10.9, 14.4, 9.1, 17.1, 10.9, 29.8, 19.1, 37.9. Compare these two batches in all the ways used in exhibit 2.

10b) Panel A gives, for two elections, percent of the major party vote for Republican presidential candates in eight southwestern states. Make schematic plots for:

- ◇ the 1960 data.
- ◇ the 1964 data.
- ◇ the change (swing) from 1960 to 1964.

Discuss your results.

10c) Choose another group of eight states, find the data, and make the same plots as for (10b).

S) SOURCE

Richard M. Scammon. America at the Polls: A Handbook of Presidential Election Statistics, 1920–1964. Pittsburgh: University of Pittsburgh Press (1965) 521 pp.

exhibit **11** of chapter 4: data and problems

Seasonal snowfall in Buffalo, New York and Cairo, Illinois, from 1918–19 to 1937–38 (inches)

A) DATA

	Buffalo	Cairo
1918–19	25.0	1.8
1919–20	69.4	4.5
1920–21	53.5	13.9
1921–22	39.8	4.0
1922–23	63.6	1.2
1923–24	46.7	6.8
1924–25	72.9	7.2
1925–26	79.6	11.5
1926–27	83.6	6.2
1927–28	80.7	0.4
1928–29	60.3	11.5
1929–30	79.0	12.4
1930–31	64.8	11.3
1931–32	49.6	2.9
1932–33	54.7	7.4
1933–34	71.8	2.7
1934–35	49.1	1.6
1935–36	103.9	14.1
1936–37	51.6	5.4
1937–38	81.6	3.0

P) PROBLEMS

11a) The snowfalls in Buffalo, New York and Cairo, Illinois for the 20 consecutive winters from 1918–19 to 1937–38 are given in Panel A. Make schematic plots for both places, for this time period.

11b) What light do these two batches throw on how they should be expressed?

S) SOURCE

Report of the Chief of the Weather Bureau, 1918–1919 to 1934–1935, and U.S. Meteorological Yearbook 1935 to 1938.

exhibit **12** of chapter 4: data and problems

Consistency of drape measurements

A) DRAPE, measured by AREA in SQUARE INCHES

Sample A	Sample B
28.92	50.04
28.82	49.94
28.96	40.08
28.89	50.27
28.96	50.03
28.85	50.06
28.97	50.00
28.89	49.99
29.00	49.75
28.99	49.94

P) PROBLEMS

12a) In 1950, Chu, Cummings, and Teixeira gave the results of drape tests in panel A. Adjust schematic summaries to equal medians. Does expression in square inches seem to give equal spread?

12b) Compare log H-spreads with log medians. What expression is suggested? Repeat (12a) for this expression.

S) SOURCE

C. C. Chu, C. L. Cummings, and N. A. Teixeira, 1950. "Mechanics of elastic performance of textile materials. Part V: A study of factors affecting the drape of fabrics--The development of a drape meter," **Textile Research Journal 20**: 539–548.

exhibit **13** of chapter 4: data and problem

Microdetermination of carbon monoxide in air

A) PARTS PER MILLION of CARBON MONOXIDE

Sample	Standard*	Trial results †
A	90.5	95,96,92,102,103,93,101,92,95,90
B	184.6	184,202,215,204,195,201,201,169,182,192,
C	44.8	40,54,42,49,64,62,50,67,64,43,
D	320	261,279,281,278,269,264,266,261,266,276,
E	244.7	215,214,197,216,215,208,226,208,216,214,
F	25.8	26,23,25,25,21,22,27,27,21,25,
G	66.2	56,55,61,57,60,57,65,55,60,61,
H	137.8	128,119,119,123,117,122,127,121,122,119,
I	137.8	155,142,146,149,149,146,152,159,

* I_2O_5 method
† Beckman–McCullough method, in order of analysis

P) PROBLEM

13a) In 1948, Beckman, McCullough, and Crane gave the results in panel A of microdeterminations of carbon monoxide in air. Adjust schematic summaries to equal medians.

S) SOURCE

A. O. Beckman, J. D. McCullough, and R. A. Crane, 1948. "Microdetermination of carbon monoxide in air," **Analytical Chemistry 20**: 674–677.

exhibit **14** of chapter 4: data and problem

Polarographic determination of aluminum

A) % Al_2O_3--oxygen-containing samples--in 0.01%

Certified value	Sample	Determinations
416	Limestone	**3**** | 92,94 **4** | 16,36
196	Silica brick	**19*** | 02336 **20*** | 04
191	Magnetite ore	171,175,177,190,210
189	Soda-lime glass	159,161,214,214
103	Iron ore	**10*** | 336799 **11*** | 236678
6.7	Dolomite	56,56 (in 0.001%)

B) % Al_2O_3--alloys--in 0.01%

113	Manganese bronze 62	**11*** | 444468
097	Manganese bronze 62b	**8*** | 9 **9*** | 4799
106	Nitralloy steel 106	**9*** | 5 **10*** | 88 **11*** | 01
107	Nitralloy steel 106a	**10*** | 08 **11*** | 022
026	High silicon steel	**25*** | 18 (in 0.001%) **26*** | 009 **27*** | 0

P) PROBLEM

14a) In 1950, Willard and Dean gave the results in panel A on the measurements of aluminum content. Find residuals. How should they be pooled? Combine them and discuss the results.

S) SOURCE

H. H. Willard and J. A. Dean (1950). "Polarographic determination of aluminum: Use of an organic reagent." Analytical Chemistry 22: 1264–1267. Table II on p. 1266.

exhibit **15** of chapter 4: data and problem

Kjeldahl ultramicrodetermination of nitrogen

A) The BATCHES--hundredths of microliters of 0.01N Hcl for a 10.42 microliter sample

Sample	Material	Determinations
A	Acetanilide	**6**** 60,78,81,
		7** 03,57,60,72,
B	Acetanilide	**6**** 13,37,39,46,
		7** 47,59,75,
C	$C_{14}H_{22}N_2O_2S$	**6**** 25,63,67
		7** 32,43,63,
D	$C_{13}H_{20}N_2O_2S$	**6*** 56,59,65,

P) PROBLEM

15a) In 1950, Kuck, Kingsley, Kinsey, Sheehan, and Swigert gave the results in exhibit 15 on the ultramicromeasurement of nitrogen. Calculate and pool residuals. Can these data show how they should best be expressed? If not, why? If yes, show how.

S) SOURCE

J. A. Kuck, A. Kingsley, D. Kinsey, F. Sheehan, and G. F. Swigert 1950. Kjeldahl ultramicrodetermination of nitrogen: Applications in the industrial laboratory. Analytical Chemistry 22: 604–611.

exhibit **16** of chapter 4: data and problems

Precipitation of platinum sulfide by various methods

A) AMOUNTS of PRECIPITATE--calculated as milligrams of platinum

Method	Platinum recovered from 10.12 mg Pt
Treadwell and Hall	**10.2*** 89
	10.3* 13334
Gilchrist and Wichers	**10.2*** 2334789
Hillebrand and Lundell	**10.2*** 25
	10.3* 03688

P) PROBLEMS

16a) In 1942, Geoffrey Beall ("The transformation of data from entomological field experiments so that analysis of variance becomes applicable," *Biometrika* **32**: 243–262) counted the number of *Phlegethontius quinquemaculata* (an insect) per plot after various insect control treatments. Slight simplifications of his results include (letters are treatments): A; 10, 7, 20, 14, 14, 12, 10, 23, 17, 20, 14, 13. B; 11, 12, 21, 11, 16, 14, 17, 17, 19, 21, 7, 13. C; 0, 1, 7, 2, 3, 1, 2, 1, 3, 0, 1, 4. D; 3, 5, 12, 6, 4, 3, 5, 5, 5, 5, 2, 4. E; 3, 5, 3, 5, 3, 6, 1, 1, 3, 2, 6, 4. F; 11, 9, 15, 22, 15, 16, 13, 10, 26, 26, 24, 13. What choice of expression is indicated if we take the counts for each treatment as a batch?

16b) In 1950, Jackson and Beamish gave the results in panel A on the precipitation of platinum sulfide. Compare the quality of the three methods. Find the residuals. Should they be pooled?

16c) In 1952, S. P. Hersh and D. J. Montgomery ("Electrical resistance measurements on fibers and fiber assemblies," *Textile Research Journal* **22**; 805–818) gave these resistivities (in 10^9 ohm-cm^2/cm) for the fibers indicated (at the relative humidities indicated in parentheses): Nylon monofilament--340 denier (60%): 12, 13, 12, 12, 16, 13, 12, 13. Nylon monofilament--30 denier 64%): 32, 26, 26, 29, 29, 41, 32. Nylon two--3 denier (85%): 1.00 (six times), 0.95 (three times), 1.10, 0.90. Human hair ~0.001 diameter (85%): 5.7, 6.8, 6.2, 6.2. Wool--Columbia 58's (85%): 0.095, 0.092, 0.089, 0.075, 0.050, 0.087. Make the best graphical presentation and comparison you can. (What do you think "resistivity," "denier," and "relative humidity" mean?)

S) SOURCE

D. S. Jackson and F. E. Beamish, 1950. "Critical examination of platinum sulfide precipitation," **Analytical Chemistry 22: 813–817.**

exhibit **17** of chapter 4: data and problems

Head breadths of termites (in hundredths of mm)

A) The DATA

	Nest 668	Nest 670	Nest 672	Nest 674	Nest 675
Small soldiers	227.3	247.9	249.4	244.7	245.6
	233.2	260.3	245.7	238.8	262.6
	237.5	261.3	245.2	251.5	263.3
	237.3	255.7	239.6	244.5	248.7
	231.8	237.7	277.9	231.2	241.0
Large workers	214.2	231.5	235.9	234.9	236.0
	213.9	246.3	234.4	230.0	247.8
	222.5	248.5	239.4	243.6	253.9
	235.2	247.7	236.4	242.3	246.1
	226.9	233.4	226.9	230.2	231.1

P) PROBLEMS

17a) In 1948, an anonymous correspondent Query 60, *Biometrics* **4**: 213–214) gave weights (in grams) of ducks (at age six weeks) after various protein supplements as follows (protein source in parentheses): (Horsebean) 179, 160, 136, 227, 217, 168, 108, 124, 143, 140. (Linseed oil meal) 309, 229, 181, 141, 260, 203, 148, 169, 213, 257, 244, 271. (Soybean oil meal) 243, 230, 248, 327, 329, 250, 193, 271, 316, 267, 199, 171, 158, 248. (Sunflower seed oil meal) 423, 340, 392, 339, 231, 226, 320, 295, 334, 322, 297, 318. (Meat meal) 325, 257, 303, 315, 380, 153, 262, 242, 206, 344, 258. (Casein) 368, 390, 379, 260, 404, 318, 352, 359, 216, 222, 283, 332. What expression is indicated?

17b) Make an effective graphical presentation and comparison of the data given in (17a). (What do you think "sunflower seed oil meal" is?)

17c) In 1909, Warren gave the head measurements on termites in panel A. Find residuals. Should they be combined? If not, why? If yes, show how.

S) SOURCE

Ernest Warren, 1909. "Some statistical observations on termites, mainly based on the work of the late Mr. G. D. Haviland." Biometrika: 6: 329–347.

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Plots of relationship 5

We have learned something about scratching down batches--sets of similar numbers. We found more things to do--saw that more things could be learned about data by doing them--than we might have expected. We turn now to the use of simple plots, to the plotting of y against x, and find, again, more things to do and more gains from doing them than we might expect.

The most quoted passage of all the Sherlock Holmes stories concerns the unusual behavior of the dog in the night time. Holmes called attention to its unusualness--Watson couldn't see what was unusual--Holmes pointed out that the dog did nothing.

The moral is clear--we ought to judge each occurrence against the background of--or a background derived from--other "nearby" occurrences. We do not ordinarily think of "zero" as unusual--yet a 24-hour period when no one died anywhere in the world, or a winter with no snow at all in the White Mountains, would be recognized by each of us as strikingly unusual.

As in the large, so in the small. Once we have plotted some data, and found out how it behaves generally, our next step is to look at each of its elements--at each "point"--and ask how it seems to deviate from the general behavior of all the "points". To do this it very often helps to make a new plot, one that focuses on such deviations--in brief, "it very often pays to plot residuals".

As we have begun, so will we continue. Here, as well as later, we will be repeatedly engaged in splitting up data according to one version or another of the key relation

given = fit PLUS residual.

Here the fit is our current description--always incomplete, always approximate--of the overall behavior of the data. Each individual observation is split up into a sum of this fit and what is left over, called

a residual.

Residuals are our main tool in going further. They are to the data analyst what powerful magnifying glasses, sensitive chemical tests for bloodstains, and delicate listening devices are to a story-book detective. They permeate all sorts of data analysis and appear in many guises.

The most conventional plots follow a simple pattern. A quantity, habitually shown on a vertical scale, which we think of as

a response.

Another quantity, habitually shown on a horizontal scale, which we call

a factor

which is usually

a circumstance

and which we think of as possibly explanatory or descriptive. The data--in whole or part--will consist of pairs of numbers of the form

(factor, response)

which will be plotted as points using these scales.

Most plots--at least in the popular press--concentrate on the fit. Too often, however, such plots are only to remind us that the anticipated relationship is there--that, for example, the population of the U.S.A. still increases. As such they are usually matters of elementary exposition rather than analyses. They show us "the big picture" that we already knew about. (In the sense of Linus in the familiar comic strip, they are just "security blankets".) At other times, such plots "in the large" are to show us the unexpected--either an unsuspected relationship or an unsuspected strength or weakness of an anticipated relationship. For these purposes, plots "in the large" are part of data analysis. They tell us of investigation's successes, perhaps very effectively, but they still need to be supplemented by pictures of residuals--pictures that tell us whether there is yet more to investigate.

For us the most useful plot will be one that might reveal the unexpected or the unobvious. Sometimes a plot "in the large" will do this. Usually, however, it is the plot of residuals that has the greatest use and the greatest impact.

review questions

How ought we judge occurrences or numbers? What is a "residual"? What is the key relation involving residuals? What is a fit? Is it final, complete, or exact? What is a "response"? A "factor"? A "circumstance"? How are they usually plotted? How is data involving one factor and one response usually written down? Plotted? When are plots "in the large" useful? When are they part of data analysis? How do we ask whether there is more to investigate? What kind of plot is most likely to help us?

5A. How to plot *y* against *x*

Plots are important. Great differences in ease of construction and great differences in effectiveness of use depend on apparently minor, purely technical procedures. The remarks in this section are intended to finish getting you over most of the elementary hurdles. NOW GO BACK and read the parts of section 2C, dealing with "tracing paper", "scale values," and "plotting without graph paper". All this still applies here.

choice of ruling

If plotting is to be easy, you want graph paper with at least three different thicknesses of ruling:

- ◇ light lines for "units".
- ◇ medium lines for "fives".
- ◇ heavy lines for "tens".

A wide variety of graph papers can be obtained with these characteristics. (Some papers have extra heavy lines for "twenties". You will have to learn for yourself whether these help you more than they get in your way.)

DO NOT use papers ruled in "fours" and "eights", or in "sixes" and "twelves" for plotting data given in ordinary (decimal) numbers. (For monthly data, of course, "twelves" in one direction do help.)

If you want to plot fast, easily, and accurately, avoid "dime-store" sheets with only two thicknesses of ruling and, above all, avoid quadrille sheets with only one. (Quadrille sheets are very useful, for almost everything except plotting graphs.) (If you need to save money, see below.)

choice of scale units

When you come to plot, you must choose units. Don't try to make one step of ruling (light, medium, or heavy) equal to 3 units, 7 units, 0.03 units, 0.007 units, or any such uncomfortable number. Stick to one step being 1, 2, or 5 times a power of ten. (One square = 20,000 or one step = 0.05 are examples that can work out quite well.)

You will find it hard enough to learn to be a fast and accurate plotter with two three-speed gear shifts (1, 2, or 5 across the graph and 1, 2, or 5 up the graph).

If you need to use an abnormal scale--three units to the square, for example--convert your numbers into plotting units--as by dividing by 3--by slide rule, hand arithmetic, or what have you, BEFORE you start to plot. It will take less time, overall, to make a good picture.

consequences of our purpose

Throughout this account we shall be interested in graphs to be looked at, not to be used to find numbers. Our graphs are means for looking at the data, not stores of quantitative information. This means:

- ◇ we will want to keep our eyes on the points.
- ◇ we will usually **not** connect one point to the "next" one. (We are likely to draw in fitted lines or curves.)
- ◇ we will want to suppress the rulings of graph paper, at least from our mind's eye.

◇ we will need only a few numbered ticks along the axes, horizontal or vertical, of the final picture.

◇ we will want to use symbols large enough to stand out (and if we need more than one kind, different kinds should be both clearly different and--usually--almost equally noticeable).

There cannot be too much emphasis on our need to see behavior.

We must play down or eliminate anything that might get in the way of our seeing what appears to go on.

kinds of grids

The use of tracing paper makes it possible for any of us to really arm himself with an easy-graphing capability of broad scope. A tracing pad and one sheet of each of 20 kinds of graph paper puts us in business to do many things. (Actually, it usually seems worthwhile to have a reasonable number of sheets, perhaps in the form of a pad, of at least one or two of the most used kinds of graph paper. Many will wish to first plot on the graph paper and then copy on the tracing paper.)

"Ordinary" graph paper frequently comes 8 small squares to the inch (beware; it may be lined in "4's" or "8's" which you should never, never use--unless what you are plotting comes in eighths of an inch, or, like stock market prices, in eighths of a dollar), 10 to the inch, 12 to the inch (beware, it may be ruled in "6's" or "12's"), 20 to the inch, and 10 to the centimeter. The writer likes the coarser rulings at least as well as the fine; it is up to you to learn your own preference.

Beware of 12-to-the-inch rulings, since too many are *ruled* in "sixes" and "twelves"; but bear in mind that 12-to-the-inch ruled in fives and tens--like Codex Nos. 31,253 and 32,253, for example--is a very good base for graphs where the final product is a typed page. (Typewriters like 12 to the inch.) It has little special advantage for most other purposes, although it is quite satisfactory.

Semilogarithmic (uniform scale one way, log scale the other) and full logarithmic (log scales both ways) graph papers come in a variety of scales and patterns. To have a reasonable selection is to save time and encourage inquiry. One CAN always look up the logs and then plot on uniform-scale graph paper, but WILL one? If chapter 3 is at hand, probably yes. (We rarely get much more than two-decimal accuracy out of a plot on log paper, so exhibit 2 of chapter 3 will usually do as well as special graph paper.) Otherwise? As we saw above, logs are often the way to make data reveal itself.

A variety of other graph papers are useful, as we shall see later. We note here that paper with two square-root scales is available (Codex No. 31,298 or No. 32,298) and that trilinear or isometric paper--with three sets of rulings at 120° to one another--is often quite useful. (It is available from various

manufacturers.) Reciprocal (one-way) paper is available both from Codex and from Keuffel and Esser.

shape of plot

Naturally enough, graph paper usually comes in the same general shape and size (letter size) as ruled writing pads and typing paper. Clearly it can be used in two ways:

- ◇ narrow edge at the top of the plot, or
- ◇ broad edge at the top.

By analogy with the use of ruled writing pads, it is natural to make plots with the narrow edge up. Full use of the area then usually makes a plot that is taller than it is wide.

There are some purposes for which this is a good shape. The ever-faster-rising curve of early growth is an example.We can quite well use taller-than-wide plots for such simple pictures, most of which tell us that we haven't yet made an analytically useful plot. Sometimes, indeed, such simple plots can be clearer when they are taller than wide, rather than wider than tall.

Most diagnostic plots involve either a more or less definite dependence that bobbles around a lot, or a point spatter. Such plots are rather more often better made *wider* than tall. Wider-than-tall shapes usually make it easier for the eye to follow from left to right.

Perhaps the most general guidance we can offer is that smoothly-changing curves can stand being taller than wide, but a wiggly curve needs to be wider than tall (sometimes after a smooth part has been taken out).

When a plot is made wider than tall, convention says it should be turned in the direction illustrated by many plots in this chapter, even if this makes a down-to-up legend upside down when we first see the plot as we turn the pages.

ticks and numbers along axes

We use marks along axes for two quite different purposes: (1) to plot the points; (2) to look at them. Different purposes call for different techniques, and the graph-paper–tracing-paper combination makes separate techniques easy. If one begins by plotting on the graph paper, it helps to have many numbers and ticks on the scale. (Remember to have the horizontal scale **above** the plot, and the vertical scale to the left*--both out from under the plotting hand.) To look effectively at the traced result, it helps to have ONLY a few numbers and ticks. Exhibit **1** shows five pairs of vertical axes, one in each pair for plotting and one for looking. Notice the technique, in each PLOTTING scale, of:

- ◇ putting 4 ticks--or dots--between consecutive numbers (often useful: NOT to be religiously followed; sometimes no tick between, or one tick between, is better).

---

* To the right, for southpaws.

◇ putting a tick--or dot--for every one or two steps of some digit in what is plotted (sometimes it pays to do this at every five, instead).

(Some even standardize on dots for each one and ticks for each two.) All this is focused on making it as easy as possible to find where to put the point. Doing less than this slows down our plotting and wastes both effort and temper.

Notice also the technique, in each LOOKING scale, of using:

◇ only four or five ticks.

◇ only two or three numbers.

Doing more than this distracts our attention from what we ought to see. (If the scale is irregular, we may need more ticks and values. Scales for dates, where individual values like 1066, 1776, or 1929 are well remembered, often deserve more ticks and values.)

exhibit **1** of chapter 5; illustrative

Pairs of scales, showing the difference between scales for plotting (marked "draw") and for looking (marked "show")

Clearly, separating plotting from looking can be a great help. Three cheers for tracing paper or transparent plastic.

review questions

What is the minimum for well-ruled graph paper? What is optional? What choices of scale units are reasonable? What should we do if we need an unreasonable choice? Should we connect our points with lines or curves? Do graph-paper grids help us to plot points? To see what our points look like? How many ticks and numbers for looking? For plotting? Where should we put the scales for plotting? How do we use tracing paper? What are some conveniently available kinds of graph paper? Who should use graph paper with heavy 4's and 8's? With heavy 6's and 12's? What shapes of plot are desirable? Does this book follow the rule? How can we plot without graph paper?

5B. Looking at subtraction

Undoubtedly, the form of graphical representation we have been most exposed to in school is one involving two variables called x and y, in which y is said to **depend on** x.

In data analysis, a plot of y against x may help us when we know nothing about the logical connection from x to y--even when we do not know whether or not there is one--even when we know that such a connection is impossible.

Before we can make full use of such plots, we need to understand--in terms both of doing and of feeling--certain things about such a plot, including:

- ◇ how to subtract one "curve" from another.
- ◇ how to find a numerical formula for a straight line drawn on graph paper.
- ◇ what effect subtracting different--two or more--straight lines from the same data points has.
- ◇ how to try to re-express either y or x, or both, so as to make the data appear more nearly straight.
- ◇ why graph paper--on which we plot easily--expresses the essence of how points represent numbers much better than the kind of picture we will usually want to look at--which has axes with ticks and numbers.

When we have x alone, subtraction is simply and easily represented by sliding arrows along. Subtracting 3 from 5 is a matter of drawing an arrow starting at +3 and extending to +5, and then sliding this arrow until it starts at 0. Its new endpoint, +2, is the result of subtraction. Exhibit **2** shows this example and three others that involve various combinations of minus signs.

In dealing with data we usually deal with subtraction of y's rather than of x's. For a hypothetical ABC Corporation, in 1960, sales were 44 million and

expenses were 32 million. We can easily subtract expenses from sales to find profit before taxes of 12 million, as shown in exhibit **3**. Again, we are sliding the arrow to put its base at zero. This time we have to be careful to keep it at 1960.

In this exhibit and the next--for clarity of what we are doing, rather than for clarity of result--we have used two time scales placed side by side. Ordinarily we would use one time scale and slide each arrow along a single vertical line--as for the broken arrows on the left sides of these exhibits.

Exhibit **4** retains the pattern and shows sales, expenses, and profits for 12 consecutive years. To avoid confusing detail, the arrows and their sliding are shown for only three years, 1951, 1957, and 1960.

exhibit **2** of chapter 5: illustrative

Four examples of subtraction

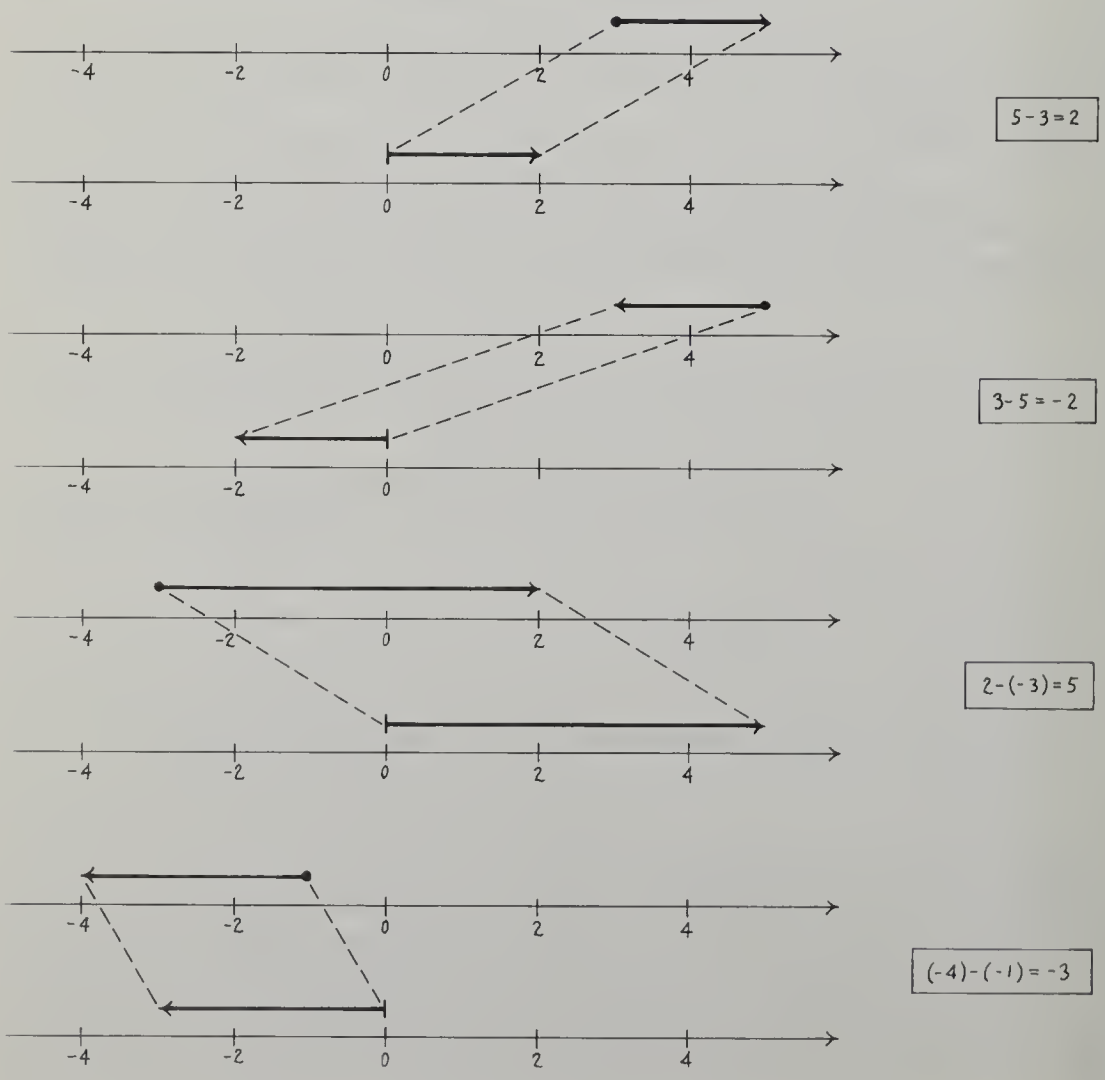


exhibit **3** of chapter 5: ABC Corp.

The ABC Corporation in 1960

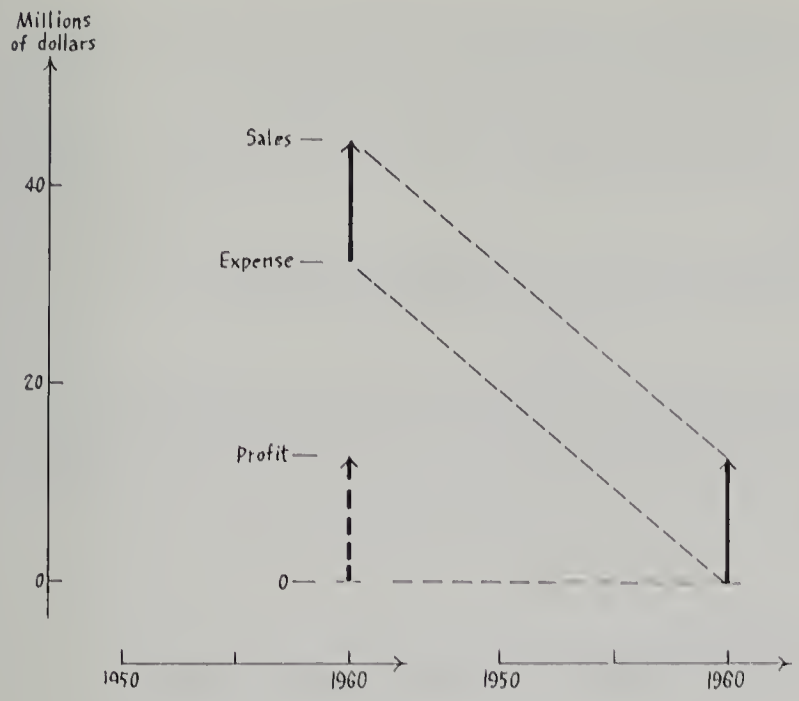
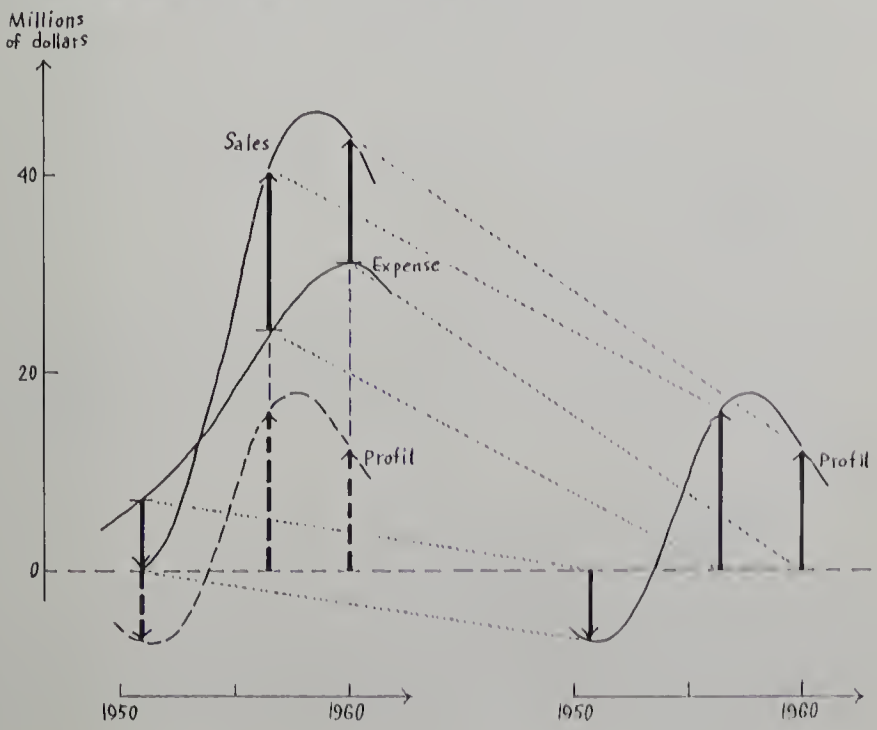


exhibit **4** of chapter 5: ABC Corp.

Twelve years of the ABC Corporation



The case where we will be most concerned with graphical subtraction is the case where we solve the basic relation

$$\text{data} = \text{incomplete description PLUS residual}$$
$$= \text{fit PLUS residual}$$

for the residual, finding

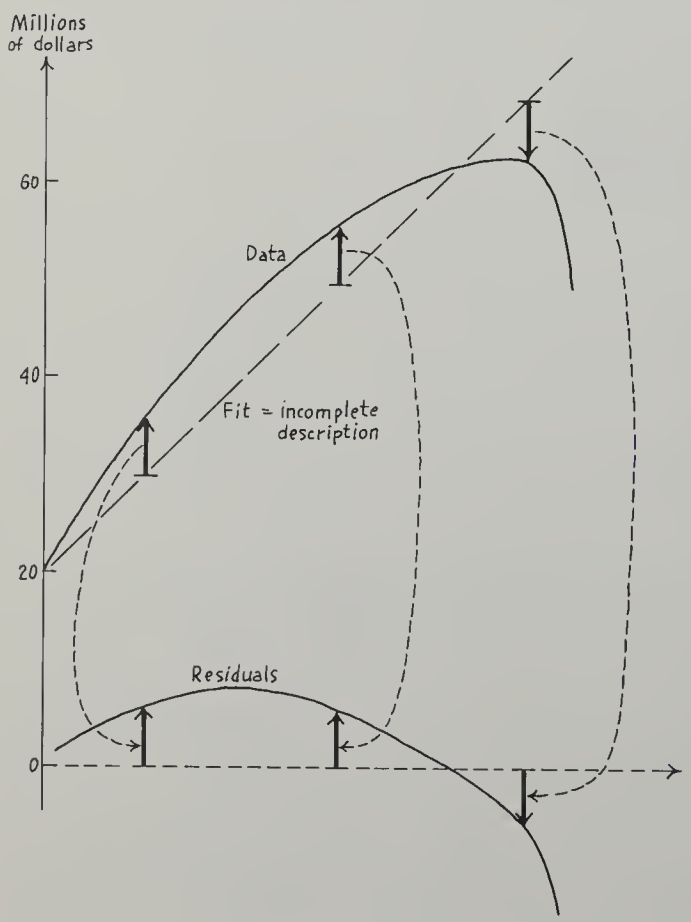
$$\text{residual} = \text{data MINUS incomplete description.}$$

untilting

Exhibit **5** shows an example of this where the incomplete description is a straight line. Here we have slid the vertical arrows down (or up) the vertical on which they lie. Again we have shown only three arrows of the many possible.

exhibit **5** of chapter 5: illustrative

Subtraction of an incomplete description from data to form residuals



It is natural to think--and speak--of such a subtraction of a straight-line partial description as an *untilting*. Natural and useful. Yet such use of words seems likely to distort our thinking a little if we are not wary. Untilting suggests, at least to some of us, a rigid motion in which the final curve comes from the initial curve by a rigid motion--a rotation about some point. This is quite wrong.

There are many ways to see this. An easy one is to look at the length of the curve between the two crossings of the straight line. This distance is obviously greater for "data" than for "residuals". Thus we do not have rigid motion.

A convenient picture involves a deck of cards, on whose edges--the edges on one side--we have drawn both "data" and "partial description". Now let us:

- ◇ clamp the deck together and saw off its bottom edge on a slant--a slant that is parallel to, because it is a constant vertical distance from, the partial description.
- ◇ unclamp the deck and strike it on a table to line up its new bottom edges horizontally.
- ◇ clamp the deck again.

The marks for "partial description" will now lie along a horizontal line, since they are a constant distance from the new bottom edges. If we call this upper line zero, the marks for "data" will now show us the "residuals".

The sliding of the cards with respect to one another exactly corresponds to sliding the various vertical arrows with respect to one another. This is a proper mechanical picture of graphical subtraction of one y from another. It works--rigid motion does not.

Of course, all this works for curved fits as well as for straight ones. (At least if we can make a curved cut in our deck of cards.)

review questions

Why would we want to subtract one curve from another? A curve from some point? How does subtracting a straight line behave? Does it have anything to do with rotation? What is a mechanical model for subtracting a straight line?

5C. Subtracting straight lines

If we have taken our data, plotted it, and drawn a straight line through it, and now we wish to use the straight line as an incomplete description, our next task is to subtract the line from the data. Sometimes, as in exhibit 4, we can do the subtraction graphically. Often, however, this is more work than we like.

finding the line

To do the subtraction by arithmetic, we have to turn the straight line into numbers. The easy way to do this is to choose two points on the line--let us call them (x_1, y_1) and (x_2, y_2)--read off their coordinates, and say that an equation for the line is

$$y = y_1 + b(x - x_1)$$

where the slope, b, of the line is given by

$$b = \frac{y_2 - y_1}{x_2 - x_1}$$

Clearly, where $x = x_1$, we have $x - x_1 = 0$ and the equation gives $y = y_1$. When $x = x_2$, the second term on the righthand side of the equation is

$$\frac{y_2 - y_1}{x_2 - x_1}(x_2 - x_1) = y_2 - y_1$$

and we have

$$y = y_1 + (y_2 - y_1) = y_2$$

as we should.

To make the arithmetic of using such an equation easy, we should choose x_1 so that the values of $x - x_1$ are as simple as possible. To make finding the equation easy, we should choose x_2 so that $x_2 - x_1$ is a simple number. (Of course, we have to balance the advantages of simplicity against the advantages of getting well out toward the end.) There need never be great difficulty in turning a straight line into an equation.

an example

Exhibit **6** shows a plot of the population of England and Wales at every decade from 1801 to 1931. An eye-fitted line has been drawn in and the necessary simple arithmetic--shown at the lower right--performed.

In carrying out the eye-fitting of a line and its conversion into numbers it is important:

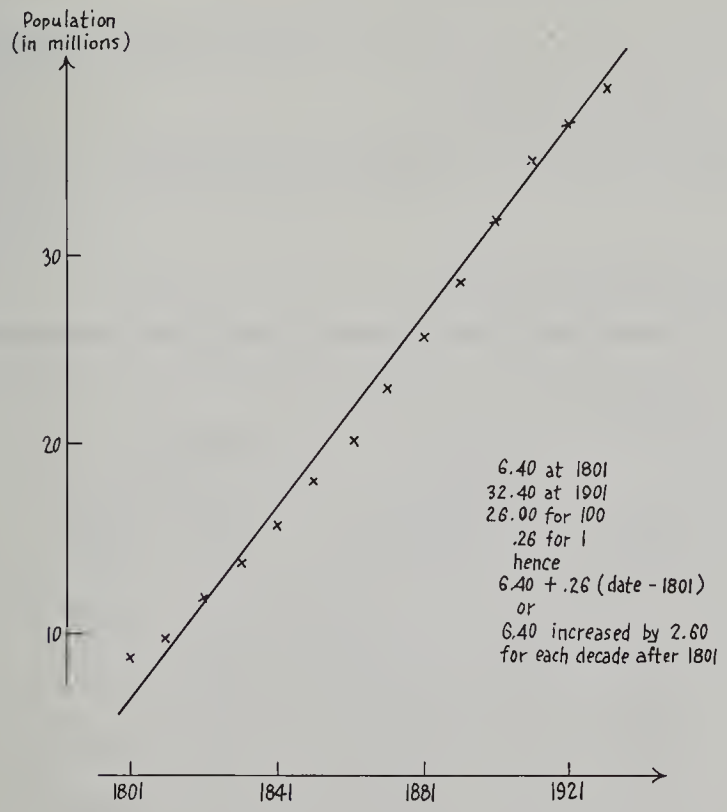
- ◇ to eye-fit the line on a picture without excessive background--use the tracing paper (or transparent plastic) version WITHOUT graph paper underneath.
- ◇ to find two points through which to pass the line, put the transparent version back on the graph paper, and locate the points with its aid.

It takes two different looks at a point-swarm to do all we want do.

Exhibit **7** shows the calculation of the residuals, not only from the line of exhibit 6 (line 1) but also from another line (line 2). (The rest of this exhibit will be discussed in a moment.)

exhibit **6** of chapter 5: England and Wales

Population in millions at successive censuses (1801–1931)



When, as in this example, the x-values step along in steps of constant size, one easy way to find the y-values is to begin at the low end and add the constant difference. For line 1, this means beginning with 6.40 and adding 2.60 repeatedly.

When the steps in x are not all the same (with the possible exception of short gaps), we have to do a little more arithmetic.

subtracting different straight lines

If we look at data, and draw what looks like a reasonable straight line through it, we are not likely to draw exactly the line that will serve us best. What are the consequences of a somewhat unsatisfactory line?

- what will it do to the residuals?
- how hard will it be to move to a better choice?

An easy way to approach both these questions is to ask what happens if we subtract first one straight line from y and then a second straight line from the

residuals thus formed. Algebra is easier than geometry here. If we begin by subtracting $a + bx$ we will form

$$y - (a + bx)$$

as our residual, where (x, y) is the data point and $a + bx$ is some fit or other. We have just seen examples of this (in exhibit 7). If, from the residuals thus formed, we subtract $A + Bx$, we will form

$$[y - (a + bx)] - (A + Bx) = y - [(a + A) + (b + B)x].$$

exhibit **7** of chapter 5: England and Wales

The population of England and Wales, with residuals from various lines (populations in millions)

A) DATA and CALCULATIONS

		line 1*		line 2*			supplementary line*	
year	pop'n	fit	resid	fit	resid		fit†	resid‡
1801	8.89	6.40	2.49	6.	2.89	s	1.73	1.16
11	10.16	9.00	1.16	8.5	1.66	t o p	1.64	.02
21	12.00	11.60	.40	11.	1.00		1.56	−.56
31	13.90	14.20	−.30	13.5	.40	h e r	1.48	−1.08
41	15.91	16.80	−.89	16.	−.09	e	1.39	−1.48
1851	17.93	19.40	−1.47	18.5	−.57	u n t i l	1.30	−1.87
61	20.07	22.00	−1.93	21.	−.93		1.22	−2.15
71	22.71	24.60	−1.89	23.5	−.79		1.14	−1.93
81	25.97	27.20	−1.23	26.	−.03	a f t e r	1.05	−1.08
91	29.00	29.80	−.80	28.5	.50		.96	−.46
1901	32.53	32.40	.13	31.	1.53		.88	.65
11	36.07	35.00	1.07	33.5	2.57	e x h	.80	1.77
21	37.89	37.60	.29	36.	1.89		.71	1.18
31	39.95	40.20	−.25	38.5	1.45	8	.62	.83

*** Line 1: 6.40 + .26 (date − 1801); Line 2: 6 + .25 (date − 1801). Supplementary line: 1.73 − .0085 (date − 1801).**
† One more decimal kept in additions.
‡ These residuals are from the fit of the supplementary line to the residuals from line 2.

The result of two subtractions--of $a + bx$ and $A + Bx$--is always the same as the result of a single subtraction, of

$$(a + A) + (b + B)x$$

--of a subtraction of the

sum of the two lines.

Those who wish can show the same fact geometrically.

This fact about subtracting lines goes far toward answering our two questions:

◇ if we subtract one line and, on looking at the residuals, find them still tilted, we are free to draw a line among these residuals and subtract it further, thus finding new residuals. The new residuals correspond to subtracting a single line--the sum of the two actually subtracted. We need not go back and start again. This is particularly handy when the residuals are much smaller numbers than are the data.

◇ if we subtract an unsatisfactory line, and discover that we have done this by looking at the residuals, we could always correct this by a further subtraction. Accordingly, our first residuals will differ from the better residuals by being somewhat tilted. Since slight tilts do little to hide what we are looking at the residuals for--evidence of further structure or of unusual values--it will rarely be necessary to do the second subtraction (unless we want to publish the residuals). We can see what we need to see in the slightly tilted residuals. And, if we want to find the equation of the better line, we can draw a correction line and sum the expressions of the original and correction lines.

Subtracting lines is simple and convenient in many ways.

back to the example

Exhibit **8** shows the residuals from line 2 of exhibit 7, plotting them against date. One eye-fitted line is shown. (Clearly, there could be considerable debate about which line to fit to this sequence of points.) This is naturally called a

supplementary line

since it is fitted to the residuals from a first fit. The result is

$$1.73 - .0085(\text{year} - 1801).$$

Since the residuals fitted came from the line

$$6 + .25(\text{year} - 1801)$$

the 2nd residuals--visible in exhibit 8 as deviations from the line, given

numerically in the righthand column of exhibit 7--are residuals from the sum of these two lines, namely

$$(1.73 + 6) + (-.0085 + .25)(\text{year} - 1801)$$

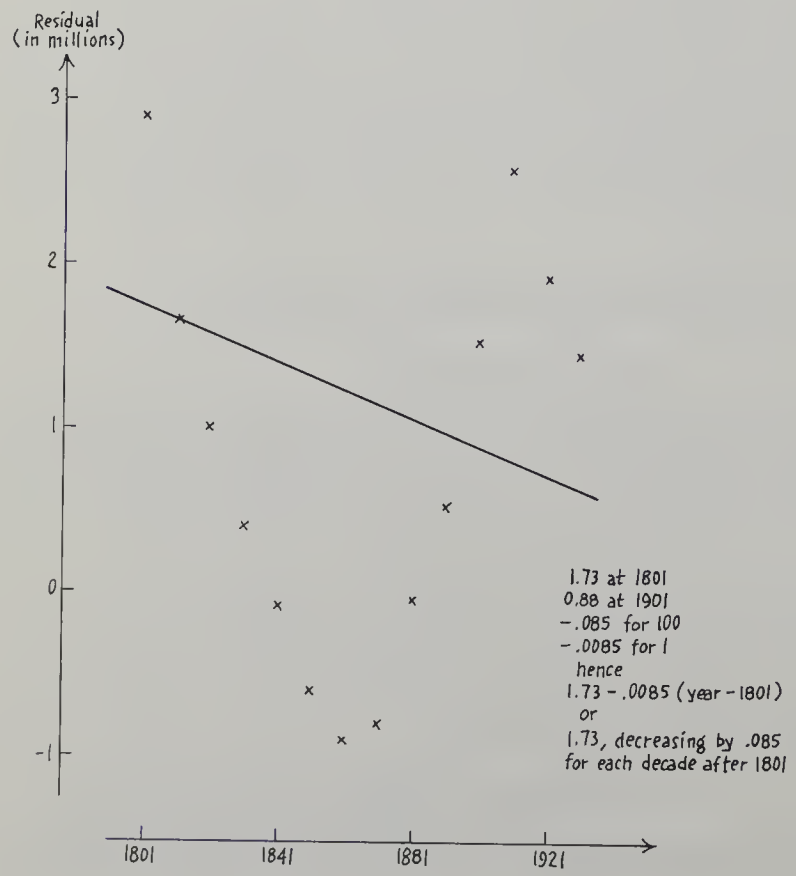
or, simplified

$$7.73 + .2415(\text{year} - 1801).$$

Taking out first a preliminary line and then a supplementary one often, as in this example, helps us keep our (hand-done) arithmetic simpler. If we use "easy numbers" in the preliminary fit, we can often do the two fittings for less than the price of one. In addition--in fact a more important consideration--we get a look at a picture of some residuals, something that can have a variety of advantages, and often does.

exhibit **8** of chapter 5: England and Wales

The residuals--from line 2 of exhibit 7--plotted against date and eye-fitted with a line.



review questions

How does one fit a straight line to two points? How does this help us in eye-fitting a line? What did we choose as an example? What happens if we subtract two lines, one after the other, from either some points or a curve? Can this make our arithmetic simpler? Why?

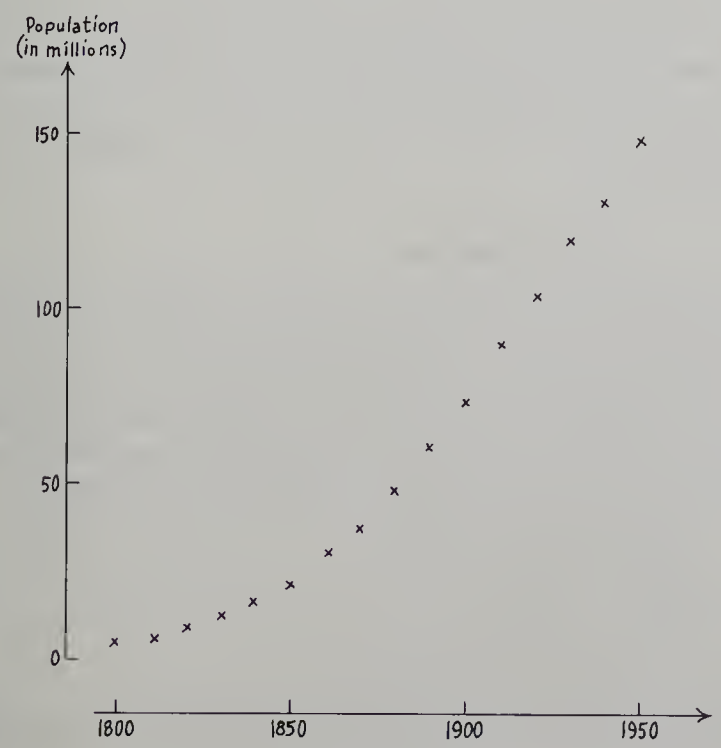
5D. Plotting the population of the U.S.A.

Many people would think that plotting y against x for simple data is something requiring little thought. If one only wishes to learn but little, it is true that a very little thought is enough, but if we want to learn more, we must think more.

A convenient example of how thinking more lets us learn more is given by the population of the U.S.A. as counted by the census every ten years from 1800 to 1950. Exhibit **9** shows the "little thought" version, in which millions of people are plotted against date. What can we see in this graph?

exhibit **9** of chapter 5: U.S.A. population

Population of the U.S.A. (in millions; linear scale)



For something like the first half of the period the curve is hollow upward, so that over this period the population was growing at a steadily increasing rate. In fact, it might have been growing at something like a constant percentage each decade. For something like the second half of the period, the growth seems to approximate a straight line. Beyond this, the value for 1940 seems to be somewhat low.

All this is helpful. If we had never before looked at the population of the United States as a function of time, we would have rightly felt that we had learned quite a lot from exhibit 9. But once we have come this far, must we stop? Let us use what we have so far learned to help us look more deeply into the growth of the U.S.A.'s population.

In the present instance this is easy to do. What do we have as bases for further steps? Two things come in sight in exhibit 9:

- ◇ the early years were years of accelerated growth, possibly at a constant percent per year.
- ◇ in the later years, the population grew by about the same number of people each decade.

We can check up on these appearances and, more importantly, try to use them to go further.

To check up on constant-percent growth, the easy thing is to look up the logs of the population sizes, and then plot them against date. (If we didn't expect to go further, we could just plot the raw values on semilogarithmic graph paper instead.) Exhibit **10** shows the result of making such a plot. The earlier part of the plot now looks quite straight, even if we put our eye close to the paper so that we can look right along this hypothetical line. This looks very much as if constant percentage growth per decade is a good description of U.S.A. population growth in the early 1800's. Let us keep this appearance in mind, and plan to come back to it.

the later decades

Before going further with the early decades, however, we shall turn to the apparent linearity of population growth seen in the original graph, exhibit 9, for the later decades. Exhibit **11** shows the result of drawing in a comparison line. To find an equation for this line, note that in 1870 the line has a height of about 35--35 million people--and in 1950 it has a height of somewhat less than 150, say 147. the line through (1870, 35) and (1950, 147) has slope

$$\frac{147 - 35}{1950 - 1870} = \frac{112}{80} = 1.4$$

and thus the equation is

$$y = 35 + 1.4(x - 1870)$$

This calculation is made by simple steps in the form of writing down words and

numbers in the lower right of the exhibit. We recommend such a form (in general with "at" a value of x rather than "in" a date) whenever we eye-fit a line.

This line follows the data quite well, leaving us with a confirmed feeling about both approximate straightness and the dip in 1940. Need we stop here? Surely not. The straight line is an incomplete description of how the data behaves in later years. One of the great arts of data analysis consists of subtracting out incomplete descriptions and examining the residuals that are left. Let us do just this.

coming to details

Exhibit **12** shows the residuals from the line of exhibit 11, for the period from 1800 to 1950. (For example, at 1880 the census population is 50.2 millions and the fit is 35 + 1.4(10) = 49, so that the residual is +1.2 millions.) The detailed behavior of the population in the later half of this period is now fairly well revealed. The earlier half of the period is, however, telling us little. (Especially since the comparison line

$$35 + 1.4(\text{date} - 1870)$$

gives rather large residuals before 1840, making the comparison in earlier years unlikely to be helpful.) If we gave up looking at the early half, and

exhibit **10** of chapter 5: U.S.A. population

Population of the U.S.A. (in millions; logarithmic scale)

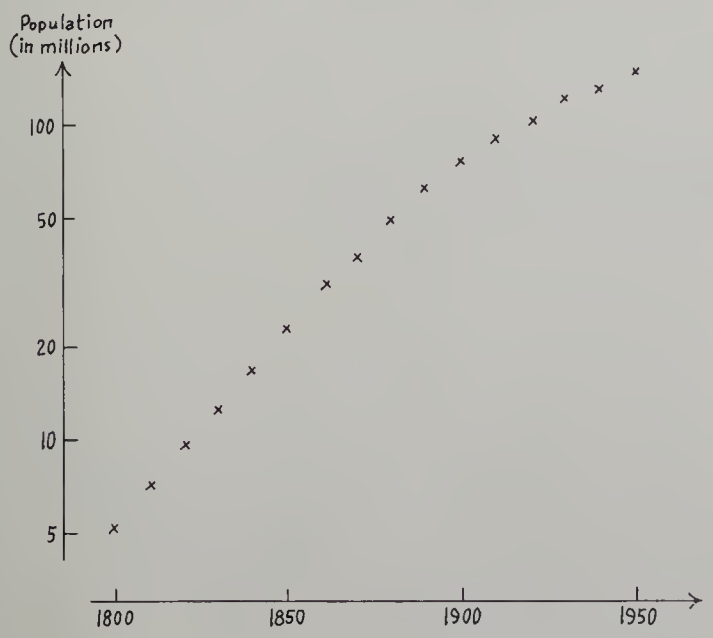


exhibit **11** of chapter 5: U.S.A. population

Population of the U.S.A. (linear scale with comparison line)

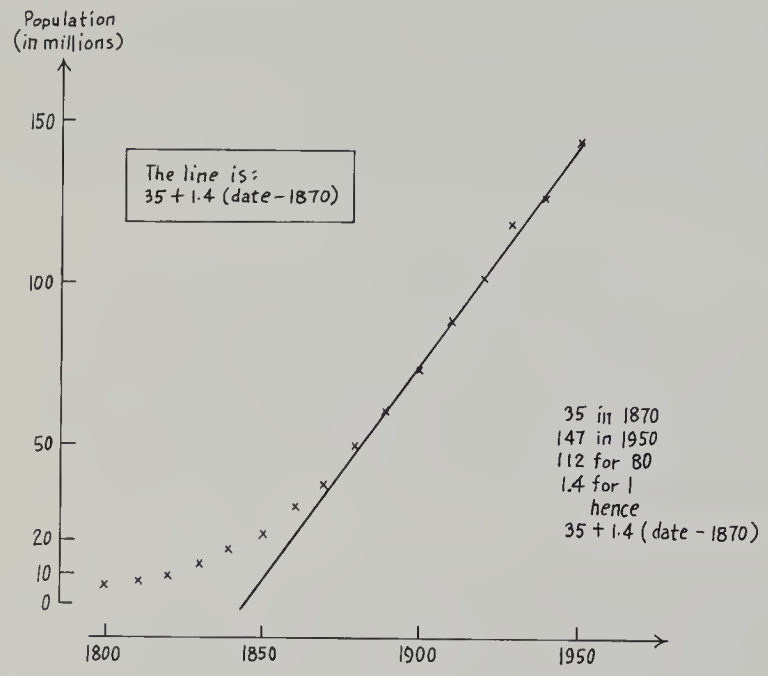
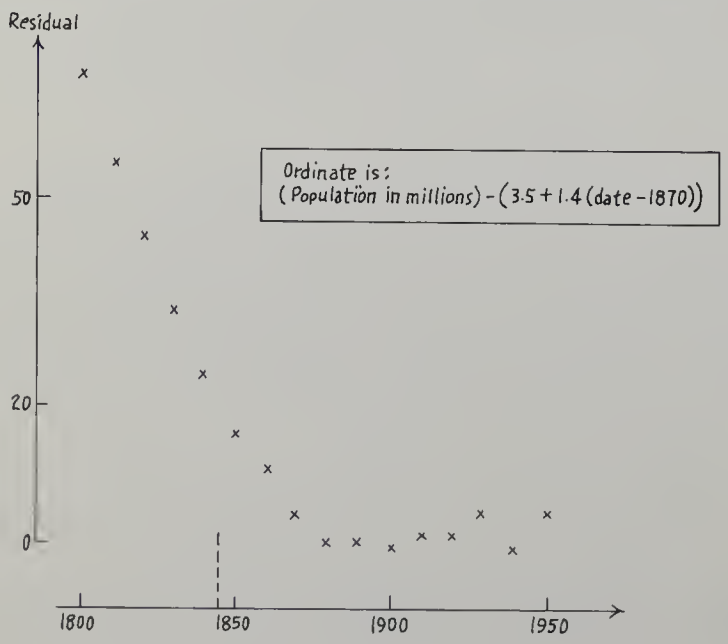


exhibit **12** of chapter 5: U.S.A. population

Population of the U.S.A. (residuals = deviations from the specified straight line)



focused on the later half, we could put the data under a very much more powerful microscope. Why not do just this?

Exhibit **13** shows the same values as the righthand side of exhibit 12 but at 15 times the vertical scale. We can now see that the 1940 population was about five million less than would fit smoothly into the adjacent values. (Why do you think this happened?) And more, for we can now see that 1920 was a couple of millions low also (unless 1930 is thought to be unduly high).

Our magnifying glass is now working at full capacity, at least until we identify some further partial description and arrange to subtract it out also. To learn more about the later years of U.S.A. population growth, we would need either to get year-by-year estimates or to study the mechanisms that are involved.

the earlier details

So much for the later years--what of the earlier ones? We left unfinished business when we said that the lefthand side of exhibit 10 seemed quite straight. We can attack this logarithmic straightness in the early years just as we attacked the linear straightness in the later ones.

exhibit **13** of chapter 5: U.S.A. population

Later population of the U.S.A. (expanded deviations from the specified straight line)

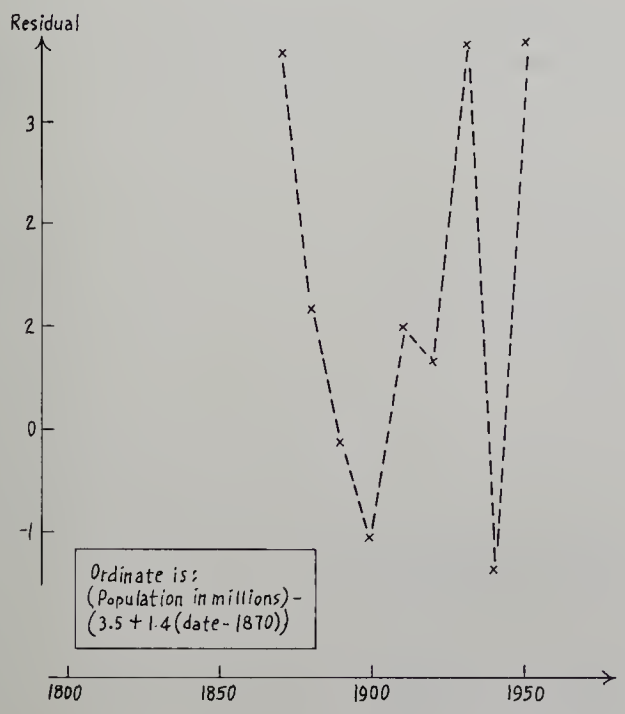


Exhibit **14** shows the result of drawing a straight line on exhibit 10. It is encouraging enough for us to hasten to the residuals, shown in exhibit **15**. (In 1880, a population of 50.2 millions gives a log of 7.70, while the fit is 6.75 + .012(80) = 7.71, so that the residual is −0.01. More decimals were used in calculating the points for exhibit 15.) As we ought to have expected, the residuals prove useful in the early years but of very dubious value in the later ones. We can again afford to use the magnifying glass on the relatively flat section, as is done in exhibit **16**.

Exhibit 16, once we realize that ±0.01 in log is about ±2.3% in size, gives us a quite delicate view of U.S.A. population growth during the nineteenth century. After 1860, population growth was not as fast as before. Moreover, 1800 appears to have been additionally depressed by 3 or 4 per cent. Why? Again, we have gone as far with our microscope as seems reasonable without further inputs.

Experts believe many of the detailed fluctuations now so clear to us are due to variations in completeness of the census, rather than to changes in population growth. Clearly, the data cannot contradict the experts. The fluctuations are in the numbers, whatever their source. It is worthwhile to find them, whether they tell us about the growth of U.S. population or the deficiencies of U.S. censuses.

exhibit **14** of chapter 5: U.S.A. population

Population of the U.S.A. (logarithmic scale with comparison line)

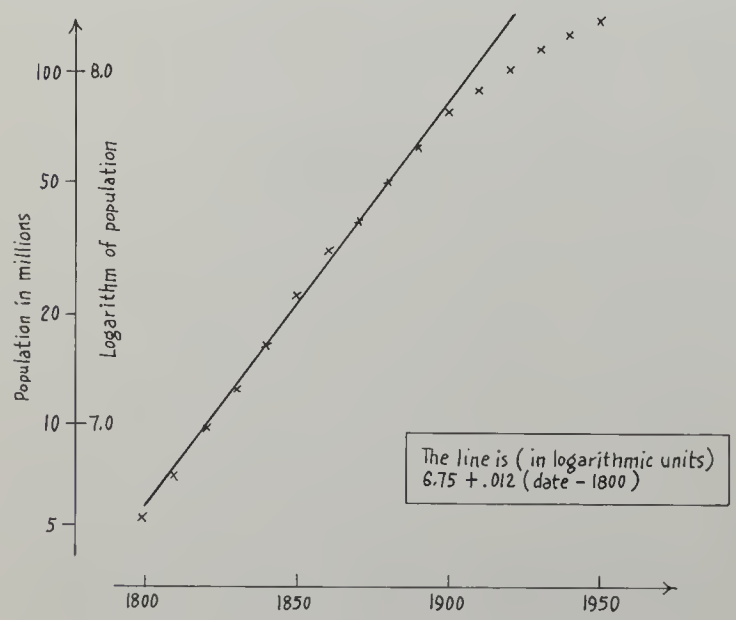


exhibit **15** of chapter 5: U.S.A. population

Population of the U.S.A. (deviations from logarithmic straight line)

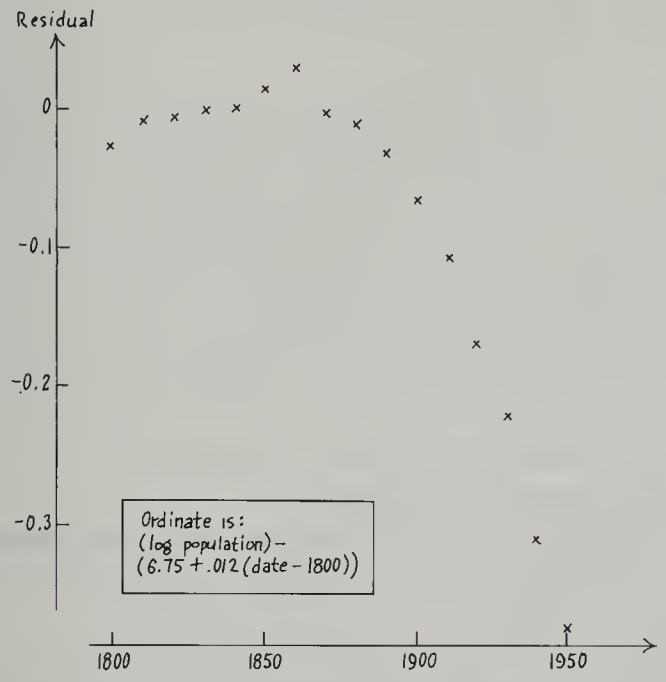
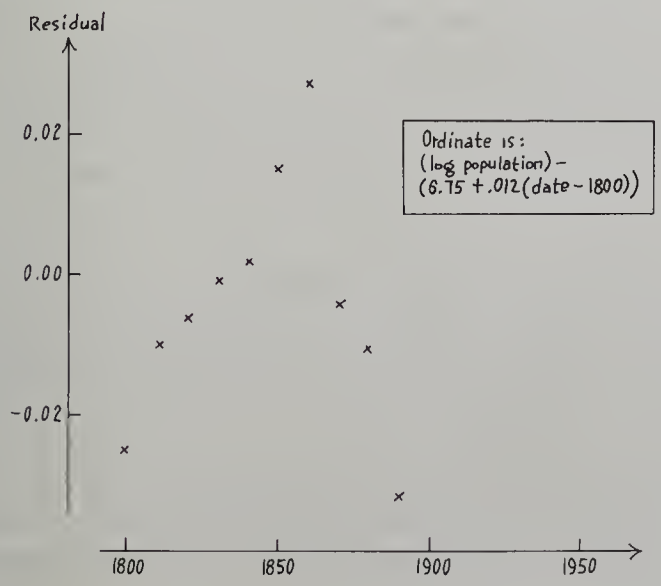


exhibit **16** of chapter 5: U.S.A. population

Earlier population of the U.S.A. (expanded deviations from given logarithmic line)



If we had to choose a set of pictures to summarize U.S.A. population growth as completely as we now can, we would probably choose to show all four of the following:

- ◇ exhibits 14 and 11, to show the general patterns of growth.
- ◇ exhibits 16 and 13, to show local behavior.

Together, these four would be responsive to the request: Make useful plots of U.S. population against date. (If we were population specialists, we would know about logistic functions and be able to fit a single incomplete description all the way from 1800 to 1950. This would simplify the plotting of a single graph of residuals, and would probably allow us to summarize the situation in two plots, one showing the fit and the other the residuals.)

What are the lessons to be learned from this example? Not merely that thought can help us see deeper. We have seen specific examples of very general principles, including these:

- ◇ **choosing scales to make behavior roughly linear always allows us to see local or idiosyncratic behavior much more clearly.**
- ◇ **subtracting incomplete descriptions to make behavior roughly flat always allows us to expand the vertical scale and look harder at almost any kind of remaining behavior.**

Whatever the data, we can try to gain by straightening or by flattening. When we succeed in doing one or both, we almost always see more clearly what is going on.

review questions

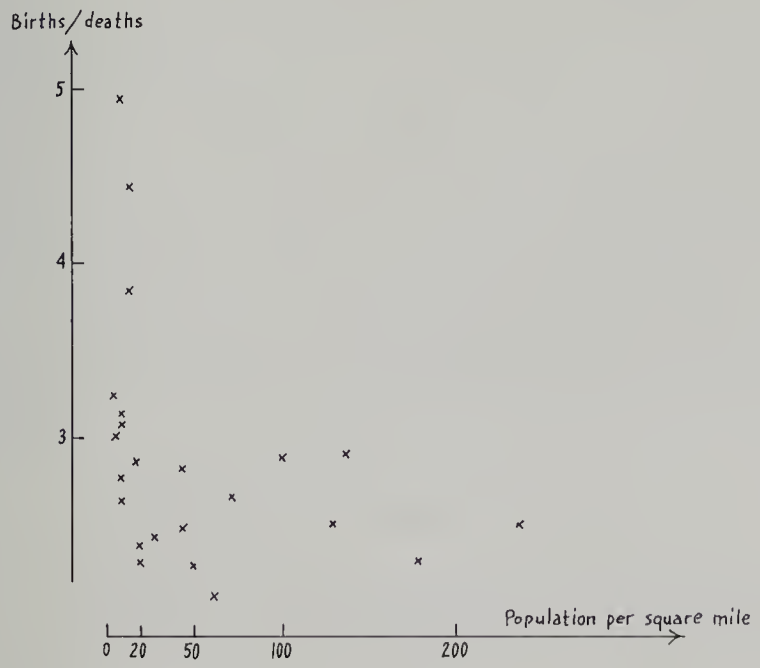
Can we plot without thinking? How much will we learn? What can we see from exhibit 9? How can we use what we see? What does exhibit 11 tell us to do next (go forward as far as you can)? Did we do it? Where? What does exhibit 14 tell us to do next (go forward as far as you can)? Did we do it? Where? Which pictures would we choose to "tell all" about the U.S. population? Why? What are two important lessons to be learned from this example?

5E. Plotting the ratio of births to deaths

The *County and City Data Book* of the U.S. Bureau of the Census contains much varied information. In particular, the 1962 edition gives for each state the number of live births for 1959, the number of deaths for 1959, and the density of population for 1960. One who believed in the "wide open spaces" might feel that the ratio of births to deaths would appear to be influenced by the density of population, at least if the South and the Atlantic coast states were set aside. A plot of the ratio of births to deaths against

exhibit **17** of chapter 5: births and deaths

Births/deaths and population density



population density for the remaining states appears as in exhibit **17**. About all we can say from this plot is that the point spatter seems crudely L-shaped. The use of a linear scale for population per square mile has squeezed so many of the states up against the vertical axis that we can't be sure what is going on. If we are to see what, if anything, is going on, we must adjust the left-to-right scale so that the states are less jammed together.

Exhibit **18** shows the result of using a log scale for population density. We now see that the three states with unusually high ratios of births to deaths have low, but not very low densities, and do not appear to be typical of very low-density states. Looking only at the other states, there may be a faint tendency for a higher ratio of births to deaths to go with lower population density. (By setting aside the three states, we could double the vertical scale for the rest, putting this tendency under a slightly stronger microscope. Doing this teaches us almost nothing new, as the reader may verify.)

another try

If we are to give up on population density, what next? The three unusual states (and their birth/death ratios) were: New Mexico (4.95), Utah (4.46), and Arizona (3.83). Looking again in the *County and City Data Book*, we discover

exhibit **18** of chapter 5: births and deaths

Births/deaths and population density by states (density on logarithmic scale)

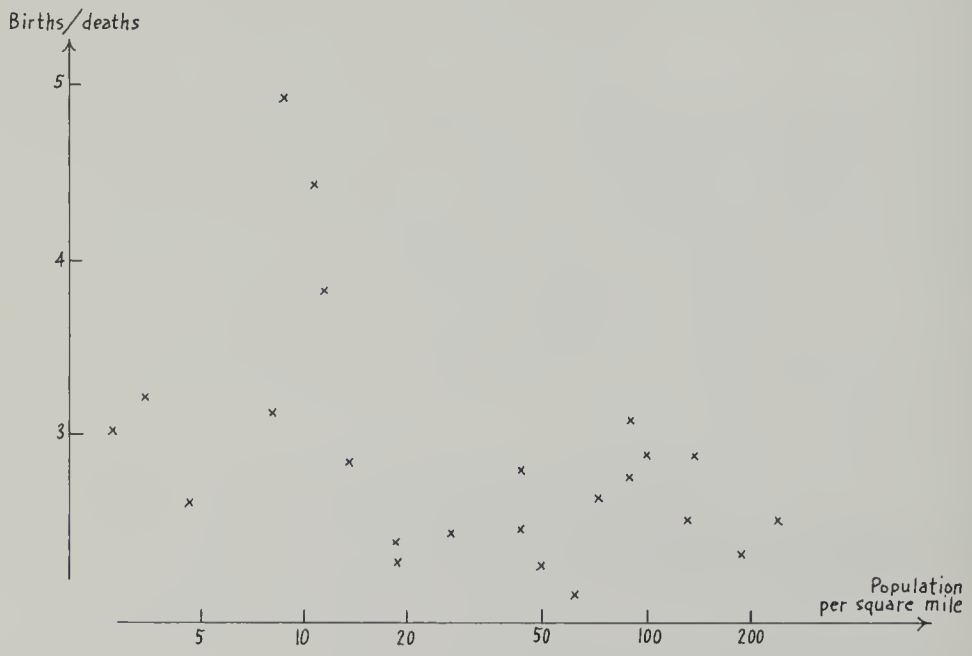
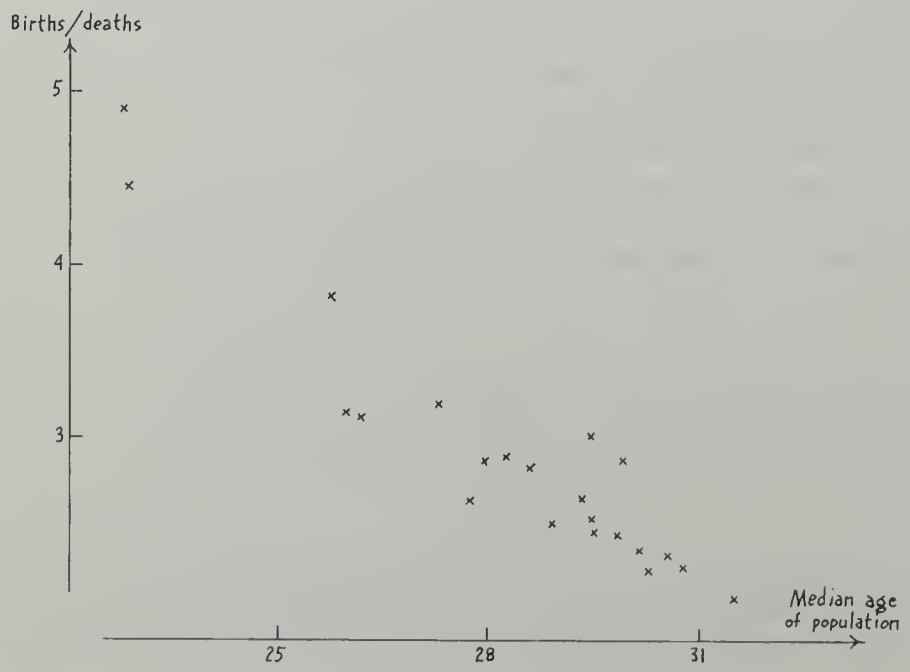


exhibit **19** of chapter 5: births and death

Births/deaths and median age by state



that all three have young populations, for instance, as measured by median age. Thus it is natural to plot births/deaths against median age.

Exhibit **19** shows the result. Clearly, median age does a much better job of appearing to explain the ratio of births to deaths than did population density, though the apparent dependence is, of course, not perfect. If we are to go further, we will need to find and subtract out some partial description of this apparent dependence.

When we "put eye to paper" and look along the point spatter in exhibit 19, we seem to see a definite tendency to curvature (hollow above). If we could eliminate this, we could reasonably compare the individual points with a straight line. How might we approach such a simpler partial description?

One thing to try is changing the down-to-up scale. It is easy to see that using squares to fix this scale would make curvature worse, so we may as well try going in the opposite direction by using logs, a choice which is attractive because of the more symmetric way it treats births and deaths. Notice how the identities

$$\log\left(\frac{\text{births}}{\text{deaths}}\right) \equiv \log \text{births} - \log \text{deaths}$$

$$\equiv -(\log \text{deaths} - \log \text{births}) \equiv -\log\left(\frac{\text{deaths}}{\text{births}}\right)$$

exhibit this symmetry.

Exhibit **20** shows the result, complete with a convenient comparison line. The point spatter is now much more nearly straight. When we plot the corresponding residuals against median age and identify the more extreme states, we find the results shown in exhibit **21**.

going to the map

The two states with notably high residuals are adjacent to one another on the map. The four states with middle median age and notably low residuals also touch one another. Clearly, we need to see the residuals on a map.

Exhibit **22** shows the residuals spread across a map of the United States. The roughly regular structure of these residuals is rather easily seen--adjacent states are clearly more often similar than are distant ones. (Might the somewhat surprisingly positive residual for Illinois be due to the unusual size of Chicago?) To go further here requires either:

- ◇ more careful allowance for the age of the population, or
- ◇ more knowledge about the mechanisms affecting birth and death rates in general. (Would state-by-state information on economic conditions help?)

Indeed, both of these are likely to be needed!

exhibit **20** of chapter 5: births and deaths

Births/deaths (logarithmic scale) and median age by state

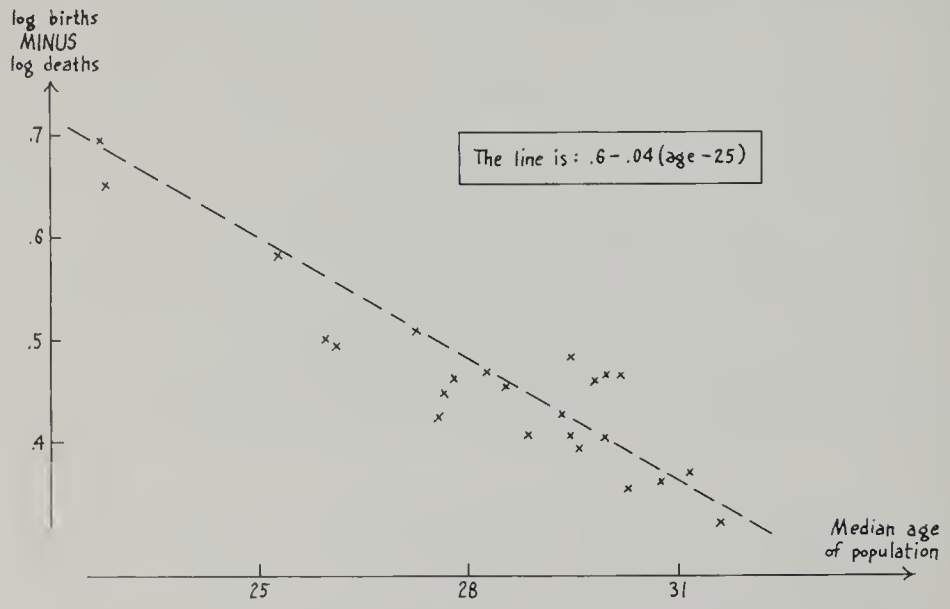
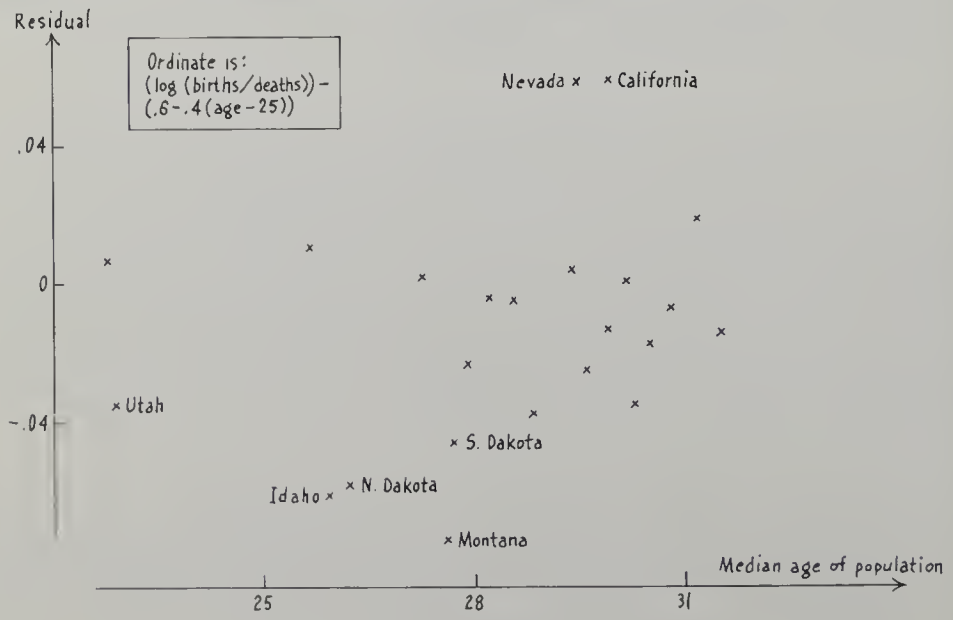


exhibit **21** of chapter 5: births and deaths

Residuals of log $\frac{\text{births}}{\text{deaths}}$ against median age by state



In this example, we have again seen the same main points as in the previous one:

- ◇ **changing scales to make dependences roughly linear usually helps.**
- ◇ **flattening by subtraction makes it much easier to see what is going on at more subtle levels.**

The fact that our dependences were approximate rather than exact did not alter these main points at all.

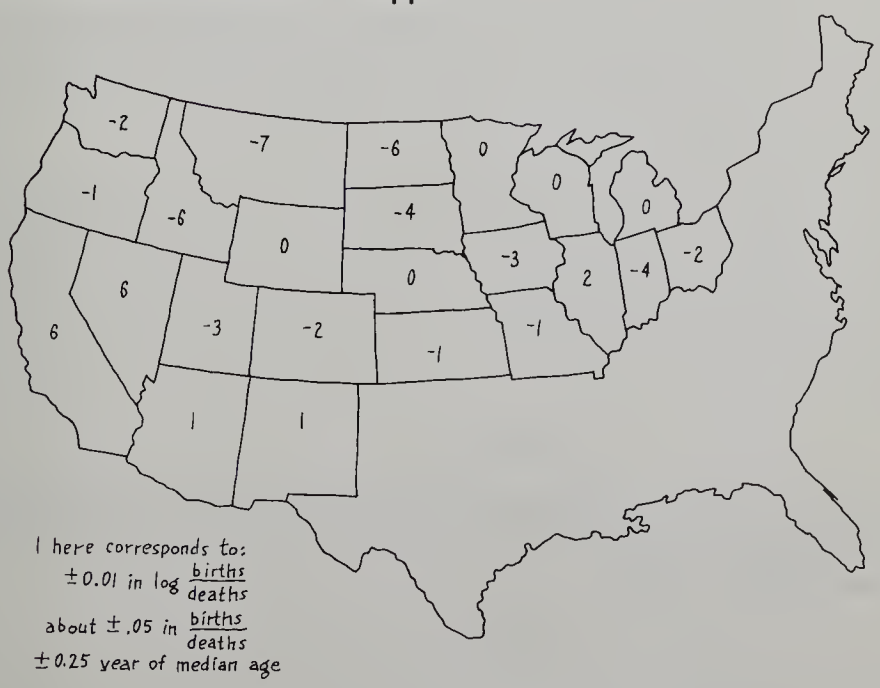
Approximate dependence did, however, bring in one new aspect, as we saw in exhibits 17 and 18:

- ◇ **the usefulness of changing scales to reduce confusion caused by crowding.**

In both examples, it is clear that we never expect the data to be ON the line--only that it might, if we are lucky, be NEAR the line. Once we put the first example under the microscope, what we saw is not intrinsically different from the other example. If we had contracted out the 1850 census to two

exhibit **22** of chapter 5: births and deaths

The residuals of exhibit 17 mapped



different contractors, we would have gotten two different numbers for the population of the U.S.A. The potential values of the U.S.A. population--those we might reasonably have found--do not lie ON a curve; they merely lie QUITE NEAR one. We have happened not to buy more than one census value at a time. This makes the first example LOOK a little different, but will not keep us, in the next chapter, from interchanging the axes used for date and population. Once we face the uncertainties of "what the numbers might have been," almost all data is at best "just NEAR a line or curve".

review questions

Where did we get the data for the example of this section? What did our first try teach us? Why did it pay to use a log scale for population density? What did we try next? Why? How well did it work? Why did we try a log scale for births/deaths? What two exhibits combine to tell the story (no map yet)? Why did we go to a map? How well did it work?

5F. Untilting defines "tilt"

We are now well aware of how much we can gain by flattening our picture, because this lets us expand its vertical scale. It is well to have things straight before flattening them; we can then "blow up" the picture even more, but flattening of even unstraightened pictures can help too.

We want a procedure that flattens the data out, whether or not it is straight. Here "flattening" must refer to the general run of the data, and not to its detailed behavior. Something that looks like a plausible set of residuals from a straight line is "flattened", though it may not be flat.

Exhibit **23** shows a rather extreme example of curvature, and exhibit **24** the results of "flattening" exhibit 23. Clearly, exhibit 24 is far from being flat, so "flattening" is a poor term. Equally, it is poor use of words to talk about "the slope" of exhibit 23, since there is a very small slope toward the left of the plot and a very large slope to the right.

Tilt is a short word, and conveys an appropriate feeling. We shall say that exhibit 24 is

untilted,

and that the slope of the line whose subtraction converts exhibit 23 into exhibit 24--thus clearing away an important part of what is going on so that we can see better what remains--is the

tilt

of exhibit 23.

exhibit **23** of chapter 5: illustrative

A tilted set of data

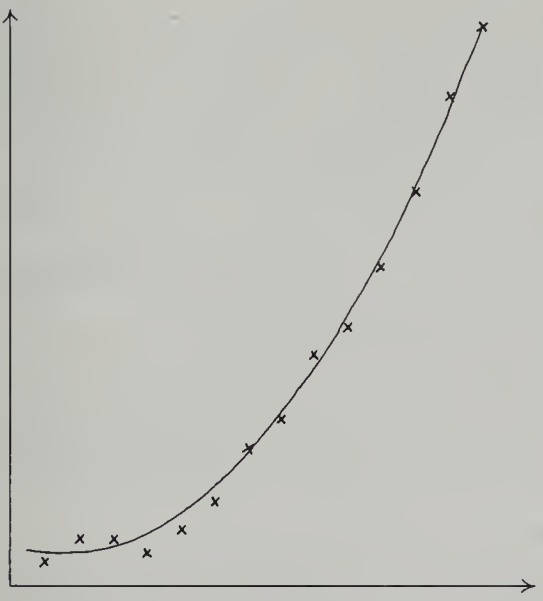
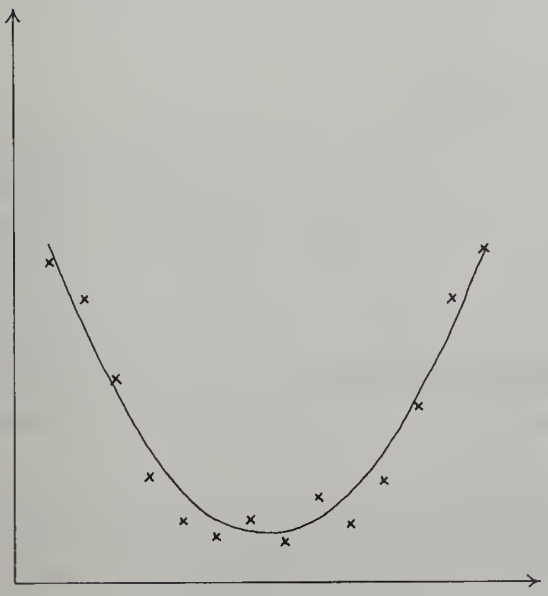


exhibit **24** of chapter 5: illustrative

The data of exhibit 23 untilted



Thus, we take the question "How much is y tilted against x?" to mean: "What value of b leaves $y - bx$ apparently untilted?" Equally, we shall take the b thus found as our assessment of tilt, and the values of $y - bx$ as what we have been able to do about freeing y of tilt against x.

There can be a variety of different--often only slightly different, usually not very much different--definitions of "apparently untilted". For each such definition there is a--comparably slightly different--value of b and thus both an assessed value for the tilt and a set of apparently untilted values. These differ--from definition to definition--by only comparably slight amounts.

The existence of such alternatives--and the need to choose between them--usually bothers us not at all.

In the great majority of cases, our concern with tilt is like the woodsman's concern with bushes and fallen trees on a trail he plans to use:

- ◇ we may be concerned to know that there appears to be some tilt (though we may have known that there would be a tilt in this particular direction long before we collected the data).
- ◇ we may even want to know how much tilt there seems to be.
- ◇ **we are almost certain,** whatever else, **to want to clear the tilt out of our way.**

We have already, in specific instances, cleared our way by using an eye-judged tilt to flatten our graphs. We frequently need to do something similar in fixing numbers for future analysis. Especially because we are going to use any tilt we assess in such a way, we will meet our needs quite well enough if we do a reasonably good job of assessing our tilt. (We need not worry as to in what way, if any, our assessment might be **best possible**--or even whether anyone can define "best possible" sensibly.)

review questions

What is a tilt? Why use a special word? What is it to be untilted? How do we define tilt? Are there many definitions, or few, or only one definition of "untilted"? Do we need a "best possible" tilt? Can you define a "best possible" tilt?

5H. How far have we come?

In this chapter we have met plots of y against x. Perhaps we have even come to know them a little.

The state of our progress is not measured by the specific techniques we have seen or understood--though that kind of progress is essential. Rather, our progress is measured by our acceptance of such propositions as these:

1. **Graphs are friendly.**

2. **Arithmetic often exists to make graphs possible.**

3. **Graphs force us to note the unexpected; nothing could be more important.**

4. **Different graphs show us quite different aspects of the same data.**

5. **There is no more reason to expect one graph to "tell all" than to expect one number to do the same.**

6. **"Plotting y against x" involves significant choices--how we express one or both variables can be crucial.**

7. **The first step in penetrating plotting is to straighten out the dependence or point scatter as much as reasonable.**

8. **Plotting y^2, $\sqrt{y}$, $\log y$, $-1/y$ or the like instead of y is one plausible step to take in search of straightness.**

9. **Plotting x^2, $\sqrt{x}$, $\log x$, $-1/x$ or the like instead of x is another.**

10. **Once the plot is straightened, we can usually gain much by flattening it, usually by plotting residuals (with regard to the partial description implied by the straight line we may not have quite drawn in yet).**

11. **When plotting point scatters, we may need to be careful about how we express x and y in order to avoid concealment by crowding.**

In particular, we have learned--or been reminded--how to:

- ◇ understand about subtracting one curve from another.
- ◇ use two points to find an equation for a straight line.

Our two examples differed in one way. The years of census are fixed by law, and at each a population is measured. It is rather easy to try to think of the points as having been selected from a curve with one population at each possible date. Births/deaths, on the other hand, whether compared with population density or with median age of population, provides a much more symmetric situation. The boundaries of states are fixed by law, and for each state two things happen, an x and a y. There is no possibility of assuming that all the data are ON a curve; the most we can hope is that the data are NEAR a line or curve.

5P. Additional problems

See exhibits **25**, **26**, **27**, **28**, **29**, **30**, **31**, **32**, **33**.

exhibit **25** of chapter 5: data and problems

Data for the 15 smallest counties of 3 states (from 1962 *County and City Data Book*)

A) SOUTH CAROLINA

		Population (1960)				Families (1959)	Local Gov't (1957)	
(*)	Name	Total	% less than 5 yrs school†	% at least 65	per sq. mile	% less than $3000 income	% budget for education	Total budget ($1000's)
2424	McCormick	8,629	28.2	8.8	23	59.8	62.0	566
2130	Allendale	11,362	34.1	7.9	27	60.1	63.1	1082
2051	Jasper	12,237	37.1	7.5	19	60.2	79.7	1082
2050	Calhoun	12,256	26.1	7.9	33	68.2	76.6	659
1852	Saluda	14,554	17.0	9.7	33	50.6	68.9	656
1753	Edgefield	15,735	23.1	8.0	33	55.3	70.1	785
1720	Bamberg	16,274	26.6	8.3	41	58.5	63.1	1340
1622	Hampton	17,425	31.8	7.2	31	58.0	57.8	1312
1608	Barnwell	17,659	23.2	7.4	32	47.5	82.9	1565
1405	Fairfield	20,713	30.8	7.8	30	54.2	56.8	1596
1356	Abbeville	21,417	23.0	8.7	42	42.1	56.4	1717
1339	Lee	21,832	31.7	6.4	53	68.6	75.8	1671
1211	Dorchester	24,383	23.1	6.6	43	49.6	67.9	1684
1092	Colleton	27,816	29.3	7.5	27	57.9	46.7	3090
1066	Marlboro	28,529	29.6	6.8	59	58.3	63.4	1948
(Whole state)		((34,262))	(20.3)	(6.3)	(79)	(39.5)	(63.4)	––

State has 46 counties.

B) GEORGIA

3077	Echols	1,876	28.0	8.5	4	55.6	59.5	122
3051	Quitman	2,432	34.8	9.3	14	70.0	79.5	205
3028	Glascoch	2,672	35.6	56.4	19	61.1	67.9	252
2979	Webster	3,247	34.5	8.7	17	71.2	73.3	225
2978	Schley	3,256	23.0	10.3	20	67.4	69.7	221
2970	Taliaferro	3,370	27.9	13.4	17	68.5	48.8	391
2950	Dawson	3,590	22.4	8.6	17	64.7	71.3	293
2934	Long	3,874	22.7	7.5	10	60.1	52.0	408
2875	Towns	4,538	11.3	10.2	27	63.7	33.4	724
2674	Baker	4,543	33.2	8.9	15	74.1	56.4	622
2873	Clay	4,551	26.8	10.5	20	66.6	59.2	417
2832	Lanier	5,097	30.1	7.5	31	57.4	64.5	686
2810	Charlton	5,313	26.5	6.2	7	44.0	49.1	703
2808	Heard	5,333	22.7	10.7	18	56.4	57.1	652
2805	Wheeler	5,342	29.4	9.5	18	63.7	54.4	418
(Whole state)		((12,038))	(17.6)	(7.4)	(68)	(35.6)	(45.0)	––

State has 159 counties.

→

exhibit **25** of chapter 5 (continued)

C) ALABAMA

2201	Coosa	10,726	18.4	10.7	18	51.5	39.5	1174
2179	Cleburne	10,911	21.5	9.4	19	52.4	48.2	1088
2034	Clay	12,400	14.4	13.0	21	54.0	48.3	1347
1945	Bullock	13,462	32.7	11.7	22	69.4	42.8	1727
1934	Greene	13,600	38.0	9.8	21	74.0	57.9	1514
1876	Lamar	14,271	14.7	11.0	24	51.4	37.5	2507
1870	Bibb	14,357	22.7	9.6	23	54.4	54.7	1405
1828	Winston	14,858	17.3	9.8	24	53.8	55.7	1072
1824	Crenshaw	14,909	23.9	11.2	24	69.5	51.4	1332
1796	Henry	15,286	27.8	9.0	27	63.8	56.0	1313
1794	Washington	15,372	23.3	7.8	14	51.7	54.5	1353
1790	Lowndes	15,417	37.2	9.2	22	72.1	55.4	1243
1729	Fayette	16,148	16.9	10.8	26	54.7	43.0	1425
1715	Cherokee	16,303	16.6	8.9	27	49.1	53.7	1498
1628	Perry	17,358	28.7	10.5	24	69.2	49.4	1689
(Whole state)		((25,738))	(16.3)	(8.0)	(64)	(39.1)	(45.8)	--

State has 67 counties.

* **National rank by population.**
† **Of those 25 or over.**
( ) **Median.**

P) PROBLEMS

25a) Panels A to C contain selected information about the 15 smallest counties in South Carolina, Alabama, and Georgia. Plot

y = % with less than five years schooling

against

x = % with less than $3000 income

for at least two states. Continue the analysis. Comment.

S) SOURCE

1962 ***County and City Data Book.***

exhibit **26** of chapter 5: data and problems

Some problems

26a) The following data was obtained in preparing a standard curve for the determination of formaldehyde by the addition of chromatropic acid and concentrated sulphuric acid and the reading of the consequent purple color on a Beckman Model DU Spectrophotometer at 570 mμ.

Amount of CH_2O Used	Optical Density
0.1	0.086
0.3	0.269
0.5	0.446
0.6	0.538
0.7	0.626
0.9	0.782

Analyze graphically, using at least two graphs. Comment. (Bennett & Franklin, p. 216; from Roberts.)

26b) The relation between the amount of β-erythrodine dissolved in water and the turbidity of the solution--as read on a colorimeter--is not quite as simple. Some data gives:

Concentration (in mg/ml)	Colorimeter reading
40	69
50	175
60	272
70	335
80	390
90	415

Analyze graphically. Comment. (Bennett & Franklin, p. 217, from Wois-lawski.)

26c) Find two different collections of (x, y) points that interest you, and make useful plots.

S) SOURCE

See exhibit 27.

exhibit **27** of chapter 5: data and problem

Carbon content of 36 clays measured directly and estimated indirectly

A) DATA

Clay #	Direct measurement	Indirect estimate
1	1.53	2.46
2	0.87	1.54
3	0.28	0.70
4	0.27	−0.40
5	3.07	4.82
6	0.25	0.30
7	0.25	0.64
8	0.29	0.78
9	0.12	0.12
10	1.50	2.36
11	1.31	2.14
12	0.31	0.08
13	0.14	−0.01
14	2.98	4.53
15	6.84	9.94
16	2.15	3.68
17	1.35	1.84
18	0.40	0.97
19	4.18	6.14
20	0.22	0.52
21	0.38	0.40
22	0.24	0.46
23	1.79	2.80
24	0.58	2.09
25	6.55	9.68
26	2.54	4.08
27	1.43	2.80
28	2.74	3.93
29	6.08	8.22
30	0.75	0.28
31	0.16	0.35
32	5.06	7.49
33	0.86	1.41
34	0.16	−0.50
35	11.43	15.80
36	0.19	0.18

exhibit **27** of chapter 5 (continued)

P) PROBLEM

27a) The amount of carbon in a clay can be measured directly by heating the clay until all the carbon compounds are burned, collecting the carbon dioxide thus formed, and measuring its amount. The amount of carbon can be estimated by combining the amounts of its constituents in a suitable standard way. The results of such measurements on 36 clays from South Devonshire, England, are given in exhibit 27. Analyze graphically. Comment.

S) SOURCE

C. A. Bennett and N. L. Franklin 1954, Statistical Analysis in Chemistry and the Chemical Industry. John Wiley, New York. Table 6.3 on page 218.

exhibit **28** of chapter 5: data and problems

Percentage Democratic in 12 presidential elections for 24 Northeastern and Central States (percentage Democratic of major party vote)

A) DATA

	1920	1924	1928	1932	1936	1940	1944	1948	1952	1956	1960	1964
Colorado	37.7	27.8	34.4	57.0	61.9	48.7	46.6	52.7	39.3	39.5	45.1	61.6
Connecticut	34.5	30.9	45.9	49.4	57.8	53.6	52.7	49.2	44.1	36.3	53.7	67.9
Delaware	43.0	38.9	33.9	48.8	54.9	54.8	54.6	49.4	48.1	44.7	50.8	61.1
Illinois	27.3	28.4	42.6	56.8	59.2	51.2	51.7	50.4	45.0	40.4	50.1	59.5
Indiana	42.3	41.2	39.9	56.0	57.5	49.3	47.1	49.6	41.4	39.9	44.8	56.2
Iowa	26.4	23.0	37.8	59.1	56.0	47.8	47.7	51.4	35.8	40.8	43.3	62.0
Kansas	33.4	27.7	27.3	54.8	53.9	42.7	39.4	45.4	30.7	34.3	39.3	54.6
Maine	30.2	23.3	31.1	43.6	42.8	48.8	47.5	42.7	33.8	29.1	43.0	68.8
Maryland	43.3	47.7	42.6	63.1	62.7	58.8	51.9	49.3	44.2	40.0	53.6	65.5
Massachusetts	28.9	28.5	50.5	52.1	55.1	53.4	52.9	55.9	45.6	40.5	60.4	76.5
Michigan	23.4	14.8	29.1	54.1	59.2	49.8	50.5	49.1	44.2	44.2	51.0	66.8
Minnesota	21.6	11.7	41.4	62.3	66.6	51.9	52.8	58.9	44.4	46.2	50.7	63.9
Nebraska	32.6	38.5	36.4	64.1	58.4	42.8	41.4	45.8	30.8	34.5	37.9	52.6
New Hampshire	39.7	36.7	41.2	49.3	50.9	53.2	52.1	47.1	39.1	33.9	46.6	63.9
New Jersey	29.6	30.6	40.0	51.0	60.1	51.8	50.7	47.7	42.5	34.6	50.4	66.0

→

exhibit **28** of chapter 5 (continued)

New York	29.5	34.3	48.8	56.7	60.2	51.8	52.5	49.5	44.0	38.7	52.6	68.7
North Dakota	18.9	12.7	44.8	71.3	69.2	44.7	45.8	45.4	28.6	38.2	44.5	58.1
Ohio	39.8	28.9	34.7	51.5	60.8	52.2	49.8	50.1	43.2	38.9	46.7	62.9
Pennsylvania	29.2	22.6	34.2	47.1	58.2	53.5	51.4	48.0	47.0	43.4	51.2	65.2
Rhode Island	33.9	37.9	50.3	56.0	56.8	56.8	58.7	58.2	49.1	41.7	63.6	80.9
South Dakota	24.5	21.2	39.4	64.9	56.0	42.6	41.7	47.6	30.7	41.6	41.8	55.6
Vermont	23.5	16.7	33.0	41.6	43.4	45.1	42.9	37.5	28.3	27.8	41.4	66.3
West Virginia	43.9	47.1	41.3	55.1	60.7	57.1	54.9	57.6	51.9	45.9	52.7	67.9
Wisconsin	18.5	17.9	45.3	67.0	67.8	50.9	49.1	52.3	38.8	38.1	48.1	62.2

P) PROBLEMS

Panel A gives the % Democratic vote in each of 24 northeastern and central states for 12 presidential elections, 1920 to 1964. Plot the following:

28a) 1964 against 1956

28b) 1960 against 1920

28c) 1952 against 1932

28d) Any one against any other that you think will have a close relationship.

exhibit **29** of chapter 5: data and problems

More problems

29a) Determination of ethylene chlorohydrin. (264, 270) translates as: "With 26.4 milligrams of ethylene chlorohydrin present, 27.0 milligrams were found". Data-sets (6): (264, 270), (595, 594), (1173, 1183), (1777, 1780), (2355, 2370), (3578, 3576). SOURCE: K. Uhrig 1946. Determination of ethylene chlorohydrin. *Industrial and Engineering Chemistry, Analytical Edition* **18**: 369 only. Table 1 on page 369. PROBLEM: Choose a plot that is likely to be revealing by thinking hard. Explain the reasons for your choice. Make the plot.

29b) Polarographic behavior of ions containing vanadium. (94, 35) translates as: "For a concentration of vanadite ion of 0.094 millimoles per liter, the anodic diffusion constant in microamperes was 0.35 microamperes". Data-sets (8): (94, 35), (278, 98), (508, 178), (880, 309), (1548, 563), (1840, 696), (352, 1285), (505, 1813). (Last point: 5.05 millimoles/liter, 18.13 microamperes.) SOURCE: J. J. Lingane 1945, "Polarographic characteristics of vanadium in its various oxidation states," *J. Amer. Chem. Soc.* **67**:182–188. Table I on page 186. PROBLEM: Make useful plots of diffusion current against vanadite concentration. →

exhibit **29** of chapter 5 (continued)

29c) Amount of desired product in a chemical reaction after different reaction times and under different circumstances. (1; 32, 54; 87, 159, 226) translates as: "In run 1, the amount of desired product in moles per liter was 0.032 after 80 minutes, 0.054 after 160 minutes, 0.087 after 320, 0.159 after 640, and 0.226 after 1280 minutes." Data-sets (16 runs under 16 different sets of conditions): (1; 32, 54; 87, 159, 226), (2; 147, 234; 343, 342, 203), (3; 48, 108; 225, 346, 420), (4; 232, 390; 556, 634, 416), (5; 37, 38; 172, 200, 239), (6; 179, 283; 405, 342, 216), (7; 86, 133; 259, 398, 508), (8; 309, 514; 722, 764, 389), (9; 74, 99; 200, 309, 249), (10; 253, 343; 391, 284, 75), (11; 133, 271; 430, 580, 494), (12; 508, 756; 842, 570, 115), (13; 96, 158; 276, 339, 230), (14; 308, 444; 467, 249, 29), (15; 228, 372; 579, 691, 539), (16; 626, 880; 895, 434, 58). SOURCE: G. E. P. Box and W. G. Hunter 1962, "A useful method for model-building," *Technometrics* **4**: 301–318. Table 1 on page 304. PROBLEM: Make useful plots of concentration at 640 minutes against concentration at 160 minutes.

29d) Make useful plots for one or more other pairs of reaction times. (Data-sets in problem (29c) above)

exhibit **30** of chapter 5: data and problems

Yet more problems

30a) Analysis of samples for chrysanthenic acid. (0, 23) translates as: "When 0 micrograms of synthetic racemic chrysanthenic acid were added, the colorimeter scale reading was 23". Data-sets (13): (0, 23), (5, 32), (10, 40), (20, 54), (40, 86), (60, 118), (80, 146), (100, 179), (120, 212), (140, 240). (160, 272), (180, 300), (200, 330). SOURCE: A. A. Schreiber and D. B. McClellan 1954. Estimation of microquantities of pyrethroids. *Analytical Chemistry* **26**: 604–607. Table I on page 605. PROBLEM: Make useful plots of colorimeter reading against amount of chrysanthenic acid.

30b) Residual strength of 8-oz. cotton duck attacked by 4 different kinds of fungus. (3; 97, 105; 103, 101) translates as: "After 3 hours of incubation, the strengths--referred to initial strength = 100--of the sample exposed to Thielaria was 97, that exposed to Humicola was 105, for Chaetomium was 103, for Myrothecium was 101." Data-sets (24): (3; 97, 105; 103, 101), (6; 98, 106; 101, 105), (9; 95, 107; 99, 95), (12; 96, 105; 95, 95), (15; 97, 106; 90, 100), (18; 98, 102; 91, 97), (21; 97, 101; 78, 98), (24; 97, 90; 74, 93), (27; 90, 81; 71, 82), (30; 96, 78; 71, 76), (33; 89, 73; 65, 67), (36; 88, 69; 58, 64), (39; 89, 63; 53, 59), (42; 86, 59; 47, 54), (45; 82, 55; 44, 50), (48; 79, 53; 44, 42), (51; 73, 52; 42, 41), (54; 73, 41; 40, 40), (57; 73, 42; 40, 39), (60; 68, 41; 39, 35), (63; 59, 36; 38, 37), (66; 57, 37; 37, 33), (69; 57, 31; 35, 34), (72; 55, 34; 36, 31). SOURCE: E. Abrams 1950, "Microbiological deterioration of cellulose during the first 72 hours of attack," *Textile Research J.* **20**: 71–86. Table 2 on page 75. PROBLEM: Plot helpful curves for loss of strength from at least two kinds of fungus.

→

exhibit **30** of chapter 5 (continued)

30c) Rapid analysis for caffeine. (257, 131) translates as: "For a caffeine concentration of 0.257 milligrams in 100 milliliters, the average optical density was 0.131." Data-sets (20): (257, 131), (498, 262), (506, 265), (514, 263), (747, 384), (760, 393), (770, 396), (996, 512), (1013, 518), (1027, 523), (1245, 633), (1266, 643), (1284, 650), (1494, 760), (1519, 768), (1541, 775), (1798, 903), (2054, 1040), (2311, 1160), (2568, 1290). SOURCE: N. H. Ishler, T. P. Finucaine, and E. Borker 1948, "Rapid spectrophotographic determination of caffeine," *Analytical Chemistry* **20**:1162–1166. Table 1 on page 1162. PROBLEM: Make useful plots of optical density against caffeine concentration.

exhibit **31** of chapter 5: data and problems

Still more problems

31a) Survival of automobiles and trucks in use by a public utility. (0h, 990) translates as: "After 1/2 year, 0.990 of all vehicles were still in service." Data-sets (8): (0h, 990), (1h, 972), (2h, 944), (3h, 895), (4h, 784), (5h, 679), (6h, 593), (7h, 497). SOURCE: S. A. Krane 1963, "Analysis of survival data by regression techniques," *Technometrics* **5**: 161–174. Table on page 168. His source: H. A. Cowles, Jr., 1957. Prediction of mortality characteristics of industrial property groups. Ph.D. Thesis, Iowa State University. PROBLEM: Make helpful plots of fraction surviving against age.

31b) Heat and entropy contents of a sodium silicate. (400, 3080, 885) translates as: "The increases from 'room temperature' (298.16°K) to an absolute temperature of 400°K were 3,080 calories per mole for the heat content of Na_2SiO_3 and 8.85 calories/degree/mole for its entropy content." Data-sets (17): (400, 3080, 885), (500, 6300, 1604), (600, 9650, 2214), (700, 13190, 2760), (800, 16910, 3256), (900, 20730, 3708), (1000, 24700, 4124), (1100, 28770, 4511), (1200, 32940, 4874), (1300, 37210, 5216), (1361, 39870, 5416), (1361, 52340, 6332), (1400, 54010, 6453), (1500, 58390, 6748), (1600, 62570, 7024), (1700, 66850, 7284), (1800, 71130, 7528). (The last point, at 1800°K, gives a heat content change of 71,130 calories/mole and an entropy content change of 75.28 calories/degree/mole.) SOURCE: B. F. Naylor 1945, "High-temperature heat contents of sodium metasilicate and sodium disilicate," *J. Amer. Chem. Soc.* **67**: 466–467. Table II on page 467. PROBLEM: Make helpful plots of the increase in heat content against temperature.

31c) For Naylor's data (immediately above) make helpful plots of increase in entropy content against temperature.

exhibit **32** of chapter 5: data and problems

And yet more

32a) Equilibrium splitting of plutonium tribromide by water (gases at high temperature). (911, 153) translates as' "For an absolute temperature of 911°K, the observed equilibrium constant was 0.0153/atmosphere." Data-sets (11): (911, 153), (914, 156), (919, 149), (920, 163), (882, 246), (876, 282), (875, 247), (883, 243), (815, 704), (817, 502), (816, 692). SOURCE: I. Shift and N. R. Davidson 1949. Equilibrium in the vapor-phase hydrolysis of plutonium tribromide. Paper 6.24, at pages 831–840 of *The Transuranium Elements*, edited by Seaborg, Katz, and Manning. National Nuclear Energy Series IV-14B. McGraw Hill. Table 2 on page 835. PROBLEM: Make useful plots of equilibrium constant against temperature. Which three of the 11 data-sets do you think the authors rejected?

32b) Sales of Swiss bond issues since World War II. (46, 527) translates as: "In 1946, total sales of Swiss bonds––governmental and private––were 527 million francs". Data-sets (23): (46, 527), (47, 276), (48, 472), (49, 342), (50, 174), (51, 434), (52, 333), (53, 249), (54, 242), (55, 492), (56, 613), (57, 1148), (58, 827), (59, 686), (60, 890), (61, 1023), (62, 1124), (63, 2091), (64, 2503), (65, 2523), (66, 2292), (67, 2446), (68, 2648). SOURCE: *Swiss Statistical Abstract*, issued by the Swiss Credit Bank, November 1969. (Title also in French and German.) Table on page 46. PROBLEM: Make useful plots based on the data from 1950 to 1968.

32c) Comparison of two ways of measuring the water content of samples of the sea bed. (0 to 3; 76, 76) translates as: "For a sample from 0 to 3 inches below the surface of the sea level, measurement of % water by drying in an oven gave 76%, measurement by analyzing for chloride––and using the known concentration of chloride in deep sea water––gave 76%". Data-sets (14): (0 to 3; 76, 76), (3 to 6; 68, 72), (6 to 9; 69, 69), (9 to 12; 67, 67), (12 to 15; 60, 64), (15 to 18; 62, 62), (18 to 21; 60, 60), (21 to 24; 58, 59), (24 to 27; 57, 57), (27 to 30; 55, 56), (30 to 33; 55, 55), (33 to 36; 55, 55), (36 to 39; 53, 54), (39 to 42; 54, 54). SOURCE: L. J. Anderson 1948, "Conductometric titration of chloride in sea water and marine sediments," *Analytical Chemistry* **20**: 618–619. Table II on page 619. PROBLEM: Find differences between water by chloride and water by oven. Make a stem-and-leaf display of these differences. Comment on the appearance of this display. Find and plot residuals from a straight-line fit of water content against depth, separately for each method of finding water content. What do you conclude about the two methods of measuring water?

exhibit **33** of chapter 5: data and problems

Median ages of urban and rural populations and estimated colonial population

A) MEDIAN AGE, URBAN and RURAL POPULATIONS––at U.S. Censuses

Year	Median age	Urban population	Rural population
1950	30.4	88,927,464	61,769,897
40	29.5	74,923,702	57,245,573
30	27.1	68,954,823	53,830,223
20	26.1	54,157,973	51,552,647
10	24.9	41,998,932	49,973,334
1900	23.8	30,159,921	45,834,654
1890	22.9	22,106,265	40,841,449
80	21.6	14,129,735	36,026,048
70	20.6	9,902,361	28,656,010
60	20.2	6,216,518	25,226,803
50	19.5	3,543,716	19,648,160
40	17.9	1,845,055	15,224,398
30	17.2	1,127,247	11,738,773
20	16.5	693,255	8,945,198
10	15.9	525,459	6,714,422
1800	15.7	322,371	4,986,112
1790	15.9	201,655	3,727,559

Median age = A90 Median age of white males
Urban population = A195 Population of "urban territory"
Rural population = A206 Population of "rural territory"

B) ESTIMATED POPULATION OF AMERICAN COLONIES

Year	Estimated population
1780	2,780,369
70	2,148,076
60	1,593,625
50	1,170,760
40	905,563
30	629,445
20	466,185
10	331,711
1700	250,888
1690	216,372
80	151,507
70	111,935
60	75,058
50	50,368
40	26,634
1630	4,646

→

exhibit **33** of chapter 5 (continued)

P) PROBLEMS

33a) Analyze the median ages of panel A carefully.

33b/c) Analyze the urban/rural populations of panel A carefully.

33d) Take logs of the populations in panel B, and compare them with the extension of the fit given in text for 1790 to 1860.

33e) Fit the data of panel B, re-expressing it if necessary.

S) SOURCES

Historical Statistics of the U.S. Colonial times to 1957. Washington 1960.
(Panel A entries are from series A90, A195, A206 as indicated.

Straightening out plots (using three points) 6

chapter index on next page

We are now sure that we want to first straighten out and then flatten out plots. Straightening out is important--in the language of the opening of Chapter 4, a "big deal"--we want to learn to do it as easily as we reasonably can. This chapter is devoted to techniques and examples.

The sort of re-expression that concerns us is--as we have said--almost entirely re-expression of amounts, including large counts. (Here, counts that are never smaller than 3 are surely "large", and others may be.) These are the kinds of numbers where it is natural to take **powers, roots,** and **logs.**

In dealing with them, we will want to be sure that our origins are reasonably chosen. It is only for amounts measured from reasonable origins that we are likely to get full value from changing to a power, a root, or a logarithm.

Powers and roots of amounts are again amounts. Logs of amounts are **balances.** For these purposes--as for so many others--large counts are merely a special kind of amount. Accordingly, (nonzero) powers and roots of large counts are amounts, while logs of large counts are balances.

In thinking about our problems of re-expression--which need not be the same sort of thing as analyzing the data involved--we need to think about whether x--or y--varies much or little. So long as we deal with amounts, the natural way to make comparisons is by ratios--including percents. Thus we are interested in such facts as:

$$\frac{\text{largest } x}{\text{smallest } x} = 3,$$

All x are within $\pm 50\%$ of a middle value,

or

$$\frac{\text{largest } x}{\text{smallest } x} = 1.1,$$

largest x = smallest x plus 10%.

When we work in logs, we are concerned with balances, not amounts, and the natural way of expressing spread is either in log units or--occasionally--by the ratios into which **differences** in log units can be re-expressed.

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review questions

What will we try to do in this chapter? What sort of re-expression concerns us? Need we bother about choice of origin? What is important about how much x--or y--varies? When it is an amount or count? When it is a balance?

6A. Looking at three points

We have seen various illustrations of straightening out data. So far, either these have come from simple rational considerations, or else they have come--"out of the air"--with little apparent reason for the choice. How are we to behave when we have some other kind of data? Do we have to try all possible combinations of an expression of y with an expression of x on the whole data? Or can we save most of the effort this would involve?

The purpose of these changes of expression is to straighten out the data. If the data looks curved, in some overall way, we can make this obvious by picking out three representative points. For the early growth of the U.S.A. population, for example, we could choose the points corresponding to 1800, 1850, and 1890. These three points are:

(1800, 5.3)
(1850, 23.2)
(1890, 62.9)

A simple way to see that three points do not lie on a single straight line is to find the slopes of the straight lines through the first pair and the second pair, respectively; this gives:

$$\frac{23.2 - 5.3}{1850 - 1800} = \frac{17.9}{50} = 0.36$$

and

$$\frac{62.9 - 23.2}{1890 - 1850} = \frac{39.7}{40} = 0.99$$

which are quite different. The second slope is greater, so the curve is hollow upward. (Draw yourself a sketch.)

If any pair of choices of expression is going to straighten out the early portion of the U.S.A. population curve, these same choices will have to do a reasonably good job in straightening out these three points. We can save a lot of effort by screening our pairs of expressions on these three points. We will then need to try only the one best--or perhaps the few best--on the whole data.

We can often ease the task a little by choosing the spacing of the three points to simplify matters further. Had we chosen

(1810, 7.2)
(1850, 23.2)
(1890, 62.9)

we could have compared

$$16.0 = 23.2 - 7.2$$

with

$$39.7 = 62.9 - 23.2$$

since both changes in x are the same. ($1850 - 1810 = 40 = 1890 - 1850$).

Sometimes we can go almost this far by picking points so that changes in x have a simple ratio--1 to 2, 1 to 3, 3 to 2, etc.--instead of being equal.

Even trying all reasonably possible pairs of expressions on three points is an effort. Can we save some of this by looking into the direction in which changing an expression shifts curvature?

review questions

What relation ought three selected points have to all the available points? How do we ask three points about curvature? How can we make asking easier?

6B. Re-expressing *y* alone

Exhibit 19 of chapter 3 shows various expressions of y plotted against y. We see at once that the higher curves are hollow upward, while the lower curves are hollow downward.

To say that a curve is **hollow upward** means that if we take three points on the curve, the middle point is **below** the line joining the other two. Similarly, **hollow downward** means that the middle point is **above** the line joining the other two. Since we are trying to get a middle point **onto** the line joining the outer two, these are just the sort of facts that matter to us.

We can say more about the simple ladder of ways of expression, which includes

$$y^3$$
$$y^2$$
$$y$$
$$\sqrt{y}$$
$$\log y$$
$$-\frac{1}{\sqrt{y}}$$

$$-\frac{1}{y}$$

$$-\frac{1}{y^2}$$

$$-\frac{1}{y^3}$$

than we just have. We stated one special case of the following:

◇ **if one expression is straight, those above it are hollow upward, those below it are hollow downward.**

Exhibit 19 of chapter 3 has already shown us that this is true when y is straight. Exhibit 20 of chapter 3 shows us that this is true when log y is straight. The appearance of both those exhibits makes it plausible that the statement is true if any expression of the simple ladder is straight. (The doubting reader should try to use the following two statements to construct a general proof.)

◇ **if our three points are hollow UPward, we look further DOWN the ladder for straightness.**

◇ **if our three points are hollow DOWNward, we look further UP the ladder for straightness.**

These must be true. Consider the first: If a new expression is to be straight, and our present expression is hollow upward, the present expression has to be higher up the ladder than the new expression. To find the new expression, we must move down the ladder from the present expression.

So far as re-expressing y goes, the rule is simple:

◇ **move on the ladder as the bulging side of the curve points.**

U.S. population again

Applying this to the early U.S. population--since we have seen that the curve is hollow upward--moves us down the ladder. Let us try this, trying $-1/y$ first.

Turning to exhibit 6 of chapter 3 we can find

$$-1/5.3 = -0.188$$
$$-1/23.2 = -0.043$$
$$-1/62.9 = -0.016$$

The three points and the two slopes are now

$$(1800, -0.188)$$
$$(1850, -0.043)$$
$$(1890, -0.016)$$

$$\frac{-0.043 - (-0.188)}{1850 - 1800} = \frac{0.145}{50} = 0.0029$$
$$\frac{-0.016 - (-0.043)}{1890 - 1850} = \frac{0.027}{40} = 0.0007$$

The slope for the first pair of points is now four times that for the second pair. When we used y, that for the second was three times that for the first. We would like equal slopes, so it is natural to try next about halfway from y to $-1/y$. Thus we are led to try $\log y$. Turning to exhibit 3 of chapter 3, we see that

$$\log 5.3 = 0.72$$
$$\log 23.2 = 1.37$$
$$\log 62.9 = 1.80$$

so that the three points and two slopes are

$$(1800, 0.72)$$
$$(1850, 1.37)$$
$$(1890, 1.80)$$

$$\frac{1.37 - 0.72}{1850 - 1800} = \frac{0.65}{50} = 0.013$$
$$\frac{1.80 - 1.37}{1890 - 1850} = \frac{0.43}{40} = 0.011$$

Now the slopes agree with each other rather well. We should now go ahead--calculating $\log y$ and plotting x and $\log y$ for either many more or all of the points. Exhibits 7 and 10 of chapter 5 have already shown us how well this choice works.

fitting lines to three points

Once we have our three points fairly well on a line, we may as well fit a line to them. The three points just fixed offer a reasonable example.

To fit a line to three reasonably-spaced points, we usually do well to fit a slope to the two endpoints, and then take the mean of the three adjusted

values to find the constant. For (1800, 0.72), (1850, 1.37), (1890, 1.80), this leads to

$$\frac{1.80 - 0.72}{1890 - 1800} = \frac{1.08}{90} = 0.012$$

and since it seems easy to work with

$$x = \text{date} - 1800$$

we form the three values of

$$y - .012(\text{date} - 1800)$$

namely

$$\begin{aligned} 0.72 - .012(0) &= 0.72 \\ 1.37 - .012(50) &= 0.77 \\ 1.80 - .012(90) &= 0.72 \end{aligned}$$

for which the mean is 0.74 to two decimals. (The agreement of the two 0.72's is an important check on our arithmetic.) Thus our fit is

$$\text{Population(log of millions)} = 0.74 + 0.012(\text{date} - 1800)$$

Three points can take us a long way. If they are well chosen, they can do very well for us.

review questions

How do three points indicate hollow upward? Hollow downward? In which direction are expressions hollow that fall above (on the ladder of ways of expression) a straight expression? Those below? How do these rules help us in straightening the growth of the U.S. population? How do we fit a line to three (reasonably spaced) points? Why did we not use the median of the three adjusted values? Should we be surprised that 0.012 (in the 3-point fit) falls between 0.011 and 0.013 (for the two pairs of points)?

6C. Re-expressing *x* alone

We now know how to be guided in choosing a better expression for y. What if we wish to leave y alone and re-express x?

If we flip our picture over, interchanging the y- and x-axes, we convert one problem into the other. (We have to look through the back side of the paper after the flip, but a straight line stays a straight line.) Since it is now the same problem, all the same arguments apply.

This means that if the curve bulges toward large x and we are to re-express x, we ought to move x up the ladder, while if the curve bulges toward small x, we should move x down the ladder.

back to the population of the U.S.

Exhibit **1** shows three points from the population curve after the interchange of x and y. We see that the bulge is toward larger x (upward on this interchanged plot), so if we are to re-express x we would have to move upward on the x ladder.

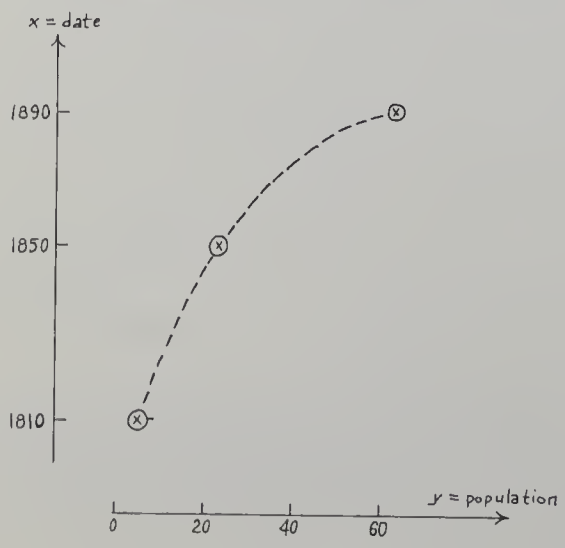
Should we try re-expressing x in this situation? No, because we can see that re-expression is unlikely to help us. As amounts, 1810, 1850, and 1890 are very similar, the outer values differing only a few percent from the middle one.

Re-expressing a variable that changes by only a few percent rarely gets rid of more than a barely detectable curvature.

One way to make the x's less alike as amounts would be to figure dates from some origin later than the birth of Christ. To figure from 1776 would be quite unrealistic, since there had been much immigration before that date. To figure from 1600 would be more realistic, but even this would be neglecting a substantial Amerind population.

exhibit **1** of chapter 6: U.S. population

The early population of the U.S., *x* and *y* interchanged



Let us try "date − 1600" anyway. An example should be useful, so long as it is not quite foolish. To keep our numbers of convenient size, we may as well measure "date − 1600" in centuries.

Our new x values for the last 3 points are 2.1, 2.5, and 2.9. They now vary by about ±15%, which is much more than before. We have some hope, but we cannot expect to find it easy to take care of substantial curvature, such as we still face. Since the change from date in years to [(1/100)(date) − 16] is a trivial re-expression--one that involves only multiplication by, and addition of, constants, our need to move up the x ladder is unchanged.

Let us begin by going to the cube of the new x. We find

$$(2.1)^3 = 9.261$$
$$(2.5)^3 = 15.625$$
$$(2.9)^3 = 24.389$$

The three points are:

$$(9.3, 7.2)$$
$$(15.6, 23.2)$$
$$(24.4, 62.9)$$

and the two slopes are:

$$3.2 = \frac{23.2 - 7.2}{15.6 - 9.3}$$
$$4.5 = \frac{62.9 - 23.2}{24.4 - 15.6}$$

The slope is still larger for the second interval, though only by about 2 to 1 rather than by 3 to 1. We have made progress, but clearly need to go further still.

Trying x^6, which is easy to find by squaring x^3, gives

$$(85, 7.2)$$
$$(244, 23.2)$$
$$(501, 62.9)$$

and

$$.102 = \frac{23.2 - 7.2}{244 - 85}$$
$$.114 = \frac{62.9 - 23.2}{591 - 244}$$

This is a lot closer, but we are not there yet.

Trying x^8 gives

$$\begin{aligned}&(\ 378,\ \ 7.2)\\&(1526,\ 23.2)\\&(5002,\ 62.9)\end{aligned}$$

and

$$.0139 = \frac{23.2 - 7.2}{1526 - 378}$$

$$.0114 = \frac{62.9 - 23.2}{5002 - 1526}$$

We seem at last to have gone too far.

Trying x^7 should come quite close. We find

$$\begin{aligned}&(\ 180,\ \ 7.2)\\&(\ 610,\ 23.2)\\&(1725,\ 62.9)\end{aligned}$$

$$.0372 = \frac{23.2 - 7.2}{610 - 180}$$

$$.0357 = \frac{62.9 - 23.2}{1725 - 610}$$

Agreement is now fairly good.

If we are to re-express

$$(\text{year} - 1600),$$

we find that our three points suggest the use of either

$$(\text{year} - 1600)^7$$

or, equivalently,

$$\left(\frac{\text{year} - 1600}{100}\right)^7.$$

We ought at least look at the results of doing this. Exhibit **2** shows the numbers, exhibit **3** the gross picture, exhibit **4** some differences. As either exhibit 2 or exhibit 4 shows, the deviations of the U.S. population, expressed in millions, from

$$0.72 + 0.036\left(\frac{\text{date} - 1600}{100}\right)^7$$

are less than 150,000 from 1800 to 1830 and from 1870 to 1890, with an "extra" million or two counted in 1840, 1850, and 1860.

exhibit **2** of chapter 6: U.S. population

The results of re-expressing x in dealing with U.S. population in the nineteenth century

Date	z $\frac{(\text{date} - 1600)^7}{100}$	y Population (in millions)	.036z	Diff.
1800	128	5.31	4.61	0.70
1810	180	7.24	6.48	0.76
1820	249	9.64	8.96	0.68
1830	340	12.87	12.24	0.63
1840	459	17.07	16.52	1.55
1850	610	23.19	21.96	1.23
1860	803	31.44	28.91	2.53
1870	1046	38.58	37.66	0.92
1880	1349	50.16	48.56	1.60
1890	1725	62.95	62.10	0.85
1900	2187	76.0	78.7	−2.7
1910	2751	92.0	99.0	−7.0
1920	3436	105.7	123.7	−18.0

exhibit **3** of chapter 6: U.S. population

The early U.S. population plotted against $z = \left(\frac{\text{date} - 1600}{100}\right)^7$

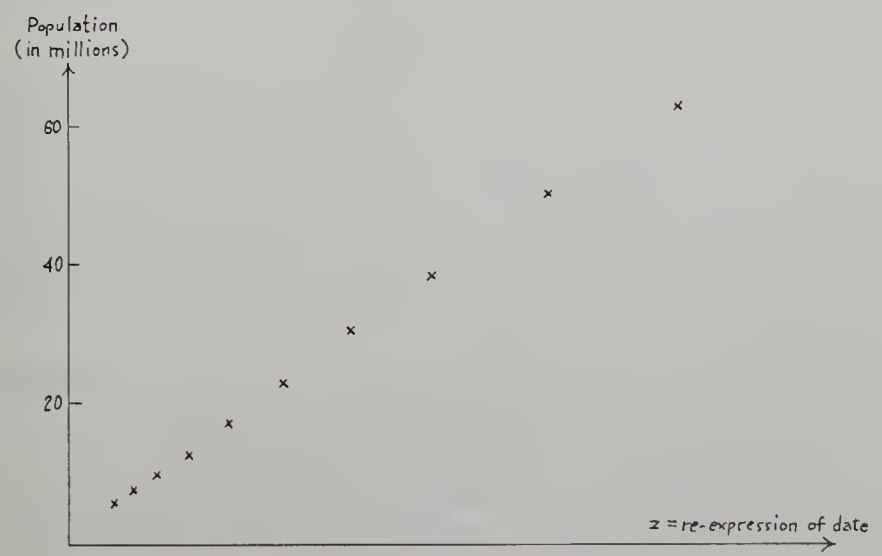
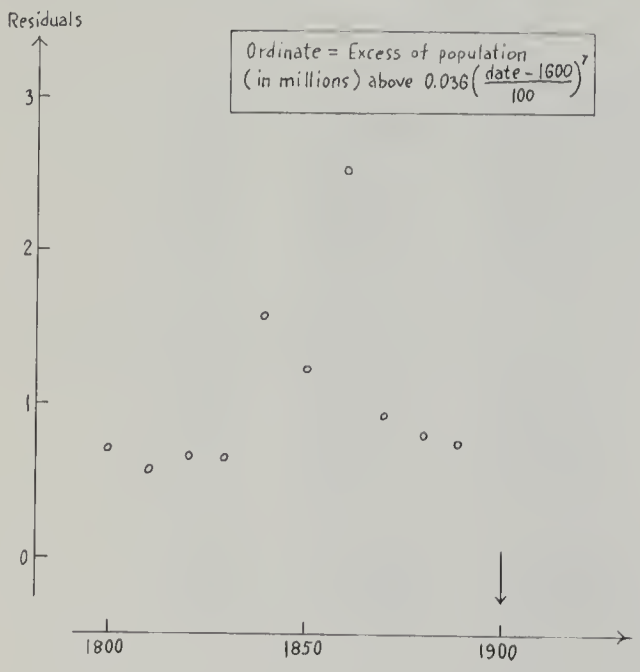


exhibit 4 of chapter 6: U.S. population

The result of flattening exhibit 3



a caveat

So far, either as quality of fit or ability to set residuals in clear view goes, plots such as exhibits 3 and 4 are as good as the plots of the last chapter, which related log population to date in years. If our purpose is to examine residuals--as it so often is--either plot is effective and useful.

When then should we make a distinction between the two plots? Surely we do need to make a distinction when we want to find an easily communicated description. That the U.S. population grew about 2.8% per year from 1800 to 1890 is relatively easy both to communicate and to understand. This is a major advantage, since we cannot say the same of the y vs. x^7 relationship. (Both, of course, apply over a limited span of years. Both fail--faster and faster--as we move on beyond 1900.) For communication, there is no doubt that log y vs. x is a more useful description.

Like any good fit, either the log y vs. x or the y vs. x^7 plot is subject to dangers of overvaluation. We see that each fits closely, though they cannot be exactly alike. Particularly if we have found only one of the two, there is a very natural tendency to convert "a good fit" into "this is how it had to be" or "a basic law of population growth". **One example of a close fit, by itself, is far**

from representing evidence for such strong statements. The fact that we have found two close fits of quite different form emphasizes our need to learn to avoid this sort of jumping at conclusions. **Conversely, we can make many good uses of a close fit, whether or not it is "a basic law".**

review questions

If we are to re-express x, which way ought we move on the ladder? Why must this be so? What happens if we exchange x and y axes in the U.S. population example? If x varies by only a few percent, what then? What trials did we make in re-expressing dates? Do we expect only one choice of re-expression to straighten a given set of points out thoroughly? What if we find several? Can we infer a "basic law" from one close fit? Why is one straightening of U.S. population growth easier to communicate than the other?

6D. A braking example

Let us next look at an example where re-expressing x seems to be the natural way to make the data more orderly and more describable.

Exhibit **5** shows the speed and distance to stop for 50 cars. Exhibit **6** plots the data. We could fit most of the data points with a straight line. However, a fitted line would give zero stopping distance at a speed between 5 and 10 miles per hour. The one thing we are sure of in this example is that zero stopping distance goes with zero speed, and vice versa. We didn't have to test cars to know that--or to put a point at (0, 0). We must face curvature, and try to eliminate it.

Three reasonable points to take are, then,

$$(0, 0)$$

and the two marked by large x's in exhibit 6,

$$(15, 35) \qquad \text{and} \qquad (25, 90).$$

The two slopes are

$$\frac{35 - 0}{15 - 0} = 2.3 \qquad \text{and} \qquad \frac{90 - 35}{25 - 15} = 5.5.$$

To re-express x, since the curve bulges toward large x, we ought to move toward x^2, x^3, etc. Trying x^2 gives

$$(0, 0) \qquad (225, 35) \qquad (625, 90)$$

with slopes of

$$.15 = \frac{35 - 0}{225 - 0} \qquad \text{and} \qquad .14 = \frac{90 - 35}{625 - 225}$$

Clearly, using x^2 is a reasonable try.

Exhibit **7** shows the plot of y against x^2, which now looks quite straight. If we take

$$(0, 0) \quad \text{and} \quad (600, 80)$$

as representative points, we are led to try

$$y - .133x^2$$

as a reasonable flattened quantity.

Exhibit **8** shows the result. It is far from wonderful, but seems to be reasonably flat, although the behavior for x^2 near 0 does not fit too well with the known point at (0, 0).

using our knowledge

We need to try something further. We believe in $y = 0$ when $x = 0$. Perhaps we should use this belief in choosing what to plot. How can we do this?

exhibit **5** of chapter 6: braking distances

Speed and distance to stop

Speed, x (mph)	Distance to stop, y (in feet)
4	2, 10,
7	4, 22,
8	16,
9	10,
10	18, 26, 34,
11	17, 28,
12	14, 20, 24, 28,
13	26, 34, 34, 46,
14	26, 36, 60, 80,
15	20, 26, 54,
16	32, 40,
17	32, 40, 50
18	42, 56, 76, 84,
19	36, 46, 68,
20	32, 48, 52, 56, 64,
(21)	
22	66,
23	54,
24	70, 92, 93, 120,
25	85,

S) SOURCE

Mordecai Ezekiel, 1930. Methods of Correlation Analysis. New York, John Wiley. Table 11, page 41.

Also Table 10, page 43, of 2nd edition 1943. Table 4.1, page 45 of 3rd edition 1959, by Mordecai Ezekiel and Karl A. Fox, has similar but different data.

exhibit **6** of chapter 6: braking distances

Plot of exhibit 5s data

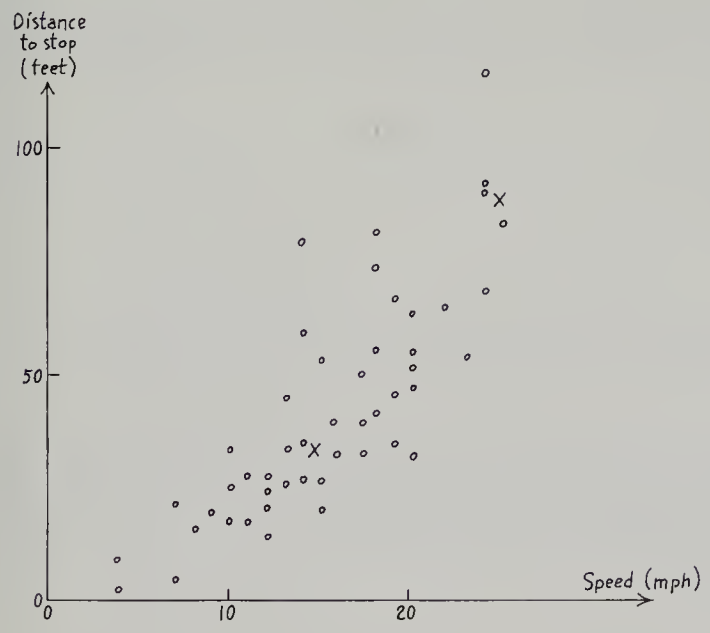
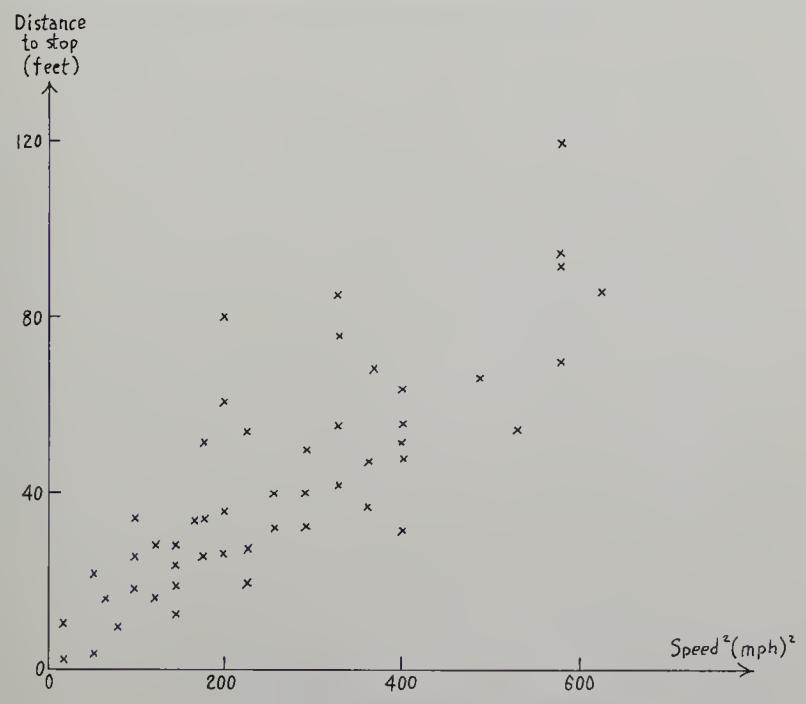


exhibit **7** of chapter 6: braking distances

Speed^2 and distance to stop for 50 motor cars



One way is to plot y/x--rather than y--against x. Any finite value for y/x will make y go to zero as x goes to zero. Thus any reasonable behavior of our y/x vs. x plot will lead to a fit that makes $x = 0$ go with $y = 0$.

Exhibit **9** gives the values of y/x. Exhibit **10** plots y/x against x. The behavior of the point cloud is quite reasonable, so we select two representative points as shown. They are

$$(5, 1.4) \qquad \text{and} \qquad (25, 3.7).$$

The corresponding line is

$$\frac{y}{x} = .115x + .8.$$

The residuals are written down in exhibit **11** and plotted in exhibit **12**.

We now see:

◇ 10 wandering points on the high side--9 between 1.1 and 1.9, one very high--these presumably represent bad brakes or slow-responding drivers.

◇ a mass of other points which, if taken by themselves, seem quite level but placed about 0.5 too low.

exhibit **8** of chapter 6: braking distances

Flattening by use of $y - 1.33x^2$

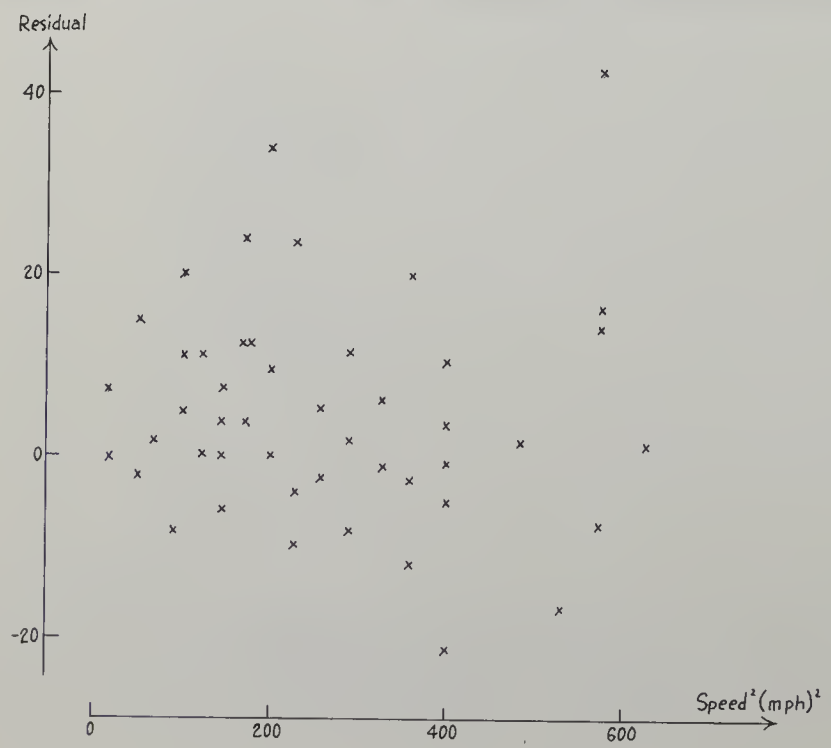


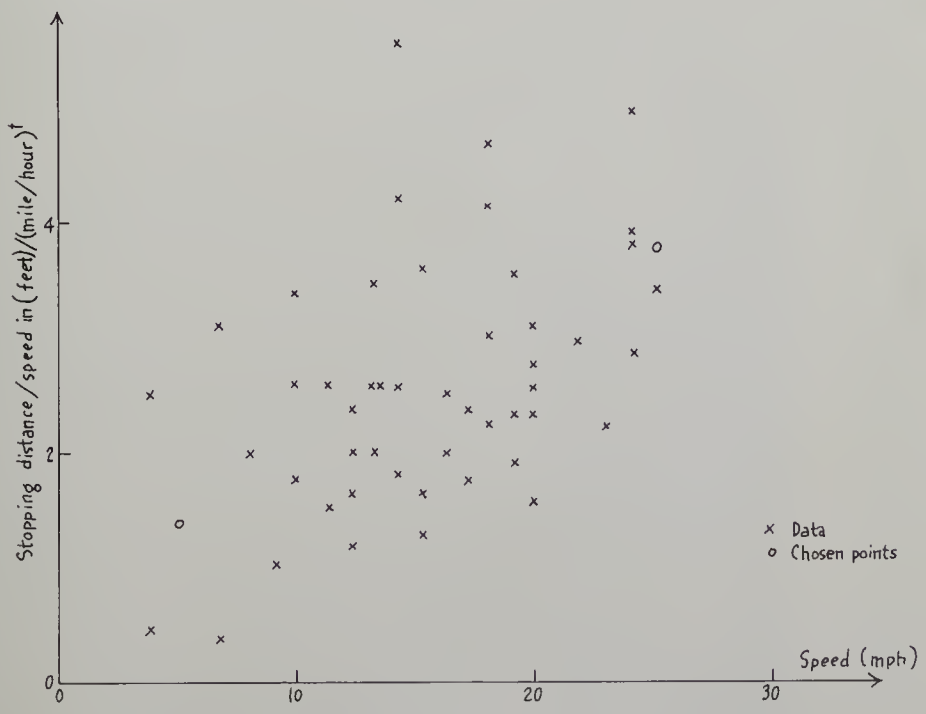
exhibit **9** of chapter 6: braking distances

The values of y/x

x	y/x
4	0.5, 2.5
7	0.6, 3.1
8	2.0
9	1.1
10	1.8, 2.6, 3.4
11	1.6, 2.6
12	1.2, 1.7, 2.0, 2.3
13	2.0, 2.6, 2.6, 3.5
14	1.9, 2.6, 4.3, 5.7
15	1.3, 1.7, 3.6
16	2.0, 2.5
17	1.9, 2.4, 3.0
18	2.3, 3.1, 4.2, 4.7
19	1.9, 2.4, 3.6
20	1.6, 2.4, 2.6, 2.8, 3.2
(21)	
22	3.0
23	2.3
24	2.9, 3.8, 3.9, 5.0
25	3.4

exhibit **10** of chapter 6: braking distances

y/x against x



† 1 foot/(mile/hour) = $\frac{30}{44}$ seconds = $\frac{15}{22}$ second

Thus, our final description can run as follows:

40 points fairly well described by:

$$\frac{y}{x} = .115x + .8 - .5$$

or

$$y = .115x^2 + .3x,$$

accompanied by 9 points with y greater than this expression by about $2x$ and one point greater by about $4x$.

Careful and repeated analysis can lead to effective description.

review questions

What example did we use in this section? What fact were we entitled to be sure of? How does exhibit 6 behave? What three points is it natural to choose? What re-expression do they lead to? How effective does this re-expression seem to be? What did we then think of trying? Why was it natural to think of it? How did the resulting plots look? Do there seem to be stray values? Are you surprised?

exhibit **11** of chapter 6: braking distances

The values of $(y/x) - 0.115x - 0.8$

x	$0.115x + 0.8$	$y/x - 0.115x - 0.8$
4	1.3	−0.8, 1.2,
7	1.6	−1.2, 1.5,
8	1.7	0.3,
9	1.8	−0.7,
10	2.0	−0.2, 0.6, 1.4,
11	2.1	−0.5, 0.5,
12	2.2	−1.0, −0.5, −0.2, 0.1,
13	2.3	−0.3, 0.3, 0.3, 1.2,
14	2.4	−0.5, 0.2, 1.9, 3.3,
15	2.5	−1.2, −0.8, 1.1,
16	2.6	−0.6, −0.1,
17	2.8	−0.9, −0.4, 0.2,
18	2.9	−0.6, 0.2, 1.3, 1.8,
19	3.0	−1.1, −0.6, 0.6,
20	3.1	−1.5, −0.7, −0.5, −0.3, 0.1,
(21)		
22	3.3	−0.3
23	3.4	−1.1
24	3.6	−0.7, 0.2, 0.3, 1.4
25	3.7	−0.3

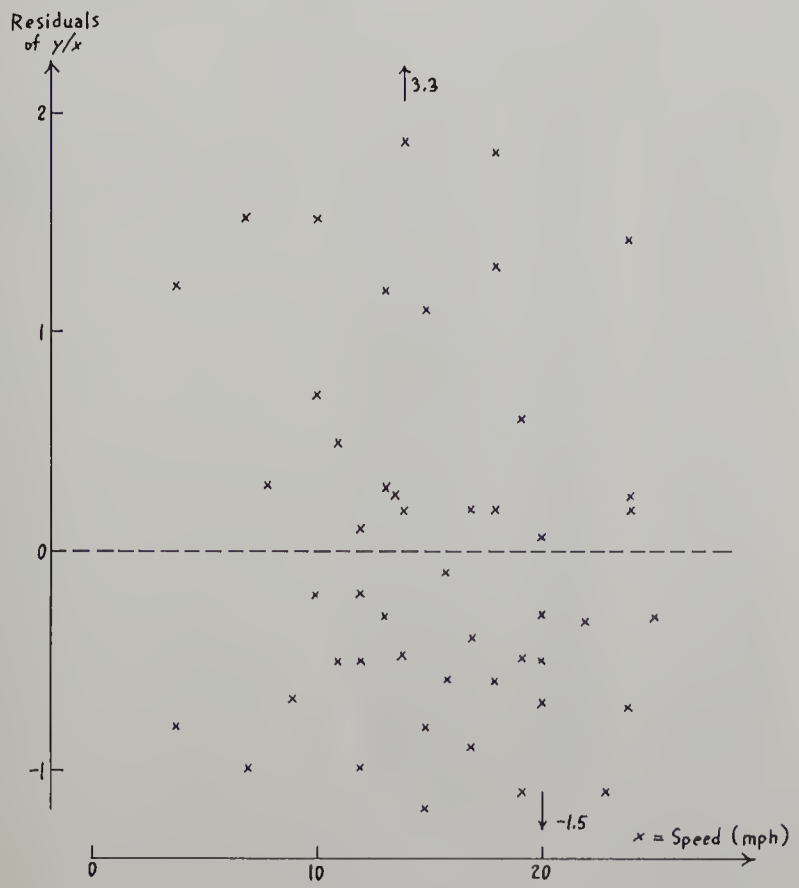
6E. The vapor pressure of H_2O

The vapor pressure of water or ice is the highest pressure of water vapor (steam, if you like) that can exist in equilibrium with the water or ice at a given temperature. Its values are well known, and of considerable practical importance. In terms of temperatures Celsius (once called centigrade, giving 0°C = 32°F and 100°C = 212°F) and pressures in mm of mercury (760 mm is one standard atmosphere), these vapor pressures at different temperatures are given in exhibit **13.**

Just a look at the first two columns of this table shows that a direct plot of p against t will show us only a very rapidly rising pressure. (If 139893.20 can be plotted, even 760.00 can hardly be seen.) Accordingly, we plan to do something to at least partly straighten out this plot. Taking the logarithm of p looks as if it might help.

exhibit **12** of chapter 6: braking distances

Plot of $(y/x) - 0.115x - 0.8$ against x



Column (3) of exhibit 13 contains the values of log p. This looks conceivably plottable, so we proceed to exhibit **14,** which can be looked over, but is far from straight.

Another step needs to be taken. We could try to find some empirical approach, but we can go more directly to our goal if we recall that plotting log pressure against the reciprocal of the absolute temperature is rather common in physical chemistry. (Or if we use the closing example of the next section.) Here, one kind of absolute temperature is the Celsius temperature increased by about 273.1°C.

Column (4) of exhibit 13 contains values of $-1000(1/T)$, where $T = t + 273.1$°C. (The factor 1000 has been used to avoid unsightly and confusing zeros.) The result of plotting log p against column (4) of exhibit 13 is shown in exhibit **15.** At last we have a reasonably good straight line, one worth calculating residuals about.

Column (5) of exhibit 13 contains values of $\log p - 2.25(-1000/T)$ which, as we see, is reasonably constant. Subtracting 8.8 clearly gives residuals well

exhibit **13** of chapter 6: vapor pressure

The vapor pressure of H_2O and some associated quantities

t(°C)	p(mm Hg)	log p	(4)	(5)
−40	0.105	−.9788	−4.2900	8.6737
−20	0.787	−.1040	−3.9510	8.7858
0	4.5687	.6598	−3.6617	8.8986
20	17.363	1.2396	−3.4118	8.9162
40	54.865	1.7393	−3.1939	8.9256
60	148.88	2.1728	−3.0021	8.9275
80	354.87	2.5501	−2.8321	8.9223
100	760.00	2.8808	−2.6802	8.9113
120	1489.14	3.1729	−2.5439	8.8966
140	2710.92	3.4331	−2.4207	8.8797
160	4636.00	3.6661	−2.3089	8.8611
180	7520.20	3.8762	−2.2070	8.8420
200	11659.16	4.0667	−2.1137	8.8225
220	17395.64	4.2404	−2.0280	8.8034
240	25100.52	4.3397	−1.9489	8.7847
260	35188.00	4.5464	−1.8758	8.7670
280	48104.20	4.6822	−1.8080	8.7502
300	64432.80	4.8091	−1.7449	8.7351
320	84686.80	4.9278	−1.6861	8.7215
340	109592.00	5.0398	−1.6311	8.7097
360	139893.20	5.1458	−1.5795	8.6997

(4) = $-1000(1/T)$, where $T = t + 273.1$°C
(5) = $\log p - 2.25(-1000/T)$

exhibit **14** of chapter 6: vapor pressure

Vapor pressure of H_2O and temperature (logarithmic scale for pressure, which is in mm Hg)

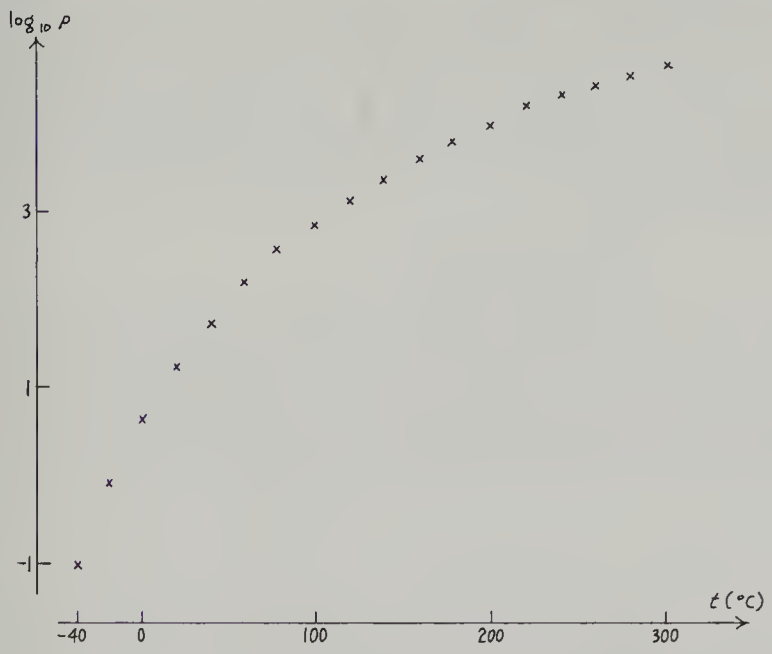
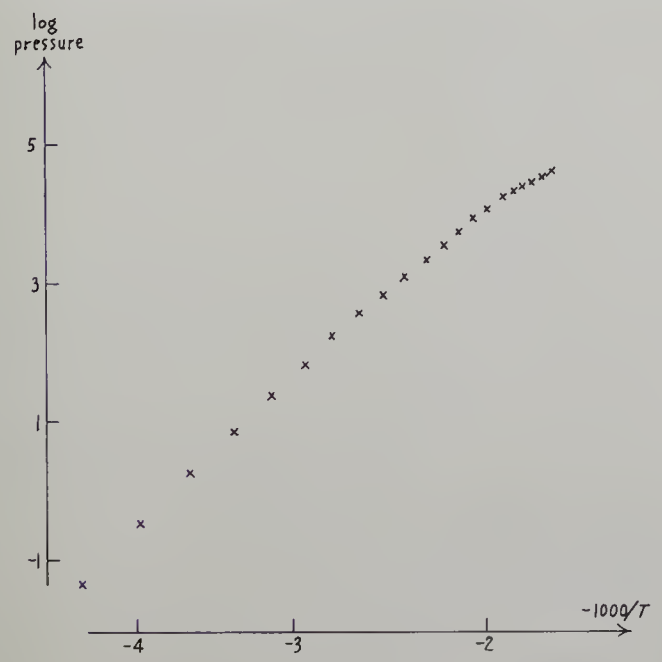


exhibit **15** of chapter 6: vapor pressure

Vapor pressure of H_2O and temperature (logarithm of pressure, reciprocal of absolute temperature)



worth plotting. Exhibit **16** shows the results. The most striking result is the apparent split of our curve into two parts with a corner at the third point from the left. What might this mean?

When we recall that this point comes at 0°C (= 32°F), which is the freezing point of water, we see that a break at this point is quite reasonable. Below 0°C we are dealing with the vapor pressure of solid water (ice), while above this point we are dealing with the vapor pressure of liquid water.

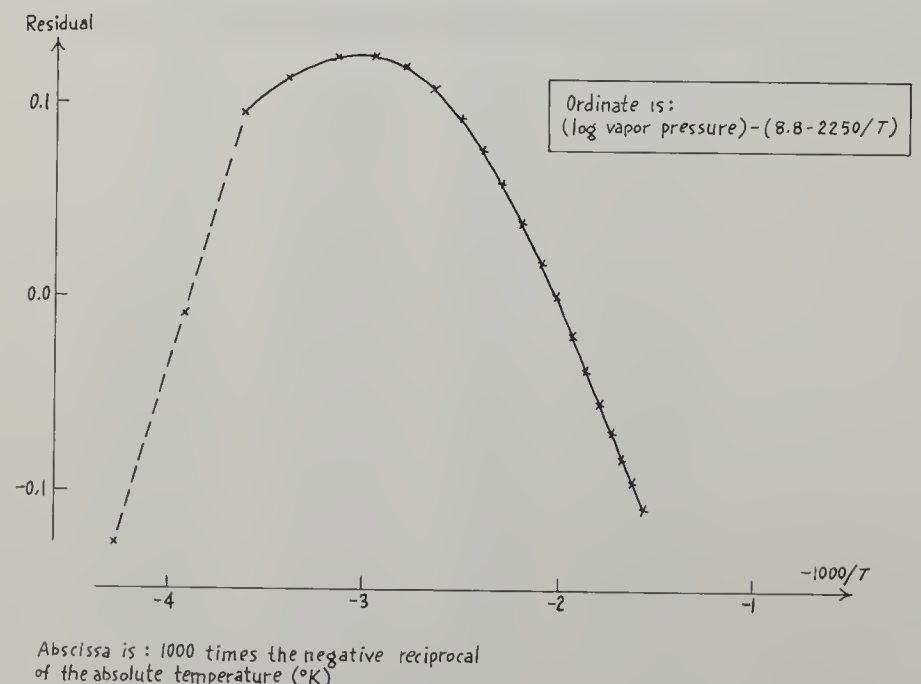
Note that this fact was not forced upon us by our data until we both straightened and flattened the plot.

review questions

What data did we look at in this section? Could we plot it at all in its original form? Did we re-express y? Why? Was this enough? Could we go further? How? How well did we then do? Could we see anything new when we looked at the residuals? Did such a look start off any ideas?

exhibit **16** of chapter 6: vapor pressure

Residuals of log vapor pressure from a straight line in $-1/T$ (water)



6F. Re-expressing the second variable

As we have just seen, it will not always be wise--or satisfactory--to change only one of the two expressions. Sometimes it pays to re-express **both** x and y. In such a situation we may need to do "cut and try". However, once we have agreed to try a specific expression for one of the variables, be it x or y, we can use the same rules to narrow down our choice of the second expression.

As an example, let us take three points from the example just treated in the last section, and see how we might have been guided. From exhibit 13 we have

$$\begin{aligned}&(0, \quad 4.5687)\\&(100, \quad 760.00)\\&(200, \ 11659.16).\end{aligned}$$

Clearly, the middle point is below the line that joins the other two--and also to the right of that line. If we are to re-express y we should look toward log y and $-1/y$. Trying logs gives

$$(0, 0.6598) \qquad (100, 2.8808) \qquad (200, 4.0667)$$

with slopes of

$$\frac{2.8808 - 0.6598}{100 - 0} = \frac{2.2210}{100} = .02221$$

and

$$\frac{4.0667 - 2.8808}{200 - 100} = \frac{1.1859}{100} = .01186.$$

Thus the middle point is now above and to the left of the line. We could try going only as far as $\sqrt{y}$, which is about halfway to log y. If we do this, say by using exhibit 5 of chapter 3, we find

$$\begin{aligned}&(0, \quad 2.12)\\&(100, \ 27.6)\\&(200, \ 108)\end{aligned}$$

and we see that $\sqrt{y}$ goes nowhere nearly far enough. To go on to

$$\sqrt{\sqrt{y}} = y^{1/4} = \sqrt[4]{y}$$

is easy. If we do this, we find

$$\begin{aligned}&(0, \quad 1.44)\\&(100, \ 5.3)\\&(200, \ 10.4)\end{aligned}$$

for which the slopes are

$$\frac{5.3 - 1.44}{100 - 0} = \frac{3.86}{100} = .0386$$

$$\frac{10.4 - 5.3}{200 - 100} = \frac{5.1}{100} = .0510$$

so that we see that we are not yet far enough. Those who love to repeat square roots will now try

$$\sqrt{\sqrt[4]{y}} = y^{1/8} = \sqrt[8]{y},$$

finding

(0, 1.20)
(100, 2.32)
(200, 3.20)

for which the slopes are

$$\frac{2.32 - 1.20}{100 - 0} = \frac{1.12}{100} = .0112$$

$$\frac{3.20 - 2.32}{200 - 100} = \frac{.88}{100} = .0088.$$

This time we have gone too far. As a result, if we are to change only y, we should try something between $y^{1/4}$ and $y^{1/8}$, perhaps $y^{1/6}$ or $y^{1/7}$.

another try

This is, however, not our only reasonable choice. Going to log y did much to straighten out our three points. True, it went a little too far, but perhaps we could do something by keeping log y and re-expressing x. Before we do this, however, we should stop and think for a moment. As we have written it, x is temperature in °C (freezing water = 0°C, boiling water = 100°C). If we are to re-express simply, we ought not to tie our zero to a property of so special a substance as water. (We are looking at water's vapor pressure, but water has a vapor pressure below 0°C, too.) Rather we should figure our temperatures so that we run from the so-called absolute zero, which is just below −273°C.

Our intermediate starting point then should be

(273.1, 0.6598)
(373.1, 2.8808)
(473.1, 4.0667)

with slopes of .02221 and .01186, and a middle point above and to the left of the line. If we are to re-express x, we should move toward $\log x$ or $-1/x$. Let us try $\log x$. We find:

$$(2.44, 0.6598)$$
$$(2.57, 2.8808)$$
$$(2.67, 4.0667)$$

$$\frac{2.8808 - 0.6598}{2.57 - 2.44} = \frac{2.2210}{.13} = 17.1$$
$$\frac{4.0667 - 2.8808}{2.67 - 2.57} = \frac{1.1859}{.10} = 11.8$$

Using logs has helped, but not enough. Let us try $-1/x$, for which we have

$$(-.00366, 0.6598)$$
$$(-.00268, 2.8808)$$
$$(-.00211, 4.0667)$$

$$\frac{2.8808 - 0.6598}{-.00268 - (-.00366)} = \frac{2.221}{.00098} = 2266$$
$$\frac{4.0667 - 2.8808}{-.00211 - (-.00268)} = \frac{1.186}{.00057} = 2080$$

for which the slopes agree better--have a ratio nearer 1--than with any other combination so far tried. Accordingly

$$\log y \quad \text{and} \quad -1/x$$

where x is absolute temperature, seems to be a good choice. As we know from the last section, it proves to be one.

review questions

Does it ever pay to re-express both x and y? Do we have to do this blind? How is guidance for re-expressing the second coordinate to be found? How much is it like guidance when only one coordinate is re-expressed? If we tackle the vapor pressure of water, re-expressing only y, what are we led to? If we pick $\log y$, what are we led to for x?

6G. Wise change of origin as a preliminary

We have already seen two examples where a change of origin was part of a sensible approach to straightening: (1) the early population of the U.S. as a function of time (where a time origin at the birth of Christ--1600 years

before the beginning of European immigration to North America--was far from sensible--once we thought about it! Why 1600 and not 600 or 2600?) and (2) the vapor pressure of water as a function of centigrade temperature. (Why should we take the origin at freezing?)

The same sort of need to think about a sensible origin arises in other problems, sometimes with regard to x and sometimes with regard to y. In one broad class of cases, the question arises because the amount we measure is the sum of contributions, some of which change slowly if at all, while the remaining contribution changes in such a way that re-expression could flatten the plot if this contribution could be measured separately.

The natural approach to this sort of data is to form

observation MINUS background

where "background" is a constant chosen to allow for the slowly changing components. We are not likely to be able to pick the value for "background" either on the basis of general insight or on the basis of doubtfully-related historical data. We expect to learn about it from the same set of data that we are trying to flatten. We are likely to approach choosing a plausible value for "background" by trying various values for it and seeing which one leads to data that can be more thoroughly flattened by further re-expression.

Observation of radioactive decay yields many problems of this sort. Any single kind of radioactive atom decays on a steady percentage basis--so many percent is gone every so many days, years, or millennia. Many processes of isolating--or making--some one kind of radioactive atom also isolate--or make--one or more other kinds. If these other kinds decay--but decay more slowly--their presence can often be adequately allowed for by a constant background.

a radioactive decay example

In 1905, the study of radioactive substances was in its infancy. Meyer and von Schweidler reported the relative activities for an experimental object set out in exhibit **17.** If we plot activity against time, the result is exhibit **18,** which is far from being straight. It is again natural to take logarithms, especially since, in simple radioactive decay, the logarithm of activity should decrease linearly with time.

Exhibit **19** shows the behavior of log activity against time. A noticeable curvature remains. In this situation the most plausible source of curvature is contamination by some other radioactive substance that decays much more slowly than the one of central interest. If such there be, it will not have contributed more than two units of activity (since at 45 days the observed activity is down to 2.1 units).

It is reasonable to explore the consequences of assuming the presence of 1.0 to 1.5 units of activity from such a contaminant by, successively:

◇plotting against time either

$$\log(\text{activity} - 1.0)$$

or

$$\log(\text{activity} - 1.5)$$

which could represent the log of the rapidly-decaying activity (and which are given in columns (4) and 5) of exhibit 17).

◇fitting a straight line, perhaps roughly.

◇plotting the residuals.

exhibit **17** of chapter 6: radioactivity

The decay of radioactivity and associated quantities

time (days)	activity (relative)	(3)	(4)	(5)	(6)	(7)
0.2	36.0	1.556	1.544	1.538	1.551	1.545
2.2	26.0	1.415	1.398	1.389	1.471	1.466
4.0	23.1	1.364	1.344	1.334	1.476	1.474
5	18.9	1.276	1.253	1.241	1.418	1.416
6	17.8	1.250	1.225	1.212	1.423	1.422
8	14.7	1.167	1.137	1.121	1.401	1.401
11	13.4	1.127	1.093	1.076	1.456	1.461
12	11.3	1.053	1.013	.991	1.409	1.411
15	8.5	.929	.875	.845	1.370	1.370
18	5.9	.771	.690	.643	1.284	1.273
26	5.0	.699	.602	.544	1.460	1.454
33	3.4	.531	.380	.279	1.469	1.434
39	2.4	.380	.146	−.046	1.433	1.319
45	2.1	.322	.041	−.222	1.526	1.353

(3) = log(activity)
(4) = log(activity − 1.0)
(5) = log(activity − 1.5)
(6) = .033t + (column 4)
(7) = .035t + (column 5)

S) SOURCE

S. Meyer and E. von Schweidler, 1905. Sitzungsberichte der Akademie der Wissenschaften zu Wien, Mathematisch–Naturwissenschaftliche Classe, p. 1202 (Table 5).

exhibit **18** of chapter 6: radioactivity

The data of exhibit 17

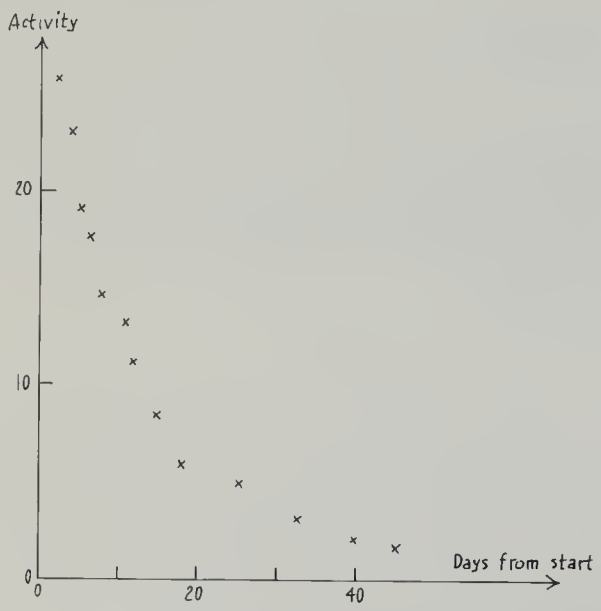
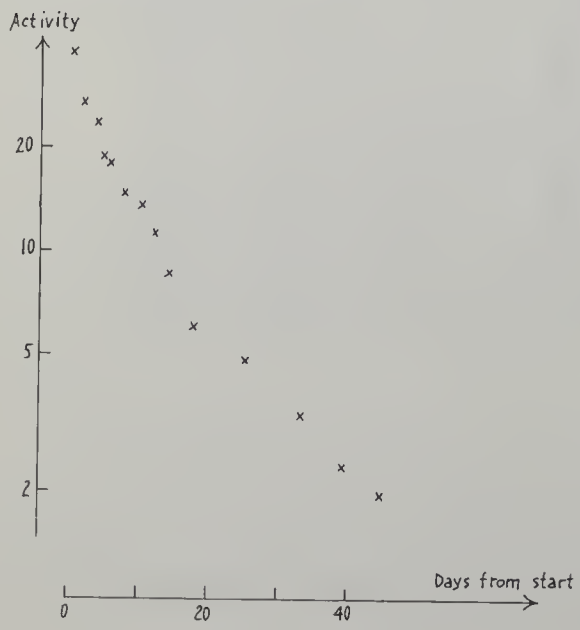


exhibit **19** of chapter 6: radioactivity

The same data on a log scale



The results are shown in exhibit **20.** (Columns (6) and (7) of exhibit 17 give these residuals increased by 1.45, calculated as ".033t + (column 4)" and ".035t + (column 5)", respectively.) Of the two choices, allowance for 1.0 unit of contamination seems to give a more nearly horizontal set of residuals--to lead to a closer fit.

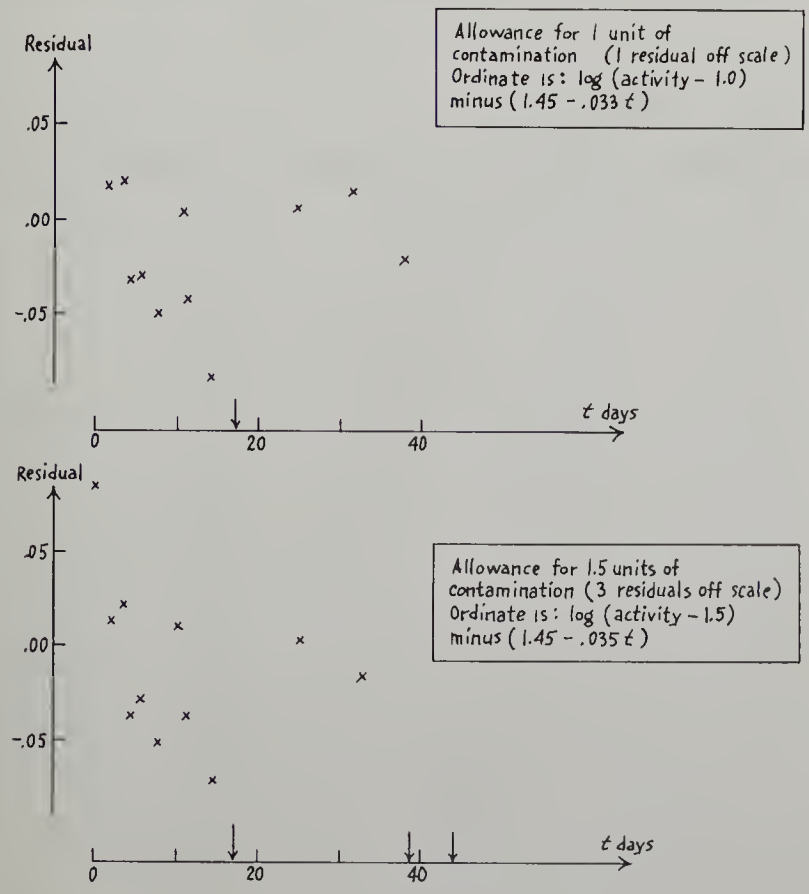
(Perhaps the residuals for 1.0 do trend upward a little. The reader may wish to try 1.1 or 1.2.)

review questions

Ought we expect to make changes of origin? What are three examples? What is a background? When we measure radioactive substances, whether made or isolated, are backgrounds common? What did a plot of raw activity against raw time show? What two things did we do then? Did it all work out well?

exhibit **20** of chapter 6: radioactivity

Residuals from decay line allowing for long-lived contamination (two versions)



6H. How far have we come?

This chapter has been devoted to guidance about how to re-express x and y with a view to straightening a plot. This problem arises most simply when one or both of x and y is an amount--or, really just a special case of an amount, a large count. The natural re-expressions are by powers, roots, and logs.

To keep computation down, we routinely begin with three well-chosen points. The basic rule of thumb is then simple:

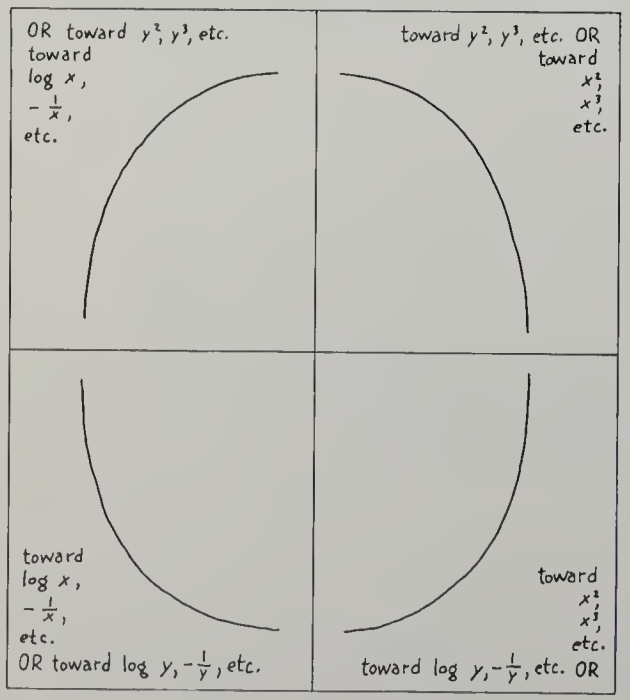
◇ **move on the ladder of expressions in the direction in which the curve bulges.**

Exhibit **21** shows the four possible cases, and the natural steps along both x- and y-ladders. (We can move along either or both.) We try new expressions on more of the data only when the three points have already responded well to them.

In certain cases, a fresh choice of origin--before going to powers, roots, or logs--is valuable. Sometimes the new choice is a matter of common sense, sometimes of how flat the final result proves to be.

exhibit **21** of chapter 6: indicated behavior

How to move for each variable separately; the four cases of curve shape



As we use this technique, we will need to remember that:

◇ straightening by re-expressing x is not the same as re-expressing y.

◇ just because re-expression makes things quite straight, there is no guarantee that we have found a new natural law.

We are now ready:

◇ to approach the analysis of (x, y) data in terms of re-expression followed by straight-line fitting (perhaps in two or more steps) and the examination of residuals.

◇ to turn to another class of important problems.

6P. Additional problems

See exhibits **22 through 26**.

exhibit **22** of chapter 6: data and problems

Three examples of radioactive decay

A) DATA

Days*	Activity†	Days*	Activity†	Days*	Activity†
0.8	6.70	0.8	2.82	0.8	2.05
2.8	6.40	2.8	2.34	1.0	2.03
6.9	5.70	4.8	1.90	1.8	1.79
8.9	5.10	6.8	1.80	2.1	1.77
13.1	4.30	11.8	1.34	3.9	1.54
15.2	4.00	13.9	1.24	5.9	1.35
16.8	3.95	16.8	1.03	7.1	1.29
20.1	3.40	19.8	1.00	9.1	1.23
20.8	3.40	23.8	0.80	12.1	1.01
20.9	3.20	31.8	0.55	12.9	0.96
21.9	3.20	32.5	0.11	16.9	0.82
31.1	2.42			19.9	0.68
36.8	2.30			22.8	0.59
43.8	2.11			25.9	0.51
49.8	2.00			33.8	0.40
53.8	1.99			44.1	0.32
58.8	1.98				
65.9	1.90				
73.8	1.80				
87.1	1.65				

*** Since start.**
† In volts/minute.

→

exhibit **22** of chapter 6 (continued)

P) PROBLEM

22a) In a later paper, Meyer and Von Schweidler reported the decay of radioactivity for three samples as in panel A. Analyze at least two of these. Comment.

S) SOURCE

S. Meyer and E. von Schweidler 1907. "Untersuchungen über radioaktive Substanzen. VIII Mitteilung: Über ein radioaktives Produkt aus dem Aktinium." Sber. Ak. Wiss. Wien. Math–Nat. Classe 116 IIA1, pp. 315–322 (especially 316–317).

exhibit **23** of chapter 6: data and problems

Vapor pressure of mercury

A) DATA

Temperature (°C)	Pressure (mm Hg)
0	0.0004
20	0.0013
40	0.006
60	0.03
80	0.09
100	0.28
120	0.8
140	1.85
160	4.4
180	9.2
200	18.3
220	33.7
240	59.
260	98.
280	156.
300	246.
320	371.
340	548.
360	790.

P) PROBLEMS

23a) The vapor pressure of mercury is stated to be as in panel A. Analyze graphically. Comment.

→

exhibit **23** of chapter 6 (continued)

23b) Given the three points found partway through section 6F,

$$(0, 0.6598)$$
$$(100, 2.8808)$$
$$(200, 4.0667),$$

what change in expression of this (once new) y will come close to straightening out these points?

23c) Apply the result of the last problem to the data of exhibit 13, fit a straight line, and plot the residuals. How do the results compare with exhibit 16?

23d) Apply the re-expression

$$y \text{ becomes } y^{1/6}$$

(suggested by the analysis of section 6E) to the data of exhibit 13, fit a straight line, and find residuals. How do the results compare with exhibit 16? Complete the graphical analysis. Comment.

exhibit **24** of chapter 6: data and problems

More problems

24a) Rates of mortality from breast cancer in different latitudes. (50; 1025, 513) translates as: "In latitude 50°N the mortality index for breast cancer is 102.5 and the mean annual temperature is 51.3." Data-sets (16): (50; 1025, 513), (51; 1045, 499), 52; 1004, 500), (53; 959, 492), (54; 870, 485), (55; 950, 478), (56; 886, 473), (57; 892, 451), (58; 789, 463), (59; 846, 421), (60; 817, 442), (61; 722, 435), (62; 651, 423), (63; 681, 402), (69; 673, 318), (70; 525, 340). SOURCE: A. J. Lea, 1965, "New observations on distribution of neoplasms of female breast in certain European countries," *British Medical J.* **1** (for 1955): 486–490. Table II on page 489. PROBLEM: Can we straighten this plot? (Mortality index against mean annual temperature.)

24b) Plasticity of wool: slow stretching of a single fiber. (1, 321) translates as: "After 1 minute under load, a fiber 53.3 microns in average diameter (coefficient of variation of diameter = 5.4%) had stretched 32.1% beyond its unloaded length." Data-sets (34): (1, 321), (3, 330), (5, 334), (8, 337), (16, 342), (32, 348), (50, 352), (110, 361), (240, 377), (440, 394), (740, 413), (1310, 442), (1460, 449), (1630, 458), (1900, 469), (2090, 478), (2760, 499), (2950, 505), (3080, 509), (3460, 519), (4280, 540), (4970, 556), (5720, 572), (6000, 579), (6320, 586), (7120, 600), (7360, 604), (7540, 607), (8520, 623), (9020, 629), (9230, 633), (9950, 643), (10260, 647), (10680, 654). (The last data set is for 10,680 minutes under load and 65.4% stretch.) SOURCE: O. Ripa and J. B. Speakman, 1951. "The plasticity of wool." *Textile Research J.* **21**: 215–221. Table I on page 217. PROBLEM: How can we straighten the plot?

→

exhibit **24** of chapter 6 (continued)

24c) Amount of interstitial space in young chickens. ("Interstitial space" is a volume defined by the amount of thiocyanate ion taken up outside the blood within 10 minutes after injection and measured by its ratio to blood volume.) (1, 52) translates as: "For chickens 1 week old, the interstitial space accounted for 52% of body weight." Data-sets (8): (1, 52), (2, 42), (3, 39), (4, 38), (6, 37), (8, 36), (16, 25), (32, 22). SOURCE: W. Medway and M. R. Kare, 1959, "Thiocyanate space in growing fowl," *Amer. J. of Physiology* **196**: 873–875. Table 1 on page 874. PROBLEM: How can we straighten the plot?

24d) A more detailed look at interstitial volume (1, 55, 52) translates into: "At age 1 week, the average body weight of 6 chickens was 55 grams, of which the interstitial space was 52%." Data-sets (8): (1, 55, 52), (2, 108, 42), (3, 175, 39), (4, 242, 38), (6, 372, 37), (8, 527, 36), (16, 1137, 25), (32, 1760, 22). SOURCE: As for problem (24c). PROBLEM: How can we straighten the plot? How does the straightness here compare with that for problem (24c)? What expression of amount (not percent) of interstitial space in terms of weight corresponds to our fit?

exhibit **25** of chapter 6: data and problems

Some more problems

25a) Vapor pressure of a boron analog of mesitylene. (130, 29) translates as: "At a temperature of 13.0°C, the vapor pressure of B-trimethylborazole was 2.9 millimeters of mercury." Data/(sets) (13): (130, 29), (195, 51), (225, 85), (272, 103), (318, 146), (384, 213), (457, 305), (561, 514), (644, 745), (714, 1002), (805, 1437), (857, 1769), (915, 2169). SOURCE: E. Wiberg, K. Hertwig, and A. Bolz, 1948, "Zur kenntnis der beiden symmetrischen Trimethyl-borazole ("anorganisches Mesitylen")," *Zeitschrift für Anorganische Chemie* **256**: 177–216. Table on page 191. PROBLEM: How can we straighten this plot? Check your answer!

25b) Effects of small amounts of biotin on the mobility of a microorganism. (5E–7, 1354) translates as: "At a biotin concentration of 5×10^{-7}, the mobility of *Lactobacillus casei* was 1.354 units". Data-sets (7): (0, 1415), (5E-7, 1354), (1E-6, 1311), (5E-6, 1230), (1E-5, 1234), (5E-5, 1181), (1E-4, 1188). SOURCE: V. R. Williams and H. B. Williams, 1949, "Surface activity of biotin," *Journal of Biological Chemistry* **177**: 745–750. PROBLEM: How can we straighten the plot?

25c) Electric current produced by heating aluminum phosphate. (880, 1) translates as: "At a temperature of 880°C––an absolute temperature of 880 + 273 = 1153°K––positive electrification produced a current of 1 unit, where 1 unit = 2×10^{-9} amperes". Data-sets for run A (8): (880, 1), (950, 4), (970, 7), (995, 15), (1030, 35), (1030, 35), (1055, 49), (1110, 126). Data-sets for run B (9): (1036, 1), (1088, 8), (1135, 5), (1160, 8), (1195, 15), (1230, 34), (1245, 35), (1295, 74), (1330, 168). SOURCE: A. E. Garrett, 1910, "Positive electrification due to heating aluminium phosphate," (London, Edinburgh, and Dublin) *Philosophical Magazine* **20**: 571–591. Table on page 581. PROBLEM: How can we straighten the plots?

→

exhibit **25** of chapter 6 (continued)

25d) Growth-promoting effect of a purine for a deficient strain of red bread-mold. (0, 112) translates as: "When 0 moles of guanine per mole of adenine were included with 0.1 milligrams of adenine in 25 milliliters of basal medium, the dray weight of mycelium produced by a purine-deficinet strain of *Neurospora* was 11.2 milligrams". Data-sets (9): (0, 112), (0.25, 135), (0.59, 152), (0.75, 185), (1, 196), (1.5, 203), (2, 203), (2.5, 243), (3, 224). SOURCE: J. L. Fairley, Jr., and H. S. Loring, 1949, "Growth-promoting activities of guanine, guanosine, guanylic acid, and xanthine for a purine-deficient strain of *Neurospora*," *Journal of Biological Chemistry* **177**: 451–453. Table I on page 453. PROBLEM: How can we straighten the plot?

exhibit **26** of chapter 6: data and problems

Still more problems

26a) Measurement of a certain impurity in DDT; change of scale factor with temperature. (21, 248) translates as "At 21°C, the rate of crystallization (in microns per 5 minutes) is 24.8 times the log of the percent of this particular impurity." Data-sets (14): (21, 248), (22, 308), (23, 388), (24, 465), (25, 569), (26, 678), (27, 806), (28, 959), (29, 114), (30, 139), (31, 168), (32, 202), (33, 236), (34, 270). SOURCE: W. McCrone, A. Smedal, V. Gilpin, 1946, "Determination of 2,2, bis-p-chlorophenyl-1,1,1-trichloroethane in technical DDT: A microscopical method," *Industrial and Engineering Chemistry* **18**: 578–582. Table IV on page 582. PROBLEM: How can this plot be straightened?

26b) Demand deposits in post-office savings accounts in Switzerland. (37, 458) translates as: "In 1937, there were 458 million francs in post-office savings accounts". Data-sets (29): (37, 458), (38, 498), (39, 523), (40, 643), (41, 701), (42, 787), (43, 839), (44, 927), (45, 1001), (46, 1079), (47, 1007), (48, 1033), (49, 1090), (50, 1125), (51, 1212), (52, 1248), (53, 1334), (54, 1393), (55, 1443), (56, 1720), (57, 1720), (58, 1896), (59, 2050), (60, 2268), (61, 2643), (62, 3140), (63, 3353), (64, 3513), (65, 3810). SOURCE: *Swiss Statistical Abstract*, issued November 1969 by the Swiss Credit Bank. (Title also in French and German.) Tables at pages 24 and 25. PROBLEM: How can we best straighten a plot for this data? Make the fit and plot the residuals. Summarize all results.

26c) Revenue passenger miles on U.S. passenger airlines. (37, 412) translates as: "In 1937, there were 412,000 revenue passenger miles on U.S. domestic scheduled airlines". Data-sets (24): (37, 412), (38, 480), (39, 683), (40, 1052), (41, 1385), (42, 1418), (43, 1634), (44, 2178), (45, 3362), (46, 5948), (47, 6109), (48, 5981), (49, 6753), (50, 8003), (51, 10566), (52, 12528), (53, 14760), (54, 16769), (55, 19819), (56, 22362), (57, 25340), (58, 25343), (59, 28269), (60, 30514). (The last translates as: "In 1960, 30,514,000 revenue passenger miles".) SOURCE: Robert G. Brown, 1963. *Smoothing, Forecasting and Prediction of Discrete Time Series*, Prentice-Hall. Table C.7 on page 427. His source: *F.A.A. Statistical Handbook of Aviation*. PROBLEM: How can this plot be straightened? Plot residuals from a well-chosen straight-line fit.

26d) Find, from other sources, 2 batches of (x, y) data-sets that interest you and deserve to have their plots straightened by the methods of this chapter. Straighten the plots. Plot the residuals.

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Smoothing sequences 7

"Why do I want to smooth?" This is not an unlikely question in a reader's mind when he or she first comes to this chapter. A few examples may show why it often helps to smooth.

rainfall, wheat, and gold

Exhibit **1** shows the annual precipitation in New York City, year by year, from 1872 to 1958. What we see is mainly a blur--quite spread out but apparently horizontal. Can we look under this blur and see anything more?

Exhibit **2** shows the production of wheat in the U.S., year by year, from 1897 to 1958. What we see here is a rising blur, perhaps with some peaks on it. Can we look under this blur and see anything more?

Exhibit **3** shows the production of gold in the U.S. (Alaska included), year by year, from 1872 to 1956. This is less blurred, with various ups and downs moderately clear. Can we see all that we would like to see here? Can we unhook our eyes from the irregularities?

smoothing

Most of us could do a fair job of looking at exhibit 3 and drawing a "smooth" curve "through" the points. Doing a similar thing with exhibit 2 would be more difficult, and doing anything useful about exhibit 1 by eye alone would be very hard indeed.

smoothing gold, wheat, and rainfall

The techniques we will learn in this chapter, when applied to exhibit 3, give the result shown in exhibit **4.** While this "smooth" curve has some wiggles left that we might--or might not--like to smooth off, in general it does a pretty good job of showing us how gold production has behaved, details set aside. We may feel that the 1873–4 peak should be shown as going higher, for example, but for many (if not most) purposes, this "smooth" tells us well about the general run of gold production in the U.S. The value of smoothing --here, as elsewhere--is the clearer view of the general, once it is unencumbered by detail.

The same techniques applied to the wheat-production data in exhibit 2 gives the "smooth" of exhibit **5.** Now we see not only the general rise but some swings up and down. The low patch in the middle thirties and the high patch(es) in the teens and the late 40's are now quite striking. We see quite a lot more from the smooth than we could from the data.

Finally, exhibit **6** shows a "smooth" of the data of exhibit 1. Where once we could see almost nothing, we can now see a lot. Periods of wetness in 1884 to 1904 and 1934 to 1953 seem to be surrounded by dry spells, some of which last 25 to 30 years. If the water supply of New York City were your concern, you would learn much from exhibit 6 that would be hard to gain from exhibit 1.

exhibit **1** of chapter 7:
N.Y.C. precipitation

Annual precipitation in New York City, 1872–1958––actual

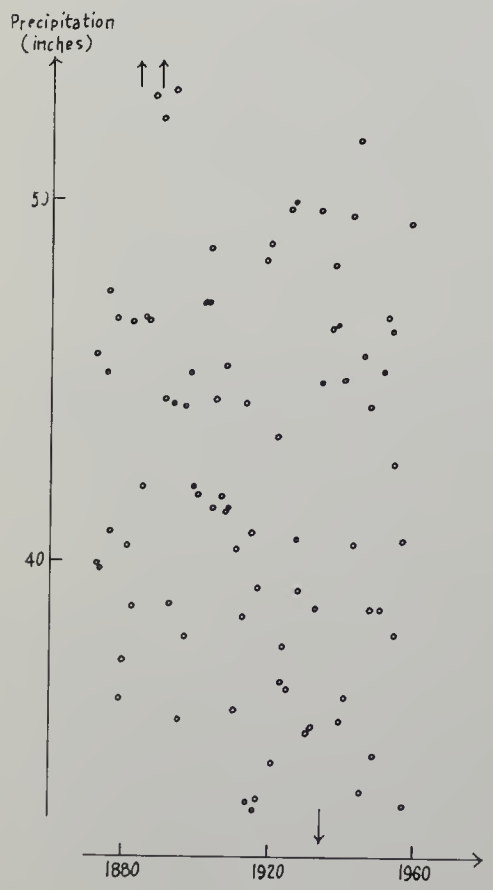
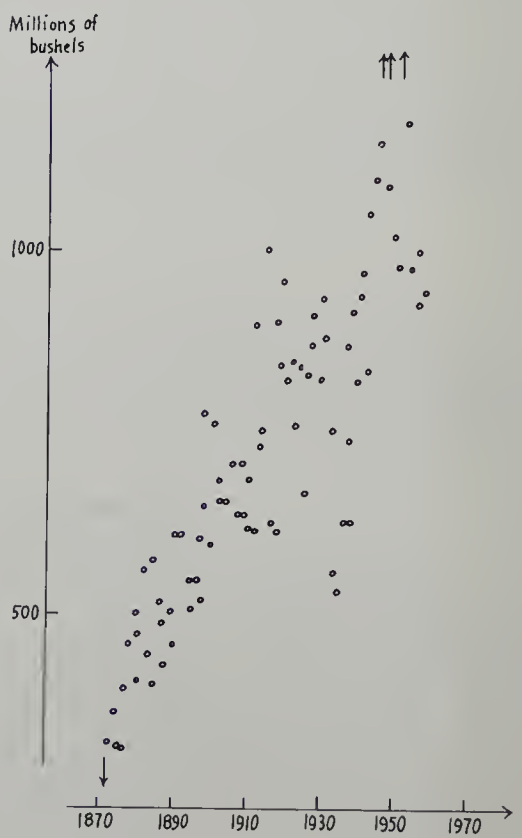


exhibit **2** of chapter 7:
wheat production

U.S. annual wheat production, 1872–1957––actual (in millions of bushels)



what approach?

Now that we understand something of what we want to do, how do we proceed? In earlier chapters, we gave straight lines a special place, first by working hard to straighten plots, and then by emphasizing residuals from lines and working with schematic plots based on them. Straight lines are as important as their treatment in those chapters would suggest.

But not everything *is* a straight line. We need techniques that will help us see what the data are saying, even when they don't follow a straight line.

One approach is to try to climb the polynomial ladder. (Bah, it isn't straight; try a quadratic! Double bah, it isn't quadratic; try a cubic--and so on.) Occasionally this works well--especially when (1) the curve is nearly

exhibit **3** of chapter 7:
gold production

U.S. gold production, 1872–1956 -- actual (in millions of fine troy ounces)

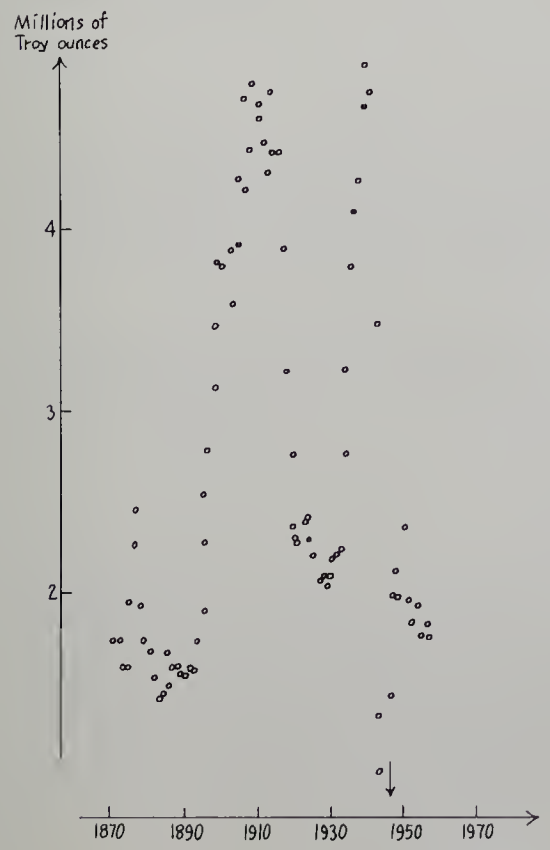
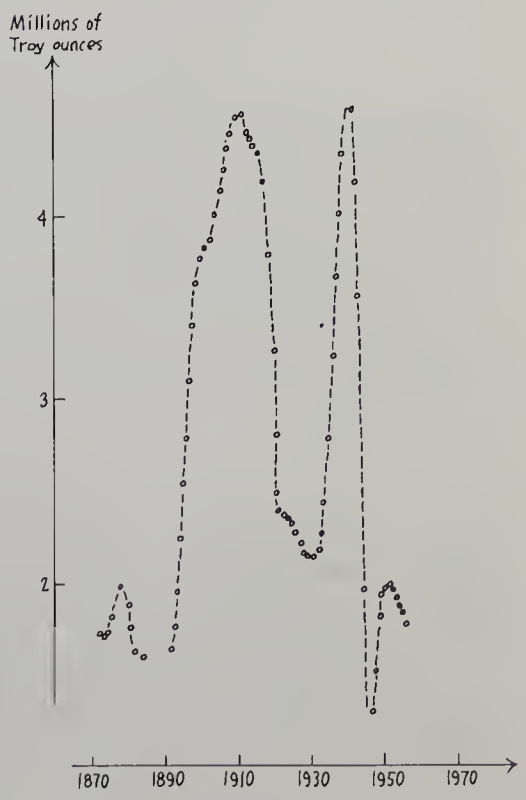


exhibit **4** of chapter 7:
gold production

U.S. gold production, 1872–1956 --smoothed (in millions of fine troy ounces)



straight or (2) there is some principle that points to a particular kind of polynomial. These cases are nice--when they happen. Counting on their happening to you, however, is unsafe.

While it was profitable to start with a family of straight lines and then help the data choose one as "the fit", this is not a good pattern for smoothing. Often the data can be usefully thought about in terms of some relatively smooth curve--even though it would be difficult, unpleasant, or just impossible to specify a family of relatively smooth curves from which the one we use came. If we have no selected family of curves, how can we go ahead?

The two keys are reliance on nearby values for guidance and a number of steps of smoothing. At each step, we compare each value with a few adjacent values--often only 2 or 3--and alter it as they suggest. Continuing this process with steps of various kinds, we may hope that the result gets smoother and smoother. (It may tend to drift away a little, too. We will learn to deal with this when we come to chapter 16.)

Techniques that lead us to "smoothings" of the data which we can think of as reflecting smooth curves can be useful, particularly since their use preserves much of our flexibility. Because we have smooth curves in mind, the process is called smoothing. As a fitting process, it leads to the general relation

data = fit PLUS residuals

which here can be written

given data = smooth PLUS rough

Since the **"smooth"** is intended to be smooth, we will usually show its points **connected.** Sometimes we join the points of the smooth sequence with line segments, obtaining a chain of line segments. Sometimes we just draw a "smooth curve" through these points. Since the **"rough"** is rough, we show its residuals as **separated points.**

The simplest kind of (x, y) data has x proceeding in order by equal steps, as when we have a value every year--or every mile. We can smooth such data without any specific reference to the values of x--our attention can be focussed on the

sequence

made up of the successive values of y. For equal x-spacing we start with a given sequence--and from it we find two sequences, a smooth--a sequence of smooth values--and a rough--a sequence of residuals. This is the simplest and neatest case, and we can learn to deal with it effectively in a simple way.

In this chapter we learn about simple and easy smoothing, leaving elegance of appearance, which is often valuable, to a later chapter and to other accounts.

In this chapter, where we deal only with sequences, we can forget about the x's; similarly, we will often use "date" to describe "when" an input value belongs. We do this when the input values come once a year. We would do it equally freely if they came once every million years, once every 7.543 days, or once every tenth of a second.

We might well do it, also, when the input values come once a mile down a road from a given starting point. So long as we have equal spacing, we usually may as well use the words "time" and "date", for what we do--and how well

exhibit **5** of chapter 7:
wheat production

U.S. annual wheat production, 1872–1957--smoothed (in millions of bushels)

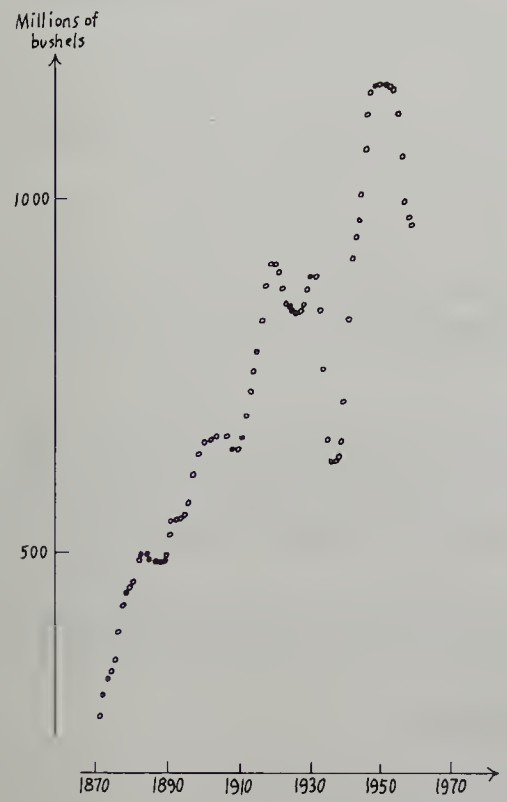
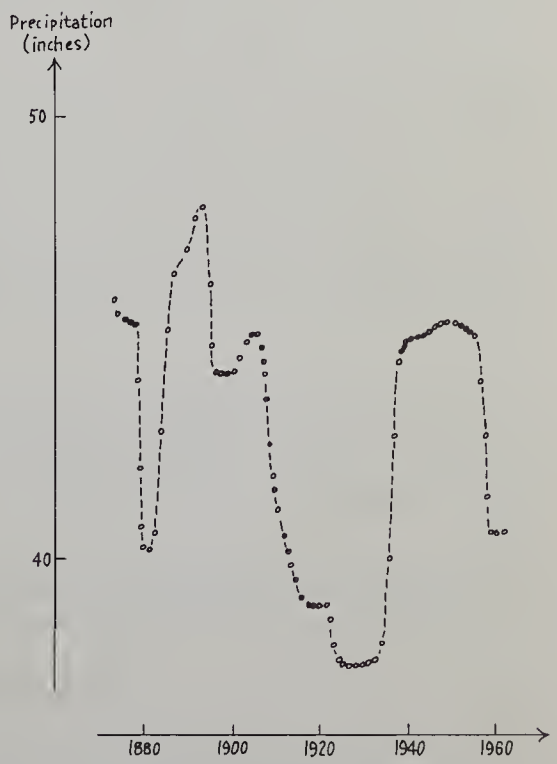


exhibit **6** of chapter 7:
N.Y.C. precipitation

Annual precipitation in New York City, 1872–1958--smoothed



it serves us--will both be very close to what would happen if we did have values equally spaced in time. The words "time series" are often used in similar situations. Since their use often goes with assumptions of more detailed structure for the data and/or use of more complex techniques, we will stick to "sequence", which is equally precise.

review questions

What examples did we start with? How did each set of data look? What did their smoothing do for us? What do we want to look under? What do we want to unhook our eyes from? Are straight lines important? Are they always good enough? What about polynomials? Can we look for a fit without explicitly specifying what we are fitting? What is the role of repeated steps? Of adjacent values? What is smoothing? What is a sequence? What is a smooth? What is a rough?

7A. Medians of 3

Given a sequence of numbers like this:

4, 7, 9, 3, 4, 11, 12, 1304, 10, 15, 12, 13, 17

we easily see that a reasonable smoothing of this sequence will increase slowly from near 5 to near 15, paying as little attention as reasonable to 1304.

1304 may be real, indeed very important, but it has nothing to do with a smooth curve. It ought to generate a very large residual, and leave the smooth fit as nearly alone as we can conveniently arrange to have it do. Whether OR not it is an error, we do NOT want it to disturb the smooth fit.

Let us take the values 3 at a time, rearrange each set of three in order of value, and take their median, as in exhibit **7.** The first three have the same order in value as in sequence, so little happens. The 2nd, 3rd, and 4th in sequence order have a different order when put in order of value. The 7, which is 2nd (middle) in value--and thus the median--is not the middle one in sequence.

It is easy to learn to look at three successive values in a sequence, find the middle one in order of value, and write it down--across from the middle one in sequence order. (A moving finger, covering up the values on one side or another of the three being medianed, often helps.) Exhibit 7 shows how this works.

Because the blocks whose medians we take can be thought of as sliding along, such sequences of medians are often called moving medians or

running medians

We have made a good start on smoothing this particular sequence, but we have yet to learn what to do about the missing values at the ends.

exhibit **7** of chapter 7: illustrative

Overlapping blocks of three, their medians, and the result

A) The INPUT

4, 7, 9, 3, 4, 11, 12, 1304, 10, 15, 12, 13, 17

B) EXAMPLES of RUNNING MEDIANS--we will never bother with this much detail again

Three successive values						Median
As given (sequence order)			In order (value order)			
4	7	9	4	7	9	7
7	9	3	3	7	9	7
9	3	4	3	4	9	4
3	4	11	3	4	11	4
4	11	12	4	11	12	11

C) The "3" SMOOTH and its ROUGH

Given	Smooth	Rough
4	?	?
7	7	0
9	7	2
3	4	−1
4	4	0
11	11	0
12	12	0
1304	12	1292
10	15	−5
15	12	3
12	13	−1
13	13	0
17	?	?

Examples: 7, 7, 4, 4, 4, etc., as in **(B)**. $0 = 7 - 7$, $2 = 9 - 7$, $-1 = 3 - 4$, etc.

P) PROBLEMS

7a) Smooth the smooth of (C) above by running medians of 3. Is the result smoother? Why/why not?

7b) Smooth the rough of (C) above by running medians of 3. Comment on the result.

7b2) Smooth the result of (7b) again by running medians of 3. Comment on the result.

repeated medians

We are now ready to see what happens when we take repeated medians of 3. Starting with an extension of the same example by two more items (20 and 24), the first two smoothings by 3's are:

4 7 9 3 4 11 12 1304 10 15 12 13 17 20 24
? 7 7 4 4 11 12 12 15 12 13 13 17 20 ?
? ? 7 4 4 11 12 12 12 13 13 13 17 ?

In this instance, after the final smoothing by 3's--here the 2nd, since nothing more will happen if we smooth by 3's again--the behavior of the smoothed sequence is put together from these kinds of (overlapping) parts: steady upward, flat tops (of length at least 2), steady downward and flat bottoms (of length at least 2). All such parts will remain untouched by further median smoothing by 3's, so we know we were right to say that we were done.

exhibit **8** of chapter 7: coal production; suspended deposits

Bituminous coal production (in millions of net tons per year) in the United States, 1920–1968, and deposits in suspended banks (in millions of dollars) in the United States, 1921–1967

Year	Coal production	3*	3R	Suspended deposits	3R
1920	569	?			
1	416	422		196	?
2	422	422		111	189
3	565	424		189	189
4	484	520		213	189
25	520	520		173	213,194
6	573	520		272	194
7	518	518		194	194
8	501			139	194
9	505			230	230
30	468			853	230
1	382	382		1691	853,116
2	310	334		716	1691,853
3	334	334		3599	716
4	359	359		37	37
35	372			10	11
6	439			11	11
7	446			11	11
8	349			19	
9	395			13	
40	461			34	
1	511			5.9	
2	583			3.7	
3	590			1.7	
4	620			6.2	

→

exhibit **8** of chapter 7 (continued)

Year	Coal production	3*	3R	Suspended deposits	3R
45	578			.40	
6	534			0	
7	631			0	
8	600			.17	
9	438			0	
50	516			.04	
1	534			3.1	
2	467			1.4	
3	457	457		44.4	
4	392	457	457	2.9	
55	467	467	467	6.5	
6	500	493	493	11.9	
7	493	493		12.9	
8	410			6.3	
9	412			2.0	
60	416			8.0	
1	403			7.5	
2	422			1.2	
3	459			23.3	
4	467			22.0	
65	512			45.9	
6	534	534		0.7	
7	552	545		11.8	
8	545	?			

*** "3" means here "smoothed by medians of 3".**

P) PROBLEMS

8a) Complete the "3" column for coal production.

8a2) Would anything happen if this "3" column were smoothed by medians of 3 again? Why or why not?

8a3) What would a "3R" column look like for coal production?

8b) Complete the "3R" column for suspended deposits.

8c) Find a different sequence of data (at least 20 entries long) that interests you and do smoothing by medians of 3.

8c2) Complete the result to "3R" smoothing.

8d/8d2) As problem 8c/8c2, but for at least 50 entries.

S) SOURCES

Coal Production; ***World Almanac,*** **1946 page 635, 1957 page 715, 1965 page 721, 1970 page 149, 1969 page 128.**

Their source: Bureau of Mines.

Suspended deposits: ***World Almanac;*** **1931 page 306, 1965 page 749, 1970 page 99.**

Their source: Federal Reserve System.

notation

We write "3" for a single smoothing by medians of 3 and "3R" for repeated smoothing by medians of 3, a smoothing continued until there is no further change.

a comment

Notice that we are now well into the pattern suggested in the opener for this chapter:

- ◇ we are using repeated steps.
- ◇ each step is applied separately to each value.
- ◇ what happens to a value in a step depends only on that value and a few neighboring values (here one on each side).

bituminous coal; suspended payments

Exhibit **8** gives the data, and part of the smoothing, for forty-odd years of: (a) U.S. bituminous coal production and (b) total deposits for those U.S. banks suspending payments in any year. For coal production we show the given values, part of a "3" smooth by medians of 3, and a smaller part of a "3R" smooth by repeated medians of three. For the suspended deposits, we show the given values, and the condensed calculation of its "3R" smooth. Notice, in the suspended-deposits example, at 1925, how we write "213,194" to mean that smoothing by 3's gave 213, while resmoothing by 3's gave 194. Notice also that, in these examples, once we have done a first smooth by medians of 3, going on to 3R involves little effort (and causes little change).

review questions

What is a running median? How do we calculate one? What partial behaviors cause values to be unchanged by running medians of 3? What behaviors ensure change? What happens when running medians of a given length are repeated? Are repeated many times? How can our fingers help? How does what we do fit the pattern announced in the opener of this chapter?

7B. Eye resmoothing

How can we make as effective use as possible of the 3R smooths we have just learned to make?

Such smooths are much better behaved than the raw data. If we had only the raw data, and wanted to tell ourselves about what it seemed to say, we could try to draw a smooth curve "through" it. It is much easier to draw a smooth curve through the points of our 3R smooth than through the raw data. Exhibit **9** shows what can happen when we do draw a smooth curve through the 3R? smooth of the bituminous coal data.

judgment = choices

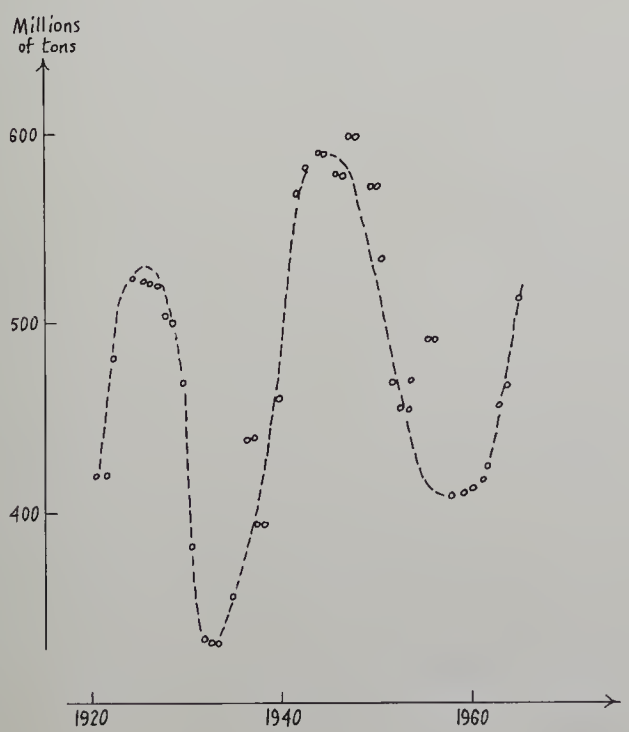
Once we bring in judgment, we have choices. Exhibits **10** and **11** show two more eye smooths of the same 3R smooth we saw in exhibit 9. In exhibit 10 we have chosen the long view--one so long that 10-year-long peaks are not of interest. In exhibit 11 we have gone in the opposite direction, following the 3R-smooth points **more** closely than in exhibit 9 rather than less.

Different users are always likely to prefer different choices. A great advantage of judgment is its ability to meet different needs. (The corresponding disadvantage is shown by the arguments among those who want to make different judgments.) Those who eye-smooth their 3R smooths are using a minimum of arithmetic and a maximum of judgment.

Those who want to use more arithmetic and somewhat less judgment can use their judgment to pick out one of the smooths that we shall learn to do in later sections of this chapter (or even what we will learn when we come to Chapter 16).

exhibit **9** of chapter 7: coal production

A moderate eye smooth of the numerical NE smooth of exhibit 3 (the first of three eye smooths of the 3R? smooth)



review questions

Is there a place for eye-fitting combined with numerical fitting? Why/why not? Can we use snap judgment for arithmetic? Why/why not?

7C. Looking ahead

We now know how to do the very simplest and easiest smooths. Let us look to see how much better we could do if we wanted to, and how we can build easily on what we now know.

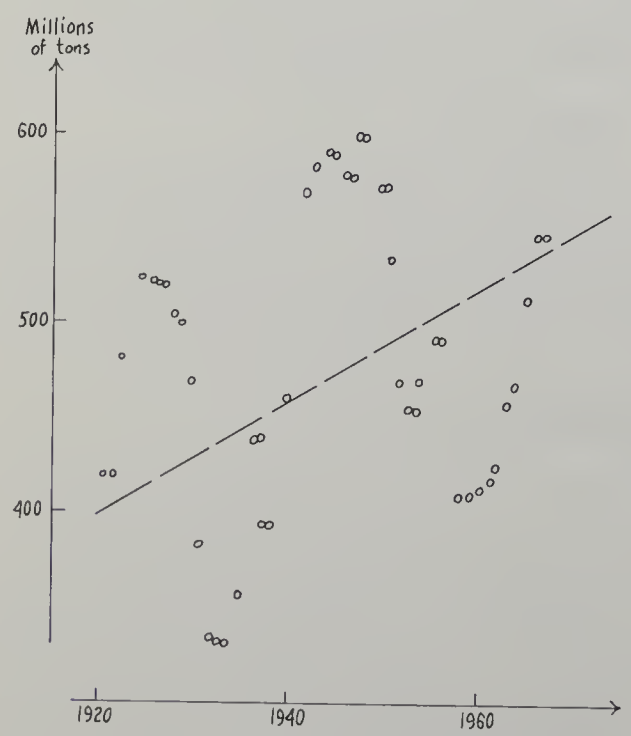
a comparison of four pictures

Exhibit **12** pictures, for the bituminous-coal example,

- ◇ the raw data (in NW corner).
- ◇ the data smoothed as in section 7A (3R? smooth, in NE corner).

exhibit **10** of chapter 7: coal production

A very general eye smooth of the NE smooth of exhibit 3 (the second of three eye smooths of the 3R? smooth)



◇ the data smoothed as of the end of section 7E (3RSS smooth, in SW corner).

◇ the data smoothed quite heavily, as in section 7F (3RSSH3RSSH3 smooth, in SE corner).

Here the "?" in "3R?" refers to our postponed treatment of end values, to be settled in section 7D, the "S" in "3RS" to splitting, which will be explained at the end of section 7F, and "H" to "hanning", which will be explained in section 7G.

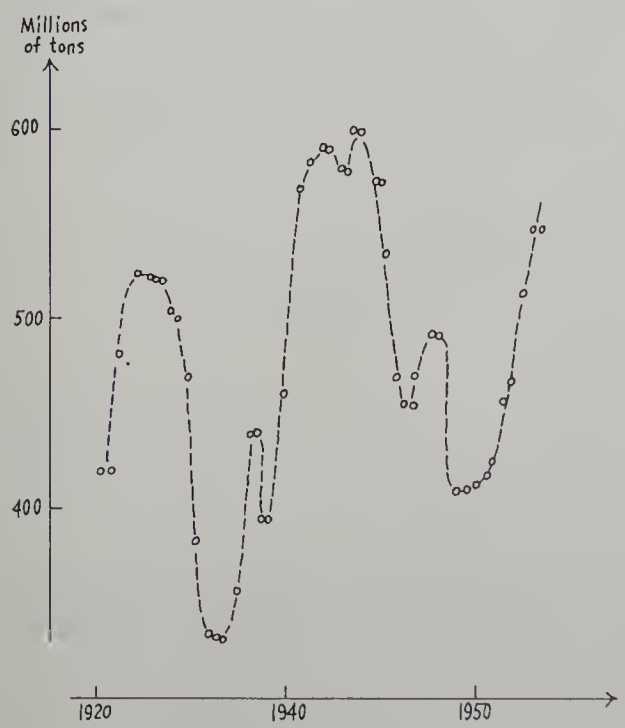
These pictures deserve careful looking at and careful thinking over. Notice:

◇ that, as we go from NW to NE to SW to SE, the general behavior becomes clearer.

◇ that, as we go back any step, we can see, by looking carefully, that whatever we saw in the later picture can in fact be seen in the earlier one--but with greater effort.

exhibit **11** of chapter 7: coal production

A detailed eye smooth to the NE smooth of exhibit 3 (the third of three eye smooths of the 3R? smooth)



◇ how hard it is to see the general behavior in the NW version, which is so irregular as to make attempts to join the points unproductive.

◇ how MANY actual details are gone in even the NE version, while essentially ALL details are gone in the SE version.

◇ how close the eye-smoothing of exhibit 9 came to the final (SE) version.

This example shows clearly many of the main points about smoothing. The analyst who needs the actual details cannot depend on the smooth (though he can often do well with smooth and rough combined). The analyst who needs general behavior naturally turns to a smooth (as often he MUST). The purpose of the smooth is to show off general behavior. It is NOT to show most of the actual details. And, at least in its present form, it is NOT to give an impression of variability, NOT to show how precise we ought to think the smooth is. (We make it smooth to ease the eye's task and keep the eye from getting stuck on trivia. We often want it to be smooth enough so that if we compared several different smooths, all reasonable enough, they would look quite different.)

blurring, advance notice

Fortunately for those who do not want to go to more complicated arithmetic, and who find-eye smoothing "too indefinite," there is another way to unhook the viewer's eye from the minor wiggles of a less-than-polished smooth. We can do this by blurring the smooth, something easily done by replacing points by short vertical lines of appropriate length. Exhibit **13** shows a blurring of the SW version of exhibit 3, in comparison with the more highly polished SE version (of the same exhibit). Clearly we have gained by blurring a lot--probably most--of what additional arithmetic could bring us.

We will want to do such blurring with a view to the size of the rough. Details will wait until we reach section 7E below.

agenda

What do we now know we want to learn? Surely these things:

◇ how to deal with end values (7D).

◇ how to blur (7E).

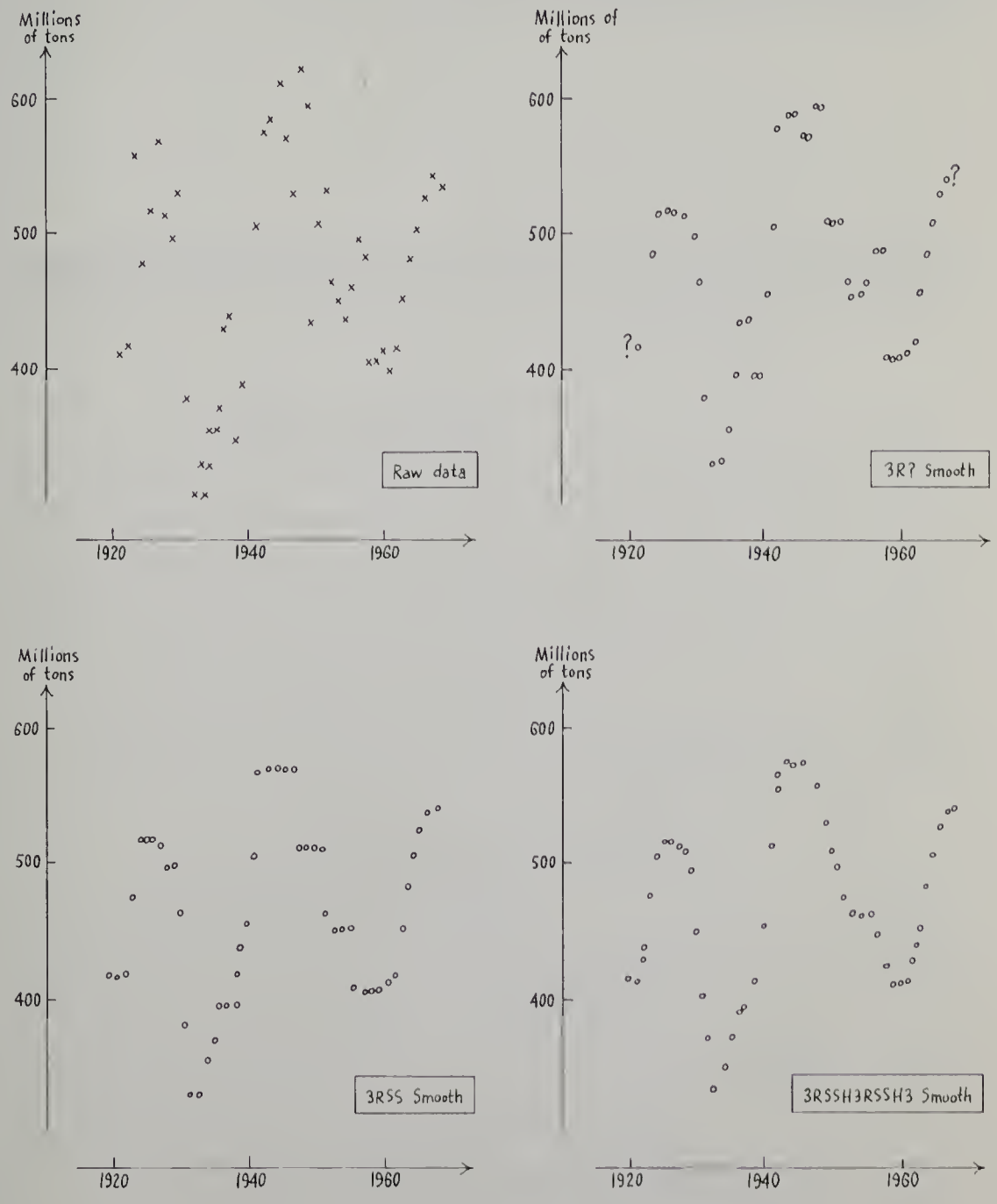
◇ how to deal with the flat peaks and valleys only two entries wide (noticeable in exhibit 7, for instance) (7F).

◇ how to do a still smoother smooth (7G).

In addition, we will eventually come, in chapter 16, to ways for following ups and downs more closely, thus leaving little systematic behavior in our roughs and thus getting still better smooths. In this chapter, however, we will learn no further new principles, only techniques for doing a smoother job than 3R? does. (In the optional sections, chapter 7^{+}, these same principles will be used more thoroughly.)

exhibit **12** of chapter 7: coal production

Raw data and three smooths--U.S. bituminous coal production in millions of tons (data in exhibit 8)

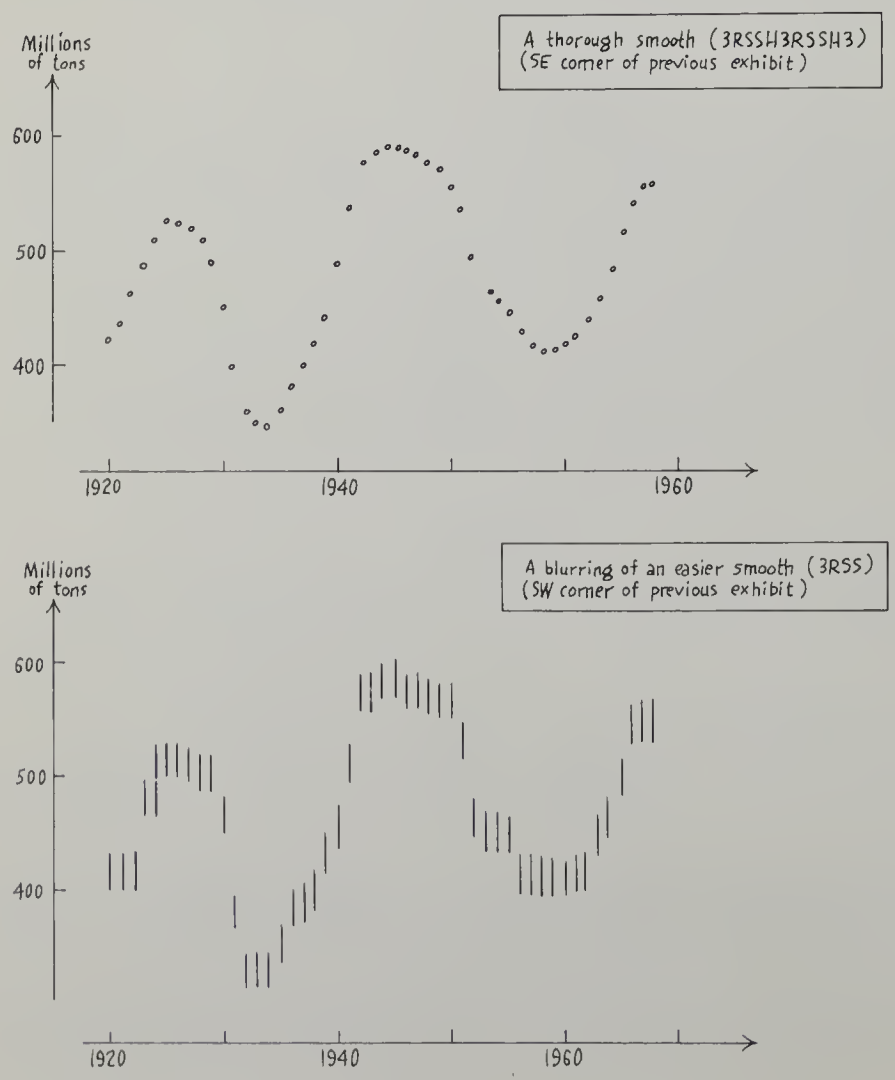


review questions

What happened when we compared various smooths of the same data? What example did we choose? What did we show as ways to unhook eyes from wiggles and corners? What do we need to learn now?

exhibit **13** of chapter 7: coal production

Two different ways to unhook the viewer's eyes from the wiggles and corners of a moderately easy smooth



7D. Copying-on--and more, usually.

When we introduced repeated medians of 3 (in section 7A) we were purists, and let the question-marks creep in from each end. We did this so we could talk about one idea, repeated medians of 3, by itself. In the real world we will not let the ?'s creep in; instead we will "copy on" the end values. Thus the example on page 212 really looks like this:

4 7 9 3 4 11 12 1304 10 15 12 13 17 20 24
4 7 7 4 4 11 12 12 15 12 13 13 17 20 24
4 7 7 4 4 11 12 12 12 13 13 13 17 20 24

This will at least keep our sequences from shrinking, but is it enough? Occasionally, yes, when we think an end value may have a special message; but usually no. So we need some end-value smoothing. We turn now to the simplest end-value smoothing that seems to do what we need.

end-value smoothing

We have now to be slightly complicated. Exhibit **14** shows some pictures. All show the same two end input values and the same five smoothed values. What we are to do uses only the extreme end input value and the next two smoothed values. The NW panel emphasizes the actual end value with a square box. The NE panel emphasizes the next-to-end smooth, also with a square box. The SW panel emphasizes the result of straight-line extrapolation of the next-to-end and next-but-one-to-end smoothed values--extrapolation NOT just to the end, but one step BEYOND.

The SE panel emphasizes not only all three of the points emphasized in the other panels but especially the median of these three, which is what we take as our smoothed value at the end.

for those who prefer verbal arithmetic

A more arithmetic way to put it is this:

◇ the change from the end smoothed-value to the next-to-end smoothed-value is between 0 and +2 times the change from the next-to-end smoothed-value to the next-to-end-but-one smoothed value.

◇ subject to this being true, the end smoothed-value is as close to the end input-value as possible.

This means that we can look at two differences:

end input-value MINUS next-to-end smoothed-value

and

next-to-end smoothed value MINUS next-but-one-to-end smoothed value

and if the first is between 0 and +2 times the second, we can copy on.

Otherwise we make

end smoothed-value MINUS next-to-end smoothed value

either zero or two times

next-to-end smoothed-value MINUS next-but-one-to-end smoothed value.

3R or 3R′

Since copying-on during the 3's, followed by end-value smoothing after completing the 3's, is the most usual case, we will use 3R to label it, saving 3R′

exhibit **14** of chapter 7: purely illustrative

End-value smoothing: parts and result

to label the use of copying-on only (during repeated smoothing by medians of 3).

a comment

Notice that both copying-on and end-value smoothing again represent steps where a value is, at most, compared with a few nearby values.

review questions

How were we purists in 7A? What do we do in the real world? What is copying on? Do we allow smoothing by medians of 3 to shorten our sequences? Is copying-on usually enough? What else might we do? How often? When not?

7E. Blurring the smooth--and setting the fences

We want next to fix the lengths of the vertical lines we use to blur our smooth. Clearly these lines should go equally far above and below our 3R smoothed values. It seems natural to use the sizes of the residuals as a basis. But how?

In our schematic plots (chapter 2), we used a box to cover half the points. So let us start by going up-and-down far enough to cover half the rough. This means going as far as the median of the sizes of the residuals. Panels A and B of exhibit **15** show stem-and-leaf displays of the sizes of residuals for the two smooths of the bituminous-coal production that have concerned us most. Panel C calculates the ordinary medians, hinges, and eighths. Are we happy?

modified hinges, medians, sizes, etc.

For the rough that goes with the more highly polished smooth, whose rough had only 8 zeros out of 49, we can be fairly happy. To know that half the roughs have sizes of 11 or less is helpful. For the 3RSS rough--the one that goes with the 3RSS smooth--we are far from happy. Being told that the median size is 7 is some help--we get the idea, correctly, that many, many residuals are zero; but we really don't get an idea how large the residuals are. Perhaps we should modify our definitions.

One thing we could do easily would be to treat each

exact zero

as some fraction of a value, not a whole one. The easiest fraction is 1/2. Since it seems to work, we will use it. The median sizes of residual for the two examples are now 13 (instead of 11) and 12h (instead of 7). Both are informative.

We indicate such modification by putting a star, *, in front of each and any of M, H, E, . . . , f, F, #, etc. that we calculate. We then call these

*-letter values.

exhibit **15** of chapter 7: coal production

Looking at sizes of roughs, here also sizes of residuals

A) The SE (polished) ROUGH––"Z" means zero

49	**Z**	ZZZZZZZZ
41	**0**	1112223333445778
25	**1**	13337
20	**2**	34
18	**3**	0145
14	**4**	1469
10	**5**	36
8	**6**	289
5	**7**	88
3	**8**	1
2	**9**	0
1	**H**	(147)

B) The SW (3RSS) ROUGH––"Z" means zero

49	**Z**	ZZZ. . . (18 in all). . . ZZZ
31	**0**	2444677
24	**1**	002233
18	**2**	445
15	**3**	06
13	**4**	44
11	**5**	1336
7	**6**	05
	7	
5	**8**	118
	9	
2	**H**	(147, −138)

C) SIMPLE MEDIANS, HINGES, etc., of SIZES

(for the SE)		(for the SW)	
#49	(all)	#49	(all)
M25	11	M25	7
H13	44	H13	44
E 7	68	E 7	60

D) MODIFIED MEDIANS, HINGES, etc., of SIZES

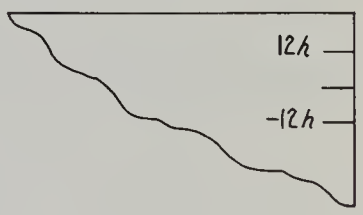
(for the SE)		(for the SW)	
*#45 = 41 + 4		*#40 = 31 + 9	
*M23	13	*M20h	12h
*H12	46	*H10h	52
*E 6h	68h	*E 5h	73

→

From now on the use of *-medians (and other *-letter values) will be standard with sizes of residuals. That is why the vertical lines of exhibit 10 went up-and-down ±12h--the *M of the sizes of residuals is 12h.

proper laziness

Only one who loves either arithmetic or self-torture will make this kind of picture by adding 12h to and subtracting 12h from each smoothed value, and then plotting these two ends of each vertical line. It is much easier to do this:



- ◇ plot all the smoothed values.
- ◇ mark the edge of a piece of paper or card with three marks 12h apart.
- ◇ put the center mark on each plotted point in turn, and draw the vertical between the endpoints. (If you use first graph paper and then tracing paper, as you should, it is usually easier to put only center and endpoints on the graph paper, and draw the lines only on the tracing paper.)

This keeps the work to a reasonable level.

exhibit **15** of chapter 7 (continued)

E) PARTS of STEM-and-LEAF DISPLAYS for the RESIDUALS--not their sizes

(for the SE)		(for the SW)	
–0	874433221	**–0**	7642
–0	331	**–1**	320
–2	4	**–2**	4
–3	40	**–3**	6
–4	4	**–4**	4
–5		**–5**	6
–6	2	**–6**	50
L	(–78)	**L**	(–138)

Note: These are needed for some of the problems.

P) PROBLEMS

15a/b Make up full stem-and-leafs for the residuals (SE for (a), SW for (b) and find (i) their simple medians, hinges, etc., including fences, and (ii) their modified medians, hinges, etc., including fences.

15a2/b2) Find what values of residuals are (i) outside and which are (ii) far out.

15c) Find raw and modified medians of the residuals of the 3R? smooth (found when exhibit 8 was completed; note that residual at each end is zero).

outside values

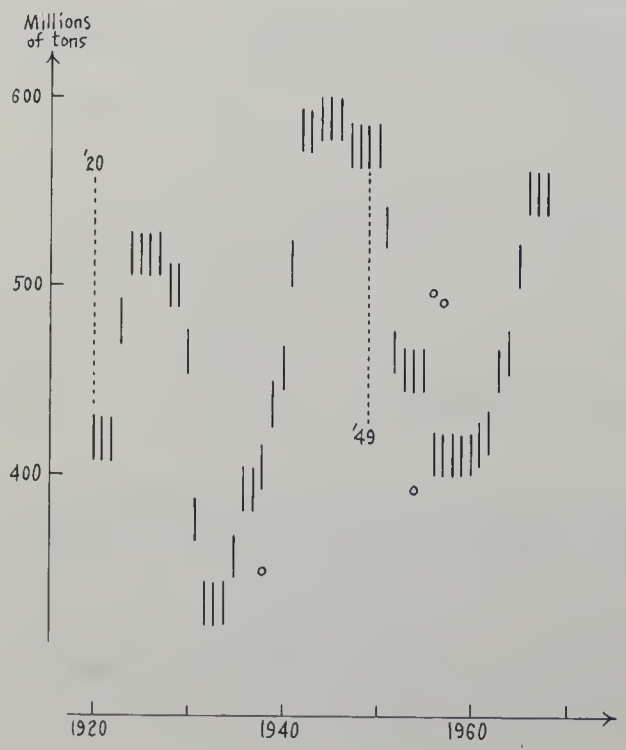
If we are to make good use of our roughs to support our smooths, just using the modified median of their sizes to fix how far our blurring lines extend is not enough. We also need to identify outside--and far out--points.

The problems of exhibit 15 included doing this for our two usual examples, using both simple hinges and modified hinges. With fences hung on simple hinges, the situation for the 3RSS example (SW version) is bad. With 18 zeroes, the hinges were at 10 and 0. Of 31 nonzero values, **only 8** were **not** outside, and 16 were far out. To call for identification and special treatment for 19 out of 49 residuals--or, if you prefer, 19 out of the 27 nonzero ones--is to dilute unusualness till it is not useful. (If we try to do the same with the 3R? rough, we will not find it much better.)

As we had to do with medians of sizes, so too will we have to do with fences. We shall use *-fences--fences hung on *-hinges--routinely in dealing with 3R roughs.

exhibit **16** of chapter 7: coal production

Outside and far out points added to the blurred smooth of exhibit 4 (the 3RSS smooth)



For the 3RSS case, then, we have *-hinges at −8h and 24, so that the *-fences are: first at −59 and 74, second at −107h and 123h, leaving 5 values outside and 2 values far out. Exhibit **16** shows these 7 added to the blurred smooth of exhibit 10.

When both blurring and outside points appear, we refer to the

smooth supported by the rough.

Notice that we have used a moderately weak circle for outside values, and an only moderately strong string of dots for far out values, and that we have tagged (here dated) only the far out values. This has been done so as to leave the individual points somewhat noticeable, without letting them interfere too much with the general impressions to be gained from the blurred smooth itself.

review questions

What earlier case guided us in deciding how broadly we should blur? What aspect of rough and smooth do we use to fix the breadth of the blur? What are modified hinges? Modified median sizes? Other modified letter values? (Does it make sense to modify a median? Why/why not? How do we show that letter values have been modified? How can we save effort in showing blurs? How do we hang fences when there are many zeroes? Where can we best show outside and far out points? What does "the smooth supported by the rough" mean? How can we keep outside (and far out) points from distracting our attention from the rest of the rough?

7F. Splitting peaks and valleys

The most annoying aspect of the kinds of smooths we are now getting is probably the little flat-topped hills and valleys, each 2 points wide, that are scattered through our smooth. How can we conveniently deal with these "bumps"? A simple approach is to treat each such pair as if they were end values--first copying on, and then using end-value smoothing--and then smooth once more by 3R. After the first 3R, then, we first break the result up into subsequences and then bring these subsequences back together and use 3R again to smooth over roughness at the joints.

Panels A and B of exhibit **17** illustrate, on some pieces of the 3R smooth of the bituminous-coal production data, just how such a calculation goes. Panel C carries out the calculation for a 25-year stretch of that data.

smoothing short subsequences

The last subsequence, formed by splitting the 457 2-wide valley, contains only two entries: 457, 467. It is too short to think of smoothing, though there is no reason not to use the followup 3R in the hope of pulling the subsequence smooths together. The first subsequence has three values: 468, 382, 334. These

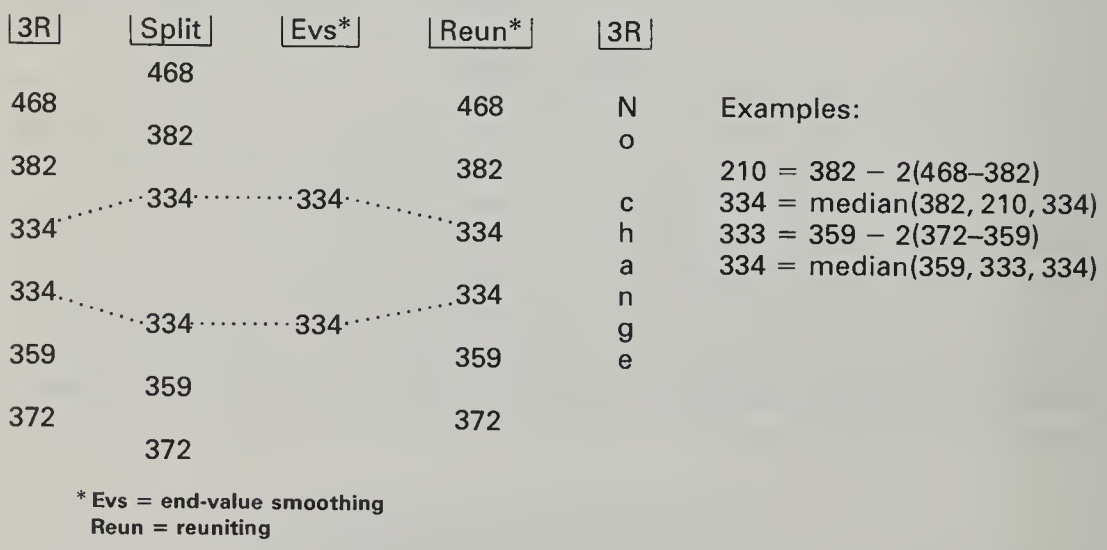
are unaffected by the first 3R. We are willing to continue with end-value smoothing, for which we have the fewest number of entries that will work. In the case of these three, nothing happens.

If we had had 468, 454, 334, end-value smoothing would have given 468, 454, 426. If we had had 468, 354, 334, it would have given 394, 354, 334. For a subsequence of three entries, end-value smoothing will change at most one.

exhibit **17** of chapter 7: bituminous coal

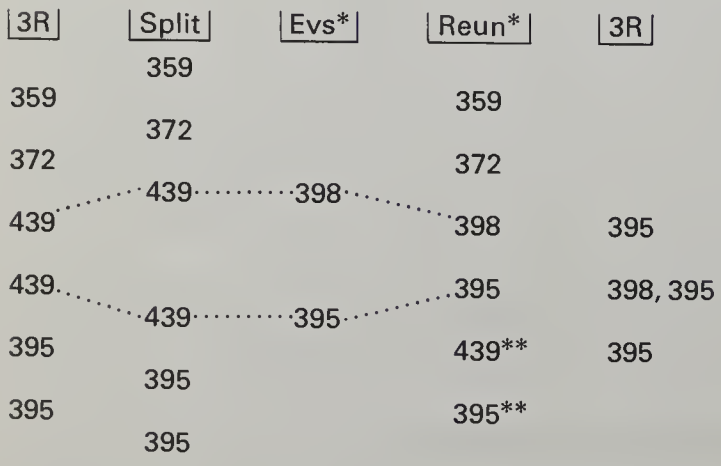
Splitting the peaks and valleys illustrated on the 3R smooth

A) A VALLEY



* Evs = end-value smoothing
Reun = reuniting

B) A HILL



** These values come from 395, 395 being a valley, and corresponding calculations not shown in this panel.

→

exhibit **17** of chapter 7 (continued)

C Above included in a 25 YEAR STRETCH––working style

		1st split		2nd split	
Year	3R	Raw	3R	Raw	3R
1930	468		468		
1	382		382		
2	334	334	334	334	334
3	334	334	334	334	334
4	359		359		
35	372		372		
6	439	398	395		
7	439	395	395		
8	395	439	398		
9	395	395	439		
1940	461		461		
1	511		511		
2	583		583		
3	590	590	583		
4	590	578	590		
45	578	590	590		
6	578	600	590		
7	600	578	578		
8	600	516	516		
9	516		516		
1950	516		516		
1	516		516		
2	467		467		
3	457	457	457	457	457
4	457	457	457	457	457
55	467		467		

Notice:

1. Use of] to mark 2-wide peaks and valleys.
2. Use of end-value smoothing to find all values in the "Raw" columns.
3. Repetition of the S process exactly twice.
4. That, in this example, splitting the second time had no effect.

P) PROBLEMS

17a) Extend this calculation to the entire 3R smooth for this data, finding a 3RSS smooth.

17b) The same for the suspended bank deposits example.

Notice that splitting is planned to be done exactly twice. Accordingly, we symbolize its use, after 3R, by writing

3RSS

as the label of our new smooth.

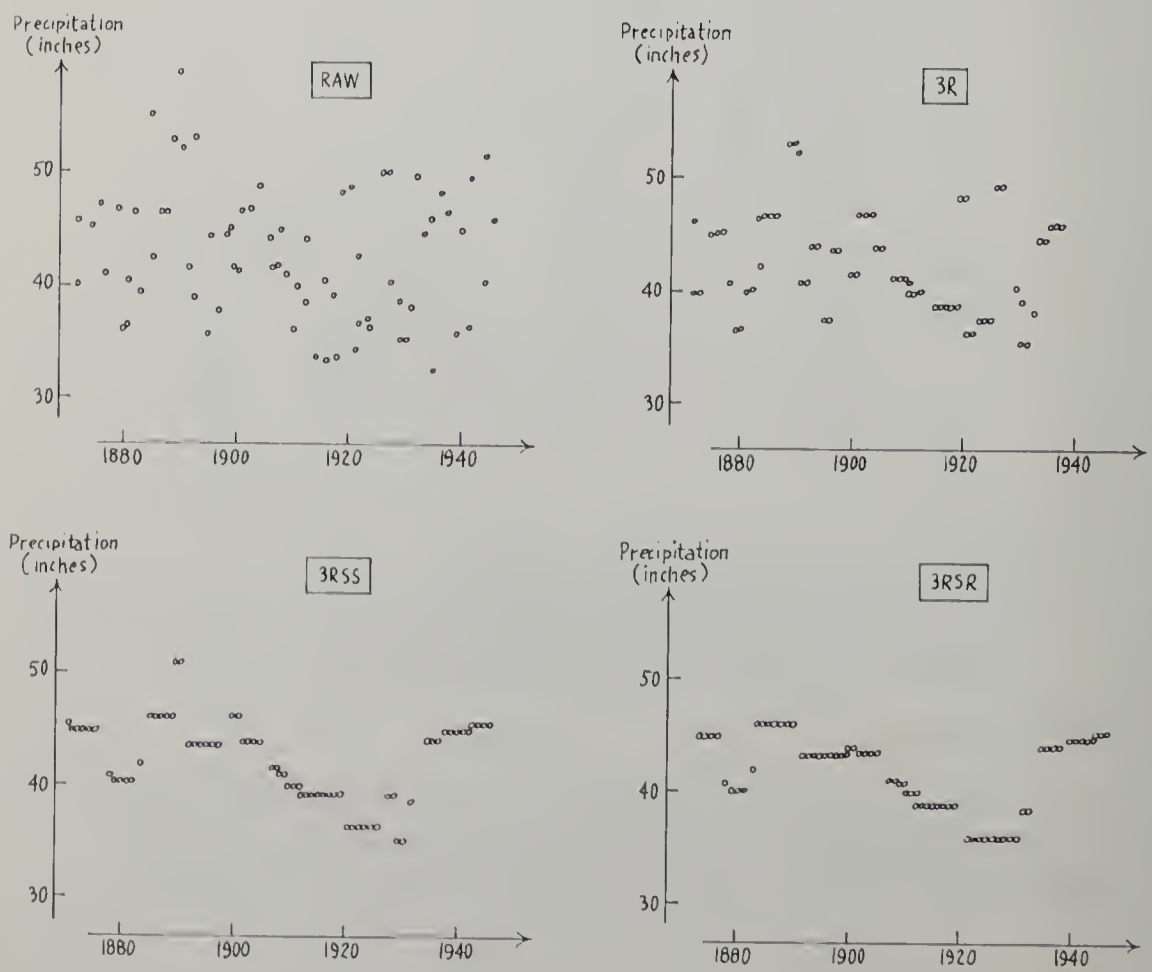
going farther (optional)

Two uses of "splitting" is one good standard--in many examples it seems to be a good place to stop. Sometimes, however, we can gain by going farther, or not so far.

When we do go farther, it sometimes seems useful to "split it to death" just as we are used to "3-ing it to death" when we do 3R. When we do, we

exhibit **18** of chapter 7: N.Y.C. precipitation

The 1872–1945 portion of the precipitation record and three of its smooths compared



write "SR". Exhibit **18** shows, for the New York City precipitation example of the opener (exhibits 1 and 6) of this chapter, a comparison between

- ◇ the given data up to 1945 (NW corner).
- ◇ a 3R smooth (NE corner).
- ◇ a 3RSS smooth (SW corner).
- ◇ a 3RSR smooth (SE corner).

In this example, it clearly is worthwhile to go all the way to SR.

Notice also that, just as 3R smoothings are often the same as 3 smoothings would be, so SR smoothings are often the same as SS smoothings would be.

a comment

We are still proceeding step-by-step, comparing each value with nearby values. (We have to look at 4 adjacent values to identify a peak or valley that is only 2 points wide. We have to look at only 3 values to make the end-value adjustment.)

review questions

What problem concerns us in this section? How do we approach it? What happens when a 2-wide peak is adjacent to a 2-wide valley? Which end-value rule do we use? What do we do after "splitting"? How many times do we do all this? What do we use to tag the new smooth?

7G. Hanning

A steadily increasing sequence, such as

4, 5, 6, 8, 11, 74, 78, 79, 80, 81

will not be affected by the most powerful smoothers we have so far found it necessary to use--3RSS or 3RSR. (So long as the values increase [or decrease] steadily, the median by value of three adjacent numbers will be the same as the median in sequence. Thus "3R" will have no effect and "S" will not be called on to do anything.)

We may not find such a sequence smooth enough for our task. How then are we to make it smoother? A little thought leaves us with the belief that nothing simpler than old-fashioned arithmetic will do. Surely, however, we want to keep the effort involved as small as we can. The one piece of arithmetic that is simplest for us is that involved in taking the median of a batch of 2, 4, or some other even count of numbers, namely

$$\frac{\text{(one number)} + \text{(another number)}}{2}$$

How can we use this in our smoothing?

Exhibit **19** shows us two ways to reach the same result. In panels A and B we take the means of adjacent values, not once but twice. (Notice that it is natural to write the results of the first step half-way between the lines we used for the start.)

In panels C and D, designed to avoid half-lines, we do

$$\frac{\text{(one number)} + \text{(another)}}{2}$$

twice, also, but in a different pattern. We start by forming

skip means

in which the two values combined lie one line above and one line below the line on which the skip mean is entered. Thus 11 and 78, no matter what

exhibit **19** of chapter 7: illustrative

Introduction to hanning (or how to use $(a + b)/2$ to make things smoother)

A) MEANS OF PAIRS OF ADJACENT **B)** The SAME REPEATED NUMBERS

Start	Mean of adjacent	Start	Mean of adjacent	Again
4		(4)		
	4h		4h	
5		5		5
	5h		5h	
6		6		6
	7		7	
8		8		8
	9h		9h	
11		11		26
	42h		42h	
74		74		59
	76		76	
78		78		77
	78h		78h	
79		79		79
	79h		79h	
80		80		80
	80h		80h	
81		(81)		

Examples:

4h = (4 + 5)/2, 5h = (5 + 6)/2

7 = (6 + 8)/2, 6 = (5h + 7)/2

8 = (7 + 9h)/2, 26 = (9h + 42h)/2

→

separates them in the start, come out with a skip mean thus:

11
xy 44h
78

We head a column of skip means with ">".

Once we have a full set of skip means, we set about forming means of the last two values in each line, so that 74 and 44h always give 59, as in the pattern

tu *vw*
74 44h 59
za *bc*

no matter what comes above or below them in either column. Again the only difficulty--here not at all serious--is end values. With only one value in the

exhibit **19** of chapter 7 (continued)

C) ISOLATED SKIP MEANS

In	>
4	
5	5
6	
8	8h
11	
74	44h
78	
79	
80	80
81	

D) A FULL SET of SKIP MEANS followed by LINE MEANS

In	>	H	(copy on)
4			(4)
5	5	5	
6	6h	6	
8	8h	8	
11	41	26	
74	44h	59	
78	76h	77	
79	79	79	
80	80	80	
81			(81)

Examples:

5 = (4 + 6)/2, 8h = (6 + 11)/2
44h = (11 + 78)/2, 80 = (79 + 81)/2, . . .
5 = (5 + 5)/2, 6 = (6 + 6)/2, 8 = (8 + 8h)/2
26 = (11 + 81)/2, 59 = (74 + 44h)/2, . . .

P) PROBLEMS

19a) Hann the result of (17a), which was a 3RSS smooth of the bituminous coal production example.

19a2) "3" the result of (19a).

19a3) Smooth the result of (19a2), using 3RSSH3 again.

19a4) Find the rough, corresponding to the result of (19a2), and smooth it with 3RSSH3; comment on the result.

line, about all we can do sensibly is to copy on (as shown in parentheses). We head a column of such means with "H", for hanning.

Notice that, except for end values, the two procedures--that of panel B, and that of panel D--give the identical result. That means that we can leave the choice among them to the reader. Printers and those who like to save on paper--or the cost of books--will be likely to prefer the skip and line means of panel D, with its avoidance of half lines. The writer has decided that he finds this approach easier to use by hand, also. So all examples from here on will use the > (for skip mean), H (for line mean) approach. (Readers need not follow this choice.)

The name hanning is after Julius von Hann, an Austrian meteorologist of the last century, who liked to smooth weather information--temperatures, pressures, etc.--in this way.

comments

Again we add a new kind of step that involves only a few adjacent values --three for hanning. Again we apply this step separately to each and every value (end values treated specially). Again we assemble a variety of such steps to make a smoother that is better than any one individual step could provide.

If we fear the influence of stray values, as we almost always should, we would not dare hann the raw data. An isolated stray point, if hanned directly, will spread and become three stray points. While they will not stray as far, their three-ness is likely to trouble us. We trust to the medians of 3R--and the short peak and short valley removal of SS--to bring most strays to values that will not trouble us too much, **before** we hann.

going a little farther

Especially in some of the uses in the next chapter, we find that 3RSSH has some residual local roughnesses that catch our eye when we wish this didn't happen. An easy thing to do is one more smoothing by running medians of 3. We naturally call the result 3RSSH3. (See problem (19a2).)

going a lot farther

Anything we can do once, we can do twice. So having applied 3RSSH to the given data, we can apply 3RSSH to the 3RSSH-smooth. The result is a 3RSSH3RSSH smooth, and we can sprinkle another "3" on the tail end, thus making 3RSSH3RSSH3, if we wish. (See problem (19a3).)

what of the rough?

Problem (19a4) asked us to smooth the rough for the 3RSSH3 smooth of the coal-production example. Those who did this found that this smooth did not come out to be zero. Clearly, this means that there is more that we should

subtract from the rough before we look at the rough in too much detail. When we come to chapter 16 we will do just that.

Sometimes the change in the smooth from being careful about what was left in the rough is important. Often it is not. We have had enough details in this chapter to make it wise for us to postpone such matters.

review questions

What kind of sequences that escape 3RSS smoothing might we want to smooth further? How did we decide to tackle them? Are there alternative ways to hann? What are their advantages/disadvantages? Can you describe one in detail? how do we head which columns? Why is it called "hanning"? How does hanning fit into the general pattern of this chapter? Would we dare to hann raw data? How can we go a little farther? A lot farther? What are we postponing to Chapter 16?

7H. How far have we come?

This chapter has introduced us to the smoothing of sequences. Sometimes such smoothing will let us see regularities where without it there was only a fog. More often, perhaps, smoothing lets us see easily what we might have seen by patient and detailed examination and re-examination.

A sequence from which to start, and a smooth found from it, implies the existence of a rough, satisfying

$$\text{input} = \text{smooth} + \text{rough}$$

the current version of

$$\text{data} = \text{fit} + \text{residual}.$$

We have postponed to chapter 16 any detailed consideration of roughs. (This has, in particular, let us keep our smoothing techniques simpler.)

Running medians of 3 have been our prime tool. (Surely nothing could be simpler for hand computation.) Almost always we used them repeatedly, until no further change occurs.

We have a wide choice as to how thoroughly we smooth a sequence. If we smooth it too hard, we may miss detail we wanted to see. If we don't smooth it enough we may find our eyes hung up on irrelevant detail. What is "too much" or "not enough" will depend upon the specific example.

In particular, we are ready:

- to use, but not always, end-value smoothing, thus finding a smooth for the extreme ends of the sequence, where running medians of 3 do not apply.
- to blur the smooth carefully, thus detaching our eyes from the smooth and saving us from wanting or needing to smooth further.

◇ to use end-value smoothing to help us deal with the narrowest peaks and valleys (those made up of only two points) that running medians of three leave behind (splitting).

◇ to use hanning, conveniently represented in terms of split means and line means, to smooth still further.

Optional sections for chapter 7 7+

chapter index on next page

The three sections that follow take the considerations of chapter 7 somewhat further. Exhibit numbers and section letters are continued.

7I. Breaking a smooth

Sometimes the given values of a sequence show a clear break. When this happens we have a choice:

- ◇ we may smooth across the break.
- ◇ we may treat the data as two sequences, applying the end-value smoothing on either side of the break.

Which we do depends on our purposes and our understanding of the data.

If we believe that there was a very unusual change between two adjacent values--whether from an extra-heavy dose of whatever drives the values of our particular sequence up and down OR from an isolated event that happens quite rarely--we are not likely to want to smooth across the gap with any smoother more polishing than 3RSS. If we go on to 3RSSH or 3RSSHS, we will surely find the values next to the supposed break pulled toward each other. Indeed, if we use only 3R, we may at times see this happen. To have these adjacent values pulled toward each other is a way to hide the "break". If we think the break is "real", we usually do not want to hide it.

If, on the other hand, we think the possible "break" is only a somewhat unusually large change, we will not want to preserve a break--rather we will want to smooth over it.

Whether we want to show a "break" or a smoother appearance will also be affected by whether a noticeable break in the smooth will seriously distract attention from other aspects that we particularly need to see--or to show to others--in this particular picture.

presidential New Hampshire

Exhibit **20** starts from the Presidential vote in the two northernmost of the counties of New Hampshire that adjoin the state of Maine, noting the

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percent of the vote of each county for the Democratic candidate in each election from 1896 to 1972. The difference in percent Democratic, Coos MINUS Carroll, starts small, rises somewhat in 1916 to 1924, and breaks upward in 1928, and then remains close to 25%. The 3R column of exhibit 20 shows the result of the broken smooth. (The parenthetic entries in the "(unbroken)" column show what would have happened had we not broken the data before smoothing.)

Exhibit **21** shows the broken smooth and hints at the unbroken version. Clearly we have a wide choice between appearances. We can have a wide gap--about 15% change--or we can have a smooth which shows almost no unusual change at all. Which should we prefer (as amateur psephologists!)?

It seems to be a tradition among political scientists to stress critical changes in individual elections. Such a tradition would stress the change between '24 and '28 and thus lead to a preference for the broken smooth. Those who go this way would presumably describe behavior in two intervals: 1896–1924 and 1928–1972.

exhibit **20** of chapter 7+: Coos–Carroll

Percent Democratic vote for President in two counties of New Hampshire and a split smooth of their difference

A) DATA, BROKEN, SMOOTH, and ROUGH--"Z" is an exact zero

	% Demo. for Pres.					Unbroken	
	Coos	Carroll	Diff	3R	Rough	3R	(Rough)
1896	29.75	28.38	1.37	.82	.55		
1900	41.33	40.54	.79	.82	−.03		
	39.61	38.79	.82	.82	Z		
1908	39.25	37.73	1.52	.82	.70		
	40.52	43.12	−2.60	1.52	−4.12		
1916	51.99	46.65	5.34	5.34	Z	(5.34)	(Z)
	44.40	35.01	9.39	6.55	2.84	(6.55)	(2.84)
1924	39.65	33.10	6.55	6.55	Z	(9.39)	(−2.84)
	43.11	22.37	20.74	20.74	Z	(16.94)	(3.80)
1932	52.14	35.20	16.94	20.74	−3.80	(20.74)	(−3.80)
	55.67	33.26	22.41	22.41	Z	(22.41)	(Z)
1940	60.30	33.66	26.64	26.45	.19		
	58.36	31.91	26.45	26.64	−.19		
1948	52.29	23.22	29.07	26.64	2.43		
	44.03	17.39	26.64	26.64	Z		
1956	33.86	14.53	19.33	26.64	−7.31		
	57.28	20.39	36.89	26.08	10.81		
1964	71.09	45.01	26.08	30.15	Z		
				26.08			
	53.37	23.22	30.15	26.08	4.07		
1972	37.45	21.54	15.91	26.08	−10.17		

→

A hard look at the broken smooth, however, makes us wonder. Its pattern can be generally described as follows:

- ◇ an initial flat at about 0.8%.
- ◇ a rise from 0.8% to 6.6%.
- ◇ a "jump" from 6.6% to 20.6%.
- ◇ a rise from 20.6% to 26.5%.

exhibit **20** of chapter 7⁺ (continued)

B) The ROUGH--from the BROKEN smooth

1	**H**	(10.81)
2	**4**	07,
	3	
4	**2**	84, 43
	1	
7	**0**	55, 70, 19,
(7)	**Z**	Z, Z, Z, Z, Z, Z, Z
	−0	03, 19
	−1	
	−2	
4	**−3**	80,
3	**−4**	12,
2	**−L**	(−7.31, −10.17)

#16h (zeroes/2)

		Z	
M	8h		
H	4H	1.56	−2.00
E	2h	3.46	−5.72
X	1	10.81	−10.17
	f	6.90	−7.34
	F	12.24	−12.68

outside: 10.81 and −10.17
far out: none
almost out: −7.31

(sizes)

3	**H**	(10.81, 7.31, 10.17)
5	**4**	07, 12,
6	**3**	80
8	**2**	84, 43
	1	
(6)	**0**	55, 70, 19, 03, 19
7	**Z**	Z, Z, Z, Z, Z, Z, Z,

median = 1.56
(at depth 8h)

C) The ROUGH from the UNBROKEN smooth

1	**H**	(10.81)
2	**4**	07
3	**3**	80
5	**2**	84, 43
	1	
8	**0**	55, 70, 19
(5)	**Z**	Z, Z, Z, Z, Z,
7	**−0**	03, 19
	−1	
5	**−2**	84,
4	**−3**	80,
3	**−4**	12,
2	**L**	(−7.31, −10.17)

#17h (zeroes/2)

		Z		
M	9			
H	5	2.43	−2.84	5.27
E	3	3.80	−4.12	
X	1	10.81	−10.7	
	f	10.33	−10.74	
	F	18.24	−18.64	

outside: 10.81
far out: none
almost out: −10.17

P) PROBLEMS

20a1) Find a pair of counties in another state whose difference in % Democratic shows a comparable shift.

20a2) Carry through the analysis parallel to that above.

Of a total rise of about 26%, almost half occurred before '24 or after '28. Accordingly, it is not at all unreasonable to use the unbroken smooth and describe behavior in **three** intervals: 1896–1910, 1910–1938, 1938–1972. This choice is somewhat supported by the appearance of the raw data, shown in exhibit **22.** (Thinking about a change spread over 28 years is quite different from trying to think about a change over 4 years!)

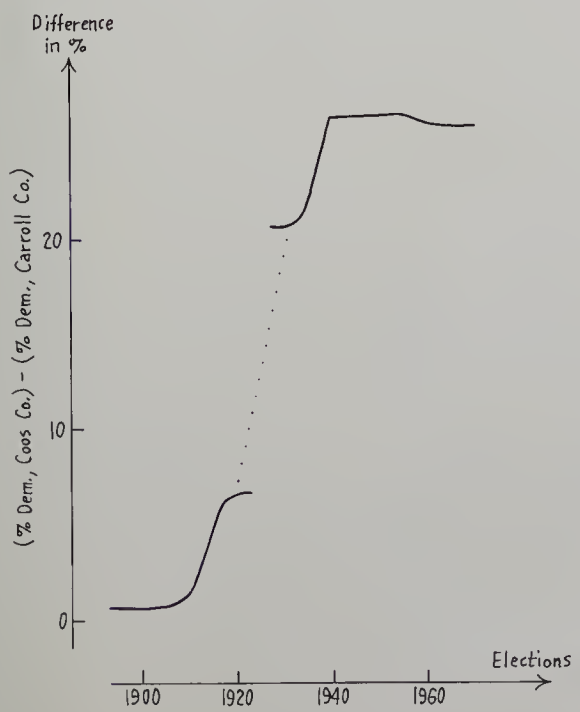
Exhibits **23** and **24** show the result of bringing in the rough to assist both broken and unbroken smooths. The naive looker can surely be pardoned for seeing no strong case for a break.

back to suspended deposits

For an example in which a broken smooth is suggested by the data AND turns out to be very much in order, we return to the total deposits in banks suspending payments in the U.S. each year, where, up to 1933, the amounts were large and generally increasing, but thereafter quite small. We use started logs, starting all values by 1 million dollars.

exhibit **21** of chapter 7^{+}: Coos–Carroll

Broken smooth of the difference in Presidential Democratic in the voting of two adjacent New Hampshire counties (unbroken smooth in faint dots)



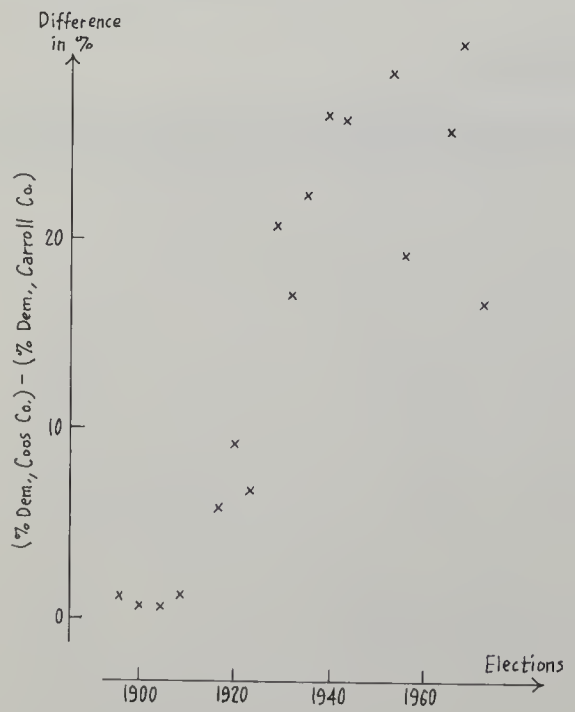
Breaking the sequence between 1933 and 1934 gives the values shown in the "3R/broken" column of exhibit **25.** It would seem that the break is quite worthwhile.

When we look up the facts, we find not only that "up to 1933" is different from "1934 on"--because of the establishment of the Federal Deposit Insurance Corporation--but that 1933 is different from both before and after--because of banks not allowed to reopen after the bank holiday.

This leads us to the "double break", found on the extreme right of exhibit 25, where we break once between '32 and '33 and once between '33 and '34. The resulting analysis is shown--rough supporting the smooth--in exhibit **26,**

exhibit **22** of chapter 7+: Coos–Carroll

The raw data: difference in % Democratic for President, Coos County MINUS Carroll County, 1896–1972



which uses one length of bar for all dates, and exhibit **27,** which also shows outside and far outside points.

review questions

What might we do when the given values of a sequence seem to show a break? How do we decide? To what example did we turn? What choice were we faced with? What might subject-matter experts have preferred? To what second example did we then turn? Did we really have a choice? Why/why not? Do we ever need double breaks? How close to one another might double breaks reasonably be?

exhibit **23** of chapter 7^{+}: Coos–Carroll

The broken smooth supported by its rough

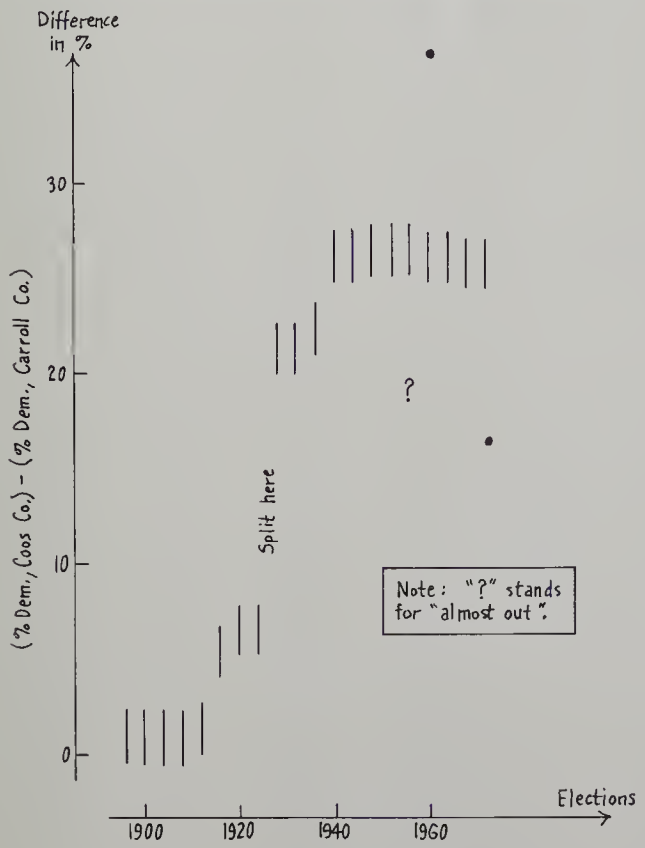


exhibit **24** of chapter 7⁺: Coos–Carroll

The unbroken smooth supported by its rough

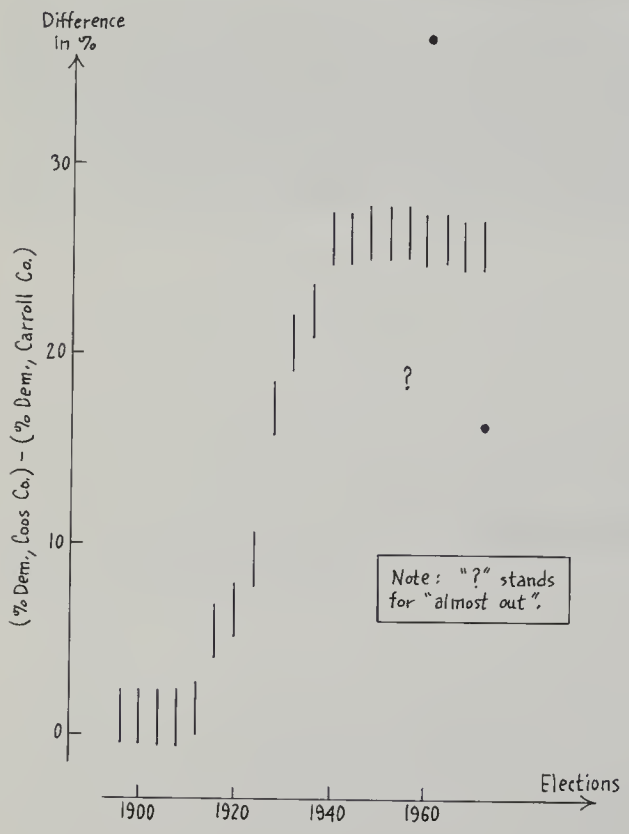


exhibit **25** of chapter 7+: suspended deposits

Split smooth of 100 log (*c* + suspended deposits) for *c* = 1 million dollars

A) The DATA SMOOTHED

year	suspended bank deposits millions	same + 1	log	broken 3R	rough (Z = zero)	3R double break	rough	(outside)
1921	196	197	229	229	1			
22	111	112	205	225	−23	s		
23	189	190	228	228	Z	a		
24	213	214	233	228	5	m		
25	173	174	224	233, 229	−5	e		
26	272	273	244	229	15			
27	194	195	229	229	Z			
28	139	140	215	229	−14			
29	230	231	236	236	Z	236	Z	
30	853	854	293	293	Z	293	Z	
31	1691	1692	323	293	30	293	30	(F)
32	716	717	286	323	−37	293	−7	
33	3599	3600	356	356	Z			
34	37	38	158	108	50			(F)
35	10	11	104	108	−4			
36	11	12	108	108	Z			
37	11	12	108	108	Z			
38	19	20	130	115	15			
39	13	14	115	130, 115	Z			
40	34	35	154	115	39	t		(F)
41	5.9	6.9	84	84	Z	h		
42	3.7	4.7	67	67	Z	e		
43	1.7	2.7	43	67	−24			(f)
44	6.2	7.2	86	43	43			(F)
45	.4	1.4	15	15	Z	s		
46	0	1.0	0	0	Z	a		
47	0	1.0	0	0	Z	m		
48	.17	1.17	7	0	7	e		
49	0	1.0	0	2	−2			
50	.04	1.04	2	2	Z			
51	3.1	4.1	61	38	23			(f)
52	1.4	2.4	38	61, 59	−21			(f)
53	44.4	45.4	166	59, 61	105			(F)
54	2.9	3.9	59	88	−29			(f)
55	6.5	7.5	88	88	Z			
56	11.9	12.9	111	111	Z			
57	12.9	13.9	114	111	3			
58	6.3	7.3	86	86	Z			
59	2.0	3.0	48	86	−38			(F)
60	8.0	9.0	95	93	2			
61	7.5	8.5	93	93	Z			
62	1.2	2.2	34	93	−59			(F)
63	23.3	24.3	139	136	3			
64	22.0	23.0	136	139, 136	Z			
65	45.9	46.9	167	136	31			(f)
66	0.7	1.7	23	111	−88			(F)
67	11.8	12.8	111	111	Z			

→

exhibit **25** of chapter 7⁺ (continued)

B) SOME INFORMATION about SIZE of ROUGH

	*Hspr	*Espr	*M	*H	*f = 4(*M)	*F = 7(*M)
'21–'33	7	20	5	14		
'34–'67	19	68	11	38		
'21–'32	7	20	5	14		
all (split once)	14	60h	4h	34		
all (split twice)	13	56h	4h	30	18	31h

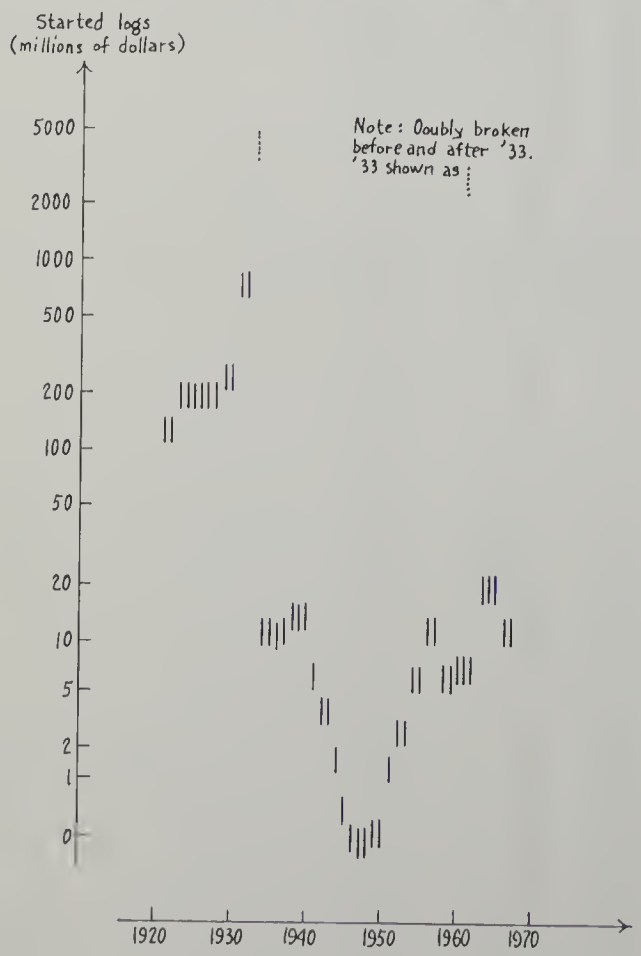
P) PROBLEMS (teasers)

25a1) Try to find another sequence that needs splitting almost as much as this one.

25a2) Analyze it with splitting.

exhibit **26** of chapter 7⁺: suspended deposits

Deposits in banks suspending payment, U.S., 1921–1967, rough supporting smooth; vertical scale = started logs, start = 1 million dollars



7J. Choice of expression

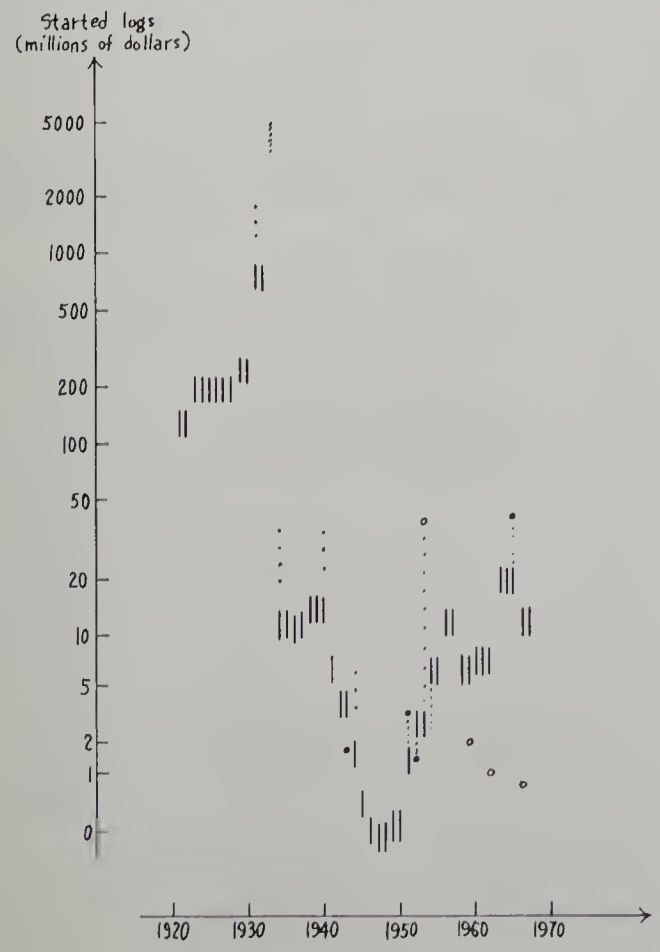
There are four places where the choice of expression might matter in the kind of smoothing we are so far using:

- ◇ first, in the main body of our 3R smoothing.
- ◇ second, in using the end-value rules.
- ◇ third, in calculating residuals, and in picturing the rough.
- ◇ fourth, in picturing the smooth.

Let us look at these four in turn.

exhibit **27** of chapter 7': suspended deposits

Deposits in banks suspending payments, U.S. 1921–1967, vertical scale in millions of dollars (started logs, start = 1 million dollars). Rough supporting smooth and outside points shown.



first, 3R

In the main body of our 3R smoothing, choice of expression ordinarily makes no difference at all. So long as one choice rises whenever another does --and falls when it falls--the same "date" will be the one to provide the "median by value" for any set of three adjacent "dates". So our two smooths will be related by exactly the same change of expression as the values they come from. Exhibit **28** gives a simple example. Notice (in panel B) that, for example, wherever the raw smooth is 64, the root smooth is 8 and the log smooth is 1.8, just as these numbers would be related for input values. So we can forget about choice of expression--so far as 3R goes.

second, end values

End values--and splitting--may be a different story. At the foot of exhibit 28, if we use raw or root values, end-value smoothing squeezes the end-value in (from 1024 to 160, or from 32 to 16, respectively). If we use log values, such smoothing does not squeeze in 3.0 at all.

third, residuals as a basis of choice

When we look only at residuals--at the rough--our main concern is usually with answers to three questions:

- ◇ about how big are the residuals?
- ◇ roughly how does this size change--as "time" changes, or as the size of the smoothed values change, or both?
- ◇ which individual residuals look unusual, and roughly how unusual?

We want to answer these questions as well as we can. What effect will the choice of expression have on this?

Consider, as a simple example, the third line from the top and the fifth line from the bottom of exhibit 28. For raw expression, they give:

residual =	input −	smooth	
768 =	1024 −	256	(3rd line)
48 =	64 −	16	(−5th line)

for root expression, they give

16 =	32 −	16	(3rd line)
4 =	8 −	4	(−5th line)

and for log expression, they give

0.6 =	3.0 −	2.4	(3rd line)
0.6 =	1.8 −	1.2	(−5th line)

In this instance, we see residuals of very different size for raw expression--ones much more nearly of a size for root expression--and equal-sized ones for log expression. Why?

Because the input (and smoothed) values were of a very different size near one end than near the other. So far as residuals go, changing from raw to root to log expression is like introducing a microscope of varying magnification, one where differences between little things are enlarged more than differences between large things. (To do this may be either good or bad.)

What *would* we like to do in our imaginary case? What can choice of expression do for us, either here or in general? Mainly it can make the answers to our three questions as simple as we can arrange to have them. How can such answers become simpler? Rather easily:

- ◇ if one expression makes the size of the residuals change less than another: (i) the first question--how big?--will be easier to answer AND its answer will tell us more; (ii) the answer to the second question--How changing?--can be much simpler.

exhibit **28** of chapter 7^{+}: illustrative

A simple example of 3R smoothing different choices of expression

			Raw		Root		Log	
raw	root	log	in	3R	in	3R	in	3R
4	2	.6	4	?	2	?	.6	?
64	8	1.8	64	64	8	8	1.8	1.8
1024	32	3.0	1024	256	32	16	3.0	2.4
256	16	2.4	256	256	16	16	2.4	2.4
16	4	1.2	16	256	4	16	1.2	2.4
256	16	2.4	256	64	16	8	2.4	1.8
64	8	1.8	64	64	8	8	1.8	1.8
16	4	1.2	16	64	4	8	1.2	1.8
64	8	1.8	64	64	8	8	1.8	1.8
4096	64	3.6	4096	64	64	8	3.6	1.8
16	4	1.2	16	64	4	8	1.2	1.8
64	8	1.8	64	16	8	4	1.8	1.2
16	4	1.2	16	16	4	4	1.2	1.2
4	2	.6	4	16	2	4	.6	1.2
64	8	1.8	64	64	8	8	1.8	1.8
1024	32	3.0	1024	?	32	?	3.0	?

P) PROBLEMS

28a) Invent an example of your own and work it through to show that the same sort of thing happens.

◇ if one expression makes the residuals more symmetrically spread around zero, it is likely that fewer will seem unusual, and the answer to the third question will be simpler.

We seek the same kinds of simplicity for our residuals here that we sought in a seemingly quite different situation in chapter 4.

comparing expressions

We can use the same techniques that we have already used to compare one expression with another:

◇ plotting size of residual against fitted value to look for stability of size.

◇ finding hinges, eighths, etc., to look for symmetry.

We leave this to problems (see exhibit **29**).

Sometimes we can profit by half-simplicity, by having the behavior of the rough simple, not in the data as a whole, but only in each of two or three sections.

fourth, picturing the smooth

If we are to bring the rough to support the smooth--by blurring and showing outside points--we will ordinarily want to use the same expression for the smooth that made the behavior of the rough simple. (Otherwise we will get into blurring by bars of different length, for example, something that we can of course face when the different expression of the smooth is really worth the trouble.)

If we want to use a different expression for the smooth, we can do this easily enough so long as we confine ourselves to

◇ the points of the smooth, and

◇ the outside--and far out--points.

These can be re-expressed without difficulty.

counted data

As usual, the analysis of counted data follows the same pattern as that of other data, subject always to:

◇ we look first at either the logs or the (square) roots of counts.

◇ if we start with roots of counts, we frequently drift in the direction of logs of counts.

◇ if we go to logs, and have small counts, we tend to start the logs slightly --taking logs of "count + start" rather than of "count" alone--with +1/6, +1/4, and +1 likely initial choices (we should be prepared to revise our initial choice once this seems indicated).

exhibit **29** of chapter 7+: data and problems

Some problems on choice of expression (from looking at the roughs [residuals] alone) (H = hundreds, K = thousands, M = millions, T = tens, TM = tens of millions, c = counts, b = barrels, lb = pounds, t = tons)

A) FIRST SECTIONS (to 1825 or 1855) of FIRST GROUP OF SEQUENCES--see panel 1 for identification

Year	H541 (T#)	Q167 (KT)	U1/(U10) (MS)	Y369 (S)	Year	C89 (K#)	C90 (K#)	C91 (K#)	C92 (K#)	C94 (K#)	W76 (#)
1790	58	—	20	—	1820	7.7	2.4	3.6	.02	1.0	—
1	76	—	19	18.6	1	5.9	3.2	1.5	.02	.4	—
2	66	—	21	18.4	2	4.4	1.2	2.3	.03	.1	—
3	68	—	26	18.6	3	4.0	1.1	1.9	.01	.2	—
4	67	4	33	17.5	4	5.0	1.3	2.3	.02	.2	—
95	61	3	48	17.5	5	8.5	2.1	4.9	.02	.4	—
6	57	2	59	17.6	6	9.8	2.3	5.4	.03	.5	—
7	59	1	51	16.8	7	16.7	4.2	9.8	.03	.4	—
8	60	1	61	15.8	8	24.7	5.4	12.5	.06	1.9	—
9	62	6	79	15.2	9	12.5	3.2	7.4	.03	.6	—
1800	65	3	71	15.7	1830	7.2	1.2	2.7	.02	2.0	—
1	73	3	93	15.1	1	13.0	2.5	5.8	.04	2.4	—
2	87	3	72	14.2	2	34.1	5.3	12.4	.33	10.2	—
3	104	12	56	13.1	3	29.1	4.9	8.6	.19	7.0	—
4	114	12	78	14.2	4	57.5	10.5	24.5	.07	17.7	—
05	120	6	96	13.2	35	42.0	9.0	20.9	.07	8.3	—
6	131	11	102	11.7	6	70.5	13.1	30.6	.47	20.7	8
7	145	9	108	10.4	7	71.0	12.2	28.5	4.0	23.7	7
8	153	5	22	9.5	8	34.1	5.4	12.6	.11	11.7	17
9	164	4	52	8.1	9	64.1	10.3	24.0	.38	21.0	10
1810	175	4	67	7.4	1840	80.1	2.6	39.4	.21	29.7	19
1	185	5	61	6.4	1	76.2	16.2	37.8	.23	15.3	21
2	196	3	39	5.9	2	99.9	22.0	51.3	.39	20.4	11
3	215	3	28	7.2	3	49.0	8.4	19.7	1.8	14.4	8
4	212	1	7	10.0	4	74.7	14.4	33.5	1.3	20.7	20
15	212	1	53	11.9	45	109.3	19.2	44.8	1.0	34.4	12
6	215	1	82	14.7	6	146.3	22.2	51.8	2.0	57.6	19
7	226	5	88	13.9	7	229.1	23.3	105.5	1.3	74.3	21
8	230	17	93	11.3	8	218.0	35.2	112.9	1.1	58.5	14
9	242	32	70	19.2	9	286.5	55.1	159.4	3.5	60.2	17
1820	258	36	70	9.5	1850	308.3	51.1	164.0	1.6	78.9	20
1	282	28	65	10.0	1	369.5	51.5	221.3	2.4	72.5	17
2	299	49	72	9.1	2	362.5	40.7	159.5	4.1	145.9	20
3	314	41	75	8.6	3	361.6	37.6	162.6	3.4	141.9	26
4	330	33	76	8.3	4	405.5	58.6	101.6	4.2	215.0	35
1825	342	35	100	7.5	1855	187.7	47.6	49.6	1.8	71.69	41

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exhibit **29** of chapter 7+ (continued)

B) SECOND SECTIONS (1921 to 1956) of FIRST GROUP of SEQUENCES

Year	H541 (T#)	Q167 (Kt)	U1 (MS)	Y369 (K#)	C89 (K#)	C90 (K#)	C91 (K#)	C92 (K#)	C94 (#)	W76
1921	6289	***	4560	220.9	625.4	51.1	28.4	22.8	6.8	3963
2	6444	* 1	3931	208.6	216.4	25.2	10.8	14.6	17.9	4455
3	6522	* 8	4269	199.6	307.9	45.8	15.7	34.2	48.3	4133
4	6604	* !	4753	186.2	364.3	59.5	17.1	35.6	75.1	4723
5	7066	***	5272	177.1	148.4	27.2	26.7	16.8	46.1	5347
6	6830	42	5017	167.3	155.6	25.5	24.9	16.8	50.4	5103
7	7171	46	5142	155.5	168.4	23.7	28.5	16.9	48.5	4918
8	7248	55	5776	146.1	158.5	20.0	25.3	16.2	45.8	5218
9	7245	57	5491	149.0	158.6	21.3	19.9	17.4	46.8	5921
1930	7319	40	4013	131.5	147.4	31.0	23.4	6.9	26.6	6085
1	7247	83	2918	135.4	61.9	9.1	7.3	3.1	10.4	6897
2	7301	73	2434	156.1	20.6	2.1	.5	.9	27	7376
3	7153	102	2061	179.5	12.4	1.0	.3	.5	1.9	7170
4	7254	108	2202	214.1	17.2	1.3	.4	.6	4.4	6489
5	7320	98	2304	225.6	22.8	1.4	.4	.7	5.2	5980
6	7346	146	2495	263.8	23.5	1.3	.4	.6	6.3	5734
7	7387	129	3407	282.8	31.9	1.7	.5	1.0	10.9	5638
8	7507	125	3107	286.3	41.5	2.3	1.1	1.4	17.2	5776
9	7590	132	3192	309.0	63.9	3.1	1.2	1.2	33.5	6338
1940	7360	137	4030	323.2	50.4	6.2	.8	1.3	21.5	6148
1	7683	157	5153	367.1	26.5	7.7	.3	1.1	4.0	5311
2	7838	152	8081	537.1	11.1	.9	.1	.4	2.2	3943
3	7979	153	13028	999.8	4.9	1.0	.2	.2	.2	2625
4	8046	169	15345	1452	4.5	1.3	.1	.3	.2	2564
5	8084	191	10897	1849	5.9	3.0	.4	.2	.2	2112
6	8430	187	9996	1905	52.9	33.6	1.8	1.3	2.6	1656
7	8568	194	14674	1792	83.5	23.8	2.6	4.9	13.9	1617
8	8651	193	12967	1721	103.5	26.4	7.5	6.1	19.4	1984
9	8293	180	12160	1695	129.6	21.1	8.7	6.7	55.3	3105
1950	8936	146	10816	1697	199.1	12.8	5.8	5.7	128.6	4408
1	9066	182	15672	1653	149.5	14.9	3.1	5.5	87.8	4888
2	9180	194	15262	1650	193.6	22.2	3.5	5.4	104.2	5635
3	9152	193	15827	1667	82.4	16.6	4.3	5.5	27.3	4331
4	9223	182	15136	1670	92.1	16.7	4.7	5.5	33.1	4433
5	9313	187	15563	1660	110.6	15.8	5.2	5.2	29.6	4065
6	9445	189	19124	1622	156.9	19.0	5.6	5.7	44.6	6646
7	—	191	20989	1580	169.6	24.0	8.2	6.2	60.4	6282

exhibit **29** of chapter 7+ (continued)

C) FIRST SECTIONS (to 1919) of SECOND GROUP of SEQUENCES--see panel J for identification.

Year	D711 (#)	D772 (#)	J241 (in)	K77 $/mo	K78 $/mo	M178 (M1b)	M179 ($/b)	Q145 (#)	Q184 (Ht)	W74 (T#)	X113 (TM$)	X118 (M$)
1882	353	38	—	—	19	3.2	2.25	—	940	86	—	—
3	372	55	—	—	—	4.2	2.15	—	1102	102	—	—
4	341	50	—	—	19	4.0	2.10	—	840	115	—	—
85	486	67	—	—	19	4.2	1.95	—	481	77	—	—
6	1073	210	—	—	—	4.5	1.95	—	306	59	—	—
7	836	299	—	—	19	6.9	1.95	—	240	95	—	—
8	540	163	—	—	19	6.5	1.95	—	338	83	—	—
9	662	173	—	13h	19h	6.8	1.67	—	400	72	—	—
1890	1039	318	—	13h	19h	7.8	2.09	—	786	89	—	—
1	867	334	—	13h	20	8.2	2.13	293	1055	84	—	—
2	693	261	—	13h	20	8.7	2.91	376	606	82	—	—
3	783	257	104	14	20	8.0	1.96	299	371	90	—	—
4	865	206	114	12h	18h	8.4	1.73	324	287	93	—	—
95	810	217	93	12h	18h	8.7	1.60	170	268	111	—	—
6	547	297	101	—	—	9.5	1.57	181	396	144	2	96
7	680	193	95	—	—	11.0	1.61	222	219	162	2	87
8	645	236	86	13h	19	12.3	1.62	221	239	180	5	104
9	1014	471	114	14	20	15.8	1.43	239	688	214	8	98
1900	931	414	101	—	—	17.2	1.09	249	722	175	16	124
1	1413	1016	101	—	—	20.1	.99	282	830	173	10	143
2	1604	1051	92	15h	22	25.8	1.21	345	759	64	12	139
3	1778	1200	69	—	—	39.9	1.24	355	670	54	15	169
4	944	964	73	—	—	31.7	.88	441	514	55	11	170
05	942	800	64	—	—	40.1	.94	537	1193	49	8	225
6	—	—	69	18h	26	51.0	1.13	359	323	62	9	239
7	—	—	61	—	—	52.2	1.11	610	444	59	18	314
8	—	—	72	—	—	52.9	.85	381	709	76	13	202
9	—	—	75	22	28	66.7	.81	253	272	68	7	219
1910	—	—	67	21	28	77.8	.89	324	234	64	5	311
1	—	—	53	21h	28	79.5	.84	299	237	100	5	344
2	—	—	73	22	29h	85.9	.81	283	231	134	6	374
3	—	—	76	22h	30	89.5	1.00	350	271	168	5	410
4	403	253	83	22h	29h	87.3	.93	232	150	171	7	458
15	770	312	72	22h	30	82.7	86	199	186	154	5	486
6	2036	721	78	25	33	95.4	1.10	239	376	174	4	561
7	2268	799	82	31	40h	91.3	1.35	301	525	150	15	1029
8	1869	584	83	37h	48h	71.3	1.60	471	883	121	154	2221
1919	2036	609	74	143	56	86.1	1.71	273	1778	152	91	2517

→

exhibit **29** of chapter 7+ (continued)

D) REMAINDER (from 1920) of SECOND GROUP of SEQUENCES

Year	D771 (#)	D772 (#)	J241 (in)	K77 $/mo	K78 $/mo	M178 (M1b)	M179 ($/b)	Q145 (#)	Q184 (Ht)	W74 (T#)	X113 (TM$)	X118 (M$)
1920	2038	622	90	51	65	97.1	2.02	229	2080	248	26	4540
1	1501	373	100	33h	44h	96.0	1.89	205	1507	327	40	3598
2	563	208	10	33	43h	118.6	1.76	203	570	161	16	1805
3	721	308	71	37h	47h	137.2	1.90	143	131	193	30	2218
4	537	244	79	38	49	147.5	1.81	153	32	267	18	1644
25	537	219	71	38h	49	159.0	1.77	176	56	282	18	1887
6	478	206	72	395	50	164.2	1.71	152	50	260	24	2157
7	273	240	83	39h	50	174.0	1.62	88	66	239	23	2227
8	222	226	73	39h	50	178.1	1.57	91	114	318	27	3364
9	373	382	50	40	51	172.0	1.48	114	128	290	38	3628
1930	284	207	69	37h	48	160.8	1.44	61	186	271	30	2891
1	447	221	90	28h	38	128.4	1.11	46	266	294	45	2358
2	560	162	93	20h	29	81.4	1.01	27	521	294	43	2513
3	926	533	88	18	25h	64.8	1.33	51	259	241	56	1516
4	717	835	82	20	28	76.6	1.54	38	9	292	174	848
35	760	945	80	22	30h	76.2	1.51	30	19	386	82	620
6	256	1083	65	24	32h	114.6	1.51	41	7	456	114	751
7	1410	2728	76	27h	36h	115.7	1.48	34	—	514	67	776
8	776	1385	61	27	36	108.2	1.48	81	—	503	60	601
9	699	1411	75	27	36	125.1	1.47	40	—	559	79	632
1940	753	1243	78	27h	37h	132.9	1.36	83	—	614	82	598
1	1535	2138	65	34h	49h	170.4	1.47	48	—	649	75	637
2	1423	943	59	45h	59	187.8	1.53	122	—	373	184	128
3	1906	565	59	59	77	129.5	1.57	278	—	223	803	626
4	2146	808	73	71	91	95.6	1.59	267	—	291	1951	785
45	1956	946	84	79	101	107.8	1.63	156	—	351	2438	866
6	2238	1617	82	86	108	172.1	1.72	128	—	278	1341	1073
7	1707	1102	77	92	117	190.4	1.90	79	—	210	136	1164
8	1737	780	89	99	124	207.7	2.18	59	—	397	218	1353
9	1682	781	74	99	121	209.3	2.30	37	—	445	230	1386
1950	2559	919	102	99	121	231.0	2.35	180	—	472	380	1680
1	2102	888	89	113	137	244.6	2.54	150	—	416	633	2078
2	2447	839	69	119	146	254.8	2.54	24	—	296	612	2453
3	2825	745	92	122	151	264.3	2.67	49	—	271	394	2603
4	1726	588	86	120	151	278.4	2.76	30	—	254	589	2478
55	2154	844	81	123	154	310.2	2.86	24	—	271	541	2826
6	1821	744	79	128	161	325.6	3.05	57	—	298	553	3460
7	1730	751	72	133	168	—	—	15	—	236	362	5248

→

exhibit **29** of chapter 7+ (continued)

I) IDENTIFICATION of FIRST GROUP of SEQUENCES

C––(see below)

H541 Membership of Methodist Church (in thousands) (pp. 228–229)

Q167 Gross tonnage of documented vessels engaged in whale fisheries (in thousands of tons) (pp. 444–445)

U1 Value of U.S. Exports (gold, silver, and merchandise) (in millions of dollars) (pp. 537–538). [series U10 used for 1790–1820.]

W––(see below)

Y369 Per capita public debt (in dollars) (pp. 720–721) [calculated from Y368 and A2 for 1791–1850.]

C89 Total Immigrants (in thousands per year) (pp. 56- 57)

C90 Immigrants (in thousands per year) from Great Britain (pp. 56–57)

C91 Immigrants (in thousands per year) from Ireland (pp. 56–57)

C92 Immigrants (in thousands per year) from Scandinavia (pp. 56–57)

C94 Immigrants (in thousands per year) from Germany (pp. 56–57)

W76 Patents issued to residents of foreign countries (pp. 607–608)

J) IDENTIFICATION of SECOND GROUP of SEQUENCES (units are those used above)

D771 Number of work stoppages with major issue "wages and hours" (page 99)

D772 Number of work stoppages with major issue "union organization" (page 99)

J241 Annual total precipitation (in inches as rain) at the Weather Bureau Office, Tatoosh Island, Washington.

K77 Average wages (in dollars per month) for farm workers; when board included (pp. 280–281).

K78 Average wages (in dollars per month) for farm workers: when board NOT included (pp. 280–281).

M178 Shipments of cement (in millions of barrels) (page 364)

M179 Average values of same (in dollars per barrel) (page 364)

Q145 Number of passengers killed on railroad trains (page 437)

Q184 Merchant vessels built and documented on the New England Coast (in thousands of tons) (pp. 448–449)

W74 Number of design patents granted (in tens of patents) (pp. 607–608)

X113 U.S. Government deposits in all commercial banks (in tens of millions of dollars) (page 632)

X118 Other liabilities of all commercial banks (in millions of dollars) (page 632)

→

suspended bank counts

While banks come in different sizes, the data we have been using on deposits in suspended banks is not very far from being counted data. And we can have real counted data by going to the same sources and finding the **number** of banks that suspended payment in each year. Exhibit **30** has the analysis when the started logs are started by one count. Exhibit **31** shows the resulting picture, with the rough supporting the smooth. Clearly we have shown the picture for bank suspensions at least as clearly by using COUNTS of suspending banks as we did earlier using AMOUNTS of suspended deposits (in millions of dollars).

review questions

In what four places in the use of smoothing might choice of expression have made a difference? Which of these matter? When? How may residual behavior guide our choice of expression? How many general questions do we expect to ask about the rough? How can choice of expression simplify answers to these questions? How do we compare one choice of expression with another? What do we gain/lose by expressing the smooth with a choice different from the one that makes the rough simply behaved? What about smoothing sequences of counts? What did we take as an example? How is it related to an earlier example using amounts? What do we think is the moral?

exhibit **29** of chapter 7⁺ (continued)

P) PROBLEMS

29a/b/c/d/e/f/g/h/i/j) Smooth and examine rough for re-expression, starting from H541/A167/U1(10)/Y369/C89/C90/C91/C92/C94/W76.

29k/l/m/n/o/p/q/r/s/t/u/v) Do the same, starting with D771/D772/J241/K77/K78/M178/M179/Q145/Q184/W74/X113/X118.

29w) Do the same for some expression of D771 divided by D772.

29x) Take the difference of K78 and K77 as indicating the value of board (as provided to farm workers) and analyze it.

29pq2) Plot the rough of M178 against the rough of M179; discuss.

29y, on) In the earlier, 1945, version of the source, such sequences as A109, E113, and E114 are a few of those that could be considered.

S) SOURCE

Historical Statistics of the United States, Colonial times to 1957. Washington, 1960. Column headings are the labels used in the source.

exhibit **30** of chapter 7⁺: suspended banks

Split smooths of log (1 + number of banks suspending payment)

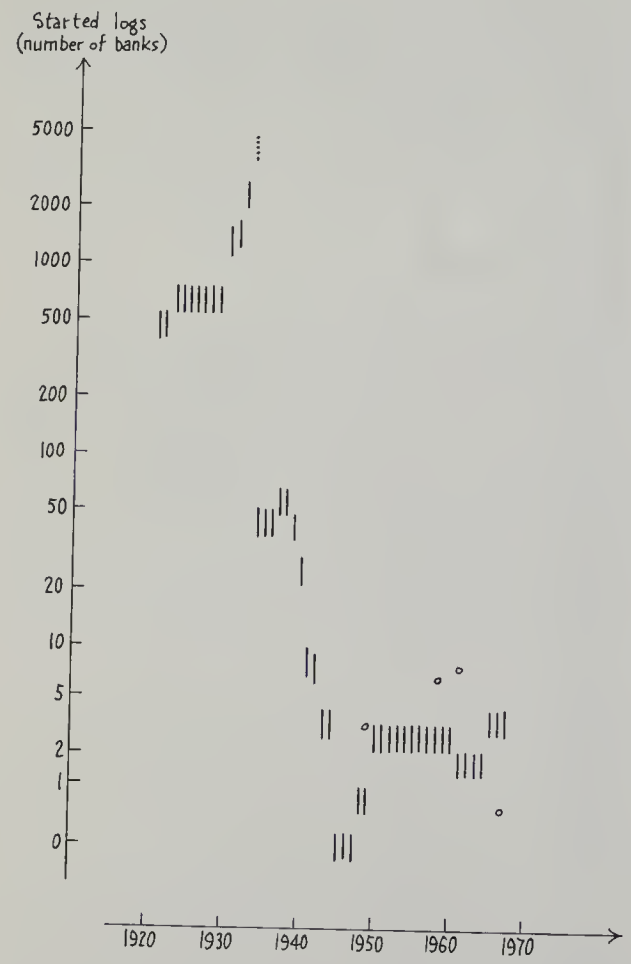
banks suspending							same split	
year	#	#+1	log (#+1)	3R split	rough	out-side	double split	rough
1921	505	506	270	270	Z			
22	367	368	257	270	−13			
23	648	649	281	281	Z			
24	776	777	289	281	8			
25	618	619	279	289, 283	−4			
26	976	977	299	283	16			
27	669	670	283	283	Z			
28	499	500	270	282	−12			
29	659	660	282	282	Z		282	
30	1352	1353	313	313	Z		313	Z
31	2294	2295	336	316	20		316	20
32	1456	1457	316	336	−20		316	Z
33	4004	4005	360	360	Z		360	
34	57	58	176	165	11			
35	34	35	154	165	−11			
36	44	45	165	165	Z			
37	59	60	178	175	3			
38	55	56	175	175	Z			
39	42	43	163	163	Z			
40	22	23	136	136	Z			
41	8	9	95	100	−5			
42	9	10	100	95	5			
43	4	5	70	70	Z			
44	1	2	30	30	Z			
45	0	1	0	0	Z			
46	0	1	0	0	Z			
47	1	2	30	0	30			
48	0	1	0	30	−30			
49	4	5	70	30	40	f		
50	1	2	30	60	−30			
51	3	4	60	60	Z			
52	3	4	60	60	Z			
53	4	5	70	60	10			
54	3	4	60	70, 60	Z			
55	4	5	70	60	10			
56	3	4	60	60	Z			
57	3	4	60	60	Z			
58	8	9	95	60	35	f		
59	3	4	60	60	Z			
60	2	3	48	60	−12			
61	9	10	100	48	52	f		
62	2	3	48	48	Z			
63	2	3	48	48	Z			
64	8	9	95	90	5			
65	7	8	90	90	Z			
66	1	2	30	70	−40	f		
67	4	5	70	70	Z			

S) SOURCE

The World Almanac 1969/ 144(1929–67), 1954/753(26–52), 1931/306(21–29), 1937/ 288(1921–1935)

exhibit **31** of chapter 7⁺: suspended banks

Number of banks suspending payments in the U.S., 1921–1967 (vertical scale = started log, start = 1 bank; see exhibit 30 for numbers).



7K. A 2-section example

Let us turn back to suspended deposits as given in exhibit 8. We need something like logs to make the sizes of residuals about the same. Exhibit **32** lists the 3R smooth of the logs, and its rough, and gives the stem-and-leaf and modified letter values--both for residuals and for their **sizes**--doing this first for all years and then for two groups of years. Clearly, the residuals are bigger for the second group of years, the comparison being

	*-Spreads of residuals		*-Letter values of sizes	
	*Hspr	*Espr	*M	*H
1921–44	20	57	8	26h
1948–67	60	140	33	98

The apparent ratio of sizes varies from about 3:1 to about 4:1.

exhibit **32** of chapter 7+: suspended deposits

The 3R smooth of the logs of suspended deposits and the corresponding residuals (deposits in 100 log of millions of dollars)

A) The SMOOTH and the ROUGH

Date	Susp. deposits	3R	Rough	Size	f or F
1921	229	228	1		
2	205	228	−21	23	
3	228	228	Z		
4	233	228	5	5	
25	224	233, 229	−5	5	
6	243	229	14	14	
7	229	229	Z		
8	214	229	−15	15	
9	236	236	Z		
30	293	293	Z		
1	323	293	30	30	
2	285	323, 293	−8	8	
3	356	285	71	71	f
4	157	157	Z		
35	100	104	−4	4	
6	104	104	Z		
7	104	104	Z		
8	128	111	17	17	
9	111	128, 111	Z		
40	153	111	42	42	
1	77	77	Z		
2	57	57	Z		
3	23	57	−34	34	
4	79	23	56	56	

→

exhibit **32** of chapter 7⁺ (continued)

(Panel A continued)

Date	Susp. deposits	3R	Rough	Size	f or F
45	−40	−40	Z		
6	L	L	?		?
7	L	L	?		?
8	−77	L	H		F
9	L	−140	L		F
50	−140	−140	Z		
1	49	15	34	34	
2	15	49,46	−31	31	
3	165	46,49	116	116	F
4	46	81	−35	35	
55	81	81	Z		
6	108	108	Z		
7	111	108	3	3	
8	80	80	Z		
9	30	80	−50	50	
60	90	88	2	2	
1	88	88	Z		
2	8	88	−80	80	f
3	137	134	3	3	
4	134	137,134	Z		
65	166	134	32	31	
6	−15	107	−122	122	F
7	107	107	Z		

L = "lower still" entered as the log of zero
F = "f" for analysis of panel E

B) STEM-and-LEAF for ALL residuals--?'s not counted

```
 3   H | (71, H, 116)((?))
 4   5 | 6
 5   4 | 2
 8   3 | 041
     2 |
10   1 | 47
14   0 | 9322
19   Z | 0000000000000000000
12  −0 | 4584
 8  −1 | 5
    −2 |
 7  −3 | 415
    −4 |
 4  −5 | 0
 3   L | (L, 3, −80, −122)((?))
```

→

exhibit **32** of chapter 7⁺ (continued)

C) *LETTER VALUES WITHOUT and WITH ?'s for ALL residuals

*#36	(zeros/2, no?)		
*M18h		Z	
*H 9h	15h		−11h
*E 5	42		−35

		40	
*f	55h		−51h
*F	95h		−91h

outside: 71, −80
far out: h, 116, −122, L

*#38	(zeros/2, ?in)			
*M19h†		Z		
*H10†	17		−15	32
*E 5h†	49		−42h	91h

		48	
*f	65		−63
*F	113		−111

outside: 71, −80
far out: (?), H, 116, −122, L, (?)

† **Since counts in (B) omit ?'s, reduce by one unit before using (B)'s counts.**

D) STEM-and-LEAF and *-LETTER VALUE for SIZES of ALL RESIDUALS

8	**H**	(71, 116, H, 80, 122, L, ?, ?)
10	**5**	06
11	**4**	2
16	**3**	01245
19	**1**	457
27	**0**	12334558
18	**z**	000000000000000000

*-median = 11 (at depth 19h, since * = 38h)

E) STEM-and-LEAF and LETTER-VALUE for RESIDUALS for 1921–45 and 1948–67

1921–45			1948–67		
1	**H**	(71)	2	**H**	(H, 116)
2	**5**	6		**5**	
3	**4**	2		**4**	
4	**3**	0	4	**3**	42
	2			**2**	
6	**1**	47		**1**	. . .
8	**0**	15	7	**0**	323
11	**z**	00000000000	7	**z**	000000
6	**−0**	458		**−0**	
3	**−1**	5		**−1**	
2	**−2**	3		**−2**	
1	**−3**	4	7	**−3**	15
	−4			**−4**	
	−5		4	**−5**	0
	L		3	**L**	(L, −80, −122)

→

exhibit **32** of chapter 7⁺ (continued)

*#19h	(zeros/2)			
*M10		Z		
*H 5h	15h		−4h	20
*E 3	42		−15h	57

	30		
*f	45h		−34h
*F	75h		−64h

outside: 56, 71
far out: none

*#16h	(zeros/2)			
*M 8h		Z		
*H 4h	17h		−42h	60
*E 2h	75		−65	140

	90		
*f	107h		−132h
*F	197h		−222h

outside: 116
far out: H, L

F) STEM-and-LEAF and *-LETTER VALUES for SIZES of residuals in 1921–44 and 1948–67

1921–45			1948–67		
1	**H**	(71)	5	**H**	(H, 116, 80, 122, L)
2	**5**	6	6	**5**	0
3	**4**	2		**4**	
5	**3**	04	10	**3**	1245
6	**2**	3		**2**	
9	**1**	457		**1**	
14	**0**	14558	13	**0**	233
12	**Z**	00000000000	7	**Z**	0000000
*median = 8 (at depth 10)			*median = 33 (at depth 9)		
*H = 26h (at depth 5h)			*H = 98 (at depth 5)		

P) PROBLEMS

32a) Make stem-and-leaf displays, and find *M and *H for the sizes of residuals for periods 1921–33 and 1934–45. How do they compare with each other and with 1948–67? How many groups seem to be needed? What are they?

32b) (Triple value) Repeat Panels (A), (B), and (F) for started logs, using 100 log(1 + suspended deposits in millions).

32c) (After (b)) Do (32a) for this new analysis.

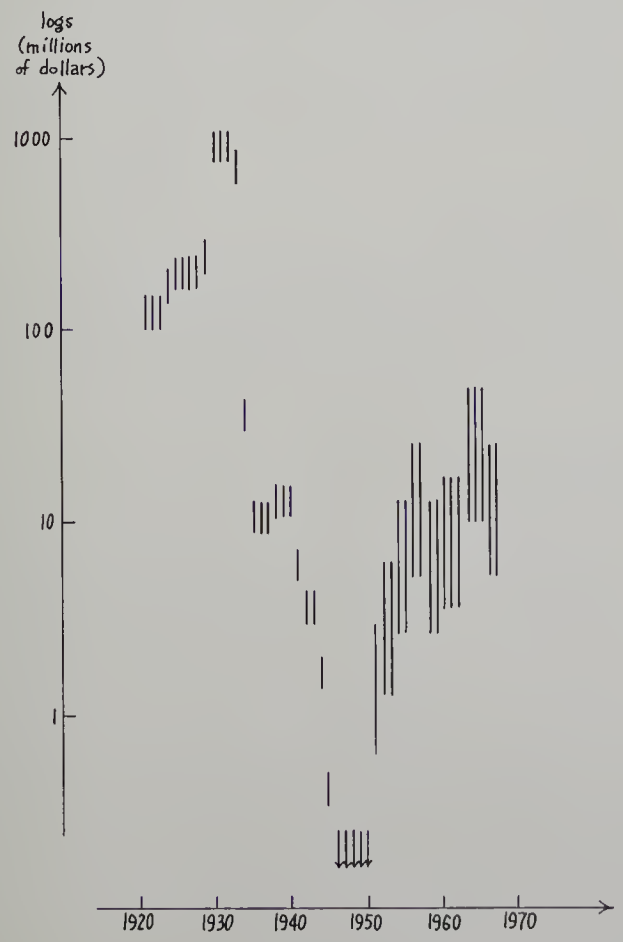
Exhibit **33** shows the corresponding plot, with the rough supporting the smooth. Here we have used two lengths of bars, one for each group of years. The result is quite effective.

review questions

What did we take as an example? How many sections seemed needed? How did the sizes behave?

exhibit **33** of chapter 7⁺: suspended deposits

Deposits in banks suspending payment, U.S., 1921–1967 (rough supporting smooth: two median sizes, one up to 1945, one from 1968 on, 5 values off scale)



7M. How much more may we have learned?

We have extended what we knew at the end of chapter 7 in a variety of ways. We are now ready:

- ◇ to split a sequence whenever this is clearly indicated.
- ◇ to think hard about whether or not to split in a specific instance.
- ◇ to use our usual techniques for looking at residuals to judge which expression of the data yields the more reasonable smooth.
- ◇ to do all this equally freely for counted data.
- ◇ to assess size of rough and thence calculate a blurring, separately in two or more sections of the sequence (two or more subsequences).
- ◇ to show blurrings involving two or more sizes of bar.

We now understand more clearly:

- ◇ that re-expression does alter the smooth, but not by a tremendous amount, so that not bothering to re-express is often sensible.
- ◇ which steps in the smoothing are most subject to alteration by re-expression.

Parallel and wandering schematic plots

8

chapter index on next page

Now that we are able to smooth sequences with moderate ease and effectiveness, we have the essential tool for extending to batches of points in the plane--to (x, y) pairs--some of the techniques we used to summarize batches of numbers. To do this, we will need to replace a number for a median by something that tells us about the middle of one variable, say y, for different values of the other, here x. Similarly, we will have to replace numbers by curves--by "traces" as we will say--in the roles played by hinges and (inner) fences. The ideas of outside and far-out values will clearly go over with little change to outside and far out points, once we have fences. The idea of adjacent points will need a small amount of thought, but will in fact go over quite easily.

This chapter will close with the simplest analog for (x, y) points of a schematic plot for values, which we will call a **wandering schematic plot.** In chapter 9, then, we can go on to other analogs, for (x, y) pairs, of schematic plots for values.

review questions

What will we be up to in this chapter? What tool are we now ready to use? What will we need to replace median with? Hinges? Fences? Outside and far-out values? Adjacent values? How far will we get in this chapter? How is this chapter related to the next?

8A. Parallel schematic plots

One of the simplest things we can do to look at a collection of (x, y) pairs is:

- ◇slice the data up according to its x-values.
- ◇find a schematic plot for the y-values in each slice.
- ◇set these plots up side by side.

Exhibit **1** sets out, in two different sorted ways, the paired usages of electricity and gas during a winter for 152 townhouses at Twin Rivers, New Jersey. The data was collected as part of a study of energy consumption, in which differences in energy use were to be compared with structural aspects of the individual buildings and behavioral aspects of their inhabitants. Panels A and B give the data arranged in order, once by increasing electrical use and once by increasing gas use. This prepares us to start from either coordinate.

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exhibit of **1** of chapter 8: Twin Rivers

The 152 (ElUse, GaUse) pairs (winter energy consumption in therms; ElUse = therms of electricity; GaUse = therms of gas; 1 therm approximately 30 kilowatt hours and exactly 100,000 btu)

A) The DATA ordered by ElUse--(ElUse; GaUse) pairs

7 (89; 424) (108; 388) (119; 778) 129; 780) (134; 832, 1016) 143; 988)
14 (152; 1186) (153; 612) (156; 668) (157; 1108) (159; 752) (163; 676) (167; 968)
21 (170; 854) * (171; 670) (175; 530) (177; 810, 766) (178; 1008) (180; 958)
28 (182; 686) (186; 986) (188; 824) (189; 802) (190; 1014, 1016) (191; 1078)
34 (192; 978, 568) * (193; 1144) (195; 838) (196; 1298) (198; 612)
41 (204; 584, 908) (206; 1102) (207; 756, 810) (209; 1028) (211; 610)
49 (214; 1044) (217; 1128, 922, 980) * (218; 830, 812) (224; 1038, 1022)
56 (225; 832) (227; 938) (229; 742) (231; 1152) (233; 844, 1030) (234; 790)
62 (235; 886, 1064) (238; 860) (239; 922) * (240; 912) (241; 892)
71 (242; 1068, 860) (243; 756) (244; 1068, 788) (245; 658, 876) (246; 992, 774)
9 (247; 700) (252; 1058, 1020, 1184, 1026) * (254; 1050) (255; 840, 912, 862)
72 (256; 1060) (257; 1018) (258; 870, 824, 840, 1062) (259; 1062) (262; 812)
64 (263; 1000) (264; 922) (265; 982) * (266; 684, 816, 754) (267; 748) (268; 956)
56 (270; 928, 746) (271; 1068, 834) (272; 1086) (273; 988, 824 (274; 940)
48 (275; 1074, 958) (276; 686) * (277; 868, 852) (279; 1158) (280; 754, 782)
40 (281; 958) (283; 796) (284; 1030) (285; 1012) (287; 1070, 788, 1096)
33 (288; 654) (290; 892) (291; 1004) * (292; 798, 766) (293; 942)
27 (294; 926, 1120) (295; 882) (297; 1144) (298; 814) (299; 1214) (300; 766)
20 (301; 974) (302; 924) (305; 1064) (306; 900) (315; 956) * (316; 828)
14 (319; 1024) (323; 892) (324; 932) (325; 662) (333; 1150) (334; 946)
8 (347; 732) (348; 956) (352; 720) (367; 762) (371; 1088) (422; 1222)
2 (429; 1002) (435; 860)

Note: (134; 832, 1016) stands for (134; 832) and (134; 1016), etc.

B) The DATA ordered by GaUse--(GaUse; ElUse) pairs

6 (388; 108) (424; 89) (530; 175) (568; 192) (584; 204) 610; 211)
13 (612; 198, 153) (654; 288) (658; 245) (662; 325) (668; 156) (670; 171)
20 (676; 163) (684; 266) (686; 276, 182) (700; 247) (720; 352) (732; 347)
28 (742; 229) (746; 270) (748; 267) (752; 159) (754; 280, 266) (756; 243, 207)
36 (762; 367) (766; 300, 292, 177) (775; 246) (778; 119) (780; 129) (782; 280)
44 (788; 287, 244) (790; 224) (796; 283) (798; 292) (802; 189) (810; 177, 207)
52 (812; 218, 262) (814; 298) (816; 266) (824; 188, 258, 273) (828; 316)
60 (830; 218) (832; 225, 134) (834; 271) (838; 195) (840; 255, 258) (844; 233)
68 (852; 277) (854; 170) (860; 238, 435, 242) (862; 255) (868; 277) (870; 258)
76 (876; 245) (882; 295) (886; 235) (892; 290, 323, 241) (900; 306) (908; 204)
76 (912; 240, 255) (922; 217, 239, 264) (924; 302) (926; 294) (928; 270)
68 (932; 324) (938; 227) (940; 274) (942; 293) (946; 334) (956; 348, 315, 268)
60 (958; 281, 180, 275) (968; 167) (974; 301) (978; 192) (980; 217) (982; 265)
52 (986; 186) (988; 273, 143) (992; 246) (1000; 263) (1002; 429) (1004; 291)
45 (1008; 178) (1012; 285) (1014; 190) (1016; 190, 134) (1018; 257) (1020; 252)
38 (1022; 224) (1024; 319) (1026; 252) (1028; 209) (1030; 284, 233) (1038; 224)
31 (1044; 214) (1050; 254) (1058; 252) (1060; 256) (1062; 259, 258)
25 (1064; 305, 235) (1068; 244, 271, 242) (1070; 287) (1074; 275) (1078; 191)
17 (1086; 272) (1088; 371) (1096; 287) (1102; 206) (1108; 157) (1120; 294)
11 (1128; 217) (1144; 297, 193) (1150; 333) (1152; 231) (1158; 219) (1184; 252)
4 (1186; 152) (1214; 299) (1222; 422) 1298; 196)

Note: (612; 153, 198) stands for (612; 153) and (612; 198), etc. →

exhibit **1** of chapter 8 (continued)

P) PROBLEMS––suitable for sharing among a group.

1a) Insert *'s in panel B to make batches of 15 and 16.

1b/c/d/e/f/g/h/i/j/k) Find the *y*-letter values for the corresponding slice thus formed.

1l) Prepare a table like exhibit 2 for this case.

1m) Prepare a picture like exhibit 3 for this case.

1n) Prepare a picture like exhibit 4 for this case.

S) SOURCE

Personal communication from Professor R. H. Socolow. (Data from a Princeton University School of Engineering study of energy use and energy saving.)

exhibit **2** of chapter 8: Twin Rivers

Letter values and (inner) fences for the GaUse batches defined by the *'s in exhibit 1

#	x = ElUse		y = GaUse							ElUse
	min	max	lo out	lo adj	lo H	M	hi H	hi adj	hi out	med
15	89	to 170	×××	388	672	780	978	1186	×××	152
15	171	to 192	×××	530	726	824	997	1078	×××	186
15	193	to 217	×××	584	783	922	1073	1298	×××	207
15	218	to 239	×××	742	831	886	1026	1152	×××	231
16	240	to 252	×××	658	781	902	1042	1184	×××	245
15	254	to 265	×××	812	851	922	1034	1062	×××	258
16	266	to 276	×××	684	751	881	973	1086	×××	271
15	277	to 291	×××	654	792	892	1021	1158	×××	284
15	292	to 315	××	766	848	926	1019	1214	×××	298
15	316	to 435	×××	662	795	932	1013	1222	×××	347

P) PROBLEMS

2a/b/c/d/e) Write out the stem-and-leaf for the *y*-values (GaUse) in the 1st/2nd/3rd/4th/5th slice and verify the letter values above.

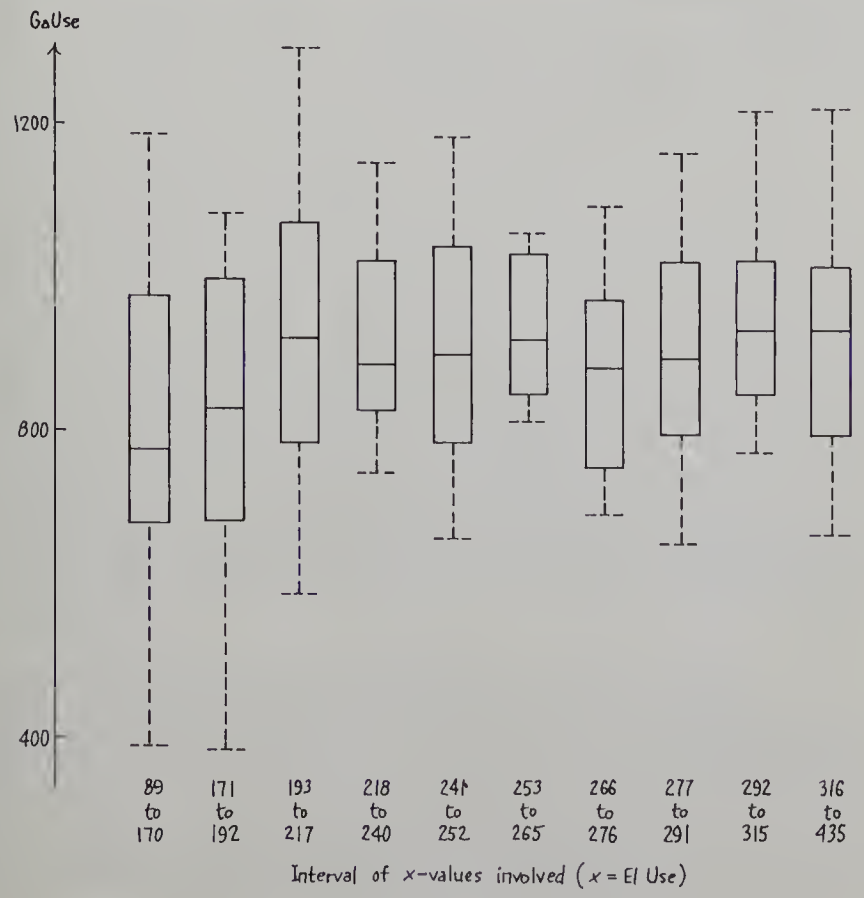
2f/g/h/i/j) Do the same for the *x*-values (ElUse) of the 6th/7th/8th/9th/10th slice.

In panel A of exhibit 1, the asterisks divide the 152 points into ten slices, each of 15 or 16 points. When we calculate letter values for y in each slice separately, we obtain the results set down in exhibit **2**. Plotted as a set of parallel schematic plots, we get the picture of exhibit **3**.

The fact that we have to write in numbers for the x-intervals defining the slices shows that we are not yet fully graphic. Exhibit **4** centers each schematic plot at the medians of the slice it represents, and shows, as light dotted lines, the cuts that define the slices. It also provides a crossing of line segments where the x-median and y-median of each slice cross each other. A hard look at this particular picture shows us one of its most important weaknesses: a lot of detail near the median of x, where nothing very interesting is likely to happen; little detail for extreme x, where something interesting is much more likely.

exhibit **3** of chapter 8: Twin Rivers

Parallel schematic plots (regrettably equally spaced)



While a first thought might be to go to slices of equal width in x, a little thought shows that this would tend to make the number of points in the end slices impractically small. A reasonable compromise is often to use

letter cuts

to:

◇start with the letter values of x, moving them a shade inward where necessary to avoid any value exactly on a cut. (The median may well be moved outward a shade in both directions.)

Panel C of exhibit **5** shows the letter values and letter cuts for x, going out as far as B on each side. We are left with 4 slices of count 3 or less. Were we interested only in medians, slices of 3 would serve us, perhaps well. If we want hinges as well, we need at least 5 values. Accordingly, panel D shows the mark-up using C, D, . . . , D, C.

exhibit **4** of chapter 8: Twin Rivers

Parallel schematic plots (spaced to fit *x*-medians)

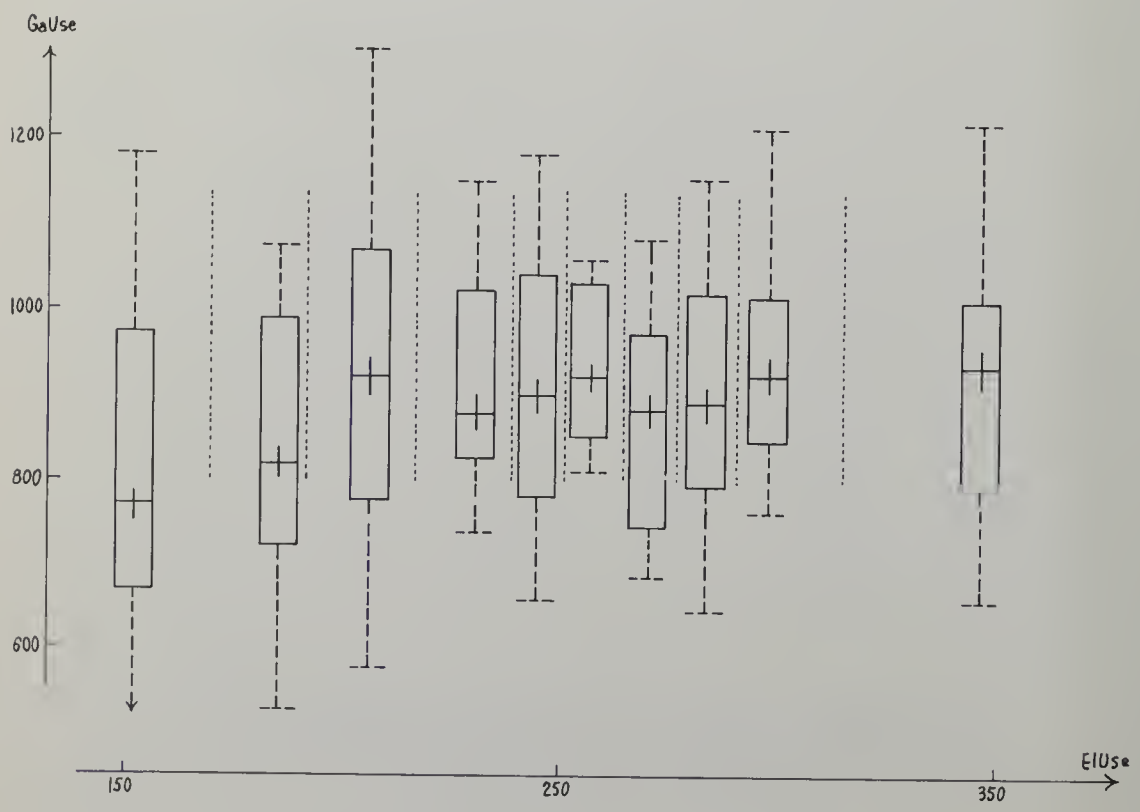


exhibit **5** of chapter 8: Twin Rivers

Continuation of exhibit 1

C) LETTER VALUES and LETTER CUTS

#152		letter values		letter cuts			slice counts			
M	76h	253		M	252h	253h	HM	37	38	MH
H	38h	207	283h	H	207h	283h	EH	20	19	HE
E	19h	177h	301h	E	177h	301h	DE	9	9	ED
D	10	156	333	D	156h	332h	CD	4	5	DC
C	5h	134	359h	C	134h	359h	BC	3	2	CB
B	3	119	422	B	119h	421h	–B	3	3	B–

Note that (1) each letter cut is at an h, the median having been split and moved outward if needed to reach an h, the other letter points having been moved inward when needed to reach an h (thus 177h and 301h were not changed). (2) Each piece has been tagged with the two letters, in order, of the cuts that define it––though we will often just use serial numbers. Note also that it would be a very helpful option to move the hiC cut inward one value to obtain at least 3 values in slice #11 (CB).

D) Panel A MARKED UP according to PANEL C––only to C's, with loD moved 1 in.

7 (89; 424) (108; 388) (119; 778) (129; 780) (134; 832, 1016) **C** (143; 988)
13 (152; 1186) (153; 612) (156; 668) (157; 1108) **D** (159; 752) (163; 676)
20 (167; 968) (170; 854) (171; 670) (175; 530) (177; 810, 766) **E** (178; 1008)
27 (180; 958) (182; 686) (186; 986) (188; 824) (189; 802) (190; 1014, 1016)
34 (191; 1078) (192; 978, 568) (193; 1144) (195; 838) (196; 1298) (198; 612)
41 (204; 584, 908) (206; 1102) (207; 756, 810) **H** (209; 1028) (211; 610)
50 (214; 1044) (217; 1128, 922, 980) (218; 830, 812) (224; 1038, 1022) (225; 832)
56 (227; 938) (229; 742 (231; 1152) (233; 844, 1030) (234; 790)
64 (235; 886, 1064) (238; 860) (239; 922) (240; 912) (241; 892) (242; 1068, 860)
72 (243; 756) (244; 1068, 788) (245; 658, 876) (246; 992, 774) (247; 700)
9 (252; 1058, 1020, 1184, 1026) **M** (254; 1050) (255; 840, 912, 862) (256; 1060)
71 (257; 1018) (258; 970, 824, 840, 1062 (259; 1062 (262; 812) (263; 1000)
63 (264; 922) (265; 982) (266; 684, 816, 754) (267; 748) (268; 956)
56 (270; 928, 746) (271; 1068, 834) (272; 1086) (273; 988, 824) (274; 940)
48 (275; 1074, 958) (276; 686) (277; 868, 852) (279; 1158) (280; 754, 782)
40 (281; 958) (283; 796) **H** (284; 1030) (285; 1012) (287; 1070, 788, 1096)
33 (288; 654) (290; 892) (291; 1004) (292; 798, 766) (293; 942) (294; 926, 1120)
25 (295; 882) (297; 1144) (298; 814) (299; 1214) (300; 766) (301; 974) **E**
19 (302; 924) (305; 1064) (306; 900) (315; 956) (316; 828) (319; 1024)
13 (323; 892) (324; 932) (325; 662) **D** (333; 1150) (334; 946) (347; 732)
7 (348; 956) (352; 720) **C** (367; 762) (371; 1088) (422; 1222) (429; 1002)
1 (435; 860)

P) PROBLEMS––suitable for sharing among a group

5a) Insert letter cuts in panel B, moving them as desirable.

5b/c/d/e/f/g/h/i/j/k) Find the *y*-letter values for the corresponding slice thus formed.

5l) Prepare a table like exhibit 6 for this case.

5m) Prepare a picture like exhibit 7 for this case.

The y-letter values for these slices are as in exhibit **6** giving the picture of exhibit **7**. We have now paid the price of exploring the tails in more detail. Because we have 3 batches of the minimum size--5--for sensible schematic plots, we expected greater irregularity. And we received it. On the other hand, instead of having the extreme schematic plots coming in toward the middle of the x values as far as 170 for the low extreme slice and as far as 316 for the high one, the extreme slices now before us extend to 134 below and 367 above. Clearly, we are looking near the extremes in more detail.

Whichever choice we make--exhibit 4, exhibit 7, or the result of some other cutting--we now need some smoothing, to which we will turn in the next section.

exhibit **6** of chapter 8: Twin Rivers

Letter values (HMH) and adjacent values for the batches defined by the letter cuts of exhibit 5

A) CALCULATIONS

	ElUse		GaUse							ElUse
	min	max	lo out	lo adj	lo H	M	hi H	hi adj	hi out	median
6	89	to 134	×××	388	424	779	832	1016	×××	124
5	143	to 157	×××	612	668	988	1108	1186	×××	153
8	159	to 177	×××	530	673	759	832	968	×××	170h
20	178	to 207	×××	568	779	933	1015	1293	×××	192
37	209	to 252	×××	610	830	922	1030	1184	×××	235
38	254	to 283	×××	684	816	891	1000	1158	×××	267h
19	284	to 301	×××	654	806	942	1050	1214	×××	292
9	302	to 325	one	828	892	924	956	1024	one	316
5	333	to 352	×××	720	732	946	956	1150	×××	347
5	367	to 435	×××	762	860	1002	1088	1222	×××	422

The values outside are 662 and 1064. The H-spreads are, in order of appearance; 408, 440, 159, 236, 200, 184, 244, 64, 224, 228. The outside values are associated with the strikingly low H-spread of 64.

P) PROBLEMS

6a) Divide the pairs of panel A of exhibit 1 into slices of successive sizes 3, 5, 9, 17, 33, 18, 33, 17, 9, 5, 3.

6b/c/d/e/f/g/h/i/j/k (uses 6a)) Find as many analogs of the values in Panel A above for the 1st/2nd/3rd/.../10th slice as you can.

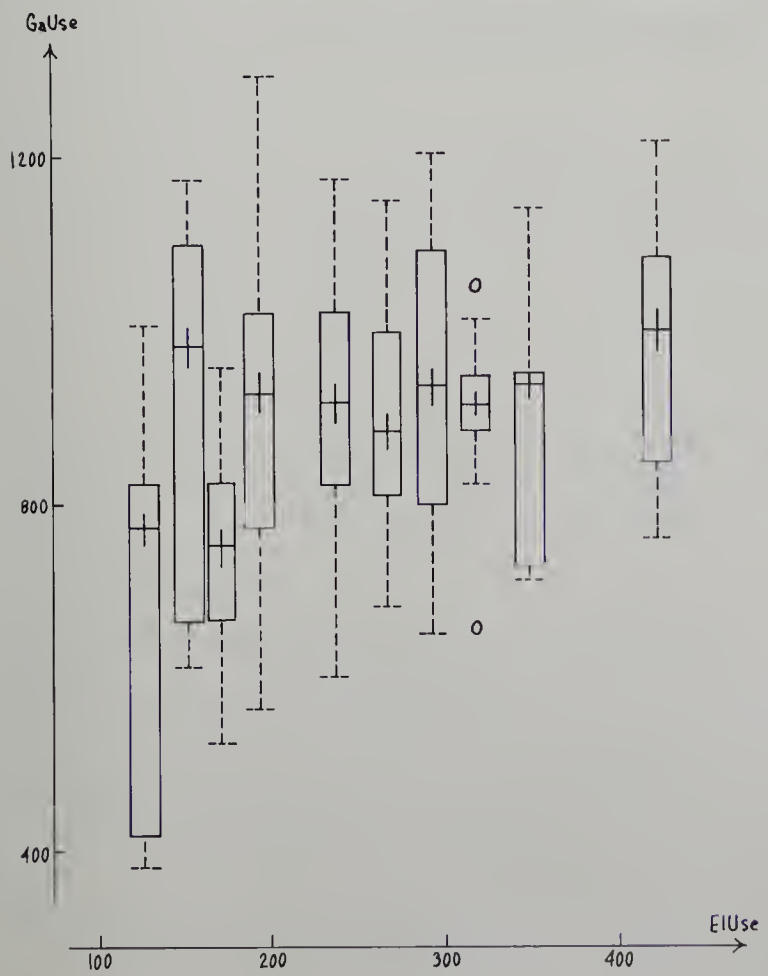
6l (uses all earlier)) Combine these results in an analog of exhibit 7.

review questions

What is one of the simplest things we can do with (x, y) pairs? What are its three parts? What example did we try this on? What was the result? (Try to give yourself a description in words!) How did we make this more graphic? What seemed to be a major weakness? What did we decide to do about this? What are letter cuts? How did we use them? What was the result? What do we now need?

exhibit **7** of chapter 8: Twin Rivers

Parallel schematic plots (using letter cuts)



8B. Smoothing the broken median

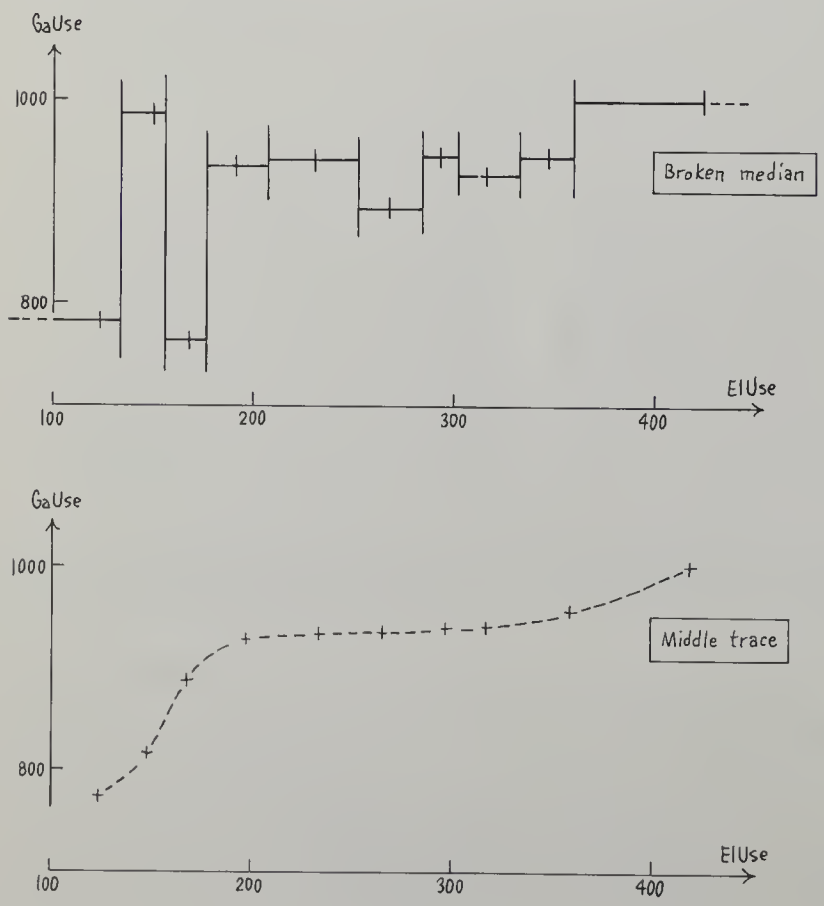
If we want to summarize the "middle" of gas use as a function of electrical use, we can put together slice medians and letter cuts from exhibits 5 and 6, as shown in the top panel of exhibit **8**. While it is quite easy to say how we got this

broken median

it is also easy to see that most of what we have is "wiggle" and that our eye is distracted from any general trends or bends that may be present. Clearly we need to smooth something.

exhibit **8** of chapter 8: Twin Rivers

Broken median--and the corresponding middle trace



The easiest things to smooth are the

cross-medians

the places where an x-median crosses a y-median for the same slice. (These have been shown as + in exhibits 4, 7, and 8.) We want at least initially to smooth lightly at the ends. After all, any extreme slice may be trying to tell a special story. So we plan to copy extremes on, and do only 3R′SS, where the ′ in R′ reminds us that we are copying on.

Exhibit **9** has the essentials of the arithmetic; the lower part of exhibit 8 has the picture. Notice that, for reasons to be discussed in an optional section (8I), the sequence of x-medians is also smoothed. (Since they are in increasing order anyway, hanning will be the first--and only--part of the smooth that affects them.)

exhibit **9** of chapter 8: Twin Rivers

Smoothing the cross-medians of exhibit 6

A) SMOOTHING

Slice	ElUse		GaUse				
#	median	H	median	3R′	3R′SS	3R′SSH	3R′SSH3
1	124	124	779	779	T	779	T
2	153	150	988	779	h	815	h
3	170h	171	759	922	e	886	e
4	192	197h	933	922		922	
5	235	232h	922	922	s	922	s
6	267h	265h	891	922	a	922	a
7	292	292	942	924	m	928	m
8	316	318	924	942	e	938	e
9	347	358	946	946		959	
10	422	422	1002	1002		1002	

P) PROBLEMS

9a) Plot the broken median based on exhibits 1 and 2.

9b) Write out the smoothing for that case.

9c) Plot the middle trace based on this smoothing.

9d) Show the two broken medians (from exhibits 1 and 2, and from exhibits 5 and 6) in the same picture. Discuss.

9e) Show the two middle traces in the same picture. Discuss.

9f) Take the results of problems 6b to k and carry out the analog of panel A. Plot the resulting median trace.

We call the result of smoothing a broken median a

middle trace

whatever set of cuts was used to fix the slices and whatever smooth was used. It represents a good smooth idea of how the "middle" of y seems to behave as x changes. In it we take advantage of the ideas at hand: breaking into slices, medians, smoothing to get something that traces a possible middle--something reasonably smooth but not constrained to any such particular family of shapes as all quadratics, all exponentials, or all sine waves. We will feel free to also call it a

median trace

whenever the context makes this term more natural.

review questions

What is an easy way to summarize the median of y as a function of x? What is a broken median? How well does it serve us? What are its main deficiencies? What are we going to need to do? What are cross-medians? What are we going to do with them? Which section will discuss why we smooth x's as well as y's? What do we think about extreme slices? How does this affect our choice of smoother? What smoother do we choose? What happened when we tried it? What is a middle trace? What is a median trace?

8C. Smoothing broken hinges

We are ready to draw a broken median, and smooth the cross-medians to find a middle trace instead. Can we gain by doing something similar for hinges? Only if we are careful.

Exhibit **10** shows the broken hinges that result from taking letter cuts and y-hinges from exhibit 6--with x-medians crossed and x-hinges shown, for further reference, by little dots. Exhibit **11** shows smoothings over slices for upper and lower GaUse hinges (y-hinges)--and for ElUse hinges (x-hinges).

Exhibit **12** plots the smooths, which are candidates for being "hinge traces", together with circles showing where the median trace went. Two things are now relevant, one to be observed and the other to be remembered:

◇particularly after the middle trace has been firmly drawn in exhibit 12, our attention focusses more on the spacing between the traces than on the exact location of the individual traces.

◇we are still unsettled as to whether to plot where (smoothed) y-hinges cross (smoothed) x-medians, or whether we should define

cross-hinges

as the intersection of each y-hinge with an appropriate x-hinge. (Which one would be appropriate?)

exhibit **10** of chapter 8: Twin Rivers

Broken hinges

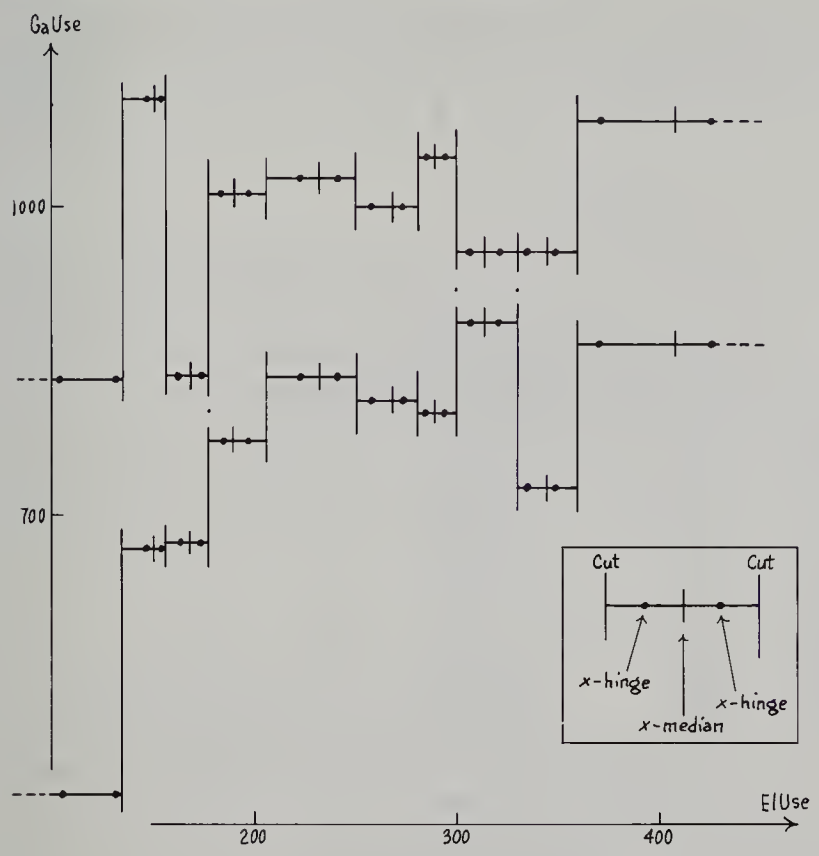


exhibit **11** of chapter 8: Twin Rivers

Smoothing the *y*-hinges and *x*-hinges for (GaUse, ElUse) (mostly based on exhibit 6)

A) UPPER HINGE––see exhibit 9 for smooth of ElUse median

ElUse median	GaUse upper hinge				
H3	raw	3R'	3R'SS	3R'SSH	3R'SSH3
124	832	832		832	832
150	1108	832		878	878
171	832	1015		969	969
197h	1015	1015		1015	1015
232h	1030	1015		1015	1015
265h	1000	1015		1011	1011
292	1050	1000		996	996
318	956	956	970	978	996
358	956	956	970	1000	1000
422	1088	1088		1088	1088

→

What are we to do about the two questions thus posed?

review questions

What would we like to try to do? What picture did we start with? What list of things did we smooth? What did we plot? What observation then arose? About what were we still unsettled?

exhibit **11** of chapter 8 (continued)

B) LOWER HINGE

EIUse median	GaUse lower hinge				
H3	raw	3R′	3R′SS	3R′SSH	3R′SSH3
124	424	424	T	424	T
150	668	668	h	608	h
171	673	673	e	698	e
197h	779	779		760	
232h	830	816	s	804	s
265h	816	816	a	816	a
292	806	816	m	814	m
318	892	816	e	827	e
358	732	860		849	
422	860	860		860	

C) SMOOTHING the EIUse HINGE––*x*-hinges

	lower EIUse hinges			upper EIUse hinges		
Count	raw	H	H3	raw	H	H3
6	108	108	T	134	134	T
5	152	144	h	156	155h	h
8	165	167h	e	176	177	e
20	188h	191h		201	205h	
37	224	223h	s	244	241	s
38	258	257	a	275	272h	a
19	287h	285	m	296	297h	m
9	306	308h	e	323	322h	e
5	334	336		348	362	
5	371	371		429	429	

P) PROBLEMS

11a/b/c) Do the calculation of panels A/B/C for the numbers of exhibit 2.

11d (uses a, b, and c)) Make a plot like exhibit 12 based on the results of 11a, 11b, and 11c.

11e/f/g) As 11a/11b/11c for the slicing of problem 9a.

11h (uses e, f, and g)) As (11d) for the results of (11e), (11f), and (11g).

8D. Dealing with the two questions

Our basic difficulty in the second question thus posed is "which x-hinge goes with a chosen y-hinge?"

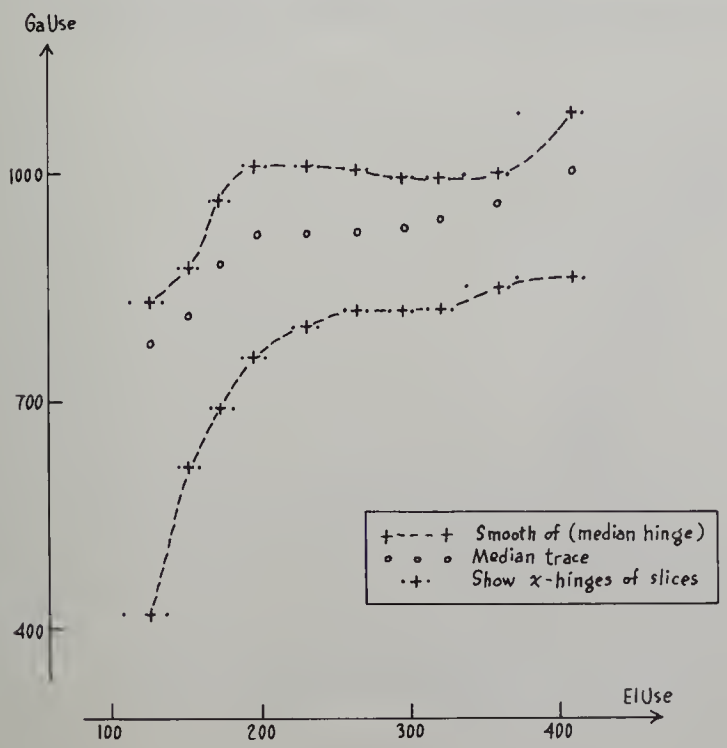
Exhibit **13** shows what happens when the given points lie along a smooth curve. Since all the points are ON this curve, the curve itself ought--except for undesired byproducts of smoothing--to be not only the middle trace but also both hinge traces. The only way to get a good start on this is to define the

cross-hinges

as the intersection of upper hinge with upper hinge--and lower hinge with lower hinge--WHEREVER the curve--or the sequence of cross-hinges--is RISING (as on the left of exhibit 13) BUT as the intersection of upper hinge with lower hinge--and of lower hinge with upper hinge--WHEREVER the curve--or the sequence of cross-hinges--is FALLING (as on the right of exhibit 13). Where the curve is sensibly flat, it does not matter which choice we make, so that, while either can be used, it is often reasonable to compromise and use the x-median rather than either x-hinge.

exhibit **12** of chapter 8: Twin Rivers

Smooths of broken hinges (and the middle trace)



If we follow this lead, we will make similar matchings whenever a "hinge trace" appears to be rising--or be falling--or be nearly level. In the Twin Rivers example, exhibit 12 suggests that we may as well take both hinge traces as rising throughout. Exhibit **14** shows the result of doing this, with middle trace solid, and the "hinge traces" thus defined dashed. The dots show the (x-median, y-hinge) points, which would otherwise have defined the "hinge traces". In this example the "hinge traces" are brought appreciably together.

smoothing the separations

The other thing we may want to do is to focus on the separations between "hinge trace" and middle trace, rather than on the position of the hinge traces themselves. The way to do this is clearly to begin by finding the corresponding

letter-value differences

here "H-to-M", defined as

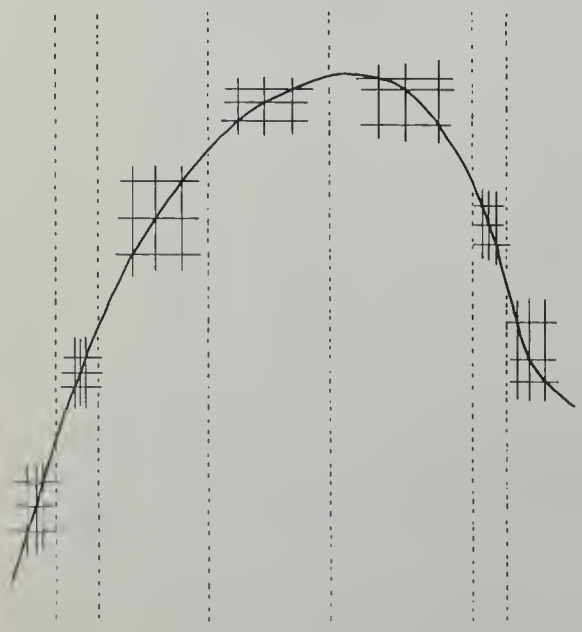
median MINUS lower hinge

and "M-to-H", defined as

upper hinge MINUS median

exhibit **13** of chapter 8: illustrative

Assumed cuts, slice medians and slice hinges for a batch of points lying on a smooth curve



and then:

◇smoothing the letter-value differences.

◇combining them with the middle traces to produce smoothing of separated "hinge traces".

◇smoothing once more, perhaps only with 3, to remove any newly-created irregularities.

Exhibit **15** gives the details for our example. Notice the use of 3R (end-value rule) instead of 3R′ (copying on).

Exhibit **16** plots the result, USING THE X-HINGES CHOSEN ABOVE, thus taking action about both the questions toward which this section is directed. The results are good enough, and the issues well-enough handled, for these traces to deserve being called

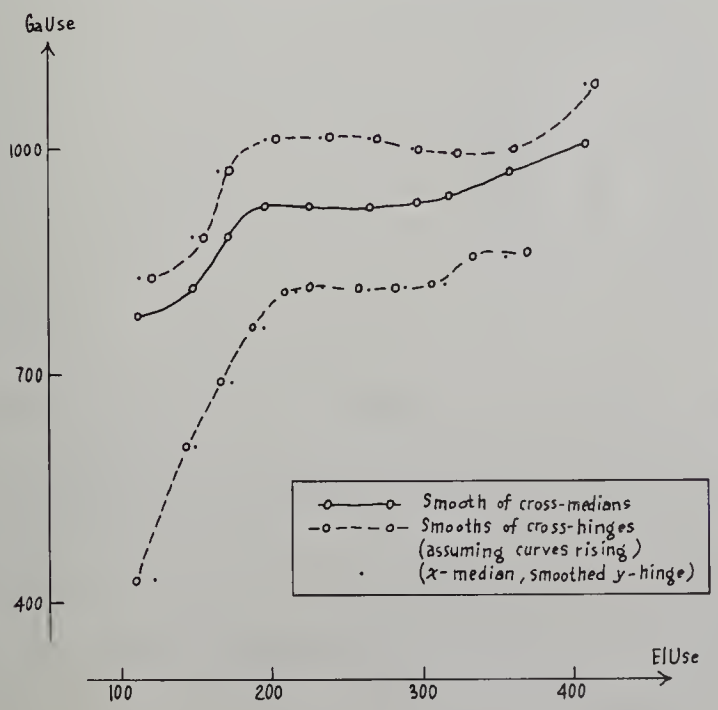
hinge traces.

review questions

What is the difficulty about y-hinges? What example did we look at? What did it tell us? What are cross-hinges? What do we do where the sequence of

exhibit **14** of chapter 8: Twin Rivers

Using the x-hinges for the smoothed broken hinges, shown with middle trace



cross-hinges is rising? What where it is falling? What real example did we then consider? What did we do? What was the result?

To what problem did we then turn? What are letter-value differences? Which ones did we turn to? How did we use them? What did we add together? How did we show the result? What is a hinge trace?

exhibit **15** of chapter 8: Twin Rivers

Smoothing and recombining the hinge–median differences (based mainly on exhibit 6)

A) SMOOTHING HINGE–MEDIAN DIFFERENCES

	GaUse HtoM					GaUse MtoH				
(count)	raw	3R′	3R′SS	3R′SSH	3R′SSH3	raw	3R′	3R′SS	3R′SSH	3R′SSH3
(6)	355	355	T	355	T	53	53		53	T
(5)	320	320	h	288	h	120	73		76	h
(8)	86	156	e	181	e	73	108		99	e
(20)	154	92		108		190	108		108	
(37)	92	92	s	92	s	108	108		108	s
(38)	75	92	a	92	a	109	108		108	a
(19)	136	92	m	103	m	108	108		108	m
(9)	32	136	e	126	e	32	32	108	102	e
(5)	214	142		140		10	32	86	92	
(5)	142	142		142		86	86		86	

B) RECOMBINING SMOOTHED SEQUENCES

GaUse lower hinge				GaUse upper hinge			
mid	HtoM	lo H	3	mid	MtoH	hi H	3
779	355	424		779	53	832	T
815	288	523		815	76	891	h
886	114	705		886	99	985	e
922	108	814		922	108	1030	
922	92	830		922	108	1030	s
922	92	830		922	108	1030	a
928	103	825		928	108	1036	m
938	126	812	819	938	102	1040	e
959	140	819		959	92	1051	
1002	142	860		1002	86	1088	

P) PROBLEMS

15a/b/c) Do the calculations of panel A/B/C starting from the numbers of exhibit 2.

15d(abc)) Prepare a picture like exhibit 16 based on the results of (15a), (15b), and (15c).

15e/f/g(b)) Do the calculations of panel A/B/C starting from the numbers of problems (6b) to (6k).

15h(efg)) Prepare a picture like exhibit 16 based on the results of (15e), (15f), and (15g).

8E. Wandering schematic plots

We now have analogs of most of the parts of a schematic plot:

- ◇the hinges become hinge traces.
- ◇the median becomes a middle trace (median trace).
- ◇and the outside points?
- ◇and the adjacent points?

What we need next are fences, so as to help us distinguish outside and adjacent points.

In this case, the simplest calculation seems to be the best. We fix convenient values of x, read off (from the graph paper original) the corresponding y's according to the hinge traces, and then proceed as usual. Exhibit **17** shows the arithmetic, making it clear that no one of the 152 points in the Twin Rivers example has even a faint chance of being outside. Thus all that remains is to locate the adjacent points and represent them appropriately.

exhibit **16** of chapter 8: Twin Rivers

At last, hinge traces--and the middle trace

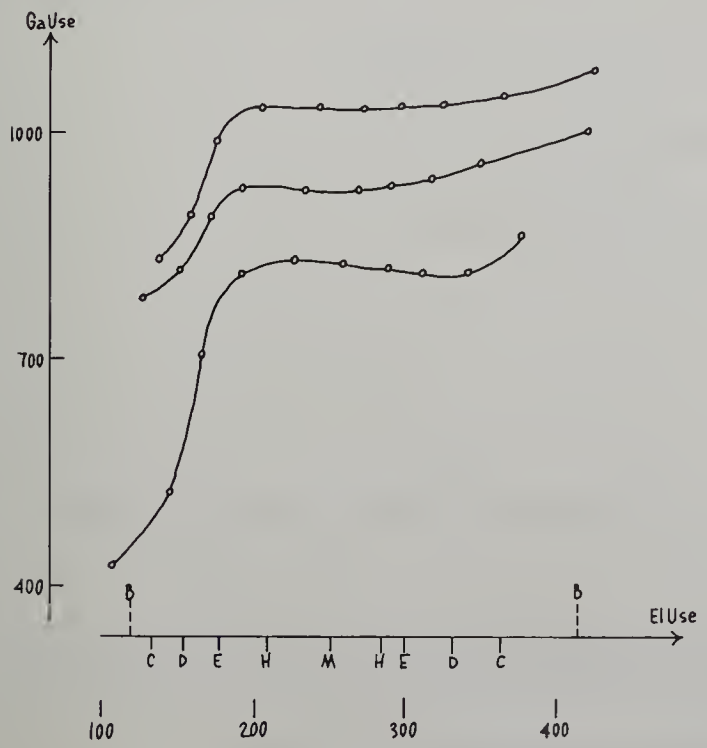


Exhibit **18** shows:

◇hinge traces and median trace.

◇each point adjacent in its slice.

◇each (non-outside) point in the two extreme slices.

◇a dashed-line convex polygon surrounding all these points just mentioned.

exhibit **17** of chapter 8: Twin Rivers

The fences

A) CALCULATION of FENCES

	GaUse		GaUse H-spread					GaUse		
ElUse	Hinge · traces		raw	3R	S	H	3	Step	inner	fences
125	560	835	245	245	T	245	T	348	212	1183
150	640	875	235		h	236	h	354	286	1229
175	750	980	230		e	228	e	342	408	1322
200	825	1033	208	215		218		327	398	1360
225	830	1045	215	214	s	214	s	321	509	1366
250	828	1042	214		a	214	a	321	507	1363
275	828	1042	214		m	216	m	324	504	1366
300	823	1045	222		e	220	e	330	493	1375
325	820	1048	228	224		224		336	484	1384
350	826	1050	224	224		224		336	490	1386
375	(868)	1056	(188)	224		224		336	(532)	1392

Note: Further smoothing of the fences thus found is optional

B) POINTS MOST NEARLY OUTSIDE--see exhibit 2

Slice	Low extreme	High extreme	Near fence
1	(108, 388)		below 148
2			
3		(196, 1298)	above 1322
4			
5			
6			
7			
8			
9			
10			

Other points omitted as having no chance of approaching fences. [In first slice, (89, 424) was the only other possibility.]

P) PROBLEMS

17a(15d*)) Use the graph-paper original of problem 15d to find the numbers needed for the calculation of panels A and B. Compare your result with that found above.

17b(15h*)) Do the same from the graph-paper original of problem 15h.

If we omit the points, keeping the

adjacent polygon

dashed and the hinge and median traces solid, we will obtain what we may as well call

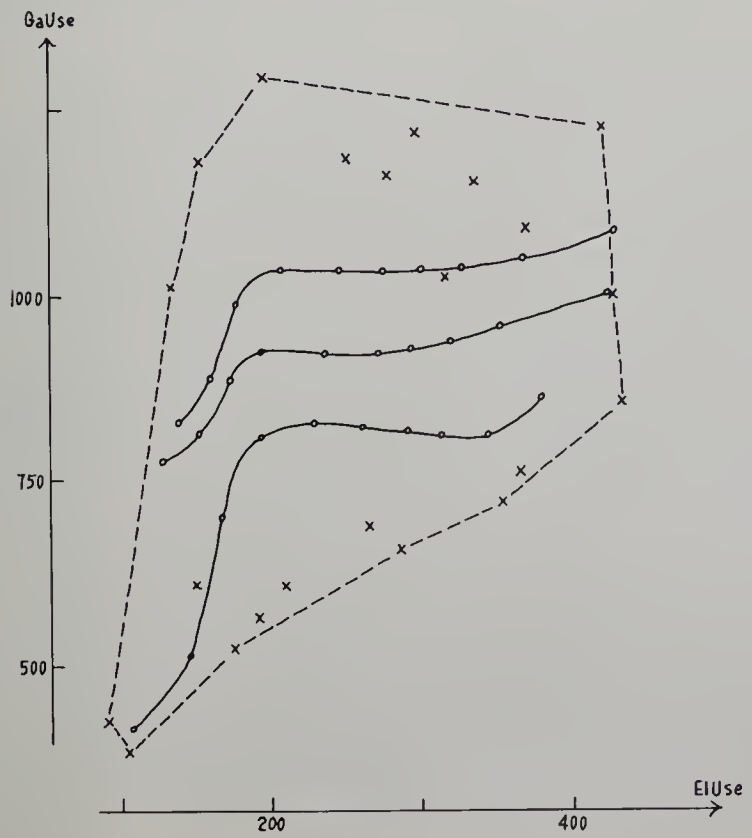
wandering schematic plot

as shown in exhibit **19**. This example is of course rather special, in that there are no outside points.

Here we have made the adjacent polygon convex. In other circumstances, for instance where all the traces and hence the fences were high at both ends and low in the middle, so that a crescent-shaped adjacent polygon would clearly make good sense, we do not require that the adjacent polygon be convex. Reasonable judgment should be used, always being sure that the adjacent polygon includes all points not outside.

exhibit **18** of chapter 8: Twin Rivers

Hinge and median traces--candidate points--adjacent polygon



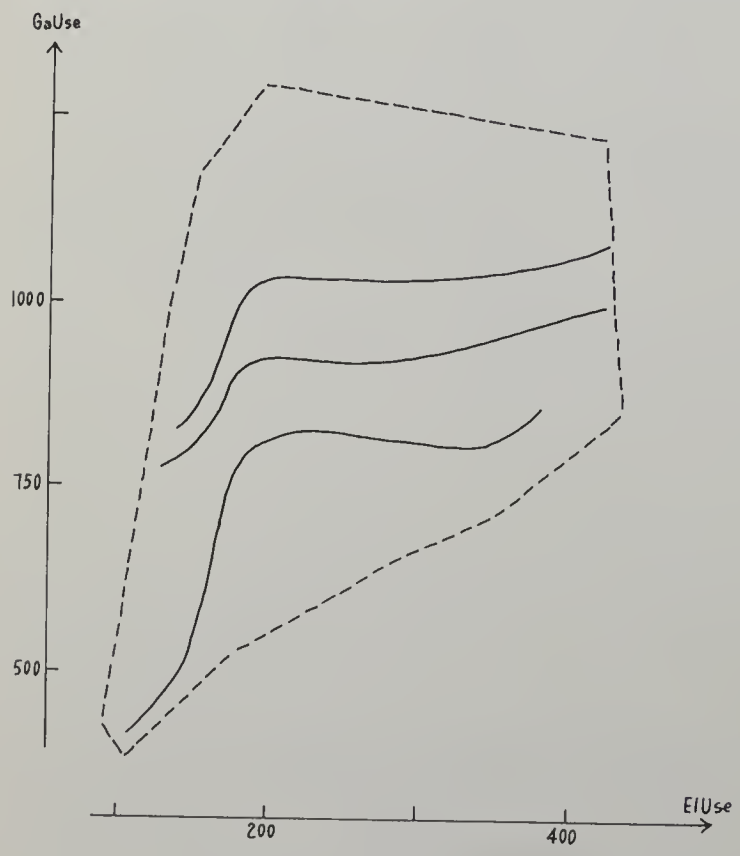
interpretation?

What interpretation can we place upon this particular wandering schematic plot? The largest part of each trace is roughly horizontal. At the left, however, there is tilt, strongest for the lower hinge trace. (At the right, there is a weak indication of tilt, also, but hardly strong enough to persuade us of its importance or reality.) This is just what we ought to expect, if

- ◇for most townhouses there was no connection--not even a statistical one--between gas and electrical usages.
- ◇a few townhouses had strikingly low values of both usages.

exhibit **19** of chapter 8: Twin Rivers

The wandering schematic plot



And how might this last have come about? Perhaps a few residents took winter vacations, turning the heat down and the lights off. Perhaps something else happened.

review questions

What parts for a schematic plot had we when this section began? Which did we need? Did this lead us to search for fences? What seemed the simplest approach? What did we do? How did things come out in the Twin Rivers example? What list of plausible possibilities for adjacent points did we consider? With what did we surround them? Was this always to be convex? Why/why not? What is a wandering schematic plot? What are its pieces? What is an adjacent polygon?

8F. A more demanding example: Governor's salary and bank deposits

As so often happens, it is not large batches of points that call for attention in the construction of wandering schematic plots. Rather it is small batches that need special care. We devote this section to a batch of 50, one for each of the U.S.A.'s 50 states. So far as we can see, there is no way to gather any larger batch, for there are no political divisions elsewhere that have been operating in similar political and economic conditions. So all we can do is to analyze these 50 points with considerable care.

first analysis

Exhibit **20** sets out, for about 1969, the total deposits in banks and the governor's salary for each of the 50 states, continuing with convenient multiples of the logarithms of each. As an amount with greatest/least close to 300, we should surely expect to start with logs of bank deposits. Since the corresponding ratio is only 5 for governor's salary, we could hardly be sure that logs were needed for this variable, but we will find them helpful.

Exhibit **21** continues with an arraying of the points, first (BaDep, GoSal) in BaDep order, then GoSal sorted within slice, with the medians and hinges displayed by circling. (The other two columns will be discussed below.) Here we have sliced to our own convenience, making slices of 5, 5, 9, 12, 9, 5, and 5. (The column of state names will help us to identify outside and adjacent states.)

When slicing for convenience, we often

- ◇choose slice counts of 5, 9, 17, etc., when minimizing hand arithmetic, since these choices make letter depths without halves.
- ◇choose slice counts of 4 (if only a median is wanted), 8, 16, 32, . . . , when looking for a little more numerical stability in the letter values, since these choices gives h's to as many letter depths as possible, thus more often using means of two observations.

exhibit **20** of chapter 8: governor's salary

Bank deposits and governor's salary for the 50 states ca. 1969

A) DATA--and logs (a and b indicate how further digits would break ties. a < b)

	Bank deposits		Governor's salary, 1969/70	
	Billions	100 $\log_{10}$	thousands	1000 $\log_{10}$ (raw/10)
Alab	4.287	63	25	398
Alas	.430	−37	32	505
Ariz	2.856	46	27.5	439
Ark	2.625	42b	10	0
Cal	45.422	166	44.1	644
Colo	3.886	59	20	301
Conn	9.328	97	35	544
Del	1.456	16	35	544
Fla	11.542	106	36	556
Georg	6.529	81	42.5	628
Haw	1.380	14	33.5	525
Idaho	1.146	6	17.5	243
Ill	31.587	150	45	653
Ind	9.603	98	25	398
Iowa	5.806	76b	30	477
Kans	4.412	64	20	301
Kent	4.652	67	30	477
Louis	5.708	76a	28.374	453
Maine	1.862	27b	20	301
Maryl	5.512	74	25	398
Mass	20.006	130a	35	544
Mich	19.234	128	40	602
Minn	9.088	96	27.5	439
Miss	2.698	43	25	398
Mo	10.933	104	37.5	574
Mont	1.400	15	23.25	366
Nebr	3.094	49	18	255
Nev	.956	−2	25	398
NH	1.855	27a	30	477
NJ	15.776	120	35	544
NM	1.217	9	20	301
NY	121.905	209	50	699
NC	6.326	80b	35	544
ND	1.362	13	18	255
Ohio	20.094	130b	40	602
Okla	4.854	69	25	398
Ore	3.969	60	28.5	455
Penn	29.648	147	45	653
RI	2.609	42a	30	477
SC	1.986	30	25	398
SD	1.326	12	18	255
Tenn	6.221	79	18.5	267
Tex	23.475	137	40	602
Utah	1.696	23	22	342

→

exhibit **20** of chapter 8 (continued)

Verm	.994	−0	25	398
Virg	6.876	84	30	477
Wash	6.283	80a	32.5	512
WVa	2.298	36	25	398
Wisc	8.489	93	25	398
Wyo	.703	−15	20	301
(Name)		(BaDep)		(GoSal)

S) SOURCE

The 1970 World Almanac: Bank deposits (from FDIC) at page 91 and governor's salaries at page 7.

exhibit **21** of chapter 8: governor's salary

Exhibit 20 continued

B) WORKING ARRAYS--circled values are slice medians and slice hinges

BaDep	GoSal		GoSal MINUS 3(BaDep)			Name
(sorted overall)	BaDep order	sorted in slices	BaDep order	sorted in slices		BaDep order
−37	505	243	616	225		Alas
(−15)	301	(301)	346	(346)		Wyo
(−2)	398	(398)	404	(398)	slice 1	Nev
(−0)	398	(398)	398	(404)		Verm
6	243	505	225	616		Ida
9	301	255	274	216		NM
(12)	255	(255)	219	(219)		SD
(13)	255	(301)	216	(274)	slice 2	ND
(14)	525	(366)	483	(321)		Haw
15	366	525	321	483		Mont
30						
16	544	0	496	−126		Del
23	342	301	273	220		Utah
(27)	477	(342)	396	(269)		NH
27b	301	398	220	273	Slice 3	Maine
(30)	398	(398)	308	(290)		SC
36	398	398	290	308		WVa
(42)	477	(477)	351	(351)		RI
42b	0	477	−126	396		Ark
43	398	544	269	496		Miss

→

Exhibit **22** completes the calculation of a middle trace and two hinge traces, as seen in exhibit **23**. We cannot be too happy with the crossings of these curves. (Nor would we be happy to plot the smoothed GoSal hinges at BaDep medians rather than at BaDep hinges.) The plausible cause of trouble is the steepness of the traces toward the righthand side.

exhibit **21** of chapter 8 (panel B continued)

46	439	255	301	30		Ariz
49	255	267	108	108		Nebr
(59)	301	(301)	124	(109)		Colo
(60)	455	(301)	275	(124)		Ore
63	398	398	209	176	Slice 4	Alab
(64)	301	(398)	109	(191)		Kans
(67)	477	(398)	276	(209)		Kent
69	398	439	191	225		Okla
(74)	398	(453)	176	(249)		Maryl
(76a)	453	(455)	225	(275)		Louis
76b	477	477	249	276		Iowa
79	267	477	30	301		Tenn
80a	512	398	272	104		Wash
80b	544	398	304	119		NC
(81)	628	(439)	385	(151)		Ga
84	477	477	225	225	Slice 5	Virg
(93)	398	(512)	119	(253)		Wisc
96	439	544	151	262		Minn
(97)	544	(544)	253	(272)		Conn
98	398	574	104	304		Ind
104	574	628	262	385		Mo
106	556	544	238	154		Fla
(120)	544	(544)	184	(184)		NJ
(128)	602	(556)	218	(212)	Slice 6	Mich
(130)	544	(602)	154	(218)		Mass
130b	602	602	212	238		Ohio
137	602	602	191	72		Tex
(147)	653	(644)	212	(146)		Penn
(150)	653	(653)	203	(191)	Slice 7	Ill
(166)	644	(653)	146	(203)		Cal
209	699	699	72	212		NY

→

untilted analysis

Since the righthand side slope is close to 3, it is natural to consider working with

GminB = GoSal MINUS 3 BaDep

exhibit **22** of chapter 8: governor's salary

Exhibit 21 continued

C) SMOOTHING of GoSal and BaDep MEDIANS

median GoSal					median BaDep	
raw	3R′	S	H	3	raw	H
398		T	398	T	−2	−2
301	398	h	398	h	13	13h
398		e	398	e	30	35
398		s	424	s	65h	63h
512		a	494	a	93	95
556		m	569	m	128	125
653		e	653	e	150	150

D) SMOOTHING of LOW HINGES

HtoM GoSal					loH		loH BaDep	
raw	3R	S	H	3	comb	3	raw	H
97	56		56	T	342	T	−15	−15
46	56		56	h	342	h	12	9
56			56	e	342	e	27	31h
97	73	56	46	s	380	s	59h	57
73		18	26	a	468	a	81	85h
12			13	m	556	m	120	117
9			9	e	644	e	147	147

Note: Values of "comb" are final smooths of "median GoSal" (see panel C) MINUS final smooths of "HtoM GoSal".

E) SMOOTHING of HIGH HINGES

MtoH GoSal					hiH		hiH BaDep	
raw	3R	S	H	3	comb	3	raw	H
0	65	T	65	T	463		−0	0
65	65	h	65	h	463		14	17h
79	65	e	63	e	461	463	42	43
56	56	s	56	s	480		75	72
32	46	a	46	a	539		97	100
46	32	m	26h	m	598		130	131
0	4	e	4	e	657		166	166

Note: Values of "comb" are final smooths of "median GoSal" (see panel C) PLUS final smooths of "MtoH GoSal".

The two further columns of exhibit 20, and exhibit **24**, have all the necessary arithmetic, including, in panel G, the conversion of GminB back to GoSal. Exhibit **25** shows the median and hinge traces for GminB, according to panel F (of exhibit 23), against BaDep (using panels C to E of exhibit 22). In this exhibit, since the general trend is downward, we have paired low hinges for each of GaUse and ElUse with high hinges for the other. There is now no tendency for the traces to cross toward the right (although a crossing has appeared on the left).

exhibit **23** of chapter 8: governor's salary

Median and hinge traces as computed directly (slopes of 2 and 3 shown for comparison)

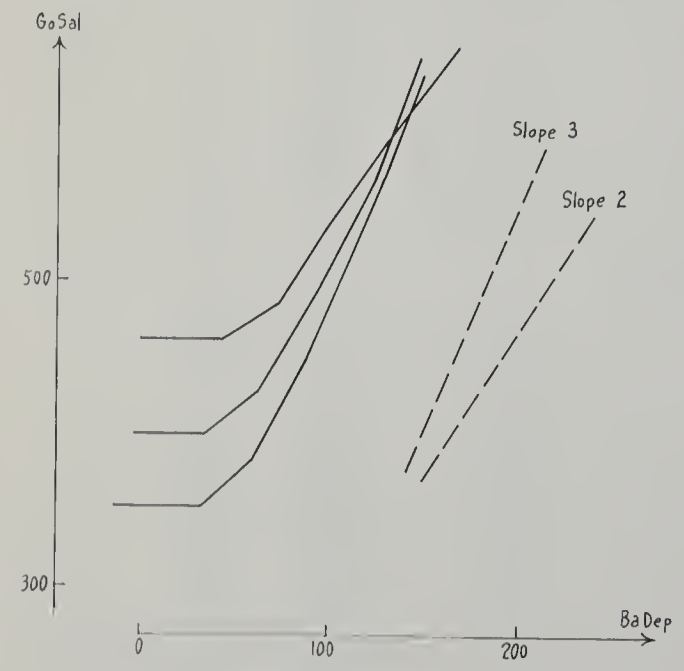


exhibit **24** of chapter 8: governor's salary

Exhibit 22 continued

F) SMOOTHING of GoSal MINUS 3BaDep--called below GminB

median					HtoM					lo H	
raw	3R'	S	H	3	raw	3R	S	H	3	comb	3
398	398	T	398	T	52	52	52	52	T	346	T
274	290	h	313	h	55	52	52	52	h	261	h
290	274	e	273	e	21	55	52	50	e	220	e
200	253	s	248	s	83h	83h	45	47	s	201	s
253	212	a	222	a	102	83h	45	45	a	177	a
212	212	m	207	m	28	45	45	45	m	162	m
191	191	e	191	e	45	45	45	45	e	146	e

→

exhibit **24** of chapter 8 (continued)

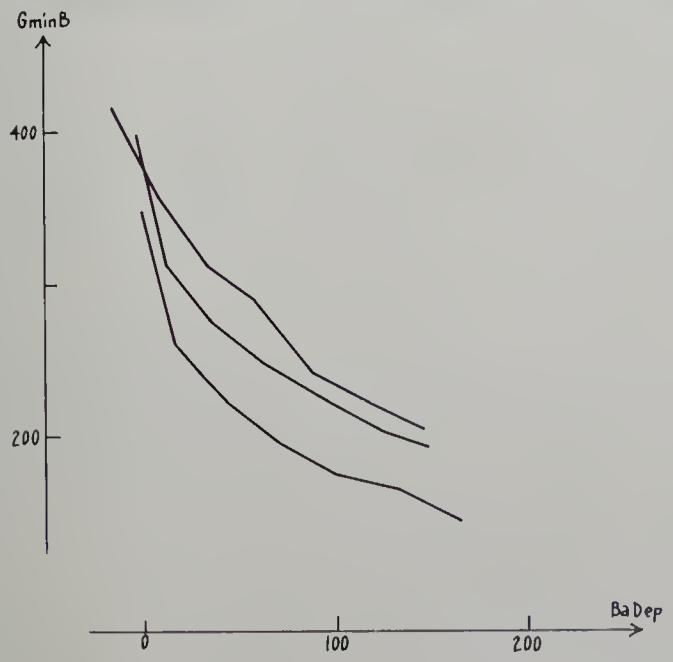
MtoH					hi H	
raw	3R	S	H	3	comb	3
6	19		19		417	
47	47		40		353	
61	47		44	40	313	
62	33		33		281	
19	19		20		242	
26	12		14		221	
12	12		12		203	

G) REASSEMBLING SMOOTH of GoSal––as GminB PLUS 3 BaDep

median				loH GoSal				hiH GoSal			
med 3BaDep	med GminB	sum	3	hiH 3BaDep	loH GminB	sum	3	loH 3BaDep	hiH GminB	sum	3
−6	398	392		0	346	346		−45	417	372	T
40h	313	353h	378	52h	260	312h	346	27	353	380	h
175	273	378		129	220	349		94h	313	407h	e
190h	248	438h		216	196	412		171	281	451	s
285	222	507		300	174	474		256h	242	498h	a
375	207	582		393	162	555		351	221	572	m
450	191	641		498	146	644		441	203	644	e

exhibit **25** of chapter 8: governor's salary

Median and hinge traces for GminB against BaDep



The result of plotting the complete conversion back to GoSal (see panel G) is shown in exhibit **26**.

combined analysis

We are now prepared to choose final hinge and median traces. For the three leftmost slices, the general run of GoSal seems about level. For these slices, then, the results of exhibits 20 and 21 seem appropriate. For the three rightmost slices, the general run of the GoSal values is slanted steeply upward. For these slices, the results of exhibits 22 and 24 seem appropriate. For the middle slice we could use either, and will take both. Combining then gives the values in panel A of exhibit **27** and the picture in exhibit **28**.

Proceeding to calculate fences in exhibit **29**, and to plot possibly outside and possibly adjacent points (based on panel B of exhibit 27), we finally reach the wandering schematic plot of exhibit **30**. Clearly, Arkansas is far out. We have shown New York as outside, although we did not have enough data to draw fences for BaDeps larger than 160. In fact, New York is outside for BaDep alone.

exhibit **26** of chapter 8: governor's salary

The hinge and median traces of exhibit 25 translated back in terms of GoSal

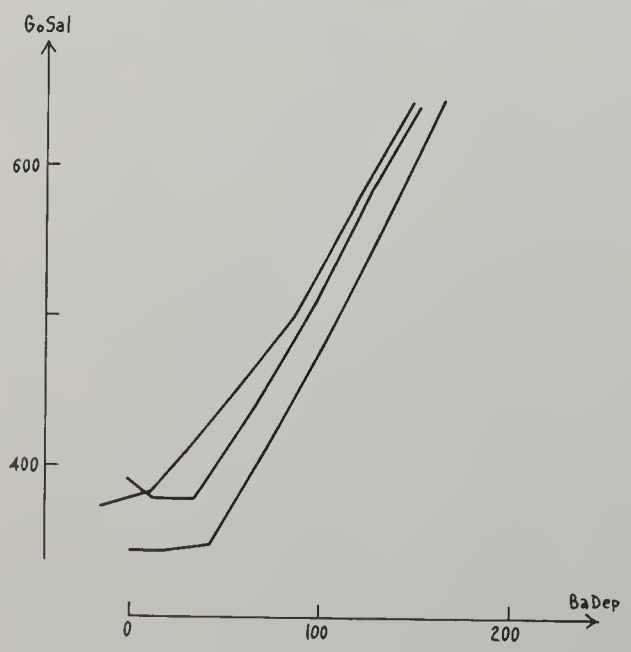


exhibit **27** of chapter 8

Values needed in plotting

A) CHOSEN TRACES

	median trace			lo hinge trace			hi hinge trace		
	BaDep	GoSal	3	BaDep	GoSal	3	BaDep	GoSal	3
1	−2	398	T	−15	342		0	463	
2	13h	398	h	9	342		17h	463	
3	35	398	e	31h	342		43	463	
4	63h	427	433	72	380	396	57	480	466
4	63h	438h	433	72	412	396	57	452	466
5	95	507	s	100	474		85h	498h	
6	125	553	a	131	555		117	572	
7	150	641	m e	166	644		147	644	

B) POSSIBLY INTERESTING STATES

	BaDep	GoSal	
Ark	42	0	outside
Tenn	79	267	outside
NY	209	699	outside
Alaska	−37	505	slice 1
Wyo	−15	301	slice 1
Nev	−2	398	slice 1
Verm	−0	398	slice 1
Ida	6	243	slice 1
Tex	137	602	slice 7
Penn	147	653	slice 7
Ill	150	653	slice 7
Cal	166	644	slice 7
(NY	209	699)	slice 7

	BaDep	GoSal	
Hawaii	14	525	other high in slices 2 to 6
Del	16	544	other high in slices 2 to 6
Iowa	76	477	other high in slices 2 to 6
Ga	81	574	other high in slices 2 to 6
Mich	128	602	other high in slices 2 to 6
SD	12	255	other low in slices 2 to 6
ND	13	255	other low in slices 2 to 6
Ark	42	0	other low in slices 2 to 6
Nebr	49	255	other low in slices 2 to 6
Ind	80	398	other low in slices 2 to 6
NJ	120	544	other low in slices 2 to 6
Mass	130	544	other low in slices 2 to 6

P) PROBLEMS

27a) Can anything useful be said about the differences between "high" states and "low" states?

exhibit **28** of chapter 8: governor's salary

Hinge and median traces--and fences--combining the two delineations (e.g., exhibits 23 and 26)

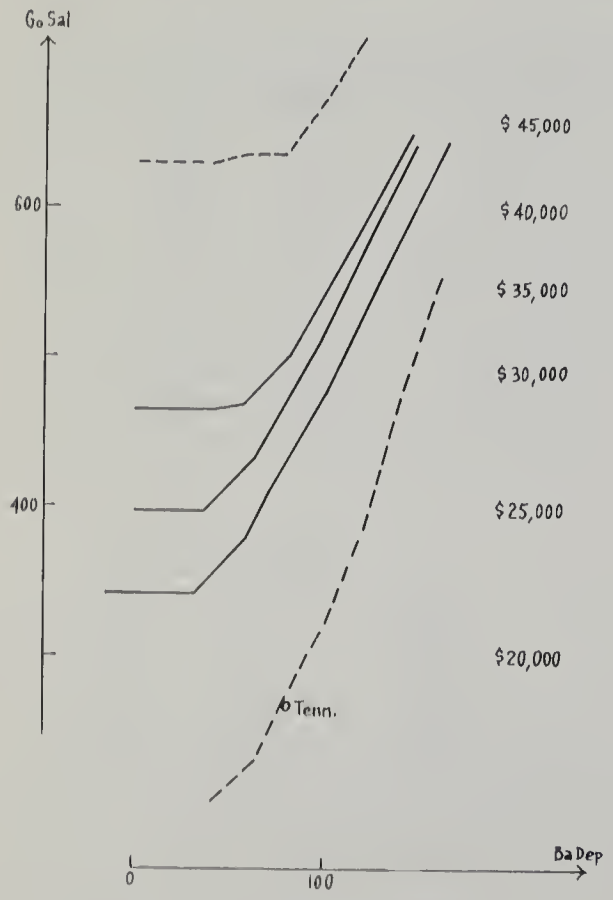


exhibit **29** of chapter 8: governor's salary

Calculation of fences and spread

A) CALCULATION of FENCES

						inner fences			
BaDep	loH	hiH	Hspr	3	step	lo	3	hi	3
0	358	463	111		116	186		629	
20	352	463	111		166	186		629	
40	362	463	101	102	153	209		616	629
60	382	484	102	101	152	230		636	
80	412	498	86	92	138	274		636	
100	450	542	92	86	129	321		671	
120	508	592	84		126	382		718	
140	570	635	65		98	472		733	
160	628	(675)	(47)		(71)	(557)		746	

→

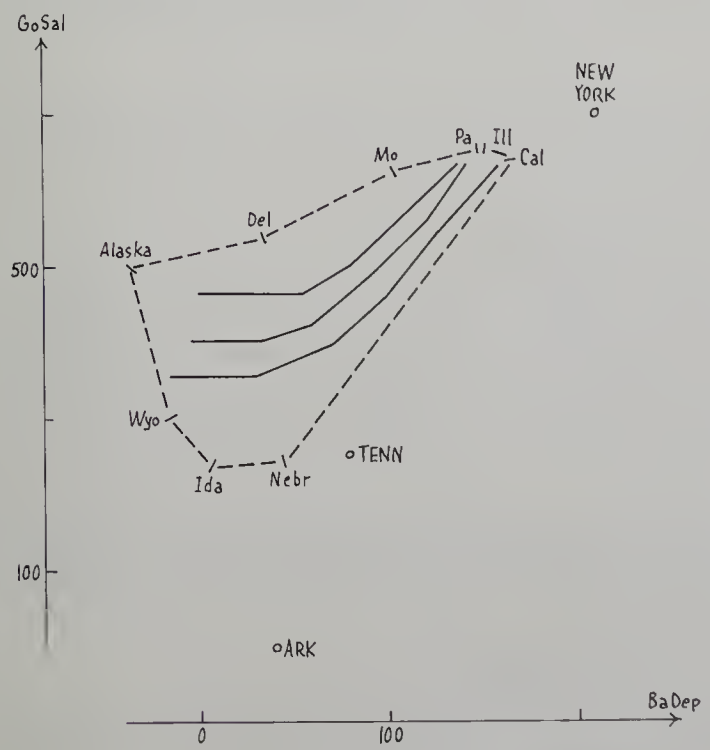
, exhibit **29** of chapter 8 (continued)

B) HINGES and SPREAD in RAW (thousands of) DOLLARS

BaDep	loH	hiH	Hspread	3
0	22.5	29.0	6.5	6.5
20	22.5	29.0	6.5	6.5
40	23.0	29.0	6.0	6.4
60	24.1	30.5	6.4	6.4
80	25.8	31.5	5.7	6.4
100	28.1	34.8	6.6	6.6
120	32.2	39.1	6.9	6.6
140	37.2	43.1	5.9	5.9
160	42.5	(47.3)	(4.8)	(4.8)

exhibit **30** of chapter 8: governor's salary

The wandering schematic plot that results (based on exhibits 28 and 29)



review questions

Will more care in making wandering schematic plots be needed for large batches? For small ones? What example did we consider in this section? How large a batch? Could it have been bigger? Why/why not? Did we need to take logs? Of the values of one variable? Of those of both variables? Why/why not? What did we do? What kind of slices did we take? What are particular choices for slice sizes? Why are they desirable?

What happened when we plotted hinge and median traces? Why might this have happened? What was the plausible way out? What did we do? How did it work? If we subtract before finding hinge and middle traces, what must we do afterwards? Is combining the result of two such analyses reasonable? Did we do it? How did we like the results?

Did we calculate fences? Were any states outside? Far out? How did the wandering schematic plot come out?

8G. Further questions/analysis in the example

We can--and should--take the example further in a number of ways.

time changes

Of the five states on the upper boundary of the adjacent polygon, two (Alaska and Delaware) had raised their governor's salary in the previous year, while two more (Illinois and Missouri) had done this a year earlier. Of the four states on the lower boundary, three (California, Idaho, and Wyoming) raised their governor's salary in the next year, as did both states (Tennessee, New York) that were outside but not far out. Arkansas remained a far out exception.

If we had used smoothed values of governor's salaries, suitably smoothed over years, we would have had an appreciably tighter picture for our wandering schematic plot.

vertical width

In addition to the wandering schematic plot, we also want to look at the behavior of the vertical width as BaDep changes. (With the very different slopes in different parts of the range, we dare not hope to "look" at exhibit 30 and see how the width changes.) Panel A of exhibit 29 shows a hinge-spread decreasing from 111 to 65 (or 47). This is in logarithmic units, and we wonder about behavior in raw dollars.

Panel B expresses the "hinges" in raw dollars. We find a quite constant hinge-spread, amounting to $6,500, over most of the range. (Since only eight states have BaDep's above 120, we do not have to pay too much attention to the apparent drop at the upper end.)

slopes and levels

For small BaDep, the slope of all three traces is close to zero and the level of the middle trace is just about $25,000.

For large BaDep's, as exhibit 25 shows, the slope of GoSal on BaDep is close to 3. Taking account of units, this indicates that, for high BaDep, governor's salary varies about as the 0.3 power of bank deposits. A plausible multiplying constant turns out to be 31, so that, on the high side

$$\text{governor's salary} \quad \text{about} \quad 31\,(\text{bank deposits})^{0.3}$$

is a reasonable fit.

consolidated summary

We can now summarize the situation for governor's salary vs. bank deposits as follows:

- ◇for low bank deposits, governor's salary seems to center near $25,000.
- ◇for high bank deposits, governor's salary seems to center near 31(bank deposits)$^{0.3}$.
- ◇the transition is moderately abrupt, and could be taken either as a dogleg or as somewhat smoothed out.
- ◇in raw dollars, the vertical width seems quite constant, with an H-spread of $6,500.
- ◇most states near the upper margin had just raised their governor's salaries.
- ◇most states low outside or near the lower margin raised their governor's salary during the next year.

moral

What we have learned from this example is that

- ◇it is not safe to trust our middle-trace and hinge-trace calculations when these traces are too steep, where steepness relates to slice width. (If we had had 500 states, we would have had little trouble with steepness--except for the end slices.)
- ◇we can avoid such difficulties by subtracting a simple function of x to make matters less steep.
- ◇we can combine results from different subtractions when the slope differs in different intervals.

We could have subtracted something that would work fairly well across the whole range, had we wished. Two candidates would be:

- ◇the greater of 398 and 188 + 3(BaDep).
- ◇the original middle trace interpolated between the final smoothed points (and extrapolated outside them) to give a definition everywhere it is needed.

We leave the exploration of these possibilities to the reader (see exhibit **31**). Exhibit **32** offers data for further problems.

exhibit **31** of chapter 8: Texas data

Data on the male and female populations of the 254 counties of Texas

MaPop = $100 \log_{10}$ (male population in thousands)
FePop = $100 \log_{10}$ (female population in thousands)

A) EXAMPLES of CALCULATION

County	Male Population			Female Population		
	raw	$\log_{10}$	MaPop	raw	$\log_{10}$	FePop
Anderson	13,397	4.13	113	14,765	4.17	117
Andrews	6,887	3.84	84	6,563	3.82	82
Angelina	19,568	4.29	129	20,246	4.31	131

B) The (MaPop, FePop) PAIRS, ORDERED on MaPop

(−94, −95), (−45, −55), (−31, −40), (−27, −27), (−25, −29), (−23, 023 and −28), (−9, −11), (−6, −6), (1, −2), (3, 0 and 5), (5, −10), (9, 4), (10, 12), (12, 10 and 11), (14, 14), (15, 14), (16, 15 and 18), (17, 12 and 18), (19, 17 and 20), (21, 20), (24, 22), (25, 26), (26, 25 and 27), (27, 17 and 26 and 28), (28, 26), (29, 27 and 29 and 31), (30, 29), (31, 32), (32, 32 and 35), (34, 31 and 36), (35, 35), (36, 32), (37, 37), (39, 40 and 41), (40, 41), (41, 39 and 42), (43, 44 and 44 and 44), (45, 48), (46, 46 and 48), (47, 49), (48, 47 and 49), (49, 49 and 51), (50, 49 and 49 and 52), (51, 49 and 30), (52, 49 and 49 and 52), (54, 52), (55, 54 and 57), (56, 47 and 56 and 57 and 58 and 59), (58, 59 and 61), (59, 59 and 59 and 60 and 60 and 60 and 61 and 61), (60, 63), (61, 64 and 64), (62, 59 and 62), (63, 65 and 66), (64, 63 and 66), (66, 64 and 67 and 68), (67, 64), (68, 66), (69, 62 and 71 and 71), (70, 70), (71, 70 and 71), (72, 71 and 71), (73, 72 and 73 and 74), (74, 74 and 74 and 76 and 76), (75, 75), (76, 78), (77, 79), (80, 75 and 77 and 79), (81, 80), (82, 82), (83, 81 and 83 and 84), (84, 82 and 82 and 84), (85, 85 and 87), (86, 88), (87, 96 and 88 and 88), (88, 87 and 91), (89, 93), (90, 91), (91, 91 and 93 and 94), (92, 90 and 93 and 93 and 93 and 95), (93, 93 and 93 and 95), (94, 95 and 96 and 96), (96, 93 and 97 and 98), (97, 97 and 97 and 98 and 101 and 104), (98, 98 and 98 and 99), (99, 100 and 101 and 101 and 101), (100, 101 and 102 and 102), (101, 96 and 101 and 102 and 103 and 104), (104, 100 and 104 and 105 and 106 and 107), (106, 104 and 108 and 109), (107, 105 and 108 and 111), (109, 96 and 106 and 109), (110, 108), (111, 111), (113, 113 and 117), (114, 116), (115, 117), (116, 119), (118, 97), (119, 120), (120, 115 and 120), (121, 123), (122, 125 and 125), (123, 122 and 124 and 125 and 125), (124, 123), (125, 127), (127, 126), (128, 128 and 131), (129, 131), (131, 137 and 130), (133, 128), (135, 135), (136, 137 and 139), (138, 132), (146, 149), (149, 153), (152, 154), (153, 155), (160, 156), (162, 165), (166, 165), (170, 170), (171, 164), (178, 175), (180, 178), (184, 185), (187, 188 and 189), (189, 189), (195, 196), (202, 203), (204, 209), (208, 210), (220, 219), (242, 244), (253, 254), (266, 269), (279, 280).

C) Some COUNTY NAMES and MAP REFERENCES

Name	Map	MaPop	FePop	Name	Map	MaPop	FePop
King	E9	−45	−55	Kerr	J9	91	94
Kenedy	N12	−31	−40	Caldwell	J12	92	95
Roberts	A8	−27	−27	Beaver	G4	96	93
Borden	F7	−25	−29	Eastland	G10	97	101
McMullen	L11	−23	−28	Limestone	H13	97	104
Glasscock	G7	−22	−29	Houston	H15	101	96

→

exhibit **31** of chapter 8 (panel C, continued)

Jeff Davis	I3	−9	−11	Falls	H13	101	104
Real	H7	3	0	Cooke	D12	104	107
Oldham	B6	5	−10	Willacy	O12	104	100
Edwards	J8	9	4	Hill	I2	106	109
Kinney	K8	11	6	Brown	G10	107	111
Culberson	G3	17	12	Bee	L12	109	106
Lipscomb	A9	24	22	Walker	I14	109	96
Hudspeth	G2	27	17	Valverde	J7	110	108
Ellis	F13	32	35	Anderson	G14	113	117
Crockett	I7	34	31	Kaufmann	T14	116	119
Zapata	N10	36	32	Coryell	H12	118	97
Delta	E14	45	48	Kleberg	N12	120	115
Dallam	A6	51	49	Lamar	D14	122	125
Brewster	I4	52	49	Navarro	G13	122	125
Cochran	E6	52	49	Hunt	E14	128	131
Orange	J17	56	47	Fort Bend	J14	133	128
Trinity	H15	56	59	Harrison	F16	134	137
Camp	E15	58	61	Denton	E12	136	139
Hardeman	D9	60	63	Brazos	I14	138	132
Childress	C9	61	69	Bowie	E16	146	149
Hamiton	H2	61	64	Webb	M9	149	153
Yoakum	E6	62	59	Brazoria	K15	160	156
Stephens	F10	63	66	Smith	E15	162	165
Bailey	D6	67	64	Bell	H12	171	164
Coleman	G10	78	81	Potter	B7	178	175
Pecos	H5	79	76	Cameron	O12	187	189
Gaines	F6	80	77	Travis	J12	202	203
Irion	H8	80	75	Nueces	M12	204	209
Zavala	L9	81	80	Jefferson	J16	208	210
Winkler	G5	84	82	El Paso	G1	220	219
Red River	D15	88	91	Tarrant	F12	242	244
Robertson	H14	89	83	Dallas	F13	266	269
				Harris	J14	279	280

P) PROBLEMS

31a) Go through the analysis of GoSal against BaDep, subtracting max(398, 188 + 3(BaDep)) before finding middle and hinge traces. Compare your results with those in earlier exhibits.

31b) Go through the analysis of GoSal against BaDep, subtracting an inter (extra)polation of the final smooth of panel C of exhibit 22 before finding (new) middle and hinge traces. Compare your results with those in earlier exhibits.

31c) Analyze the data in panel B above, using the methods of this chapter. Discuss your results.

31d) Analyze the data in panel C above, using the methods of this chapter.

31cd2) Compare these analyses. Were the 77 values enough? Do you think 77 taken "at random" would be enough? Why/why not?

S) SOURCE

Census of Population: 1960; Volume I. Characteristics of the Population; Part 45. Texas, pages 45–181 to 45–244. U.S. Department of Commerce, Bureau of the Census, 1963.

exhibit **32** of chapter 8

Time of high tide (24-hour clock), time since last high tide (in minutes over 12 hours) and height of tide in tenths of feet above mean low water) predicted for the 704 high tides at Portland, Maine 1975.

A) EXAMPLES

For 24 Dec 75, high tides of 9.2 and 9.3 feet were forecast for 0248 (2:48 am) and 1506 (3:06 pm). For 25 Dec 75 a high tide of 9.3 feet was forecast for 0.343 (3:43 am). Thus (1506, 34, 93) and (−343, 37, 93) appear below, where 34 = 1506 − 1448 with 1448 = 0248 − 1200, and 37 = 0343 − 0306, with 0306 = 1506 − 1200.

B) DATA TRIPLES

(0003, 32, 83) (0004, 32, 84) 0006, 13, 91) (0006, 26, 108)
(0008, 8, 103) (0009, 13, 90) (0009, 23, 86) (0010, 39, 92)
(0011, 27, 109) (0012, 8, 94) (0014, 12, 108) 0014, 21, 88)
(0024, 20, 109) (0024, 36, 88) (0025, 28, 84) (0025, 37, 97)
(0028, 7, 99) (0033, 8, 83) (0033, 33, 104) (0037, 18, 109)
(0038, 30, 84) (0041, 9, 101) (0041, 14, 107) (0041, 16, 89)
(0041, 21, 87) (0041, 32, 91) (0041, 33, 101) (0043, 34, 84)
(0049, 37, 94) (0050, 25, 86) (0053, 39, 98) (0056, 48, 103)
(0057, 39, 96) (0059, 41, 94) (0059, 29, 104) (0101, 25, 108)
(0102, 11, 103) (0103, 33, 83) (0107, 34, 89) (0110, 9, 90)
(0112, 7, 99) (0113, 19, 87) (0113, 29, 84) (0115, 10, 90)
(0117, 18, 87) (0117, 23, 105) (0124, 31, 104) (0125, 5, 96)
(0127, 28, 82) (0127, 27, 84) (0126, 41, 93) (0130, 9, 94)
(0134, 19, 104) (0137, 38, 96) (0140, 10, 85) (0141, 8, 91)
(0144, 51, 99) (0145, 40, 87) (0146, 35, 81) (0147, 11, 101)
(0148, 12, 87) (0149, 16, 87) (0150, 29, 84) (0152, 10, 98)
(0152, 23, 105) (0155, 13, 90) (0155, 33, 99) (0155, 37, 91)
(0156, 20, 83) (0201, 11, 97) (0208, 29, 80) (0211, 5, 91)
(0211, 43, 188) (0213, 26, 100) (0214, 40, 85) (0218, 47, 103)
(0219, 12, 93) (0219, 16, 100) (0224, 9, 87) (0227, 27, 83)
(0229, 18, 86) (0229, 21, 100) (0231, 16, 83) (0232, 37, 80)
(0233, 38, 84) (0235, 43, 90) (0239, 24, 80) (0240, 37, 98)
(0242, 16, 89) (0243, 16, 98) (0245, 11, 92) (0247, 23, 101)
(0248, 37, 92) (0253, 37, 92) (0256, 13, 96) (0256, 34, 78)
(0301, 10, 87) (0306, 39, 88) (0306, 42, 84) (0309, 26, 82)
(0312, 14, 83) (0312, 16, 92) (0342, 19, 85) (0312, 32, 94)
(0314, 37, 100) (0315, 15, 94) (0317, 21, 80) (0324, 39, 82)
(0325, 40, 80) (0327, 30, 77) (0328, 24, 95) (0333, 18, 88)
(0336, 45, 85) (0338, 37, 97) (0341, 14, 87) (0343, 21, 93)
(0343, 37, 93) (0346, 24, 95) (0350, 38, 77) (0356, 13, 83)
(0356, 18, 93) (0356, 28, 82) (0401, 16, 79) (0404, 21, 85)
(0405, 43, 87) (0405, 45, 81) (0406, 25, 76) (0412, 18, 91)
(0413, 30, 96) (0415, 18, 88) (0415, 37, 88) (0415, 39, 80)
(0421, 34, 75) (0426, 95, 82) (0432, 22, 88) (0432, 31, 90)
(0439, 35, 96) (0441, 15, 83) (0441, 47, 81) (0442, 35, 94)
(0447, 25, 91) (0447, 26, 82) (0449, 23, 90) (0150, 16, 79)
(0450, 42, 77) (0455, 22, 76) (0500, 21, 92) (0502, 23, 86)
(0503, 32, 74) (0505, 44, 79) (0506, 41, 90) (0506, 46, 82)

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exhibit **32** of chapter 8 (panel B, continued)

(0510, 37, 80)	(0517, 21, 91)	(0518, 29, 92)	(0521, 41, 74)
(0523, 22, 84)	(0523, 41, 84)	(0530, 46, 85)	(0538, 23, 89)
(0539, 36, 86)	(0541, 18, 80)	(0541, 23, 84)	(0542, 32, 94)
(0544, 46, 79)	(0546, 34, 96)	(0547, 20, 77)	(0551, 27, 75)
(0553, 31, 89)	(0554, 21, 87)	(0555, 46, 80)	(0559, 35, 73)
(0603, 41, 80)	(0604, 22, 88)	(0605, 28, 80)	(0607, 26, 91)
(0607, 37, 94)	(0613, 45, 80)	(0623, 23, 92)	(0623, 24, 90)
(0623, 40, 76)	(0625, 19, 82)	(0631, 41, 90)	(0634, 45, 81)
(0639, 21, 79)	(0641, 23, 87)	(0644, 23, 92)	(0644, 25, 77)
(0644, 26, 94)	(0645, 40, 80)	(0647, 26, 98)	(0647, 32, 74)
(0647, 41, 84)	(0655, 26, 82)	(0655, 41, 85)	(0656, 34, 81)
(0649, 19, 85)	(0700, 36, 88)	(0700, 39, 75)	(0706, 29, 98)
(0707, 20, 92)	(0713, 30, 92)	(0717, 42, 80)	(0724, 18, 82)
(0725, 42, 81)	(0727, 21, 90)	(0728, 83, 96)	(0735, 21, 80)
(0737, 18, 91)	(0737, 29, 77)	(0737, 42, 81)	(0740, 34, 82)
(0743, 37, 76)	(0745, 21, 84)	(0745, 26, 84)	(0745, 31, 94)
(0748, 24, 97)	(0749, 20, 106)	(0752, 43, 83)	(0753, 36, 92)
(0756, 37, 78)	(0759, 18, 86)	(0803, 35, 88)	(0804, 24, 103)
(0809, 20, 97)	(0812, 37, 82)	(0815, 18, 83)	(0815, 34, 93)
(0820, 37, 87)	(0822, 23, 81	(0823, 16, 90)	(0823, 26, 103)
(0826, 26, 85)	(0826, 32, 79)	(0828, 13, 87)	(0828, 19, 86)
(0830, 19, 86)	(0830, 29, 98)	(0833, 16, 97)	(0833, 37, 82)
(0833, 39, 78)	(0840, 15, 95)	(0846, 16, 102)	(0846, 44, 100)
(0847, 14, 101)	(0849, 15, 90)	(0849, 40, 84)	(0850, 41, 83)
(0859, 18, 106)	(0859, 30, 84)	(0902, 18, 85)	(0902, 40, 88)
(0905, 19, 103)	(0906, 20, 87)	(0907, 13, 89)	(0907, 26, 82)
(0910, 8, 90)	(0910, 32, 80)	(0911, 11, 91)	(0911, 33, 94)
(0913, 35, 94)	(0916, 22, 108)	(0921, 37, 82)	(0924, 35, 84)
(0926, 15, 103)	(0926, 29, 100)	(0931, 23, 95)	(0932, 14, 88)
(0938, 25, 106)	(0939, 13, 103)	(0939, 37, 89)	(0940, 25, 86)
(0942, 19, 86)	(0943, 40, 85)	(0943, 14, 89)	(0944, 17, 91)
(0944, 26, 105)	(0947, 39, 84)	(0950, 6, 93)	(0952, 15, 109)
(0955, 37, 82)	(0955, 40, 88)	(0957, 9, 91)	(1000, 20, 107)
(1002, 31, 101)	(1006, 37, 86)	(1006, 37, 94)	(1007, 17, 42)
(1007, 30, 85)	(1013, 15, 89)	(1014, 8, 95)	(1016, 20, 88)
(1019, 5, 93)	(1019, 13, 91)	(1019, 16, 107)	(1020, 22, 87)
(1021, 33, 101)	(1025, 29, 85)	(1027, 3, 95)	(1027, 21, 111)
(1027, 32, 95)	(1029, 37, 85)	(1036, 36, 85)	(1037, 10, 92)
(1037, 29, 106)	(1042, 13, 109)	(1043, 39, 88)	(1046, 27, 87)
(1048, 15, 90)	(1049, 17, 90)	(1051, 11, 92)	(1051, 26, 107)
(1053, 21, 109)	(1053, 31, 90)	(1054, 6, 95)	(1055, 4, 94)
(1055, 23, 87)	(1056, 37, 94)	(1057, 15, 113)	(1105, 32, 86)
(1107, 4, 98)	(1109, 17, 110)	(1110, 35, 86)	(1112, 8, 92)
(1112, 35, 100)	(1115, 32, 100)	(1117, 18, 113)	(1121, 14, 90)
(1121, 14, 91)	(1121, 25, 88)	(1122, 6, 93)	(1127, 29, 106)
(1127, 35, 88)	(1127, 35, 88)	(1127, 27, 88)	(1130, 25, 87)
(1131, 5, 94)	(1131, 10, 107)	(1132, 5, 95)	(1129, 36, 94)
(1140, 24, 110)	(1142, 35, 86)	(1144, 22, 109)	(1146, 9, 91)
(1146, 36, 92)	(1147, 31, 87)	(1148, 4, 99)	(1148, 14, 111)
(1153, 13, 91)	(1153, 22, 89)	(1156, 19, 89)	(1157, 7, 93)
(1200, 17, 110)	(1200, 37, 89)	(1202, 36, 98)	(1204, 27, 87)
(1204, 31, 104)	(1208, 5, 92)	(1208, 19, 113)	(1209, 5, 96)

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exhibit **32** of chapter 8 (panel B, continued)

(1212, 37, 81) (1219, 9, 104) (1219, 32, 103) (1220, 11, 90)
(1221, 34, 87) (1225, 11, 91) (1225, 19, 89) (1225, 30, 87)
(1227, 36, 97) (1228, 19, 88) (1230, 5, 93) (1230, 24, 111)
(1232, 38, 90) (1233, 9, 99) (1236, 25, 107) (1237, 12, 107)
(1241, 29, 86) (1244, 6, 90) (1246, 38, 91) (1250, 7, 96)
(1251, 37, 94) (1253, 20, 108) (1254, 13, 88) (1254, 30, 106)
(1259, 9, 91) (1259, 18, 89) (1259, 18, 110) (1301, 28, 87)
(1305, 8, 99) (1305, 24, 86) (1305, 37, 87) (1307, 14, 106)
(1310, 33, 98) (1311, 8, 92) (1315, 34, 100) (1318, 11, 98)
(1320, 39, 87) (1321, 8, 88) (1321, 32, 84) (1322, 23, 109)
(1328, 10, 102) (1329, 28, 102) (1331, 18, 85) (1333, 39, 85)
(1334, 7, 95) (1336, 19, 88) (1336, 26, 86) (1336, 32, 86)
(1336, 40, 93) (1342, 27, 83) (1342, 40, 90) (1347, 23, 103)
(1347, 30, 106) (1350, 38, 87) (1352, 15, 104) (1355, 9, 91)
(1355, 10, 94) (1400, 10, 85) (1403, 17, 102) (1403, 36, 92)
(1406, 41, 85) (1407, 37, 83) (1408, 34, 101) (1411, 16, 96)
(1411, 22, 82) (1415, 19, 87) (1415, 27, 86) (1415, 34, 84)
(1416, 21, 105) (1421, 16, 96) (1422, 14, 88) (1424, 13, 96)
(1424, 32, 95) (1426, 31, 81) (1427, 13, 93) (1427, 40, 94)
(1434, 42, 85) (1440, 27, 104) (1443, 16, 81) (1443, 25, 97)
(1443, 42, 87) (1445, 12, 88) (1445, 13, 90) (1451, 16, 98)
(1451, 40, 82) (1453, 24, 79) (1456, 25, 85) (1456, 37, 82)
(1457, 18, 66) (1500, 41, 86) (1501, 21, 97) (1504, 35, 100)
(1506, 18, 93) (1512, 16, 87) (1515, 33, 78) (1516, 23, 100)
(1520, 14, 90) (1522, 16, 91) (1522, 35, 88) (1522, 39, 95)
(1527, 42, 81) (1528, 19, 77) (1536, 12, 83) (1538, 26, 101)
(1538, 31, 88) (1541, 16, 89) (1541, 24, 84) (1543, 29, 89)
(1543, 31, 76) (1543, 42, 81) (1547, 20, 85) (1554, 18, 92)
(1554, 42, 82) (1557, 42, 81) (1601, 33, 100) (1604, 26, 91)
(1607, 24, 89) (1608, 18, 86) (1610, 37, 77) (1620, 23, 95)
(1621, 16, 85) (1621, 25, 74) (1623, 39, 96) (1625, 20, 89)
(1626, 40, 84) (1626, 45, 79) (1631, 25, 83) (1633, 18, 78)
(1633, 32, 82) (1634, 38, 80) (1639, 25, 74) (1639, 43, 90)
(1640, 19, 85) (1642, 27, 91) (1644, 18, 88) (1649, 36, 83)
(1656, 44, 83) (1658, 17, 88) (1701, 46, 77) (1703, 31, 98)
(1709, 19, 87) (1710, 31, 86) (1712, 30, 86) (1715, 43, 78)
(1718, 31, 72) (1722, 17, 81) (1722, 35, 97) (1723, 42, 78)
(1724, 21, 84) (1724, 29, 83) (1727, 32, 81) (1728, 22, 91)
(1730, 24, 88) (1733, 23, 76) (1733, 44, 77) (1741, 41, 93)
(1742, 40, 74) (1747, 22, 86) (1749, 26, 95) (1750, 20, 90)
(1759, 41, 79) (1800, 43, 87) (1805, 21, 85) (1800, 27, 98)
(1806, 43, 76) (1814, 19, 90) (1815, 24, 84) (1818, 36, 82)
(1818, 37, 73) (1818, 37, 79) (1819, 32, 82) (1821, 22, 86)
(1821, 35, 85) (1821, 43, 81) (1822, 19, 79) (1824, 31, 99)
(1835, 23, 89) (1837, 30, 89) (1840, 46, 76) (1843, 20, 90)
(1843, 36, 97) (1847, 43, 77) (1855, 21, 94) (1855, 24, 92)
(1903, 40, 92) (1906, 19, 86) (1906, 21, 84) (1906, 41, 77)
(1906, 43, 77) (1909, 22, 98) (1914, 35, 81) (1917, 17, 90)
(1917, 22, 94) (1919, 23, 79) (1919, 38, 75) (1924, 24, 100)
(1924, 29, 74) (1924, 40, 81) (1924, 40, 86) (1929, 42, 85)
(1935, 18, 89) (1940, 34, 90) (1941, 28, 101) (1941, 42, 76)

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exhibit **32** of chapter 8 (panel B, continued)

(1943, 18, 95)	(1949, 42, 82)	(1954, 11, 90)	(1954, 17, 83)
(1956, 19, 94)	(1957, 29, 95)	(1957, 33, 79)	(1959, 24, 84)
(2000, 20, 85)	(2001, 43, 98)	(2007, 40, 77)	(2009, 13, 94)
(2009, 17, 98)	(2009, 24, 80)	(2015, 30, 75)	(2017, 24, 99)
(2017, 40, 79)	(2022, 19, 102)	(2023, 35, 93)	(2025, 40, 81)
(2029, 17, 90)	(2030, 41, 86)	(2034, 35, 78)	(2038, 12, 90)
(2038, 18, 100)	(2038, 23, 104)	(2041, 19, 87)	(2041, 37, 93)
(2044, 11, 94)	(2044, 29, 82)	(2046, 20, 85)	(2046, 37, 88)
(2049, 16, 94)	(2054, 26, 81)	(2054, 31, 98)	(2057, 27, 104)
(2100, 37, 78)	(2012, 12, 77)	(2102, 13, 98)	(2012, 13, 100)
(2012, 34, 71)	(2111, 38, 95)	(2113, 27, 103)	(2115, 13, 103)
(2115, 16, 90)	(2118, 8, 93)	(2118, 11, 89)	(2118, 29, 81)
(2118, 31, 100)	(2118, 38, 81)	(2123, 21, 85)	(2126, 40, 87)
(2124, 18, 98)	(2129, 16, 107)	(2129, 23, 86)	(21312, 20, 105)
(2136, 29, 81)	(2137, 13, 94)	(2137, 38, 94)	(2140, 35, 95)
(2144, 34, 79)	(2148, 22, 108)	(2148, 37, 80)	(2150, 34, 100)
(2152, 20, 98)	(2155, 16, 104)	(2156, 9, 92)	(2157, 7, 91)
(2158, 16, 88)	(2158, 26, 83)	(2200, 5, 96)	(2203, 31, 91)
(2204, 9, 103)	(2206, 23, 87)	(2206, 28, 106)	(2206, 35, 82)
(2208, 24, 106)	(2214, 30, 83)	(2216, 10, 102)	(2219, 13, 107)
(2219, 12, 94)	(2219, 13, 107)	(2219, 40, 89)	(2224, 34, 81)
(2225, 23, 108)	(2226, 10, 102)	(2227, 30, 82)	(2229, 37, 94)
(2232, 12, 90)	(2232, 15, 91)	(2232, 32, 102)	(2233, 8, 94)
(2233, 20, 86)	(2235, 6, 98)	(2237, 16, 110)	(2240, 4, 99)
(2240, 21, 87)	(2242, 35, 101)	(2243, 16, 108)	(2248, 32, 82)
(2251, 32, 83)	(2252, 9, 102)	(2252, 33, 97)	(2256, 10, 94)
(2258, 21, 110)	(2259, 32, 106)	(2303, 10, 105)	(2303, 36, 83)
(2304, 27, 95)	(2305, 10, 92)	(2306, 10, 107)	(2307, 2, 96)
(2307, 19, 80)	(2307, 19, 88)	(2316, 6, 97)	(2316, 25, 86)
(2316, 25, 109)	(2321, 39, 94)	(2322, 29, 106)	(2323, 6, 102)
(2326, 14, 110)	(2326, 32, 83)	(2327, 32, 84)	(2331, 10, 93)
(2333, 18, 109)	(2334, 37, 99)	(2335, 8, 100)	(2337, 7, 93)
(2337, 16, 89)	(2337, 25, 85)	(2340, 19, 89)	(2343, 34, 101)
(2344, 37, 85)	(2347, 5, 98)	(2347, 20, 111)	(2349, 32, 104)
(2350, 28, 86)	(2351, 12, 107)	(2354, 8, 104)	(2355, 8, 95)

P) PROBLEMS (all large)

32a) Analyze the (time, interval) pairs, using the methods of this chapter. Discuss your results.

32b) Analyze the (time, height) pairs, using the methods of this chapter. Discuss your results.

32c/d/e/f/g/h) Analyze only those (time, height) pairs where the next-to-last digit of the time is 0/1/2/3/4/5.

32i/j/k/l/m/n/o/p/q/r) Analyze only those (time, interval) pairs where the last digit of the time is 0/1/2/3/4/5/6/7/8/9.

S) SOURCE

Tide Tables, 1975, East Coast of North and South America, including Greenland, pp. 32–33. U.S. Department of Commerce, National Oceanic and Atmospheric Administration. 1974.

review questions

What happened when we looked up changes in governor's salary? What did this suggest about smoothing over years? Did we dare look at the wandering schematic plot to "see" vertical width directly? Why/why not? How did vertical width seem to behave in logarithmic units? In raw dollars? What seemed to be the main conclusions about this example?

Do we believe it is generally safe to work with middle and hinge traces based directly on the data? When/when not? What do we do to avoid trouble? Can we plan to combine? What other sorts of things might we have subtracted in this example? Would we be free to do similar things generally?

8H. How far have we come?

This chapter started with parallel schematic plots, and progressed to wandering schematic plots. It is our first major stage in providing summaries for batches of (x, y) points that can help us to understand:

- ◇the broad pattern of distribution of most points,
- ◇the existence and location of unusual points,

just as schematic plots did for us when dealing with batches of x-values (or of y-values).

We are now ready to:

- ◇slice up an (x, y) batch according to its x-values.
- ◇do this using letter cuts, slice sizes selected by judgment, or slice sizes selected from such sequences as 5, 9, 17, 35, . . . , or 8, 16, 32, . . .
- ◇find medians and hinges, within each slice, separately for both x-values and y-values.
- ◇to replace, for a while, the y-hinges by letter-value differences and to smooth the results, obtaining a median trace and two letter-difference traces; and then, by addition and subtraction, two hinge traces.
- ◇to be careful which x-hinge is matched with which y-hinge for each slice.
- ◇to be prepared to go further, as in section 8F, when necessary.
- ◇to go on, finding fences, outside and far-out points, and adjacent polygons, thus being ready to draw wandering schematic plots.

We now understand that it may be reasonable to:

- ◇copy on in smoothing medians, but use the end-value rule in smoothing letter-differences or hinges.
- ◇to subtract from y known functions of x before finding median and hinges, restoring what was subtracted in due course.

◇to combine results from different subtractions before finalizing median and hinge traces.

If we have covered optional section 8I (below), we understand why:

◇we expect to smooth x whenever we smooth y.

8I. The need to smooth both coordinates (optional)

In section 8B, we came across--for the first time--the need to smooth a sequence of points in the plane. In such a case, we work with two matched sequences, one of the first coordinate, one of the second. If, as was the case in section 8C, the sequence of first coordinates comes to us in order, we might think it would be enough to smooth only the second coordinate.

How might we test such a view? Under what circumstances can we say "Any reasonable smooth of this pair of sequences must do thus-and-so!"? Surely if the initial points are on a straight line, it would be hard indeed to argue for smoothed points that do not fit on the same straight line.

Exhibit **33** shows arithmetically what can happen when the data all lie on a single straight line, here

$$y = 2x$$

and we smooth only the second, y, coordinate, using 3R'SSH or 3R'SSH3. (What would have happened with 3R or 3RSS?) Exhibit **34** shows the picture. Clearly, we go off the line! And, as a result, to most of us it would be clear that the sequence of points was rougher than before. (More nearly evenly spaced, perhaps, but rougher.)

There is no escape. When we smooth one coordinate, we are almost always wise to smooth the other coordinate in just the same way.

exhibit **33** of chapter 8: illustrative data

Smoothing one of two coordinates

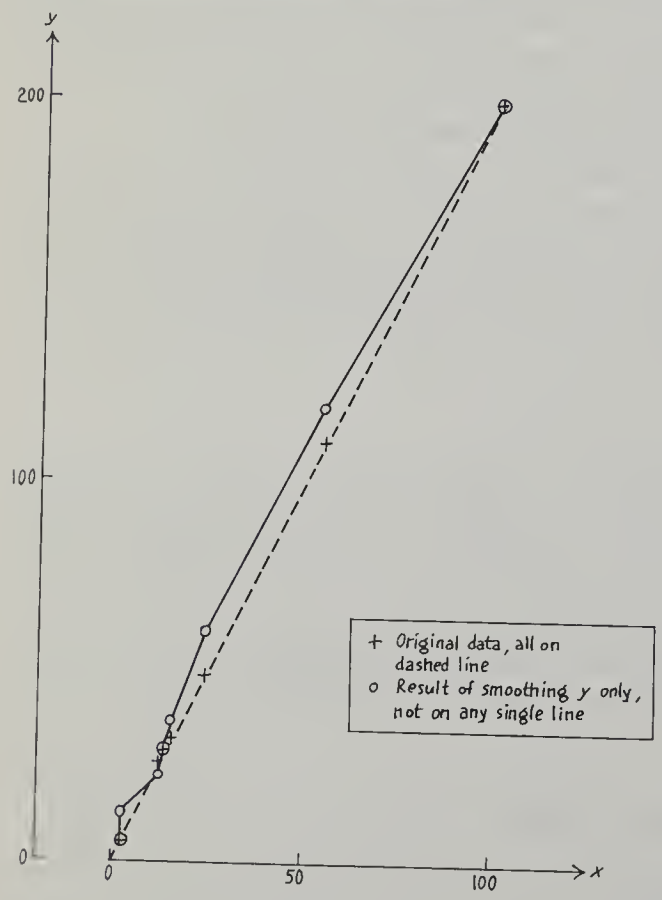
	y, the second coordinate					y minus $2x$
x	raw	3R'	3R'SS	3R'SSH	3R'SSH3	final smooth
0	0	T	T	0	T	0
1	2	h	h	2	h	0
2	4	e	e	8h	e	4h
12	24			20		−4
14	28	s	s	28	s	0
16	32	a	a	36	a	4
26	52	m	m	62	m	10
56	112	e	e	119	e	7
100	200			200		0

review questions

Why is there a problem? What test can we impose on a smoother of points? What was our example like? What happened? What must we conclude?

exhibit **34** of chapter 8: illustrative only

How smoothing only one coordinate can roughen a point sequence



Delineations of batches of points 9

chapter index on next page

If we want to be as effective in summarizing batches of points as schematic plots have made us in summarizing batches of numbers, we need to do rather more than we did in the last chapter. Wandering schematic plots tell us quite a lot about y, but we could be told more. And we need to be told about x at the same time.

Actually, we will find that we already have all the tools we need. We can carry further some of the things we already do; we can display some of the things we usually calculate. So this will be a short chapter.

review questions

Are wandering schematic plots effective enough in summarizing batches of points? Why/why not? What might we do? Will it take new tools? A long chapter?

9A. E-traces and D-traces

Our hinge, or H-traces were found, in section 8D, by starting with the identities

$$H_{lo} = M - (\text{H-to-M})$$

$$H_{hi} = M + (\text{M-to-H})$$

and then

- ◇smoothing M to a middle trace.
- ◇smoothing H-to-M--and, separately, M-to-H
- ◇recombining (by addition or subtraction) to get smoothed H's, which we then take as H-traces.

If we want E-traces, we can start from

$$E_{lo} = H_{lo} - (\text{E-to-H})$$

$$E_{hi} = H_{hi} + (\text{H-to-E})$$

smooth E-to-H--and, separately, H-to-E--and combine the results with the H-traces we already had found. The results are naturally taken as E-traces.

Taking further steps going on to D-traces--thence to C-traces and beyond--introduces nothing new, just more arithmetic of the same kind. So let us plan to do this.

review questions 309

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review questions

Just how do we find an upper E-trace? A lower E-trace? An upper D-trace? How are these calculations related to one another? Do we have to learn anything new? Will we plan to go ahead?

9B. Simple delineation--Twin Rivers again

Let us go back to the Twin Rivers example of the last chapter, doing two things:

- ◇adding E-traces and D-traces over all that part of the data for which they are well defined.
- ◇adding the vertical letter cuts.

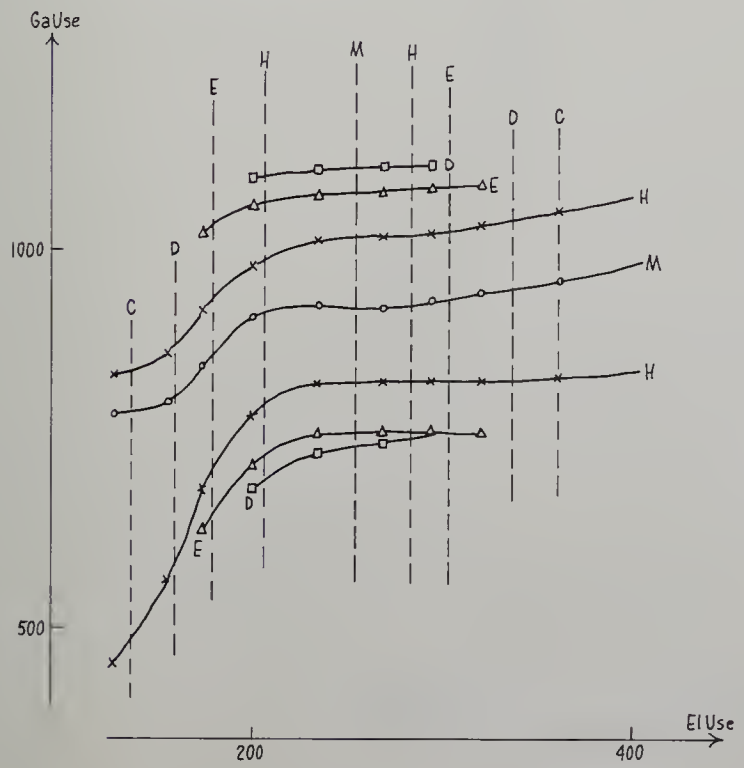
The result, shown in exhibit **1**, is appropriately called a full--or simple--

delineation

of the batch. We now see the same things we saw (in exhibit 14 of chapter 8) earlier, but now even more clearly presented.

exhibit **1** of chapter 9: Twin Rivers

A full delineation of the (ElUse, GaUse) scatter (GaUse = gas usage, ElUse = electric usage)



an engineering example

In their book, Daniel and Wood (C. Daniel and F. S. Wood, 1971, *Fitting Equations to Data*, Wiley–Interscience) discuss quite carefully the fitting of suitable equations to 105 observed values of heat of solution for 14 cements. If we turn to their pages 255 and 256, we find 104 (fitted value, residual) pairs (one observation has been set aside as discrepant) which, when delineated, look as shown in exhibit **2.**

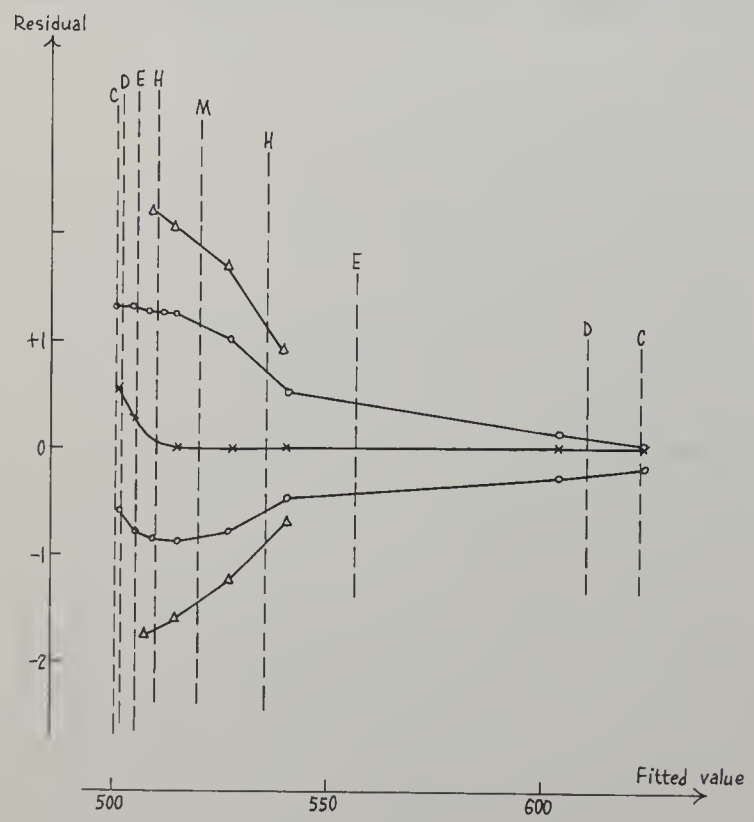
We see several things more or less clearly:

- ◇a squeezing-together of the fitted values at the low end (toward 500) (seen from the vertical lines at the letter cuts).
- ◇a wedging-together of the residuals--a decrease in spread--as the fitted values increase (seen from the H-traces).
- ◇perhaps a tendency to high residuals for the very lowest fitted values (seen from the H- and M-traces).

The first of these was to be expected from the nature of what was being done. As the cement samples aged, their heat of solution was known to decrease,

exhibit **2** of chapter 9: heats of solution

Residuals delineated against fitted values for the heats of solution of 104 cement samples



flattening out to a constant value for each cement, these constant values being close to equality. With data taken in equal time steps, such squeezing together at small heats is only to be expected.

The strikingly larger residuals at the lower fitted values clearly deserve an explanation. We do not have one handy.

The upward kink is interesting, but it would be more so if we were surer that it was not an accident.

review questions

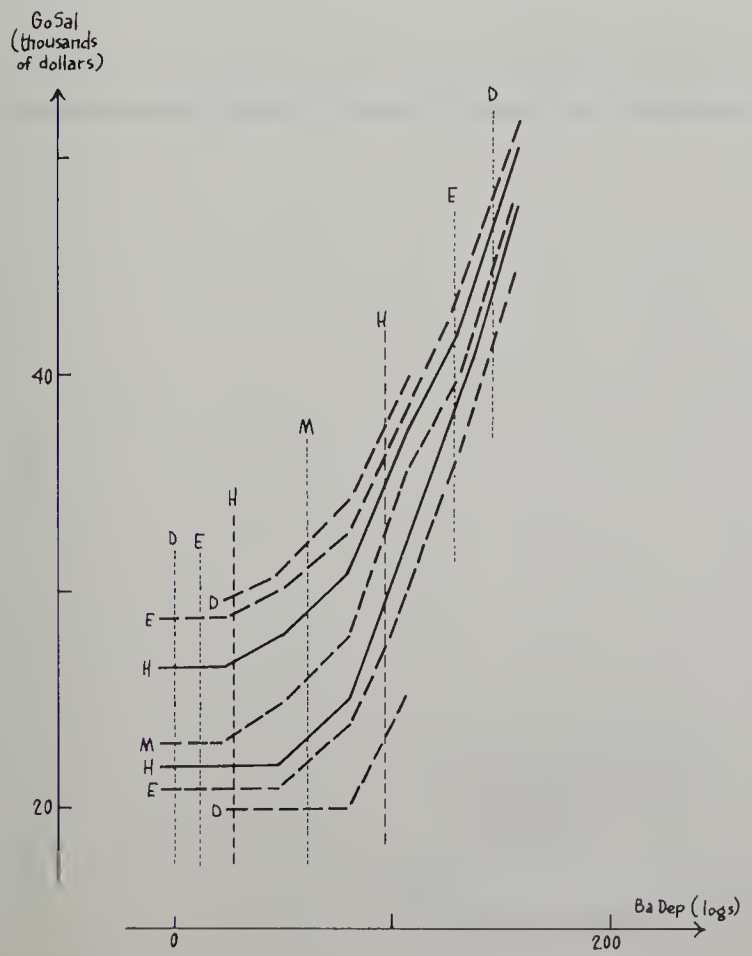
What did we add? How symmetrically or unsymmetrically does the result trace x and y? (Be specific.) What do we call it? How well did it work in the example?

9C. Reduced and schematic delineations

Exhibit **3** shows a full delineation of (raw) governor's salary, 1969, against (log) bank deposits, 1968. (Recall that we used log governor's salary in chapter

exhibit **3** of chapter 9: governor's salary

A full delineation of governors' salary, 1969, against log bank deposits, 1968



8.) Notice that we have used lighter marking, · · · ·, and ----, for the verticals than for the traces, —— and – – – –. The reason for this is simple; straight lines, especially when parallel, seize the eye more effectively than do broken lines. If we are to give the verticals and the traces equal chance to seize the eye, we must favor the traces in how we show them.

Some, perhaps many, will find exhibit 3 somewhat too detailed. What can we do for them? What detail can we strip out?

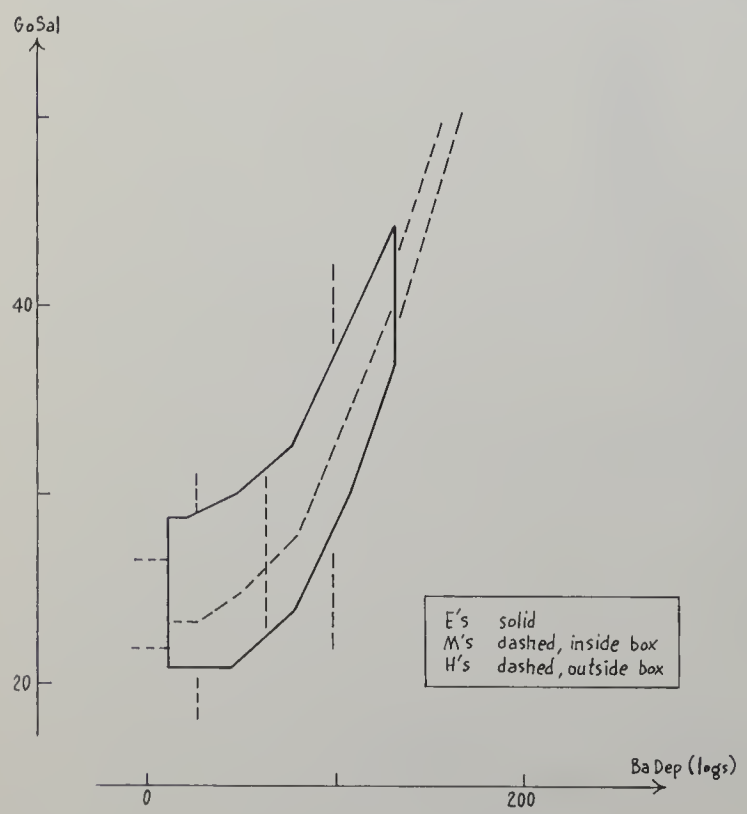
Exhibit **4** shows the result of one such stripping out. The E-verticals and E-traces have been used to make a box. The M-vertical and M-trace have been shown dashed, ONLY WITHIN the E-box. The H-verticals and H-traces have been shown dashed, ONLY OUTSIDE the E-box. This

reduced delineation

does a quite reasonable job of showing us, schematically, what the scatter of 50 (log bank deposits, Governor's salary) points are like. What more can we ask?

exhibit **4** of chapter 9: governor's salary

Reduced delineation of governors' salary, 1969 against log bank deposits, 1968



schematic (x, y)-plots

Our schematic plots for x alone used fences, outside and adjacent points. In the last chapter (8), we introduced y fences and identified y-outside and adjacent points. We need, in addition, only look at the x-values--since the letter cuts are vertical lines, looking at x's from (x, y) pairs is now no different from looking at x's alone. Thus providing x-fences is simple.

In this governor's salary example, we have chosen to take our human prerogative seriously, and treat Alaska distinctively. Exhibit **5** shows all nonoutside points, with x's for those we are going to treat as adjacent. (Except for Alaska, which appropriately enough appears on the NW of this plot, this means exactly the points we would have taken as adjacent if Alaska were excluded.)

Exhibit **6** shows the completed

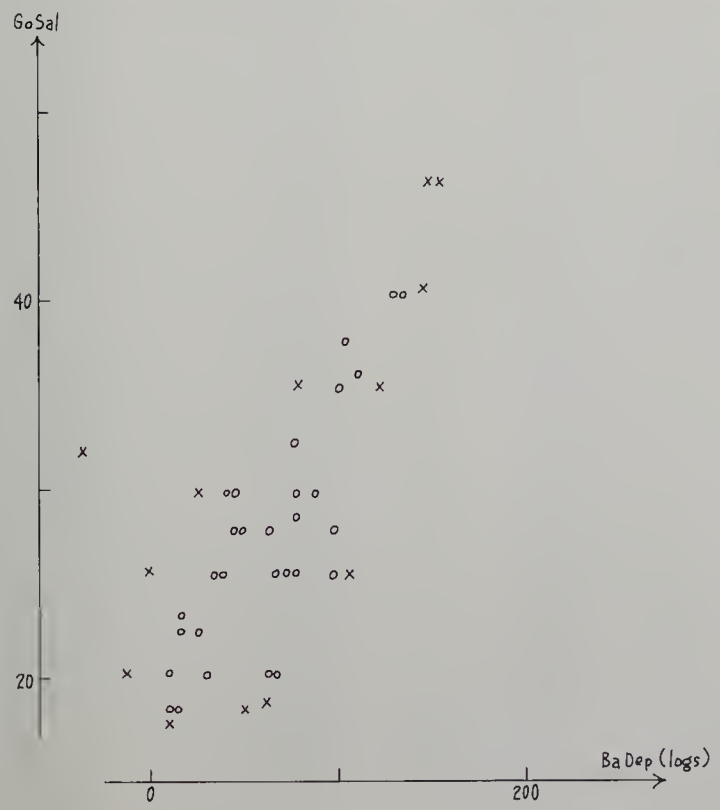
two-way schematic plot

What about the states found outside?

◇Delaware is a geographically small state and probably does considerable banking in Philadelphia and Baltimore.

exhibit **5** of chapter 9: governor's salary

All not-outside points, with the chosen rim points shown by ×'s



◇Hawaii and Alaska are noncontiguous--whether greater independence or mainland banking accounts for their location on this plot is questionable.

◇Arkansas, with a 10,000-dollar salary, clearly expected its governor to gain his living some other way.

◇New York was outside on bank deposits alone.

◇Georgia offers a problem--who knows the answer?

◇Massachusetts is formally outside, but very close to the E-box--probably an accident.

◇California is close to the fence, but below--for a state that large, perhaps a somewhat higher salary is to be expected. (It was raised the next year.)

(See also comments in Section 8G.)

Exhibit **7** sets some problems.

exhibit **6** of chapter 9: governor's salary

The completed schematic plot

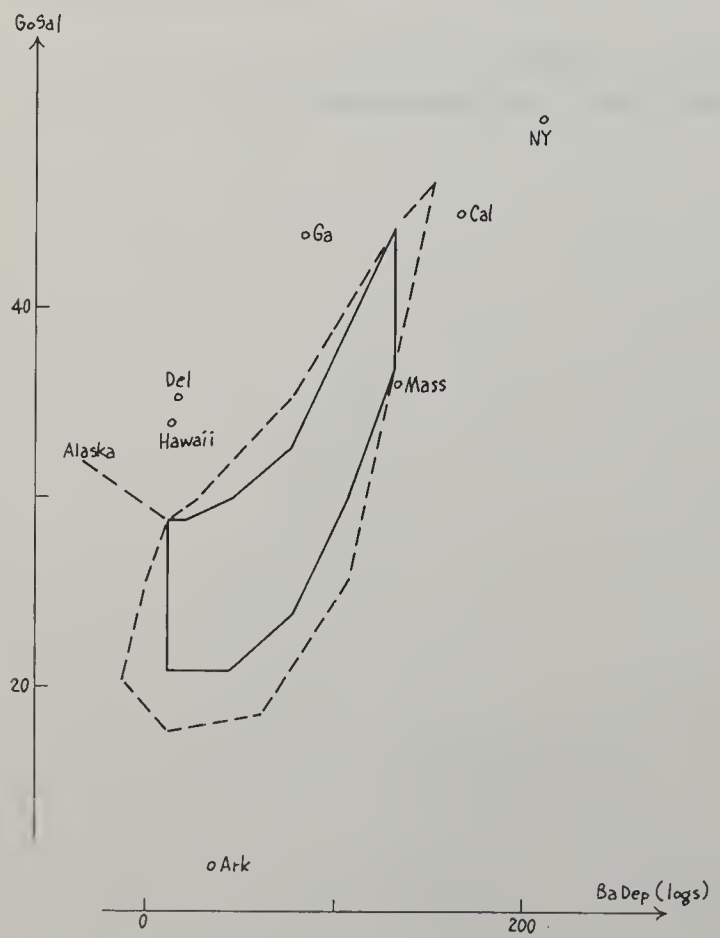


exhibit **7** of chapter 9: data and problems

Some problems using schematic plots

A) MARRIAGES and MOTOR VEHICLE DEATHS by STATE, 1958

	Marriages	Motor vehicle deaths	*x*	*y*
Ala	24,444	852	39	93
Ariz	9,917	508	0	71
Ark	15,574	444	19	65
Calif	96,330	3510	98	155
Colo	14,688	396	17	60
Conn	16,879	251	23	40
Del	2,311	84	−64	−8
DC	8,094	62	−9	−21
Fla	35,243	1134	55	105
Ga	45,863	956	66	98
Ida	9,520	270	−2	43
Ill	82,860	1886	92	128
Ind	41,226	1056	62	102
Iowa	25,101	598	40	78
Kans	15,481	554	19	74
Kent	26,095	789	42	90
La	21,447	844	33	93
Maine	7,104	205	−15	31
Maryl	41,403	505	62	70
Mass	45,959	590	66	77
Mich	53,662	1375	73	114
Minn	23,410	707	37	85
Miss	36,198	548	56	74
Mo	33,976	975	53	99
Mont	6,160	193	−21	29
Nebr	10,637	346	3	54
Nev	5,755	190	−24	28
NH	7,078	101	−15	0
NJ	39,113	754	59	88
NM	5,850	408	−23	61
NY	124,513	2118	110	133
NC	27,228	1081	43	103
ND	4,306	155	−37	19
Ohio	65,479	1812	82	126
Okla	33,444	667	52	82
Oreg	9,798	448	−1	65
Penna	64,529	1654	81	122
RI	5,653	74	−25	−13
SC	38,550	610	59	79
SD	5,662	240	−25	38
Tenn	27,947	719	45	86
Tex	89,702	2342	95	137
Utah	6,741	191	−17	28
Vt	3,376	77	−47	−11
Va	36,588	861	56	94
Wash	27,966	573	45	76

→

exhibit **7** of chapter 9 (continued)

WVa	14,213	387	15	59
Wisc	25,073	822	40	91
Wyo	2,945	137	−53	14

x = −400 plus 100 times log of number of marriages, 1958.

y = −200 plus 100 times log of number of motor vehicle deaths, 1958.

B) NUMBERS OF STAMENS and PISTILS for 268 EARLY FLOWERS of *Ranunculus ficaria*

Number of pistils, y	Number of stamens, x																	
	18	20	22	24	26	28	30	32	34	36	37	38	(Sum)
≤6											**(37, 2)**		(1)
8							**1**						(1)
10		1			1								(2)
12			2		1								(3)
14		3	1	2	1	1	3	1	1									(13)
16		1	1	1	4	1	1	1			1	1						(12)
18		4	3		1	2	4	1	2	3	1	1						(22)
20			1	2	4	3	7	4	4	5	2	1		1	1			(35)
22	1			2			1	5	3	5	4	5	3	2				(31)
24					2	2	1	4	2	3	1	2	5	2	1			(25)
26				1		2	4	3	1	7	1	3	2	1		2		(27)
28						1	2		5	4	4	1	1	1	1	1		(21)
30				2		1	1	2	2		1	3	4	2	1			(19)
32				1				2	2			2	4	1	1			(13)
(Sum)	1,	6,	8,	9,	16,	12,	22,	26,	26,	38,	14,	23,	20,	20,	13,	7,	1,	4,	1,	1	268

→

review questions

For which familiar example did we present a full delineation? How did we differentiate the lines and traces? Why did we do this? What is a reduced delineation? How did it seem to work in this example? What were the essentials in a schematic plot (or a *y*-scatter)? How can we imitate a "middle half" box using parts of a full delineation? What is an E-box? How hard is it to find fences? *x*-wise? *y*-wise? How did we find the latter? What did we include in a finished schematic plot? Can we reasonably make modifications based on human judgment? Why/why not? What modifications did we make in our example? What did the schematic plot seem to tell us?

9D. What our schematic plots and delineations have missed

This section of chapter 9 is intended to show us something of what we have missed--missed because our summaries are designed to miss it/them.

pattern of salaries

As a schematic device, we expect delineation to smooth out irregularities, and show us the smoothed-out picture. It does as we expect. Exhibit **8** shows the detailed distribution of Governor's salaries, one stem per thousand dollars; this points out to us what we already knew, that most salaries (30 out of 50) are in multiples of 5,000 dollars. (And of the 20 that remain, 8 are other multiples of 2,500 dollars and 8 other multiples of 2,000 dollars.)

exhibit **7** of chapter 9 (continued)

P) PROBLEMS

7a) Delineate *x* of panel A against *y*. Comment.

7b) Delineate *y* of panel A against *x*. Comment

7c) Delineate number of stamens of panel B against number of pistils. Comment.

7d) Delineate number of pistils of panel B against number of stamens. Comment.

7e) (uses a and b) Compare the results of 7a and 7b. Comment.

7f) (uses c and d) Compare the results of 7c and 7d. Comment.

S) SOURCES

For A: The World Almanac for 1960. Pages 306 and 308.

For B: L. H. C. Tippett, 1952. *The Methods of Statistics,* 4th edition. New York, John Wiley.

exhibit **8** of chapter 9: governor's salary

The details of governors' salaries, 1969

A) The EXPANDED STEM-and-LEAF

GoSal

10 0

17 h
18 888h
19
20 00000
21
22 2
23 q
24
25 5555555555
26
27 hh
28 xh
29
30 00000
31
32 2h
33 h
34
35 55555
36 6
37 h
38
39
40 000
41
42 h
43
44 d
45 55

50 0

Note that 32 of 50 end in 0 or 5
Note that 888h after **18**| means 18, 18, 18h, and so on.
(q = and a quarter, x = and .374, d = and .1)

B) The THOUSANDS DIGIT

(15)	**0***	000000000000000
	0·	
	1*	
	1·	
(2)	**2***	22
(2)	**2·**	hh
	3*	
(2)	**3·**	qh
	4*	
(1)	**4·**	d
(17)	**5***	55555555555555555
	5·	
(1)	**6***	6
	6·	
	7*	
(4)	**7·**	hhhh
(3)	**8***	888
(3)	**8·**	hxh
	9*	
	9·	

P) PROBLEM

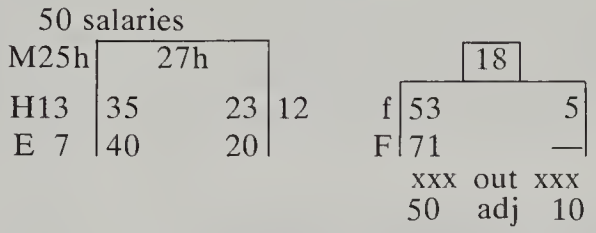
8a) Find the corresponding data for another year and make the corresponding analysis.

S) SOURCE

The 1970 World Almanac, page 57. (Not for the same date as in the examples of chapter 8.)

Any smooth-looking picture that shows us some aspects of our (BaDep, salary) swarm must--because it is smooth--fail to show us this lumpiness in governor's salaries. This may be good--and often is. It may also be bad--and sometimes is. It is an essential characteristic of any smooth schematic picture, whether we like it or not. We just need to know that it happens.

The same thing, of course, happens with a one-way schematic plot for these same 50 salary values, where the letter values are:



so that the one-way schematic plot is perfectly bland, with no indication of any granularity.

A similar thing would, of course, happen if our swarm had diagonal stripes instead of horizontal stripes--or if any of a variety of detailed patterns had occurred. We also need to know that such things happen. We also need to know that we shouldn't let their possibility worry us whenever it is reasonable for us to take a thoroughly schematic view.

review questions

What do we expect delineation to show us? What do we want it to overlook? What about governors' salaries? Can any smooth summary tell about such lumps? In particular, what about an ordinary schematic plot for governors' salaries alone? Are there other kinds of lumping that a delineation should and would miss? What are some examples?

9E. Three variables at once--or more

handling more variables

So long as we have only two variables--and want only to delineate one against the other--the extended stem-and-leaf techniques in which the leaves are "(trailing digits of x, all of y)" is quite reasonable. Once we have several variables, and may want to delineate any one against any other, the thought of copying down all the values many times is worse than just boring. We need another approach.

Our problem is to sort data-sets, at least as pairs, probably as wholes; to do this easily; to divide into slices; to sort each slice (on a second variable), and find its letter values. What is easy to sort, to divide into parts, and to re-sort? One obvious answer is a deck of index cards, $3'' \times 5''$ for modest numbers of variables, and $4'' \times 6''$ or $5'' \times 7''$ for larger numbers of variables.

index cards

Exhibit **9** shows how this could be done, for instance with the 88 unincorporated places over 25,000 population found in the 1960 U.S. Census. We write the names and units of the quantities we want to work with in easily recognizable places on a master 3 × 5 card, and then put identification on each detail card, together with the values of the chosen variables, each in the place corresponding to its name on the master card.

We can now easily shuffle (unimportant), sort (in order according to any chosen one of the eight variables), and make piles of these cards, carrying out all the operations needed to form slices on any of the 8 variables and then to order any slice on any of the 7 variables remaining. Exhibit **10** offers some data and problems.

exhibit **9** of chapter 9: unincorporated places

Illustration of master 3 × 5 card and two of the 88 unincorporated places in the 1962 *County* and *City Data Book* (8 selected quantities)

U.S. rank (pop, 1960)	Land area (sq. mi)	% nonwhite 1960
% housing units with automobile	MASTER	% foreign-born
% housing units with > 1.01 persons/room	% completed less than 5 yrs school	% native with a foreign-born parent

37	8.6	5.5
92.0	Altadena, Calif.	10.0
3.9	1.8	18.4

78	8.4	1.5
91.4	Inglewood, Tenn.	0.5
4.7	2.8	1.6

Data and problems

A) The DATA--variables as in exhibit 9

Place	U.S. rank	Land area (sq mi)	% non white	% foreign born	% native with for. parent	% <5 years school	% Housing units ≥1.01 pers/room	% Housing units using automobile
CALIFORNIA								
Altadena	37	8.6	5.5	10.0	18.4	1.8	3.9	92.0
Arden-Arcade	6	22.0	0.8	3.6	13.9	0.9	6.7	97.7
Bell Gardens	79	2.5	1.4	2.3	6.8	6.6	21.4	87.3
Carson	42	8.4	2.5	4.5	11.9	4.9	23.1	95.5
Castro Valley	44	6.4	0.7	5.7	18.8	3.0	6.5	94.2
East Los Angeles	1	7.9	3.8	20.3	34.0	19.3	22.8	74.6
Florence-Graham	41	3.0	45.8	8.4	13.6	14.4	23.8	71.3
Lancaster	84	11.3	0.6	2.8	11.2	1.5	8.3	93.7
Lennox	57	2.4	1.4	6.5	13.8	3.1	10.7	86.7
South San Gabriel	82	3.9	0.8	7.9	18.1	6.4	12.4	87.6
Temple City	54	5.1	0.4	7.3	17.7	2.1	3.9	91.9
West Hollywood	66	1.9	0.6	24.0	30.0	4.2	1.7	81.3
CONNECTICUT								
East Hartford	31	18.2	0.8	7.8	27.3	3.9	10.7	82.0
Enfield	56	33.2	0.7	7.7	26.1	5.9	9.9	89.5
Fairfield	25	29.9	0.6	9.4	30.3	2.9	5.1	93.7
Greenwich	18	47.6	2.2	13.0	25.4	4.4	4.5	91.6
Hamden	36	24.8	1.5	9.7	32.1	4.0	3.3	90.8
Manchester	34	27.5	0.4	10.7	28.9	4.0	5.2	88.4
Stratford	27	18.5	3.0	8.9	31.6	4.8	6.5	93.1
Wallingford	62	41.6	0.1	9.1	26.7	4.5	9.6	90.3
West Hartford	10	21.2	0.4	11.7	31.1	2.9	1.8	91.5
West Haven	32	10.9	2.0	9.1	31.1	5.5	6.1	87.5

exhibit **10** of chapter 9 (panel A, continued)

Place	U.S. rank	Land area (sq mi)	% non white	% foreign born	% native with for. parent	% <5 years school	% Housing units ≥1.01 pers/room	% Housing units using automobile
FLORIDA								
Brownsville	40	10.5	15.4	1.1	2.8	8.2	17.9	87.6
HAWAII								
Kailua-Lanikai	87	5.4	40.1	3.6	17.6	3.7	18.9	94.3
MARYLAND								
Bethesda	13	14.3	1.4	6.0	13.7	0.8	2.0	84.8
Catonsville	43	6.2	7.3	3.1	10.7	3.8	3.9	88.6
Dundalk	4	10.6	8.2	2.5	11.4	4.6	10.8	83.5
Essex	48	5.7	0.6	1.8	8.4	4.7	10.2	88.3
Parkville-Carney	74	6.1	...	3.3	12.0	3.5	4.8	91.7
Silver Spring	7	9.3	1.7	5.7	18.2	1.2	3.3	94.4
Wheaton	15	9.2	1.4	3.3	13.5	1.4	9.3	98.6
MASSACHUSETTS								
Arlington	23	5.1	0.3	12.1	32.1	2.8	4.4	87.2
Belmont	68	4.6	0.3	13.7	31.7	3.2	2.4	89.2
Braintree	58	14.3	0.2	8.0	28.3	1.9	6.8	95.0
Brookline	16	6.8	1.0	19.5	35.8	3.1	2.8	72.0
Framingham	30	24.1	0.6	8.5	25.0	3.8	6.2	84.8
Lexington	73	16.5	0.5	9.5	28.7	3.4	4.1	97.0
Methuen	71	22.5	0.3	12.7	34.8	7.0	6.1	85.6
Milton	81	12.9	0.1	11.3	32.1	1.9	1.9	93.0
Natick	67	15.1	0.5	8.2	26.1	2.5	6.2	89.9
Needham	86	12.5	0.3	8.2	25.9	1.9	1.9	95.4
Watertown	39	4.2	0.2	16.7	35.0	6.2	5.9	80.8
Wellesley	83	9.9	0.4	7.8	20.2	1.9	1.3	94.1
Weymouth	24	17.7	0.3	6.6	25.2	1.9	8.7	89.2

NEW JERSEY								
Cranford	80	5.0	3.2	7.1	25.5	2.6	3.2	93.3
Delaware	55	24.4	1.9	4.6	19.2	3.2	4.2	96.4
Edison	29	31.0	3.3	7.9	26.4	4.5	9.0	93.3
Ewing	77	15.4	6.6	6.5	22.2	5.3	7.2	93.2
Hamilton	9	40.3	2.8	7.1	25.2	6.1	6.1	91.2
Middletown	38	54.9	2.6	5.8	19.2	3.0	7.3	95.4
New Hanover	69	24.4	9.2	3.6	15.3	0.7	18.2	85.9
North Bergen	33	5.6	0.3	17.7	34.5	6.8	6.5	75.4
Parsippany-Troy Hills	88	23.7	2.6	10.8	25.0	6.2	7.1	97.7
Pennsauken	51	10.4	4.5	4.9	21.0	4.6	4.1	89.4
Teaneck	35	6.1	4.2	11.8	33.0	2.8	2.8	90.2
Union	21	9.2	6.0	12.2	33.0	4.4	3.1	92.9
Wayne	64	25.3	0.3	7.5	25.9	3.0	5.4	97.4
Woodbridge	5	24.1	1.6	8.1	30.4	5.1	8.2	93.5
NEW YORK								
Baldwin	60	4.1	0.6	8.0	27.2	2.0	2.6	90.7
Cheektowaga-Northwest	19	6.3	0.1	6.1	25.3	2.6	8.5	92.4
East Meadow	26	6.7	2.0	6.7	31.6	2.5	5.8	94.7
Eggertsville	28	10.9	0.2	6.6	23.2	1.2	3.1	84.7
Elmont	61	3.2	2.5	12.0	34.4	6.4	7.5	89.4
Franklin Square	53	2.8	0.2	11.5	34.0	4.2	3.4	93.1
Hicksville	22	6.7	0.4	6.7	27.6	2.9	7.1	95.6
Levittown	8	6.6	0.3	5.8	26.7	1.7	10.1	95.6
Massapequa	52	5.4	0.3	7.2	28.3	2.6	4.2	96.6
Oceanside	59	5.0	0.7	8.8	31.4	2.6	2.5	94.7
Plainview	72	3.6	0.4	6.3	33.7	1.7	2.6	98.5
Tonawanda	3	17.6	0.3	6.5	22.2	1.7	7.4	94.0
Wantagh	50	5.5	0.3	6.4	27.5	2.0	3.8	97.8
NORTH CAROLINA								
Kannapolis	49	27.6	11.6	0.2	0.3	15.7	12.0	81.2
PENNSYLVANIA								
Abington	14	16.3	4.8	5.7	20.2	2.3	2.7	92.7

exhibit **10** of chapter 9 (panel A, continued)

Place	U.S. rank	Land area (sq mi)	% non white	% foreign born	% native with for. parent	% <5 years school	% Housing units ≥1.01 pers/room	% Housing units using automobile
Bristol	12	16.9	4.0	3.4	17.1	2.3	14.5	95.6
Cheltenham	45	9.0	2.4	7.8	27.7	2.6	1.5	90.4
Falls	65	27.2	0.7	2.5	16.3	1.8	13.3	97.7
Haverford	17	10.0	1.3	6.7	21.6	2.8	2.0	93.8
Hempfield	63	84.4	1.6	3.8	17.5	7.8	11.4	90.2
Lower Merion	11	24.0	4.9	7.3	19.7	2.9	1.4	89.2
Middletown	75	20.7	0.4	3.9	17.0	1.1	5.5	95.9
Millcreek	70	35.6	0.4	2.9	15.6	3.8	7.8	93.3
Mount Lebanon	47	6.4	0.2	3.7	18.3	1.1	1.3	89.3
Penn Hills	20	20.7	4.9	4.3	19.3	2.9	7.0	93.3
Ridley	46	5.5	1.7	4.0	18.6	3.9	6.5	91.8
Ross	85	13.8	1.4	4.0	18.9	1.8	5.5	94.4
Springfield	76	6.9	9.3	4.7	20.5	1.8	2.9	97.1
Upper Darby	2	8.5	0.2	6.7	23.3	2.8	2.7	64.8
TENNESSEE								
Inglewood	78	8.4	1.5	0.5	1.6	2.8	4.7	91.4

P) PROBLEMS

10a) (Group) Share the 88 unincorporated urban places out among the group, so each does his/her share of making up index cards according to the pattern of exhibit 12 and his/her share of checking the correctness of cards made up by another. (It may help to make 8 copies of each card, so there can be 8 sets of 88.)

10a2) Divide into 8 subgroups. Let each subgroup take the 88 cards, sort in order according to their variable, and divide into slices.

10a3) Within each subgroup assign other variables to persons and let each person, in turn, sort each separate slice on his/her variable and record the corresponding letter values.

10a4) Let each person find, smooth and backadd (or subtract) the letter differences for his/her variable when in his/her group's slices.

10a5) Let each person make the corresponding delineation.

how many data sets?

What happens if we want to work with more than two variables? The first question is: How many data sets will we need? To suggest an answer we turn back to batches of single values and batches of pairs.

We would not be too far out if we said that for batches of numbers:

- ◇20 usually gives useful letter values.
- ◇10 is likely to do so.
- ◇50 does quite well.

(more may help more), while for batches of (x, y) pairs:

- ◇100 usually gives a useful delineation.
- ◇50 is likely to do so.
- ◇250 does quite well.

(more may help more). Thus, for batches of (x, y, z) values, where we want to look at the three values together, we might expect that:

- ◇500 usually gives a useful "three-way".
- ◇250 is likely to do so.
- ◇1250 does quite well.

If we are going to play with numbers this large by hand, even with only 3 variables, the index-card technique just described is surely called for--and our desire to have a computer do the work mounts rapidly.

And what will we do with our 500-odd cards when we have filled them out? Something strictly analogous to what we did for (x, y) pairs, which can be described in these words: Arrange in order on x, letter-cut on x, treat each slice the way you would have treated a batch of y's.

For our 500-odd (x, y, z)'s, what is the analog? Clearly: Arrange in order on x, letter-cut on x, treat each slice (of count at least $5 \times 5 = 25$) the way you would have treated a batch of (y, z) pairs (this clearly means "delineate" them). The result of all this is a (y, z)-delineation for each x-slice.

How can we best look at these delineations? Side-by-side? Or one above another? Surely not--either of these arrangements tends to confuse the slicing variable x with one or the other of the variables in the delineation. What does this leave? Two possibilities, both reasonable:

If we are willing to get some moderately stiff cards, and put one delineation on each, we can riffle the pack of cards, and watch the delineation move as we hasten from slice to slice. When we do this, we are reminded that we have smoothed the delineations within each slice, but we have as yet done no smoothing across slices.

So our next step is to take our delineations--more precisely to take the first cuts, and the smoothed cross-medians which define the delineations--and smooth them across slices. The result is a new set of points in each slice suitable for drawing its delineation. These delineations will now relate much more smoothly to their neighbors.

And what if we don't like low-grade imitations of movies--and refuse to riffle cards--how can we display our delineations? The best choice seems to be to put them on a diagonal. Perhaps we show one delineation for every slice, perhaps fewer. (One choice is to take medians between adjacent, nonoverlapping pairs of delineations.)

(If we were so rash as to try four variables, we could slice twice before delineating. We might use a NW–SE diagonal for our first variable and a NE–SW diagonal for the other.)

The brave may want to try the group problem in exhibit **11.**

exhibit **11** of chapter 9: problems

Problem set for brave groups

11a (group)) The 1960 U.S. Census found 676 cities of population 25,000 or more. The 1970 Census found 840. Pick out a few hopefully interesting variables (presumably from among those in some County and City Data Book) and make up a deck of index cards (as per exhibit 9) for either all 1960 cities or all 1970 cities.

11a2 (group)) Order and then slice the deck on one of these variables.

11a3 (subgroups)) Delineate a third variable on a second variable within each of these slices.

11a4 (different subgroups)) Smooth across slices.

11a5 (subgroups)) Construct the smoothed delineations.

11a6 (group)) Compare the resulting delineations, either by fitting or diagonally, or both.

11b (subgroups)) Take the data of exhibit 32 of chapter 8 and divide it into 8 three-hour sections--or 12 2-hour sections--starting at midnight. Delineate the height of tide on the interval since last high tide for each of these sections.

11b2 (different subgroups)) Smooth, around the clock, the results thus obtained.

11b3 (subgroups)) Construct the smoothed delineations.

11b4 (group)) Compare/combine the results and make any further analysis needed.

11c/11c2/11c3/11c4) Do the same, starting at 0100 instead of midnight.

11d (uses (11b4) and (11c4)) Compare the results of (11b4) and (11c4).

9H. How far have we come?

This chapter has been devoted to expanding the wandering schematic plots we learned to make in the last chapter to displays that tell us more about the joint behavior of x and y in an (x, y)-scatter.

We are now ready:

- ◇to find E-traces, D-traces, and the like.
- ◇to combine traces and cuts into a delineation.
- ◇to convert a full delineation into a reduced one.
- ◇to start from a full delineation and construct an (x, y)-schematic plot.

We now understand more clearly:

- ◇what delineation can do for us.
- ◇what delineation and schematic plots miss--as they should.
- ◇how we might go on to the analysis of three or more variables.

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4★		340
5★		342
	10C	
	10D	
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	10E	
7		349
	10F	
8		352
9★		354
10		356
11★		357
	10G	
12★		359
	10H	

Using two-way analyses 10

It is now high time for us to look at two-way analyses, both because of their importance, and because of the diversity and richness of analysis they introduce us to. Once we have two or more handles on the same set of data--as in a two-way table--there is much more flexibility in what we do. As a result, there can be many alternative approaches instead of one or perhaps two. A few of these will appear in this chapter.

The bases of a two-way table (implied "of responses") are:

- ◇one kind of response.
- ◇two kinds of circumstances, one of each kind occurring together for every observation.

If we have birth-rates, for example, for various combinations of age of mother with some social situation (country, date, relative income level, and number of living children are each possibilities), we have a two-way table of responses, perhaps with age of mother tagging columns while number of living children tags rows.

If we have yields of wheat, for example, for various combinations of named variety of wheat with amount and kind of fertilizer application, we have a two-way table of responses--with, say, variety tagging columns while fertilizer amount and kind tags rows. If we have hardnesses for metal specimens having various combinations of composition with heat treatment, we have a two-way table of responses--with, say, composition tagging columns while heat treatment tags rows, and so on.

This chapter and the next are organized together so that the rate of flow of new ideas:

- ◇is moderately uniform.
- ◇is slow enough to be bearable.

At the same time, repetition will help these new ideas to be more firmly grasped and understood. Many things that seem like loose ends in the present chapter will be tied up in the next. This means that we will not necessarily explain in the present chapter how to carry out the analyses we discuss. Some of that will have to be left for the next chapter.

On to the examples!

review questions

Does it make a difference if we have two or more handles on a set of data? Why/why not? What are the essentials of a two-way table of responses? How does a "response" differ from a "circumstance"? What are some examples of two-way tables that you have made up for yourself? What were the three examples given above? Why is this chapter closely related to the next? How is this relationship unusual? What is hoped for the rate of new ideas for the two chapters? For the tying up of loose ends?

10A. Two-way residuals; row-PLUS-column analysis

Panel A of exhibit **1** shows the mean monthly temperatures for 7 months --from July to January--and 3 places in Arizona--Flagstaff, Phoenix, and Yuma. When we look at this table it is clear that:

- ◇ Yuma is somewhat warmer than Phoenix.
- ◇ both are much warmer than Flagstaff.
- ◇ it gets cooler from July to January.

The last surprises no one. The first two surprise no one who knows about Arizona, but may help the future visitor.

Is this all that such a table can tell us? Surely not.

We have already made good use of (one-way) residuals. How can we adapt the idea here?

Panel B shows one way to start--fitting a median value for each place and forming the corresponding residuals. With the 7 mean monthly temperatures at each place falling steadily, the easiest reasonable fit is the value for the middle month, October. This is exactly what has been used in Panel B.

The residuals, as always, have been found by subtraction, as in $19.1 = 65.2 - 46.1$, $19.3 = 90.1 - 70.8$, $18.2 = 94.6 - 76.4$. We now see that the change of the seasons changes temperatures by about the same amount--from month to month--in each of the three places. As simple a fact as this should have some use in letting us see deeper.

single and double lines

In panel B we have begun a new standard:

double lines between things that add up.

We will use single lines--or, if we are over-careful, dashed lines--to separate values from some kind of summary. Thus, the first row values of panel A and their median has

65.2 90.1 94.6 | 90.1

with a single line, while if we go on to residuals

65.2 90.1 94.6 | 90.1 || −24.9 0.0 4.5

with a double line between the 90.1 and the residuals, things that add up (90.1 + (−24.9) = 65.2, 90.1 + 0.0 = 90.1, 90.1 + 4.5 = 94.6) in this case, give the original values.

The righthand side of panel B shows the medians for (i) the place fits and (ii) the residuals (after place fits) for each month. (If we were not going slowly, this single line and column of numbers would have been set up close to the main part of panel B.)

exhibit **1** of chapter 10: Arizona temperatures

Two-way residuals (cooling down Arizona)

A) The MEAN MONTHLY TEMPERATURES––in °F

	Flagstaff	**Phoenix**	**Yuma**
July	**65.2**	**90.1**	**94.6**
Aug	**63.4**	**88.3**	**93.7**
Sept	**57.0**	**82.7**	**88.3**
Oct	**46.1**	**70.8**	**76.4**
Nov	**35.8**	**58.4**	**64.2**
Dec	**28.4**	**52.1**	**57.1**
Jan	**25.3**	**49.7**	**55.3**

B) FITTED PLACE VALUES, their RESIDUALS––and some MONTH MEDIANS

fit	46.1	70.8	76.4	70.8
July	**19.1**	**19.3**	**18.2**	**19.1**
Aug	**17.3**	**17.5**	**17.3**	**17.3**
Sept	**10.9**	**11.9**	**11.9**	**11.9**
Oct	**0**	**0**	**0**	**0**
Nov	**−10.3**	**−12.4**	**−12.2**	**−12.2**
Dec	**−17.7**	**−18.7**	**−19.3**	**−18.7**
Jan	**−20.8**	**−21.1**	**−21.1**	**−21.1**

C) The MEDIANS of panel B taken out; SECOND RESIDUALS

	eff 70.8	−24.7	0	5.6
July	**19.1**	**0**	**.2**	**−.9**
Aug	**17.3**	**0**	**.2**	**0**
Sept	**11.9**	**−1.0**	**0**	**0**
Oct	**0**	**0**	**0**	**0**
Nov	**−12.2**	**1.9**	**−.2**	**0**
Dec	**−18.7**	**1.0**	**0**	**−.6**
Jan	**−21.1**	**.3**	**0**	**0**

Note the appearance of "eff" for "effect," used for values obtained from repeated fitting, whether or not these are fits to the original data. →

exhibit **1** of chapter 10 (continued)

D) ONE set of FITS restored

		Flagstaff	Phoenix	Yuma
eff	**70.8**	**−24.7**	**0**	**5.6**
July	19.1	**0**	**.2**	**−.9**
Aug	17.3	**0**	**.2**	**0**
Sept	11.9	**−1.0**	**0**	**0**
Oct	0	**0**	**0**	**0**
Nov	−12.2	**1.9**	**−.2**	**0**
Dec	−18.7	**1.0**	**0**	**−.6**
Jan	−21.1	**.3**	**0**	**0**
fit	0	**46.1**	**70.8**	**76.4**

Examples:
$$65.2 = 70.8 + 19.1 - 24.7 + 0$$
$$65.2 = 0 + 19.1 + 46.1 + 0$$
$$90.1 = 70.8 + 19.1 + 0 + .2$$
$$90.1 = 0 + 19.1 + 70.8 + .2$$
$$94.6 = 70.8 + 19.1 + 5.6 + (-.9)$$
$$94.6 = 0 + 19.1 + 76.4 + (-.9)$$
$$63.4 = 70.8 + 17.3 - 24.7 + 0$$
$$63.4 = 0 + 17.3 + 46.1 + 0$$

(The "0" appearing in the 2nd, 4th, . . . lines is the "0" in the lower left, outside the double lines.)

E) EXERCISES to be done by EVERY READER (replaces PROBLEMS)

1a) Take medians out of the data in panel A in the other order, first by rows and then by columns. Did you get the same answer? Is the difference enough to matter?

1b) Instead of adding place fits, as in panel D, add month fits to the display of panel C. Where did you put the month fits? Why? What value did you put in the corner?

1c) Can you add BOTH place fits and month fits to panel C at the same time? If you can, what value could go in the lower right corner outside the double lines?

1d) Write out the expressions of these given values––52.1, 57.0, 57.1––in terms of residuals and effects.

1e) 88.3 occurs twice as a given value. Write out the two different expressions in terms of residuals and effects.

S) SOURCE

Climatography of the United States No. 60–2. Climates of the states. Arizona. U.S. Weather Bureau, Washington, D.C., September 1959.

effects, common, eff, all

Panel C goes on to use these medians as further fits, finding second residuals. Here "effect", usually abbreviated "eff" appears for the first time. We will use effect or eff routinely for the results of repeated fitting procedures. In panel C, −24.7 is a column effect, often abbreviated "col eff", while 19.1 is a "row eff".

Usually, as here, an effect is an allocation of part of each of several observed values to a circumstance or set of circumstances. It may be that what we have allocated really belongs to the circumstance, or it may not. But we can be sure that if what is allocated is larger than what is left over, we have made it easier to look at, and understand, what is going on.

The number taken out of each and every data value, here 70.8, is naturally called "common" and abbreviated "all". It is, if you like, an allocation to whatever circumstances are common to all the data--or to all but a few strays.

Thus the overall "verbal arithmetic" for panel C is

65.2 =	70.8	+	(−24.7)	+	19.1	+	0.
data =	common	PLUS	col eff	PLUS	row eff	PLUS	residual

Panel D goes on to show how we can include one set of fits along with all the effects, thus having a choice both in what we look at and in how we think of the given values as broken down. Taking this still further is placed on the reader's shoulders in panel E.

Notice that we have put the effects above and to the left of the residuals--and let the fits stay to the right and below. We have done this so that (i) the scanning eye is likely to see the effects first and (ii) the effects are as close to the tags as possible. Doing this betrays a bias--a belief that, on average, the effects will be more used--more often looked at and interpreted--than the fits. It is right to accept such a bias, if well-founded, as a general policy. It would be a mistake to feel that such a path should always be followed.

We routinely put effects first, but we feel free to

- ◇ omit the fits.
- ◇ omit the fits and put effects last.

Indeed, we even feel free to turn the display around--fits first, effects last--or to omit the effects and keep the fits. We can--and should--do whatever the special case seems to call for. The display, including its labels, should match the need that calls it forth. Exhibit **2** shows some alternative displays of the particular

row-PLUS-column

analysis of the 21 Arizona mean temperatures that we worked toward in exhibit 1.

a comment

We are, of course, up to our old trick of seeking a simple partial description--a partial description that may be easier to comprehend--a partial description that, once subtracted away, will leave us able to look more effectively at what has so far not been described.

review questions

What example did we start with? What did we find first, and what did we do with these values? What did we do next? What was the result? How do we use double lines? Single lines? What is an effect? What is a common term? What does "all" refer to? What do "col eff" and "row eff" refer to? What kind of residuals concern us here? What is the overall "verbal arithmetic"? How do we arrange the blocks in the all-purpose form? Of what kind of analysis is this a display? Can we use other forms of display of this analysis? Why/why not? What is the connection between a fit and a partial description?

exhibit **2** of chapter 10: Arizona temperatures

Some alternative displays for a particular row-PLUS-column analysis (data from exhibit 1)

A) The ALL-PURPOSE form

	eff	**Flagstaff**	**Phoenix**	**Yuma**	**fit**
eff	70.8	−24.7	0	5.6	0
July	19.1	**0**	**.2**	**−.9**	89.9
Aug	17.3	**0**	**.2**	**0**	88.1
Sept	11.9	**−1.0**	**0**	**0**	82.7
Oct	0	**0**	**0**	**0**	70.8
Nov	−12.2	**1.9**	**−.2**	**0**	58.6
Dec	−18.7	**1.0**	**0**	**−.6**	52.1
Jan	−21.1	**.3**	**0**	**0**	49.7
fit	0	46.1	70.8	76.4	(−70.8)

B) The SAME with its PLUS nature EMPHASIZED

	eff	**Flagstaff**	**Phoenix**	**Yuma**	**fit**
eff	70.8	−24.7	0	5.6	0
July	19.1	**0**	**.2**	**−.9**	89.9
Aug	17.3	**0**	**.2**	**0**	88.1
Sept	11.9	**−1.0**	**0**	**0**	82.7
Oct	0	**0**	**0**	**0**	70.8
Nov	−12.2	**1.9**	**−.2**	**0**	58.6
Dec	−18.7	**1.0**	**0**	**−.6**	52.1
Jan	−21.1	**.3**	**0**	**0**	49.7
fit	0	46.1	70.8	76.4	(−70.8)

→

10B. The row-PLUS-column fit

To get the fit we must leave out the residual. These words have different meanings when we want to find numbers for the fit and when we want to display its components. Exhibit **3** works with the same example. To get numbers for the fit, it is often easiest to use

fit = given MINUS residual

as has been done in panel A.

Panel B shows a check that the fit is actually of the form

all + row eff + col eff.

This check, which does NOT check that this is the fit we WANT--only that it is of "row-PLUS-column" FORM, that it is a fit of the kind we thought we were getting--is both important and useful.

Panel C shows the components of the fit. Notice (1) that the panel of residuals is left blank and (2) that this panel is now bordered with single lines

exhibit **2** of chapter 10 (continued)

C) An EFFECTS-only FORM

	eff	**Flagstaff**	**Phoenix**	**Yuma**
eff	70.8	−24.7	0	5.6
July	19.1	**0**	**.2**	**−.9**
Aug	17.3	**0**	**.2**	**0**
Sept	11.9	**−1.0**	**0**	**0**
Oct	0	**0**	**0**	**0**
Nov	−12.2	**1.9**	**−.2**	**0**
Dec	−18.7	**1.0**	**0**	**−.6**
Jan	−21.1	**.3**	**0**	**0**

D) A FITS-only FORM

	fit	**Flagstaff**	**Phoenix**	**Yuma**
fit	(−70.8)	46.1	70.8	76.4
July	89.9	**0**	**.2**	**−.9**
Aug	88.1	**0**	**.2**	**0**
Sept	82.7	**−1.0**	**0**	**0**
Oct	70.8	**0**	**0**	**0**
Nov	58.6	**1.9**	**−.2**	**0**
Dec	52.1	**1.0**	**0**	**−.6**
Jan	49.7	**.3**	**0**	**0**

P) PROBLEMS

2a/b/c/d/e/f) Write down the same display in a way different from (i) all those above and (ii) those written down for earlier problems of this string, and discuss the pros and cons of the way chosen.

(double lines are left outside). Both of these indicate that we are NOT adding in the residuals.

Notice carefully how much--and how little--the check checks. Notice too that the difference of two row-PLUS-column fits is a row-PLUS-column fit. Thus we can check the difference of two fits in the same way.

have we learned more?

It is natural to ask if this fit has taught us more than the three rather trivial statements we found after looking at panel A of exhibit 1. What more can we learn from the full two-way analysis? At least these things:

◇the largest residual, 1.9, is small compared to differences in place effects and most differences between month effects. We can hardly fail to say that a "place-PLUS-month" fit has done a quite good job.

exhibit **3** of chapter 10: Arizona temperatures

Displaying and finding the fit (data from exhibit 1)

A) VALUES of the FIT--found as given MINUS residual

	Flagstaff	Phoenix	Yuma
July	65.2	89.9	95.5
Aug	63.4	88.1	93.7
Sept	58.0	82.7	88.3
Oct	46.1	70.8	76.4
Nov	33.7	58.6	64.2
Dec	27.4	52.1	57.7
Jan	25.0	49.7	55.3

B) CHECKING the values of the FIT

Phoenix minus Flagstaff	Yuma minus Phoenix
24.7	5.6
24.7	5.6
24.7	5.6
24.7	5.6
24.9*	5.6
24.7	5.6
24.7	5.6

Note that this check is only a check of whether what we have written down is a possible fit--not that it is the fit we want.

*** This value, unequal to the others, shows an error. Going back to exhibit 1, we find 35.8 MINUS 1.9 should be 33.9, not 33.7 (as in panel A above). We will make this correction in all that follows.**

→

◇we now see explicitly that Flagstaff is about 25°F cooler than Phoenix, while Yuma is 5 or 6 degrees warmer than Phoenix.

◇we have a sequence of month effects--the rounded values are 19, 17, 12, 0, −12, −19, −21--that show a steady decrease from one month to the next--first slow, then fast, then slow--that is close to symmetric around October (changes from September to October and from October to November are almost twice as large as those between other pairs of adjacent months).

◇the large residuals are up for Flagstaff in November and December and down for Flagstaff in September and July.

Surely this is much more than we can gather easily from panel A. **We have lifted both lids**--the effects of season and the effects of place. Once both are lifted together we can **see quite a lot of what is left**--in this case, mainly that what is left is relatively small and, beyond that, which are the larger residuals, what is their sign, and roughly how large they are.

exhibit **3** of chapter 10 (continued)

C) the COMPONENTS of the FIT displayed

eff	70.8	−24.7	0	5.6	0
July	19.1				89.9
Aug	17.3				88.1
Sept	11.9				82.7
Oct	0				70.8
Nov	−12.2				58.6
Dec	−18.7				52.1
Jan	−21.1				49.7
fit	0	46.1	70.6	76.4	(−70.8)

Some ways of finding the value of the fit in the upper left corner are:

$$70.8 - 24.7 + 19.1 = 65.2$$
$$0 + 46.1 + 19.1 = 65.2$$
$$0 - 24.7 + 89.9 = 65.2$$
$$(-70.8) + 46.1 + 89.9 = 65.2$$

P) PROBLEMS

3a/b/c) Write out the value of the fit in the upper-right/lower-left/lower-right corner in each of the four ways illustrated in panel C and also as "given MINUS residual".

3d) Do the same for a noncorner entry of your choice.

AN analysis--not THE analysis

In dealing with row-PLUS-column fits we are deliberately flexible, just as we were earlier in dealing with

value = fit PLUS residual.

When "fit" was a typical value--or a straight line--we were prepared to have more than one alternative fit, each with its own set of residuals. Things are not different with row-PLUS-column fits.

Panels A and B of exhibit **4** show two (of the very many possible) alternatives to the row-PLUS-column analysis shown, in various forms, in exhibit 2. The differences in the three analyses are illustrated by the three breakdowns for July in Flagstaff.

$$65.2 = (70.8) + (19.3) + (-24.7) + (-.2) \quad \text{exhibit 2}$$
$$65.2 = (70.8) + (19.1) + (-24.7) + (0) \quad \text{A of exhibit 4}$$
$$65.2 = (70.8) + (18.7) + (-24.4) + (.1) \quad \text{B of exhibit 4}$$

The differences are small but do not vanish.

We will have, in the next chapter, some suggestions about which row-PLUS-column analyses to use in routine exploratory work. But these will only be suggestions. Anything that conforms to

value = common + row eff + col eff + residual

is a row-PLUS-column analysis. Whether it meets our needs in a particular case is up to us--after we look at it carefully and critically.

exhibit **4** of chapter 10: Arizona temperatures

Two alternative row-PLUS-column analyses (in all-purpose form) of the same data (starts at exhibit 1)

A) ANOTHER row-PLUS-column ANALYSIS--compare with panel D of exhibit 1

	eff	**Flagstaff**	**Phoenix**	**Yuma**	**fit**
eff	70.8	−24.7	0	5.6	0
July	19.1	0	.2	−.9	89.9
Aug	17.3	0	.2	0	88.1
Sept	11.9	−1.0	0	0	82.7
Oct	0	0	0	0	70.8
Nov	−12.4	2.1	0	.2	58.4
Dec	−18.7	1.0	0	−.6	52.1
Jan	−21.1	.3	0	0	49.7
eff	**0**	**46.1**	**70.8**	**76.4**	**(−70.8)**

→

Thus there can be arguments about whether the analysis of panel B of exhibit 4 is as generally useful as the simpler-appearing analysis of exhibit 2, but there is no doubt that panel B shows clearly that the residuals can all be squeezed between −0.8°F and +0.8°F, and one fit that does this. Is this what we want to do? If it is, fine. If not, down with this analysis––go back to exhibit 2 or try again!

review questions

How do we find the fit? The values of the fit? How do we check the values of the fit? How often should we check the fit? What does/does not the check check? Did we learn more from our analysis than we learned from our first unguided look at the data? What form of display is nearest the all-purpose form in general usefulness? What did we learn from the analysis that we had not already learned? Should we be flexible in which fits we consider row-PLUS-column fits? In which fit we look at? Why/why not? How do we judge which fit to use in a specific instance? Do we necessarily prefer a fit whose residuals are squeezed together?

exhibit **4** of chapter 10 (continued)

B) STILL ANOTHER row-PLUS-column analysis of the SAME DATA

eff	**70.8**	**−24.4**	**−.2**	**5.4**	**0**
July	18.7	.1	.8	−.3	89.5
Aug	17.4	−.4	.3	.1	88.2
Sept	11.4	−.8	.7	.7	82.2
Oct	0	−.3	.2	.2	70.8
Nov	−11.4	.8	−.8	−.6	59.4
Dec	−18.5	.5	0	−.6	52.3
Jan	−21.0	−.1	.1	.1	49.8
eff	**0**	**46.4**	**70.6**	**76.2**	**(−70.8)**

E) EXERCISES to be done by ALL READERS (replaces PROBLEMS)

4a) Write out a row-PLUS-column analysis of this same data different (in residuals, not just in form of display) from the three so far illustrated.

4b) Write out still another such.

4c) Comment on your feelings about choosing among the five row-PLUS-column analyses of this single set of data.

exhibit **5** of chapter 10: Arizona temperatures

Alternating bolding applied to four forms of display (same analysis as panel A of exhibit 2; data in exhibit 1)

A) WITH "eff" and "fit" DRAWN TOGETHER––and alternating bolding

	Flagstaff	**Phoenix**	**Yuma**	**eff**	**fit**
July	**–.2**	**0**	**–1.1**	19.3	90.1
Aug	**–.2**	**0**	**–.2**	17.5	88.3
Sept	**–1.0**	**0**	**0**	11.9	82.7
Oct	**0**	**0**	**0**	0	70.8
Nov	**2.1**	**0**	**.2**	–12.4	58.4
Dec	**1.0**	**0**	**–.6**	–18.7	52.1
Jan	**.3**	**0**	**0**	–21.1	49.7
eff	–24.7	0	5.6	**70.8**	**0**
fit	46.1	70.8	76.6	**0**	**–70.8**

B) OMITTING "fit" and MOVING "eff"

	Flagstaff	**Phoenix**	**Yuma**	**eff**
July	**–.2**	**0**	**–1.1**	19.3
Aug	**–.2**	**0**	**–.2**	17.5
Sept	**–1.0**	**0**	**0**	11.9
Oct	**0**	**0**	**0**	0
Nov	**2.1**	**0**	**.2**	–12.4
Dec	**1.0**	**0**	**–.6**	–18.7
Jan	**.3**	**0**	**0**	–21.1
eff	–24.7	0	5.6	**70.8**

C) WITH ONE fit GIVEN but KEPT IN PARENTHESIS

	Flagstaff	**Phoenix**	**Yuma**	(fit)	eff
July	**–.2**	**0**	**–1.1**	(90.1)	19.3
Aug	**–.2**	**0**	**–.2**	(88.3)	17.5
Sept	**–1.0**	**0**	**0**	(82.7)	11.9
Oct	**0**	**0**	**0**	(70.8)	0
Nov	**2.1**	**0**	**.2**	(58.4)	–12.3
Dec	**1.0**	**0**	**.6**	(52.1)	–18.7
Jan	**.3**	**0**	**0**	(49.7)	–21.1
eff	–24.7	0	5.6	**(0)**	**70.8**

10C. Some points of technique

use of bolding

Here is a good place to remind the reader that **bolding** is not something confined to the pages of books. We early recommended the use of multiple colors--or at least ink contrasted with pencil--by those preparing even the simplest displays by hand. Even those who stick to pencil--or to one color of ink--can always make a respectably clear impression on their lookers by encircling (individually or in blocks) those figures that might well be printed **bold.**

As our displays become a little more diversified and detailed, it becomes more and more important to use bolding--and to use it well. Exhibit 5 shows how--in any of a variety of forms of display--we can seize the eye a little, emphasizing the block structure by alternating bolding so that blocks of entries that are side by side or top by bottom are always different--one bold, one not.

exhibit **5** of chapter 10 (continued)

D) As B, but with BOLD and NONBOLD interchanged

	Flagstaff	**Phoenix**	**Yuma**	**eff**
July	−.2	0	−1.1	**19.3**
Aug	−.2	0	−.2	**17.5**
Sept	−1.0	0	0	**11.9**
Oct	0	0	0	**0**
Nov	2.1	0	.2	**−12.4**
Dec	1.0	0	−.6	**−18.7**
Jan	.3	0	0	**−21.1**
eff	**−24.7**	**0**	**5.6**	70.8

P) PROBLEMS

5a/b/c/d) Imitate, respectively, panel A/B/C/D above for the analysis of panel A, exhibit 4.

5e/f/g/h) Copy, respectively, panel A/B/C/D above, using FOUR COLORS of ink or pencil for the numbers to emphasize the separateness of its parts.

5a2/b2/c2/d2) Do the same for the analysis of panel A of exhibit 4.

zeros and decimal points

We have found it convenient, in our displays, to distinguish between exact zeros and near zeros. The difference between a very small residual and a residual that is exactly zero

- ◇is entirely unimportant so far as the meaning of that residual--or the given value from whence it came--is concerned.
- ◇is useful as a matter of technique, in showing us both something about how that particular analysis was done and something about the small-scale distortions of residuals associated with our method of fitting.

We have shown values with decimal points, contrary to our usual practice. So long as we do this, we can make the occurrence of exact zeros quite clear. All we need do is line the exact zeros up with the decimal points.

If, as often ought to be the case, we have no decimal points, it is easy to identify the exact zeros by some other character. Two choices are natural:

- ◇the letter "z" for "zero,"
- ◇the check-mark "√" for "the fit is exact."

We should feel free to use either.

review questions

How do we use bolding? Rigidly or flexibly? What replaces bolding in hand work? How did we make exact zeros distinctive? For what purposes is an exact zero different from a near zero? For what purposes not? Do we have alternative markings for "zero"? What are they?

10D. Row -TIMES-column analysis

We learned earlier that it is often not wise to analyze our numbers in the form they came to us--whether from others or from our first writing of them down--roots, logs, and, occasionally, (negative) reciprocals ought by now to be not only commonplace but automatically considered. This is at least as true for row-PLUS-column analyses as for the simpler sorts of analyses we have looked at.

Panel A of exhibit **6** gives the expenditures for five kinds of personal consumption at five-year intervals from 1940 to 1960. Even a hasty look at the values shows that "row-PLUS-column" is not a good approach, at least so long as raw values are to be analyzed. Food and tobacco rose 64 billion--from 22 to 86--while private education and research rose only 3 billion--from 0.64 to 3.64.

Since both rises were by similar factors of about 4 or 5, however, we may expect to do better by taking logs. Panel B does just this, while panels C to E go on with a "rows-PLUS-columns analysis of LOGS." Looking at this analysis, and at the stem-and-leaf of its residuals in panel F, we see that

exhibit **6** of chapter 10: personal consumption

Selected personal consumption expenditures (for the U.S.)

A) The DATA--in billions of dollars

	1940	**1945**	**1950**	**1955**	**1960**
Food and tobacco	22.2	44.5	59.6	73.2	86.8
Household operation	10.5	15.5	29.0	36.5	46.2
Medical care and health expenses	3.53	5.76	9.71	14.0	21.1
Personal care	1.04	1.98	2.45	3.40	5.40
Private education and research	.641	.974	1.80	2.60	3.64

B) LOGS of EXPENDITURES--log billions of dollars

Food	1.35	1.65	1.77	1.87	1.94
Household	1.02	1.19	1.77	1.87	1.94
Medical	.55	.76	.99	1.15	1.32
Personal	.02	.30	.39	.53	.73
Private E/R	−.19	−.01	.26	.41	.56

C) A FIRST STAGE in the ANALYSIS of the LOGS in B--one fit removed

Food	−.42	−.12	0	.10	.17	1.77
Household	−.44	−.27	0	.10	.17	1.46
Medical	−.44	−.23	0	.16	.33	.99
Personal	−.37	−.09	0	.14	.34	.39
Private E/R	−.45	−.27	0	.15	.30	.26

D) ONE row-PLUS-column ANALYSIS of the LOGS in B--using effects

Food	.02	.11	0	−.04	−.13	.78
Household	0	−.04	0	−.04	−.10	.47
Medical	0	0	0	.02	.03	0
Personal	.07	.14	0	0	.04	−.60
Private E/R	−.01	−.04	0	.01	0	−.73
eff	**−.44**	**−.23**	**0**	**.14**	**.30**	**.99**

E) ANOTHER row-PLUS-column ANALYSIS of the LOGS in B

	1940	1945	1950	1955	1960	**eff**
Food	.02	.11	0	−.04	−.13	**.78**
Household	.04	0	.04	0	−.06	**.43**
Medical	0	0	0	.02	.03	**0**
Personal	.03	.10	−.04	−.04	0	**−.56**
Private E/R	−.01	−.04	0	.01	0	**−.73**
eff	**−.44**	**−.23**	**0**	**.14**	**.30**	**.99**

→

exhibit **6** of chapter 10 (continued)

F) STEM-and-LEAF of RESIDUALS in E

```
      .1 |
 2    .1* | 10                25 residuals
      .0 |
13    .0* | 24304002130       M13 |    0    |
12   -.0* | 1004044400        M 7 | -.01 .02 | .03
 2   -.0 | 6
 1   -.1* | 3
     -.1 |
```

adj = −.04 (four)
adj = +.04 (twice)

out = −.06 (1960 Household)
out = +.10 (1945 Personal)
out = +.11 (1945 Food)
far = −.13 (1960 Food)

	.04h	
f	−.05h	+.06h
	one	two
F	−.10	+.11

G) THE row-TIMES-column ANALYSIS corresponding to panel D

Food	**1.05**	**1.29**	**1**	**.91**	**.74**	6.00
Household	**1**	**.91**	**1**	**.91**	**.80**	2.94
Medical	**1**	**1**	**1.05**	**1.07**	**1**	**1**
Personal	**1.17**	**1.38**	**1**	**1**	**1.10**	.25
Private E/R	**.98**	**.91**	**1**	**1.02**	**1**	**.186**
	.36	**.59**	**1**	**1.38**	**2.00**	**9.80**

MULTIPLY one entry of EACH of 4 KINDS to obtain each original VALUE

Examples:

(finding entries) log 1.05 = .02, log 1.29 = .11, log 6.00 = .78, log.36 = −.44,
log .59 = −.23, log 9.80 = .99.
(combining entries) (1.05)(.36)(6.00)(9.80) = 22.2
(1.29)(.59)(6.00)(9.80) = 44.5

H) Row-TIMES-column ANALYSIS corresponding to panel E

						eff
Food	**1.05**	**1.29**	**1**	**.91**	**.74**	**6.0**
Household	**1**	**.91**				**2.7**
Medical						**1**
Personal						**.28**
Private E/R						**.186**
eff	**.36**	**.59**	**1**	**1.38**	**2.00**	**9.8**

MULTIPLY one ENTRY of EACH of 4 KINDS to obtain each original VALUE.

I) THE row-TIMES-column FIT of H––in billions of dollars

	1940	1945	1950	1955	1960
Food	**21.**	**35.**	**59.**	**81.**	**118.**
Household	**9.5**	**15.6**	**26.**	**36.**	**52.**
Medical	**3.5**	**5.8**	**9.8**	**13.5**	**19.6**
Personal	**.98**	**1.6**	**2.7**	**3.7**	**5.4**
Private E/R	**.65**	**1.07**	**1.8**	**2.5**	**3.6**

→

◇the fit is moderately good, only the residual for 1940 medical care and health expenses being outside (other largish residuals are food in 1945 and 1960 and personal care in 1945).

◇the year effects increase fairly steadily, more rapidly between 1940 and 1950 than between 1950 and 1960.

◇the category-of-expenditure effects are quite large.

◇the three largest residuals are listed in panel F. When we review them in the context of the full analysis, we notice, first, that the residuals for food and tobacco fall steadily from 1945 to 1960––which might leave us wondering how much the recent rise in food prices is a return to a 1945 distribution of expenditures––and second, a much lower value for medical care and health expenses in 1940, a fact we might well want to understand.

exhibit **6** of chapter 10 (continued)

J) The RESIDUALS from H in %

Food	**5%**	**29%**	√	**−9%**	**−26%**
Household	√	**−9%**			
Medical					
Personal					
Private E/R					

K) The RESIDUALS from H in BILLIONS of DOLLARS––rounded to match H

Food	1	10	√	−8	−31
Household	1	−.1	.3	.5	−5.8
Medical	√	√	√	.5	1.5
Personal	.06	.4	−.3	−.3	√
Private E/R	−.01	−.10	√	.1	.04

P) PROBLEMS

6a) Complete the entries in H.

6b) Make a third row-PLUS-column analysis of the logs in B, and convert it to a row-TIMES-column analysis of the values in A.

6c) Check that the entries of G reproduce at least three of the values of A. (Set out both what is to be multiplied and what multiplication gives.)

6d) Complete panel J.

6e) Discuss the meaning of the residuals in panel E.

6f) Discuss the meaning of the residuals in panel K.

6g) Compare your discussions in (6c) and (6d).

S) SOURCE

1962 World Almanac and Book of Facts, page 756.
Their source: Office of Business Economics, U.S. Department of Commerce.

If our audience is willing to be concerned with LOGS of EXPENDITURES, panels E and F present most of the story. But what if they do not? What if billions of dollars is the only form in which they can think?

All we can sensibly do is to undo the logarithms AFTER the ANALYSIS. Consider 1960 food as an example: The breakdown set out in panel E is

$$1.94 = (.99) + (.78) + (.30) + (-.13)$$

Now $.99 = \log 9.8$, $.78 = \log 6.0$; $.30 = \log 2.0$, and $-.13 = \log .74$, so the corresponding breakdown for billions of dollars is

$$86.8 \approx (9.8)(6.0)(2.0)(.74),$$

which we may wish to write as

$$86.8 \approx (9.8)(6.0)(2.0) \text{ down } 26\%.$$

The

row-TIMES-column

analyses corresponding to the row-PLUS-column analyses in panels D and E are set out in panels G and H. These tell the truth, but not strikingly enough to make an adequate effect on most people. How can we go further?

Panel I shows the actual numbers obtained by multiplying up the fit. For some, this will make a clearer impression than the year effects and component fits of panel G.

Panel J shows the residuals from this fit as percent. Some will find this presentation--which focuses on factors of change--quite satisfactory. Others will feel that it is essential to focus on amounts of change. For them, panel K is one way to set out the residuals effectively.

Once the raw data is better fitted by row-TIMES-column, the alternatives for displaying the answer are numerous--and the choice among them is often important, almost entirely as a matter of reaching the reader.

In a row-TIMES-column analysis, as the reader has probably already noticed, we can play the "exact 1" game when we have decimal points, in the same way we played the "exact 0" game for row-PLUS-column analyses. With trivial changes, the discussion at the end of the last section applies here also.

review questions

What example did we start with? Does a row-PLUS-column analysis of the given values seem sensible? What might we do to make a PLUS analysis reasonable? Did it work? Can we convert a PLUS analysis of logs to a TIMES analysis? Of what? How do we use triple lines? How do we distinguish TIMES analyses in how we display them? When we want to be particularly explicit? When we want to be particularly explicit about PLUS analyses? How can we show what is left over after a TIMES fit? Do we have a choice? (Be explicit.) What did we conclude about our example?

10E. Looking at row-PLUS-column fits and their residuals

Our next task is to learn to look more effectively--through pictures--first at row-PLUS-column fits, and then at their residuals. For the fits, we need a way to make a picture that shows them, for example in the form

row effect PLUS column fit

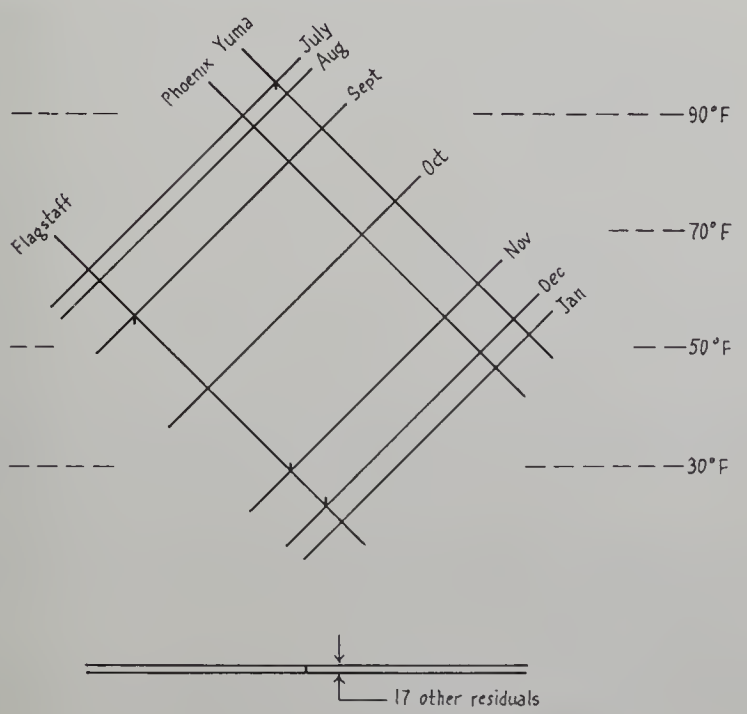
easily and clearly for each combination of row and column. As soon as there are more than a few each of both rows and columns, any attempt to do this along a single line will surely lead to crowding and confusion. We need to spread out the points that represent the different values of the fit "sidewise." As it happens, we can gain simplicity of picture at the same time. The only thing it really costs us is a very transient discomfort--one direction in our pictures will not have any meaning that we need (or want) to understand. We must learn to forget one direction in such pictures--something that is really quite easy to do.

the two-way plot

Exhibit 7 shows us the full picture corresponding to panel B of exhibit 4. We now have three columns--Flagstaff, Phoenix, and Yuma--and seven

exhibit 7 of chapter 10: Arizona temperatures

Another row-PLUS-column analysis (based on panel B of exhibit 4); behavior of the fit



rows--the months from July to January. We can easily see that, at least so far as the fit goes, Yuma in January is warmer than Flagstaff in October--and that Flagstaff in July is close to Phoenix in October. The whole behavior of the fit is before us, set out clearly.

Even though this picture concentrates on the fit, we ought not to neglect the residuals. We need to know (1) something about how large the residuals are, generally, and (2) where and how big the largest ones are. At four of the intersections in exhibit 7--Flagstaff in September, November, and December, and Yuma in July--we have drawn short vertical lines. The lengths of these lines represent the sizes of the corresponding residuals, so that the heights of their far ends represent, not the fit, but

data = fit PLUS residual.

We have shown only a few residuals as explicitly as this for two reasons:

- ◇if we try to show more than a few residuals, the picture tends to become so confused at to lose value instead of gaining it.
- ◇in the case at hand, even the larger residuals are so small that the vertical lines we show are about as short as we can appreciate. Adding shorter lines would not help us.

Instead of showing the other 17 residuals individually, we have discharged our obligation to them by showing the stripe at the bottom of exhibit 7--and pointing out that all 17 starting at some center line (we would show the center time if the residuals were even a little larger) will be ended within the outer lines.

Notice also how the one-scale-forget-the-other character of this--or any other--two-way plot is emphasized by a "successive water-line" picturing of the vertical scale. This means dashed horizontal lines on both sides of the plot and **no** formal scale in either direction.

We will learn in the next chapter how we can make such a picture, which we will call

a two-way plot

for any rows-PLUS-columns fit.

TIMES fits

And what if we want a picture of a row-TIMES-column fit? There are two answers:

- ◇make a picture of the corresponding row-PLUS-column fit to the logs--you can put either both scales or only the unlogged scale on it. (Doesn't work if the fit has negative values.)
- ◇turn to the early sections of chapter 12.

Either or both may meet your need.

looking at residuals

We also want to look at residuals--especially after the largest residuals have been shown life-size, as in exhibit 7, however; only the rough size of each residual, rather than its precise value, concerns us. It will be enough to mark each one well enough to reveal which of the following it is:

very positive (⧺) or (⧻)
large positive (+)
medium positive (+)
small (either sign) (·)
medium negative (∘)
large negative (○)
very negative (◎)

If our table is even moderately large, rules of thumb can help us choose what sizes of residual get which mark. In our present example, with 12 of the 21 residuals zero, we can do better by using judgment. Accordingly, we will call the residual of +2.1 very (positive)--those of −1.0, 1.0, and 1.1 large--and the other 5 not zero medium.

The single most useful way to look at the residuals is to plot these coded characters "on top" of the intersections of the plot of fits. This leaves them "labelled" in many useful ways, including:

- ◇ by the names of their row and column.
- ◇ by the (rough) numerical values of their row and column.
- ◇ by the (rough) value of the corresponding fit.
- ◇ and by combinations of such.

This diversity of possible labels corresponds to a diversity of ways in which we can approach these coded residuals, thus making it easier for us to notice many of the kinds of structure that residuals not too infrequently show. Exhibit **8** shows what happens for the analysis of panel B of exhibit 4. We now can see (1) an apparent excess of negative residuals in July and August and (2) a clear tendency for Flagstaff to be cooler in summer and warmer in winter than the fit indicates. (We can see each of these if we study panel B of exhibit 4 carefully, but exhibit 8 thrusts them on our attention. We should not rely on careful examination, if we can avoid the need for it.)

All that we have left to the next chapter about making such a

two-way plot of residuals

(given the two-way plot of fit) is the standard choice of which residuals to show with which symbol. Such a standard choice is often convenient, but ought not to be allowed to override good judgment.

other postponements

We also leave to the next chapter:

◇the possible condensation of residuals, as when we decide that January and February are so close as to be best treated together.

review questions

Do we want to look at row-PLUS-column fits? Why? Can we use pictures where one direction is important but the other is to be forgotten? What do two families of parallel lines do? Why does this help? To what example did we return? What did the picture show us? What about large residuals? Other residuals? Have we yet learned to make a two-way plot for a fit? Why/why not? What about pictures for TIMES fits?

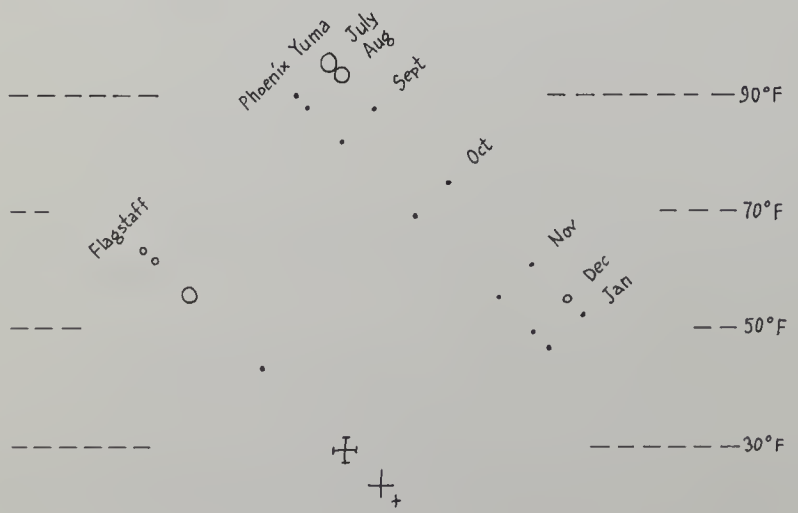
Do we want to look further at residuals? Why? In how much detail? What coding do we use? Where do we plot them? What kinds of labeling are thereby provided? What have we postponed to the next chapter?

10F. Fitting one more constant

Not infrequently we want to try to do just a little more than fit row-PLUS-column. Experience suggests that the most frequently useful fit of just a little

exhibit **8** of chapter 10: Arizona temperatures

Residuals of panel B of exhibit 4 marked to show size and direction



more is the fit of a constant times the product of the row effects and column effects already fitted. Two ways to write such a fit are

$$\text{all} + \text{row} + \text{col} + \frac{\text{row} \cdot \text{col}}{\text{div}}$$

and

$$\text{all} + \text{row} + \text{col} + k \cdot \frac{\text{row} \cdot \text{col}}{\text{all}}$$

where div (for "divisor") is a constant, as is k.

We will meet these last fractions

$$\frac{(\text{row})(\text{col})}{\text{all}}$$

often, almost always as a basis for some comparison, so we will call their values

comparison values.

One advantage of working with these values--rather than with the (row)(col) products alone--is that they have the same "dimension" as the effects. If we were to change from °F to °C in this example, all four kinds of quantities, "the fit", "row", "col", and "all", would have to be multiplied by 5/9 (after subtracting 32°F from both "the fit" and "all"). As a result, we could go on using the same numerical constant multiplying (row)(col)/all, which expression, like "the fit", would also have been multiplied by 5/9. (If we had chosen to work with "constant'(row)(col)", then, because (row)(col) would have been multiplied by $(5/9)^2$, we would have had to *divide* constant' by 5/9. Similar things, of course, happen when we change feet into inches--or make other changes of unit.

Exhibit **9** shows more monthly mean temperatures, this time warming on the East Coast (instead of cooling in Arizona), together with a row-PLUS-column analysis. Panel C of exhibit 9 sketches, in tabular form, what a two-way plot of the residuals would show, namely:

- ◇whenever row effect and column effect are of the SAME sign, the corresponding residual is either small or NEGATIVE.
- ◇whenever row effect and column effect are of OPPOSITE signs, the corresponding residual is either small or POSITIVE.

Having made the row-PLUS-column fit--and looked below it--we have seen a very clear and prominent behavior that is not yet described. It is up to us to go further--to do something more in this obvious case and to learn how to do something that we can try when matters are less obvious.

The simplest combination of row effect and column effect that behaves one way when the signs are the same--and the other when they are different--is the product

(row effect) TIMES (column effect).

exhibit **9** of chapter 10: East Coast

Warming up on the East Coast

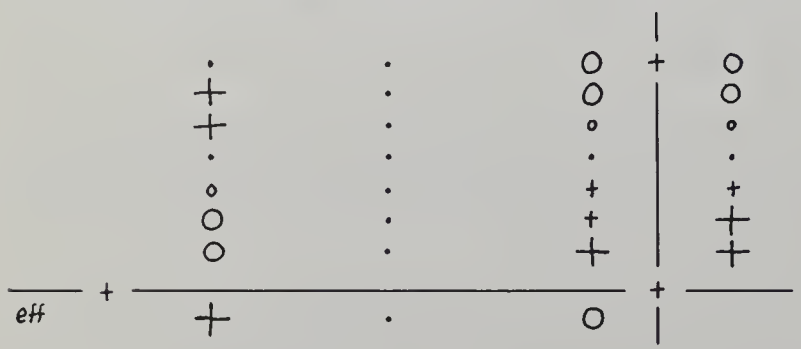
A) The DATA--monthly mean temperatures in °F

	Laredo	Washington	Caribou
Jan	**57.6**	**36.2**	**8.7**
Feb	**61.9**	**37.1**	**9.8**
Mar	**68.4**	**45.3**	**21.7**
Apr	**75.9**	**54.4**	**34.7**
May	**81.2**	**64.7**	**48.5**
June	**85.8**	**73.4**	**58.4**
July	**87.7**	**77.3**	**64.0**

B) ONE row-PLUS-column ANALYSIS of A--in 0.1°F

				eff
Jan	**0**	**1**	**−77**	−183
Feb	**33**	**0**	**−76**	−173
Mar	**16**	**0**	**−39**	−91
Apr	**0**	**0**	**0**	0
May	**−50**	**0**	**35**	103
June	**−91**	**0**	**47**	190
July	**−111**	**0**	**64**	229
	215	0	−193	**544**

C) The EFFECTS and RESIDUALS of B in CODED FORM



D) The COMPARISON VALUES

Laredo	Washington	Caribou
−72	0	65
−68	0	32
−36	0	32
0	0	0
41	0	−37
75	0	−67
91	0	−81

Examples: −72 = (215)(−183)/544 = (215/544)(−183),
−68 = (215/544)(−171)

→

Dividing every product by "all" will not change this. So our comparison values are appropriate tools or pointers.

Panel D sets out the comparison values, which clearly behave in the same general way as the residuals except for (1) a change of sign, (2) a possible difference in scale.

Clearly we ought to look at the possibility of extending our basic fit of

$$\text{common} + \text{row effect} + \text{column effect}$$

which we shall usually write as

$$\text{all} + \text{row} + \text{col}$$

to the form

$$\text{all} + \text{row} + \text{col} + k\frac{(\text{row})(\text{col})}{\text{all}}$$

where k is a constant to be determined.

The natural way to look for a value of k is to plot

$$\text{basic residual} = \text{data} \quad \text{MINUS} \quad (\text{all} + \text{row} + \text{col})$$

against

$$\text{comparison value} = \frac{(\text{row})(\text{col})}{\text{all}}.$$

Exhibit **10** does just this. The result is called a

diagnostic plot.

We see a very clear and strong dependence of basic residual on comparison value--with a suggestion of curvature, the negative slope appearing less pronounced on the left of zero than on the right. Trial with a transparent ruler or triangle shows us that

$$k = -1$$

exhibit **9** of chapter 10 (continued)

P) PROBLEMS--longish and hardish

9a) Make up a table for which the residuals tend to have the same sign as the comparison values, and imitate all panels above.

9b) Do the same for a table in which the residuals are roughly −2 times the comparison values.

9c) Do the same for a table in which the residuals are roughly +3 times the comparison values.

9d) Find a set of real data, not used in this book, where the residuals are clearly related to the comparison values, and imitate all panels above.

is not a bad compromise. Accordingly, exhibit **11** shows the residuals after including

$$-1.0\,\frac{(\text{row})(\text{col})}{\text{all}}$$

in the fit. Panel B of exhibit 11 shows how much the residuals have been drawn together by adding just one more constant to our fit.

Clearly, we have gained a surprising amount by going to an

$$\text{all-PLUS-row-PLUS-col-PLUS (constant)}\,\frac{(\text{row})(\text{col})}{\text{all}}$$

fit, which we will abbreviate as a

row-PLUS-col-PLUS-one fit.

It is safe to use this abbreviation, since no other way of adding a single constant to a row-PLUS-col fit is known that is nearly as often nearly as effective as the one we are learning about here.

exhibit **10** of chapter 10: East Coast

Plotting basic residuals against comparison values (data from exhibit 9)

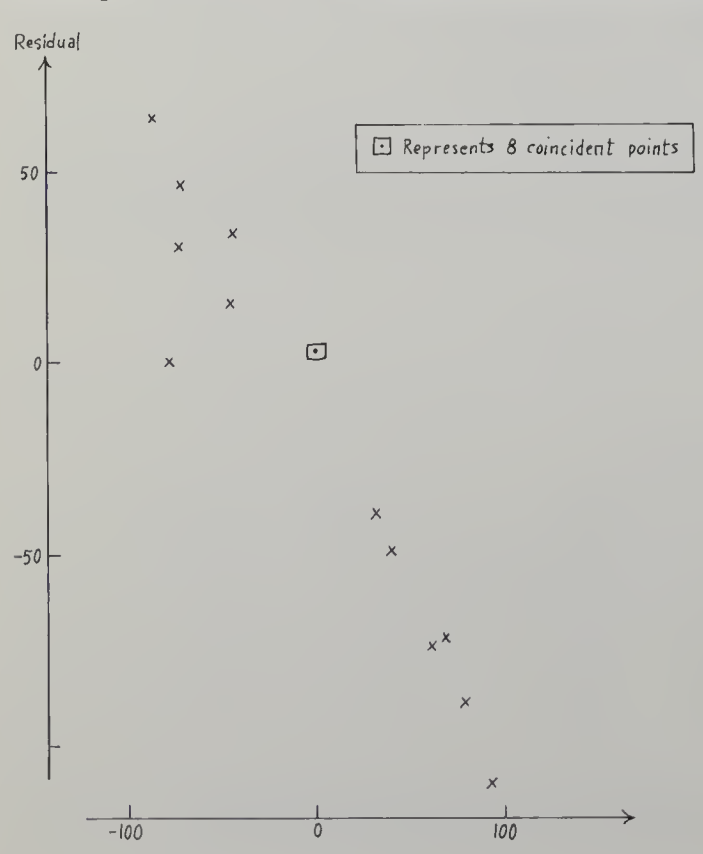


exhibit **11** of chapter 10: East Coast

The analysis with one additional constant (monthly mean temperatures in 0.1°F, data in exhibit)

A) The ANALYSIS

	residuals			values of −(row)(col)/544			
	Laredo	**Wash'n**	**Caribou**	**Laredo**	**Wash'n**	**Caribou**	eff
Jan	−72	1	−12	72	0	−65	−183
Feb	−35	0	−15	68	0	−61	−173
Mar	−20	0	−7	36	0	−32	−91
Apr	0	0	0	0	0	0	0
May	9	0	−2	0	0	37	103
June	−16	0	−20	−75	0	67	190
July	−20	0	−17	−91	0	81	229
eff	215	0	−193				544

Notes

1. The entries in the new panel are −1.0 times the comparison values found from the row, column, and common effects given. (Values in panel D of exhibit 9, for instance.)
2. The "PLUS-one" block can be omitted, and only the formula for its values given. Doing this saves space but decreases the viewer's understanding. Since there will almost always be room to put such a block above or below (if not to the left or right) of the residual block, it should almost always appear.

B) The RESIDUALS as a BATCH--before and after--rounded to 10's

(B of 9)		(A of 11)	
0+			
s	6		
f	5		
t	323		
0*	00100	**0***	000000
−0*	0000	**−0***	0111000
t	4	**t**	222222
f	5	**f**	4
s		**s**	7
−0+	988		
−1*	1		

P) PROBLEMS

11a1) Look at the residuals of panel A. Can you see any structure? What seems to be going on?

11a2) (Teaser) Can you suggest a further fit that might help?

review questions

If we want to fit just a little more--after a row-PLUS-column fit--what do we consider fitting? What is a comparison value? Why is it more convenient than just row effect times column effect? What example did we choose? What pattern did the row-PLUS-column residuals show? How was this related to the comparison values? How did we look? What is a diagnostic plot? What value of k did we choose? Did this improve the analysis?

10G. Converting PLUS to TIMES; re-expression

We have just seen--in the previous section--an example where the natural PLUS-one fit seems to be close to

$$\text{all} + \text{row} + \text{col} - \frac{(\text{row})(\text{col})}{\text{all}}.$$

Much more frequent is the case where the natural PLUS-one fit seems to be close to

$$\text{all} + \text{row} + \text{col} + \frac{(\text{row})(\text{col})}{\text{all}}$$

where the additional term has the same size but the opposite sign. A hard look at this formula--and a pinch of elementary algebra--shows us that it is identical in value with

$$\text{all}\left(1 + \frac{\text{row}}{\text{all}}\right)\left(1 + \frac{\text{col}}{\text{all}}\right),$$

as any of us can easily verify by multiplying out the latter form.

Thus, PLUS-one fits with coefficient +1.0 are just another way to find row-TIMES-column fits.

consumption again

It is now natural to return to the example of exhibit 6, where we skipped over a row-PLUS-column fit because it appeared so foolish. Exhibit **12** begins by setting out a row-PLUS-column analysis for this data--one in which the residuals are, of course, catastrophically large and systematic. Every reader should do the "everyone" problems (panel E), which will make him think hard. (*Hint.* compare the factors of increase from 1940 to 1960 for the various items of expenditure.)

A row-TIMES-column fit for a response--provided all row effects and all column effects are positive--is the same thing as a row-PLUS-column fit for the log of the response. (The log of a product is the sum of the logs, so we need only take logs of both sides.) Accordingly, finding a coefficient near +1.0 for a PLUS-one fit suggests, as one alternative, replacing the original reponse by its logarithm (unless there are negative values).

there could be others

And we would not be surprised if other coefficients suggested other re-expressions, presumbly--at least crudely--in accordance with the scale we

learned about in chapter 3, namely

$-1 \rightarrow$ square

$-0.5 \rightarrow$ take 3/2 power

$0 \rightarrow$ leave alone

$0.5 \rightarrow$ take square root

$1 \rightarrow$ take log

$1.5 \rightarrow$ take reciprocal square root

$2 \rightarrow$ take reciprocal.

exhibit **12** of chapter 10: personal consumption

The data of exhibit 5 examined anew.

A) A row-PLUS-column ANALYSIS of the data of **panel** A of **exhibit** 5

	1940	1945	1950	1955	1960	fit
Food	−31.2	−11.1	0.0	9.3	15.8	59.6
Household	−12.3	−9.5	0.0	3.2	5.8	29.0
Medical	.00	.00	.00	.00	.00	9.71
Personal	4.77	3.48	.00	−3.34	−8.44	2.45
Private	5.02	3.12	.00	−3.49	−9.55	1.80
eff	−6.18	−3.95	.00	4.29	11.39	

Note: This is the analysis that was clearly so foolish that we skipped over it in exhibit 6 above

E) PROBLEMS THAT EVERY READER SHOULD DO

12a) Convert panel A into an analysis in terms of effects.

12b) Construct the table of comparison values.

12c) Plot the basic residuals against the comparison values. This plot indicates what value or values of A?

12d) Find the residuals for the corresponding row-PLUS-column-PLUS-one fit or fits.

12e) Find the residuals for the row-PLUS-column-PLUS-one fit with coefficient = +1.0. Discuss what you have found––do these residuals make sense? Why or why not?

12f) Write the analysis with coefficient = +1.0 in the rows-TIMES-columns form. Compare with the result of completing panel G of exhibit 6.

P) FURTHER PROBLEMS

12g) Take the row-TIMES-column analysis obtained by completing panel G of exhibit 6 and write it as a row-PLUS-column-PLUS-one analysis.

12deg) Compare the results of (12g) with those of A, (12d), and (12e) above.

This is, indeed, a pretty good guess--good enough to be a quite helpful start when we do need something other than raw values or log values.

review questions

What kind of PLUS-one fit is fairly common? What is "all + row + col + (row)(col)/all" equivalent to? What is a row-PLUS-column-PLUS-one fit with $k = +1.0$ equivalent to? Can we convert PLUS to TIMES? How do logs let us go back to PLUS? Can we interpret other values of k? How? How often do you expect to want to try?

10H. How far have we come?

This chapter is our introduction to the richness of analysis possible when our data have two or more handles. All the techniques in earlier chapters used at most one handle, beyond the spread inherent in all data. Even middling and delineation used only the values of x to do something about the values of y. In this chapter we have dealt with the simplest case of one response and two families of circumstances, where all or nearly all combinations of a circumstance of one family with a circumstance of the other is represented by a response. This chapter needs to be paired with chapter 11, which follows, to give us a working set of tools.

We are now ready:

- ◇to use double lines to separate blocks of values that fit together by addition--a technique that we can often use to our advantage in the calculations of chapters 7, 8 and 9.
- ◇to check that a fit is really of the form row-PLUS-column by differencing both by rows and by columns, and then checking the differences for constancy in the other direction.
- ◇to emphasize exact zeros without confusion. In addition to using "z" we are now prepared to use "√" and to line up unadorned zeros with nearby decimal points.
- ◇to use triple lines to separate--and thus tie together--blocks of numbers that are to be combined by multiplication.

We are also ready:

- ◇to look at row-PLUS-column analyses, given as a table, in terms of either effects or fits.
- ◇to look at forget-one pictures (preferably with water-line scales) and focus our attention on (a) vertical position and (b) interlinking of points, giving no explicit attention to horizontal position.
- ◇to look at pictures of coded residuals.

◇to look at any row-TIMES-column fit, given as a table, and to wonder whether taking logs and reducing the table to a table for a row-PLUS-column fit to the log of the response is a good idea.

◇to find comparison values, one for each cell, as row-effect TIMES column-effect *divided by* common term, and to plot residuals against comparison values in diagnostic plots.

◇to fit a tilt to a diagnostic plot, thus converting our basic row-PLUS-column fit into a PLUS-one fit of the form

$$\text{all} + \text{row} + \text{col} + k \cdot \frac{(\text{row})(\text{col})}{(\text{all})}.$$

◇to convert PLUS-one fits with a constant k near +1.0 into TIMES fits.

◇when a PLUS-one fit helps, to add an extra block to our table of results to show what has been fitted.

We now recognize:

◇that there can be two-way residuals, and that finding them can be very useful.

◇the nature and flexibility of two-way fits.

◇the words "row", "column", "stripe" (for either row or column), "row effect", "column effect", "column fit", "row fit", and "common".

◇that, besides row-PLUS-column analyses, we can have row-TIMES-column analyses and row-PLUS-column-PLUS-one analyses.

We find ourselves:

◇well started on taking two-way tables apart.

◇taking things apart into more than two terms, so as to learn much more about it.

We are still looking for better and more thorough ways:

◇to find incomplete descriptions.

◇to lift incomplete descriptions out of the way so we can see what lies under them.

Those that we have newly found are even more helpful, when they can be applied, than those we learned earlier.

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EXHIBITS		PAGES
	11A	
1★		364
2★		367
3★		368
4★		369
5★		370
	11B	
6★		372
	11C	
7★		375
8		377
9		378
	11D	
10★		379
11		380
12		381
13★		381
	11E	
14		383
15		384
16		385
17★		385
18★		386
19		387
20		388
21★		389
	11F	
22		390
23		391
24★		393
25		394
26		395
27★		396
	11G	
	11H	

Making two-way analyses 11

As we pointed out in the opening to the last chapter (10, that is), these two chapters are side-to-side rather than end-to-end. (Thus, section 11A will have much more to do with section 10A than it will with 10E, 10F, and 10G, for example.) In this chapter we will usually concentrate on:

- ◇the mechanisms of making our analyses, with some attention to
- ◇the verbal arithmetic of what is going on.

As always, let us emphasize that we are doing exploratory data analysis, so that the techniques we illustrate are chosen to combine:

- ◇relative ease and simplicity of use.
- ◇adequate effectiveness (which may still leave them quite a way from the most stringent methods).
- ◇consistency with the methods already learned for use in other situations.

We assert that the results these methods give are quite "good" enough for exploration--which need not mean that they are well chosen to settle confirmatory questions.

review questions

How is this chapter related to the last? What properties have we sought to combine in our techniques? Are these techniques being advertised for use in settling confirmatory questions?

11A. Taking medians out

We opened the discussion of two-way tables--at the beginning of the previous chapter (exhibit 1 of chapter 10)--by taking medians out first by column (by place) and then by row (by month) and then stopping. This gave us--in that example--a quite respectable row-PLUS-column analysis. It is too much to expect that things will always be quite so easy.

Exhibit **1** starts with counts of numbers of an insect, *Leptinotarsa decemlineata*, each the sum of the counts for two plots treated alike, for all combinations of 4 treatments and 6 areas of the field chosen to be likely to be relatively homogeneous. (Agricultural experimenters call these areas "blocks", but we need this word for blocks of figures in this section.)

Panel A shows the counts and, in parentheses, treatment (row) medians. Panel B adopts the single-line–double-line convention of the last chapter, taking out the row medians and stating the column medians of what is left. It also drops the headings and stub for the duration of the calculation.

controlling errors

Panel C carries this two more steps. It also introduces an important practical device––circling the negative values. It is well known to those who do hand arithmetic (or hand algebra) that minus signs are not obtrusive enough. In the writer's experience, more errors in simple hand arithmetic can be traced to misreading––or insufficiently noticing––a sign than to all other causes com-

exhibit **1** of chapter 11: insect counts

Repeated median removal, illustrated for numbers of *Leptinotarsa decemlineata* for two plots combined

A) The DATA––and a beginning

	Area						
Treat	1	2	3	4	5	6	(median)
1	492	410	475	895	401	330	(442)
2	111	67	233	218	28	18	(89)
3	58	267	283	279	392	141	(273)
4	4	1	53	14	138	11	(12)

B) ONE FULL STEP––and more medians

492	410	475	895	401	330	442	50	−32	33	453	−41	−112
111	67	233	218	28	18	89	22	−22	144	129	−61	−71
58	267	283	279	392	141	273	−215	−6	10	6	119	−132
4	1	53	14	138	11	12	−8	−11	41	2	126	−1
							7	−16	92	68	39	−92

Example: 50 = 492 − 442

C) Taken TWO MORE STEPS––and negatives circled

492	410	475	895	401	330	492	50	(−32)	33	453	(−41)	(−112)
111	67	233	218	28	18	89	22	(−22)	144	129	(−61)	(−71)
58	267	283	279	392	141	273	(−215)	(−6)	10	6	119	(−132)
4	1	53	14	138	11	12	(−8)	(−11)	41	2	126	(−1)
							7	(−16)	37	68	39	(−92)
53	(−6)	6	395	(−70)	(−10)	(−10)	43	(−16)	(−4)	385	(−80)	(−20)
(−3)	(−24)	89	43	(−118)	3	18	15	(−6)	107	61	(−100)	21
(−189)	43	6	(−29)	113	(−7)	(−33)	(−222)	10	(−27)	(−62)	80	(−40)
(−19)	1	0	(−70)	83	87	4	(−15)	5	4	(−66)	87	91
−11	−2	6	7	6	−2							

→

bined. Especially in the sort of calculations done in this section, it is often helpful to make the signs much more noticeable. The simple way to do it is to circle negative values, as we have done in panel C.

The careful reader will have noticed that the median in the third column of the second block of panel B is an error--92 = $\frac{1}{2}$(41 + 144) having been put where 37 = $\frac{1}{2}$(33 + 41) belongs. In panel C, when 92 was taken out, the result was

(−59)

52

(−82)

(−51)

and since the three negative values were so noticeable, the error was detected and corrected. We must expect arithmetic errors when we do hand calculation. We need to arrange to (i) make fewer errors and (ii) catch the effects of most that we make, but it would be nonsense to have to work so carefully as not to make any.

Of course, if we can have a computer do our arithmetic for us, we can diminish errors greatly, as well as saving much effort. If the program has been carefully checked out--not always a simple task--we have only to worry about errors in getting the numbers in--and about the very infrequent printing error or machine breakdown.

subtraction

With the circle-negatives rule, subtraction can be much safer. As we go from block 2(NE) to block 3(SE) of panel C, we need to take out 7 from the first column. This means lowering each value by 7, a process which sounds easy. The most difficult aspect of lowering is that this makes the size of positive

exhibit **1** of chapter 11 (continued)

E) EXERCISES to be done by ALL READERS

The logs (in 0.01) of the counts in panel A are:

269	261	268	295	260	252
204	183	237	234	145	126
176	243	245	245	259	215
60	0	172	115	214	104

1a) Imitate panel B, starting from the log counts.

1a2) Imitate panel C.

S) SOURCE

Geoffrey Beall, 1942, "The transformation of data from entomological field experiments so that the analysis of variance becomes applicable," Biometrika 32, 243–262. Table 4 on page 245.

numbers smaller, but the size of negative numbers larger. Here again the circled negatives help us keep our arithmetic straighter.

back to the example

At the end of panel C, we have taken 3 steps--and are ready to take a fourth. The largest size of a median taken out has been 442 in step 1, 92 in step 2, 33 in step 3, and will be 11 in step 4. It would appear progress is being made. But it looks as though we might do well to go a little farther. (Some may like to stop here.)

Exhibit **2** takes repeated median removal as far as it can go with this example (unless we give up taking medians to the nearest integer and start using fractions.) The largest median taken out at the successive steps is now 442, 92, 33, 11, 6, 2, 2, 2, 1, 1. In view of the sizes of the residuals remaining --much, much larger than 1 or 2--it is quite unlikely that we needed to go "to the end". We could have stopped earlier.

a rule of thumb

It is often both convenient and sparing of effort to take

four steps.

(Some speak of this as "four steps of median polish"; others prefer to say "two cycles", since in four steps we have taken out row and column medians twice each.) This is a reasonable guideline, not an immutable law. If we take four steps, we need to at least find the medians we would use in the next step, to convince ourselves--if this is right--that we can stop with four steps in our particular instance.

the fit

Having the residuals and the starting values, we can find the fit. Exhibit 3 of chapter 10 has already illustrated finding and checking the fit, but exhibit **3** here does this again to show how the checking can be laid out so as to be easier and more effective. Panel A finds what we hope is the fit. Panel B starts to check it by looking at column-to-column differences in each row. Between the first and second columns, to begin with, we have $414 - 428 = -14$, $89 - 103 = -14$, $222 - 236 = -14$, and $-2 - 12 = -14$, so all four of these differences are -14. We write (-14) at the top of the space between these columns, and put a $\surd$ on every row where the difference is -14. Similarly with (61) from column 2 to column 3. When we go from column 3 to column 4, we find three differences of 32, and one that isn't. So we make only 3 $\surd$'s. And so on. The last check column, headed (90), has the differences from the last column to the **first** column.

Panel C does the same for differences between adjacent rows, as well as adjacent columns. It also puts the first-MINUS-last checks both before the first and after the last.

exhibit **2** of chapter 11: insect counts

Repeated median removal taken further (initial stages from exhibit 1)

A) The CALCULATION

492	410	475	895	401	330	442	50	−32	33	453	−41	−112
111	67	233	218	28	18	89	22	−22	144	129	−61	−71
58	267	283	279	392	141	273	−215	−6	10	6	119	−131
4	1	53	14	138	11	12	−6	−11	41	2	126	−1
							7	−16	37	68	39	−92

53	−6	6	395	−70	−10	−10	43	−16	−4	385	−80	−20
−3	−24	89	43	−118	3	18	15	−6	107	61	−100	21
−189	43	6	−29	113	−6	−33	−222	10	−27	−62	80	−39
−19	1	0	−70	83	87	4	−15	5	4	−66	87	91
−11	−2	6	7	6	−2							

64	−4	0	388	−76	−8	−2	66	−2	2	390	−74	−6
8	−22	83	36	−124	5	6	2	−28	77	30	−130	−1
−178	45	0	−36	107	−4	−2	−176	47	2	−34	109	−2
−8	3	−6	−63	77	89	−2	−6	5	−4	−61	79	91
							−2	2	2	−2	2	−2

70	−2	2	394	−74	−2	−2	68	−4	0	392	−76	−4
2	−32	73	30	−134	−1	2	4	−30	75	32	−132	1
−174	45	0	−32	107	0	√	−174	45	0	−32	107	0
−4	3	−6	−59	77	93	√	−4	3	−6	−59	77	93
−1	√	1	−1	2	√							

71	−2	1	395	−76	−2	√	71	−2	1	395	−76	−2
3	−32	72	31	−136	−1	1	2	−33	71	30	−137	−2
−173	45	−1	−31	105	0	√	−173	45	−1	−31	105	0
−3	3	−7	−58	75	93	√	−3	3	−7	−58	75	93
							√	√	√	√	√	−1

√	71	−2	1	395	−76	−1
√	2	−33	71	30	−137	−1
√	−173	45	−1	−31	105	1
√	−3	3	−7	−58	75	94

P) PROBLEMS

2a) Draw a picture of this exhibit on a separate sheet of paper, and then draw a line showing how the calculation flows from block to block.

2b) Find a set of data of interest to you of about the same size and do the corresponding calculations.

In panel C it is easy to see that most of the values are tied firmly together with √'s. The two exceptions, 77 in the bottom row and 137 in the last column, stand out clearly. What does it seem that these should be? We have

$$59 + 32 = 91 \qquad 285 - 139 = 146$$
$$61 - (-30) = 91 \qquad 13 + 133 = 146$$
$$315 - 224 = 91 \qquad -78 - (-224) = 146$$
$$507 - 416 = 91 \qquad 236 - 90 = 146$$

exhibit **3** of chapter 11: insect counts

Finding and checking the fit

A) FINDING--after four steps

		starting from						residuals			
492	410	475	895	401	330	64	−4	0	388	−76	−8
111	67	233	218	28	18	8	−22	83	36	−124	5
58	267	283	279	392	141	−178	45	0	−36	107	4
4	1	53	14	138	11	−8	3	−6	−63	77	89

		fit(?)			
428	414	475	507	477	338
103	89	150	182	152	13
236	222	283	315	285	137
12	−2	59	77	61	−78

Examples: 428 = 492 − 64
414 = 410 − (−4)

B) CHECKING--one way, differences in parentheses

	(−14)		(61)		(32)		(−30)		(−139)		(90)
428	√	414	√	475	√	507	√	477	√	338	√
103	√	89	√	150	√	182	√	152	√	13	√
236	√	222	√	283	√	315	√	285		137	
12	√	−2	√	59		77		61	√	−78	√

C) CHECKING--the second way also, just to be sure

	(90)		(−14)		(61)		(32)		(−30)		(−139)		(90)
(416)		√		√		√				√		√	
	√	428	√	414	√	475	√	507	√	477	√	338	√
(−325)		√		√		√		√		√		√	
	√	103	√	89	√	150	√	182	√	152	√	13	√
(133)		√		√		√		√		√			
		236	√	222	√	283	√	315	√	285		137	
(−224)		√		√		√				√			
	√	12	√	−2	√	59		77		61	√	−78	√
(416)		√		√		√				√		√	

P) PROBLEMS

3a) (uses (2b)) Check the final step of problem (2b).

3b) Find another analysis of interest to yourself, and check it.

Whichever way we approach these values, we find that 77 should have been 91 and that 137 should have been 146.

Turning back to panel A, we find that we had reached 77 as

$$14 - (-63) = 77$$

and 137 as

$$141 - (-4) = \text{(should have made) } 145.$$

(That minus sign again!)

If we are to have a possible fit, we need to change the residual of −63 to one of −77 and the residual of −4 to one of −5. We will then have

$$14 - (-77) = 91$$

$$141 - (-5) = 146$$

and we will be as in panel A of exhibit **4**.

Panel B of exhibit 4 shows how the block values for the fit can be broken down, first into a row of column effects, and then into a column of row effects and a common term.

Exhibit **5** offers us a variety of sets of data to play with.

review questions

How did we find residuals at the start of the last chapter? What example did we begin with here? What did we do to it? How did we arrange the arithmetic? What previously introduced convention did we make use of? What new convention did we add? Can you draw a picture showing the order of calculation of the blocks of numbers in exhibit 2? Why is circling negative numbers useful? In general? In thinking of taking out −5? How many steps might we ordinarily take? How many cycles? How do we calculate values for the fit? How do we check the results? How do we take a table of row-PLUS-column values apart?

exhibit **4** of chapter 11: insect counts

Breaking down the fit (numbers from exhibit 3, corrected as in text)

A) RESIDUALS and FIT

residuals						fit					
64	−4	0	388	−76	−8	428	414	475	507	477	338
8	−22	83	36	−124	5	103	89	150	182	152	13
−178	45	0	−36	107	−5	236	222	283	315	285	146
−8	3	−6	−77	77	89	12	−2	59	91	61	−78

→

exhibit **4** of chapter 11 (continued)

B) BREAKING IT DOWN

Taking the median out of one row:

12 −2 59 91 61 −78 | 36 || −24 −38 23 55 25 −114

Taking one of these results out of the corresponding column

428		452		258
103	less −24 gives	127	with median 194 giving	−67
236		260		66
12		36		−158

Result

194	−24	−38	23	55	25	−114
258						
−67						
66						
−158						

C) The row-PLUS-column ANALYSIS

Treat	−194	**1** −24	**2** −38	**3** 23	**4** 55	**5** 25	**6** −114
1	258	64	−4	0	388	−76	−8
2	−67	8	−22	83	36	−124	5
3	66	−178	45	0	−36	107	−5
4	−158	−8	3	−6	−77	77	89

P) PROBLEMS

4a) (uses (2b)) Break the fit of problem (2b) down in two different ways.

4b) Find another fit of interest to yourself; break it down.

exhibit **5** of chapter 11: data and problems

Problems on repeated medians.

A) WHEAT DRYING DATA––Ratio of dry to wet weights for wheat grain

	NITROGEN APPLIED			
Block	None	Early	Middle	Late
1	.718	.732	.734	.792
2	.725	.781	.725	.716
3	.704	1.035	.763	.758
4	.726	.765	.738	.761

→

exhibit **5** of chapter 11 (continued)

B) A SMALL NEBRASKA VOTING EXAMPLE––percent Democratic of major party vote

Mark	'20	'28	'36	'44	'52	'60	County
D0	46.7	59.0	54.6	34.5	18.9	23.6	Grant
D1	33.7	23.7	39.9	23.8	14.9	13.8	Hooker
D2	40.4	27.2	50.5	38.8	19.7	25.1	Thomas
D4	34.9	26.6	51.6	40.4	23.0	21.2	Blaine
D5	25.4	15.1	43.3	27.2	17.7	22.5	Loup
D6	29.2	16.6	48.4	31.3	19.5	22.6	Garfield
D7	31.9	35.4	57.5	44.2	33.7	36.0	Wheeler

Note: 7 counties in a row from west to east just north of the center of Nebraska.

C) YIELDS OF SUGAR BEET––tons

	Fertilizer Applied		
Block	None	PO_4 only	PO_4 NO_3
1	2.45	6.71	6.48
2	2.25	5.44	7.11
3	4.38	4.92	5.88
4	4.35	5.23	7.54
5	3.42	6.74	6.61
6	3.27	4.74	8.86

D) BIOLOGICAL VALUE OF PROTEINS––as calculated from rat growth

Pair of rats	Biological value	
	Raw	Roasted
1	61	55
2	60	54
3	56	47
4	63	59
5	56	51
6	63	61
7	59	57
8	56	54
9	44	62
10	61	58

P) PROBLEMS

5A) Take medians of the data of panel A repeatedly.

5b) The same for panel B.

5c) The same for panel C.

5d) The same for panel D.

S) SOURCES

Data of panel A: W. G. Cochran 1947. "Some consequences when the assumptions for the analysis of variance are not satisfied," Biometrics 3: 22–38, especially Table I at page 28.

Data of panel B: Richard M. Seammon, 1965. *America At the Polls.* A Handbook of American Presidential Election Statistics, 1920–1964. Pages 77–78.

Data of panel C: G. W. Snedecor 1946. *Statistical Methods,* 4th edition, Iowa State College Press, page 332.

Data of panel D: H. H. Mitchell, W. Burrough, J. R. Beadles, 1936, "The significance and accuracy of biological value of proteins computed from nitrogen metabolism data." Journal of Nutrition 11: 257–274.

11B. Alternative organizations of the arithmetic

We have fixed one reasonable route to a row-PLUS-column fit:

◇first repeatedly remove medians (usually at least four steps) to find residuals.

◇from initial values and residuals, find the fit, and check that it is row-PLUS-column, correcting residuals as necessary. (If these corrections leave the residuals too far askew, some further steps of median removal may be necessary.)

◇decompose the block of values of fit into column effects, row effects, and common.

This route is fairly nearly self-checking; moreover, errors can usually be corrected for without knowing where or how they occurred. This is a good safe procedure, and relatively easy to carry out.

Of course it is not the only reasonable procedure. Instances of procedures with different aims will come before us when we reach section 11J (in Chapter 11^+). One instance of a procedure with the same goals and results--one differing only in how we do the bookkeeping--is worth noting here.

another pattern

Exhibit **6** goes through the successive steps, with a full effects-only analysis at each stage. One can always compute this way--sometimes (as we will see when we reach section 13A) we find this the procedure of choice.

The general pattern is simple:

◇the median to be removed is ADDED to the corresponding main effect and SUBTRACTED from all the corresponding residuals.

Clearly, if we do the arithmetic correctly, such "sweepings out" do not alter the sum of effect and residual.

exhibit **6** of chapter 11: insect counts

Repeated median removal in another format (initial numbers from exhibit 2, medians in parentheses)

A) START

0	0	0	0	0	0	0	(0)
0	492	410	475	895	401	330	(442)
0	111	67	233	218	28	18	(89)
0	58	267	283	279	392	141	(273)
0	4	1	53	14	138	11	(12)

→

exhibit 6 of chapter 11 (continued)

B) AFTER STEP 1

0	0	0	0	0	0	0
442	50	−32	33	453	−41	−112
89	22	−22	144	129	−61	−71
273	−215	−6	10	6	119	−131
12	−8	−11	41	2	126	−1
(181)	(7)	(−16)	(37)	(68)	(39)	(−92)

Examples:

442 = 0 + 442
50 = 492 − 542
89 = 0 + 89
22 = 111 − 89

C) AFTER STEP 2

181	7	−16	37	68	39	−92	(22)
261	43	−16	−4	385	−80	−20	(−10)
−92	15	−6	107	61	−100	21	(18)
92	−222	10	−27	−62	80	−39	(−33)
−169	−15	5	4	−66	87	91	(4)

Examples:

181 = 0 + 181
261 = 442 − 181
−92 = 89 − 181
. . . . . .
7 = 0 + 7
43 = 50 − 7
15 = 22 − 7
. . . . . .

D) AFTER STEP 3

203	−15	−38	15	46	17	−114
251	53	−6	6	395	−70	−10
−74	−3	−24	89	43	−118	3
59	−189	43	6	−29	113	−6
−165	−19	1	0	−70	83	87
(−8)	(−11)	(−2)	(6)	(7)	(6)	(−2)

Examples:

203 = 181 + 22
−15 = 7 − 22
−38 = −16 − 22
. . . . . .
−114 = −92 − 22
251 = 261 − 10
53 = 43 + 10
. . . . . .

E) AFTER STEP 4

195	−26	−40	21	53	23	−116	(−2)
259	64	−4	0	338	−76	−8	(−2)
−66	8	−22	83	36	−124	5	(6)
67	−178	45	0	−36	107	−4	(−2)
−157	−8	3	−6	−77	77	89	(−2)

Examples: (Make your own) →

review questions

Describe our first overall process for making a row-PLUS-column analysis. What is another pattern of calculation?

11C. Making the core of a two-way plot

Making the kind of two-way plot of a fit we found helpful in the last chapter is surprisingly easy. All we have to do is to write the **fit** as a sum of two parts, as in any of these examples:

all + col PLUS row

all + row PLUS col

($\frac{2}{3}$ all + col PLUS $\frac{1}{3}$ all + row)

exhibit **6** of chapter 11 (continued)

F) AFTER STEP 5

193	−24	−38	23	55	25	−114
257	66	−2	2	390	−74	−6
−60	2	−28	77	30	−130	−1
65	−176	47	2	−34	109	−2
−159	−6	5	−4	−75	79	91
	(−2)	(2)	(2)	(−2)	(2)	(−2)

Examples: (Make your own)

P) PROBLEMS

6a) Write down the analysis after Step 6.

6a2) After Step 7.

6a3) After Step 8.

6a4) After Step 9.

6b) Write out 7 further examples of the changes made in panel B, choosing values that involve every row and every column.

6c) Do the same with 7 further values for panel C.

6d) And for panel D.

6e/f) Write out 10 examples, for values involving every row and every column––including the effect rows and columns––for panel E/F, respectively.

such that one part depends only on the column and the other only upon the row.

We then take these parts as (rectangular) coordinates. When we do this, the resulting points--one for each combination of a row and a column--make up a rectangular pattern. More precisely, they are represented by all the intersections of one family of horizontal lines and one family of vertical lines.

Exhibit **7** shows what happens numerically for the fit of panel B of exhibit 4. Exhibit **8** shows the resulting picture. The upper panel shows four vertical lines, corresponding to the four treatments, and six horizontal lines, corresponding to the six areas on which the experiment was repeated. The lower panel also shows a few of the lines whose equations are

one part PLUS the other part EQUALS constant.

exhibit **7** of chapter 11: insect counts

Breaking the fit in two parts (initial numbers from exhibit 4)

A) The FIT--in terms of common and effects

		A	B	C	D	E	F
	194	−24	−38	23	55	25	−114
1	258						
2	−67						
3	66						
4	−158						

B) ONE BREAKING IN TWO

	−24	−38	23	55	25	−114
452						
127						
260						
36						

Examples:

452 = 194 + 258
127 = 194 + (−67)
260 = 194 + 66
36 = 194 + (−158)

C) ANOTHER

	170	156	217	249	219	80
258						
−67						
66						
−158						

Do your own examples!

Since the fit is the sum of these parts, they are also lines

$$\text{sum} = \text{constant}$$

or

$$\text{fit} = \text{constant}.$$

We now have all the combinations of rows and columns plotted in such a way that the curves

$$\text{fit} = \text{constant}$$

are parallel straight lines.

To make a two-way plot instead, all we have to do is to twist the graph paper by 45° under the tracing paper before we make the tracing. Exhibit **9** shows the result. Here, as we will usually do, we have shown the level lines

$$\text{fit} = \text{constant}$$

as "water-levels" on both sides of the picture.

review questions

What is the first step in making a two-way plot? Do we have any choices? Why/why not? What do we plot on the graph paper? What does the main pattern look like? What do we add? How do we make a tracing? How do we show fit = constant?

exhibit **7** of chapter 11 (continued)

D) YET ANOTHER

	70	56	117	149	119	−20
358						
33						
166						
−58						

Examples:

358 = 258 + 100
33 = −67 + 100
. . . . . .
70 = −24 + 94
56 = −38 + 94
. . . . . .

P) PROBLEMS

7a) Write out 3 further splittings into two parts.

7b) Imitate exhibit 8, starting from panel C. How does the result differ from exhibit 8?

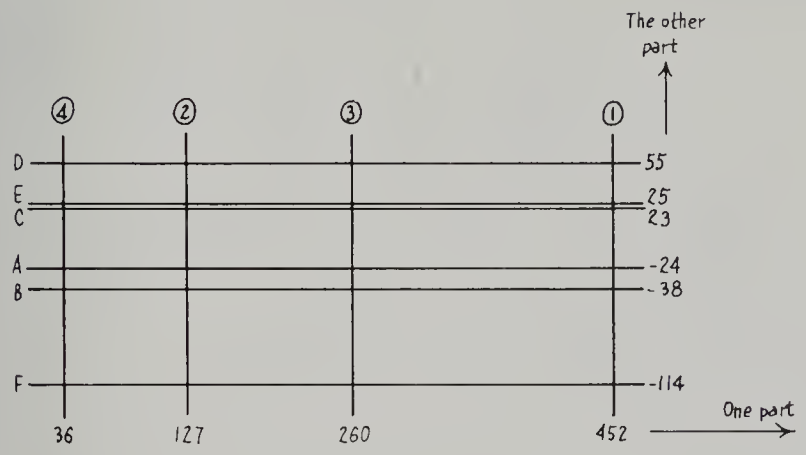
7c) Same for panel D.

7d) (Uses (7a)) Same for one of the further splittings used as an answer to (7a).

exhibit **8** of chapter 11: insect counts

Two stages in constructing a two-way plot of a fit (numbers from exhibit 7)

A) The TWO PARTS plotted by LINES



B) Some lines "ONE PART plus THE OTHER PART = CONSTANT" added

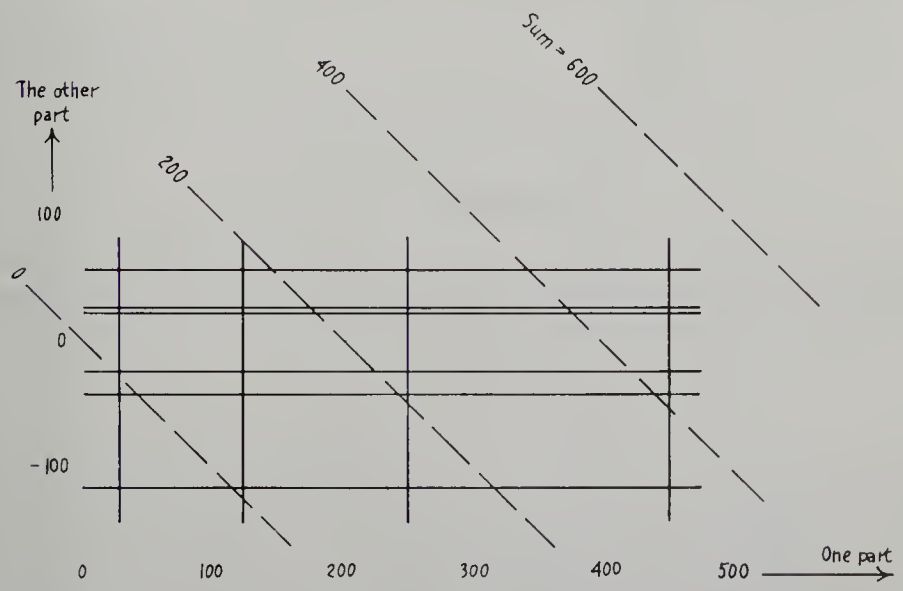
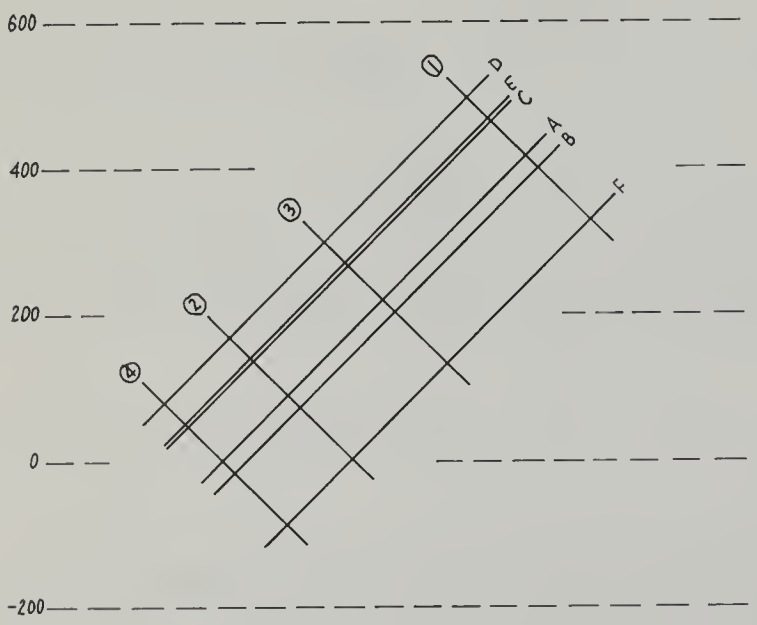


exhibit **9** of chapter 11: insect counts

The bare two-way plot (numbers from exhibit 7)



11D. Going on with the residuals

the residuals

To complete this plot, we need to turn to the residuals. Exhibit **10** has stem-and-leaf and letter-value displays. We see that we have two outside negative and one far out positive. We now need to plot

actual = fit PLUS residual

for these three.

Just how we do this depends a little on how large the residuals are. Consider the residual of −124, which occurs at 2E, which is plotted as (127, 25). We want a point whose total is 124 lower. The easiest ones are (3, 25), found from 127 − 124 = 3, and (127, −99), found from 25 − 124 = −99. On the graph paper--before the 45° twist--these are directly to the left and directly below the fit. On the two-way plot, then, they will lie off at ±45°. We would rather go straight down, which we can do easily by subtracting $\frac{1}{2}(124) = 62$ from each original coordinate, finding (65, −37) from 127 − 62 = 65 and 25 − 62 = −37. Exhibit **11** shows how this looks on the graph paper.

For a big residual, like this one, it is easiest to begin by halving and applying one-half to each coordinate. For a little residual--as +5 would have been in this example--it is easier to go down each coordinate separately,

exhibit **10** of chapter 11: insect counts

Summarization of residuals

A) The RESIDUALS as they fall--rows and columns reordered

Treat	D	E	C	A	B	F
1	388	−76	0	64	−4	−8
3	−36	107	0	−178	45	−5
2	36	−124	83	8	−22	5
4	−77	77	−6	−8	3	89

B) STEM-and-LEAF

1	H	388
2	**10**	7
	9	
4	**8**	39
5	**7**	7
6	**6**	4
	5	
7	**4**	5
8	**3**	6
	2	
	1	
12	**0**	0835
12	**−0**	068485
	−1	
6	**−2**	2
5	**−3**	6
	−4	
	−5	
	−6	
4	**−7**	76
	−8	76
	−9	
2	L	−124, −178

C) LETTER-VALUES

#24 residuals

M	12h	0		
H	6h	54	−15	69
E	3h	86	−76	
	2	107	−124	
	1	388	−178	

104

f	158	−119	out = −124, −178
	xxx	two	far = 388
F	262	−223	
	one	xxx	adj: −77 and 107

D) For SPECIAL TREATMENT

	Parts of fit		endpoints		actual
+388 at 1 D	(452, 55)	→	(840, 55)	or (452, 443)	or (646, 249)
−124 at 2 E	(127, 25)	→	(3, 25)	or (127, −99)	or (65, −37)
−178 at 3 A	(260, −24)	→	(82, −24)	or (260, −202)	or (171, −113)

P) PROBLEMS

10a) (uses (5a)) Summarize the residuals with which you ended problem (5a).

10b) (uses (5b)) Same for (5b).

10c) (uses (5c)) Same for (5c).

10d) (uses (5d)) Same for (5d).

connecting the two points thus found by a line, and then--when making the tracing-paper version--drawing a line from the lit to the middle of this line.

Exhibit 11 also shows how the schematic plot for the other 21 residuals is made. We draw a diagonal line--at exactly 45°--for the center line. (It is easiest if this goes through the corners of the large squares.) Then we choose two or three convenient points along it, and step off--either horizontally or vertically--the adjacent and hinge values, here −76, −15, 54, and 107. We can then draw the ends of the box and the joins to the adjacent lines, and are ready to trace the result at the foot of the two-way plot.

Exhibit **12** is the result. We now see that the residuals are so large compared to what is described by the lit that the usefulness of this fit is close to being in serious doubt. (In part dealt with by one of the problems of exhibit **13**.)

exhibit **11** of chapter 11: insect counts

Two constructions on the graph paper (numbers from exhibit 10)

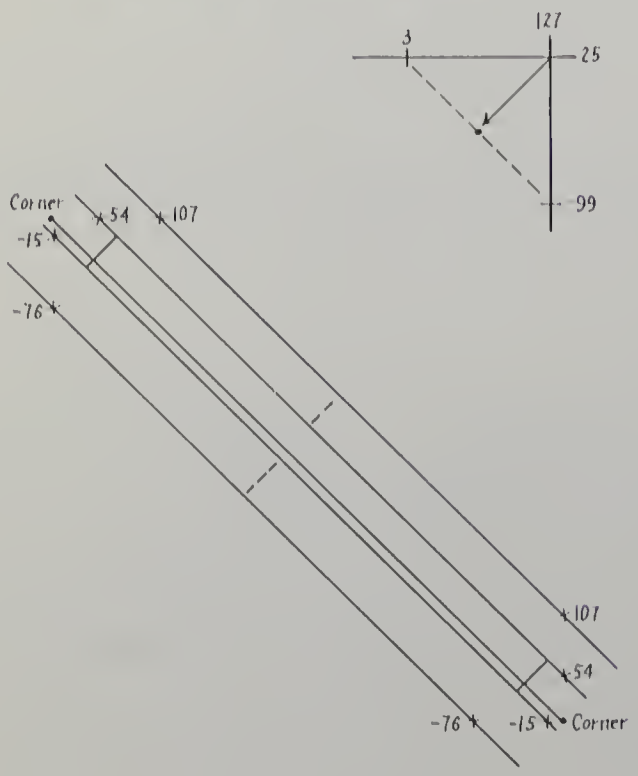


exhibit **12** of chapter 11: insect counts

The completed two-way plot (see exhibits 9, 11, etc.)

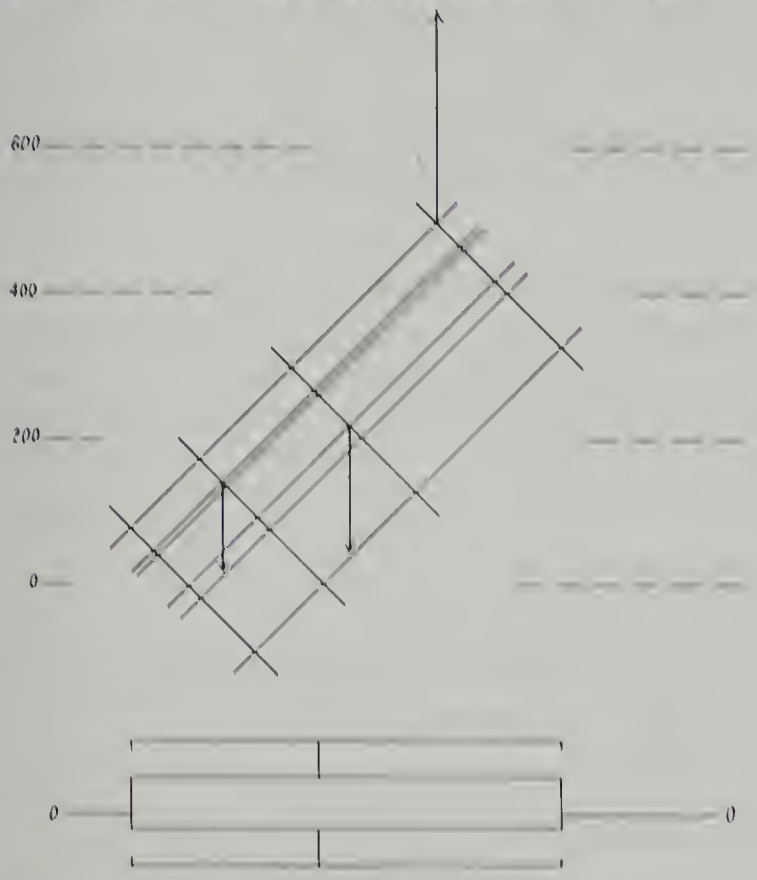


exhibit **13** of chapter 11: data and problems

Problems on two-way plots

A) AN ANALYSIS OF MEAN MONTHLY TEMPERATURES at SEVEN EAST COAST PLACES--in °F

	Laredo	Baton Rouge	Atlanta	Wash.	Boston	Portland	Caribou	off
Jan.	3.8	5.4	2.9	0.3	1.1	3.3	−8.0	−20.5
Feb.	6.3	6.8	3.2	−0.6	−2.8	4.3	−8.7	−18.7
Mar.	4.8	3.6	1.2	0.4	2.4	1.9	−4.9	−10.7
Apr.	3.0	0.7	0.9	−0.6	−2.1	−1.2	−1.1	−1.4
May	−1.0	−1.8	−0.1	−0.4	−0.8	−0.1	3.4	7.9
June	−4.4	−3.8	−0.4	1.1	0.6	1.4	5.3	15.9
July	−6.2	−6.0	−2.3	1.3	1.9	3.7	7.2	19.6
Aug.	−5.0	−5.3	−2.1	0.5	2.3	3.4	6.1	18.5
Sept.	−3.4	−2.6	−0.4	0.6	1.0	1.3	3.4	12.6
Oct.	−0.7	−0.8	−1.7	−1.1	1.4	1.0	2.0	2.9
Nov.	−6.2	−0.3	−2.2	−0.6	1.8	0.9	0.1	−8.8
Dec.	2.9	3.8	0.7	−0.7	−0.2	−1.9	−4.9	−17.7
off	18.2	11.4	6.1	0.3	−5.4	−11.8	−18.9	−56.1

→

review questions

What do we do with all residuals? Which do we show separately on the two-way plot? How do we find where to plot large residuals? What if those to be plotted are small? What kind of schematic plot do we make for the other residuals? How do we choose the spacings between lines? How do we plot them? What does the final two-way plot look like? In general? For our example? What does this lead us to think in the example?

11E. Coding residuals; condensing fits and residuals

For our two-way plots of residuals we use coding, using a standard set of seven symbols to indicate roughly the value of the residual. We have already seen--toward the end of section 10E--how we use these symbols, plotting them over an invisible two-way plot of the fit. We also saw how it can be quite reasonable to use judgment in deciding what range of values is converted into each symbol. We left to this chapter the statement of a standard coding--something to be used either when there is no urge to use judgment or as a start, as a basis from which judgment can begin.

Exhibit **14** sets out the standard cutting points, both generally and for the analysis we have been using as an example. In the example, we have 12 ·'s, as

exhibit **13** of chapter 11 (continued)

P) PROBLEMS

13a) Make the core of a two-way plot of the fit in panel A above.

13a2) Complete the two-way plot.

13b) (uses (5a)) Make the core of a two-way plot of the fit of problem (5a).

13b2) Complete the two-way plot.

13c/c2) (uses (5b)) Same for problem (5b).

13d/d2) (uses (5c)) Same for problem (5c).

13e/e2) (uses (5d)) Same for problem (5d).

S) SOURCE--for panel A

Climatography of the United States, No.: #; Climatography of the states: *state*; U.S. Weather Bureau

Place	Airport	#	*state*	Page
Laredo	Air Force Base	60–41	Texas	17
Baton Rouge	Ryan	60–16	Louisiana	8
Atlanta	Municipal	60–9	Georgia	7
Washington D.C.)	National	60–44	Maryland	9
Boston	Logan International	60–19	Massachusetts	7
Portland	City	60–17	Maine	8
Caribou	Municipal	60–17	Maine	8

we might expect ($12 = \frac{1}{2}(24)$ and the ·'s go from one hinge to the other). Of the six high, one is # and five are +. Of the six low, one is ⊙, one ○, and four ∘. These seem reasonable, as does the decoding into values set out in panel C.

exhibit **14** of chapter 11: general, and insect counts

The standard cutting points, and their application to the residuals displayed in exhibit 10 (draws heavily on exhibit 10)

A) The STANDARD CODE

#

U-hinge + 2 · H-spread

+

U-hinge + H-spread

+

U-hinge

·

L-hinge

∘

L-hinge − H-spread

○

L-hinge − 2·H-spread

⊙

(In case of doubt, round outward.)

B) The CUTS and COUNTS for the example--H-spread = 69

	one
192	
	xxx
123	
	five
54	
	twelve
−15	
	four
−84	
	one
−153	
	one

C) The ACTUAL RANGES used in the example

code	values	split in two
#	388	(194, 194)
+	xxx	
+	64 to 107	(32, 32) to (53, 54)
·	−8 to 45	(−4, −4) to (22, 23)
∘	−77 to −22	(−38, −39) to (−11, −11)
○	−124	(−62, −62)
⊙	−178	(−89, −89)

D) The CODED RESIDUALS arranged by ORDERED EFFECTS--as in panel A of exhibit 10

	D	E	C	A	B	F
1	#	∘	·	+	·	·
3	∘	+	·	⊙	·	·
2	·	○	+	·	∘	·
4	∘	+	·	·	·	+

Exhibit **15** shows the resulting plot. The scale at the right margin is constructed as in exhibit **16**. We again put half of each value in each coordinate. Exhibit **17** sets out some problems.

condensing

When we look at exhibit 8, it is very hard to escape taking the distinction between area E, fitted at 25, and area C, fitted at 23, as useless and confusing. The distinction between A, fitted at −24 and B, fitted at −38, might be useful --so far as this exhibit goes. But when we look at the sizes of the residuals, we doubt the value of this also.

So far as the plot of fit goes, we gain by condensing E with C--and we can live with leaving A and B separate. The plot of coded residuals can usefully involve condensing both E with C and A with B. When we condense residuals, we replace the residuals involved by their median. (We do **not** recalculate residuals from the condensed fit, if any.) Exhibit **18** shows the residuals after condensation, and their coding. Exhibit **19** shows the two-way plot of residuals.

exhibit **15** of chapter 11: insect counts

Plot of coded values (based on exhibit 14)

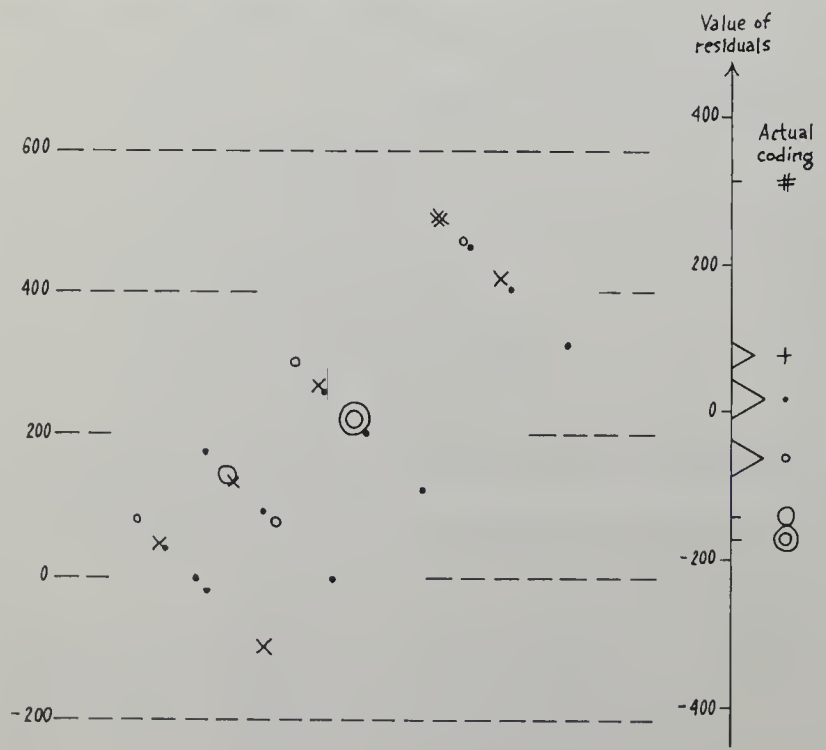


exhibit **16** of chapter 11: insect counts

Construction of scale for exhibit 15

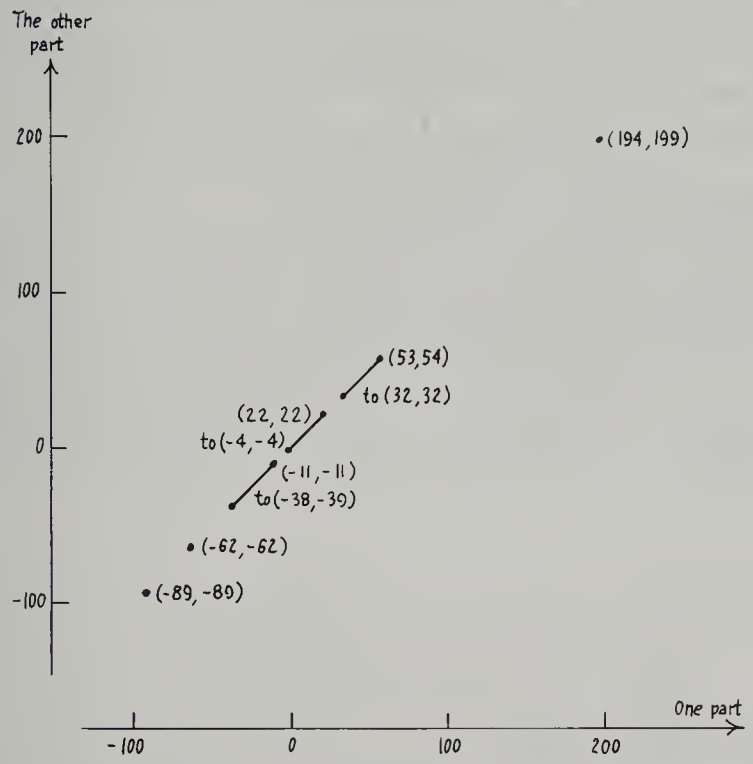


exhibit **17** of chapter 11: data and problems

More problems on two-way plots

A) AN ANALYSIS OF MEAN MONTHLY TEMPERATURES FOR FOUR ARIZONA AIRPORTS--in °F

	Flagstaff	Prescott	Phoenix	Yuma	eff
Jan	0.3	−0.3	−0.1	0.2	−19.6
Feb	−0.5	−0.5	0.2	0.7	−15.3
Mar	−0.4	−0.6	0.3	0.8	−9.8
Apr	−0.3	−0.1	−0.1	0.3	−2.2
May	−0.3	0.0	0.4	0.0	5.7
June	−0.4	0.8	0.4	−1.0	14.1
July	0.1	0.3	0.1	−0.6	20.5
Aug	−0.1	−0.2	0.0	0.1	18.9
Sept	−0.7	−0.1	0.2	0.5	13.1
Oct	−0.1	0.2	−0.2	0.1	1.6
Nov	1.2	0.4	−1.0	−0.5	−10.0
Dec	0.9	−0.3	−0.2	−0.5	−17.1
eff	−16.3	−5.7	8.5	13.8	60.9

→

exhibit **17** of chapter 11 (continued)

P) PROBLEMS

17a) Stem-and-leaf the residuals of panel A, above, and code them.

17a2) Make a two-way plot of these coded residuals.

17b) Stem-and-leaf the residuals of panel A of exhibit 13 and code them.

17b2) (Should use (13a2)) Make a two-way plot of these coded residuals.

17c/17c2) (Should use (13b2)) Same for problem (5a).

17d/17d2) (Should use (13c2)) Same for problem (5b).

17e/17e2) Should use (13d2)) Same for problem (5c).

17f/17f2) (Should use (13e2)) Same for problem (5d).

S) SOURCE

Climatography of the United States No. 60–2. Climates of the states. Arizona. U.S. Weather Bureau, Washington, D.C., September 1959.

exhibit **18** of chapter 11: insect counts

The condensed residuals for the analysis of exhibit 10

A) The RESIDUALS

Treat	D	EC	AB	F
1	388	−38	30	−8
3	−36	54	−66	−5
2	36	−21	−7	5
4	−77	36	−2	89

#16 residuals

M8h	−3h		
H4h	36	−28h	64h
E2h	71	−52	
1	388	−77	

B) The CODING

−157h	−93	−28h	36	100h	165	
◎	○	∘	·	+	+	#
xxx	xxx	four	seven	four	xxx	one
		−77	−21	36		388
		to	to	to		
		−36	30	89		

P) PROBLEMS

18a) Condense the analysis of panel A of exhibit 17.

18b) Condense the analysis of panel A of exhibit 13.

Exhibit **20** shows the bare-form condensed two-way plot of fit, where we have allowed the difference between E and C to be shown only in numbers (taking an opportunity by adding area effects and treatment fits). Note the pulling back of the "water-lines" to avoid both actual and indirect interference, with the plot and its surrounding tags.

Exhibit **21** offers some problems.

review questions

How did we decide on coding in the previous chapter? Was that a reasonable thing to do? For what purposes can we use a standard code? What standard code do we use? Do we look to see what values get what code? Can we show this too? Do we? How do we plot the coded values? Should we consider condensing in either two-way plots of fit or two-way plots of residuals? Why/why not? In which plot(s)? What can we do to show condensation in the plot of fit? In the plot of residuals? How do we find condensed residuals?

exhibit **19** of chapter 11: insect counts

Condensed two-way plot of residuals based on exhibit 18

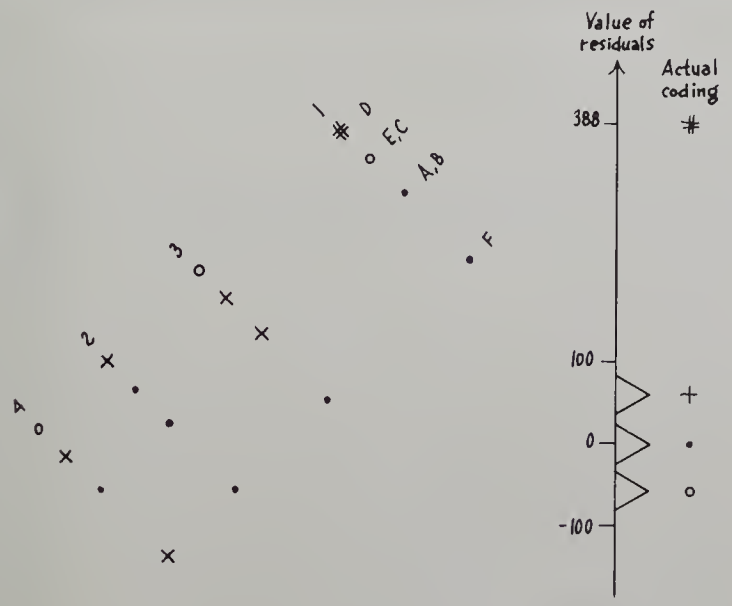


exhibit **20** of chapter 11: insect counts

Condensed two-way plot

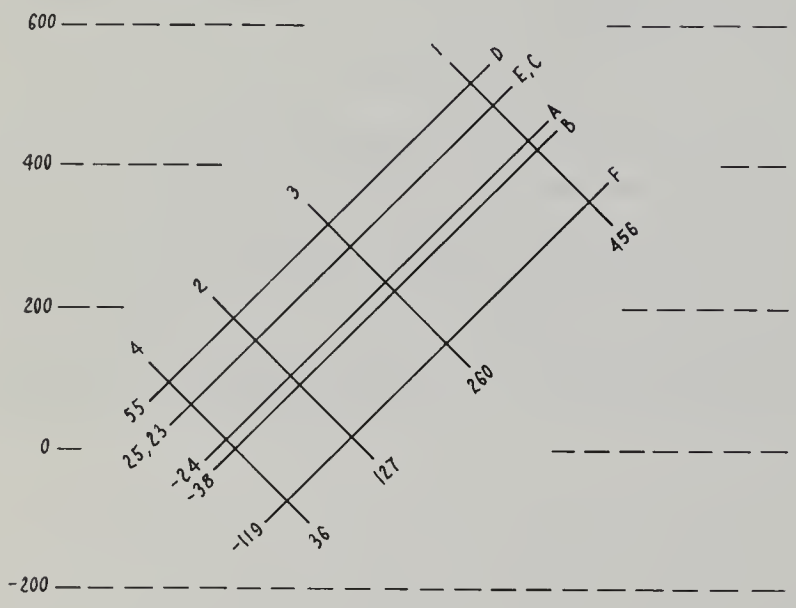


exhibit **21** of chapter 11: data and problems

Problems on condensed analyses

A) AN ANALYSIS OF A LARGER NEBRASKA PRESIDENTIAL VOTING EXAMPLE--percent Democratic among major parties

County location	Election												
	'20	'24	'28	'32	'36	'40	'44	'48	'52	'56	'6)	64	eff
I2	3.2	−0.3	−0.5	−1.6	4.2	3.6	1.3	0.8	−1.1	−2.6	1.6	5.7	1.6
H3	0.1	−2.1	0.0	2.8	2.6	−0.4	−4.0	0.3	1.3	2.3	−1.8	−2.6	−4.8
G4	2.4	3.1	−2.0	2.4	1.3	−0.1	−1.6	−0.6	−1.1	−2.0	−1.2	−0.7	−5.0
G6	0.8	−1.2	−3.5	−1.6	1.3	1.1	2.3	1.3	−2.3	0.1	2.1	0.4	−1.3
H7	−4.2	−2.9	−3.4	−5.1	3.1	0.1	1.2	1.6	2.7	2.0	2.2	2.4	0.2
G7	−6.1	−4.3	0.6	−1.6	−2.0	−0.8	0.2	0.4	3.8	2.3	1.7	5.4	0.4
E7	−3.1	4.2	3.9	1.4	−1.7	0.7	1.9	1.7	−2.7	−2.0	−2.7	−1.6	17.1
G8	3.3	3.3	2.8	0.2	−3.2	−1.2	−3.2	−1.0	1.9	−1.6	−1.5	0.6	−4.3
E8	1.9	0.7	1.2	2.4	−2.9	−2.1	−0.4	−2.6	−0.4	−0.3	1.2	1.5	−3.3
E9	2.2	−0.7	1.0	1.0	−3.2	−0.4	2.0	−2.4	−2.5	11.1	0.6	1.4	−0.6
eff	−7.7	−4.7	−8.1	23.6	16.4	−1.1	−2.4	3.6	−13.8	−9.3	−7.1	10.8	40.2

P) PROBLEM

21a) Condense the analysis of panel A above.

21b/c/d/e) (uses (5a/5b/5c/5d)) Condense the analysis of problems (5a/5b/5c/5d).

21f) Condense the analysis of panel B of exhibit 22 (following).

S) SOURCE

Richard M. Scammon, 1965, *America at the Polls*. A Handbook of American Presidential Election Statistics, 1920–1964. Pages 278–284.

11F. We can combine!

The last big idea we learned before starting on two-way tables was smoothing. What about combining the two?

presidential Connecticut

Exhibit **22** shows the percent Democratic of the major party vote for President in each of the eight counties of Connecticut in every presidential election from 1920 to 1964. Panel B shows a row-PLUS-column analysis. Clearly this analysis has done a fair amount of good--the residuals are even smaller than the county effects, which are smaller than the election effects. (The H-spreads are, respectively, 16, 60 and 116. A useless analysis, of a table with so many rows and columns, would, by contrast, have effects noticeably smaller than residuals.) If we look hard, however, we see that not everything has been pulled out that might be.

When we look at the election effects, we see that they are far from irregular. If we were to smooth them simply, say with 3RSS (see Chapter 7), the standard breakdown

$$\text{input} = \text{smooth} \quad \text{PLUS} \quad \text{rough}$$

would give a lot of smooth. (In fact, the results would be:

$$\text{smooth} = -152, -152, -65, -3, 37, 37, 26, 3, -43, -43, 35, 200,$$

and

$$\text{rough} = 0, 19, 0, 0, 26, 0, 0, 0, -15, -77, 0, 0.)$$

exhibit **22** of chapter 11: presidential Connecticut

The data, and the PLUS analysis (continued in exhibit 23)

A) The DATA

	County							
Election	Lich	Fairf	Mddx	Toll	NL	NH	Windh	Hartf
1920	32.5	30.9	33.1	31.0	34.6	36.5	37.1	35.9
24	30.0	24.5	29.9	30.3	32.1	34.4	36.6	31.4
28	36.0	43.7	39.7	39.6	43.3	50.5	48.5	46.4
1932	41.9	47.1	46.3	46.0	49.8	52.4	53.1	49.9
36	48.1	56.3	52.9	52.8	53.9	60.5	52.4	61.2
1940	46.0	50.7	49.2	50.5	54.7	55.0	55.4	56.5
44	44.4	48.9	48.6	48.5	54.8	53.9	55.3	57.3
48	41.0	43.3	47.5	46.9	51.8	50.2	53.0	54.3
1952	36.1	38.9	41.5	41.2	45.1	45.1	46.4	49.4
56	30.1	29.8	35.2	36.5	38.6	37.0	40.4	41.9
1960	46.1	46.6	50.1	48.6	51.6	58.0	57.0	58.9
64	65.8	60.8	67.5	69.0	69.1	69.1	73.5	73.0

→

Once we have noticed this, we naturally ask what would happen if we were also to smooth the eight sequences of residuals, one for each county.

Panel C--which appears in exhibit **23**--shows the result of such 3RSS smoothing. We see that there are clear trends over time in several counties. Thus, the smooth for Fairfield falls from 7 (and then 20) to −56, while those

exhibit **22** of chapter 11 (continued)

B) A county-PLUS-election ANALYSIS--IN 0.1% Democratic

484	−62	−30	−14	−18	15	31	42	50
−152	55	7	13	−4	−1	2	−3	−23
−171	49	−38	0	8	−7	0	11	−49
−65	3	48	−8	−5	−1	55	24	−5
−3	0	20	−4	−3	2	12	8	−32
63	−4	46	−4	−1	−23	27	−65	15
37	1	16	−15	2	11	−2	−9	−6
26	−4	9	−10	−7	23	−2	1	13
3	−15	−24	2	0	16	−16	1	6
−58	−3	−7	3	4	10	−6	−4	18
−120	−1	−36	2	19	7	−25	−2	5
35	4	−23	−4	−15	−18	30	9	20
200	36	−46	5	24	−8	−24	9	−4

S) SOURCE

Richard M. Scammon, 1965, *America At the Polls*. A Handbook of American Presidential Election Statistics, 1920–1964. Pages 77–78.

exhibit **23** of chapter 11: presidential Connecticut

Continuation of exhibit 22; the smoothing, *etc.*

C) The 3RSS SMOOTHS over YEARS--for each county separately

eff	Litch	Fairf	Mddx	Toll	NL	NH	Windh	Hartf
−152	55	7	8	−4	−1	2	11	−23
−152	49	7	0	−4	−1	2	11	−23
−65	3	20	20	−3	−1	12	11	−23
−3	0	20	−4	−3	−1	12	8	−6
37	0	20	−4	−1	2	12	2	−5
37			−4	−1	11	−2	1	6
26	−4	9	2	0	16	−2	1	6
3	−4	−7	2	0	16	−6	1	6
−58	−3	−23	2	4	10	−6	1	6
−58	−1	−24	2	4	7	−16	9	6
35	4	−36	2	19	−8	−24	9	5
200	14	−46	2	24	−8	−24	9	3

#96	smooths
M48h	1h
H24h	8 −4
E12h	12 −7h
D 6h	19h −23

*#93h	*smooths
*M47	2
*H24	8 −4
*E12h	12 −7h
* D6h	19h −23

→

exhibit **23** of chapter 11 (continued)

D) The CORRESPONDING ROUGHS--where smooth PLUS rough ≡ residual

0	0	0	5	0	0	0	−14	0
−19	0	−45	0	12	−6	−2	0	−26
0	0	28	−4	−2	0	43	13	18
0	0	0	0	0	3	0	0	−26
26	−4	26	0	0	−25	15	−67	20
0	5	0	−11	3	0	0	−10	−12
0	0	0	−12	−7	7	0	0	7
0	−11	−17	0	0	0	−16	0	0
0	0	16	1	0	0	0	−5	12
−62	0	−12	0	15	0	−9	1	−1
0	0	13	−6	−34	−10	54	0	15
0	22	0	0	0	0	0	0	−7

#96 roughs

M48h	0	
H24h	0	−4
E12h	13	−11
D 6h	19	−21

*#73 roughs

*M37	0		
* H19	5	−7	12
: E10	15	−12	
* D 5h	21	−25h	

E) A CODING

	−24h		−14h		−4h	4h		14h	24h	
◎		○		∘	·	+		+		#
two		eight		seven	(45)	(24)		eight	two	(smooths)
−36		−16		−5	−4	5		16	49	
to		to		to	to	to		to	to	
−46		−24		8	4	14		24	55	
six		one		sixteen	(54)	eight		seven	four	(roughs)
−25		−17		−5	−4	5		15	26	
to		to		to	to	to		to	to	
−67		−17		−8	4	13		22	54	

F) SEMI-GRAPHICAL EXAMPLES--Fairfield and Hartford counties

#												
+			20	20	20	16						
+	7	7					9					
·												
∘								−7				
○									−23	−24		
◎											−36	−46
	'20	·	**'28**	·	**'36**	·	**'44**	·	**'52**	·	**'60**	·
#												
+												
+						6	6	6	6	6	5	
·												3
∘												
○				−6	−5							
◎	−23	−23	−23									

for Litchfield, Windham, and Hartford are lower--or higher--in the middle than at the ends. Panel D has the corresponding roughs. Panel E provides a--judgment-based--coding that seems reasonably well adapted to both smooth and rough. Panel F illustrates its application to semigraphical display of the smooths. Panel G--appearing in exhibit **24**--shows an analysis into (i) common, (ii) county effects, (iii) election effects, (iv) rough election effects, (v)

exhibit **24** of chapter 11: presidential Connecticut

Continuing exhibits 22 and 23; the analysis

G) An ANALYSIS--smooth above and rough below

	484	Litch −62	Fairf −23	Toll −18	Mddx −12	NL 15	NH 29	Windh 50	Hartf 54
1920	−152	55	0	−4	6	−1	4	3	−27
24	−152	49	0	−4	−2	−1	4	3	−27
28	−65	3	13	−3	−6	−1	14	3	−27
32	−3	0	13	−3	−6	−1	14	0	−10
36	37	0	13	−1	−6	2	14	−6	−9
1940	37	−4	9	−1	−6	11	0	−7	2
44	26	−4	2	0	0	16	0	−7	2
48	3	−4	−14	0	0	16	−4	−7	2
52	−43	−3	−30	4	0	10	−4	−7	2
56	−43	−1	−31	4	0	7	−14	1	2
1960	35	4	−43	19	0	−8	−22	1	1
64	200	14	−53	24	0	−8	−22	1	−1
1920	√	0	0	0	5	0	0	−14	0
24	−25	0	−45	12	0	−6	−2	0	−26
28	√	0	28	−2	−4	0	43	13	18
32	√	0	0	0	0	3	0	0	−26
36	26	−4	26	0	0	−25	15	−66	20
1940	√	5	0	3	−11	0	0	−10	−12
44	√	0	0	−7	−12	7	0	0	7
48	√	−11	−17	0	0	0	−10	0	−0
52	−15	0	16	0	1	0	0	−5	12
56	−77	0	−12	15	0	0	−9	−11	−1
1960	√	0	13	−34	−6	−10	54	0	15
64	√	22	0	0	0	0	0	0	−7

P) PROBLEMS

24a) Write down an analysis where the county effects are combined with the smooths.

24b) Do the same including the common.

24b2) Plot the results of 24b.

smooths of residuals, and (vi) roughs of residuals. In this analysis, we took out the county medians of the smooths, transferring them to the county effects. (Transferring election medians would be a little complicated; we leave this to those intrigued by the challenge.)

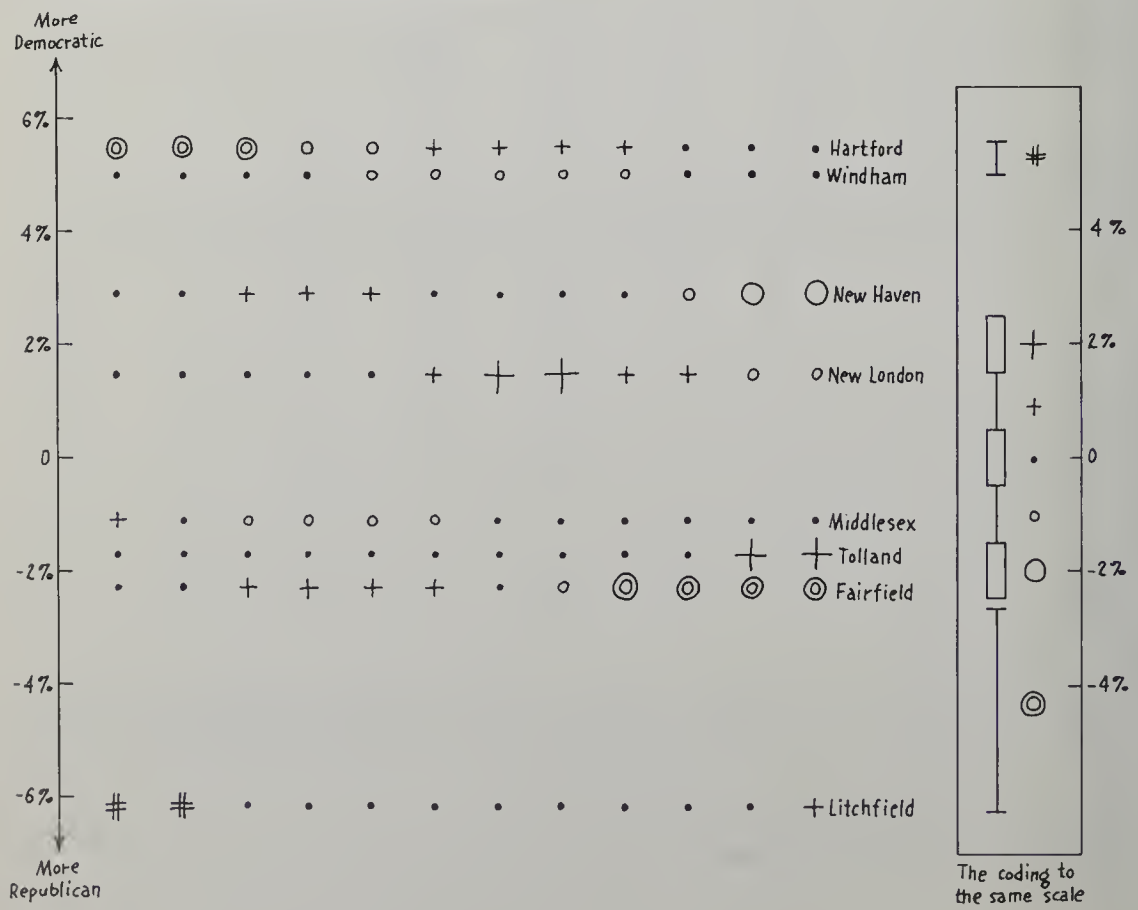
Exhibit **25** shows the smooths in coded form. The trends are now somewhat clearer than when given in "mournful numbers". The vertical coordinate has been taken to be the fitted "county effect", which seems as likely to be relevant as any numbers or ordering that we have. (Certainly much better than alphabetic order.) A corresponding plot for the roughs is left to the reader. No clear structure appeared in our plot of the roughs, so we may be near the bottom of this data.

a further opportunity

To see what is happening graphically, we would naturally think of plotting the percent Democratic against date for each of the eight counties. How well would this have done? The election effects differ among themselves by (20.0) –

exhibit **25** of chapter 11: presidential Connecticut

Smooths of exhibit 23 in coded form

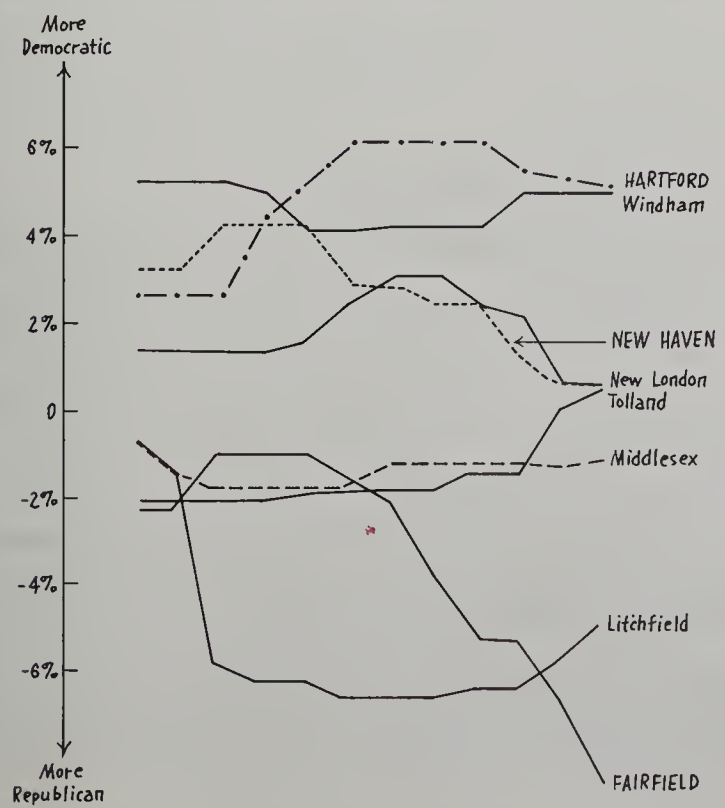


(−17.7) = 37.7%, as compared with (55) − (−65) = 12.0% for the residuals. Had we made eight separate plots, each would have shown the same general pattern (actually national in scope). Had we tried to plot the eight counties on top of one another, the actual votes would have been just irregular enough to be confusing. If we want something like a set of superposed "curves", we need to do two things: (i) take out an election effect (so that we can see smaller differences) and (ii) get rid of some of the irregularities in the behavior of the separate counties (so we can put eight "curves" on the same picture without too much confusion). In our analysis (exhibit 24), we have all the pieces we need to get a good set of curves. We need only add the county effects back to the election-wise smoothed residuals, getting a smoothed representation of the displacement of each county from a statewide election effect.

The results are plotted in exhibit **26**, which does a rather good job of telling us about the smooth parts of the county idiosyncrasies. (The three counties labelled in capital letters produced between 300 and 325 thousand votes each in the 1964 election; the **total** for the other five combined was

exhibit **26** of chapter 11: presidential Connecticut

Smooth relation of counties to state election effects (combination of two smooths)



somewhat less than 300 thousand.) The most striking things in this picture are probably:

- ◇the decline (in % Dem.) in *Litchfield County* (NW Conn.) in the twenties.
- ◇the decline in *Fairfield County* (N.Y.C. suburbs, etc.) in the forties, fifties and sixties.
- ◇lesser changes in *Hartford* (rise in the thirties) and *New Haven* (fall in the forties and fifties).

Thus, combining smoothing and two-way analysis can lead us to useful pictures accessible by neither one alone.

Exhibit **27** sets some problems.

review questions

What might we combine? What did we take for an example? What did a PLUS analysis show? What did we do next? Roughly what was the result? Did we code the result(s)? How many kinds of results were coded? How did the resulting plots appear? Did we reach a new analysis?

11G. Guidance for expression

Given some data, how should we express it for row-PLUS-column analysis? The answer for any specific set of data comes mainly by trial and

exhibit **27** of chapter 11: problems

Some combined analyses to be done (all rather large)

27a) (uses (5b)) Smooth the analysis of problem (5b) from W to E for each election. Display result.

27b) (uses (5b)) Smooth the analysis of problem (5b) over time for each county. Display result.

27c) Smooth the analysis of panel A of exhibit 13 over the cycle of the months for each airport. Display the result.

27d) Smooth the analysis of panel A of exhibit 17 likewise. Display the result.

27e) Smooth the analysis of panel A of exhibit 21 over time for each county. Display the result.

27f) Could we have smoothed this example over counties?

27g) What happens when we can smooth both ways, and do?

27k) What is the difference between smoothing "over time" or "W to E" and smoothing "over the cycle of the months"?

error. Rules of thumb about where to begin, however, can be very helpful, as can suggestions about how to decide where to go next.

We learned a little about re-expression in two-way tables at the close of the previous chapter (in section 10G). It is time to think a little more about such questions.

Good rules seem to depend most heavily upon the nature of the response, and how widely it varies. As always, we can do well by bringing in other bodies of sufficiently similar data, for:

◇**it is a rare thing that a specific body of data tells us as clearly as we would wish how it itself should be analyzed.**

amounts and counts

Amounts and counts are--by definition--nonnegative. Accordingly, we can re-express them by logs or reciprocals. Zeros may seem to offer a difficulty, but there are two ways out:

◇the easy way, where log 0 is L and $-1/0$ is L, where L is taken to be lower than any other value involved. (The fitting technique with which this chapter opened works quite well in the presence of a few L's.)

◇the careful way, where the given values are "started" by 0.1, 0.25, 0.5, 1, or more, before logging or reciprocating.

Often either works well. Sometimes the easy way is a little crude, and the careful way is worth the trouble.

If the ratio

largest amount (or count)/smallest amount (or count)

is close to 1, it will not make any appreciable difference whether we take logs (or reciprocals) or not. So, if this ratio is close to unity, we think about what we might expect to work well **if** the ratio were large, and start our analysis with that expression.

If the ratio is large, we will probably have to go to

◇logs fitted row-PLUS-column, or

◇raw values fitted row-TIMES-column.

(These two tap equivalent collections of possibilities, but using them will usually give somewhat different fits.) Hand arithmetic is usually easier when we re-express the given values as logs--and the fit is often better, too--so we usually start there.

If the ratio is intermediate in size, we ask ourselves whether we have a reasonable belief as to what ought to be preferred (in a reasonable world). If we have such a belief, we start from there--and let the diagnostic plot tell us

which way we ought to move. If we have no belief, we do a row-PLUS-column fit for the raw-values first, carrying through at least as far as a

diagnostic plot

--a plot of residuals against comparison values. Now we look to see if a slope of 1 is better than a slope of 0. It it is, we can convert our row-PLUS-column fit into a row-TIMES-column fit--but we usually do better by going to logs. If 0 is much better than 1, we stand pat. In between, we think--and look--hard.

Rarely, we will have a place for a re-expression other than logs or reciprocals. The need for this will usually be shown when two adjacent choices--from (square), raw, log, reciprocal, (reciprocal square)--show diagnostic plots with opposite tilts. Exploration and, particularly, thought are then in order.

balances

If the values in front of us include some that are negative, we cannot take logs. We can start with a row-PLUS-column fit, nevertheless, and if the diagnostic plot favors a coefficient near +1.0, we can convert this "PLUS" into a row-TIMES-column fit with ease. (We have already found the comparison values in order to make the diagnostic plot--multiplying them by +1.0 is easy. If something other than row-PLUS-column or row-TIMES-column fits is needed, exploration and thought are again the proper medicine. (See optional section 11L for an extreme example, and optional section 11I for a less extreme one.)

If we have effects that are well-balanced in sign, the row-PLUS-column fit may not be a good enough start for a row-TIMES-column fit. (Sometimes it fails miserably.) If all we really want is the TIMES fit, we can change the signs both in as many entire rows as we wish and, separately, in as many entire columns as we wish, and then make first a PLUS fit and than a TIMES fit to the new table. We can then put back the signs we took out, changing the signs of residuals accordingly.

counted fractions

We will learn some ways to re-express counted fractions when we come to chapter 15. No more here.

a helpful thougnt

We now see even more use for comparison values and diagnostic plots then we did before. It is important that PLUS-one offers us conversion from row-PLUS-column to row-TIMES-column. We ought to take essentially every fit as far as a diagnostic plot. We then always have the opportunity to choose to:

◇ask if TIMES seems better than PLUS.

◇fit one more constant.

◇or do neither, if neither seems helpful/reasonable.

review questions

Must we rely on trial-and-error in choosing an expression for row-PLUS-column analysis? Can there be guidance? Why/why not? Does any body of data usually tell us clearly about just how it ought to be analyzed? What of amounts and counts? Should we worry about logs or reciprocals of zeros? Why/why not? What if largest/smallest is close to 1? Quite large? In a middle range? Do we expect to take most PLUS analyses as far as a diagnostic plot? What two slopes play a particular role? What of balances? Counted fractions?

11H. How far have we come?

This chapter is basically a how-to-do-it chapter, filling in the techniques for making all the analyses and displays we met in chapter 10.

We are now ready:

◇to make row-PLUS-column fits by repeatedly "shifting out" medians.

◇and to call this process "median polish", organizing its calculations in either of two ways.

◇to lessen the chance of arithmetic error by circling negative values in our working calculations.

◇to arrange in a better way the check that what we have is really a row-PLUS-column fit.

◇to make a two-way plot of any row-PLUS-column fit, starting from any of various ways of dividing its values into two suitable parts, drawing horizontal and vertical lines, and turning through 45°.

◇to identify the residuals deserving of appearance in the plot of the fit--those outside--and to include them in the fit (mechanically, by adding half of each such residual to each of the two parts of the fit).

◇and to make a suitable schematic plot for the remaining residuals.

◇to use the standard code for residuals in a two-way plot of residuals.

◇and to prepare a display of the actual sizes of residuals that receive each code.

◇to use reasonable judgment followed by a diagnostic plot in choosing a re-expression of a response to be analyzed in a two-way table.

Moreover, we have seen how:

◇to condense wither a two-way plot of fit or a two-way plot of residuals.

◇to fit smoothing inside some two-way tables.

◇and to display appropriately the extra piece (or pieces) such an analysis produces.

◇to use L, taken as less than any number, for the log or reciprocal of zero, so that we have an alternative to "starting" when analyzing two-way tables with a few zeros. (And, even, occasionally, to go farther, using L for logs of negative numbers, too.)

◇to open the way to a row-TIMES-column fit by way of a row-PLUS-column one, through changing signs in several steps, each step either by entire rows or by entire columns.

We now understand much more clearly:

◇that the general maxim--it is a rare thing that a specific body of data tells us clearly enough how it itself should be analyzed--applies to choice of re-expression for two-way analysis.

◇that it is wise to take almost every two-way fit as far as a diagnostic plot.

We are now equipped to do, with relative ease, a reasonable variety of things in connection with two-way fit. The things we learned as possibilities in chapter 10 are now all things **we can do**. As we find diverse opportunities to try them out--whether or not closely related to this book--we will come to recognize more and more what they can do for us.

Optional sections for chapters 10 and 11 11+

chapter index on next page

Even though we separated "What we can do!" and "How to do it!" into two chapters, there was still more related material than seemed wise to pack into those chapters. The present chapter, then, contains a mixed grill of topics that belong close to, rather than in, chapters 10 and 11.

11I. Exploring beyond PLUS-one (extends for chapter 10)

Exhibit **28** offers, for those who wish to play, an expedition into one part of the wilderness of row-PLUS-col-PLUS-two fits.

One thing learned by those who go on this expedition is that, once we fit a PLUS-one supplement to our row-PLUS-column fit, there may be still more to gain from a further, supplementary, row-PLUS-column fit. Suppose we try this with the simple row-PLUS-column-PLUS-one fit of exhibit 11 of chapter 10.

Exhibit **29** shows both what can be gained, and something of where it leaves us. Clearly, we are likely to engage in back-and-forth fitting and refitting between "row-PLUS-column" and "(new constant)(new comparison value)" if we plan to follow this road. One step that we can take without complicating the calculation is to fit a supplementary value for the common term. Panel C of exhibit 29 shows the result.

exhibit **28** of chapter 11+: East Coast

Optional problems taking the analysis further

O) OPTIONAL PROBLEMS––all teasers

28a1) Plot the residuals of panel A of exhibit 11 of chapter 10 against the comparison values of panel D of exhibit 9 of chapter 10. What sort of very simple additional fit comes to mind?

28a2) What plot would make fitting one more constant (beyond the -PLUS-one constant) easy? Make the plot. Choose this constant. (Now you have ONE example of ONE kind of a row-PLUS-col-PLUS-two fit.)

28a3) Find the residuals for this fit. Go back and readjust each of (i) row fits and column fits, (ii) the constant *k* multiplying comparison values, to get a better row-PLUS-col-PLUS-two fit. Display its residuals. How do they compare with the residuals of exhibit 10 of chapter 10?

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37		413
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exhibit **29** of chapter 11†: East coast

A supplementary row-PLUS-column fit after a row-PLUS-column-PLUS-one fit (starts from exhibit 11 of chapter 10)

A) If we allow ourselves a SUPPLEMENTARY row-PLUS-column FIT

The data†		
−72	1	−12
−35	0	−15
−20	0	−7
0	0	0
−9	0	−2
−16	0	−21
−20	0	−17

One row-PLUS-column fit			
−53	0	0	1
−13	2	0	−2
0	0	6	0
7	−13	0	13
0	11	0	11
4	0	−8	0
0	0	−4	0
−7	13	0	−13

†**Actually, the residuals from a PLUS-one fit.**

B) WHERE ARE WE?--and where might we go?

We have now fitted

$$\text{all} + \text{row} + \text{col} - \frac{(\text{row})(\text{col})}{544} + \text{all}^* + \text{row}^* + \text{col}^*$$

where the *'s denote the supplementary fit of panel A. This is equivalent to

$$(\text{all} + \text{all}^*) + (\text{row} + \text{row}^*) + (\text{col} + \text{col}^*) - \frac{(\text{row})(\text{col})}{544}.$$

One way to neaten things up would be to find $k^\times$ in

$$(\text{all} + \text{all}^*) + (\text{row} + \text{row}^*) + (\text{col} + \text{col}^*) + k^\times \frac{(\text{row} + \text{row}^*)(\text{col} + \text{col}^*)}{(\text{all} + \text{all}^*)}$$

to do as well we can. (This we leave to panel P.)

C) The LEAST we CAN DO--two versions of a restricted (common only) fit to the residuals of exhibit 11

−65	8	−5	0
−28	7	−8	0
−13	7	0	0
7	7	7	0
−2	7	5	0
−9	7	−14	0
−13	7	−10	0
0	0	0	−7

−70	3	−10	0
−33	2	−13	0
−18	2	−5	0
2	2	2	0
−7	2	0	0
−14	2	−19	0
−18	2	−15	0
0	0	0	−2

P) PROBLEMS

29a) Find another set of data that deserves such treatment, and do the PLUS-one fit.

29a2) Continue, parallel to panel A.

29a3) Is a continuation parallel to panel C justified? Why? (Why not?) If justified, do it.

review questions

Might we go beyond-PLUS-one? What might we learn? What is the least we might do? What does this do to our example?

11J. Taking out any summary

Panel C of exhibit 1 of chapter 10 offered a row-PLUS-column analysis of the Arizona temperatures. When we check it out, we find that further steps of median removal would have no effect.

It offers a convenient starting place, however, for other kinds of removal --as would, in most other examples, the results of four steps of median removal One kind of removal that is easy to illustrate, because the arithmetic is easy, is the removal of **midextremes.** Because it is so sensitive to single exotic values, we are not likely to use it to analyze real data. (It has a place, of course, when we are constructing simple approximations to error-free tables.) We shall use it here only as an illustration--an illustration of something just about as far from the median as we can find. (If it works out all right, anything else we are likely to think of trying will work out all right, also.)

repeated midextreme removal

So let us try removing the midextreme of every row and the midextreme of every column. (Some will want to recall that the midextreme has often been called the midrange.) Exhibit **30** shows the calculation, which we have put in the same pattern that we used in section 11A for repeated median removal. The result depends a little upon where and how we start, but the three 3×7 tables that we reach, namely (in 0.1°F),

−1	6	−5	−2	6	−6	−1	6	−6
−4	−3	1	−4	4	1	−3	4	0
−8	7	7	−8	8	7	−7	8	7
−3	2	2	−3	3	2	−2	3	2
8	−8	−6	7	−8	−7	8	−8	−7
5	0	−6	5	1	−6	6	1	−6
−1	1	1	−1	2	1	−1	1	0
(panel A)			(panel B)			(panel C)		

all (happen to) have the same maximum size of residual--0.8. (If we want to use half-tenths, we might get this down to 0.7h.)

the general case

We can do this sort of calculation, removing some chosen summary value repeatedly, using any summary value we like.

repeated mean removal

Those who like lots of arithmetic can even use the mean, as in exhibit **31.** This time we take the calculations only to two steps. (They would have stopped exactly, had we used fractions of tenths.) The results obtained by calculating to one decimal are not identical. (They would have been, had we gone to fractions.) In fact, they are, in units of 0.1°F,

−1	4	−4	0	5	−4	0	4	−4
−4	1	2	−3	2	2	−3	1	2
−10	3	6	−9	4	6	−9	3	6
−3	0	3	−3	0	2	−2	0	3
10	−8	−3	11	−7	−3	11	−8	−3
6	−1	−4	6	−1	−5	6	−2	−5
−1	−1	2	−1	−1	1	0	−1	2
(panel A)			(panel B)			(panel C)		

At no location do they differ by more than a single tenth.

exhibit **30** of chapter 11+: Arizona temperatures

Taking out midextremes (extreme illustration!)

A) STARTING FROM panel C of exhibit 1 of chapter 10

0	.2	−.9	−.4	.4	.6	−.5
0	.2	0	.1	−.1	.1	−.1
−1.0	0	0	−.5	−.5	.5	.5
0	0	0	0	0	0	0
1.9	−.2	0	.8	1.1	−1.0	−.8
1.0	0	−.6	.2	.8	−.2	−.8
.3	0	0	.2	.1	−.2	−.2
				.3	−.2	−.2

−.1	.6	−.5	.2	.1	.8	−.3
−.4	.3	.1	√	−.4	.3	.1
−.8	.7	.7	√	−.8	.7	.7
−.3	.2	.2	√	−.3	.2	.2
.8	−.8	−.6	√	.8	−.8	−.6
.5	0	−.6	√	.5	0	−.6
−.1	.1	.1	−.1	−.2	0	0
√	√	√				

Examples:

$-.4 = \frac{1}{2}(.2 + .9)$

$.1 = \frac{1}{2}(0 + .2)$

$-.5 = \frac{1}{2}(-1.0 + 0)$

. . .

$.4 = 0 - (-.4)$

$.6 = .2 - (-.4)$

$-.1 = 0 - (.1)$

exhibit **30** of chapter 11+ (continued)

B) STARTING from the SAME place the OTHER way

0	.2	−.9
0	.2	0
−1.0	0	0
1.9	−.2	0
1.0	0	−.6
.3	0	0
.4	√	−.4

−.4	.2	−.5	−.2	−.2	.4	−.3
−.4	.2	.4	√	−.4	.2	.4
−1.4	0	.4	−.5	−.9	.5	.9
−.4	0	.4	√	−.4	0	.4
1.5	−.2	.4	.6	.9	−.8	−.2
.6	0	−.2	.2	.4	−.2	−.4
−.1	0	.4	.2	−.3	−.2	.2
				√	−.2	.2

−.2	.6	−.5	√	−.2	.6	−.5
−.4	.4	.2	√	−.4	.4	.2
−.8	.8	.8	−.1	−.9	.7	.7
−.3	.3	.3	−.1	−.4	.2	.2
.7	−.8	−.6	.2	.9	−.6	−.4
.5	.1	−.5	−.1	.4	0	−.6
−.1	.2	.2	−.2	−.4	0	0
√	√	.1				

−.2	.6	.6	√
−.4	.4	.1	√
−.8	.8	.7	√
−.3	.3	.2	√
.7	−.8	−.7	√
.5	.1	−.6	√
−.1	.2	.1	√

C) STARTING from the RAW DATA

65.2	90.1	94.6	79.9	−14.7	10.2	14.7
63.4	88.3	93.7	78.6	−15.2	9.7	15.1
57.0	82.7	88.3	72.6	−15.6	10.1	15.7
46.1	70.8	76.4	61.2	−15.1	9.6	15.2
35.8	58.4	64.2	50.0	−14.2	8.4	14.2
28.4	52.1	57.1	42.8	−14.4	9.3	14.3
25.3	49.7	55.3	40.3	−15.0	9.4	15.0
				−14.9	9.3	15.0

−.1	.6	−.6	.3	.2	.9	−.3
−.3	.4	.1	√	−.3	.4	.1
−.7	.8	.7	√	−.7	.8	.7
−.2	.3	.2	√	−.2	.3	.2
.8	−.8	−.7	−.1	.7	−.3	−.8
.6	.1	−.6	−.1	.5	0	−.7
−.1	.1	0	√	−.1	.1	0
√	√	√				

P) PROBLEM

30a) Start from the raw data and go the other way. Compare your result with those of panels A, B, and C.

exhibit **31** of chapter 11⁺: Arizona temperatures

Taking out means instead

A) STARTING FROM panel C of exhibit 1 of chapter 10

0	.2	−.9	−.2	.2	.4	−.7
0	.2	0	.1	−.1	.1	−.1
−.1.0	0	0	−.3	−.7	.3	.3
0	0	0	0	0	0	0
1.9	−.2	0	.6	1.3	−.8	−.6
1.0	0	−.6	.1	.9	−.1	−.7
.3	0	0	.1	.2	−.1	−.1
				.3	√	−.3

√	−.1	.4	−.4
√	−.4	.1	.2
√	−1.0	.3	.6
√	−.3	0	.3
√	1.0	−.8	−.3
√	.6	−.1	−.4
√	−.1	−.1	.2

B) STARTING from the SAME place the OTHER way

0	.2	−.9	
0	.2	0	
−1.0	0	0	
0	0	0	
1.9	−.2	0	
1.0	0	−.6	
.3	0	0	
.3	√	−.2	

−.3	.2	−.7	−.3	0	.5	−.4
−.3	.2	.2	√	−.3	.2	.2
−1.3	0	.2	−.4	−.9	.4	.6
−.3	0	.2	√	−.3	0	.2
1.6	−.2	.2	.5	1.1	−.7	−.3
.7	0	−.4	.1	.6	−.1	−.5
0	0	.2	.1	−.1	−.1	.1
				√	√	√

→

You, too, could invent your own summary, and take it out repeatedly to find yet another fit, yet another array of residuals.

review questions

What example did we choose? What did we try taking out first? What would a single very stray value have done to us? (Work an example, if need be.) How many ways did we take midextremes out of our example? How did the results compare? What did we try taking out next? What would a single **very,** very stray value have done to us? (Work an example, if need be.) How well did the results of various calculations agree? Can you take out other summaries? Which ones? In what way? Even if you invent a new summary?

11K. An example of re-expression––city killings

A simple example illustrates many of the points just stated. Exhibit **32** shows the data, a row-PLUS-column analysis, and the corresponding comparison values. Exhibit **33** shows the resulting diagnostic plot. Exhibit **34** shows an

exhibit **31** of chapter 11⁺ (continued)

C) STARTING from the RAW DATA––get out the adding machine!

65.2	90.1	94.6	83.3	−18.1	6.8	11.3
63.4	88.3	93.7	81.8	−18.4	6.5	11.9
57.0	82.7	88.3	76.0	−19.0	6.7	12.3
46.1	70.8	76.4	64.4	−18.3	6.4	12.0
35.8	58.4	64.2	52.8	−17.0	5.6	11.4
28.4	52.1	57.1	45.9	−17.5	6.2	11.2
25.3	49.7	55.3	43.4	−18.1	6.3	11.9
				−18.1	6.4	11.7
			√	0	.4	−.4
			√	−.3	.1	.2
			√	−.9	.3	.6
			√	−.2	0	.3
			√	1.1	−.8	−.3
			√	.6	−.2	−.5
			√	0	−.1	.2

P) PROBLEM

31a) Start from the raw data, and go the other way. Compare your result with that of panels A, B, and C.

exhibit **32** of chapter 11+: city killings

Numbers of murders and nonnegligent manslaughters (combined) for 1961, 1964, and 1966 in 18 large cities

A) The DATA				B) A PLUS ANALYSIS				C) COMPARISON VALUES		
City	1961	1964	1966	eff	1961	1964	1966	1961	1964	1966
New York	482	636	653	**528**	−130	0	11	−117.3	0	29.3
Chicago	365	398	510	**290**	−9	0	106	−64.4	0	16.1
Los Angeles	159	177	226	**75**	0	−6	37	−16.7	0	4.2
Detroit	136	125	214	**52**	0	−35	48	−11.6	0	2.9
Philadelphia	144	188	178	**64**	−4	16	0	−14.2	0	3.6
Baltimore	89	144	175	**36**	−31	0	25	−8.0	0	2.0
Washington	88	132	141	**24**	−20	0	3	−5.3	0	1.3
Cleveland	80	116	139	**8**	−12	0	17	−1.8	0	.4
Dallas	99	149	120	**15**	0	26	−9	−3.3	0	.8
New Orleans	52	82	113	**−26**	−6	0	25	5.8	0	−1.4
St. Louis	77	120	106	**−7**	0	19	−1	1.6	0	−.4
Akron	16	12	17	**−96**	28	0	−1	21.3	0	−5.3
Boston	26	52	58	**−56**	−2	0	0	12.4	0	−3.1
Buffalo	19	21	24	**−87**	22	0	−3	19.3	0	−4.8
Denver	32	33	39	**−75**	23	0	0	16.7	0	−4.2
Jersey City	11	17	16	**−91**	18	0	−7	20.2	0	−5.1
Kansas City	49	48	59	**−55**	20	−5	0	12.2	0	−3.1
Long Beach	10	17	20	**−91**	17	0	−3	20.2	0	−5.1
eff				**108**	−24	0	6			

P) PROBLEMS

32a) Select five comparison values that are not zero, and check that they do correspond to the fit of panel B. Show your work.

32b) Find a 3RSS middle trace for residuals against comparison values and compare with exhibit 17. Show your work.

32c) Find the numerical values of a TIMES fit to the data based on the PLUS analysis of panel B and its comparison values. Write out the fit.

32c2) Find the residuals corresponding to the fit just found, make a stem-and-leaf, and compare with the stem-and-leaf given in exhibit 38, below.

32c3) Make a diagnostic plot of these residuals against the original comparison values. Discuss the results.

32c4) Make a 3RSS middle trace for these 54 points.

S) SOURCE

World Almanac and Book of Facts for: 1963, page 310; 1966, page 307; 1968, page 903.

enlarged view of the center of exhibit 33, showing a 3RSS middle trace with the points suppressed and the two comparison lines

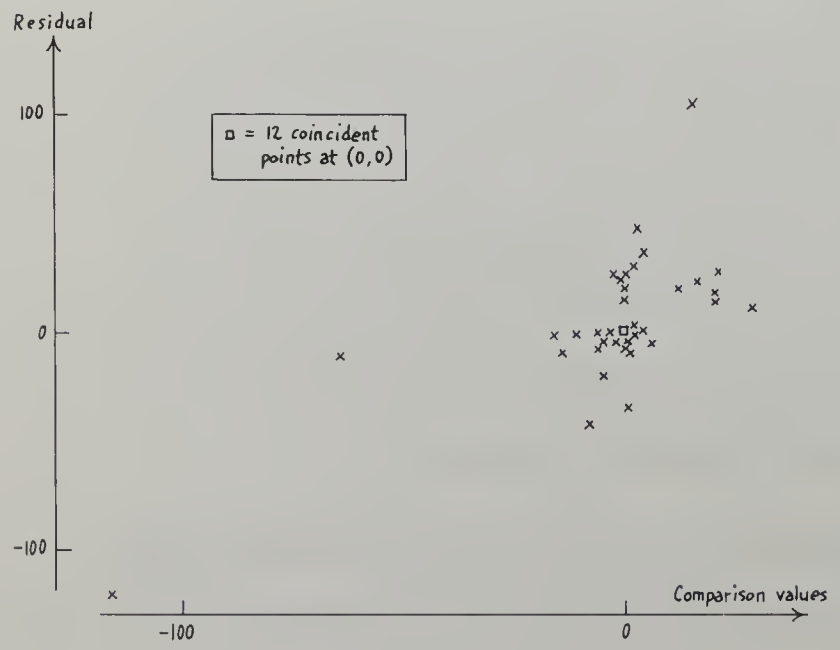
residual = zero

residual = comparison value

added. Clearly the 3RSS middle trace is closer to the later, although the dissymmetry between left and right sides is a possible worry. Next, in view of this judgment, we should go either to a row-TIMES-column fit of the counts or to a row-PLUS-column for the logs of the counts. We leave the row-TIMES-column fit based on the row-PLUS-column fit of counts to the reader (see problems (32c1/2/3)) and proceed to the logs. Exhibit **35** has log counts, a row-PLUS-column analysis, and its comparison values. Exhibit **36** shows the 3RSS middle trace for the diagnostic plot. It seems pretty quiet and uninteresting, and we can believe that we may have a satisfactory analysis.

exhibit **33** of chapter 11+: city killings

Diagnostic plot based on exhibit 32



We next want to return from log counts to raw counts, in order to make a clear comparison of residuals. Exhibit **37** has the antilogs of the PLUS fit to logs and the corresponding residuals. (These antilogs are, of course, a TIMES fit to the raw counts.) Exhibit **38** shows both stem-and-leaf and letter-value displays for three sets of residuals,

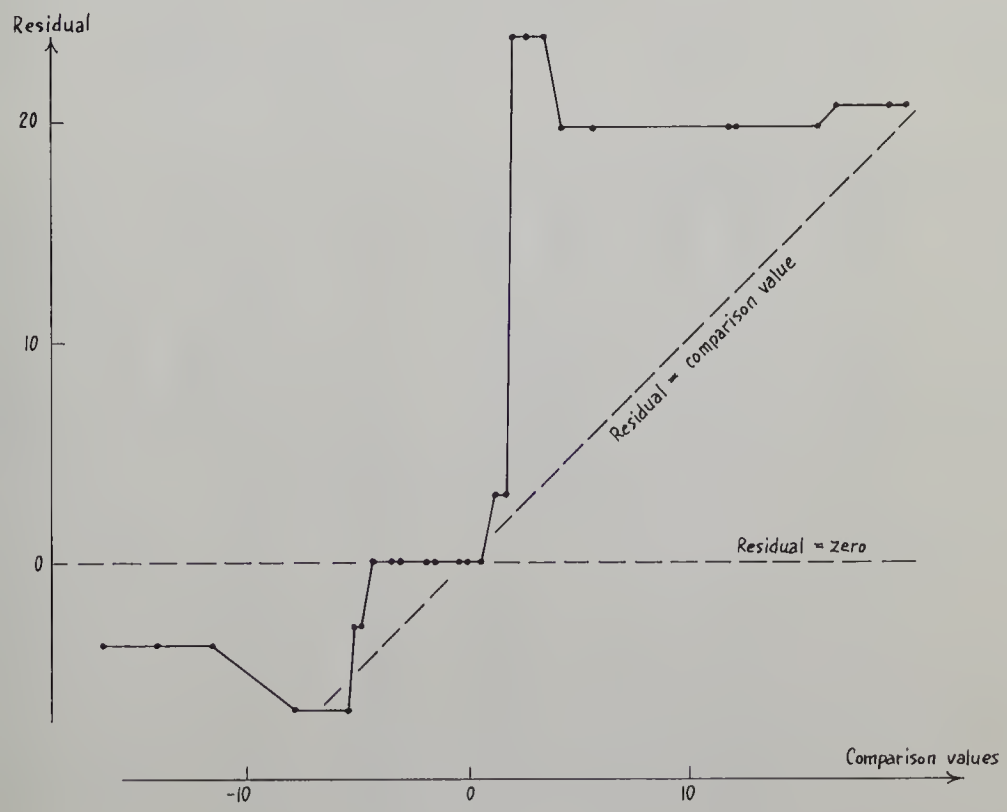
- ◇the row-PLUS-column fit to counts.
- ◇the row-TIMES-column fit based on same.
- ◇the row-TIMES-column fit from the PLUS fit to logs.

We see a steady decrease in the size of the residuals, and conclude that

- ◇it was worthwhile going from PLUS to TIMES.
- ◇the TIMES fit via the logs was even closer.

exhibit **34** of chapter 11+: city killings

Enlarged view of center of exhibit 33



This last point suggests that

- ◇the TIMES fits to raws (here counts) that is based upon a PLUS fit to raws is likely to be a good beginning.
- ◇doing it by logs instead may be worth the effort.

These considerations are, of course, already reflected in the attitudes set out in section10G.

exhibit **35** of chapter 11+: city killings

Analysis of the logs of counts

	Given values (logs*)			Row-PLUS-column				Comparison values		
City	1961	1964	1966	eff	1961	1964	1966	1961	1964	1966
New York	268	280	281	**74**	0	4	−1	**−2.93**	0	2.20
Chicago	256	260	271	**62**	0	−4	1	**−2.46**	0	1.84
Los Angeles	220	225	235	**26**	0	−3	1	**−1.03**	0	.77
Detroit	213	210	233	**10**	0	−11	6	**−.75**	0	.56
Philadelphia	216	227	225	**22**	0	3	−5	**−.87**	0	.65
Baltimore	195	216	224	**14**	−13	0	2	**−.55**	0	.42
Washington	194	212	215	**7**	−7	3	0	**−.28**	0	.21
Cleveland	190	206	214	**4**	−8	0	2	**−.16**	0	.12
Dallas	200	217	208	**6**	0	9	−6	**−.24**	0	.18
New Orleans	172	191	205	**−11**	−11	0	8	**.44**	0	−.33
St. Louis	189	208	203	**−5**	0	11	0	**.20**	0	−.15
Akron	120	108	123	**−85**	11	−9	0	**3.37**	0	−2.52
Boston	142	172	176	**−32**	−20	2	0	**1.27**	0	−.95
Buffalo	128	132	138	**−70**	4	0	0	**2.77**	0	−2.08
Denver	151	152	159	**−49**	6	−1	0	**1.94**	0	−1.46
Jersey City	104	123	120	**−88**	−2	9	0	**3.48**	0	−2.61
Kansas City	169	168	177	**−31**	6	−3	0	**1.23**	0	−.92
Long Beach	100	123	130	**−79**	−15	0	1	**3.13**	0	−2.35
eff				**202**	**−8**	**0**	**6**			

*** Unit = 0.01**

exhibit **36** of chapter 11+: city killings

3RSS middle trace for diagnostic plot of exhibit 35

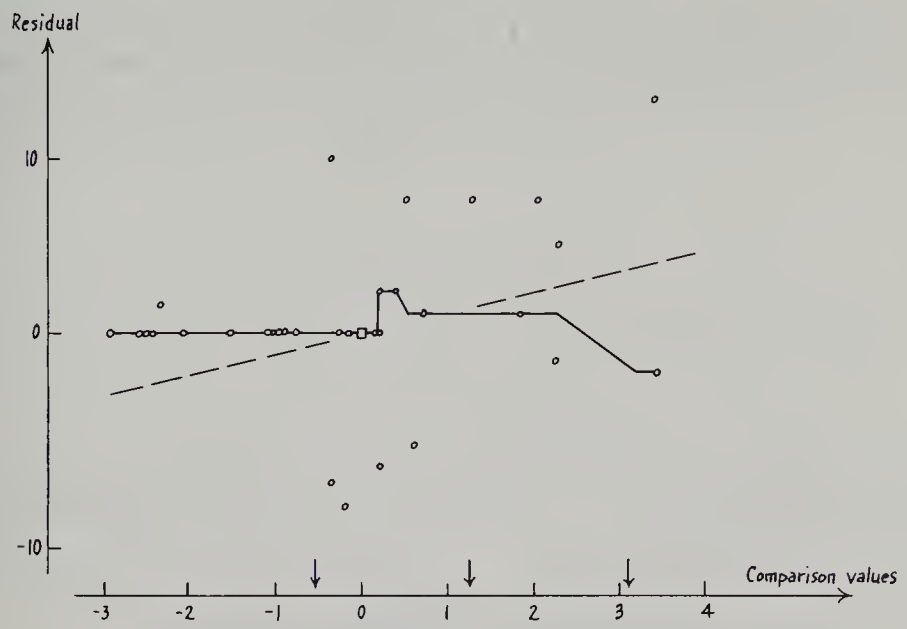


exhibit **37** of chapter 11+: city killings

Antilogs of the fit of exhibit 35--which is a rows-TIMES-column fit to counts--and the corresponding residuals

A) FIT and RESIDUALS

	fit (in counts)			residuals		
City	1961	1964	1966	1961	1964	1966
New York	479	575	661	3	61	−8
Chicago	363	436	501	2	−38	9
Los Angeles	158	191	219	1	−14	7
Detroit	135	162	186	1	−37	28
Philadelphia	145	174	200	−1	14	−22
Baltimore	120	145	166	−31	−1	9
Washington	102	123	141	−14	9	0
Cleveland	96	115	132	−16	1	7
Dallas	100	120	138	−1	29	−18
New Orleans	68	81	93	−1	27	−1
St Louis	78	93	107	−1	27	−1
Akron	12	15	17	4	−3	0
Boston	42	50	58	−16	2	0
Buffalo	19	21	24	1	0	0
Denver	32	32	39	0	1	0
Jersey City	11	14	16	0	3	0
Kansas City	43	51	59	6	−3	0
Long Beach	15	17	20	−5	0	0

review questions

What example did we try? How does the diagnostic plot look? How did we do better than just looking at its points? Where did it tell us to go next? How did this work? Could we come back? To what kind of a fit? How many fits did we compare? How did they compare? What tentative conclusions did this suggest?

exhibit **38** of chapter 11⁺: city killings

Three sets of residuals for three analyses of the counts of murders plus nonnegligent manslaughters

A) the STEM-and-LEAF DISPLAYS

	PLUS on counts	TIMES from ←	TIMES from logs
H	106,	55, 90	61
4*	8	5	
3*	7	3	
2*	0235568	366	0789
1*	1667789	026779	4
·		678	6799
0*	3	222333344	1111122334
Z	⊠⊠	⊠··	⊠:·
−0*	112334	122234	1111331
·	566799	56	58
−1*	2	0023458	446668
−2*	0	3	2
−3*	15	5	178
−4*			
L	−130,		

Note: The ±1's in the righthand display, together with most ±2's and 3's, would have been zero had the log analysis been taken to a third decimal, something that would not have been worth the trouble. (Recall that "⊠" stands for a count of 10.)

B) The LETTER-VALUE DISPLAYS

#54	PLUS on counts			TIMES from ←			TIMES from logs		
M27h		0			0			0	
H14	17	−3	20	8	−2	10	4	−3	7
E 7h	24	−8	32	21	−11	32	9	−16	25
D 4	28	−20	48	33	−15	48	27	−22	49
3	37	−31	68	45	−18	63	28	−31	59
2	48	−35	83	55	−23	78	29	−37	66
1	106	−130	236	90	−35	125	61	−38	99

11L. An unusual fit

Before we close this chapter we ought to show that row-PLUS-column and row-TIMES-column––either bare or -PLUS-one––are not all the reasonable possibilities for fitting two-way tables. We may want to start in quite a different way––or we may want to start in one of these ways and go further. (One example here; others will surface in Chapter 12.)

coastwise distances

One kind of two-way table that sometimes appears is a table of distances by sea between any two of a list of ports. Exhibit **39** sketches the simplest

exhibit **38** of chapter 11+ (continued)

C) Some SPREADS JUXTAPOSED

	PLUS on counts		TIMES from ←		TIMES from logs
H-spread	20	*	10	*	7
E-spread	32		32	*	25
D-spread	48		48		49
3-spread	68	*	63	*	59
2-spread	83	*	78	*	66
range	236	*	125	*	99

Note: "*" marks a decrease of the next column right, as compared to the next column left.

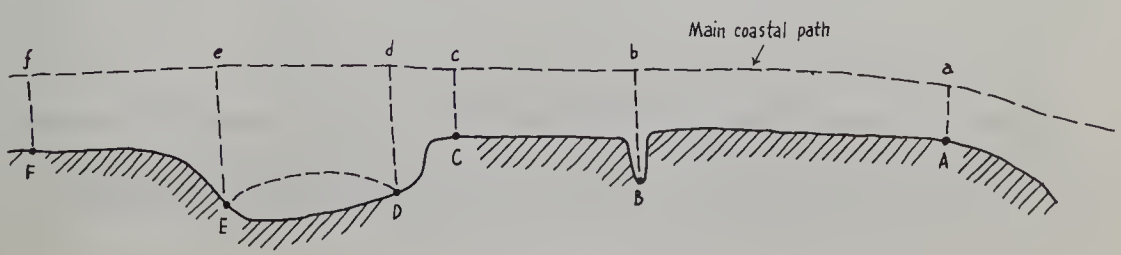
P) PROBLEMS

38A) Collect similar data for other years (preferably both earlier and later) and a similar set of cities; do an analysis parallel to exhibit 32.

38a2/a3/a4/a5/a6/a7) Continue parallel to exhibits 33/34/35/36/37/38.

exhibit **39** of chapter 11+: illustrative

The simplest situations in sea distances



situations. Capital letters--A, B, C, . . . , F--are ports, while small letters--a, b, c, . . . , f--are points on the main path of coastal traffic, where shipping for the corresponding port turns off. (In practice, things are rarely this simple.)

The sea distances between adjacent ports mainly fall into three parts, as follows:

A and B	⟨A to a⟩ + ⟨a to b⟩ + ⟨b to B⟩
B and C	⟨B to b⟩ + ⟨b to c⟩ + ⟨c to C⟩
C and D	⟨C to c⟩ + ⟨c to d⟩ + d to D⟩
D to E	(see below)
E to F	⟨E to e⟩ + ⟨e to f⟩ + ⟨f to F⟩

where the ⟨ ⟩, for this example only, indicate distances. If we are going to fit not only distances between adjacent ports but distances among other pairs, we need to do the fitting in terms of (for A and D, for example)

local A + path a to d + local D

where, for instance, local A more or less corresponds to the aA distance. If we can find, somehow, reasonable values for the "local terms", then we ought to be able to fit

distance MINUS local for row MINUS local for column

by

|path location for row MINUS path location for column|

where the vertical bars--|· · ·|--mean "take the absolute value of"--that is, "take the size of the difference with a + sign no matter what sign the difference has".

There will inevitably be residuals. Some may have simple interpretations; others may not. In the sketch of exhibit 39, the fitted distance between D and E will be

⟨D to d⟩ + ⟨d to e⟩ + e to E⟩

but the sketch shows a much shorter direct path from D to E. This surely ought to make the residual for D and E negative. Exhibit **40** gives the result of such a fitting process (but not how it was done) for eight ports on the West Coast of South America. Of the 28 pairs of ports, the residuals are no larger than 1 mile in all but eight cases. Of these eight, three suggest shortcuts from Coquimbo to ports on the Punta Arenas side, two more suggest shortcuts between adjacent ports, one is −2, and the last--+5 for Lota–Punta Arenas--is quite unexplained.

The most important morals of this example are these:

◇**while it almost always is wise to start trying a row-PLUS-column approach,** there are two-way tables that can be fitted well but only by another approach--**there is no panacea for treating all two-way tables.**

◇if we understand, even imprecisely, what is really happening in a two-way table––or in any other data structure––it is inexcusable not to use that understanding to guide our approach to its analysis. **It is wrong to expect to learn again, by making unguided approaches to each new body of data, all that we have already learned.** Rather we should use all we knew

exhibit **40** of chapter 11+: port distances

Shiproute distances between South American ports (a different kind of fit)

A) The DATA––distances in sea miles

	Ant	Arr	Cal	Coq	Iqu	Lota	P A	Val
Antofagista	x	325	215	396	224	828	1996	576
Arrea	325	x	522	702	110	1134	2301	8828
Caldera	215	522	x	196	420	628	1795	396
Coquimbo	396	702	196	x	602	455	1623	203
Iquique	224	110	420	602	x	1033	2201	782
Lota	828	1134	628	455	1033	x	1191	268
Punta Arenas	1996	2301	1795	1623	2201	1191	x	1432
Valparaiso	576	882	376	203	782	268	1432	x

B) The DATA adjusted for LOCAL DISTANCES––row and column

(local distance)	(9)	(12)	(8)	14)	(9)	(16)	(0)	(6)
Antofagista	x	304	198	373	206	803	1987	561
Arrea	304	x	502	676	89	1106	2289	864
Caldera	198	502	x	174	403	604	1787	362
Coquimbo	373	676	174	x	579	425	1609	183
Iquique	206	89	403	579	x	1008	2192	767
Lota	803	1106	604	425	1008	x	1175	246
Punta Arenas	1987	2289	1787	1609	2192	1175	x	1426
Valparaiso	561	864	362	183	767	246	1426	x

Examples: 304 = 325 − 9 − 12; 198 = 215 − 9 − 8; 502 = 522 − 12 − 8.

C) The FIT and the PATH LOCATIONS

	Ant	Arr	Cal	Coq	Iqu	Lota	P A	Val
Antofagista	x	303	200	373	205	803	1987	562
Arrea	303	x	503	676	98	1106	2290	865
Caldera	200	503	x	173	405	603	1787	362
Coquimbo	373	676	173	x	578	430	1614	189
Iquique	205	98	405	578	x	1008	2192	767
Lota	803	1106	603	430	1008	x	1184	241
Punta Arenas	1987	2290	1787	1614	2192	1184	x	1425
Valparaiso	562	865	362	189	767	241	1425	x
path	1994	2297	1794	1621	2199	1191	7	1432

Examples: 303 = |1994 − 2297|; 200 = |1994 − 1794| (Here "| |" indicates "absolute value of" = "size of".)

→

before to both guide and structure our analyses. (It is **wrong**--not quite so wrong but wrong enough--**to trust our understanding to the point where we never try approaches that our understanding tells us will not work.** Some of our understandings are wrong--we will not find out about their failure unless we probe data, at least occasionally, in ways they suggest are not useful.)

review questions

What example did we start with? What seemed to be a natural way to begin an analysis? What picture was in our mind? How might we expect the "path" terms to be further broken down? What happened when we tried this? What are the morals of this example?

exhibit **40** of chapter 11+ (continued)

D) The RESIDUALS

Antofagista	x	1	−2	0	1	0	−1	−1
Arrea	1	x	−1	0	−9	0	−1	−1
Caldera	−2	−1	x	1	−2	1	0	0
Coquimbo	0	0	1	x	1	−5	−5	−6
Iquique	1	−9	−2	1	x	0	0	0
Lota	0	0	1	−5	0	x	−9	5
Punta Arenas	0	−1	0	−5	0	−9	x	1
Valparaiso	−1	−1	0	−6	0	5	1	x

Note: Of 28 residuals, twenty are ±1, 0, or −1; two each are −2, −5, −9; one each is 5 and −6.

S) SOURCE

1963 World Almanac and Book of Facts, page 684.

P) PROBLEMS

40a) Point out a few specific numbers that show that a row-PLUS-column fit cannot fit this data well.

40b) Do the same for a row-TIMES-column fit.

T) TEASERS

Note: A teaser is a problem, methods for whose solution are not found in this book.

40c) Construct a way to fit the local distances and path locations.

40d) Would you expect a reasonable fit in this form to distances among West Indies, Caribbean, and Gulf ports? (These were in the World Almanac, too!)

11M. How much may we have learned?

This chapter has been concerned with four quite diverse topics, each of which can be read and grappled with separately:

- ◇exploring the use of a second row-PLUS-column fit after a PLUS-one fit has been made, and removed.
- ◇using summary polish to fit row-PLUS-column for any summary--including means and midextremes--and not merely for median.
- ◇an example where re-expression lets us deal more effectively with data on numbers of metropolitan killings.
- ◇an example, involving coastwise distances, where a quite different kind of supplement to a row-PLUS-column fit (really a row-MINUS-column fit) is desirable and effective.

The thrust of most of these topics is simple: What we have learned in chapters 10 and 11 is only the root--a root on which there can be many branches. And any of us may need to invent still another branch.

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Advanced fits 12

This chapter is to cap the climax so far as two-way fits go in this volume. We learned about "-PLUS-one" fits in chapter 10--but we did not learn a variety of things that can help us understand and interpret them. Such questions are discussed here, together with what two-way plots look like for such fits and how such plots can be made.

We saw one extended two-way fit at the end of chapter 11. In this chapter we will learn more about some natural extensions of row-PLUS-column fits.

review questions

Where did we learn about "-PLUS-one" fits? How much did we learn? Where will we learn more? Are there natural extensions of row-PLUS-column fits? Where will they be discussed?

12A. PLUS-one fits

When we introduced row-PLUS-column-PLUS-one fits in the form

$$(*) \qquad \text{all} + \text{row} + \text{col} + \frac{(\text{row})(\text{col})}{\text{constant}}$$

we noticed that when

$$\text{constant} = \text{all}$$

this becomes a row-TIMES-column fit. What happens in general?

A careful eye and a little algebra show that the expression (*) can be written as:

$$(\text{all} - \text{constant}) + \text{constant}\left(1 + \frac{\text{row}}{\text{constant}}\right)\left(1 + \frac{\text{column}}{\text{constant}}\right)$$

which, if we rearrange it to

$$\text{constant}\left(1 + \frac{\text{row}}{\text{constant}}\right)\left(1 + \frac{\text{col}}{\text{constant}}\right) + (\text{all} - \text{constant}),$$

is easily seen to be a

row-TIMES-column-PLUS-one fit.

We thus see that

row-PLUS-column-PLUS-one

and

row-TIMES-column-PLUS-one,

which can be written more explicitly as, respectively,

row-PLUS-column-PLUS constant times comparison value

and

row-TIMES-column-PLUS-another constant,

cover exactly the same range of possibilities. Since these two classes are **identical,** it is natural to speak--or write--simply of a

PLUS-one fit.

It will often be true, as we will see illustrated shortly, that the

row-TIMES-column-PLUS-constant

form is the easy one to understand.

the special response

In the case of no added constant, of the

row-TIMES-column

fit, the response "zero" is special. For if EITHER factor is zero, so too is the product. Thus if there are rows that go far enough, there is a row all of whose fitted responses are zero--and if there are columns that go far enough, there is a column all of whose fitted responses are zero. In a row-TIMES-column fit, this

- ◇must happen for zero.
- ◇cannot happen for any other value.

In general, if there is an added constant, given from the additive side, as we know, by

all MINUS divisor of (row) × (col),

then this is a special value in exactly the same sense: If the rows go far enough, there is a row all of whose fitted values are the same, equal to this special value--and columns similarly. Again, this

- ◇must happen for the special value.
- ◇cannot happen for any other value.

Let us turn back to the example of warming up the East Coast. There we fitted

$$54.4°\text{F} + \text{row} + \text{col} - 1.0\frac{(\text{row})(\text{col})}{54.4°\text{F}},$$

which is the same as

$$54.4°\text{F} + \text{row} + \text{col} + \frac{(\text{row})(\text{col})}{-54.4°\text{F}}.$$

Thus the special response is

$$54.4°F - (-54.4°F) = 109°F.$$

If our form for the fit were to continue to apply to very extreme cases, the following would have to happen:

- ◇if there were a month so hot that any one place had a mean temperature of 109°F, all places would have (up to small residuals) a temperature of 109°F in that month.
- ◇if there were a place so hot that any one (ordinary) month had a mean temperature of 109°F there, all months would (up to small residuals) have a temperature of 109°F at that place.

Clearly, there is no such special month--and aren't we on the East Coast glad! There almost might be such a special place--if we went farther and farther south we would expect higher temperatures and less difference between winter and summer. (Somewhere near the equator, things have to cross over, so that Southern-hemisphere summer can match Northern-hemisphere winter, and vice versa).

This is probably the easy way to think about the PLUS-one fit to the East Coast cooling down:

- ◇an "ideal equatorial" temperature of 109°F.
- ◇depression of mean monthly temperature below this by the product of a month factor and a place factor.

Exhibit **1** gives one set of these factors--and its problems ask you for several more alternative sets.

exhibit **1** of chapter 12: East Coast

The fit of exhibit 11 of chapter 10 in TIMES form

A) FIT

FIT = 109°F − (month factor)(place factor)

B) The FACTORS--one choice

Jan	72.7°F	Laredo	0.60
Feb	71.5°F	Washington	1.00
Mar	63.5°F	Caribou	1.35
Apr	54.4°F		
May	44.1°F		
June	35.4°F		
July	31.5°F		

P) PROBLEMS

1a) Find an equivalent set of factors with Laredo's factor being 1.00.
1b) Do the same for Caribou's factor = 1.00.
1c) Find an equivalent set of factors with April's factor = 1.00.
1d) Do the same with January's factor = 1.00.
1e) Do the same with July's factor = 1.00.
1f) (Teaser) Can you describe by a formula the general pair of sets of factors that reproduce the given fit?

review questions

Can we reformulate row-PLUS-column-PLUS-one analyses with k different from +1? Why/why not? What is a row-TIMES-column-PLUS-one analysis? How is it related to a row-PLUS-column-PLUS-one analysis? Why would we want to call anything just a "PLUS-one fit"? What is a special response? What is the special response for a row-TIMES-column analysis? For a PLUS-one analysis? To what example did we apply these ideas? What value was the special response? How might this be interpreted? In how many ways? Which did/didn't make sense? What form of display for the elements of the fit did we reach? Are there simple alternatives?

12B. Pictures for "-PLUS-one" fits

We are fortunate that we can make pictures of "-PLUS-one" fits that are almost as simple as those for "row-PLUS-column" fits. Instead of two sets of parallel lines, we will have:

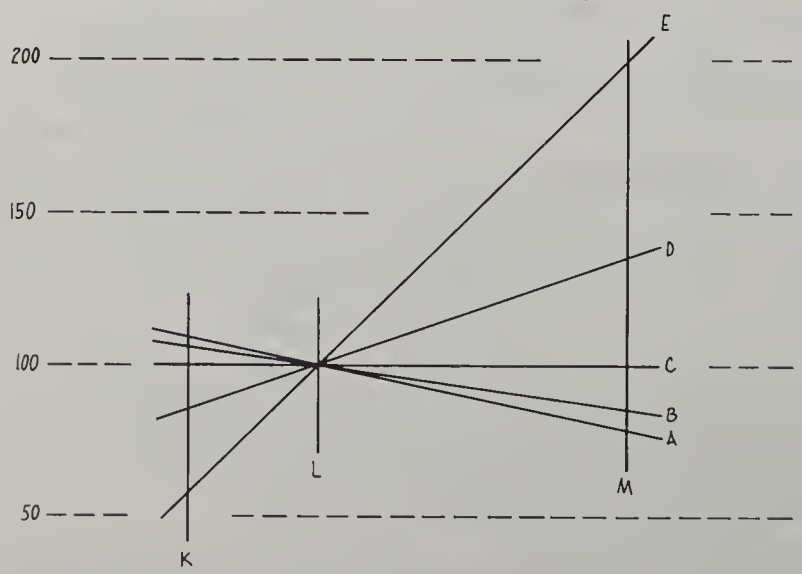
- ◇one set of parallel lines.
- ◇one set of lines with a common intersection.

We need something of this sort to take care of the special value.

Exhibit **2** shows such a picture, which is purely illustrative. (We will write down the numbers in a later exhibit.) The line L (from the parallel lines--here, by chance, vertical) and the line C (from the intersecting lines) correspond to the special value, which happens here to be 100.

exhibit **2** of chapter 12: Illustrative

A PLUS-one fit whose special value is clearly visible



It does not matter whether you choose A, B, C, D or E--or another line of the same family--the intersection with L will come at the same height, 100--because the intersection will be at their common point.

It does not matter whether you choose K or L or M--or another line of the same family--its intersection with C will come at the same height, 100--because C is horizontal.

This example is unusual in two ways:

- ◇the parallel lines are vertical.
- ◇the special value is in the picture.

Usually, the parallel lines are slanted. Often, perhaps usually, the intersection point is outside the picture--either because the special value is far from the values actually found, or because the rows actually examined are far from the special row, or because the columns are far from the special column.

another example

Let us look, then, at the East Coast temperatures again. All the real months are far from the hypothetical special month when all these places would have an equal mean monthly temperature (108.8°F!). All the places are far from the hypothetical place (near the equator, presumably) where the mean monthly temperature would be the same in every month (108.8°F again). So the intersection point will be off the picture.

Exhibit **3** shows the

PLUS-one plot

of the fit given in exhibit 13 of chapter 10, which was of the form

$$544 + \text{row} + \text{col} - \frac{(\text{row})(\text{col})}{544}$$

which is exactly the same as

$$1088 - \frac{1}{544}(544 - \text{row})(544 - \text{col}),$$

so that the point of intersection is at 1088 (in 0.1°F, hence at 108.8°F).

In particular, we see that the fitted July temperature in Caribou almost matches that for February in Laredo, and that the fitted April temperature in Caribou almost matches that for January in Washington. (The former near 65°F, the latter near 35°F.)

In exhibit 3 we chose to put places on the intersecting lines, and months on the parallel lines. This was in no way compulsory. In exhibit **4**, these choices have been interchanged. When you check the heights of corresponding points in the two exhibits, you will find them exactly the same (to the accuracy of the construction of the pictures). They do not look quite alike--indeed they stress different information, but they **contain** the same information.

exhibit **3** of chapter 12: East Coast

PLUS-one plot of the fit in exhibit 13 of chapter 10

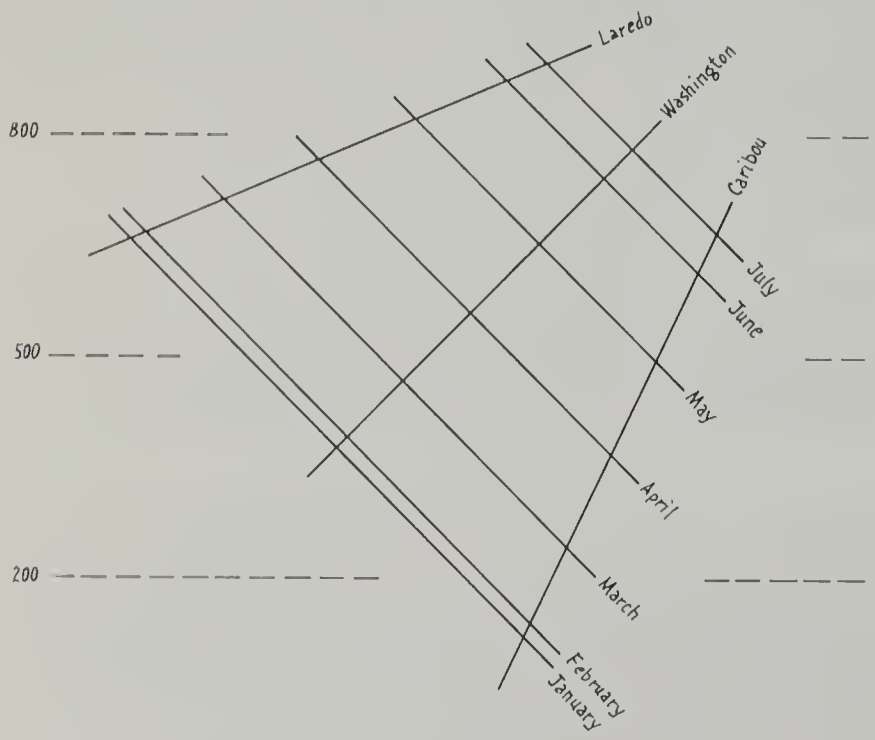
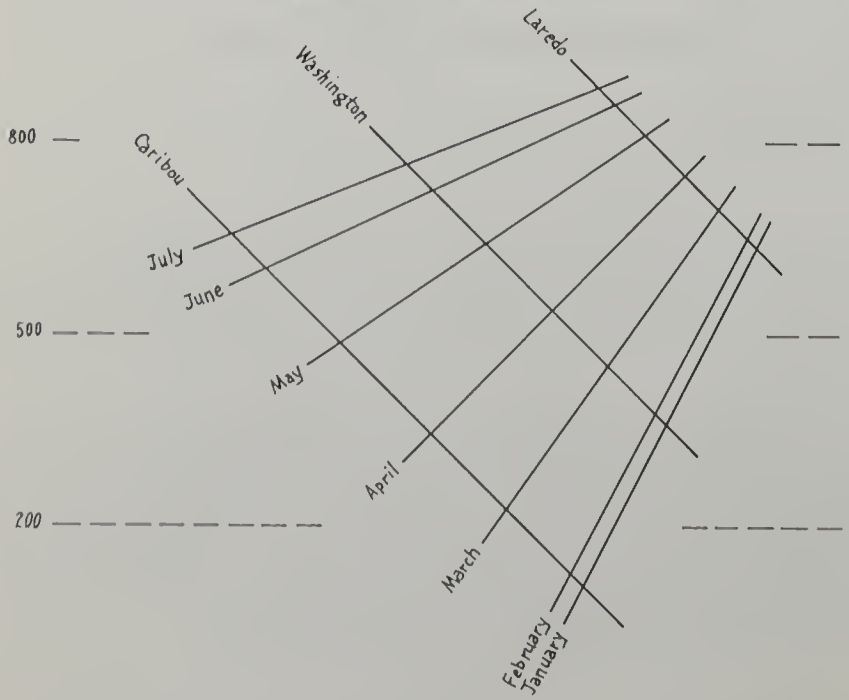


exhibit **4** of chapter 12: East Coast

PLUS-one plot of the same fit interchanging months and places



So far as stress goes, exhibit 3 urges more strongly that Caribou and Laredo differ less in temperature in July than in January, while exhibit 4 emphasizes more clearly that the January–July increase in temperature is less in Laredo than Caribou. Both facts can, of course, be read off either picture--we speak of what is stressed while both are shown. In another example, the choice of which set of lines is used for rows--and which for columns--might well be settled by which phenomenon most deserved emphasis.

-PLUS-one plot of residuals

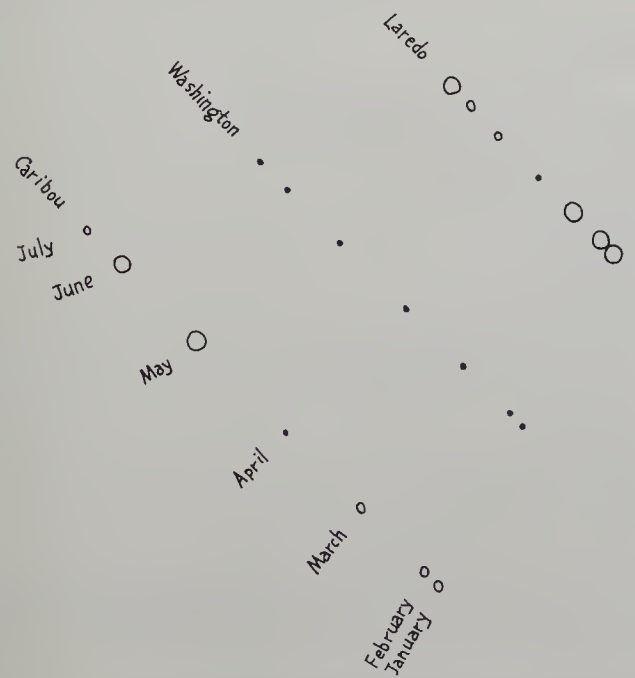
If we wish to give more attention to the residuals--especially to their pattern--we can follow the path we took in section 9E--plotting the coded values of the residuals over the grid points and omitting the grid entirely. Exhibit **5** shows the result of doing this, starting from exhibit 4. Essentially all the information about the lack of fit is preserved.

review questions

What is the essential in learning to picture PLUS-one fits? What from elementary geometry helps us do this? How? What does a PLUS-one plot involve? For what example did we draw such? Is there a choice between

exhibit **5** of chapter 12: East Coast

Plot of coded values overlying exhibit 4



distinct forms? Why/why not? Can we make a corresponding plot of coded residuals? What, if anything, is new at this stage? Which forms of PLUS-one plots of fits have which advantages?

12C. Making those pictures

Let us turn back, yet again, to the PLUS-one fit of exhibit 13 of chapter 10. The fit takes the form

$$\text{all} + \text{row} + \text{col} - \frac{(\text{row})(\text{col})}{544}$$

which we can rearrange and bracket as

$$(\text{all} + \text{col}) \quad \text{PLUS} \quad \left(\text{row} - \frac{(\text{row})(\text{col})}{544}\right).$$

Exhibit **6** shows these two terms written out.

We are going to do the same thing here that we did for row-PLUS-column plots:

- ◇break the fit into two parts.
- ◇take these parts as coordinates.
- ◇notice that

$$\text{fit} = \text{sum of coordinates}$$

and that

$$\text{sum of coordinates} = \text{constant}$$

occurs along 45° lines.

exhibit **6** of chapter 12: East Coast

A two-term form of the analysis of exhibit 11 of chapter 10

A) The ANALYSIS--incomplete, of course

	Laredo	Wash	Caribou	
Jan	−111	−183	−248	
Feb	−103	−171	−232	
Mar	−55	−91	−123	row
Apr	0	0	0	MINUS
May	62	103	140	(row)(col)/544
June	115	190	251	
July	138	229	310	
	759	544	351	all + col

P) PROBLEMS

6a) Write out the alternative two-term analysis using (all + row) and ("what it takes").

6a2) Make the analog of exhibit 7 that is based on the answer to (6a1).

◇rotate the picture so that these lines, which are also

$$\text{fit} = \text{constant}$$

are horizontal.

If "(all + col)" is one coordinate, say the horizontal one, then to each column there corresponds a vertical line. What about the other coordinate? For any fixed row

$$\text{row} - \frac{(\text{row})(\text{col})}{544}$$

reduces to

$$A + B(\text{col})$$

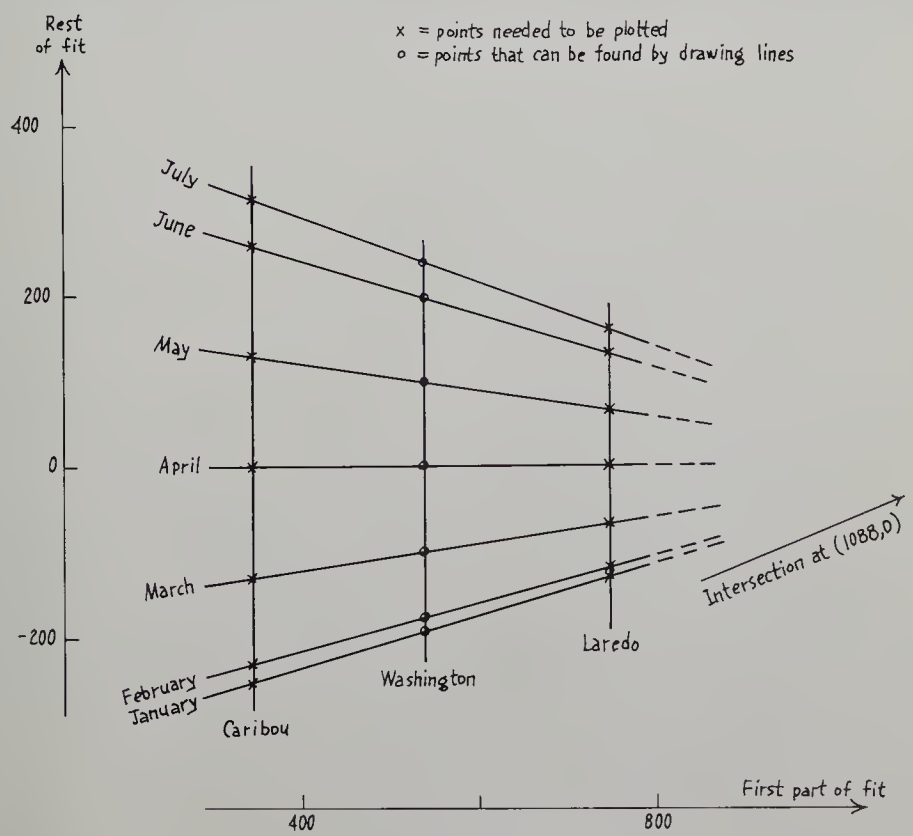
for suitable A and B. When plotted against "col" it has to give a straight line.

Thus the columns turn up as vertical lines, while the rows turn up as slanting lines. If we know one point on each vertical line, and two points on each slanting line, we can fill in the whole plot.

Exhibit **7** shows the result of plotting the terms of exhibit 6 as rectangular coordinates, with place lines and month lines drawn in. Clearly, we need have

exhibit **7** of chapter 12: East Coast

Plotting exhibit 6 on rectangular coordinates (first step toward a two-way-PLUS-one plot)



plotted only the points on the two outermost place lines--no matter how many places there were--since all the points for a given month lie along a straight line.

Exhibit **8** shows the line pattern with a few

sum of coordinates = constant

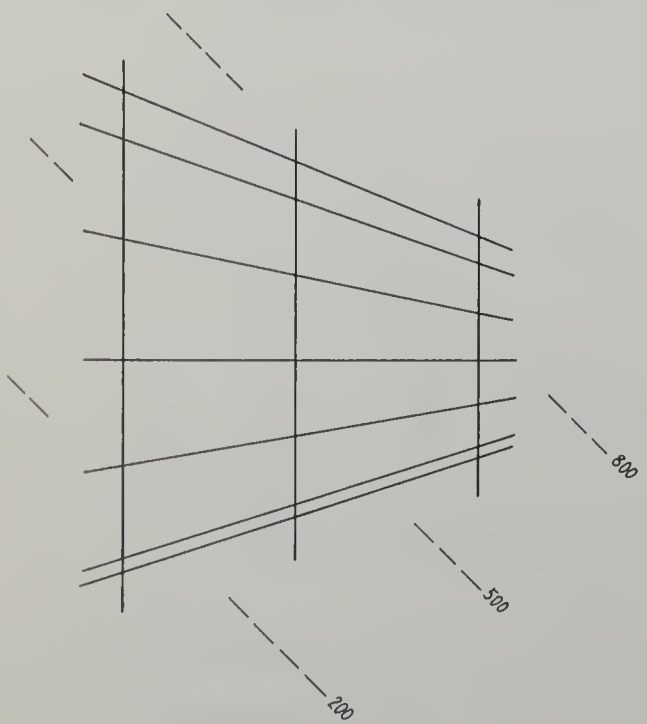
lines drawn in. Since the sum of coordinates is exactly the fit, we have only to rotate exhibit 8 through 45° in order to obtain exhibit 4.

review questions

What sorts of two-term forms of PLUS-one analysis can we use to help make PLUS-one plots? How? What example did we use?

exhibit **8** of chapter 12: East Coast

The lines of exhibit 7 with some lines "ONE PART plus THE OTHER PART = CONSTANT" indicated



12D. Sometimes we can have parallel-line plots, still

In the example at hand, where the fit can be written

$$1088 - \text{data} = \frac{1}{544}(544 - \text{row})(544 - \text{column}),$$

we can take logs, finding

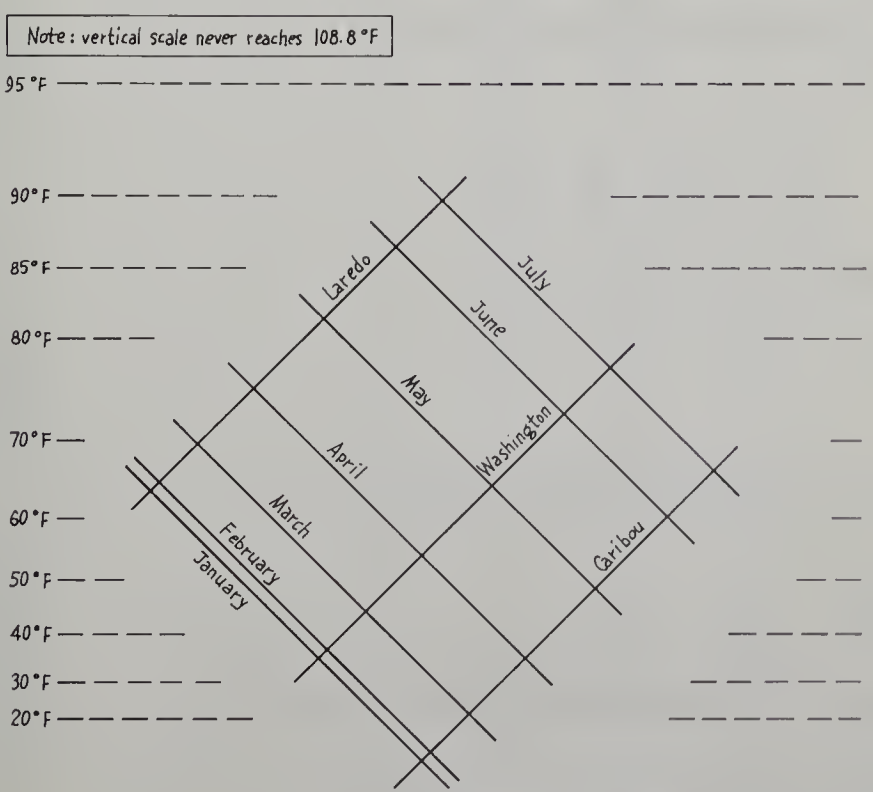
$$\log(1088 - \text{data}) = -\log 544 + \log(544 - \text{row}) + \log(544 - \text{col}),$$

which is in row-PLUS-column form, and can thus be plotted in terms of two sets of parallel lines. Exhibit **9** shows the result, when we preserve "up" by measuring log (1088 − data) from above downward.

As we learned above, making this picture, especially when a few residuals have to be shown, is not easier than making exhibit 4. Moreover, the things noted in the last section as stressed by exhibits 3 and 4 are only to be extracted from exhibit 9 by very careful consideration of the nature of the vertical scale.

exhibit **9** of chapter 12: East Coast

Two-way plot using log (1088 − temperature) as a response



If we had no intuition for--and no interest in--a temperature scale in which a change of 1°F is everywhere an equal change, we would probably find exhibit 6 the display of choice and would have to give careful attention to its scale--emphasizing, perhaps, that the difference between 20°F and 40°F appears, on this scale, closely equal to the difference between 80°F and 85°F. (Such a choice seems unlikely for this example, but possible for many others.) If, on the other hand, we are attached--for some good reason--to the °F scale, we would probably find exhibit 4 (or 3) the display of choice.

but not always

Whether or not we have a choice about turning our row-TIMES-column fit back to row-PLUS-column depends on the data. Sometimes we have the choice, but not always.

Exhibit **10** shows a simple contrived example of this. One of the simplest plus-ONE plots looks like our carlier exhibit 2, in which all five of *AL*, *BL*, . . . , *EL* plot as a single point. Not a good prospect for a rectangular grid! (We should be happy with what exhibit 2 does for us.)

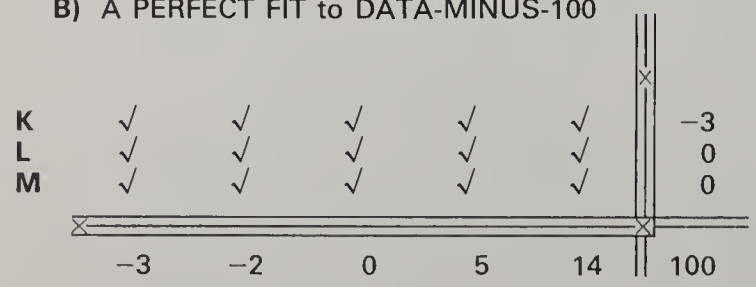
exhibit **10** of chapter 12: arbitrary example

A row-TIMES-column fit where logs do not help

A) The DATA

	A	B	C	D	E
K	109	106	100	85	58
L	100	100	100	100	100
M	79	86	100	135	198

B) A PERFECT FIT to DATA-MINUS-100



C) REMARKS

$$(\text{data} - 100) \equiv (\text{ROW effect}) \times (\text{COL effect})$$

but we cannot take logs because some effects are NEGATIVE.

P) PROBLEM

10A) Invent and analyze a similar but larger example.

review questions

Can we make a two-sets-of-parallel-lines picture of a PLUS-one fit? Why/why not? When do we get in trouble taking logs? What then?

12E. More extended fits

telephones by continent

A rather different kind of example is set out in exhibit **11**. The basic numbers are the numbers of telephones in use on 1 January of seven years in each of seven continents. (*Note.* Here Oceania includes Australia and New

exhibit **11** of chapter 12: world's telephones

The world's telephones by continent

A) The RAW COUNTS––in thousands

	1951	**1956**	**1957**	**1958**	**1959**	**1960**	**1961**
N. Amer	45,939	60,423	64,721	68,484	71,799	76,036	79,831
Eur	21,574	29,990	32,510	35,218	37,598	40,341	43,173
Asia	2,876	4,708	5,230	6,062	6,856	8,220	9,053
S. Amer	1,815	2,568	2,695	2,845	3,000	3,145	3,338
Oceania	1,646	2,366	2,526	2,691	2,868	3,054	3,224
Africa	895	1,411	1,546	1,663	1,769	1,905	2,005
MidAmer.	555	733	773	836	911	1,008	1,076

B) The LOG COUNTS––in 0.001 of log count in millions

N. Amer	1662	1781	1811	1836	1856	1881	1902
Eur	1334	1477	1512	1547	1575	1606	1635
Asia	459	673	718	783	836	915	957
S. Amer	259	410	431	454	477	498	523
Oceania	216	374	402	430	458	485	508
Africa	−48	150	189	221	248	280	302
MidAmer.	−256	−135	−112	78	−40	4	32

C) ONE row-PLUS-column ANALYSIS of the LOGS of B

								eff	fit
N. Amer	**39**	**12**	**7**	**0**	**−7**	**−14**	**−15**	1382	1836
Eur	**0**	**−3**	**−3**	**0**	**1**	**0**	**7**	1093	1547
Asia	**−111**	**−43**	**−33**	**0**	**26**	**67**	**93**	329	783
S. Amer	**18**	**2**	**9**	**0**	**−4**	**−15**	**−12**	0	454
Oceania	**−1**	**11**	**4**	**0**	**1**	**−4**	**−3**	−24	430
Africa	**−56**	**−4**	**0**	**0**	**−0**	**0**	**0**	−233	221
MidAmer.	**25**	**0**	**−12**	**−10**	**1**	**13**	**19**	−522	−68
eff	−213	−67	−32	0	27	59	81	454	

→

Zealand.) Even a glance at the raw counts indicates that we should start with logs. Where the original logs ranged from −256 to 1902, the residuals range only from −111 to 93. We have accounted for 9/10 (some would say 99/100) of the variation by our continent-and-year effects. They clearly tell most of the story.

Because we have lifted these effects out of the way, we can now see the residuals quite clearly. They now ask for us to go further. Look at the rows for North America and Asia, for instance. They show quite substantial and steady trends from year to year. Something needs to be done to take account of this. What? And how much?

At a rough glance, the steady trends of the residuals remind us of the steady trend of column effects. To look harder at this, it is natural to plot one against the other. This is done in exhibit **12**. Here we see that the North American residuals are rather well fitted by a straight line, as are those for South America.

If we look at the residuals for Asia, they fit a straight line fairly well for the five years 1957 to 1961, but not so well for 1951 and 1956. The Middle America residuals behave somewhat similarly.

If we draw eye-fitted lines on exhibit 12 and find their slopes, the corresponding multiple of the column effects of panel C of exhibit 11 are as set out in panel D. Comparing these values with the residuals of panel C shows that we can gain considerably by such an additional fit.

Panel A of exhibit **13** shows the new residuals, obtained by subtracting these additional fits, and indicates some further adjustments to row effects that now seem worthwhile. Panel B shows the analysis after this further adjustment. The fit is now of the form

$$\text{all} + \text{row eff} + \text{col eff} + (\text{row extra}) \times (\text{col eff})$$

where we have used "row extra" as a harmless name for the additional constant we are using for each row. This fit, as we will shortly see, has a reasonably simple interpretation.

exhibit **11** of chapter 12 (continued)

D) ADDITIONAL FITS as MULTIPLES of COLUMN EFFECT--from exhibit 12

								multiple
N. Amer	43	13	6	0	−5	−12	−16	−.2
Asia	−234	−74	−35	0	30	65	89	1.1*
S. Amer	21	7	3	0	−3	−6	−8	−.1
MidAmer	−85	−27	−13	0	11	24	32	.4*

*** Fits well only for later years.**

S) SOURCE

***The World's Telephones* 1961, American Telephone and Telegraph Company, pages 2 and 3.**

exhibit **12** of chapter 12: world's telephones

Plots of residual against column effect for four continents, showing four different slopes (at least for later years)

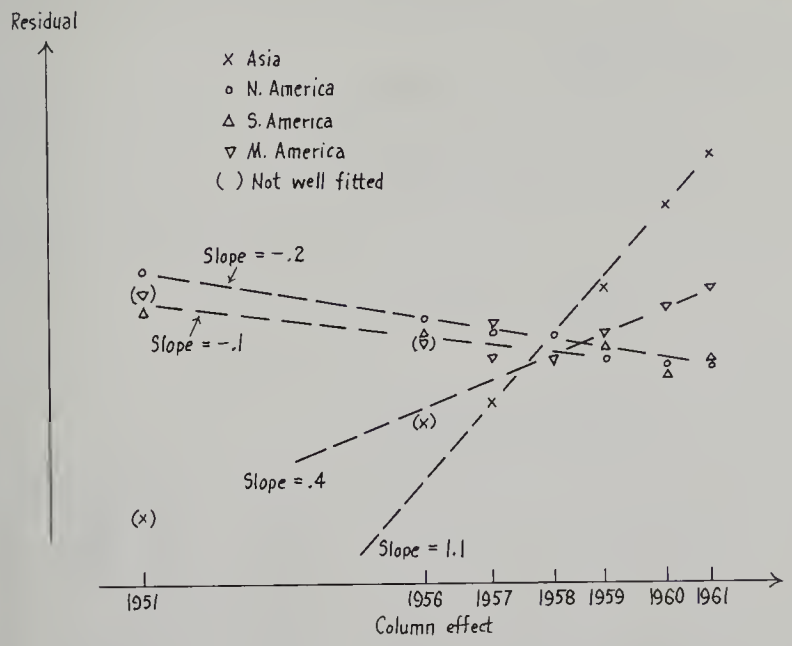


exhibit **13** of chapter 12: world's telephones

Continuation of exhibit 11

A) NEW RESIDUALS--from panels C and D of exhibit 11

	1951	1956	1957	1958	1959	1960	1961	Possible*
N. Amer	−4	−1	1	0	−2	−2	1	−1
Eur	0	−3	−3	0	1	0	7	√
Asia	123	31	2	0	−4	8	4	2
S. Amer	−3	16	6	0	−1	−9	−4	−3
Oceania	−1	11	4	0	1	−4	−3	√
Africa	−56	−4	0	0	−0	0	0	√
M. Amer	110	27	1	−10	0	−11	− 13	−10

* Possible change in row effect, to reduce residuals further.

Example: −4 = 39 − 43, −1 = 12 − 13, 0 = 0 unchanged, −3 = −3 unchanged.

→

extended row-PLUS-column fits

The kind of fit we have just made is clearly one that we can try once we have made any row-PLUS-column fit of the form

all + row eff + col eff.

Indeed, we have our choice of adding

(row extra) TIMES (col eff)

or

(col extra) TIMES (row eff)

or both.

exhibit **13** of chapter 12 (continued)

B) NEW ANALYSIS––with adjusted row effects

								eff	extra*
N. Amer.	−3	0	2	1	−1	−1	3	1381	−.2
Eur	0	−3	−3	0	1	0	7	1093	.0
Asia	121	29	0	−2	−6	6	2	331	1.1
S. Amer	0	19	9	3	2	−6	−1	2	−.1
Oceania	−1	11	4	0	1	−4	−3	−24	.0
Africa	−56	−4	0	0	0	0	0	−233	.0
M. Amer	120	37	11	0	−1	−1	0	−532	.4
	−213	−67	−32	0	27	59	81	454	

* The fit now has the form:

454 + row eff + col eff + (row extra)(col eff).

C) LETTER-VALUE displays of RESIDUAL of B

Those before Those after

#11 residuals

M	6	11		
H	3h	32	−4h	28h
E	2	120	−1	

#58 residuals

M19h	0		
H10	1	−1	2
E	5h	2h	−3

With hinges at −1 and 1, and extremes of −3 and 2h, the 38 residuals after the marks are small enough to exhibit a very good fit.

P) PROBLEMS

13a) Try a "row extra" of −.05 (instead of zero) for Oceania, and one of −.15 (instead of −.1) for South America, and complete the corresponding analysis.

13b) Display the residuals for the analysis of (13a) as in panel C above.

We can regard making a PLUS-one fit by adding

$$\frac{(\text{row})(\text{col})}{\text{div}} = \frac{(\text{row eff})(\text{col eff})}{\text{divisor}}$$

as a special case of any of these, since, if we put

$$\text{row extra} = \frac{\text{row eff}}{\text{divisor}},$$

this last addition becomes

$$(\text{row extra})(\text{col eff}),$$

while if we put

$$\text{col extra} = \frac{\text{col eff}}{\text{divisor}},$$

it becomes

$$(\text{col extra})(\text{row eff}).$$

The additions we are now considering are substantial generalizations of such "PLUS-one" additions. They leave "row extra" or "col extra" wholly at our choice, instead of restricting them to multiples of "row effect" or "col effect", respectively.

A look at panel B (of exhibit 13) shows how little "row extra" looks like a multiple of "row effect" in our case. These generalizations are not always helpful, but they can be tried--with any two-way table.

They are more likely to be useful when (1) there are several rows [or columns] which differ substantially or (2) the data is unusually well behaved. (Our example comes under (2) since we are getting a good fit to three-decimal logarithms, as panel C of exhibit 11 shows.)

interpretation of the example

How are we to interpret the "row extra" values in our example? Were they NOT there, we might have said;

> "In logarithmic form the data are rather well fitted by a steady increase from year to year, the same increase for each continent."

With the "row extra" included, we have an increase of the same shape for each continent, the amount differing.

If we write the fit as

$$\text{all} + \text{row eff} + (1 + \text{row extra}) \times (\text{col eff})$$

or, equivalently, as

$$\text{row fit} + (1 + \text{row extra}) \times (\text{col eff}),$$

we see that the fitted rates of increase for the different continents are in the ratios of the values of

1 + row extra

(in particular, 2.1 for Asia, where we fitted a row extra of 1.1, and 0.8 for North America, where we fitted a row extra of −0.2).

We can therefore describe what we have fitted in these words.

> "With the exceptions noted, the fitted increases--in the log count --for the various continents follow similar paths, the relative rates of increase being, roughly, 2.1 for Asia (after 1957); 1.4 for Middle America (after 1958); 1.0 for Europe, for Oceania (after 1957), and for Africa (after 1956); 0.9 for South America (after 1958); and 0.8 for North America."

So described, the additional fit seems both understandable and meaningful.

Note that the usual construction gives a plot with one family of parallel lines and one set of skew lines.

an interpretation

One way to look at such fits is as a mixture of PLUS and TIMES fitting. The "(1 + row extra) × (col eff)" term is clearly a TIMES fit. To this we have added "all + row eff", which is surely a PLUS fit. Further thoughts--and generalizations--are left to the reader.

review questions

What example? What expression? Why? How did a PLUS analysis work? How did we look farther? What additional fit did this suggest? How much did this help? What kind of fit do we now have? How generally can we try such a fit? How many kinds are there? How related to PLUS-one fits? How did we reformulate our fit? Was it harder/easier to understand?

12F. Simplification is sometimes possible

This same example can be used to illustrate how row-PLUS-column fits--and their extensions--can sometimes be simplified. If we set the year effects of panel B (of exhibit 13) against the years, as follows, we see a clear parallelism:

Year	1951	1956	1957	1958	1959	1960	1961
Year − 1958	−7	−2	−1	0	1	2	3
eff	−213	−67	−32	0	27	59	81

The effect is close to [30 times (year − 1958)].

Without losing much, then, we could simplify our fit by replacing

$$454 + \text{row effect} + \text{col effect}$$

by

$$454 + \text{row effect} + 30(\text{year} - 1958),$$

or, starting with the extended fit, we could replace

$$454 + \text{row effect} + (1 + \text{row extra}) \cdot (\text{col effect})$$

by

$$454 + \text{row effect} + (1 + \text{row extra})(30)(\text{year} - 1958),$$

which is equivalent to

$$\text{row fit} + (\text{row slope}) \times (\text{year} - 1958),$$

where

$$\text{row slope} = 30(1 + \text{row extra}).$$

Exhibit **14** shows, in its panel A, the analysis that the last replacement leads to. While not quite as good as panel B of exhibit 13, it does quite well for a simple fit.

exhibit **14** of chapter 12: world's telephones

Some reduced fits to the telephone data (data from exhibit 11)

A) The REDUCED FORM of the EXTENDED FIT—based on panel B of exhibit 12

	residuals							row	row
	1951	1956	1957	1958	1959	1960	1961	fit	slope
N. Amer	−5	−6	0	1	−3	2	5	1835	24
Eur	−3	−10	−5	0	−2	−1	−2	1547	30
Asia	115	14	−4	−2	−12	4	−17	785	63
S. Amer	−8	8	2	−2	−6	−12	−14	456	27
Oceania	−4	4	2	0	−2	−5	−12	430	30
Afr	−59	−11	−2	0	−3	−1	−9	221	30
M. Amer	116	27	8	0	−4	−2	−16	−78	42
(year − 1958)	(−7)	(−2)	(−1)	(0)	(1)	(2)	(3)		

Fit = (row fit) + (row slope) × (year − 1958)

P) PROBLEMS

14a/b/c/d/e/f/g) Fit a straight line against (year − 1958) by the methods of chapter 6, to the log counts for North America, South America, . . . , Middle America, respectively, being careful to omit counts for early years (if any) where fit is poor. (Usefully spread through the students of a class.)

14h) Combine the results of (14a) to (14g) into a new analysis of the form A. How much improvement do you think has been made?

→

other possibilities

We clearly were not confined to

$$\text{(constant)(year} - 1958)$$

as a replacement for

$$\text{column effect}$$

in this example. We could, for instance, have tried

$$\text{(constant)(year} - 1958) + \text{(another constant)} \times (\text{year} - 1958)^2$$

or

$$\text{(constant)(log year} - \log 1958),$$

to say nothing of many other possibilities. Where the column [or row] labels are essentially numerical--or are intimately associated with a set of numbers outside the data--we can always explore any of a variety of simplifications--either in the basic row-PLUS-column fit or in one of its extensions.

the moral

Such analyses bring to the data in each row [or column] the sort of analysis we entered into in chapters 5 and 6. They are the most obvious example of stacking modes of analysis one upon another in this book. We need to recognize--and to emphasize to ourselves--that stacking one analytical approach onto another--sometimes for several layers--is the only way we know to do a respectable job of data analysis when the data can be grasped by more than one handle.

For us, the essential novelty of two-way tables is the existence of at least two handles--rows and columns. Many sets of data have three, four, or five handles; others have ten--or even a hundred.

Learning to work with two handles is an essential part of an education in exploratory data analysis. It opens our eyes to some of the things we can do with more than one handle.

exhibit **14** of chapter 12 (continued)

14i) Compare the analysis of (14h) with that of panel B of exhibit 13. Do you think that reducing the fit (was/was not) going too far?

14j) Produce a reduced fit analysis corresponding to panel C of exhibit 11. How do you feel it compares with that of that panel?

14k) Write out the computations leading to the first row of residuals in A above. Explain in detail.

14l) Do the same for another row of your choice.

The possibility of simplification is just a case of more handles. In our telephone example we have a choice of three handles:

- ◇the name of the continent,
- ◇the name of the year,
- ◇the number of the year,

or, if we prefer, several handles:

- ◇the name of the continent,
- ◇the name of the year,
- ◇the number of the year,
- ◇the number of the year, and its square,
- ◇the logarithm of the number of the year,
- ◇and so on.

In deciding whether to simplify or nor, we are choosing between one set of two handles and another.

We need to recognize what handles are naturally available in a problem--and try to learn which handle or set of handles exposes the behavior of the data to us most usefully.

review questions

How did we simplify our fit? Was the simplified fit fairly satisfactory? To what earlier chapters is this technique related? How does its existence illustrate the potentialities of data with two or more handles? What handles did we have in the example?

12H. How far have we come?

This chapter introduces us to various extensions of the PLUS and TIMES fits, including a deeper look at PLUS-one fit. It gives us a broader hold on what we can do when we need it. Making a row-PLUS-column fit appears more and more a matter of fitting an incomplete description, one whose fitting is quite likely to disclose the need of going further.

We are now ready:

- ◇to throw back any PLUS-one fit from the form

$$\text{all} + \text{row} + \text{col} + \text{constant} \times (\text{comparison value})$$

to the form

$$(\text{constant}') + (\text{row}') \times (\text{col}')$$

(thus showing that PLUS-one fits are equally closely related to TIMES fits as to PLUS fits),

◇and to find the value of the special response, the response such that, if it occurs in one cell of a fit, it also occurs both in every cell in that column and in every cell in that row.

◇to make pictures involving one set of parallel lines and one set of radiating lines for either PLUS-one fits or their residuals, using the initial expression of the response,

◇and, by taking logs of the differences between actual response and special response, to make parallel-line pictures of such fits.

We have also seen how to go well beyond PLUS-one fits, fitting a straight line for each stripe (all rows **or** all columns) of residuals against the corresponding effects. The result can be written in various forms, and thought of, in particular, as the sum of a special PLUS fit and a general TIMES fit, thus opening the way to further generalizations.

We now understand more clearly that numerical labels for our rows--once we have fitted either row effects alone or both row effects and row extras--offer us a further opportunity, one of fitting row effects--or row extras--usefully well by some function of row labels. (Clearly we can change "row" to "column" throughout what we have just said.)

We now recognize--and can make use of--the relation of PLUS-one fits to both PLUS and TIMES fits. The same is true for generalizations combining PLUS and TIMES.

We have reached a natural terrace in our ascent of the mountain of what can be done to data once it has two simple handles. We are high enough on the shoulder of this mountain to look back and see our earlier steps in clearer perspective. We can also see that the mountain keeps on rising.

Three-way fits 13

chapter index on next page

The chapters spent on two-way fits are time and space well spent. We have learned to deal with two-way data quite well, and in a variety of ways. The arithmetic has not always been easy, but it has been feasible by hand, especially since there were comfortably usable checking techniques.

Those who were fortunate enough to have access to a suitably programmed computer could relax--those who, out in the real world, will need to get at least rough answers with pen or pencil and scratch paper can have learned to do just this.

We learned, at the end of chapter 12, a little of how more than one handle on our data makes a variety of attacks and explorations possible.

In the present short chapter, we will find an introduction to three- and more-way tables. In simple cases we really can do the arithmetic by hand, as an example will show. We do have to be more careful, working harder to get it right to begin with, since checking and salvage are not as easy.

As we deal with more more-way tables, the need for a computer goes up (and the chance that our computer is all ready to do what we want goes down). The short account here is intended:

- ◇to show what sorts of pieces turn up in such analyses.
- ◇to show that the simplest cases can be done by hand.
- ◇to encourage readers who have or are presented with three- or more-way data to analyze it simply and effectively.

review questions

What about the chapters on two-way analyses? And their arithmetic? Was there a place for a computer? To what is this chapter an introduction? What can we say about the feasibility of the required arithmetic? Is there a place for the computer? What are the aims of this chapter?

13A. Three- and more-way analyses: arrangement and tagging

The same sort of things we have done for two-way tables can be done for three- and more-way tables. Fortunately, we do not need three- or four-way sheets of paper. We can push everything into a two-way layout, provided we are willing to use more blocks.

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a psychological experiment

In 1944, Johnson and Tsao published an account of an experiment for measuring the sensitivity of individuals to changes in pull on a ring held by one finger. Measurements were made with different initial steady pulls (here called weight) and different rates of increase of the pull once increase was begun (here called rate). The persons tested included four who were blind and four who could see. The measurements recorded were the increase in pull before the subjects reported feeling a change.

We discuss only the part of the experiment involving:

- three persons, all blind, called here K, L, and M [IB(2), IIB(1), and IIB(2) in Johnson's notation].
- three initial pulls (weights), called here 1, 4, and 7 (as in Johnson 1949) and amounting to 100, 250, and 400 grams.
- four rates of increase, called here *a*, *b*, *c*, and *d* (as in Johnson 1949), amounting to 100, 200, 300, and 400 grams per minute.

This small part of the Johnson–Tsao data provides an example of convenient size, one that can show us how two-way methods can extend to three-way data. Exhibit **1** gives the sums, over two experimental sessions, called "dates" by Johnson and Tsao, of the mean of five responses on each date. (The values given have been rounded to whole numbers for simplicity; they are expressed in grams of weight.)

This table, as is almost necessary for a table put on a page, is arranged in rows and columns. The columns correspond to one variable (weight), while the rows correspond to two variables (subject and rate). In any situation involving all combinations of certain versions of three or more variables, we can think of the numbers laid out similarly, with certain variables combined into columns and others combined into rows. In Johnson's table of all 448 numbers (his table 87) for example:

- sex, sight, date, and individual are combined into columns, of which there are $2 \times 2 \times 2 \times 2 = 16$.
- weight and rate are combined into rows, of which there are $7 \times 4 = 28$.

Arrangements of this kind are the only natural way to put the numbers down. They also provide, as we shall see, a convenient way to begin their analysis.

Where we have 3, 4, . . . , 10, . . . factors, the problem of tagging them can be somewhat troublesome. Surely "row" and "column" are no longer useful. Two choices are natural:

- when we are analyzing "for real," tags linked to the way we think of the factor--such as "rate" for rate, "init" for initial pull, "subj" for person who is the subject--are likely to be best.

◇when we are analyzing to learn about analysis, it often helps to link the tag for a factor to the tags for its versions--as when KLM is used to tag the factor whose versions are K, L, or M.

a tag example

Panel B shows the first column of panel A as a two-way table, for which panel C shows a row-PLUS-column analysis. In panel D we have taken the analysis of panel C and stretched it out into a single column again. If this were

exhibit **1** of chapter 13: finger limens

The 3 × 3 × 4 = 36 basic observations

A) The DATA

Rate & subject	Initial pull (weight)		
	1	**4**	**7**
aK	39	31	26
aL	16	12	12
aM	18	13	14
bK	85	55	56
bL	32	22	20
bM	31	26	26
cK	101	84	70
cL	43	33	30
cM	42	38	40
dK	151	124	98
dL	63	51	37
dM	58	55	46

B) ONE-THIRD of ABOVE, RE-ARRANGED--data for pull 1

	Rate			
Subject	**a**	**b**	**c**	**d**
K	39	85	101	151
L	16	32	43	63
M	18	31	42	58

C) The SAME, ANALYZED

	a	**b**	**c**	**d**	eff
K	−32	−2	2	33	55
L	0	0	−1	0	0
M	4	1	0	−3	−2
	−22	−6	6	25	38

all the data, putting it in one-column form would be an empty exercise. There is more data, however. To set it all out in rows and columns, we have to divide the three factors into two groups--one for rows and one for columns. One group has to include at least two factors. (Since it is easier to have many rows than many columns, we can be almost sure that rows will have to accommodate two factors.) This means each column will need the kind of analysis we have just illustrated.

exhibit **1** of chapter 13 (continued)

D) The ANALYSIS, RE-ARRANGED

	1
aK	−32
aL	0
aM	4
bK	−2
bL	0
bM	1
cK	2
cL	−1
cM	0
dK	33
dL	0
dM	−3
aØ	−22
bØ	−6
cØ	6
dØ	25
øK	55
øL	0
øM	−2
øØ	38

P) PROBLEMS

1a) Imitate panels B, C and D with the second column of panel A.

1b) Do the same for the third column.

S) SOURCE

Palmer O. Johnson, 1949. **Statistical Methods in Research,** Table 87 on page 290.

Note how we have tagged the lines in panel D. The (two-way) residuals are tagged by their full tag--rate and subject. The subject effects are tagged with a "ø" for rate (to say that rate isn't there any more) and the tag of the subject. Similarly the rate effects are tagged with the tag of the rate and an "Ø" for the subject (to say that subject isn't there any more). Finally the common term is tagged "øØ" (to say that neither rate nor subject is there any more). Here "Ø" is a Scandinavian vowel, pronounced roughly somewhere between "uh" and "urh" and used by modern mathematicians for empty sets--sets that contain no element. We will use this same approach to tag both rows and columns in the three-way case.

review questions

Can we do for three-way tables what we have done for two-way tables? For more-way tables? What example? What three factors or kinds of circumstances? What factors in the full experiment? What example did we use to illustrate how we tag rows and columns? How did we push it around? What is ø or Ø or 0 used to mark? What expression did we begin the main analysis with?

13B. An analysis of the psychological example

Let us now turn to the three-way analysis of our $3 \times 3 \times 4$ of difference limens. We will learn below that it is good to analyze logs of weight changes rather than weight changes themselves. Accordingly, exhibit **2** opens with the logs of the values in panel A of exhibit 1. The analysis in panel B has eight blocks (2^3 here, not $2^2 = 4$, as in the two-way case) and ALL EIGHT have to be added up to get the given values. (Notice that we are using an "effects only" analysis. We could include the fits, but this might be confusing.)

For 66, the given value for ***dM7***, the eight terms in the breakdown are

0, the residual at ***d*M7**

−2, the effect at ***d*M0**

−5, the effect at ***d*Ø7**

5, the effect at **øM7**

22, the effect at ***d*Ø0**

0, the effect at **øM0**

−4, the effect at **øØ7**

50, the common term at **øØ0**

These do sum to 66, as you can easily check.

exhibit **2** of chapter 13: finger limens

The log-difference limens and a three-way analysis (entries in panel A are $-100 + 100 \log_{10}$ of entries in panel A of exhibit 1)

A) The DATA (logs)

	1	**4**	**7**
aK	59	49	42
aL	20	8	8
aM	26	11	15
bK	93	74	75
bL	51	34	30
bM	49	42	42
cK	100	92	85
cL	63	52	48
cM	62	58	60
dK	118	109	99
dL	80	71	57
dM	76	74	66

B) One THREE-Way ANALYSIS--(√ means 0)

	1	**4**	**7**	0
aK	√	2	−1	−1
aL	−2	√	4	√
aM	6	−3	√	√
bK	√	−5	√	4
bL	1	√	√	−1
bM	√	1	√	√
cK	√	2	−1	−2
cL	−1	√	√	√
cM	√	√	1	√
dK	√	−1	2	√
dL	√	1	√	√
dM	√	√	√	−2
aØ	√	√	√	−36
bØ	2	√	√	−9
cØ	−2	√	√	8
cØ	√	4	−5	22
øK	√	√	√	34
øL	2	√	√	−6
øM	−6	√	5	√
øØ	12	√	−4	50

Note that we have tried to keep our fonts straight, using **ø** for "none of abcd", **Ø** for "none of KLM", and **0** (zero) for "none of 147". Be careful to pronounce "147" as "one-four-seven"; it is NOT "one forty-seven".

C) Some EXAMPLES of the BREAKDOWN--with row labels

$$59 = \underbrace{0 + (-1)}_{\textbf{aK}} + \underbrace{0 + (-36)}_{\textbf{aØ}} + \underbrace{0 + 34}_{\textbf{øK}} + \underbrace{12 + 50}_{\textbf{øØ}}$$

$$49 = \underbrace{2 + (-1)}_{\textbf{aK}} + \underbrace{0 + (-36)}_{\textbf{aØ}} + \underbrace{0 + 34}_{\textbf{øK}} + \underbrace{0 + 50}_{\textbf{øØ}}$$

$$20 = \underbrace{-2 + 0}_{\textbf{aL}} + \underbrace{0 + (-36)}_{\textbf{aØ}} + \underbrace{2 + (-6)}_{\textbf{øL}} + \underbrace{12 + 50}_{\textbf{øØ}}$$

$$93 = \underbrace{0 + 4}_{\textbf{bK}} + \underbrace{2 + (-9)}_{\textbf{bØ}} + \underbrace{0 + 34}_{\textbf{øK}} + \underbrace{12 + 50}_{\textbf{øØ}}$$

→

We now have terms in our breakdown for, taking the above eight terms from below upwards,

◇a common term, here 50 at **øØ**0.

◇an initial pull effect (1, 4, or 7; often called an initial pull main effect), in particular 4 at **øØ7**.

◇a subject effect (K, L, or M, often called a subject main effect), in particular 0 at **øM0**.

◇a rate effect (*a*, *b*, *c*, or *d*; often called a rate main effect), in particular 22 at ***d*Ø0.**

◇a subject-initial pull effect (K1, K4, . . . , M4, or M7; often called a subject-initial pull interaction), in particular 5 at **øM7.**

◇a rate-initial pull effect (*a*1, *a*4, . . . , *d*4, or *d*7, often called a rate-initial pull interaction), in particular −5 at ***d*Ø7.**

◇a rate-subject effect (*a*K, *a*L, . . . , *d*L, or *d*M, often called a rate-subject interaction), in particular, −2 at ***d*M0.**

◇a (three-way) residual (*a*K1 to *d*M7), in particular, 0 at ***d*M7**.

In the case of ***d*M7,** whose breakdown we have just looked at, the large terms are the common and the rate effect (50 and 22), followed by two interactions (−5 and 5).

Looking at panel B of exhibit 2 as a whole, we see that the common and the three main effects are large, while the two-factor interactions do not appear much larger than the three-way residuals.

It is easy to put together detailed plots for the larger effects with box-and-whiskers (or even schematic) plots for the smaller effects. Exhibit **3** shows one such. Clearly, the three big things are:

◇the effect of rate (more weight change the faster the rate).

exhibit **2** of chapter 13 (continued)

P) PROBLEMS

2a) Write out the breakdowns for 8, 34, and 71.

2b) Is there any breakdown involving both −36 and 23? Both −6 and −6? both the −5 in line **bK** and the −5 in line **dØ**? Explain.

2c) Pick out seven entries not so far used as examples or problems and write out the breakdown, tagging as in panel C.

2d) Copy the breakdowns of panel C, tagging each value with its full identification.

2e) What kind of an effect is **aØ0**? **øL0**?

◇ the distinction between subject K (slower to respond) and subjects L and M.

◇ the effect of initial pull (slower when lower).

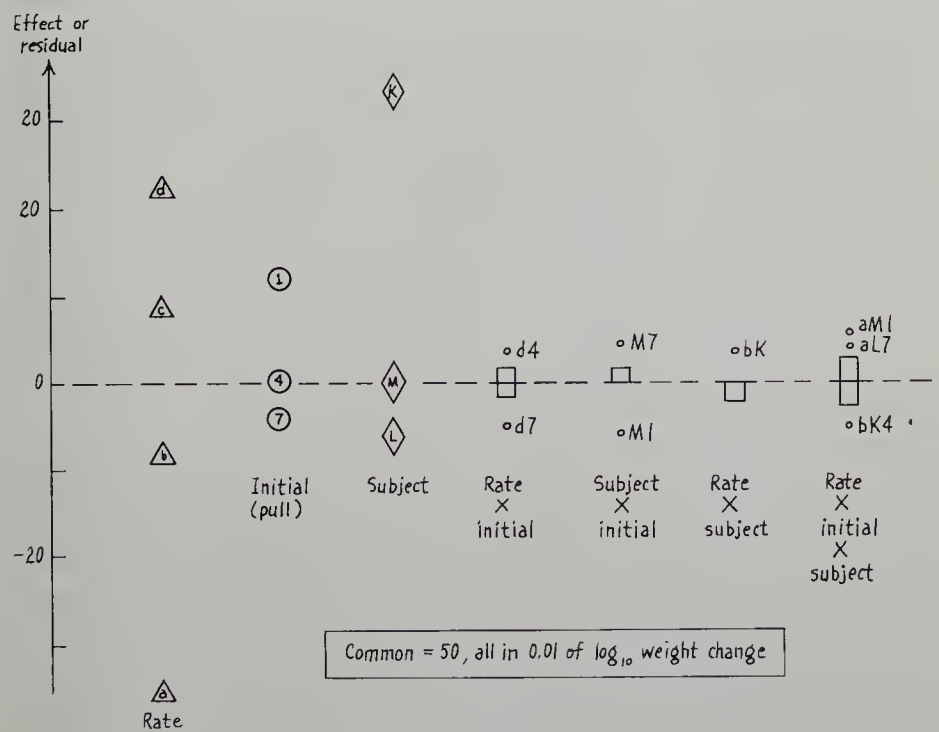
After these come some individual two-factor effects and residuals. (Among these latter, the countervailing nature of **M1** and ***a*M1** and of ***b*K** and ***b*K4** may deserve notice. Can you explain or discuss?)

review questions

What example did we take? What expression was analyzed? How many blocks do we get from a full three-way analysis? How are initial values expressed in terms of effects and residuals? How can we summarize the components of the breakdown? How many large effects did we finally have? What were they (describe)? What came next? What seemed interesting about them?

exhibit **3** of chapter 13: finger limens

Summary of the effects and residuals of the analysis of exhibit 2



13C. Making three-way analyses

Systematically analyzing a three-way set of data is straightforward. Repeated removal--in particular repeated median removal--works both well and easily. Exhibit **4** follows the details step-by-step for the example of exhibit 2.

In panel A, the separate box well below the data shows the *abcd*-medians. Their box is cut up into four parts, three of which are empty (at this stage). The value 96, tagged **K1,** is the median of 59, 93, 100, and 118, which are tagged ***a*K1, *b*K1, *c*K1,** and ***d*K1,** respectively. In the separate box, the blank tagged **Ø1,** similarly, is the median of the four blanks tagged ***a*Ø1, *b*Ø1, *c*Ø1,** and ***d*Ø1,** respectively, and so on.

Were we in a two-way situation, we would have written

59		**−37**
93	and could have	**−3**
100	converted this to	**4**
118		**22**
96		**96**

(In the two-way case, we dropped the 96, and recovered the fit later. In three- and more-way cases, it does not pay to do this. This means that we need to be more careful with our arithmetic.) In panel B, we find that this is exactly what has happened to these five entries. Step 1 consists of taking the *abcd*-medians out of the values medianed and putting them into the corresponding effects.

In panel B, as well as showing our state after step 1, we show the 147-medians (the 1, 4, 7 medians, not the "one forty-seven" medians). Here the value −37, in the ***a*K0** line, is the median of −37, −35, and −38 (tagged, respectively ***a*K1, *a*K4,** and ***a*K7**). Similarly 84, tagged **øK**, is the median of 96, 84 and 80 (tagged, respectively **øK1, øK4** and **øK7**). Of which three entries is the blank following 22, 23, 20 in the column of 147-medians the median?

In a two-way table, we would have had

−37	−35	−38	|	−37

and would have replaced this by

0	2	−1	‖	−37

Again it is just this, and its analog for the other fourteen nonblank 147-medians ("one-four-seven" not "one forty-seven"), that we do.

Panel C shows where we stand after the second step of median removal. The upper lefthand block (12 rows and 3 columns) has been shrunk till no value is larger in size than 6. Again we also show, below the main table, the medians to be used in the next step. This time all four kinds are nonblank. The −5, tagged ***d*7,** is the median of −3, −5, and −5 (tagged ***d*K7, *d*L7,** and ***d*M7,** respectively). The −36, tagged ***a*ø**, is the median of −37, −35, and −36 (tagged ***a*Kø, *a*Lø,** and ***a*Mø,** respectively). The 12, tagged **ø,** is the median of 12, 14 and 5 (tagged **øK1, øL1** and **øM1,** respectively). The 51, tagged **ø0,** is the median of 84, 43 and 51 (tagged **øK0, øL0,** and **øM0,** respectively).

Now the machinery is running at full diversity, so that the main display of panel D (found in exhibit **5**) has no blanks. (It does, of course, have some zeros.) We have now finished one cycle of median removal. This analysis is, in fact, relatively satisfactory. Since this is not always so, and since we are doing a leading example, we will continue for a second cycle of median removal.

exhibit **4** of chapter 13: finger limens (panels A, B and C)

Polishing by medians (note that blanks can be read as zeros)

A) The DATA

	1	4	7	0
aK	59	49	42	
aL	20	8	8	
aM	26	11	15	
bK	93	74	75	
bL	51	34	30	
bM	49	42	42	
cK	110	92	85	
cL	63	52	48	
cM	62	58	60	
dK	118	109	99	
dL	80	71	57	
dM	76	74	66	
aØ				
bØ				
cØ				
dØ				
øK				
øL				
øM				
øØ				

B) After STEP 1

	1	4	7	0
aK	−37	−35	−38	
aL	−37	−35	−31	
aM	−30	−39	−36	
bK	−3	−10	−5	
bL	−6	−9	−9	
bM	−7	−8	−9	
cK	4	8	5	
cL	6	9	9	
cM	6	8	9	
dK	22	25	19	
dL	23	28	18	
dM	20	24	15	
aØ				
bØ				
cØ				
dØ				
øK	96	84	80	
øL	57	43	39	
øM	56	50	51	
øØ				

147-medians

−37
−35
−36
−5
−9
−8
5
9
8
22
23
20
84
43
51

C) After STEP 2

	1	4	7	0
aK	0	2	−1	−37
aL	−2	0	4	−35
aM	6	−3	0	−36
bK	2	−5	0	−5
bL	3	0	0	−9
bM	1	0	−1	−8
cK	−1	3	0	5
cL	−3	0	0	9
cM	−2	0	1	8
dK	0	3	−3	22
dL	0	5	−5	23
dM	0	4	−5	20
aØ				
bØ				
cØ				
dØ				
øK	12	0	−4	84
øL	14	0	−4	43
øM	5	−1	0	51
øØ				

abcd-medians

K	96	84	80	
L	57	43	39	
M	56	50	51	
Ø				

KLM-medians

a	0	0	o	−36
b	2	0	0	−8
c	−2	0	0	8
d	0	4	−5	22
ø	12	0	−4	51

Accordingly, panel D (of exhibit 5) shows *abcd*-medians again. This time fifteen are zero and one is +1. Thus we have little to change in taking step 4 and going to panel E. This panel shows 147-medians (one-four-seven; also for the second time) only two of which are nonzero.

Taking step 5, and going to panel F is also easy. This panel shows **KLM** medians (again for the second time) of which two are nonzero. An easy step 7 takes us to panel G (shown in exhibit **6**).

exhibit **5** of chapter 13: finger limens

Polishing by medians continued (panel lettering continued from exhibit 4)

D) After STEP 3

	1	4	7	0
aK	0	2	−1	−1
aL	−2	0	4	1
aM	6	−3	0	0
bK	0	−5	0	3
bL	1	0	0	−1
bM	−1	0	−1	0
cK	1	3	0	−3
cL	−1	0	0	1
cM	0	0	1	0
dK	0	−1	2	0
dL	0	1	0	1
dM	0	0	0	−2
aØ	0	0	0	−36
bØ	2	0	0	−8
cØ	−2	0	0	8
dØ	0	4	−5	22
øK	0	0	0	33
øL	2	0	0	−8
øM	−7	−1	4	0
øØ	12	0	−4	51

E) After STEP 4

	1	4	7	0	147-medians
aK	0	2	−1	−1	0
aL	−2	0	4	0	0
aM	6	−3	0	0	0
bK	0	−5	0	3	0
bL	1	0	0	−2	0
bM	−1	0	−1	0	−1
cK	1	3	0	−3	1
cL	−1	0	0	0	0
cM	0	0	1	0	0
dK	0	−1	2	0	0
dL	0	1	0	0	0
dM	0	0	0	−2	0
aØ	0	0	0	−36	0
bØ	2	0	0	−8	0
cØ	−2	0	0	8	0
dØ	0	4	−5	22	0
øK	0	0	0	33	0
øL	2	0	0	−7	0
øM	−7	−1	4	0	−1
øØ	12	0	−4	51	0

F) After STEP 5

	1	4	7	0
aK	0	2	−1	−1
aL	−2	0	4	0
aM	6	−3	0	0
bK	0	−5	0	3
bL	1	0	0	−2
bM	0	1	0	−1
cK	0	2	−1	−2
cL	−1	0	0	0
cM	0	0	1	0
dK	0	−1	2	0
dL	0	1	0	0
dM	0	0	0	−2
aØ	0	0	0	−36
bØ	2	0	0	−8
cØ	−2	0	0	8
dØ	0	4	−5	22
øK	0	0	0	33
øL	2	0	0	−7
øM	−6	0	5	−1
øØ	12	0	−4	51

abcd-medians

K	0	0	0	0
L	0	0	0	1
M	0	0	0	0
Ø	0	0	0	0

KLM-medians

a	0	0	0	0
b	0	0	0	−1
c	0	0	0	0
d	0	0	0	0
ø	0	0	0	−1

exhibit **6** of chapter 13: finger limens

Polishing by medians concluded (panel lettering continued from exhibits 4 and 5)

G) After STEP 6

	1	4	7	0	medians
aK	0	2	−1	−1	0
aL	−2	0	4	0	0
aM	6	−3	0	0	0
bK	0	−5	0	4	0
bL	1	0	0	−1	0
bM	0	1	0	0	0
cK	0	2	−1	−2	0
cL	−1	0	0	0	0
cM	0	0	1	0	0
dK	0	−1	2	0	0
dL	0	1	0	0	0
dM	0	0	0	−2	0
aØ	0	0	0	−36	0
bØ	2	0	0	−9	0
cØ	−2	0	0	8	0
dØ	0	4	−5	22	0
øK	0	0	0	34	0
øL	2	0	0	−6	0
øM	−6	0	5	0	0
øØ	12	0	−4	50	0

H) SAME−−zeroes omitted

	1	4	7	0
aK		2	−1	−1
aL	−2		4	
aM	6	−3		
bK		−5		4
bL	1			−1
bM		1		
cK		2	−1	−2
cL	−1			
cM			1	
dK		−1	2	
dL		1		
dM				−2
aØ				−36
bØ	2			−9
cØ	−2			8
dØ		4	−5	22
øK				34
øL	2			−6
øM	−6		5	
øØ	12		−4	50

I) SAME−−±1 also omitted

	1	4	7	0
aK		2		
aL	−2		4	
aM	6	−3		
bK		5		4
bL				
bM				
cK		2		−2
cL				
cM				
dK			2	
dL				
dM				−2
aØ				−36
bØ	2			−9
cØ	−2			8
dØ		4	−5	22
øK				34
øL	2			−6
øM	−6		5	
øØ	12		−4	50

abcd-medians

0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0

P) PROBLEMS

6a) Take the as-initially-expressed data of panel A of exhibit 1 and carry out steps 1, 2, and 3.

6a) Continue with steps 4, 5, and 6.

6a3) Do you feel a need to go further? Make a picture analogous to exhibit 7. Summarize in words what it seems to want to tell us.

6b) Go on beyond panel (I), omitting ±2's also. Which of the three−−this, panel (H), panel (I)−−seems to you to convey the clearest and most informative picture?

6c) (teaser) Can we tell whether or not we also took the five blank medians out of the corresponding entries as part of step 2?

This panel shows not only 147-medians--all twenty of which are zero--but also *abcd*-medians--all sixteen of which are zero. Since we have just taken out all 20 KLM-medlans, we know that all 20 KLM-medians of panel G will also be zero. We can be happy--at least as doers of arithmetic--since we have come to the end of our process of median removal. Unless we want to change our analytical process, we are done. We have, in particular, reached the analysis set out in detail in panel B of exhibit 2 and pictured in exhibit 3.

In this case, two cycles of median removal have done all that median removal could do. In general, this need not happen. Most often, we think two cycles of median removal will be enough--whether or not we are at the end of the process. "Two are probably enough" is good advice, but does not absolve the analyst from looking hard to see what one or more additional cycles might do.

The remaining panels of exhibit 6--panels H and I--show the result of omitting first zeros and then also ±1's (and, in a problem, ±2's). These offer one simple way to try to show more clearly what seems to be going on in such a moderately complex situation--**taking the smallest values out of the way.**

Exhibit 7 sets some problems using other sets of data.

exhibit **7** of chapter 13: data and problems

Some three-way problems

A) CLEANING ABILITY OF DETERGENT COMBINATIONS

	A_1B_1	A_1B_2	A_1B_3	A_2B_1	A_2B_2	A_2B_3	A_3B_1	A_3B_2	A_3B_3
C_1	106	197	223	198	329	320	270	361	321
C_2	149	255	294	243	364	410	315	390	415
C_3	182	259	297	232	389	416	340	406	387

A_1, A_2, A_3 = three concentrations of a detergent;
B_1, B_2, B_3 = three concentrations of sodium carbonate,
C_1, C_2, C_3 = three concentrations of sodium carboxy–methyl cellulose.

Larger values correspond to better cleaning.

B) POROSITIES IN PACKED TUBES

	LC	LY	BC	BY	SC	SY	GC	GY
0	473	440	490	457	478	450	479	424
3	366	342	378	331	392	291	390	298
9	373	366	402	373	406	375	385	349

L, B, S, G = Lead, phosphor-Bronze, polyStyrene, Glass
C, Y = Cascaded, dropped 1 Yard
0, 3, 9 = distance from wall in tenths of an inch (in 3-inch diameter tube)
Entries are porosities in percent (in thousandths).

→

review questions

How can we tackle analyzing a three-way table? How can we lay out three-way data? Initially? After analysis? What kind of repeated removal did we try? What change did we make in the details of our calculation? On what example? How many cycles did we go through? Why did we stop there? Might we have stopped somewhere else? Why/why not?

exhibit **7** of chapter 13 (continued)

C) STRENGTHS OF CEMENT

	3p	3P	7p	7P	8p	8P	9p	9P
m	15.959	16.151	16.443	16.636	16.780	16.859	17.086	17.309
M	16.028	16.221	16.451	16.607	16.788	16.948	17.090	17.318

3, 7, 8, 9 = cure for 3 days, 7 days, 28 days, and 28 days under special conditions
p, P = old press-sheet, new press-sheet
m, M = old mould, new mould
Entries are sums of 6 logarithms of compressive strengths in kg/cm^2.

PROBLEMS (longish)

7a) Set up the data from panel A in a convenient way, and remove medians once in each of the three possible ways.

7a2) Continue the removal for a further round.

7a3) Complete the analysis.

7b) Do the same as (7a) for the data of panel B.

7b2) Continue as in (7a2)

7b3) Complete as in (7a3).

7c) Do the same as (7a) for the data of panel C.

7c2) Continue as in (7a2).

7c3) Complete as in (7a3).

S) SOURCES

(For panel A) A. J. Feuell and R. E. Wagg, 1949, "Statistical Methods in Detergency Investigations" Research 2: 334. (Used by R. L. Wine, 1964, Statistics for Scientists and Engineers, Prentice Hall, at page 492).

(For Panel B) J. C. MacRae and W. A. Gray, 1961, "Significance of the properties of materials in the packing of real spherical particles," British Journal of Applied Physics, 12: 164–72 (from their table 6, used by Wine at page 493).

(For panel C) A. Hald, 1952. *Statistical Theory with Engineering Applications*, John Wiley (see page 481).

13D. Three-way re-expression

While dealing with three-way analysis, we should take at least a brief look at re-expression--in particular at how we can try to detect a leaning toward a

row-TIMES-column-TIMES-layer

analysis once we have made a

row-PLUS-column-PLUS-layer

analysis.

Suppose that we have fitted

$$\text{all} + \text{row} + \text{col} + \text{lay}$$

where "lay" stands for "layer," a convenient word to use even if, as is the ease, we are not going to show our numbers as layers of rows crossing columns. (Even if we do not plan to show them in such a way, we may do well to think about them thus.) These are the leading terms in

$$(\text{all})\left(1 + \frac{\text{row}}{\text{all}}\right)\left(1 + \frac{\text{col}}{\text{all}}\right)\left(1 + \frac{\text{lay}}{\text{all}}\right),$$

the remaining terms of which (when we multiply out) being

$$+ \frac{(\text{row})(\text{col})}{\text{all}} + \frac{(\text{row})(\text{lay})}{\text{all}} + \frac{(\text{col})(\text{lay})}{\text{all}} + \frac{(\text{row})(\text{col})(\text{lay})}{\text{all}^2}.$$

Thus we might expect to take as comparison values:

- ◇for each two-factor interaction, a quantity formed from main effects just the way we would do for a two-way analysis.
- ◇for a three-way residual, a quantity formed by multiplying together all three of the corresponding main effects and then dividing through by the square of the common term.

We thus can look separately at each of four parts--or we can look at all of them together.

Exhibit **8** shows a three-way PLUS analysis of the original Johnson and Tsao data (from exhibit 1) and the four sets of comparison values. Exhibit **9** shows the four diagnostie plots. For each of these, if we had to choose between slope zero (residual = 0) and slope one (residual = comparison value), we

exhibit **8** of chapter 13: finger limens

A rate-PLUS-subject-PLUS-initial analysis of the Johnson and Tsao data, and the corresponding comparison values (data in exhibit 1)

A) The ANALYSIS

	1	**4**	**7**	0
aK	−7	0	2	−16
aL	0	0	0	3
aM	7	0	−3	0
bK	0	−9	1	4
bL	0	0	0	0
bM	1	0	−2	0
cK	0	4	0	−9
cL	0	0	0	0
cM	0	0	1	1
dK	4	0	−5	35
dL	0	0	0	0
dM	−3	0	1	0
aØ	−6	0	2	−18
bØ	0	0	0	−5
cØ	0	0	−1	6
dØ	2	0	−12	24
ø	11	0	−7	34
øL	0	0	0	−4
øM	−6	0	4	0
øØ	10	0	−2	31

B) Some COMPARISON VALUES

	1	**4**	**7**	
aK	−6.4	0	1.3	−19.7
aL	.7	0	−.1	2.3
aM	0	0	0	0
bK	−1.8	0	.4	−5.5
bL	.2	0	0	.6
bM	0	0	0	0
cK	2.1	0	−.4	6.6
cL	−.2	0	.1	−.8
cM	0	0	0	0
dK	8.5	0	−1.7	26.3
dL	−1.0	0	.2	−3.1
dM	0	0	0	0
aØ	−5.8	0	1.2	
bØ	−1.6	0	.3	
cØ	1.9	0	−.4	
dØ	7.7	0	−1.6	
ø	11.0	0	−2.2	
øL	−1.3	0	.3	
øM	0	0	0	

Examples: $-6.4 = (-18)(34)(10)/(31)^2$

$-5.8 = (-18)(10)/31$

$11.0 = (34)(10)/31$

$-19.7 = (-18)(34)/31$

would choose the latter. (The certainty with which we would make the choice does vary appreciably.) In this clear case, all four voices speak out for re-expression by logs.

We may also want to deal with less clear instances. In such circumstances we might like to persuade all the voices to speak together. Exhibit **10** has a stem-and-leaf combining all four sets of comparison values, and a listing of those--with near ties combined--where the comparison value seems large enough to be worth attention. Those who do problem (10a) will learn how strongly the four diagnostic plots speak out when united.

exhibit **9** of chapter 13: finger limens

The four separate diagnostic plots (to the same scale numbers from exhibit 22)

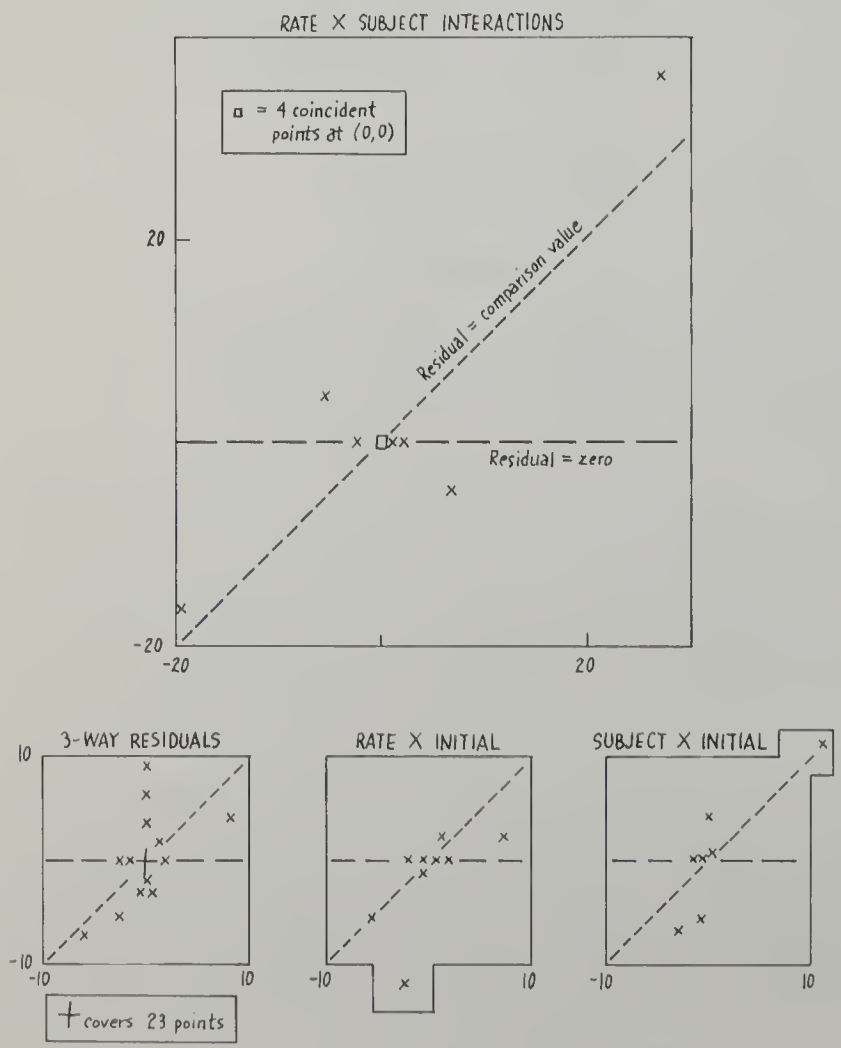


Exhibit **11** shows a second analysis of the same data--obtained in fact by mean removal--and asks, as problems, for its diagnosis.

review questions

What is the relation of TIMES to PLUS in the three-way case? (Can you forecast what happens in the four-way case?) How many kinds of comparison values are there in the three-way case? How many kinds of diagnostic plots? What example? What did the diagnostic plots indicate? Might we want to combine diagnostic plots? How might we do this? (Is there a place for 3RSS middling?)

exhibit **10** of chapter 13: finger limens

Combining and focussing the four diagnostic plots

A) STEM-and-LEAF--
FOR ALL COMPARISON
VALUES

H	11.0, 26.3
8	5
7	7
6	6
5	
4	
3	
2	13
1	239
0	12233467
Z	⊠⊠⊠::
−0	12448
−1	036678
−2	2
−3	1
−5	58
−6	4
L	−19.7,

B) Those of SIZE ≥ 1.5 ORDERED
WITH THE CORRESPONDING
RESIDUALS

comparison value(s)	residual(s)
26.3	35
11.0	11
8.5	4
7.7	2
6.6	9
2.1, 2.3	0,3
1.9	0
−1.6, −1.6, −1.7, −1.8	0, 0, −5, −12
−2.2	−7
−3.1	0
−5.5	4
−5.8	−6
−6.4	−7
−19.7	−16

P) PROBLEMS

10a) Plot the seventeen (comparison value, residual) pairs of panel B in comparison with the lines "residual = zero", and "residual = comparison value".

13E. More about the example

We have learned that the Johnson and Tsao example deserves either a TIMES analysis of the weight changes or a PLUS analysis of their logs. Can we go further?

exhibit **11** of chapter 13: finger limens

Another analysis of the same data

A) The ANALYSIS

	1	4	7	ø
aK	−.9	2.5	−1.5	−1.1
aL	−2.4	0	2.5	.4
aM	3.4	−2.4	−.9	.8
bK	1.1	−2.5	1.5	1.5
bL	.9	.3	−1.2	−1.3
bM	−2.0	2.2	−.3	−.2
cK	.0	.4	−1.3	−1.0
cL	.8	−.8	0	.2
cM	−.7	−.5	1.3	.9
dK	0	−1.3	1.4	.6
dL	.8	.5	−1.3	.8
dM	−.7	.8	.0	−1.5
aØ	.4	−1.8	1.2	−29.8
bØ	1.8	−2.4	.6	−1.8
cØ	−2.1	.7	1.4	12.7
dØ	.1	3.4	−3.3	27.1
øK	1.4	.3	−1.7	24.7
øL	1.9	−.2	−1.7	−14.8
øM	−3.3	−.2	3.3	−9.8
øØ	8.1	−2.0	−6.0	56.2

P) PROBLEMS

11a) Find the comparison values corresponding to the analysis of panel A.

11a2) Make the four diagnostic plots using these comparison values.

11a3) Combine the comparison values as in exhibit 10.

11a4) Make a unified and focussed diagnostic plot.

11b) Do (11a) for the analysis of exhibit 10.

11b2) Do (11a2) for this.

11b3) Do (11a3) for this.

11b4) Do (11a4) for this.

11c/c2/c3/c4) Do(11a/a2/a3/a4) for some other analysis as specified.

Exhibit **12** looks further at the main effects.

Panel A treats the effect of rate of change of pull. A careful look shows us that the fitted term varies quite closely with the log of the rate, the differences "fitted minus log" ranging only from −36 to −40. Let us take this for an observed fact, and return to it in a moment.

Panel B deals with the three observers present in our limited set of data. Here it looks as though sex might be relevant--though three cases are hardly enough to make any strong suggestion.

Panel C deals with the factor or initial pull, which is found to have some effect, though not a large one. Here the effect is crudely like −1/4 log(initial pull), the differences "fitted minus 1/4 log" ranging only from 10 to 12.

exhibit **12** of chapter 13: finger limens

The main effects made explicit (fitted terms from exhibit 6)

A) THE **abcd** OR "RATE" FACTOR

version	Rate (gm/min)	log rate*	fitted term	diff
a	100	0.0	−36	−36
b	200	30.1	−9	−39
c	300	47.7	8	−40
d	400	60.2	22	−38

*** These columns are −200 + 100 $\log_{10}$ (previous column).**

B) THE **KLM** OR "OBSERVER" FACTOR:

Version	Sex	fitted term
K	Male	34
L	Female	−6
M	Female	0

C) THE **147** OR "INITIAL PULL" FACTOR:

Version	Initial pull (grams)	log	fitted term	**
1	100	0.0	12	12
4	250	39.8	0	10
7	400	60.2	−4	11

**** Fitted term PLUS (1/4)(log initial pull)**

P) PROBLEMS

12a) For the modified response discussed in the text, modify exhibit 3 accordingly.

12a2) Describe in words what this new picture tells us.

Thus some discussion--interesting and possibly useful--can be made of each of the three main-effect terms.

In order to understand the first of these three it is time to note what we have been analyzing. In fact:

$$y = \text{change in pull before a change was detected,}$$
$$\log y = -100 + 100 \cdot \log_{10}(\text{change in grams}).$$

Since the latter seems to move parallel to:

$$\log \text{rate} = -200 + 100 \cdot \log_{10}(\text{grams per minute}),$$

we ought to think about the--little-changing--difference:

$$\log y - \log \text{rate} = 100 + 100 \log_{10} \frac{\text{change in grams}}{\text{grams per minute}}.$$

When we do this, we realize that

$$\frac{\text{change in grams}}{\text{grams per minute}} = \text{delay in minutes},$$

and that what we have observed can be very simply put in words as follows:

◇the time taken to detect a beginning of a change in pull is much more constant than the change in pull taken to detect such a beginning.

re-formulated responses

Once we know this, there seems little reason not to take account of this from the beginning of our analysis. We ought to do better by analyzing

log(time of detection)

rather than either the weight change for detection or the log of this weight change. Doing this alters all initial entries by an amount that is uniform for each version of *abcd*. This can mean that the residuals are unaffected, while the fit is changed in the exactly corresponding way.

Combining the common term and the rate term we have:

Version	Old fit*	Change	New fit
a	14	0.0	14
b	41	30.1	11
c	58	47.7	10
d	72	60.2	12

which leads to a common term of, say, 12 and a much smaller rate term, whose values are 2, −1, −2 and 0.

* Panel G of exhibit 6.

review questions

What happened when we looked harder at the main effects? Did this make us change the way we thought about the experiment and its data? Why/why not? Where did we come out?

13H. How far have we come?

This chapter shows us the way to generalize our analyses of two-way tables to the analysis of three-way tables, leaving the extensions to more-way tables open to those who wish to go that way. Again it is mainly a how-to-do-it chapter.

We are now ready:

◇to array three- (or more-)way data on the page, so that we can both look at and work with it effectively. As a part of this we have to (a) think of how we can effectively label rows and columns with suitable tags, using different fonts (capitals, lower case, digits, etc.) where that can help, and (b) choose a suitable symbol (ø, Ø, or zero) for the factor (= set of circumstances) that is no longer there.

◇to "median polish" such tables by "shifting out" values from all the blocks that are susceptible to taking medians over a given set of circumstances. We have to do this for each factor (= set of circumstances) at least once, often twice, and sometimes more than twice.

◇to plot a summary of all the kinds of effects--and the residuals. We do not have all the nice properties that we have become used to in the two-way case, but we can get a reasonably general picture (and perhaps be told which two-way plots will help the most).

◇to define, in the three-way case, not just a single kind of comparison value (as in the two-way case) but four kinds,

◇and to plot different blocks of the analysis against each kind, both separately and together. (We then have a choice between a PLUS-four analysis, and re-expression, among many things.)

We now understand that the obligation to simplify and interpret the blocks of our analysis, particularly (of course) the one-factor effects, is, if anything, stronger for three- and more-way tables than for two-way ones.

We now see that we can extend all that we have learned to do with two-way tables in a new direction. We see great opportunities--and we cannot help being reluctant to do all the computations by hand. Our desire--and need--for effective computer programs has to be great.

These are practical matters--often of crucial importance--but the matters of principle are untouched by their presence. We can now believe that **when data has three or more handles**--and we know how to grasp them two at a time--**there will be a way,** whether or not we see it at first, **to grasp all of them together.**

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Looking in two or more ways at batches of points 14

In chapters 8 and 9 we looked at batches of (x, y) points in a single rather dull way: First slice on x and then trace on y. Doing this produced quite helpful pictures; we learned to grasp such batches and coax useful displays out of them. The time has come for us to go on, to ask such natural questions as: Why not interchange x with y? What if we use $x + y$? Or $x - y$, or still other choices?

Sometimes using x to cut seems almost inevitable, but by no means always. If, for example, one is height and the other is weight, it is hard to say that either variable "must" come first, that it "must" be used for cutting. The present chapter explores the consequence of change, of looking in two or more ways at the same batch of (x, y) points.

review questions

What did we do earlier? In which chapters? What did this do for us? What might we now do? Are we going to do it? Is it inevitable that we use x to cut? Give an example. What are we now to do?

14A. Coordinates and level traces

When we talk about (x, y) pairs, we are thinking about objects or situations involving two numbers. Say

$$x = \text{height}$$

and

$$y = \text{weight}.$$

We naturally think of them as plotted in ordinary rectangular coordinates. And how do--or should--we think of these coordinates? Once we have this clear, we can think much more easily about as many possible coordinates, of as many different kinds, as may help us in a specific situation.

Saving of pencil and ink--avoiding confusion of what we plot: both often lead us to use "axes" or "scales" to tell us about rectangular coordinates in a finished picture. No such picture begins life with only axes or scales, however

--rather it begins on a piece of graph paper. What tells us about the x-coordinates of our points are all the vertical lines on our graph paper--both those that are there and those that could be drawn between the ones that are there. Similarly, all possible horizontal lines tell us about our y-coordinates.

The verticals that tell us about x are lines x = constant. Because x is constant, the points on any one vertical line are at the same level as far as x goes, so we should not be surprised that such lines are called

level lines

of x. What really tells us about x is the sequence of all its level lines. On graph paper we give a dense enough set of level lines for us to behave as if they were all there.

Some coordinates are constant on curves, not lines. They have

level curves

not level lines. To a mathematician, a line is a special case of curve, so he uses "level curve" as the general term. So might we, were we doing mathematics--but we're not. In the common language, a line is straight and a curve is curved, and never the twain shall meet.

We need a word to cover a straight line, a broken line, a curve, etc. We will use

trace

Thus every coordinate has its

level traces.

Not all level traces are straight lines. Not all level lines are parallel. Much surveying is done by measuring the direction of interesting points from each of two base stations. The level traces for the directions from any one point are straight lines radiating from that point. The second strip of exhibit **1** shows how these families of level lines look, alone and together.

Instead of two directions, we could try two **distances,** which will give us circles as level curves. The third strip of exhibit 1 does this for the U.S. and distances from New York and Los Angeles. We would be in a little trouble here, since there are now pairs of points both of whose coordinates are the same. But if we use distances from Sept Iles, Quebec, and Vancouver, British Columbia, as the lowest (fourth) strip shows, we can get away from this difficulty, so long as we know we are staying in the U.S.

This shows us that we can be quite free with what we use for a coordinate, even if we want to think of the points as plotted on a very definite and fixed map. All we have to do is choose two sets of level traces that, together, meet our needs, and we are practically in business.

We do have to choose what values to assign to each level trace, but the difference between two sets of choices is just a matter of re-expression. Each family of level traces by themselves corresponds to a variable that we can name, but not yet both measure and express. A family of level traces, each with a value, corresponds to something we have named, measured, and expressed.

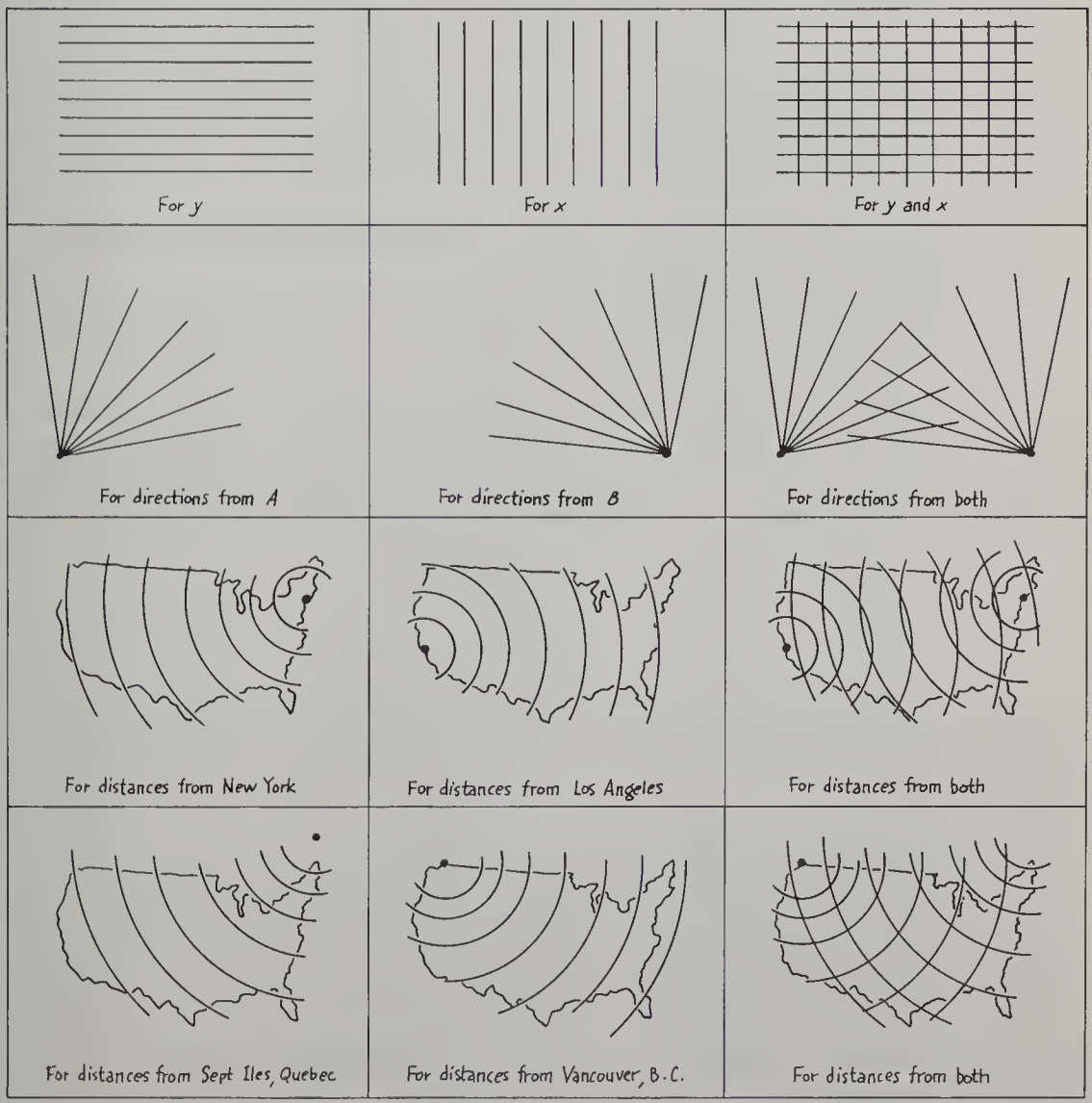
Thus, if we are concerned with the total bank deposits in a state, as we were in chapter 8, the same sets of level lines (or level curves) work for

- ◇bank deposits in dollars.
- ◇the square root of bank deposits in dollars.
- ◇the log of bank deposits in dollars.
- ◇and, of course, bank deposits in cents--or in millions of dollars.

Only the numerical labels on the level lines (or level curves) are different.

exhibit **1** of chapter 14

Level traces (several examples involving one and two sets)



review questions

What tells us about the x-coordinate of a point on graph paper? The y-coordinate? What is a level line? A level curve? A level trace? Why did we choose the term "trace"? Can we use circles as level traces? How many sets? Given a fixed map, how much is our choice of coordinates restricted? How much do we know when we know all about a family of level traces except the value taken on each? What does knowing the values add? What effect does re-expression have on the family of level traces?

14B. Different middle traces for the same slices

It is now time to look at different middle traces of the same point scatter, asking which ones are much alike and which very different.

changing the traced coordinate

The 1960 Census reported on a total of 88 unincorporated places with 25,000 or more inhabitants. We shall look at only two of the 67 columns of information provided in the 1962 *County and City Data Book*:

- ◇the % of occupied housing units with ≥ 1.01 persons per room (column 255, called x or "% crowding" below).
- ◇the % of occupied housing units with one or more automobiles (column 267, called y or "% automobiles" below).

Exhibit **2** sets out the data on slices according to x, using an alternate method, and gives corresponding medians (raw and smoothed), all this for x, for y, for $x + y$, and for $y - x$. It then gives the corresponding y values for the smooths of y, $x + y$, and $y - x$. This allows us to compare the middle traces for:

$$y \text{ against } x$$
$$x + y \text{ against } x$$
$$y - x \text{ against } x$$

which we do in exhibit **3.** We see a very similar picture for each of these three middle traces:

- ◇90% automobiles for very low % of crowding.
- ◇92 to 93% automobiles for just below and at median % crowding.
- ◇a fall in % automobiles for high % of crowding (to about 75% for the highest piece--3 of 88 places).

We need to ask **both** "Why were these traces so much alike?" **and** "Why were they different at all?"

exhibit **2** of chapter 14: unincorporated places

Sliced data and medians for x = % housing units with 1.01 persons per room, for y = % housing units with one or more automobiles, for $x+y$, and for $y-x$, (each data parenthesis contains (x, y, $x+y$, $y-x$), in that order)

A) The DATA

slice		Data	
#, count	range		
(#1, 3)	13 and 14:	(13, 941, 954, 928)	(14, 892, 906, 878)
		(13, 893, 906, 880)	
(#2, 5)	15 to 19:	(17, 813, 830, 796)	(18, 915, 933, 897)
		(19, 930, 949, 911)	(19, 954, 973, 935)
		(15, 904, 919, 889)	
(#3, 10)	20 to 28:	(20, 948, 968, 928)	(24, 892, 916, 868)
		(26, 907, 933, 881)	(25, 947, 972, 922)
		(26, 985, 1011, 959)	(27, 927, 954, 900)
		(20, 938, 958, 918)	(27, 648, 675, 621)
		(28, 720, 748, 692)	(28, 902, 930, 874)
(#4, 17)	29 to 45	(39, 920, 959, 881)	(39, 919, 958, 880)
		(33, 908, 941, 875)	(39, 886, 925, 847)
		(33, 944, 977, 911)	(41, 970, 1011, 929)
		(32, 933, 965, 901)	(42, 964, 1006, 922)
		(41, 894, 935, 853)	(31, 929, 960, 898)
		(31, 947, 978, 916)	(42, 966, 1008, 924)
		(38, 978, 1016, 940)	(29, 971, 1000, 942)
		(44, 872, 916, 828)	(45, 916, 961, 871)
		(34, 931, 965, 897)	
(#5, 19)	47 to 67:	(65, 942, 1007, 879)	(51, 937, 988, 886)
		(52, 884, 936, 832)	(65, 931, 996, 866)
		(61, 875, 936, 814)	(48, 917, 965, 869)
		(62, 848, 910, 786)	(61, 856, 917, 795)
		(62, 899, 961, 837)	(59, 808, 867, 749)
		(61, 912, 973, 851)	(65, 754, 819, 689)
		(54, 974, 1028, 920)	(58, 947, 1005, 889)
		(55, 959, 1011, 904)	(65, 918, 983, 853)
		(55, 944, 999, 889)	(67, 977, 1044, 910)
		(47, 914, 961, 867)	
(#6, 17)	68 to 99:	(83, 937, 1020, 854)	(99, 895, 994, 796)
		(96, 903, 999, 807)	(93, 986, 1079, 893)
		(68, 950, 1018, 882)	(87, 892, 979, 805)
		(72, 932, 1004, 860)	(73, 954, 1027, 881)
		(71, 977, 1048, 906)	(82, 935, 1017, 853)
		(85, 924, 1009, 839)	(75, 894, 969, 819)
		(71, 956, 1027, 885)	(74, 940, 1014, 866)
		(70, 933, 1003, 863)	(90, 933, 1023, 843)
		(78, 933, 1011, 855)	

→

Exhibit **4** shows, at an enlarged scale, the actual points and the four **median lines** for slice #8. On x, the median of these five was New Hanover, New Jersey. On y--and on $y - x$--the median of these 5 was Brownsville, Florida. On $y + x$ the median of these 5 was Bell Gardens, California. (Notice that Bristol, Pennsylvania, and Kailua-Lanikai, Hawaii, escaped being medians for these particular coordinates. It would, however, be easy to find other simple coordinates for which either of them is a median.)

With nothing to make as many as three of the median lines meet in a point, we should not be surprised that no three median lines do this. Accord-

exhibit **2** of chapter 14 (continued)

(#7, 9)	101 to 133:	(107, 867, 974, 760)	(124, 876, 1000, 752)
		(107, 889, 996, 782)	(108, 835, 943, 727)
		(102, 883, 985, 781)	(101, 956, 1057, 855)
		(120, 812, 932, 692)	(133, 977, 1110, 844)
		(114, 902, 1016, 788)	
(#8, 5)	145 to 214:	(214, 873, 1087, 659)	(179, 876, 1055, 697)
		(189, 943, 1132, 754)	(182, 859, 1041, 677)
		(145, 956, 1101, 811)	
(#9, 3)	228 to 238:	(231, 995, 1186, 724)	(228, 746, 974, 518)
		(238, 713, 951, 475)	

B) The MEDIANS

	Raw medians for				3R'SSH3 smooths for				y values from smooth of		
slice	x	y	x + y	y − x	x	y	x + y	y − x	y	y + x	y − x
#1	13	896	906	880	13	893	906	880	893	893	893
#2	18	915	933	897	19	910	928	888	910	909	907
#3	26	917	942	890	27	916	946	888	916	919	915
#4	39	931	965	898	42	917	962	884	917	920	926
#5	61	917	973	866	60	917	976	869	917	916	929
#6	78	935	1014	855	81	908	980	840	908	909	921
#7	108	883	996	781	119	890	990	778	890	871	897
#8	182	876	1087	697	176	845	990	674	845	814	850
#9	231	746	974	518	321	746	974	518	746	743	749

Examples: 893 = 906 − 13, 893 = 880 + 13, 909 = 928 − 19, 909 = 888 + 19.

P) PROBLEMS (largish)

2a) Slice the data in slices of respective sizes 5, 9, 17, 26, 17, 9, 5 and calculate the analog of panel B.

2b) Do the same for slices of 9, 9, 9, 9, 16, 9, 9, 9, 9.

ingly, the point which enters the smoothing process from this slice is not the same for y, $y + x$, and $y - x$, as exhibit **5** emphasizes.

Not the same, but not too different. No three of the four median lines intersect, but all four come reasonably close together. (Indeed, a hard look at panel B of exhibit 2 suggests that slice #8 was chosen as the piece where the failure to come together exactly was largest.)

exhibit **3** of chapter 14: unincorporated places

Middle traces for y, $y + x$, and $y - x$, all against x, plotted in (x, y) coordinates (numbers in exhibit 2)

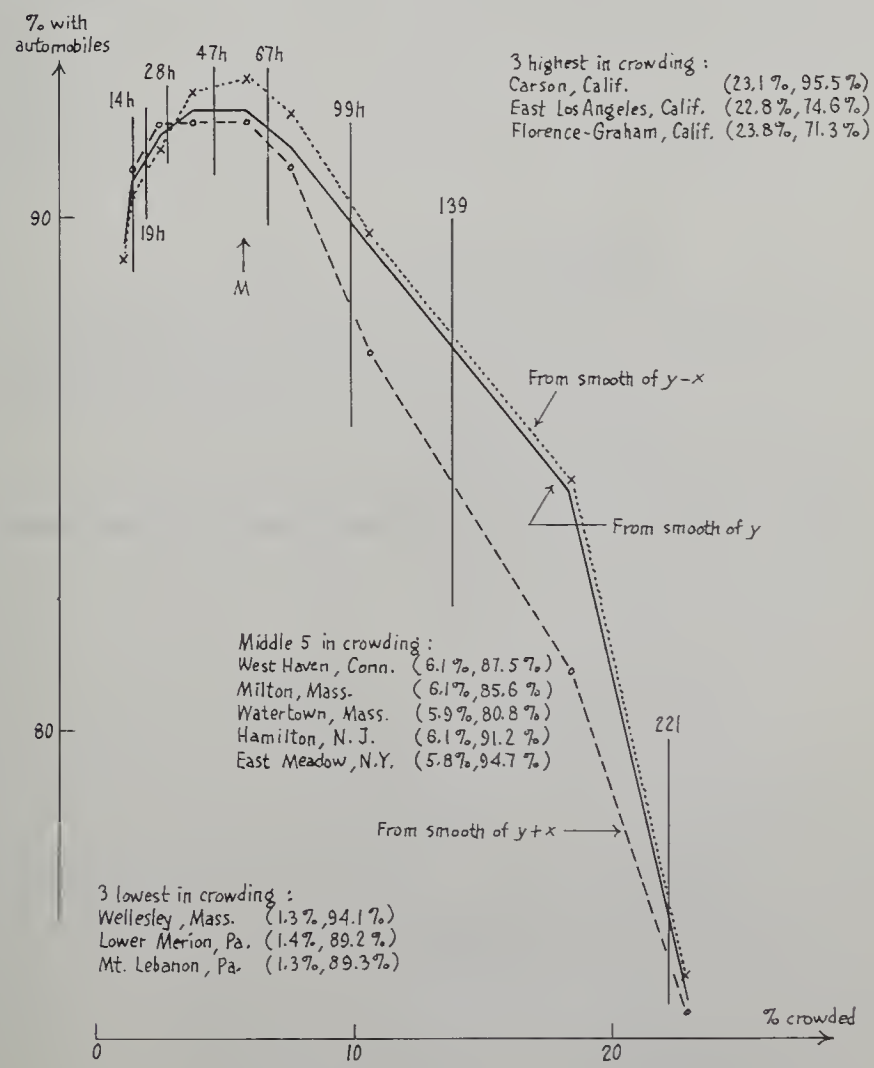


exhibit **4** of chapter 14: unincorporated places

Data points and various medians for slice #8 (numbers in exhibit 2)

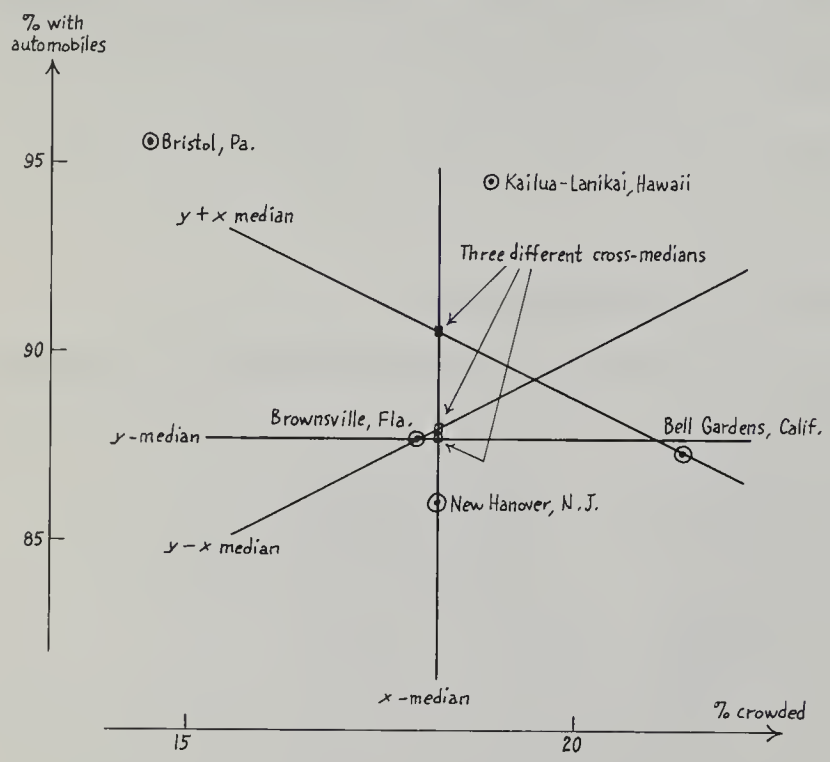
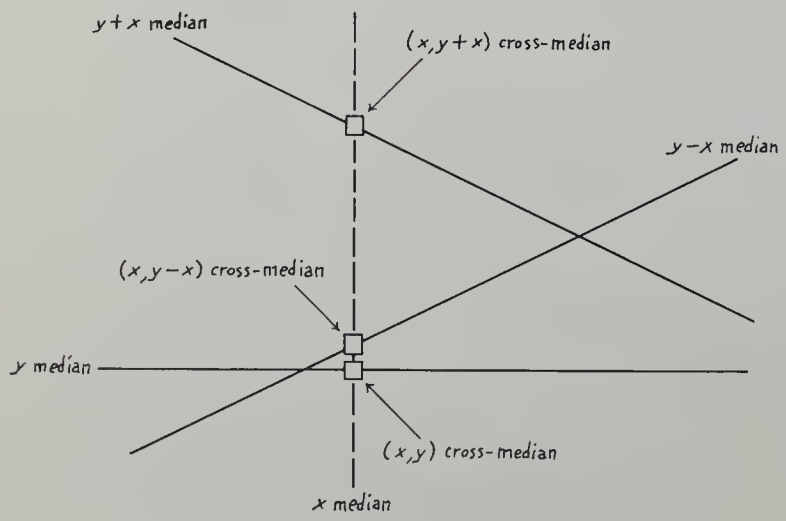


exhibit **5** of chapter 14: unincorporated places

Detail of the center of exhibit 4, showing the three alternate summary points



Thus there is no reason to expect that what we have seen here--"some difference in the middle traces, but not much"--is anything other than what we will usually see. This is fortunate, for it tells us that we will probably get away with showing only

- ◇the middle trace.
- ◇some first cuts.

If the choice of the coordinate traced against the slicing coordinate had really mattered, we would have had to show what it was like also, and this could have been a little too much for a clear picture.

review questions

Can a single point scatter have several middle traces? Why/why not? What example did we choose to illustrate the behavior of several middle traces? What did we take as the slicing variable? How did we slice? How many different tracing variables did we try? How did the three traces behave? What two questions were then important? How did we try to look at what goes on? If we draw three or more median LINES for a point scatter, will they always/never have a point in common? Why/why not? Are they likely to come close to having a point in common? What might bring them closer to this?

14C. An explanation

Why does this happen? If we look at--or think about--a piece with rather more points, the situation will become clearer. Consider slice #5, with 19 points and the following letter-value displays:

#19	x-values		
M10	61		
H5h	63h	54h	11
E3	65	51	14

#19	y-values		
M10	917		
H 5h	943	880	63
E 3	959	848	99

The spread of y is 5 to 7 times as great as that of x. This makes it almost certain that

$$\text{median } (y + x)$$

and

$$\text{median } (y + \text{median } x) = \text{median } y + \text{median } x$$

will be reasonably close to each other, since both (i) the order of the values of $y + x$ will resemble that of the values of y + median x, and (ii) the values of $y + x$ will resemble the values of y + median x.

A similar argument holds for

$$\text{median}\,(y - x)$$

and

$$\text{median}\,(y - \text{median}\ x) = \text{median}\ y - \text{median}\ x.$$

The narrower the slices, the less it can matter whether we take y, $y + x$, or $y - x$ against x. So, also, the larger the batch of (x, y) pairs, the less will it matter.

Some would like these words backed up with a picture. Exhibit **6** shows how the points in a slice shift if we move them along $y + x$ − constant (left panel) or $y - x$ = constant (center and right panels). The righthand panel shows what sort of special pattern it takes to move things "a lot", which is still not large compared to the spread of the y's in the slice.

This picture may also help to remind us that, with luck, such shifts of median (as we change the traced coordinate) will be quite irregular from slice to slice. Thus the smoothing will greatly reduce their effect on the final middle traces.

curved traces

The explanation just given works just about as well if we bring in a halving coordinate whose level traces are not straight lines. All we need is that the slopes of the level curves are moderate everywhere. (Draw yourself an analog of some of the panels of exhibit 6 using level CURVES for halving--and even one set of curves for slicing and another for halving!

Of course this argument, like most approximate arguments, is helpful only over a limited range. If, in our example, we had used $y + 25x$ or $y - 25x$, where $25x$ is now much more variable than y, we could not be as certain of what would happen.

review questions

What explanation did we consider? How well did it work? To what picture did we turn? What did it help us see? Would things be similar for halving traces that are not straight lines?

14D. Changing the slicing coordinate

We are now ready to see what happens when we change the slicing coordinate. How does middling against x compare with middling against y? What about the comparison of both of these with such alternatives as middling against $x + y$ and middling against $x - y$?

Exhibit **7** shows, in four separate panels, middle traces for the unincorporated-places example against $y + x$, x, $y - x$, and y, respectively. The differences appear more striking than the similarities. All four are shown superposed in exhibit **8,** where we can now see a tendency toward concurrence of all four middle traces near a point at about 5 or 6% crowding and 92 or 93% automobiles.

exhibit **6** of chapter 14: purely illustrative

Examples of sliding the points of a slice onto the sliced median

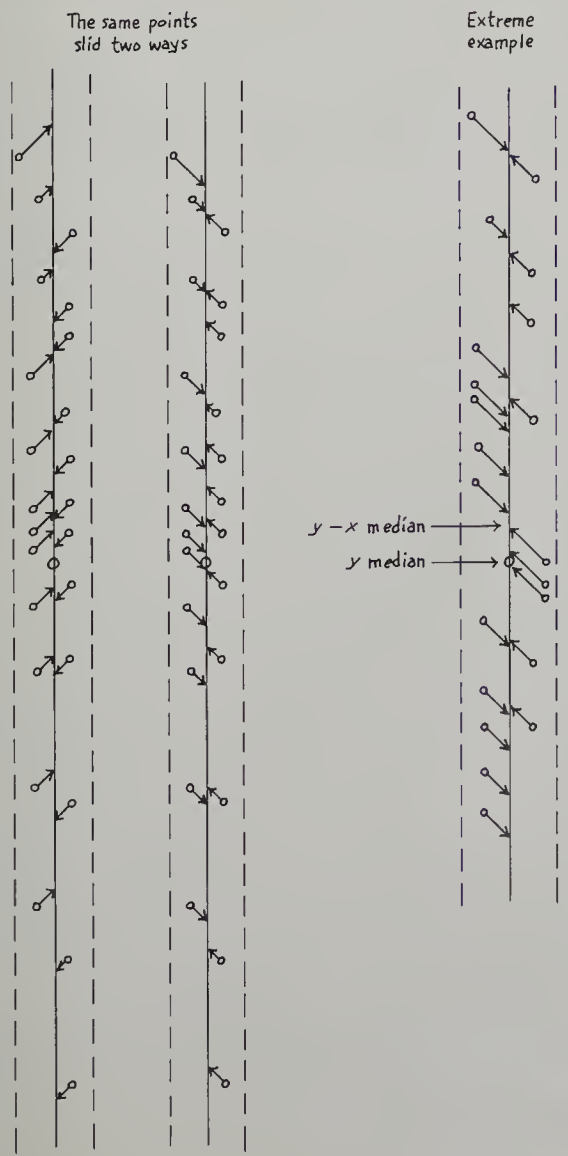
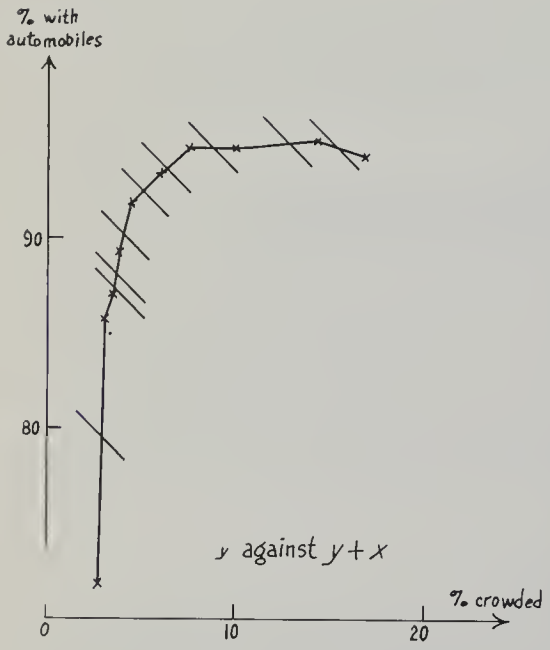
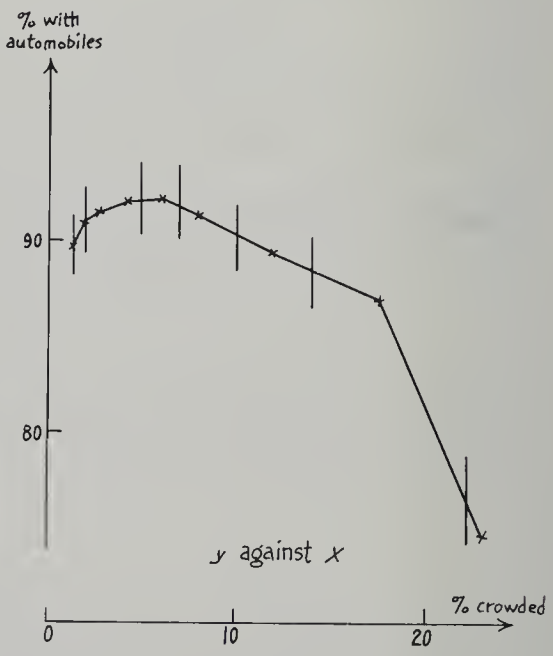
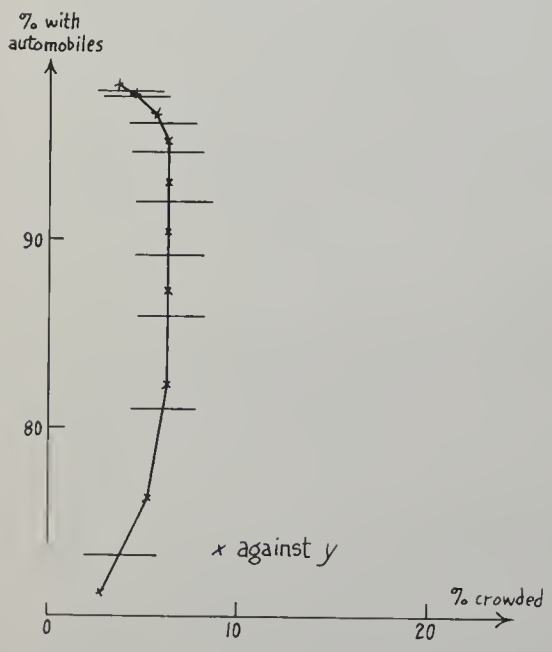


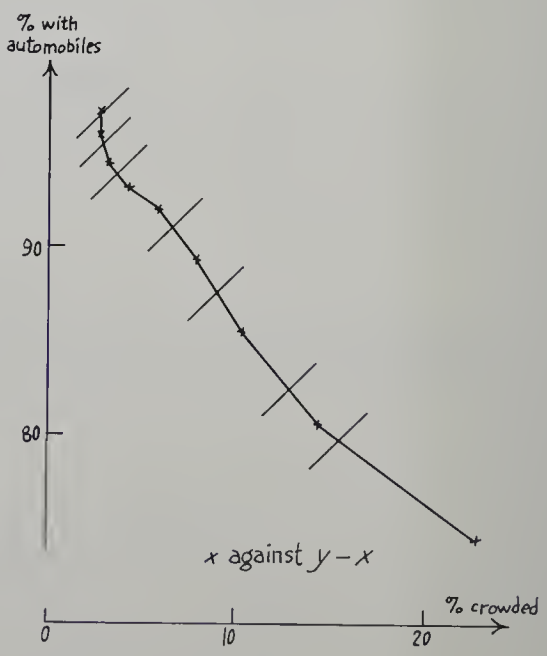
exhibit **7** of chapter 14: unincorporated places

Four different middle traces against $y + x$, x, $y - x$ and y (two "of y" and two "of x")









This concurrence will perhaps make more sense when we look at exhibit **9,** which shows the medians of $y + x$, x, $y - x$, and y, all four of which come close to a point at about the same location. This is a reasonable facsimile of a center for our particular point scatter. If there is an "almost center", it is surely not surprising that all "middle traces" run somewhere near it.

Exhibit **10** presents some problems.

review questions

What happens when we change the slicing coordinate? Is the middle trace little affected? Why/why not? Do we expect near concurrence at a point? Why/why not?

exhibit **8** of chapter 14: unincorporated places

The four middle traces of exhibit 7 superposed

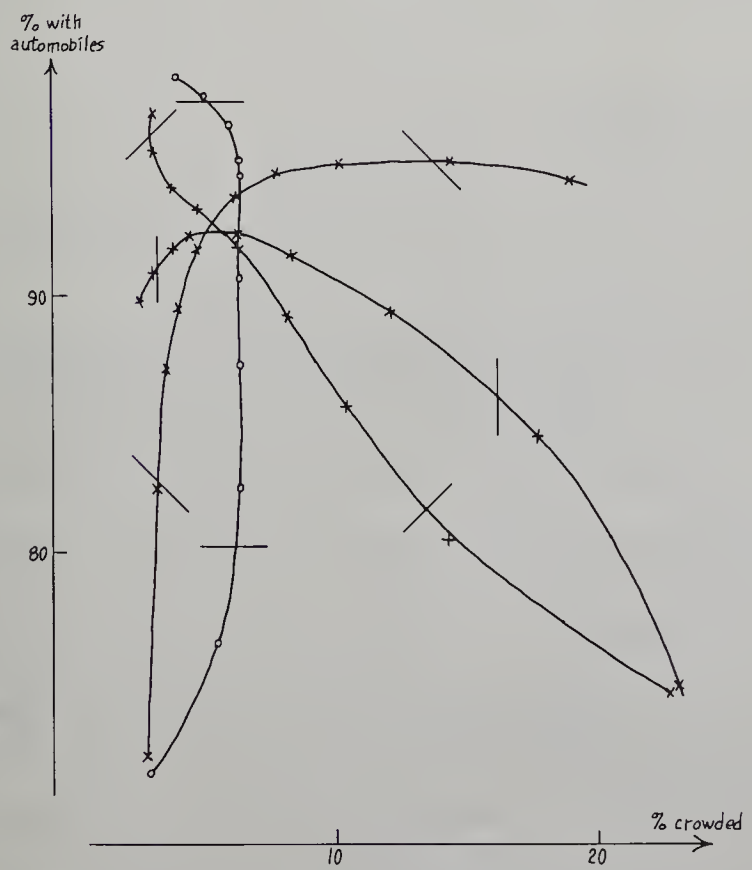


exhibit **9** of chapter 14: unincorporated places

Four medians also nearly meet at a point

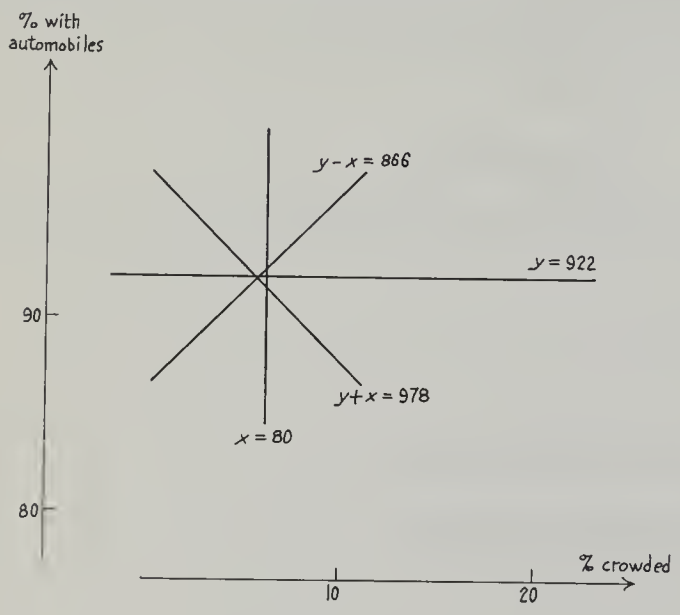


exhibit **10** of chapter 14: problems

Some further problems on changing the sliced coordinate

10a) Take the (BaDep, GoSal) data from section 8F and slice it on (i) BaDep + GoSal, (ii) BaDep − GoSal; then, for each, find the middle traces, middling the unsliced one of these two new coordinates. Plot them.

10a2) Add to the plot the results of slicing on BaDep (exhibit 21 of chapter 8) and on GoSal (problem 21 of chapter 8). How nearly do all four coincide?

10b) Plot the results of (i) slicing the Twin Rivers data on ElUse and middling on GaUse (exhibit 9 of chapter 8 and (ii) slicing on GaUse and middling on ElUse.

10c) Slice the Twin Rivers data on each of GaUse + ElUse and GaUse − ElUse, middling on the other.

10bc) Combine all four middlings on the same picture. How nearly do all four coincide?

10e) Do separate medians on BaDep, GoSal, BaDep + GoSal, BaDep − GoSal for the governor's salary data (exhibit 21 of chapter 8). Plot the four lines. How nearly do they coincide?

10e2) Do the same also for BaDep ± 1/2 GoSal. Plot all six lines. How closely do they coincide?

10f) Do the same for GaUse, ElUse, GaUse + ElUse, GaUse − ElUse for the Twin Rivers data (exhibit 20 of chapter 8).

14E. What matters?

It is not hard to see that, given any point, there is a coordinate of the form

$$cx + dy$$

for some c and d not both zero, or, if you prefer, of the form

$$x \cos \theta - y \sin \theta$$

for some θ, such that the corresponding median goes through the given point. (Those who wish to look at a not-quite-easy proof may refer to section 14I.) It is not clear that, given a reasonable point, there is a middle trace that goes exactly through that point, but it is clear that there will be one that goes close.

If we use our first cuts to fit straight lines, instead of finding middle traces, we will find the same thing: Given a point, there will be a family of first cuts whose fitted line goes close to the given point.

some morals

We have now learned that:

◇it does NOT matter very much if we change the traced coordinate--the coordinate whose middle trace we are looking at--so long as (i) we keep the set of first cuts the same, and (ii) we stick to reasonable combinations of the coordinate we started with.

◇it matters VERY MUCH indeed which coordinate we take a middle trace against--take as the slicing coordinate; by taking the right first cuts, we can make the middle trace come close to any pre-chosen point.

◇similar results hold for fitted lines.

The morals follow:

◇**the important thing to know about a middle trace is the family of first cuts that lead to it. Which coordinate was traced against this important slicing coordinate usually matters relatively little.**

◇thus we should continue to accompany each middle trace with a sufficient collection of first cuts to show us what it came from.

◇**the same thing is likely to be true for any fitted line.**

It's an old saying that "it's baloney, no matter how you slice it!", but we have now learned that "How you slice the point scatter can make all the difference in the world to your middle trace--or your fitted line!".

review questions

Given a point scatter, can you find a new point (anywhere in the plane) such that no median line goes through that point? Why/why not? Which is more important to know about a median trace--the family of slicing cuts or the family of tracing cuts? Why? What family of traces thus best supports a middle trace?

14F. Rematching and strength of relationship

Suppose, in the Twin Rivers example (exhibit 1 to 19 of chapter 8), that we had been given--not the 152 (ElUse, GaUse) pairs--but only the two stem-and-leaf displays, one for each variable. We would not have known about the actual relationship, but we could have decided on what the strongest relationship--in either direction--could have been that would be consistent with these two stem-and leafs, one for 152 electric usages, the other for 152 gas usages.

exhibit **11** of chapter 14: energy usages

Some numbers for ElUse and GaUse separately (based on exhibit 1 of chapter 8)

A) MEDIANS of LETTER SEGMENTS

Depth*		Tag		ElUse		
Point	Segment	Point	Segment	Value	Median	(**)
	(1)		1		89	(424)
1		1		89		
	(2)		1A		108	(388)
2		A		108		
	(3)		aB		119	(778)
3		B		119		
	(5)		bC		134	(832)
5–6		C		134		
	(8)		CD		152h	(828)
10		D		156		
	(15)		dE		170	(766)
19h		E		177h		
	(29)		eH		192	(958)
38h		H		207		
	(58)		hm		235	(922)
76h		M		253		
	(57h)		mh		267h	(891)
38h		H		283h		
	(29)		He		292	(942)
19h		E		301h		
	(15)		Ed		316	(924)
10		D		333		
	(8)		Dc		347	(946)
5h		C		359h		
	(4h)		Cb		369	(925)
3		B		422		
	(3)		Ba		422	(1222)
2		A		429		
	(2)		A1		429	(1002)
1		1		435		
	(1)		1		435	(860)

→

Clearly, the **strongest positive relation** would have come about when the highest gas usage was paired with the highest electrical usage, the 2nd highest with the 2nd highest, and so on to the lowest with the lowest. Equally, the **strongest negative relationship** would have paired the highest with the lowest, the 2nd highest with the 2nd lowest, and so on, to the lowest with the highest. The corresponding middle traces would have their ends as extreme as possible.

The neutral relationship--where knowing about one set of values offers no clue about the other--would correspond to a horizontal middle trace, one located at the median of the quantity taken as a response.

exhibit **11** of chapter 14 (continued)

Depth*		Tag		GaUse		
Point	**Segment**	**Point**	**Segment**	**Value**	**Median**	**(**)**
	(1)		1		388	(108)
1		1		388		
	(2)		1A		424	(89)
2		A		424		
	(3)		aB		530	(175)
3		B		530		
	(4h)		bc		576	(198)
5h		C		597		
	(8)		cD		612	(211)
10		D		658		
	(15)		dE		684	(247)
19h		E		726		
	(29)		eH		762	(266)
38h		H		789		
	(57h)		hm		839	(255)
76h		M		910		
	(57h)		mh		963	(263h)
38h		H		1016		
	(29)		He		1058	(252)
19h		E		1068		
	(15)		Ed		1096	(272)
10		D		1144		
	(8)		De		1150	(231)
5h		C		1171		
	(4h)		Cb		1185	(202)
3		B		1214		
	(3)		Ba		1214	(299)
2		A		1222		
	(2)		A1		1222	(422)
1		1		1298		
	(1)		1		1298	(196)

*** Ties are shown in the "point" column where meaningful, for example 5–6 at upper left, and allowed for in the segment column.**

**** Values in parenthesis are medians--for the other energy usage--for that segment of townhouses.**

→

When we ask how much relationship there is between the two values, we can be asking one of two quite different questions (see opener of chapter 5 for "response" and circumstance"):

- ◇ how fast does the response shift as the circumstance shifts? (A question to be answered by the tilt of a fitted line or by the, probably changing, slope of a middle trace, or by such displays as those to come next.)
- ◇ how much of the possible relationship is present? (A question partly answered by the relation of the actual middle trace to what might have happened, given the values of each variable separately--without their matching. Displays that show this more clearly will follow.)

Exhibit **11** has some arithmetic details. Exhibit **12** shows the actual GaUse middle trace (using a 3R smooth) against a background of three possible middle traces, most positive, neutral, and most negative. Exhibit **13** shows the same picture for the actual ElUse middle trace. To make the two exhibits more easily comparable in exhibit 13, we have preserved our original choice--ElUse as horizontal--and GaUse as vertical--though it would often seem more natural to interchange them, since we are now cutting on GaUse and halving on ElUse.

We see in both exhibits, actual close to most positive on the extreme lower left, but actual closer to neutral elsewhere.

dependency ratios

A natural way to take things further is to look at the ratios

$$\frac{\text{actual MINUS median}}{\text{most positive MINUS median}}$$

exhibit **11** of chapter 14 (continued)

P) PROBLEMS

11a/b Imitate panel A for upper/lower hinges instead of medians.

11a2/b2) Make a plot similar to exhibit 12 for upper/lower hinge traces instead of median traces.

11a3/b3) Make a plot similar to exhibit 13 for upper/lower hinge traces instead of median traces.

11a4/b4) Discuss the relationship of what you have just done to exhibits 12 and 13--and to itself.

which compares actual amount up with largest possible amount up (or, if this is negative, at

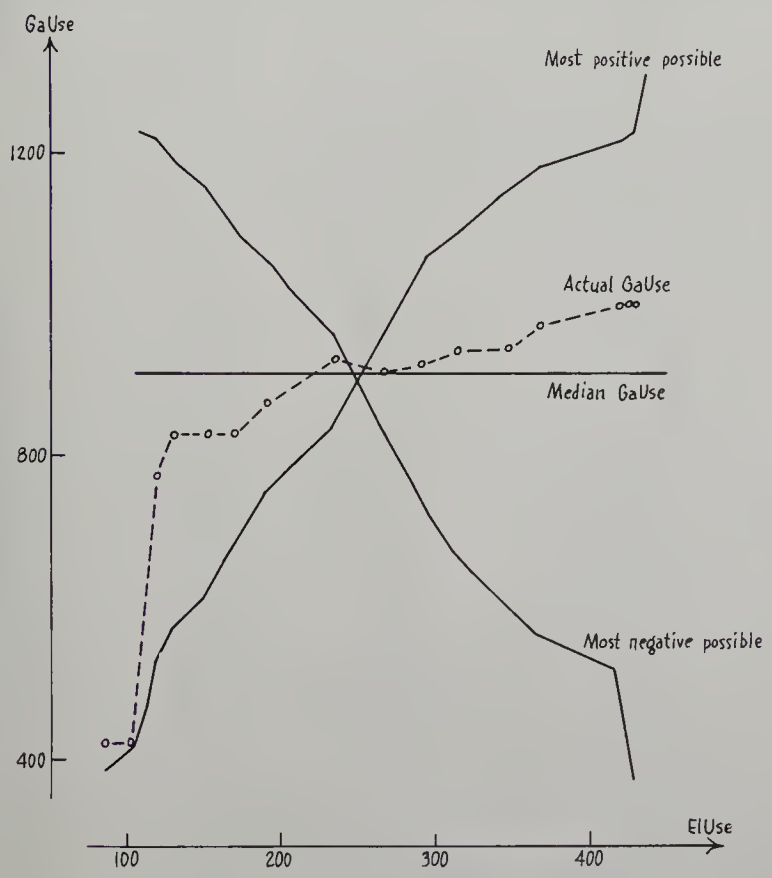
$$\frac{\text{actual MINUS median}}{\text{median MINUS most negative}},$$

which compares actual down with largest possible down, instead).

At a point where this ratio is close to +1, the actual middle trace is close to the most positive trace and the relationship is strongly positive. Where it is

exhibit **12** of chapter 14: energy usages

Actual and three potential middle traces for GaUse against ElUse (from exhibit 11)



close to −1, the relation is strongly negative. Where it is close to zero, the actual trace is close to the median and there is little sign of dependence. Hence it is natural to call each individual value a

dependency ratio,

and to call a trace reflecting its several values a

dependency trace.

Exhibit **14** finds these ratios, and then smooths them with 3R. Panel B gives the median ratios (omitting the extreme values found in the first two lines), both unweighted and weighted. We see that, if we weight, the apparent relationship is quite feeble (possibly zero) except for those two first slices.

exhibit **13** of chapter 14: energy usages

Actual and three potential middle traces for ElUse against GaUse from exhibit 11 (coordinates as in exhibit 12 rather than as natural)

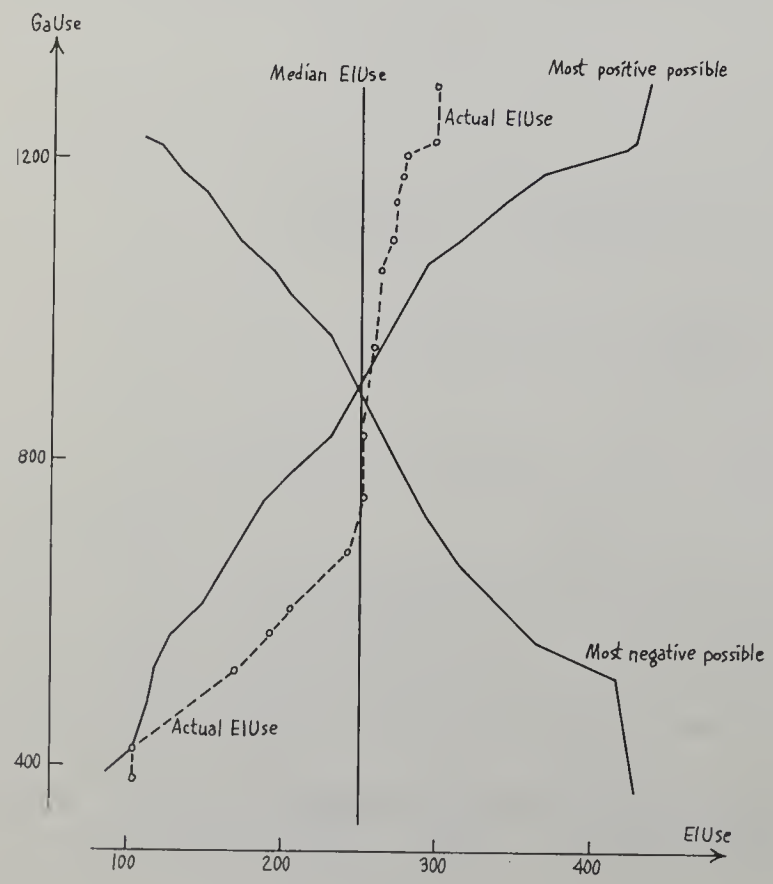


exhibit **14** of chapter 14: energy usages

Finding and smoothing dependency ratios, and making reconstituted traces (from exhibit 11)

A) CALCULATION and SMOOTHING of RATIOS

for EIUse against GaUse

	value − median		Ratio = Act/Pot		
slice	**Potent.**	**Actual**	**Raw**	**3R**	(wts)
1	−164	−145	.88	.88	(1)
1A	−145	−164	1.13	.88	(1)
AB	−134	−78	.58	.58	(1)
BC	−119	−55	.46	.46	(2)
CD	−101	−42	.42	.42	(5)
DE	−83	−6	.07	.07	(9)
EH	−61(39)	13	−.22(−.33)	−.19(−.15)	(19)
H1	−18(13)	2	−.19(−.15)	−.19(−.15)	(38)
MH	13	10h	.81	−.03	(38)
HE	39(−61)	−1	−.28(−.02)	.23	(19)
ED	63	19	.30	.23	(9)
DC	94	−22	.23	.23	(5)
CB	116(−119)	−51	−44(−.43)	.23	(2)
BA	169	46	.27	.27	(1)
A1	176	169	.96	.27	(1)
1	182(−164)	−57	−31(−.35)	.27	(1)

for GaUse against EIUse

	value − median		Ratio = Act/Pot		
slice	**Potent.**	**Actual**	**Raw**	**3R**	(wts)
1	−522	−486	.93	.93	(1)
1A	−486	−522	1.07	.93	(1)
AB	−380	−132	.35	.35	(1)
BC	−334	−78	.23	.31	(3)
CD	−298	−82	.28	.31	(4)
DE	−226	−144	.64	.28	(10)
EH	−148(148)	48	−.32(−.32)	−.17(−.23)	(19)
HM	−71(53)	12	−.17(−.23)	−.17(−.23)	(38)
MH	53(−153)	−19	−.36(−.36)	−.17(−.23)	(37)
HE	148	32	.22	.08	(11)
ED	186	14	.08	.08	(10)
DC	240	36	.15	.15	(5)
CR	275	15	.05	.15	(2)
BA	304	312	1.03	.29	(1)
A1	312	92	29	29	(1)
1	388(522)	−50	−13(−.10)	29	(1)

Examples: −164 = 89 − 253; −145 = 108 − 253; .88 = (−145)/(−164); −522 = 388 − 910; −486 = 424 − 910; .93 = (−486)/(−522); −145 = 108 − 253; −164 = 89 − 253; 1.13 =(−164)/(−145).

Note: Values in parenthesis from−−or figured from−−other extreme.

→

Exhibits **15** and **16** plot the resulting

dependency traces,

which do a good job of answering the second question above.

exhibit **14** of chapter 14 (continued)

B) MEDIAN RATIOS--omitting two first lines

	ElUse Ratio (smoothed)	GaUse Ratio (smoothed)
Raw	.23	.22
Wtd.	−.02	.17

C) RECONSTITUTED TRACES

	ElUse against GaUse Smoothed value − median				GaUse against ElUse Smoothed value − median			
slice	**ratio**	**Potent.**	**Recon.**	Value recon.	**ratio**	**Potent**	**Recon.**	Value recon.
1	.88	−164	−144	109	.93	−522	−485	425
1A	.88	−145	−128	125	.93	−486	−452	458
aB	.58	−134	−78	175	.35	−380	−133	777
bC	.46	−119	−55	198	.31	−334	−104	806
cD	.42	−101	−42	211	.31	−298	−92	818
dE	.07	−83	−6	247	.28	−226	−63	847
eH	−.15	(39)	−6	247	−.17	−148	25	935
hm	−.15	(13)	0	253	−.17	−71	12	922
mh	−.03	(13)	0	253	−.17	53	−9	901
He	.23	39	9	262	.08	148	12	922
Ed	.23	63	14	267	.08	186	15	925
Dc	.23	94	22	275	.15	240	36	946
Cb	.23	116	27	280	.15	275	41	951
Ba	.27	169	46	299	.29	304	88	998
A1	.27	176	48	301	.29	312	90	1000
1	.27	182	49	302	.29	388	113	1023

P) PROBLEMS

14a/b/c/d) Make calculations analogous to panels A, B, C for the data of (21a/b/c/d), respectively, in chapter 8.

14a2/b2/c2/d2) Make a plot like exhibit 15 for this data.

14a3/b3/c3/d3) Make a plot like exhibit 16 for this data.

14e) Make a plot of the two reconstituted traces in panel C in comparison with most positive and most negative traces.

14f) Smooth the reconstituted traces, and then repeat the ratioing and smoothing processes. Compare the result with panel A. Discuss the changes.

exhibit **15** of chapter 14: energy usage

Dependency ratio for GaUse against ElUse (from exhibit 14, 3R smoothed)

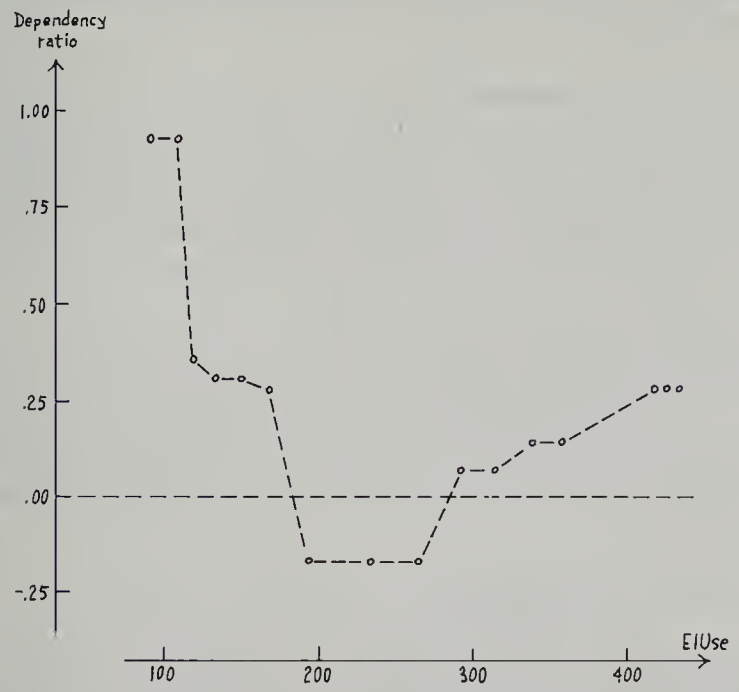
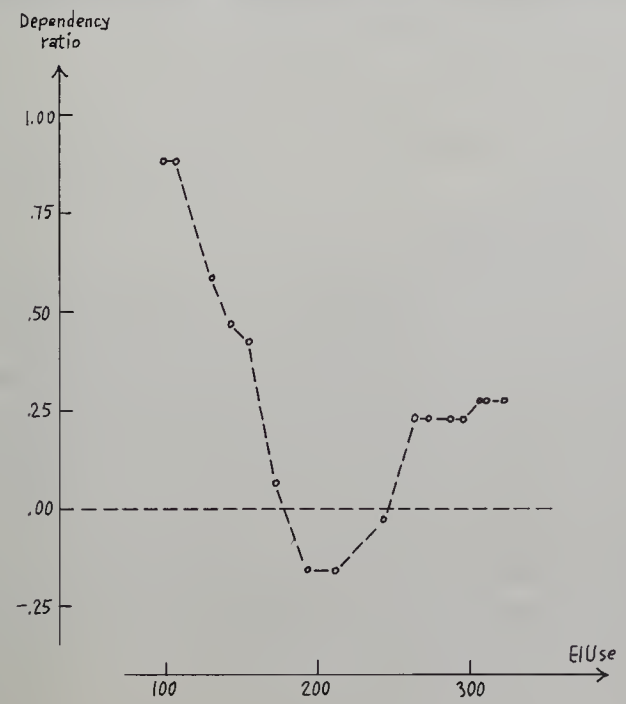


exhibit **16** of chapter 14: energy usages

Dependency ratio for ElUse against GaUse (from exhibit 14)



reconstituted traces

Panel C (of exhibit 14) illustrates the reconstitution of traces by multiplying the smoothed dependency ratios by the potential differences. This may sometimes be useful (see also problem (14f).)

a comment

If we had not already seen, in Section 14D, how greatly the choice of the slicing coordinate affects both the middle trace and the fitted line, we would have **had** to stress here **the difference between *y* middled against *x* and *x* middled against *y*.** The point is so important, however, that we are going to stress it anyway.

In most cases, we expect to find the ends of the

actual middle

"somewhere between"--either between the most positive trace and the horizontal median OR between the horizontal median and the most negative trace. Similarly, we expect the other

actual middle

found by interchanging variables to also lie "somewhere between"--but not in the SAME somewhere between. Rather, with the axes preserved, as in exhibit 13, we expect it either between the most positive trace and the vertical median or between the vertical median and the most negative trace.

Thus we expect the two middle traces to have different tilts, the former less than the tilt of the most positive possible, the latter greater. Whether we fit lines or smooth middle traces, we always expect two different tilts, etc. If we are dealing with (Adata, Bdata) pairs we expect TWO tilts:

- one for Adata against Bdata.
- the other for Bdata against Adata.

review questions

What could we do with our *x*'s and *y*'s if we did not know which *x* went with which *y*? What does rematching mean? What do we have to know? What do we do? How many potential middle traces do we find? What are they? Can we use such results as a background? Why/why not? What might "how much relationship" mean? What two questions have we emphasized? How do they differ? Have we dealt with one meaning already? Why/why not ? Can we compare actual relationship with possible relationship? Graphically? Numerically? What number or numbers might be a partial substitute? What is a dependency ratio? Can its value be greater than 1? Why/why not? Less than −1? Why/why not? Against what do we usually plot dependency ratios? How

would we interpret a dependency ratio of −0.97? Of +0.82? Of −0.05? Of +0.32? How did the dependency ratio behave in our example? What is a reconstituted trace? Do we expect "y middled against x" and "x middled against x" to be quite similar? Why/why not? Will they ever cross? Why/why not? Where do we expect to find the ends of middle traces? What if we look at lines, instead, for "y fitted to x" and "x fitted to y"? Will they agree with each other? Why/why not? Will they be parallel? Why/why not? Will they cross? Why/why not?

14H. How far have we come?

In this chapter we have looked at any (x, y) batch in two or more ways--changing either the slicing variable or the tracing variable used to lead us to a middle trace--or swapping x and y in fitting lines. We have also compared middle traces with what the actual values of x and of y, taken separately, could provide.

We are now ready:

- ◇ to insist on supporting our middle traces with adequate indications of what cuts were used to make the slices.
- ◇ to rematch x and y values, finding the most positive and most negative relationships that could have happened, given the x's and given the y's.
- ◇ to assess the strength of relationship actually present by comparison with these extremes, either graphically or numerically.

We now understand more clearly:

- ◇ what a coordinate can be.
- ◇ what to expect when we keep the slicing variable fixed and change the tracing variable.
- ◇ what to expect when we do the converse.
- ◇ why the slicing variable is MORE important, whether we are finding middle traces or fitting straight lines.
- ◇ that the straight lines for "y fitted to x" and "x fitted to y" are expected to cross and often to be quite different.
- ◇ that the same is true for the traces: "y middled on x" and "x middled on y".

14I. The ubiquity of medians (optional section)

Medians are rather different for batches of odd count or even count. We will want to take these cases separately.

odd count

Suppose we take some arbitrary point (x_{arb}, y_{arb}) and a batch of n points (x_1, y_1), (x_2, y_2), . . . , (x_n, x_n), where n is an odd number. Can we show that there is a coordinate of the simple form

$$z = cx + dy$$

with c, d not both zero, whose median passes through (x_{arb}, y_{arb})?

Note, first, that the problem depends ONLY on the directions of the points (x_i, y_i) from (x_{arb}, y_{arb}). For whether $z = cx_{arb} + dy_{arb}$ is a median of an **odd** number of points or not depends only on on what side of this line, which goes through (x_{arb}, y_{arb}), the various points lie. So we may as well draw a circle around (x_{arb}, y_{arb})--any circle with this center will do--and project the n points on the circle.

This gives us a simpler-seeming problem: Given n points on a circle, is there a diameter that halves them? This is easy to deal with if we work with directed diameters, with diameters with an arrow--something to give them left and right sides.

Once we do this we can write down:

$$\text{discrepancy} = \begin{pmatrix}\text{number of points}\\ \text{on left}\end{pmatrix} \text{ MINUS } \begin{pmatrix}\text{number of points}\\ \text{on right}\end{pmatrix}$$

for any directed diameter. Several properties of this are easy to see, including:

◇if we turn a directed diameter through 180 degrees, the new discrepancy is MINUS the original (we have interchanged left and right!).

◇if we turn the directed diameter smoothly, its discrepancy does not change unless one or both ends of the diameter hit one of the n points.

◇accordingly, discrepancies are constant on not more than n intervals, with a point of change linking each interval to the next.

◇if the discrepancy changes sign as we pass through some direction--some change point--that direction is a median. (The count of (x_i, y_i) points strictly to the left of the directed diameter in that direction is the lesser of the counts for the adjacent segments and is thus strictly less than $n/2$. The same holds on the right. There must be enough points on this directed diameter to make the count "left or on" $\geq n/2$. The same has to be true for "right or on" and the result is proved.)

◇a segment of constant discrepancy cannot have discrepancy zero. (Some direction in such a segment goes through NO point. With an odd number of points, the discrepancy must now itself be an odd number.)

These facts are enough to settle matters: There are segments of constancy. None have discrepancy zero. For a segment of one sign there is an exactly opposite segment of the other sign. Somewhere a PLUS segment must touch a MINUS segment. The linking point fixes a direction that is a median.

even count

The case of an even number of points is a little more difficult and is left to the reader as a teaser. (*Hint.* Don't try to move the points to a circle. Define for each direction, as shown by its diameter:

$$\text{lateral discrepancy} = \begin{pmatrix}\text{smallest PLUS devi-}\\ \text{ation (from diameter)}\end{pmatrix} - \begin{pmatrix}\text{least MINUS devi-}\\ \text{ation (from diameter)}\end{pmatrix}.$$

You CAN go on to show that there is still a median through the arbitrary point.

review questions

What question concerned us? How could we reduce it? To what form? To what indicating quantity did we then turn? What properties did this have? How did they combine to settle the question?

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Counted Fractions 15

We need to deal with counted fractions, like the 17/23 of 23 marbles that were blue and the 244/59,085,182 of 59,085,182 U.S. 1958 incomes that were over a million dollars. The second example shows a cutting value--we will shorten this to cut--a value that divides "below"--in our example up to one million dollars, from "above"--in our example from one million and one dollars up. We need to be concerned with counted fractions both for their own sake--we may want to point out values, compare, or otherwise analyze counted fractions--and as an essential element in describing how something is distributed.

In very many instances it is natural to have a sequence of counted fractions referring to the same information. If we can count incomes over a million, we might have counted incomes (i) over two million, (ii) over half a million, (iii) over each of a fairly long list of cuts. Something similar can happen for the 23 marbles; there may be a sequence of cuts "above" which fall:

20 marbles that are at least bluish,

17 marbles that are "blue",

13 marbles that are "very blue".

We cannot be surprised by any observed counted fraction appearing as one of a sequence of counted fractions, defined by a sequence of cutting values. (If we know the fraction of 1958 incomes above, say, each of 24 well-chosen cuts, we know a lot about how 1958 income was distributed.)

Whether we deal with counted fractions separately or as part of a sequence of such, it pays us to face up to two considerations:

- there are likely to be observed values that are "on the fence"--that EXACTLY equal a cutting value.
- where we have observed "none", it may be reasonable that further repeated observations may find an occasional instance, so that using "zero" for the count seems too drastic.

Having dealt with these two considerations, it is natural to go on and be sure that we are treating the fraction "in" and the fraction "out"--often the fraction "below" and the fraction "above"--symmetrically.

We can extend all these ideas to looking at cuts on two different variables --say, sex and occupation--at the same time.

review questions

What is a counted fraction? Do counted fractions always come alone? Why/why not? What two considerations have we to face? Should we treat the fractions "in" and "out" differently? What about cuts on two different bases at once?

15A. Started counts and counted fractions

Suppose we have 27 observations; what fraction do we assign to:

- the next-to-lowest observation?
- a cutting value such that two of the observations are below it and none equal to it?
- a cutting value such that two of the observations are below it and three equal to it?
- a cutting value so low that NO observation is below it?

These are only a few of the possibilities with which we must deal.

It is a long-used statistical practice to treat any observed value exactly equal to a cutting value as if it were half below and half above. We shall continue this tradition. Thus we will treat "two below and three equal to" as $2 + \frac{1}{2}(3) = 3.5$ below. We refer to such a value as 3.5 as a

split count

when a specific term is needed.

The desirability of treating "none seen below" as something other than zero is less clear, but also important. Here practice has varied, and a number of different usages exist, some with excuses for their being and others without. The one we here recommend does have an excuse, but since this excuse (i) is indirect and (ii) involves more sophisticated considerations, we shall say no more about it. What we recommend is adding 1/6 to all split counts, thus "starting" them. We would use "zero" only for the "count" below a cutting value that was such that NO value could be below it (and even one equal to it is essentially out of the question). Since we are **starting** a **split** count when we use

(the count below) PLUS (1/2 the count equal) PLUS 1/6

or

(the count above) PLUS (1/2 the count equal) PLUS 1/6,

it is not unreasonable to distinguish either of these as an

ss-count.

ss-fractions

Notice that

ss-count below PLUS ss-count above = total count PLUS 1/3

no matter where the cut falls. Thus, if we define

$$\text{ss-fraction below} = \frac{\text{ss-count below}}{\text{ss-count below PLUS ss-count above}},$$

we can write

$$\text{ss-fraction below} = \frac{\text{ss-count below}}{\text{total count PLUS 1/3}},$$

which, when we put in all the details for "ss-count below", can be written

$$\text{ss-fraction below} = \frac{\text{count below PLUS 1/2 count equal PLUS 1/6}}{\text{total count PLUS 1/3}}.$$

Exhibit **1** gives some examples and problems for the use of ss-fractions.

Whenever we say "fraction" in the rest of this chapter, we will mean "ss-fraction". (When counts are large, we can, of course, neglect the starts—and when there are no observations exactly equal to our cutting values, we shall not need to split any counts.)

review questions

What is the simplest way to count ordered observations? And the empty segments between them? Why may this not be a good choice? What is another way? What are some of its difficulties? What is a reasonable compromise? What is an "ss-count"? An "ss-fraction"?

exhibit **1** of chapter 15: examples and problems

The use of ss-fractions

A) EXAMPLES

observations	cut	# below	# equal	ss-count	total # PLUS $\frac{1}{3}$	ss-fraction
0, 1, 2, 2, 3, 7	$\sqrt{2}$	2	0	$2\frac{1}{6}$	$6\frac{1}{3}$	.342
0, 1, 2, 2, 3, 7	2	2	2	$3\frac{1}{6}$	$6\frac{1}{3}$	.500
0, 1, 2, 2, 3, 7	3	4	1	$4\frac{2}{3}$	$6\frac{1}{3}$	.737
0, 1, 2, 2, 3, 7	4	5	0	$5\frac{1}{6}$	$6\frac{1}{3}$	.816
.123, .456, .654, .789	.2	1	0	$1\frac{1}{6}$	$4\frac{1}{3}$	.269
.123, .456, .654, .789	.5	2	0	$2\frac{1}{6}$	$4\frac{1}{3}$	.500
.123, .456, .654, .789	.654	2	1	$2\frac{2}{3}$	$4\frac{1}{3}$	.615

→

15B. Three matched scales for counted fractions

Now we have "started" our counted fractions, how shall we express them?

Treating "those who are" and "those who aren't" symmetrically is often useful and important. This means two things:

- ◇choosing scales with a value of 0 at 50%;
- ◇arranging matters so that a swap of f with $1 - f$, where f is the fraction concerned, will change the sign, but not the size, of our expression of f.

While we are at it, since $f = \frac{1}{2}$ is to be scored as 0, we may as well plan to have a variety of choices that agree, not only exactly for $f = \frac{1}{2}$ *but quite closely for values around* $f = \frac{1}{2}$. This is a matter of convenience. It also allows ever-repeated emphasis on how similar all the expressions are "in the middle"--on how little it matters which we choose if all our fractions are near $\frac{1}{2}$. (If this is mysterious enough to bother you, turn ahead to exhibit 2, where three modes of expression are matched near 50%, though they differ widely at extreme %'s.)

Such matching costs us nothing, because whether we analyze a given set of numbers or a set of numbers each exactly three times as large as those given is almost certain to make no essential difference: When complete, the results can, in any case, be translated to a common, generally understandable scale. Trivial re-expressions are indeed trivial.

With these guideposts, what can we do that will be simple and effective? The simplest measure that re-expresses $f = \frac{1}{2}$ as 0 is the "plurality"

$$(f) - (1 - f)$$

which is the difference between the fraction of those that are and the fraction of those that aren't. If we take f and $1 - f$ in %, we call the result "folded percents", abbreviated "fcents".

exhibit **1** of chapter 15 (continued)

P) PROBLEMS

	observations	cut	ss-fraction
1a)	13, 27, 29, 35, 47, 53	29	?
1b)	7, 9, 9, 9, 11, 15, 25	9	?
1c)	13, 27, 29, 35, 47, 53	34	?
1d)	7, 9, 9, 9, 11, 15, 25	5	?
1e)	13, 27, 29, 35, 47, 53	204	?

1f) Out of 323 earthquakes, only 3 were stronger than a given cut-off. If none were **at** the cut-off, what is the ss-fraction "to the right" of the cut-off?

1g) As (1f) but with 2 **at** the cut-off.

1h) As (1f) but with only one beyond.

1i) As (1f) but with only one beyond and 2 **at** the cutoff.

exhibit **2** of chapter 15: reference table

Pluralities, folded roots, folded logarithms--alternative expressions for counted fractions (take sign of answer from head of column giving %)

A) MAIN TABLE

+	|Plur.|	|froot|	|flog|	−	+	|Plur.|	|froot|	|flog|	−
50%	use	.00	use	**50%**	**85%**	.70	.76	.87	**15%**
51	→	.02	←	**49**	**86**	.72	.78	.91	**14**
52		.04		**48**	**87**	.74	.81	.95	**13**
53		.06		**47**	**88**	.76	.84	1.00	**12**
54		.08		**46**	**89**	.78	.87	1.05	**11**
55%	use	.10	use	**45%**	**90.0%**	.80	.89	1.10	**10.0%**
56	→	.12	←	**44**	**90.5**	.81	.91	1.13	**9.5***
57		.14		**43**	**91**	.82	.92	1.16	**9**
58		.16		**42**	**91.5**	.83	.94	1.19	**8.5**
59		.18		**41**	**92**	.84	.96	1.22	**8**
60%	use	.20	use	**40%**	**92.5%**	.85	.97	1.26	**7.5%**
61	→	.22	←	**39**	**93**	.86	.99	1.29	**7 ***
62		.24		**38**	**93.5**	.87	1.01	1.33	**6.5**
63	.26	.26	.27	**37**	**94**	.88	1.02	1.37	**6**
64	.28	.28	.29	**36**	**94.5**	.89	1.04	1.42	**5.5**
65%	.30	.30	.31	**35%**	**95.0%**	.90	1.06	1.47	**5.0%**
66	.32	.32	.33	**34**	**95.5**	.91	1.08	1.53	**4.5**
67	.34	.35	.35	**33**	**96**	.92	1.10	1.59	**4**
68	.36	.37	.38	**32**	**96.5**	.93	1.12	1.65	**3.5**
69	.38	.39	.40	**31**	**97**	.94	1.15	1.74	**3**
70%	.40	.41	.42	**30%**	**97.2%**	.94	1.16	1.77	**2.8%**
71	.42	.43	.45	**29**	**97.4**	.95	1.17	1.81	**2.6**
72	.44	.45	.47	**28**	**97.6**	.95	1.18	1.85	**2.4**
73	.46	.47	.50	**27**	**97.8**	.96	1.19	1.90	**2.2**
74	.48	.50	.52	**26**	**98.0**	.96	1.20	1.95	**2.0**
75%	.50	.52	.55	**25%**	**98.2%**	.96	1.21	2.00	**1.8%**
76	.52	.54	.58	**24**	**98.4**	.97	1.22	2.06	**1.6**
77	.54	.56	.60	**23**	**98.6**	.97	1.24	2.13	**1.4**
78	.56	.59	.63	**22**	**98.8**	.98	1.25	2.21	**1.2**
79	.58	.61	.66	**21**	**99.0**	.98	1.27	2.30	**1.0**
80%	.60	.63	.69	**20%**	**99.2%**	.99	1.28	2.41	**0.8%**
81	.62	.66	.72	**19**	**99.4**	.99	1.30	2.55	**0.6**
82	.64	.68	.76	**18**	**99.6**	.99	1.32	2.76	**0.4**
83	.66	.71	.79	**17**	**99.8**	1.00	1.35	3.11	**0.2**
84	.68	.73	.83	**16**	**100.0%**	1.00	1.41	*	**0.0**

Supplementary Table →

Another reasonable name for this expression would be folded fraction. Clearly, if we fold any other expression of a fraction--by subtracting the given function of those who aren't from the same function of those who are--we will force $f = \frac{1}{2}$ to have score zero, and will make the score of 63% the negative of the score of 37%, whatever expression we chose to begin with. Indeed, if we are to have the sort of symmetry around 50% we seek, we can confine our attention to folded expressions. (Proof?)

exhibit 2 of chapter 15 (continued)

B) SUPPLEMENTARY TABLE--for flogs of fractions beyond 1% or 99%

+	|flog|	−	+	|flog|	−
99.0%	2.30	1.0%	99.80%	3.11	.20%
.1	2.35	.9	.82	3.16	.18
.2	2.41	.8	.84	3.22	.16
.3	2.48	.7	.86	3.28	.14
.4	2.55	.6	.88	3.36	.12
99.50	2.65	.50	99.90	3.45	.10
.52	2.67	.48	.91	3.51	.09
.54	2.69	.46	.92	3.57	.08
.56	2.71	.44	.93	3.63	.07
.58	2.73	.42	.94	3.71	.06
99.60	2.76	.40	99.95	3.80	.05
.62	2.78	.38	.96	3.91	.04
.64	2.81	.36	.97	4.06	.03
.66	2.84	.34	.98	4.26	.02
.68	2.87	.32	.99	4.61	.01
99.70	2.90	.30			
.72	2.94	.28	Examples:		
.74	2.97	.26			
.76	3.01	.24	99.29% gives 2.47		
.78	3.06	.22	0.37% gives −2.80		

P) PROBLEMS

2a) What is the froot corresponding to 87%? The flog?

2b) What is the % corresponding to a froot of 1.00?

2c) What is the froot corresponding to a flog of −2.48? The % corresponding?

2d) What is the % corresponding to a flog of 4.00? The froot corresponding?

2e) What is the froot corresponding to 99.975%? The flog?

2f) For what % or %'s does/do the corresponding froot and flog differ by .60? By 3.80?

While a number of more complicated expressions have been--and may well continue to be--popular, we can usually do quite well by folding some very simple expression. By analogy with all our other inquiries into good ways to express data, we find it natural to begin with a square root and a logarithm. We can use the root--or the log--of any multiple of f. If we are to match folded percents near 50%, it turns out that we should fold $\sqrt{2}f$ and $\frac{1}{2}\log_e f = 1.15 \log_{10} f$. (Here $\log_e$ indicates the so-called natural or Napierian logarithm --the logarithm to the base $e = 2.71828\ldots$). [That this works will be clear when we come to exhibit 2.]

We need names for these expressions--both formal names and working names. Clearly, *folded roots* and *folded logs* are both formal and descriptive, while *froots* and *flogs* are convenient. Specifically, then, we plan to consider (notice that we are meeting logs of roots again):

$$\begin{aligned} \text{froot} &= \sqrt{2f} - \sqrt{2(1-f)} \\ \text{flog} &= \tfrac{1}{2}\log_e f - \tfrac{1}{2}\log_e (1-f) \\ &= \log_e \sqrt{f} - \log_e \sqrt{1-f} \\ &= 1.15 \log_{10} f - 1.15 \log_{10} (1-f) \\ &= 1.15 \log_{10} (f/(1-f)) \end{aligned}$$

as alternatives to

$$\text{plurality} = f - (1-f)$$

or

$$\text{feent} = (\%\ \text{yes}) - (\%\ \text{no}).$$

Exhibit **2** sets out two-decimal values of pluralities, froots, and flogs for every % from 10 to 90 and at closer intervals for smaller and larger %'s. Notice that--to the accuracy given, which is usually sufficient--the three expressions agree exactly for fractions between 38% and 62%--and to within 0.02 for fractions between 30% and 70%. (It was just to have this happen that we put the $\sqrt{2}$'s in our froots and the $\frac{1}{2}$'s in our flogs.) Unless we have to deal with some fractions below 20% or above 80% it is likely to make only trivial differences which of these expressions we use for our counted fractions (so long as we are consistent).

froots and flogs were brought in to stretch the ends of the scale. If the ends of the scale are not needed--because there is no data there--it is most convenient and satisfactory that froots and flogs behave very, very much like pluralities or feents.

As always, it is nice to look at changes of expression graphically as well as in numbers. In this situation, where we think and write of "stretching the tails", it is unusually desirable to compare the different scales side by side and both watch and feel--to the extent that the feeling can be given--the tails stretch

more and more as we change from pluralities--first to froots and then to flogs. Exhibit **3** is intended to do just this.

review questions

How should we treat "those who are" and "those who aren't" when we re-express fractions? How can we do this easily? What is a trivial re-expression? In which of the two re-expressions are we most interested? Can we match re-expressions of fractions? How? What is "folding"? How do our matched expressions behave in the middle of the scale (near 50%)? At the ends of the scale (near 0% or 100%)? What does tail-stretching mean? Can there be a picture of it? Looking like what? Do we express ss-fractions? How? What is a froot? And a flog?

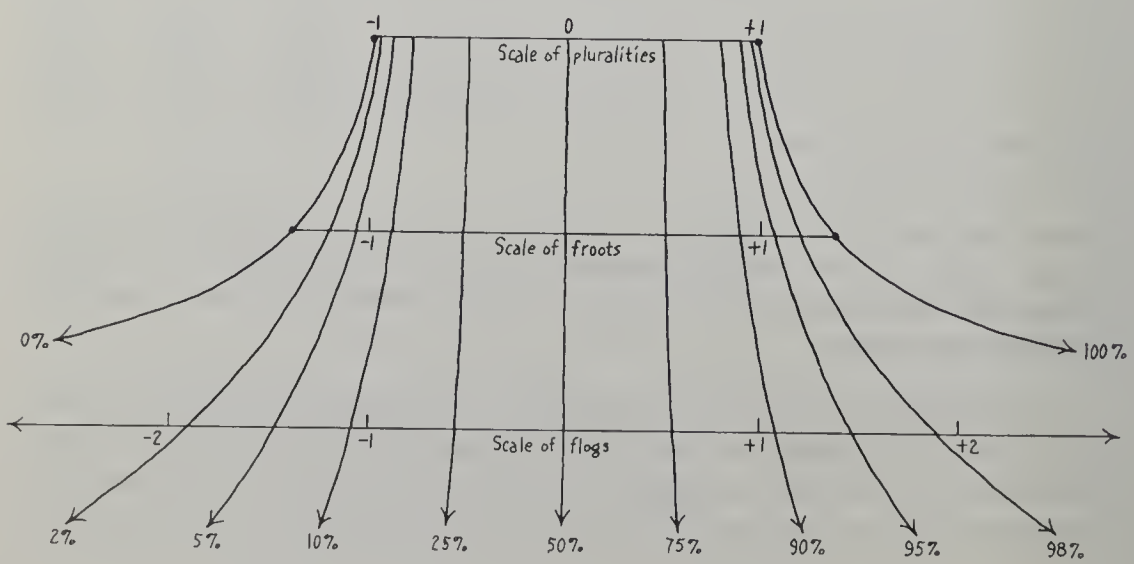
15C. Quicker calculation

Most of the time we will want to use flogs and froots for ss-fractions, for fractions of the form

$$\frac{\text{count} + \frac{1}{6}}{\text{total} + \frac{1}{3}}.$$

exhibit **3** of chapter 15: comparative display

The scales of pluralities (fcents), froots and flogs compared



We may as well do as much as we can to make the overall calculation simple.

For flogs we have

$$\begin{aligned}\text{flog} &= \tfrac{1}{2}\log_e \frac{\text{count} + \frac{1}{6}}{\text{total} + \frac{1}{3}} - \tfrac{1}{2}\log_e \frac{\text{remainder} + \frac{1}{6}}{\text{total} + \frac{1}{3}} \\ &= \tfrac{1}{2}\log_e (\text{count} + \tfrac{1}{6}) - \tfrac{1}{2}\log_e (\text{remainder} + \tfrac{1}{6}) \\ &= \log_e \sqrt{\text{count} + \tfrac{1}{6}} - \log_e \sqrt{\text{remainder} + \tfrac{1}{6}}\end{aligned}$$

(where the last form arises because the log of a ratio is the difference of the logs, and the $\pm\frac{1}{2}\log_e(\text{total} + \frac{1}{3})$ terms cancel out). Exhibit **4** gives values of $\frac{1}{2}\log_e(\text{count} + \frac{1}{6}) = \log_e\sqrt{\text{count} + \frac{1}{6}}$ for each count from 0 to 200, and every 10th count from 200 to 3090. (More detail is usually worthless.)

exhibit **4** of chapter 15: reference table

Values of $\log_e\sqrt{\text{count} + \frac{1}{6}}$ (a) for integer counts from 0 to 199, (b) for every 10th count from 200 to 3090, (c) for larger counts, (d) for half-integers ≤ 20

A) TABLE for COUNTS up to 199--see panel D for half integers up to 20

	0	1	2	3	4	5	6	7	8	9
00	−.90	.08	.39	.58	.71	.82	.91	.98	1.05	1.11
10	1.16	1.21	1.25	1.29	1.33	1.36	1.39	1.42	1.45	1.48
20	1.50	1.53	1.55	1.57	1.59	1.61	1.63	1.65	1.67	1.69
30	1.70	1.72	1.74	1.75	1.77	1.78	1.79	1.81	1.82	1.83
40	1.85	1.86	1.87	1.88	1.89	1.91	1.92	1.93	1.94	1.95
50	1.96	1.97	1.98	1.99	2.00	2.01	2.01	2.02	2.03	2.04
60	2.05	2.06	2.06	2.07	2.08	2.09	2.10	2.10	2.11	2.12
70	2.13	2.13	2.14	2.15	2.15	2.16	2.17	2.17	2.18	2.19
80	2.19	2.20	2.20	2.21	2.22	2.22	2.23	2.23	2.24	2.25
90	2.25	2.26	2.26	2.27	2.27	2.28	2.28	2.29	2.29	2.30
100	2.30	2.31	2.31	2.32	2.32	2.33	2.33	2.34	2.34	2.35
110	2.35	2.36	2.36	2.36	2.37	2.37	2.38	2.38	2.39	2.39
120	2.39	2.40	2.40	2.41	2.41	2.41	2.42	2.42	2.43	2.43
130	2.43	2.44	2.44	2.45	2.45	2.45	2.46	2.46	2.46	2.47
140	2.47	2.47	2.48	2.48	2.49	2.49	2.49	2.50	2.50	2.50
150	2.51	2.51	2.51	2.52	2.52	2.52	2.53	2.53	2.53	2.53
160	2.54	2.54	2.54	2.55	2.55	2.55	2.56	2.56	2.56	2.57
170	2.57	2.57	2.57	2.58	2.58	2.58	2.59	2.59	2.59	2.59
180	2.60	2.60	2.60	2.61	2.61	2.61	2.61	2.62	2.61	2.62
190	2.62	2.63	2.63	2.63	2.63	2.64	2.64	2.64	2.64	2.65

Table continues →

exhibit **4** of chapter 15 (continued)

B) TABLE for COUNTS from 200 to 3090--grab nearest; don't interpolate

	00	10	20	30	40	50	60	70	80	90
200	2.65	2.67	2.70	2.72	2.74	2.76	2.78	2.80	2.82	2.84
300	2.85	2.87	2.88	2.90	2.91	2.93	2.94	2.96	2.97	2.98
400	3.00	3.01	3.02	3.03	3.04	3.05	3.07	3.08	3.09	3.10
500	3.11	3.12	3.13	3.14	3.15	3.16	3.16	3.17	3.18	3.19
600	3.20	3.21	3.21	3.22	3.23	3.24	3.25	3.25	3.26	3.27
700	3.28	3.28	3.29	3.30	3.30	3.31	3.32	3.32	3.33	3.34
800	3.34	3.35	3.35	3.36	3.37	3.37	3.38	3.38	3.39	3.40
900	3.40	3.41	3.41	3.42	3.42	3.43	3.43	3.44	3.44	3.45
1000	3.45	3.46	3.46	3.47	3.47	3.48	3.48	3.49	3.49	3.50
1100	3.50	3.51	3.51	3.52	3.52	3.52	3.53	3.53	3.54	3.54
1200	3.545	3.549	3.553	3.557	3.562	3.566	3.569	3.573	3.577	3.581
1300	3.585	3.589	3.593	3.597	3.600	3.604	3.608	3.611	3.615	3.619
1400	3.622	3.626	3.629	3.633	3.636	3.640	3.643	3.647	3.650	3.653
1500	3.657	3.660	3.663	3.667	3.670	3.673	3.676	3.679	3.683	3.686
1600	3.689	3.692	3.695	3.698	3.701	3.704	3.707	3.710	3.713	3.716
1700	3.719	3.722	3.725	3.728	3.731	3.734	3.737	3.739	3.742	3.745
1800	3.748	3.751	3.753	3.756	3.759	3.762	3.764	3.767	3.770	3.772
1900	3.775	3.777	3.780	3.783	3.785	3.788	3.790	3.793	3.795	3.798
2000	3.800	3.803	3.805	3.808	3.810	3.813	3.815	3.818	3.820	3.822
2100	3.825	3.827	3.830	3.832	3.834	3.837	3.839	3.841	3.844	3.846
2200	3.848	3.850	3.853	3.855	3.857	3.859	3.862	3.864	3.866	3.868
2300	3.870	3.873	3.875	3.877	3.879	3.881	3.883	3.885	3.887	3.890
2400	3.892	3.894	3.896	3.898	3.900	3.902	3.904	3.906	3.908	3.910
2500	3.912	3.914	3.916	3.918	3.920	3.922	3.924	3.926	3.928	3.930
2600	3.932	3.934	3.935	3.937	3.939	3.941	3.943	3.945	3.947	3.949
2700	3.951	3.952	3.954	3.956	3.958	3.960	3.962	3.963	3.965	3.967
2800	3.969	3.970	3.972	3.974	3.976	3.978	3.979	3.981	3.983	3.985
2900	3.986	3.988	3.990	3.991	3.993	3.995	3.997	2.998	4.000	4.002
3000	4.003	4.005	4.007	4.008	4.010	4.011	4.013	4.015	4.016	4.018

C) For COUNTS above 3000

First cut:	Divide by	Look in	Add (or)
	10	panel	1.151 (1.15)
	100	B	2.303 (2.30)
	1000		3.454 (3.45)

Example: Given 4567, to 457 (rounded), find 3.07 at 460, add 1.15, to conclude with 4.22.

For froots, we have no equally simple device. We can often gain, however, from a table of $\sqrt{(\text{split count}) + \frac{1}{6}}$, given as exhibit **5**, since we can write

$$\begin{aligned}\text{froot} &= \sqrt{2f} - \sqrt{2(1-f)} \\ &= \sqrt{2\left(\frac{\text{split count left} + \frac{1}{6}}{\text{total count} + \frac{1}{3}}\right)} - \sqrt{2\left(\frac{\text{split count right} + \frac{1}{6}}{\text{total count} + \frac{1}{3}}\right)} \\ &= \frac{\sqrt{\text{split count left} + \frac{1}{6}} - \sqrt{\text{split count right} + \frac{1}{6}}}{\sqrt{\text{half total count} + \frac{1}{6}}}\end{aligned}$$

exhibit **4** of chapter 15 (continued)

If more accuracy is required (quite unlikely), go to more-place log table. If table of natural logs is available, use $\frac{1}{2}\log_e$ count. If only table of common logs is at hand, use $1.1513 \log_{10}$ count.

D) TABLE for HALF-INTEGERS < 20––for > 20, interpolate in panel A

	0h	1h	2h	3h	4h	5h	6h	7h	8h	9h
00	−.20	.26	.49	.65	.77	.87	.95	1.02	1.08	1.13
10	1.18	1.23	1.27	1.31	1.34	1.38	1.41	1.44	1.46	1.49

E) EXAMPLES––"remainder" is right count

left count = 43 , remainder = 28 , hence flog = 1.88 − 1.67 = .21
left count = 504 , remainder = 13 , hence flog = 3.11 − 1.29 = 1.82
left count = 1272 , remainder = 3057 , hence flog = 3.573 − 4.013 = −.440
left count = 27 , remainder = 1304 , hence flog = 1.65 − 3.585 = −1.94
left count = 18h, remainder = 32h, hence flog = 1.46 − 1.74 = −.28

Note that 504 was looked up as 500, 1272 as 1270, 3057 as 3060, and 1304 as 1300. Note further that, when one look-up has 3 decimals and one has 2, only 2 decimals are kept in the difference. Note still further that the fact that 1304 > 1300 was allowed to creep through when rounding 3.585 (to 3.59 instead of 3.58).

P) PROBLEMS

4a) What is $\log_e \sqrt{\text{count} + \frac{1}{6}}$ when count = 7? = 77? = 777?

4b) What is the flog of each of the ss-fractions for the following pairs of counts? (2, 27); (20, 270); (200, 2700). Comment.

4c) What is the flog of each of the following pairs of counts? (17, 9h); (26h, 203h); (42h, 197h).

4d) What count has a log $\sqrt{\text{count} + \frac{1}{6}}$ nearest to 2.68? To 3.566?

4e) What multiple of log (count + $\frac{1}{6}$) is log $\sqrt{\text{count} + \frac{1}{6}}$?

exhibit **5** of chapter 15: reference table

Values of $\sqrt{\text{count} + \frac{1}{6}}$ (a) for integer counts from 0 to 199, (b) for every 10th count from 200 to 3090, (c) for half-integer counts < 20

A) TABLE for COUNTS up to 199--see panel C for half-integers < 20

	0	1	2	3	4	5	6	7	8	9
00	.41	1.08	1.47	1.78	2.04	2.27	2.48	2.68	2.86	3.03
10	3.19	3.34	3.49	3.63	3.76	3.89	4.02	4.14	4.26	4.38
20	4.49	4.60	4.71	4.81	4.92	5.02	5.12	5.21	5.31	5.40
30	5.49	5.58	5.67	5.76	5.85	5.93	6.01	6.10	6.18	6.26
40	6.34	6.42	6.49	6.57	6.65	6.72	6.79	6.87	6.94	7.01
50	7.08	7.15	7.22	7.29	7.36	7.43	7.49	7.56	7.63	7.69
60	7.76	7.82	7.88	7.95	8.01	8.07	8.13	8.20	8.26	8.32
70	8.38	8.44	8.50	8.55	8.61	8.67	8.73	8.78	8.84	8.90
80	8.95	9.01	9.06	9.12	9.17	9.23	9.28	9.34	9.39	9.44
90	9.50	9.55	9.60	9.65	9.70	9.76	9.81	9.86	9.91	9.96
100	10.01	10.06	10.11	10.16	10.21	10.26	10.30	10.35	10.40	10.45
110	10.50	10.54	10.59	10.64	10.68	10.73	10.78	10.82	10.87	10.92
120	10.96	11.01	11.05	11.10	11.14	11.19	11.23	11.28	11.32	11.37
130	11.41	11.45	11.50	11.54	11.58	11.63	11.67	11.71	11.75	11.80
140	11.84	11.88	11.92	11.97	12.01	12.05	12.09	12.13	12.17	12.21
150	12.25	12.29	12.34	12.38	12.42	12.46	12.50	12.54	12.58	12.62
160	12.66	12.70	12.73	12.77	12.81	12.85	12.89	12.93	12.97	13.01
170	13.04	13.08	13.12	13.16	13.20	13.24	13.27	13.31	13.35	13.39
180	13.42	13.46	13.50	13.53	13.57	13.61	13.64	13.68	13.72	13.75
190	13.79	13.83	13.86	13.90	13.93	13.97	14.01	14.04	14.08	14.11

B) TABLE for COUNTS from 200 to 3090--grab nearest; don't interpolate

	00	10	20	30	40	50	60	70	80	90
200	14.15	14.50	14.84	15.17	15.50	15.82	16.13	16.44	16.74	17.03
300	17.33	17.61	17.89	18.17	18.44	18.71	18.98	19.24	19.50	19.75
400	20.00	20.25	20.50	20.74	20.98	21.22	21.45	21.68	21.91	22.14
500	22.36	22.59	22.81	23.03	23.24	23.46	23.67	23.88	24.09	24.29
600	24.50	24.70	24.90	25.10	25.30	25.50	25.69	25.89	26.08	26.27
700	26.46	26.65	26.84	27.02	27.21	27.39	27.57	27.75	27.93	28.11
800	28.29	28.46	28.64	28.81	28.99	29.16	29.33	29.50	29.67	29.84
900	30.00	30.17	30.33	30.50	30.66	30.82	30.99	31.15	31.31	31.47
1000	31.63	31.78	31.94	32.10	32.25	32.41	32.56	32.71	32.87	33.02
1100	33.17	33.32	33.47	33.62	33.77	33.91	34.06	34.21	34.35	34.50
1200	34.64	34.79	34.93	35.07	35.22	35.36	35.50	35.64	35.78	35.92
1300	36.06	36.20	36.33	36.47	36.61	36.74	36.88	37.02	37.15	37.28

Table continues →

exhibit **5** of chapter 15 (continued)

1400	37.42	37.55	37.69	37.82	37.95	38.08	38.21	38.34	38.47	38.60
1500	38.73	38.86	38.99	39.12	39.24	39.37	39.50	39.63	39.75	39.88
1600	40.00	40.13	40.25	40.38	40.50	40.62	40.75	40.87	40.99	41.11
1700	41.23	41.35	41.47	41.60	41.72	41.83	41.95	42.07	42.19	42.31
1800	42.43	42.55	42.66	42.78	42.90	43.01	43.13	43.25	43.36	43.48
1900	43.59	43.71	43.82	43.93	44.05	44.16	44.27	44.39	44.50	44.61
2000	44.72	44.83	44.95	45.06	45.17	45.28	45.39	45.50	45.61	45.72
2100	45.83	45.94	46.05	46.15	46.26	46.37	46.48	46.59	46.69	46.80
2200	46.91	47.01	47.12	47.22	47.33	47.44	47.54	47.65	47.75	47.86
2300	47.96	48.06	48.17	48.27	48.38	48.48	48.58	48.68	48.79	48.89
2400	48.99	49.09	49.20	49.30	49.40	49.50	49.60	49.70	49.80	49.90
2500	50.00	50.10	50.20	50.30	50.40	50.50	50.60	50.70	50.80	50.89
2600	50.99	51.09	51.19	51.29	51.38	51.48	51.58	51.67	51.77	51.87
2700	51.96	52.06	52.16	52.25	52.35	52.44	52.54	52.63	52.73	52.82
2800	52.92	53.01	53.11	53.20	53.29	53.39	53.48	53.57	53.67	53.76
2900	53.85	53.95	54.04	54.13	54.22	54.32	54.41	54.50	54.59	54.68
3000	54.77	54.86	54.96	55.05	55.14	55.23	55.32	55.41	55.50	55.59

C) TABLE for HALF-INTEGER COUNTS < 20--for > 20, interpolate in panel A

	⌊0h⌋	⌊1h⌋	⌊2h⌋	⌊3h⌋	⌊4h⌋	⌊5h⌋	⌊6h⌋	⌊7h⌋	⌊8h⌋	⌊9h⌋
00	.82	1.29	1.63	1.91	2.16	2.38	2.58	2.77	2.94	3.11
10	3.27	3.42	3.56	3.70	3.83	3.96	4.08	4.20	4.32	4.43

D) EXAMPLES--"remainder" is right count

left count = 43, remainder = 28, hence half total = 35h and
froot = (6.57 − 5.3)/5.97 = .21

left count = 504, remainder = 13, hence half total = 258h and
froot = (22.36 − 3.63)/16.13 = 1.16

left count = 18h, remainder = 32h, hence half total = 25h and
froot = (4.32 − 5.72)/5.07 = −.28

Note that 504 was looked up as 500, and that 35h, 258h, 32h, and 25h were interpolated in panel A.

P) PROBLEMS

5a) What is $\sqrt{\text{count} + \frac{1}{6}}$ when count = 7? = 77? = 777?

5b) What is the froot of each of the ss-fractions for the following pairs of counts? (2, 27); (20, 270); (200, 2700)? Comment.

5c) What is the froot of each of the following pairs of counts? (17h, 9h); (26h, 203h); (42h, 197h)?

When the total count is constant, we may be able to avoid the division and use

$$\sqrt{\text{split count left} + \tfrac{1}{6}} - \sqrt{\text{split count right} + \tfrac{1}{6}},$$

which is then a constant multiple of the froot. (Especially when table 5 is not at hand, it can help to use a slide rule efficiently and easily, as explained in section 15G.)

review questions

Can we simplify the arithmetic for re-expressing ss-fractions as froots? As flogs? Do we need to keep track of every digit in a count when we re-express it? Why/why not?

15D. Examples where careful expression clearly pays off

a 2 × 2 opinion example

Being careful in how one expresses counts or counted fractions gives us better analyses in many, many instances. It is not always easy to be sure that this is happening to us; often the improvement is not obvious. If the data contains no two portions whose analysis we can usefully compare, it is hard to be sure that we have indeed gained by care in choosing the expression to be analyzed. Even if one portion can be checked against another, we are likely to learn little, so long as everything is very nearly the same. In such a situation, the effect of a choice of expression on the two portions will be almost the same; consequently the comparison of the two for any one expression is likely to be very similar to that for any other.

To learn about choice of expression from internal comparisons--from comparisons within one set of data--we need a situation where we can compare the behavior of one portion with that of another where:

- ◇our attention is directed toward differences, so that satisfactory agreement means similar differences.
- ◇the differences that are formed--and found to be nearly the same--are differences of smaller numbers in one portion, of larger numbers in the other.

The simplest such situation, of course, is a two-way table in which both factors show substantial effects.

Let us start with a 2 × 2 example. Under the headline "Protestants shift support to Kennedy," the *Washington Post* for February 2, 1962 reported the

following Gallup poll results for November 1960 and January 1962:

	Protestants		Catholics	
Date	11/60	1/62	11/60	1/62
Kennedy	38%	59%	78%	89%
Nixon	62%	41%	22%	11%

From an engineering (i.e., practical political) viewpoint, George Gallup was correct in saying that "one of the major--reasons for the President's popularity has been his success in allaying anti-Catholic sentiment while not losing the support of his fellow Catholics." An improvement of 21% among four-fifths of the population is many more votes than an improvement of 11% among the remaining one-fifth.

But we might want to try to understand (numerically) what is really going on. We can be very sure that this situation demands scale stretching near the ends. (It is surely easier to change 38% to 42% than to change 89% to 93%.) So let us go to flogs, and look at the same data again (with f = fraction for Kennedy). The flogs are:

	Protestants	Catholics	Difference
11/60	−0.24	+0.63	+0.87
1/62	+0.18	+1.05	+0.87
Change	up 0.42	up 0.42	

so that the Protestant and Catholic changes are very similar and mutually supporting.

Understanding of what happened to John F. Kennedy's popularity between November 1960 and January 1962--a scientific matter--is much better served by the flog analysis. We can say: "JFK's popularity improved by +0.42 on the flog scale; you would have got this same answer had you asked only Catholics or only Protestants."

It is this constancy, present--in this example--in flogs but not in % or pluralities, that shows the advantages--in this example--of a choice of flogs.

We cannot count on such nice things happening when we go to flogs. Sometimes going to flogs makes things wonderful; sometimes merely good, sometimes no better, sometimes worse. On balance, though, they seem to help often enough so that the writer usually tries flogs even before percents (plain or folded) or pluralities--flogs are the first-aid bandages for counted fractions.

two-way tables of counts

We often have larger two-way tables, each cell of which contains a counted fraction. Opinion polls on several dates, reported separately for four different regions of the country, offer one example.

Two-way tables with froots or flogs as entries also arise in quite a different way. If we have a response that can reasonably be cut at two or more different places, then we can calculate a different flog or froot for each. Anticipating our next example a little, consider the counts in exhibit **6**. For each of the cuts that are feasible for the data given--Very large/Large, Large/Medium, Medium/Small, and Small/Very small--there is a flog--respectively −4.20, −2.78, −2.22, −1.76. If we had another such set of counts--as we soon will--we could write down a 2 × 4 table of flogs and analyze it. (If we had a third such set, we could have a 3 × 4 table, and so on.)

In 1954--in the course of a vigorous discussion with Gerschenkron--Landes published matched figures on the sizes of various kinds of business and industrial establishments in France and Germany as of 1906–07. Here "size of establishment" is given--by number of employees--in five broad groups. Instead of worrying how these sizes ought to be expressed in our analysis, we can begin by calling them very small, small, medium, large, and very large, and

exhibit **6** of chapter 15: illustrative

Several flogs for a single set of counts (sizes of textile establishments in France, 1906)

A) CALCULATION

Size	Count	Cum down	Cum up	Exhibit 4* down	Exhibit 4* up	flog
Very large	48					
		48	213,579	1.94	6.136	−4.20
Large	766					
		814	212,813	3.35	6.134	−2.28
Medium	1,676					
		2,490	211,137	3.910	6.130	−2.220
Small	3,636					
		6,126	207,501	4.360	6.121	−1.761
Very small	207,501					

* Exhibit 4 used thrice, table of logs used five times.

P) PROBLEMS

6a) Just how does the number 2,490 arise?

6b) Same for 212,813?

6c) And for 6.130?

6d) And for 3.35?

considering each and all of the four possible cuts. The data for all textile establishments in France in 1906 has already been flogged in exhibit 6; similar data for Germany in 1907 is given, with its flogs, in exhibit **7**. The difference between French and German flogs is of consistent sign and roughly constant size--say .4--indicating not only a consistent tendency for the German textile establishments to be somewhat larger, but a quite consistent difference, in flog terms.

The difference is consistent enough to encourage the belief that, if one more census could be found for either country,

- ◇it would show a relatively consistent difference over time.
- ◇an appropriate fraction of this time difference could be used either to adjust the French data to 1907 or to adjust the German data to 1906, so as to make a cleaner comparison of the two countries (one based on the same date).

exhibit **7** of chapter 15: textile establishments

Sizes of textile establishments in France and Germany, 1906–1907

A) DATA (for France in '06, see exhibit 6) and CALCULATIONS

	Germany, 1907		France, 1906 flog		
Count of employees	# estab	flog	flog*	diff	
	1001 up	71			
1000.5		3.77	4.20	.43	
	201–1000	1,013			
200.5		2.41	2.780	.37	
	51–200	2,748			median
50.5		1.757	2.220	.463	= .44
	11–50	4,984			
10.5		1.321	1.761	.440	
	1–10	123,768			

*** Note that, in this exhibit, we have calculated the flogs from below upwards--not from above downward--thus changing the sign in comparison with exhibit 6, whence come the French flogs.**

P) PROBLEMS

7a) What expression of establishment size gives a simple behavior of the flogs for either country or both countries?

7b) Plot the two sets of flogs against this expression of establishments.

7c) Seek out data to allow comparison of France and Germany at the same date (see text and Landes).

7d) Make such a comparison.

S) SOURCE

D. S. Landes 1954. "Some attitudes, entrepreneurship, and economic development: A comment," Explorations in Entrepreneurial History 6: 245–272. Appendix table 2 on pages 268–269.

The pessimist will ask "Did we really gain by taking flogs--what if we had stuck to %?" For the largest cut--at 1000.5 employees--the %'s are 0.022 for France and 0.053 for Germany, a difference of 0.031%. For the smallest cut--at 10.5 employees--they are 2.9% for France and 6.6% for Germany--a difference of 3.7%. One is about 1/30 of a percent, the other nearly 4 percent. Clearly there was no agreement about

Germany MINUS France

in percent (or in fcents or in plurality), although we have just seen good agreement when we work in flogs.

A few further examples are set as problems in exhibit **8**.

review questions

What was the first example of this section? How did we treat it? Why did we like/hate the re-expressions? How do you think re-expression as froots would have worked: compared to raw fractions? compared to re-expressed fractions? Can we treat larger sets of data similarly? For instance? Is this the

exhibit **8** of chapter 15: problems

Problems involving double flogs

P) PROBLEMS

8a) Brownlee gave the following counts for each of 6 grades of quality for the production on each of three shifts:

	Qual. a	Qual. b	Qual. c	Qual. d	Qual. e	Qual. f
Shift A	11	23	8	5	18	18
Shift B	17	29	10	17	7	15
Shift C	6	21	8	24	15	9

Analyze as in exhibit 7. Comment on the results.

8b) Clausen gave the following %'s of each level of definiteness of plan to return to previous employer for each of four durations of previous employment for soldiers leaving the U.S. Army in July 1945:

Years in job	Definite plans	Tentative plans	Considering return	Not considering	(total)
<1	12%	9%	20%	59%	(100%)
1–2	22%	11%	18%	49%	(100%)
2–5	31%	12%	17%	40%	(100%)
>5	49%	16%	14%	42%	(121%)??

Analyze as in exhibit 7. Comment on the results. (Notice, here and below, that we are taking the row labels as circumstances.)

→

only way in which tables full of froots or flogs arise? Why/why not? What example did we look at next? To what class of data sets did it belong? What did we do with it? How well did this work? Why do you say so? How does this behavior affect using a third census to match dates more closely?

15E. Double folding--the 2 × 2 case

We dealt in Section 12E/F with a two-way table of counts, where the counts are really responses and both ways describe circumstances. In the last section, we looked at two-way tables of counts where one way is a level of cutting the response while the other is a circumstance--or is treated as such.

exhibit **8** of chapter 15 (continued)

8c) The same authors also gave the following figures for soldiers leaving the army in September 1945:

Years in job	Definite plans	Tentative plans	Considering return	Not considering	(total)
<1	18%	12%	17%	43%	(100%)
1–2	35%	24%	19%	22%	(100%)
2–5	47%	17%	13%	23%	(100%)
>5	69%	16%	13%	7%	(100%)

Analyze as in exhibit 7. Comment on the results.

8d) (Double weight) Combine the July and September data and analyze all together as above.

8bc2) Compare the results of (8b) and (8c). Are they as you would have expected? (Consider the % not considering return.) Can you combine the effects of choice of cutting point for the two analyses? How would you analyze them?

8bcde2) Compare the analysis "all together" (8c) with the combination of the two analyses (8d). Discuss the differences/similarities.

S) SOURCES

K. A. Brownlee 1949. *Industrial Experimentation* (4th ed.) Chemical Rubber. Table on page 46.

J. A. Clausen 1950. "Studies of the postwar plans of soldiers: a problem in prediction." Chapter 15 (pages 568 to 708) of *Measurement and Prediction*, volume 4 of *The American Soldier* (ed. S. A. Stouffer), Princeton. Table on page 629.

We dealt with these by introducing a cut between each adjacent pair of levels and making a two-way table whose entries were froots or flogs, whose divisions were one in terms of circumstances and one in terms of cutting a response. It is time now to deal with two-way tables of counts, both of whose ways are levels of cutting a response--tables rightfully called "contingency tables with ordered margins".

The prototype is a double cutting of the total count into $2 \times 2 = 4$ counts, thus

$$\begin{array}{c|c} a & b \\ \hline c & d \end{array}$$

If we want to seize the information in this 2-by-2 table that is not seized by the two margins--the two sets of combined counts, for rows: $a + b$ above and $c + d$ below; for columns: $a + c$ on the left and $b + d$ on the right, which we can schematize as

$$--\frac{a+b}{c+d}-- \qquad\qquad a + c \,|\, b + d$$

(or, equivalently, by the two marginal fractions and the total)--we do well to use logs and fold twice (once each way--once horizontally and once vertically), reaching some multiple of

$$\log\frac{a}{a+b+c+d} - \log\frac{b}{a+b+c+d} - \log\frac{c}{a+b+c+d} + \log\frac{d}{a+b+c+d},$$

which can be more simply written as

$$\log a - \log b - \log c + \log d = \log\frac{ad}{bc}$$

or, possibly more revealingly, as either

$$\left(\log\frac{a}{a+b} - \log\frac{b}{a+b}\right) - \left(\log\frac{c}{c+d} - \log\frac{d}{c+d}\right)$$

or, equally, as

$$\left(\log\frac{a}{a+c} - \log\frac{c}{a+c}\right) - \left(\log\frac{b}{b+d} - \log\frac{d}{b+d}\right)$$

In practice, we don't want to divide, so we prefer

$$\log a - \log b - \log c + \log d$$

as a basis for our arithmetic.

What is the simplest behavior? Suppose we have 100 football stars and 100 senior hospital nurses, where we know 102 are male and 98 are female. If we knew nothing about football or hospitals -- and if we had to guess -- we might say to ourselves "half football players and half nurses, eh? Maybe 51 male football players and 51 male nurses (and 49 female nurses and 49 female football players)." We all know too much about football and hospitals to pay any attention to simplicity in this example, but we would take it seriously if we knew nothing.

How do our doubly folded logs behave in relation to this example? The simple version would be

Male	Female	
51	49	
51	49	with $\log a - \log b - \log c + \log d = 0$

and the likely version would be

Male	Female	
100	0	football players
2	98	nurses

for which we have trouble with $\log 0$, and thus go to

$$\begin{aligned} &\log\left(a + \tfrac{1}{6}\right) - \log\left(b + \tfrac{1}{6}\right) - \log\left(c + \tfrac{1}{6}\right) + \log\left(d + \tfrac{1}{6}\right) \\ &= 2.00 \quad - \quad (-.78) \quad - \quad (.34) \quad + \quad 1.99 \\ &= 4.43. \end{aligned}$$

In simple language, the zero value of the doubly folded difference of logs for the "simple" table announced no apparent connection between sex and occupation, while the large value for the likely table announced a very strong relation between sex and occupation.

In general we can take a zero value of the doubly folded difference as indicating "no apparent relation".

When we dealt with counted fractions we had two counts -- and we started both. Moreover, we threw in a factor of $\frac{1}{2}$ by taking logs (to base e) of the square roots of the (started) counts. It seems very natural to start all four counts here and to throw in the factor of $\frac{1}{2}$ by looking at square roots. As a result, we turn to

$$\log_e \sqrt{a + \tfrac{1}{6}} - \log_e \sqrt{b + \tfrac{1}{6}} - \log_e \sqrt{c + \tfrac{1}{6}} + \log_e \sqrt{d + \tfrac{1}{6}}$$

(This actually keeps our life simple, since we can look up each of the four terms in the table of exhibit 4. In the last example, this gives 5.10 instead of 4.43--just the factor of 1.15 again.)

The result we call

a double flog

which we often write as a

fflog.

Exhibit **9** rings some changes on data of Karl Pearson on intelligence and athletic power in school boys. However we calculate it--as a double difference of exhibit 4 contributions, as a difference of flogs one way, as a difference of flogs the other--we always get .27 for the fflog. The appearances are that "intelligence" and "athletic power" come together more frequently than the "simplest situation" of no relationship. The value of .27 shows about how much they appear to do this.

How big do we think .27 is? How are we to judge the size of flogs? One easy scale is offered by the quantitive relation between relatives (which works about as well for mental measurements as for physical). Some standard values, roughly correct, are:

great grandfather with great grandson: fflog = .15

cousin with cousin: fflog = .15

grandfather with grandson: fflog = .31

aunt with niece: fflog = .31

father with son: fflog = .67

Perhaps this scale will help you think about what a fflog of .27 means.

review questions

How many kinds of two-way tables of counts have we already looked at? Are there more? Why/why not? What kind of tables do we deal with in this section? What are they rightfully called? What is the simplest case? What are "*a*", "*b*", "*c*", and "*d*"? What is one way to seize upon the information missed by the margins? Is it natural to start the counts? To insert a factor? If yes, what factor? If not, why not? What is a double flog? A fflog? Is it easy to calculate a fflog from the four counts? Why/why not? What did we choose as an example? How did it come out? What is double folding? Why is it so called? Are there various ways to write a double flog?

15F. Double folding--larger cases

When we have more than two levels of either response--or of both responses--we have merely to work with all possible 2×2 tables made by using one of the available cuts for each response. For each paired cut, there will be four counts and a fflog. Since the paired cuts naturally fall into a two-way table, this means that we get a two-way table of fflogs to describe the apparent relationship.

exhibit **9** of chapter 15: school boys

Intelligence and athletic power in school boys

A) The DATA as COUNTS--and their fflogs

	Intelligent	**Not intelligent**	(sum)
Athletic	581	567	(1148)
Not athletic	209	351	(560)
(sum)	(790)	(918)	((1708))

fflog = 3.18 − 3.17 − 2.67 + 2.93 = .27
(Found by entering exhibit 4 with 580, 570, 210, and 350.)

B) The DATA as % ATHLETIC

	Among intelligent	**Among not intelligent**	(All) (combined)
%	73.5%	61.8%	(67.2%)
flogs	.51	.24	(.36)

Difference in flogs = .51 − .24 = .27

C) The DATA as % INTELLIGENT

	%	**flog**
Among athletic	50.6%	.01
Among not athletic	37.3%	−.26
(All combined)	(46.3%)	(−.07)

Difference in flogs = .01 − (−.26) = .27

D) The DATA RE-EXPRESSED

Total count: 1708
Intelligent: 46.3% or flog = −.07
Athletic: 67.2% or flog = .36
fflog: .27

P) PROBLEMS

9a/b/c/d) The data of exhibit 10, below, can be condensed to read, in part

(for 9a)

821	232
231	1958

(for 9b)

1018	35
1744	445

(for 9c)

1009	179
43	471

(for 9d)

2579	149
183	333

Carry through the analog of the analysis above, giving full detail.

S) SOURCE

K. Pearson 1906. "On the relationship of intelligence to size and shape of head, and to other physical and mental characters." Biometrika 5: 105–146. Condensed from table xxxiii on page 144.

exhibit **10** of chapter 15: two eyes

Calculation of double flogs for unaided distance vision in the two eyes of 3242 men aged 32–39, employed in U.K. Royal Ordnance Factories 1943–1946

A) The DATA

Right eye	Vision of left eye			
	Highest	**Second**	**Third**	**Lowest**
Highest	821	112	85	35
Second	116	494	145	27
Third	72	151	583	87
Lowest	43	34	100	331

B) Same numbers, with CORNER TOTALS

Highest			Second			Third			Lowest		
3242		1052	2190		1843	1399		2762	480		3242
	821			112			85			35	
1053		821	232		933	120		1018	35		1053
2189		231	1958		910	1279		1744	445		2189
	116			494			145			27	
1835		937	898		1543	292		1773	62		1835
1407		115	1292		300	1107		989	418		1407
	72			151			583			87	
2728		1009	1719		1766	962		2579	149		2728
514		43	471		77	437		183	331		514
	43			34			106			331	
3242		1052	2190		1843	1399		2762	480		3242

C) The FFLOGS in 0.01, FLOGS in 0.01 (at *'s), and TOTAL COUNT

Right eye	Vision of left eye						
	(Highest)	α	(second)	β	(Third)	γ	(Lowest) →	*
(Highest) α		170		120		100		−37
(Second) β		123		149		125		14
(Third) γ		93		117		172		83
(Lowest) → *		−37		14		87		3242

P) PROBLEMS

10a) Compare the single flogs for rows (right eye) and columns (left eye). What might the differences mean?

10b) Describe the 3 × 3 table of double flogs in words.

10c) What kind of numbers for the double flogs would have made a 3 × 3 table simplest to describe? Next simplest?

→

Exhibit **10** goes through this program for data on vision of the left eye and vision of the right eye, each cut at three places. Panel A shows the data. Panel B shows the result of adding up things in each of several ways-- each internal block of four values giving the corner totals for a paired cut whose crossing point is at that location. Thus, for example, the following paired cut

		β		
	821	112	85	35
α				
	116	494	145	27
	72	151	583	87
	43	34	106	331

leads to corner totals of

		β	
	933	120	
α			
	910	1279	

and hence (panel C) to a double flog of 1.20.

exhibit **10** of chapter 15 (continued)

10d1) Kendall and Stuart (Table 33.5 at page 586 of volume 2) also give counts according to unaided distance vision of 7427 women employed by the U.K. Royal Ordnance factories in 1943–46. For decreasing left-eye strengths, the counts are: right eye highest (1520, 266, 124, 66, total 1926); for right eye second (234, 1512, 432, 78, total 2256); for right eye third (117, 362, 1712, 205, total 2456); for right eye lowest (36, 82, 179, 492, total 789), with row totals (1907, 2220, 2507, 841). Parallel panel A.

10d2) Repeat the analysis of panels B and C on this data.

10d3) Compare the results for men with those for women, separately for: (i) size, (ii) single flogs, (iii) double flogs. Discuss your results.

10e) Return to problem 8b, take "years in job" as a response, as well as definitions of plans, and repeat the analysis of panels B and C.

10f) The same for the data of problem 8c.

10g) Compare the analysis of 10e and 10f. Discuss your comparison.

10h) (Double weight). The same for the data of problem 8d.

S) SOURCE

M. G. Kendall and A. Stuart 1961. *The Advanced Theory of Statistics* (3 volume edition). Table 33.3 at page 564 of volume 2.

Letting the same adding-up process extend outside the table leads to pairs of counts, as when

		β	
821	112	85	35
116	494	145	27
72	151	583	87
43	34	106	331

gives

	β	
1843	399	

corresponding to a single flog of 0.14. Thus our two-way table of double flogs is naturally bordered by a row and a column of single flogs.

If we fill in the total count where the additional row crosses the additional column, we will, it turns out, have completed giving enough information to allow reconstructing the given counts. Thus panel C is honestly an ANALYSIS of the given data, one in which information about size of collection, distribution of the separate responses, and mutual relationship has been separated.

review questions

What do we do when more than one cut is available for either response or both responses? What else does adding-up arithmetic, if used freely, give us? All put together, is this enough to reconstruct the data?

15G. Easy froots and flogs with a slide rule (optional)

Tables of froots and flogs are not always available. In any event, they require conversion of counts into fractions or percent before use. Accordingly, there can be advantages to more direct calculation, especially when a slide rule is available. This is particularly so when the froots or flogs to be found come in sets with the same total, something that will, in particular, occur whenever we are dealing with several alternative cutting points for a response.

Suppose, then, that, for some cutting point, our counts, each increased by $\frac{1}{6}$, are a and b so that

$$f = \frac{a}{a + b}$$

$$1 - f = \frac{b}{a + b}$$

$$\text{froot} = \sqrt{2f} - \sqrt{2(1 - f)}$$

$$= \sqrt{\frac{a}{(a + b)/2}} - \sqrt{\frac{b}{(a + b)/2}}$$

or

$$= \sqrt{2a/(a + b)} - \sqrt{2b/(a + b)}$$

Thus, if $a = 43s$, $b = 471s$, $a + b = 514ss$, $(a + b)/2 = 257s$ (where, as above, s stands for $\frac{1}{6}$), the corresponding froot is

$$\sqrt{\frac{43s}{257s}} - \sqrt{\frac{471s}{257s}}.$$

One setting of the slide rule (with many settings of the cursor) suffice to read out all the numbers needed for all calculations with the same total. (One resetting may be needed to overlap the other end of the scale.)

After we have set 257s on B against 1 on DF (or, if we prefer, 514ss on B against 2 on DF), against 43s on B we read 0.409 on DF and against 471s on B we read 1.353 on DF. (On a rule without a DF scale, we use the D scale BOTH TIMES instead.) Since

$$0.409 = \sqrt{\frac{43s}{257s}}$$

we have

$$\text{froot} = 0.409 - 1.353 = -0.944$$

With three or more froots to compute with the same total, this approach should be faster than using a table to convert %'s into froots, the %'s having been found with the slide rule.

For flogs on the slide rule we write

$$\begin{aligned}\text{flog} &= 1.15\left(\log_{10}\frac{a}{a+b} - \log_{10}\frac{b}{a+b}\right)\\ &= 1.15(\log_{10} a - \log_{10} b)\end{aligned}$$

and calculate according to the last line. For our example we have

$$\log 43 = 1.634$$

$$\log 471 = 2.673$$

$$\text{flog} = 1.15(1.634 - 2.673) = 1.15(-1.039) = -1.196$$

(The multiplications by 1.151 may either all be postponed to the end of the calculations--and then be done together--or (if D and L scales are both available on the body of rule) the rule can be left set for multiplication by 1.151.)

This procedure, too, is likely to be easier than using a log table but not as easy as using exhibit 4 (when available).

A good hand-held calculator is even easier.

review questions

In what form do we write the formula for froots in preparation for slide-ruling it? Which scales do we use? What do we set against what? What do we set the cursor to? Might we have to use a different scale? Which one? How would it be used? And what about flogs?

15H. How far have we come?

This chapter has focussed on effective expressions of counted fractions and the extension to two-way tables according to ordered variables.

We are now ready to:

◇recognize and use ss- (split and started) counts and fractions.

◇use any or all of three folded expressions of fractions that are matched near 50%: pluralities (fcents), froots, flogs.

◇take advantage of tables and shortcuts (and, in optional section 15G, a slide rule) in calculating these folded expressions.

We are also ready to:

◇find, and make use of, several flogs (or several values of another folded expression) for a single set of counts.

◇recognize, and use, fflogs (double flogs) as expressions of disproportionality in the 2×2 table formed by intersecting cuts (paired cuts).

◇do this for a single paired cut, several paired cuts (in a streak), or a two-way array of paired cuts.

In the process, we have come to understand more clearly the importance of:

◇seeking constancy of difference in re-expression (here, of counted fractions).

Better smoothing 16

chapter index on next page

In chapter 7 we learned to make smooths quite smooth. We left to the present chapter the question of how good were the roughs that went with our so-smooth smooths. We could do this because there we were looking to our smooths for, basically, only rather qualitative results: Did the curve seem to go up, go down, oscillate, curve one way or another? We did try to:

- ◇ look hard at the roughs that went with the smooths (though we used them to size our blurs and identify outside points).
- ◇ ask whether the smooths were as good smooths as we could easily make.

We concentrated on getting the details out of the way, on unhooking our eyes from irregularities and corners--so that we might see some sort of "big picture".

In this present chapter, we are going to find out how to do a "better" job of smoothing--how to stick much closer to the data without making the smooth much rougher. (We will need this in the next chapter.)

We did NOT postpone doing a still better job of smoothing to this chapter because this is unimportant. We postponed it because there was already enough detail in chapter 7 and we thought we could get away with postponement. Accordingly, we regard this chapter as something that belongs in the mainstream of our techniques.

Clearly, it is important to understand chapter 7 quite well before tackling the present chapter.

review questions

What did we do in chapter 7? What did we postpone? What did we not do in chapter 7 that let us postpone? What are we going to do in this chapter? Is making better smooths part of our mainstream? Why/why not? What is needed before tackling this chapter?

16A. Reroughing

We are now ready to look hard at roughs to see what they tell us about better smooths. We start with simple illustrative cases and go on from there. While hanning does do a fair amount to make smooths smoother, it has its disadvantages. Consider the sequence

. . . , 36, 51, 64, 75, 84, 91, 96, 99, 100, 99, 96, 91, 84, 75, 64, 51, 36, . . .

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(the law of whose formation is easily found by subtracting each entry from 100). If we hann this, we find, first for the skip means and then for the completed hanning:

$$\ldots, 63,\ 74,\ 83,\ 90,\ 95,\ 98,\ 99,\ 98,\ 95,\ 90,\ 83,\ 74,\ 63,\ \ldots$$

$$\ldots, 63\text{h}, 74\text{h}, 83\text{h}, 90\text{h}, 95\text{h}, 98\text{h}, 99\text{h}, 98\text{h}, 95\text{h}, 90\text{h}, 83\text{h}, 74\text{h}, 63\text{h}, \ldots$$

The corresponding rough is

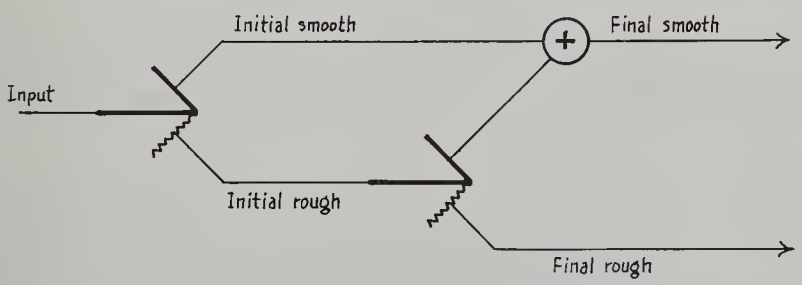
$$\ldots, \text{h}, \text{h}, \text{h}, \text{h}, \text{h}, \text{h}, \text{h}, \text{h}, \text{h}, \text{h}, \text{h}, \text{h}, \text{h}, \text{h}, \text{h}, \text{h}, \text{h}, \text{h}, \ldots$$

A solid sequence of one-halves (h's) is not a very satisfactory rough. It could easily be much more nearly centered at zero.

What we need to do is to smooth the rough, and add it back to the smooth we started with. The scheme is as shown in exhibit **1**, which also sets out the verbal algebra.

exhibit **1** of chapter 16: technique schematic

The scheme and algebra of reroughing



Since

$$\text{start} \equiv \text{smooth PLUS rough},$$

and

$$\text{rough} \equiv (\text{smooth of rough}) \text{ PLUS } (\text{rough of rough}),$$

we must have, first substituting and then redefining:

$$\text{start} = \text{smooth PLUS} \begin{pmatrix}\text{smooth}\\ \text{of rough}\end{pmatrix} \text{PLUS} \begin{pmatrix}\text{rough}\\ \text{of rough}\end{pmatrix},$$

$$\text{start} = \text{final smooth PLUS final rough},$$

where

$$\text{final smooth} \equiv \text{smooth PLUS} \begin{pmatrix}\text{smooth}\\ \text{of rough}\end{pmatrix}$$

and

$$\text{final rough} \equiv \text{rough of rough}.$$

→

If we do add together:

- ◇the smooth of the start, and
- ◇the smooth of the (first) rough,

then we get a final smooth whose rough is precisely the rough of the rough of the original sequence--hence the name

reroughing.

If the two smoothers are exactly the same, we will have applied the same smoother "twice", so that it is rather natural to think and write of

twicing

"Twicing" or use of some specified smoother twice, as in "3R,twice" is just a special case of reroughing. Specifically, "3R,twice" is just the same as "3R,reroughed with 3R". Symbolically, we represent either repetition with a comma (following the tag of the smoother used first, and preceding either "reroughed by ---" or the word "twice" as may be appropriate). Notice that stringing the names of two smoothers together without a comma means using the second to smooth still further the smooth of the first (to do resmoothing).

review questions

When does hanning slip a little? How is this revealed? What can we do about it? What is reroughing? Why is it so called? What is twicing? Why so called? How are these two represented in a tag?

16B. Some examples

Exhibit **2** shows some of the calculation for turning a 3RSS smooth of the bituminous-coal production example of chapter 7 into two alternative smooths:

exhibit **1** of chapter 16 (continued)

P) PROBLEMS

1a) Draw the corresponding diagram for resmoothing, as when 3R is followed by SSH3.

1b) Draw the corresponding diagram for resmoothing twice, as when 3R is followed by SS is followed by H3.

1c) Draw the corresponding diagram for reroughing twice, as when the result of reroughing one smoother with a second is, regarded as a unit, reroughed with a third smoother.

◇a smooth that should be tagged either 3RSSH twice or 3RSSH reroughed by 3RSSH

◇a smooth obtained by applying (H,twice) to 3RSS--note that the combination of these two is further smoothing, not reroughing (which already took place within ("H,twice").

comparison of smooths

Panel C shows, semischematically, four smooths (and the original data). This type of display does not try to imitate a graph, but rather tries to emphasize whether the sequence is going up or going down, the right-and-left displacement corresponding to this sort of pattern:

```
        high
down
down
        low
              up
              up
              up
        high
        low
              up
              up
        high
```

When we scan panel C, we notice the great irregularity of the starting sequence and the noticeable irregularity of 3RSS(H,twice). The latter does not seem to be a smooth we should bother with.

This leaves us comparing the other three smooths. Panel D shows their values at tops and bottoms. We can see there what we ought to expect:

◇as we go from 3RSS to 3RSSH, there is a tendency to pull bottoms up, and peaks down.

◇this tendency is reversed, somewhat more vigorously, as we go from 3RSSH to "3RSSH,twice".

After all, we know that hanning does tend to pull tops down and bottoms up. And we twiced things to get away from this effect, both in H and in the 3RSS to which H was applied. It looks as if, in this example, twicing--a special case of reroughing--was doing what we would like it to do.

review questions

What example did we try? What reroughings did we try? How did we compare them? What did we conclude?

Some examples of reroughing based on the 3RSS smooth of bituminous coal production (see exhibits 8 and 17 of chapter 7)

A) FINDING "3RSSH,twice"

Start	Its 3RSS	Hanning >	Hanning H	1st rough	Its 3R	Its 3RSS	Hanning >	Hanning H	Sum of Smooths	Final rough	(>)*
569	422		422	147	?	−6		−6	416	153	(416)
416	422	422	422	−6	−6	−6	−6	−6	416	0	(424)
422	422	453	438	−16	−6	−6	−6	−6	432	−10	(445)
565	484	471	478	87	−6	−6	−3	−4	474	91	(470)
484	520	502	511	−27	0	0	−3	−2	509	−25	(498)
520	520	520	520	0	0	0	2	0	521	−1	(516)
573	520	519	520	53	3	3	3	1	522	51	(520)
518	518	512	515	3	3	3	3	3	518	0	(516)
501	505	510	508	−7	3	3	3	3	511	−10	(508)
505	501	486	494	11	11	3	3	3	497	8	(484)
468	468	442	455	13	11	3	−2	0	455	13	(442)
382	382	401	392	−10	−10	−6	−2	−4	388	−6	(398)
310	334	358	346	−36	−10	−6	−6	−6	340	−30	(362)
334	334	346	340	−6	−6	−6	−4	−5	335	−1	(347)
359	359	353	356	3	−2	−2	−2	−2	354	5	(356)
372	372	377	374	−2	3	3	0	2	376	−4	(374)
439	395	384	390	49	49	3	3	3	393	46	(388)
446	395	396	396	50	49	3	3	3	399	47	(398)
349	398	417	408	−59	−39	−7	−2	−4	404	−55	(416)
395	439	430	434	−39	−39	−7	−2	−4	430	−35	(433)
461	461	475	468	−7	−7	−7	−6	−6	462	−1	(472)
511	511	522	516	−5	−5	−5	0	−2	514	−3	(515)
583	588	547	565	18	6	6	0	3	568	15	(552)
590	583	586	584	8	6	6	6	6	590	0	(581)
620	590	586	588	32	6	6	6	6	594	26	(593)
578	590	590	590	−12	−12	6	6	6	596	−18	(592)
534	590	584	587	−53	−12	6	3	4	591	−57	(582)
631	578	553	566	65	65	0	3	2	568	63	(562)
600	516	547	532	68	65	0	0	0	532	68	(542)
438	516	516	516	−78	0	0	0	0	516	−78	(524)
516	516	516	516	0	0	0	0	0	516	0	(510)
534	516	492	504	30	0	0	−2	−1	503	31	(495)
467	467	486	476	−9	−3	−3	−2	−2	474	−7	(480)
457	457	462	460	−3	−3	−3	−3	−3	457	0	(464)
392	457	457	457	−65	−3	−3	−3	−3	454	−62	(450)
461	457	434	446	21	21	−3	−2	−2	444	23	(438)
500	412	434	423	77	77	0	−2	−1	422	78	(428)
493	412	412	412	81	77	0	0	0	412	81	(417)
410	412	412	412	−2	0	0	0	0	412	−2	(412)
412	412	412	412	0	0	0	0	0	412	0	(412)
416	412	414	413	3	0	0	−4	−2	411	5	(411)
403	416	417	416	−13	−8	−8	−4	−6	410	−7	(416)
422	422	438	430	−8	−8	−8	−8	−8	422	0	(428)
459	459	444	452	7	−8	−8	−2	−5	447	12	(449)
467	467	486	476	−9	3	3	−2	0	476	−9	(478)
512	512	500	506	6	3	3	3	3	509	3	(505)
534	534	528	531	3	3	3	3	3	534	0	(527)
552	545	540	542	10	3	3	3	3	545	7	(541)
545	545		545	0	?	3		3	548	−3	(548)

Examples (1st line): 569 given, 422 given, 422 = 422, 147 = 569 − 422, −6 from end-value rule, −6 = −6, −6 = −6, 416 = (422) + (−6), 153 = 569 − 416. Note: In "Its 3R" column, → points to values changed by the second running median of 3.

*** Values in parenthesis, in last column, will be discussed in a later section.**

→

B) Finding "3RSS(H,twice)"--so sad

The 3RS	Hanning >	Hanning →	Local rough	Hanning >	Hanning →	Sum of smooths	Final rough
422		422	0		0	422	147
422	422	422	0	−8	−4	418	−2
422	453	438	−16	3	−6	432	−10
484	471	478	6	−4	1	479	86
520	502	511	9	3	6	517	−33
520	520	520	0	4	2	522	−2
520	519	520	0	2	0	521	52
518	512	515	3	−2	1	515	3
505	510	508	−3	5	3	509	−8
501	486	494	7	5	6	500	5
468	442	455	13	−2	5	461	7
382	401	392	−10	0	−5	387	−5
334	358	346	−12	−8	−10	336	−26
334	346	340	−6	−4	−5	335	−1
359	353	356	3	−4	0	356	3
372	377	374	−2	4	1	375	−3
395	384	390	5	−2	2	392	47
395	396	396	−1	−2	−2	394	52
398	417	408	−10	2	−4	404	−55
439	430	434	5	−8	−2	432	−37
461	475	468	−7	0	−4	464	−3
511	522	516	−5	6	0	516	−5
583	547	565	18	3	8	573	10
583	586	584	−1	10	5	588	2
590	586	588	2	0	0	589	31
590	590	590	0	2	0	591	−13
590	584	587	3	6	4	591	−57
578	553	566	12	−6	4	564	67
516	547	532	−16	6	−4	527	73
516	516	516	0	−8	−4	512	−74
516	516	516	0	6	3	519	−3
516	492	504	12	−4	4	508	26
467	486	476	−9	4	0	474	−7
457	462	460	−3	−4	0	456	1
457	457	457	0	4	4	459	−67
457	434	446	11	−6	4	448	19
412	434	423	−11	6	−2	419	79
412	412	412	0	−6	−2	409	84
412	412	412	0	0	0	412	−2
412	412	412	0	0	0	412	0
412	414	413	−1	0	0	413	3
416	417	416	0	−4	−2	414	−11
422	438	430	−8	4	−3	424	−2
459	444	452	7	−8	0	452	7
467	486	476	−9	6	2	474	−7
512	500	506	6	−3	2	508	4
534	528	531	3	4	2	535	−1
545	540	542	3	2	2	544	8
545		545	0		0	545	0

Examples (3rd line): 422 given, 453 = $\frac{1}{2}$(422 + 484), 438 = $\frac{1}{2}$(422 + 453), −16 = 422 − 438, 3 = $\frac{1}{2}$(0 + 6), −6 = $\frac{1}{2}$(−6 + 3), 432 = (438) + (−6), −10 = 422 (given in A) − 432.

→

exhibit **2** of chapter 16 (continued)

C) SCHEMATIC COMPARISON

(*)	3RSS	3RSSH	3RSS(H,twice)	3RSSH,twice	(start)
	422	422	422	416	569
*	422	422	418	416	416
	422	438	432	432	422
	484	478	479	472	565
	520	511	517	509	484
	520	520	522	521	520
*	520	520	521	522	573
	518	515	515	518	518
	505	508	509	511	501
	501	494	500	497	505
	468	455	460	455	468
	382	392	387	388	382
	334	346	336	340	310
*	334	340	335	335	334
	359	356	356	354	359
	372	374	375	376	372
	395	390	392	393	439
	395	396	394	399	446
	398	408	404	404	349
	439	434	432	430	395
	461	468	464	462	461
	511	516	516	514	511
	583	566	573	568	583
	583	584	588	590	590
	590	588	589	594	620
*	590	590	591	596	578
	590	587	591	591	534
	578	566	564	568	631
	516	532	527	532	600
	516	516	512	516	438
	516	516	519	516	516
	516	504	508	503	534
	467	476	474	474	467
	457	460	456	457	457
	457	457	459	454	392
	457	446	448	444	467
	412	423	419	422	500
	412	412	409	412	493
*	412	412	412	412	410
	412	412	412	412	412
	412	413	413	411	416
	416	416	414	410	403
	422	430	424	422	422
	459	452	452	447	459
	467	476	487	476	467
	512	506	508	513	512
	534	531	535	534	534
	545	542	544	545	552
*	545	545	545	548	545

→

16C. If we want things still smoother

What if "3RSSH,twice" is not smooth enough for us? As exhibit **3** shows, this might well be the case. The general appearance of this 3RSSH,twice smooth is quite smooth--if we allowed ourselves even a slightly free hand, we could make this very smooth very easily--but the details are not yet very smooth. One natural thing to do would be to hann again. Exhibit **4** shows the result--better but not excellent. Since hanning got us about half-way there, perhaps we should just skip-mean our 3RSSH,twice smooth, leaving out the line-mean. Exhibit **5** shows us that this does help, at least in this example. (The numbers are in the last column, headed (>), of panel A of exhibit 4.)

exhibit **2** of chapter 16 (continued)

D) TOPS and BOTTOMS--on lines marked "*" in panel C

3RSS	3RSSH	3RSSH,twice
422	422	416
520	520	523
334	340	335
590	590	596
412	412	412
545	545	548

P) PROBLEMS

2a) Find a sequence at least 20 long of interest to you, and smooth it with 3RSSH,twice.

2b) Do the same with a sequence at least 50 long.

2c) Do the same with a sequence suggested by your instructor.

2d) Starting with the (>) column of panel A, find the corresponding rough, smooth it by 3RSSH, add the result to (>), and plot the sum.

2e) Try 3RSS>-twice applied to the start above (leave out the → columns, and the columns of line means. How do you like the results? Does it do well or poorly in avoiding flat tops?

2e2) Draw a hand-smoothed curve through the points found in (2e). Compare with exhibit 7. Comment on the difference.

2f) Starting with the "Final rough" column of panel A, do another 3RSSH smooth and add the result back to the "sum of smooths" column. Comment.

2f2) Collect the results of the smooth of (2f) according to the values of the last H column of panel A. Comment further.

2g) Take the 3RSSH smooth (left column of panel A) on to a 3RSSH3RSSH smooth, and find its rough. Comment.

2g2) Take the result from 2g and do a 3RSSH smooth on it. What now?

exhibit **3** of chapter 16: coal production

The 3RSSH,twice smooth of bituminous coal production

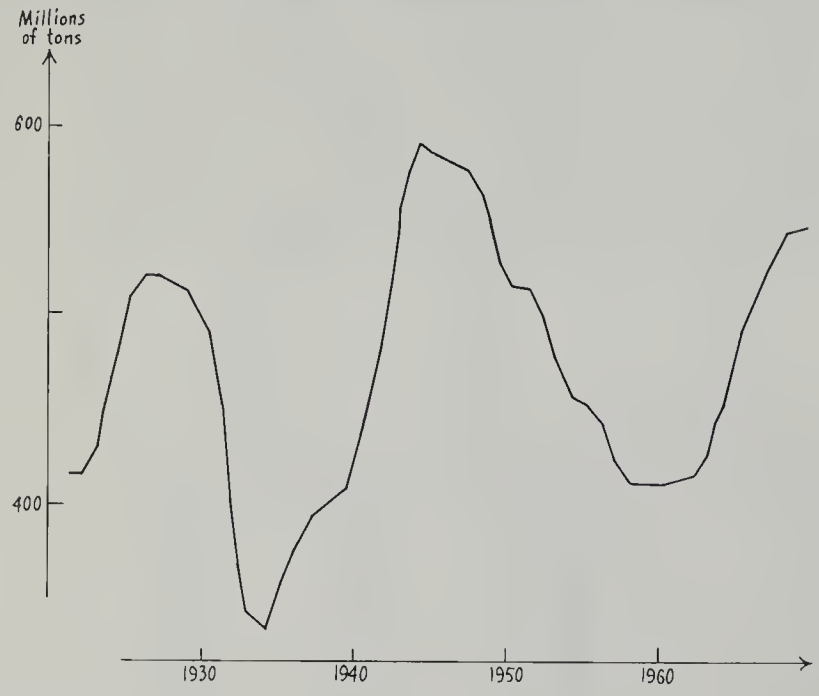
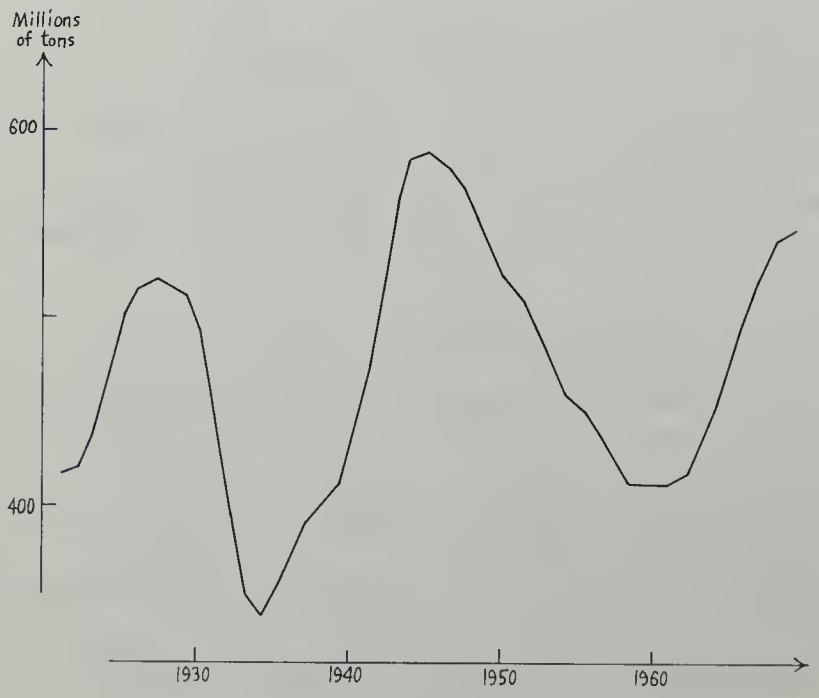


exhibit **4** of chapter 16: coal production

The result of hanning the smooth of exhibit 3



By now we have the smooth so smooth as to make us wonder how close to the data we are sticking. Whenever this worries us, it is time to rerough. Exhibit **6** shows the result of finding the rough that goes with exhibit 5, smoothing it with 3RSSH, and adding the smooth back. Now we have a highly polished smooth which we are sure has not wandered very far from the data.

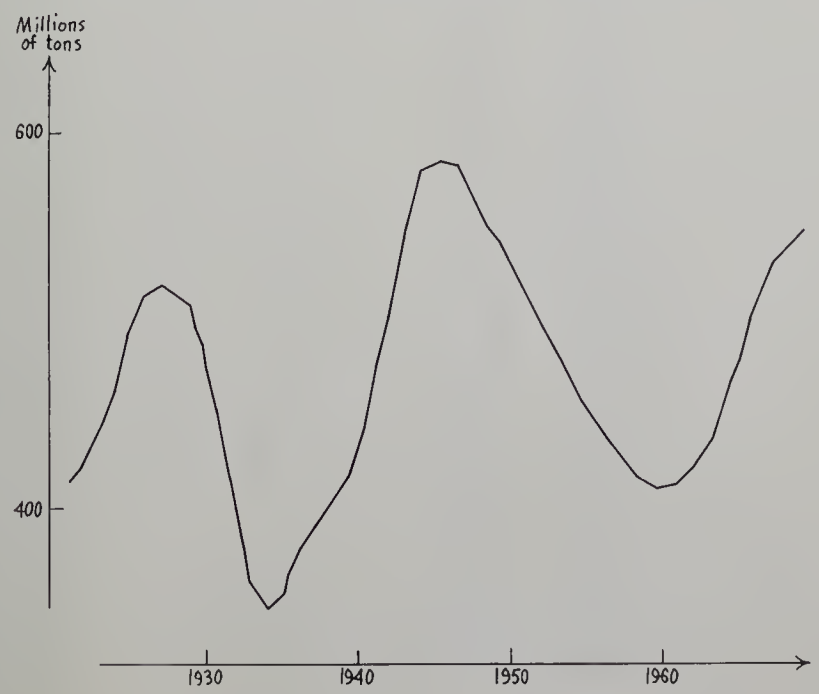
Notice the comment on exhibit 6 about drawing a smooth curve through the points, rather than using a broken line. We could have made exhibit 5 look just as smooth as--maybe even a shade smoother than--exhibit 6, had we drawn a curve through its points. The true difference between exhibit 5 and exhibit 6 is that we are surer that exhibit 6 goes as high on the hills and as low in the valleys as we would like--not that the points are smoother.

review questions

How can we make our smooth smoother? Should we worry how close we then stick to the data? What can we do?

exhibit **5** of chapter 16: coal production

The result of skip-meaning the 3RSSH,twice smooth of exhibit 3



16D. Further possibilities

new components

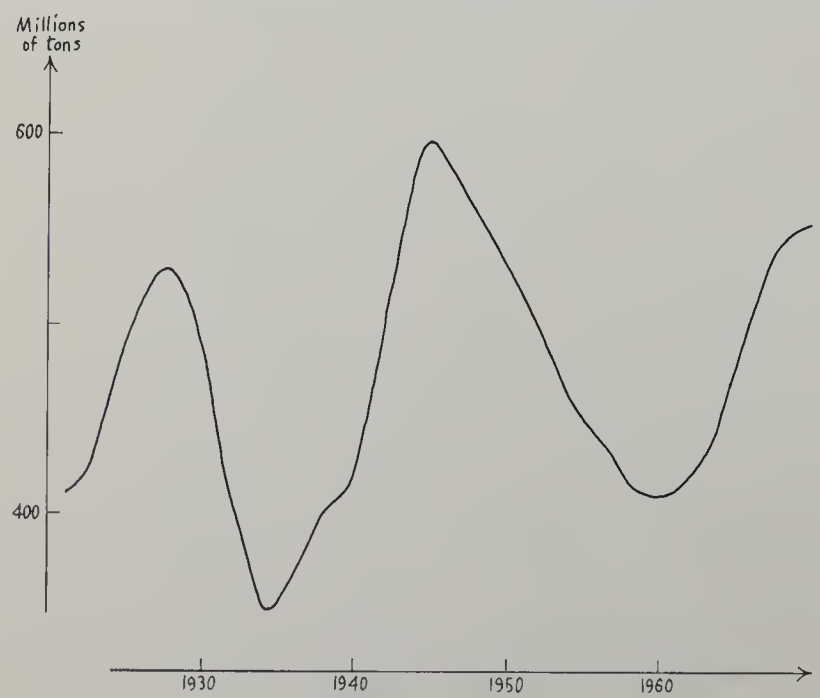
Clearly the pieces from which we have made our smoothers have close relatives. We could use running medians of 5,7,9, or what have you. We could use » for a double-skip-mean--one that looks exactly two steps back and two steps forward--and then use I for double-skip mean followed by line mean, and so on. We shall not need any of these here, though some readers may occasionally find them helpful. Note that, as we lengthen the span of the pieces of our smoothers, the need for reroughing will increase.

heavier smoothers

We learned, even in chapter 7, that one way to make a smooth still smoother was to smooth it some more. We can perfectly well begin with 3RSSH3RSSH3, if we are energetic enough, find its rough, and smooth the rough with 3RSSH3RSSH3. If arithmetic is no effort--perhaps because we have a friendly computer all programmed--we can be serious about

exhibit **6** of chapter 16: coal production

The result of reroughing the smooth of exhibit 5 with 3RSSH. (This time we have drawn a smooth curve through the points instead of a broken line.)



trying "3RSSH3RSSH3,twice" or even "(3RSSH3RSSH3,3RSSH3RSSH3), 3RSSH3RSSH3" which we naturally call "3RSSH3RSSH,thrice".

problems

Exhibit **7** presents some problems. Exhibit **8** presents some optional problems.

exhibit **7** of chapter 16: data and problems

Further data and problems on reroughing

A) DATA before 1776

Year	**Z123** (Klb)	**Z225** (Klb)	**Z254** (Hlb)	**Z350** (s/g)	**Z357** (*)
1775	1122	834	80	—	166.0
4	747	1191	302	2.17	169.7
3	721	964	2063	2.20	165.8
2	747	684	1512	2.19	161.2
1	434	1136	2829	2.19	165.6
0	550	190	859	2.19	154.0
1769	403	203	860	2.16	158.3
8	498	88	2919	2.23	166.4
7	530	44	1524	2.08	166.2
6	492	114	1190	2.23	165.4
5	336	704	1754	2.04	171.6
4	529	765	1432	2.05	172.4
3	439	647	375	2.59	173.1
2	255	2226	516	2.79	175.8
1	384	796	70	3.04	174.1
0	508	989	—	3.54	160.3
1759	696	120	—	3.94	154.7
8	563	273		3.12	159.2
7	876	369		2.74	166.0
6	223	289		2.35	172.5
5	304	241		2.27	168.9
4	130	830		2.44	168.2
3	29	451		2.47	168.0
2	4	83		2.39	166.7
1	20	162		2.51	170.6
0	63	12		2.53	171.1
1749	138	321		2.72	172.4
8	62	393		3.60	174.3
7	138	287		3.62	184.6
6	—	81		2.69	179.2
5				2.65	175.7
4		35		2.51	167.4
3		515		2.36	160.3
2		558		2.84	159.7
1		70		2.46	145.2
1740		49		1.81	160.1

→

exhibit 7 of chapter 16 (continued)

B) FIRST SECTIONS (to 1895 or 1920) of TEN SEQUENCES

Year	J16 (H/)	M259 (/)	P196 (M)	X56 (M$)	X286 (M$)	Year	D606 ($/y)	M117 (M$)	M251 (Kt)	X274 (M$)
1860	—	—	—	—	—	1884	—	4.91	—	—
1	—	—	—	—	—	5	—	5.1	—	—
2	—	—	—	—	—	6	—	6.8	—	—
3	8.2	—	—	10	—	7	—	7.6	—	—
4	8.4	—	—	147	—	8	—	8.5	—	—
5	8.9	—	—	614	—	9	—	10.3	—	—
6	15.4	—	—	695	10.5	1890	560	11.5	—	—
7	17.0	—	—	685	18.7	1	554	10.4	—	—
8	23.7	—	—	745	17.6	2	563	12.0	—	14
9	25.6	—	—	716	30.0	3	563	9.5	—	14
1870	34.0	211	16	706	32.1	4	546	9.2	—	14
1	39.8	210	20	791	17.8	1895	546	13.3	—	13
2	38.7	223	24	805	26.4	6	544	11.8	1	15
3	31.6	263	28	836	34.3	7	543	13.3	1	16
4	29.3	260	35	828	18.0	8	542	16.0	1	53
5	20.7	260	59	897	17.5	9	543	19.7	2	76
6	25.1	254	113	812	24.2	1900	548	20.5	3	99
7	18.1	244	147	818	32.3	1	549	21.8	3	99
8	35.6	260	210	814	24.9	2	562	25.4	3	124
9	41.0	389	371	1090	15.3	3	593	25.3	3	147
1880	47.3	280	533	1085	8.0	4	600	23.7	4	110
1	37.0	410	595	1364	5.8	1905	589	32.2	5	75
2	45.3	502	599	1365	5.0	6	607	36.4	7	89
3	56.6	593	844	1337	59.8	7	661	40.8	8	180
4	55.0	762	920	1233	71.1	8	667	16.0	5	130
5	60.9	574	1080	1420	126.7	9	644	39.3	15	70
6	61.6	530	1607	1459	76.0	1910	677	41.7	18	54
7	52.0	535	1865	1650	91.2	1	705	35.6	19	48
8	46.2	728	2212	1716	121.1	2	721	44.0	21	58
9	42.2	668	2413	1920	117.1	3	760	46.3	24	49
1890	40.2	733	2605	1979	130.8	4	795	34.6	29	66
1	37.6	956	3137	1974	120.1	1915	815	41.6	45	48
2	55.1	991	3282	2327	141.1	6	817	54.5	58	140
3	48.4	958	3661	1939	92.6	7	989	55.6	65	1102
4	56.6	958	3621	2228	66.3	8	1424	56.5	62	1631
5	37.3	1142	4238	2279	48.4	9	1509	44.2	64	1016
6			4967	2141	42.2	1920	1817	51.3	69	329

→

exhibit **7** of chapter 16 (continued)

C) SECOND SECTIONS (from 1921 on) of the same TEN SEQUENCES

Year	J16 (H#)	M259 (#)	P196 (M)	X56 (M$)	X280 (M$)	D606 ($/y)	M117 (M$)	M251 (Kt)	X274 (M$)
1921	43.8	1995	52,770	15142	200.6	1032	25.3	27	461
2	29.3	1984	56,413	16323	173.3	1591	37.1	37	203
3	18.9	2462	67,239	16899	386.5	1585	57.0	69	361
4	13.9	2402	73,256	18349	801.4	1570	44.3	75	232
5	11.0	2234	82,712	19912	1004.8	1597	51.3	70	205
6	10.4	2518	92,523	20644	1057.4	1613	56.9	74	239
7	10.5	2231	100,260	21778	1007.1	—	51.1	82	256
8	10.4	2176	109,131	22645	1019.1	—	52.8	105	295
9	11.6	2187	122,822	21586	935.0	—	59.9	114	417
1930	12.7	2063	124,193	23235	995.0	—	48.0	115	345
1	12.6	1463	117,402	22164	996.5	—	33.5	89	486
2	10.6	1207	106,915	17428	715.7	—	21.8	52	422
3	7.53	1064	115,087	16742	265.5	—	27.6	43	887
4	7.51	1226	130,287	19896	149.7	—	31.8	37	1797
5	3.30	1242	190,147	22477	117.2	—	35.1	60	913
6	1.21	1342	159,076	26153	100.8	—	46.3	112	1832
7	.56	1413	170,171	26716	88.1	—	52.4	146	759
8	.45	1105	171,842	26763	78.5	—	32.5	143	1459
9	.38	1078	180,828	29416	71.9	—	44.3	164	1736
1940	.35	1388	189,508	33014	66.8	—	57.0	206	1062
1	.40	1266	218,083	37273	62.9	—	65.2	309	1733
2	.28	1471	257,657	40533	59.4	—	70.6	521	2127
3	.21	1451	296,305	54590	56.9	—	71.7	920	8503
4	.16	1298	323,734	65585	54.0	—	74.0	776	20156
5	.18	1068	332,345	76534	52.1	—	67.3	495	24980
6	.14	968	350,132	80212	50.2	—	58.5	410	14249
7	.47	1158	369,763	77146	47.8	—	73.4	572	2123
8	.64	999	386,916	78753	45.2	—	74.9	623	4108
9	.68	585	385,046	7849	42.7	—	63.6	603	2742
1950	.52	643	392,025	82430	40.8	—	72.7	719	4751
1	.36	785	418,872	86589	39.1	—	79.3	837	6649
2	.46	548	435,616	92719	32.9	—	68.3	937	6454
3	.48	461	423,129	94475	36.6	—	78.8	1252	4074
4	.47	396	401,849	99358	35.5	—	59.7	1461	6770
5	.48	420	412,309	98631	34.5	—	75.3	1566	5796
6	.45	448	424,247	100820	33.5	—	75.5	2679	6059
7	.66	—	442,328	100972	32.5	—	—	—	4098

→

exhibit **7** of chapter 16 (continued)

D) TWELVE SEQUENCES at 5-year INTERVALS

	(100 log sum of 5 years*)								(sum of 5 years*)			
Year	**C97** (H log)	**C99** (H log)	**C102** (H log)	**C123**† (H log)	**C124**† (H log)	**E101** ($/b)	**E109** (R/Hp)	**E111** ($/g)	**U6** (M$)	**U9** (M$)	**U182** (M$)	**W74** (//)
1975	—	—	—	—	—	—	—	—	—	—	16	—
1800	—	—	—	—	—	1.82	10.7	2.50	—	—	−3	—
5	—	—	—	—	—	1.95	10.5	3.61	—	—	19	—
10	—	—	—	—	—	1.80	9.5	3.94	—	—	−10	—
5	—	—	—	—	—	1.56	12.5	4.48	—	—	5	—
20	—	—	—	(1.29)	(1.31)	.93	9.8	2.37	—	—	−108	—
5	1.61	2.40	—	3.74	3.35	.92	7.3	.40	—	12	6	—
30	1.53	2.20	—	3.94	3.89	1.07	5.5	.29	5	−17	−3	—
5	2.37	3.28	—	4.49	4.15	1.22	6.0	.55	11	−30	−103	—
40	1.71	2.55	—	9.76	4.59	1.06	5.5	.27	11	−13	130	—
5	2.35	2.83	—	4.80	4.77	1.04	4.8	.40	−9	0	7	44
1850	2.52	2.63	—	5.27	5.32	1.28	3.7	.33	−11	0	−14	297
5	1.32	3.50	—	5.47	5.59	2.44	4.1	.43	169	2	−213	709
60	2.64	3.54	—	5.07	5.19	1.50	3.1	.42	256	10	−66	1321
5	2.80	3.54	—	4.83	5.25	2.16	7.1	1.52	205	4	−380	2194
70	3.27	3.46	—	5.18	5.54	1.37	4.4	.43	193	70	−590	2307
5	4.19	4.43	1.15	5.17	5.57	1.40	3.4	.35	205	111	−688	4335
80	4.38	4.46	1.72	5.04	5.02	1.06	3.7	.38	−61	47	656	3196
5	4.79	5.10	.70	5.36	5.83	.86	2.3	.35	−105	52	23	4359
90	5.18	5.30	3.35	5.14	5.81	.89	3.0	.41	17	66	−776	3983
5	5.39	5.46	4.12	5.20	5.76	.60	2.1	.29	230	100	−258	4585
1900	5.41	5.56	4.33	5.07	5.65	.70	2.6	.48	118	134	1973	8757
5	5.82	5.98	4.48	5.64	6.09	1.01	1.9	.63	7	113	1613	3943
10	5.97	6.04	4.67	6.07	6.03	1.10	1.9	.68	−74	70	1091	3279
5	5.95	5.97	4.86	6.00	5.90	1.29	1.7	.46	−13	114	2709	7271
20	4.45	5.23	3.85	4.86	5.36	2.46	4.2	1.73	−535	426	17426	8458
5	4.88	5.57	4.27	5.05	5.64	1.67	2.8	1.01	−1323	49	6399	12290
30	4.15	4.92	2.58	4.99	5.27	.90	2.2	.47	−167	93	5456	13781
5	3.55	4.54	2.42	3.65	3.97	1.04	2.6	.50	2399	471	2010	15071
40	3.59	4.53	1.82	3.30	3.99	.87	2.6	.37	12955	600	4488	25455
5	3.11	2.97	2.05	2.91	3.58	1.66	2.8	.79	416	91	38354	18880
1950	3.48	4.75	2.03	3.81	4.32	2.22	6.3	.53	−4171	269	34813	18016
5	3.43	4.86	2.06	4.30	4.68	2.26	8.2	.64	−249	382	24243	12642

*** 100 times the log to the base 10 of sum for 5 years.**
† Values for every 5th year.

→

exhibit 7 of chapter 16 (continued)

E) TWELVE SEQUENCES FOR 1920–1957

Year	C123 (#)	C124 (#)	D47 (%)	J194 (in)	L147 (Mlb)	L172 (Mlb)	M215 (t)	U6 (M$)	U9 (M$)	W77 (H#)	V99 (%)	Y268 (M$)
1920	15257	83496	4.0	22.3	119	—	103	−95	26	103	9.9	104
1	32400	162859	11.9	27.5	59	—	—	667	−12	116	6.5	154
2	10529	33797	7.6	37.7	93	—	—	238	8	129	7.4	139
3	25905	86617	3.2	46.4	159	—	114	−294	−2	148	6.7	127
4	27492	112344	4.4	21.8	243	—	269	−258	36	157	7.4	103
25	16022	36610	4.0	31.2	315	—	566	134	35	138	8.2	169
6	17390	45199	1.9	31.6	287	20	658	−98	23	150	8.0	119
7	23698	55969	4.1	20.6	342	30	554	−6	21	146	7.9	101
8	24161	37904	4.4	36.8	420	30	575	392	19	141	8.2	60
9	19849	21873	3.2	38.4	652	36	395	−175	19	145	7.8	62
1930	13736	18080	8.7	26.9	449	48	334	−280	11	132	4.0	65
1	3422	4806	15.9	37.8	300	29	668	−145	−2	114	−1.5	48
2	254	1157	23.8	42.7	312	29	188	446	−6	96	−5.0	47
3	134	887	24.9	29.7	510	35	426	173	−41	91	−9.6	34
4	233	1154	21.7	321	1146	47	975	−1134	−86	114	5.2	113
35	408	1355	20.1	33.2	1168	60	1140	−1739	−336	109	5.1	212
6	324	1195	16.9	34.9	1503	64	1243	−1117	−171	107	10.0	379
7	378	1904	14.3	23.3	1140	76	1666	−1586	−80	112	7.1	306
8	609	2411	19.0	21.1	1110	66	1448	−1974	−223	102	6.1	417
9	415	2070	17.2	16.7	1241	87	2040	−3574	−71	105	7.4	361
1940	252	2120	14.6	33.0	914	100	2531	−4744	−55	98	7.0	360
1	129	732	9.9	47.5	1328	62	3125	−982	−41	85	7.5	407
2	92	493	4.7	40.0	975	52	4441	−316	−39	68	8.6	433
3	164	681	1.9	33.6	997	56	5684	−69	3	56	9.1	447
4	203	1030	1.2	27.4	1147	60	4893	845	104	60	10.0	511
45	225	886	1.9	25.7	850	89	2633	106	64	75	11.0	643
6	189	1473	3.9	37.1	531	94	2471	−311	−21	81	10.1	677
7	442	381	3.6	19.3	272	117	1472	−1866	−37	90	8.6	779
8	946	4826	3.4	19.9	373	140	1919	−1680	−58	115	7.6	899
9	933	6192	5.5	35.5	634	141	1316	−686	50	160	8.2	797
1950	3976	5693	5.0	13.9	715	176	2294	372	−103	168	8.7	706
1	4972	5481	30	25.5	329	159	2986	550	−94	174	7.8	730
2	6289	8969	2.7	32.2	14	180	3622	−684	−62	161	8.2	833
3	1538	5369	2.5	19.3	9	189	4564	−2	−86	166	7.9	891
4	1622	10061	5.0	15.4	137	207	6515	−16	−75	159	9.6	935
5	5486	17518	4.0	19.4	146	196	7810	−97	−65	182	8.1	936
6	9050	27807	3.8	19.3	70	233	7014	−106	−122	208	7.9	1171
57	4585	21826	4.3	40.1	—	—	—	−104	−147	175	8.3	1378

→

exhibit **7** of chapter 16 (continued)

I) IDENTIFICATION of SEQUENCES for PANEL A

Z123: Indigo exported from South Carolina in thousands of pounds.
Z225: Tobacco imported by England from the Carolinas in thousands of pounds.
Z254: Tea imported from England to the American colonies in hundreds of pounds.
Z350: Average price of New England rum in Philadelphia in shillings per gallon.
Z357: Average amount of Pennsylvania currency for 100 pounds sterling.

J) IDENTIFICATION of SEQUENCES for PANELS B and C

J16: U.S. homestead entries, except on ceded Indian lands, in hundreds.
M259: Number of fatal injuries in U.S. coal mining.
P196: Millions of cigarettes produced in U.S.
X56: Total liabilities of national banks in millions of dollars.
X286: Gold certificates in circulation in millions of dollars.
D606: Average annual earnings of wage earners in steam railroading in dollars per years.
M117: Coke production in millions of short tons.
M251: Primary production of aluminum (all ores) in thousands of short tons.
X274: U.S. Government deposits in all banks in millions of dollars.

K) IDENTIFICATION of SEQUENCES in PANEL D

C97: Number of immigrants from USSR and Baltic states.
C99: Number of immigrants from Italy.
C102: Number of immigrants from Turkey in Asia.
C123: Number of farm laborers and foremen (for years before 1899, **C129** "Farmers"). (As immigrants.)
C124: Laborers, excluding farm and mine. (For years before 1899, **C131** "Laborers".) (As immigrants.)
E101: Wholesale price of wheat in dollars per bushel.
E109: Wholesale price of nails in dollars per hundred pounds.
E111: Wholesale price of turpentine in dollars per gallon.
W74: Number of design patents issued.

L) IDENTIFICATION of SEQUENCES in PANEL E

C123: Number of immigrants: Farm laborers and foremen.
C124: Number of immigrants: Laborers, except farm and mine.
D47: Unemployment as a percent of the civilian labor force.
J194: Annual precipitation in inches at Beeville Experiment Station, Texas.
L147: Landed catch of sardines, Pacific Coast states, in millions of pounds.
L172: Production of canned tuna and tuna-like fish, in millions of pounds of net weight.
M215: Output of tungsten concentrates, in short tons.
U6: Excess of exports (+) or imports (−) of gold in millions of dollars.
U9: Excess of exports (+) or imports (−) of silver, in millions of dollars.
W77: Number of trade marks registered, in hundreds.
X199: Net profits of national banks as percent of total capital accounts.
Y268: Estate and gift taxes, in millions of dollars.

→

review questions

Are these close relatives of the pieces we are now accustomed to put together to make a smoother? How could we try to make our smooths still smoother? Need we allow it to get away from the data? Why/why not?

exhibit **7** of chapter 16 (continued)

P) PROBLEMS

7a/b/c/d/e) Analyze sequences **Z123/Z225/Z254/Z350/Z357** from panel A.

7f/g/h/i/j/k/l/m/n) Analyze sequences **J16/M259/P196/X56/X286/D606/M117/M251/X274** from panels B and C.

7o/p/q/r/s/t/u/v/w/x/y/z) Analyze sequences **C97/C99/C102/C123/C124/E101/E109/E111/U6/U9/U182/W74** from panel D.

7aa/ab/ac/ad/ae/af/ag/ah/ai/aj/ak/al) Analyze sequences **C123/C124/D47/J194/L147/L172/M215/U6/U9/W77/X199/Y268** from panel E.

7am) Analyze the difference of sequences **C97** and **C99**.

7an) Analyze the ratio of sequences **C123** to **C124**.

More: Other sequences in the same source that could be considered for analysis include **C101, C107, E103, E105, M225, M249, P190, Q185, Q186, X215, X274**. In the earlier (1945) version of the source (numbering unrelated), the following, among others, could be considered: **E114, F193, G102, J169, N21, P66, P67,** the ratio **P66/P67, P111,** App. 30 (Aug).

S) SOURCE

Historical Statistics of the United States: Colonial Times to 1957. **U.S. Dept. of Commerce, 1960.**

exhibit **8** of chapter 16: data and problems

Some optional problems for the energetic (and the computerized)

8a/b/c/d/e/f/g/h/i/j/k/l) Turn to the source mentioned in section S of exhibit 7 and analyze the full sequence for **C97/C99/C102/C123/C124/E101/E109/E111/U6/U9/U182/W74.**

8m/n/o/p/q) Turn to the source mentioned in section S of exhibit 7 and analyze the full sequence for **J16/M259/P196/X56/X286.**

8r) Find a long sequence of interest to yourself, and analyze it.

16H. How far have we come?

The basic idea in the more careful smoothing is that better roughs require more care in taking what might be smooth away from the rough (and putting it in the smooth). Both idea and process are somewhat complementary to what we have already done to make smooths better, namely, to search harder in the smooth for what might be rough (and then move it to the smooth).

We are now ready to:

◇clean up our roughs by smoothing them again, adding what we get out to the original smooth ("twicing", "reroughing").

We now understand more clearly that:

◇whenever we are concerned about a smooth sticking closely enough to the input, we should smooth the corresponding rough, ending up by re-roughing the smooth.

◇more powerful smoothing components are available if and when we need them.

COUNTS in BIN after BIN 17

chapter index on next page

We are now starting a group of chapters concerned with how values are distributed. This chapter approaches one of the most general cases, one where we are endeavoring to learn how the values are distributed for a sequence of counts, one for each of a system of cells or bins (such as, for example, the counts of family units with 0, 1, 2, 3, . . . children, or the counts of NBA basketball players with heights falling in each inch of height).

We want to begin by leaning only on the given data. To this end, we find that things we have already learned something about (going at least to square roots of counts, using the improved smoothing techniques of the last chapter) help us considerably.

Having a good smooth, and having had a hard look at a good rough, we will then be ready to try to make simple fits to our smooth. In this chapter--and throughout the book--we shall confine ourselves, with a few exceptions, to fits symmetrical around a peak. We find that these do well for us here.

review questions

What new area are we beginning on? How general a case do we consider in this chapter? What is it? What will we try to do first? Will earlier chapters help us? What will we try next? Under what limitations?

17A. Root smooth and root rough--bins of equal width

We are now ready to begin dealing with strings of counts. How do such strings arise? Usually because the possible values for some observation--the height of a basketball player, the occurrence of a flash of light, the number of books in a home--are divided up into intervals--often called cells or bins--and a record is kept of how many events fall into each bin. (We use "bin" instead of "cell" because it is one letter shorter and has fewer alternative meanings.)

Often the bins are thought of as equal in width. We would surely do this, for example, when we binned basketball players by the inch of height. Equal widths is the easy case to begin with, and will be the subject of this section.

The general principle still applies:

Since we are dealing with COUNTS, we always START by at least taking (square) ROOTS.

We begin with counts based on measurements.

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15		560
	17D	
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17		563
18★		564
19		566
20★		567
21		569
	17E	
22★		570
23		572
24★		573
25		576
	17F	
26★		578
27★		579
28		581
	17G	
29★		582
30		585
31		585
32		586
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lengths of forearms

Exhibit **1** shows the arithmetic for data of Pearson and Lee for the lengths of forearms of 1050 fathers. The smoothing calculations are by now standard. (See Chapters 7 and 16.) We have added two lines of zeros above and below the lines with nonzero counts; we could do this since shorter or longer forearms than observed are surely possible.

exhibit **1** of chapter 17: forearm lengths

Smoothed root and its rough for Pearson and Lee's data on forearm lengths of 1050 fathers

A) DATA, ROOTS, SMOOTHS and ROUGHS

Bin #	Lengths (inches) of forearm	Count of fathers Raw	Root	3RSS	3RSSH	1st rough	3RSSH	Back add*	"Final rough"
		0	0		0	0	0	0	0
		0	0		.2	−.2	0	.2	−.2
1	15–15.5	1	1.0		1.1	−.1	−.1	1.0	0
	15.5–16	6.5	2.5		2.6	−.1	−.1	2.5	0
3	16–16.5	17	4.1		4.4	−.3	−.1	4.3	−.1
	16.5–17	49	7.0		7.3	−.3	−.1	7.1	−.2
5	17–17.5	125.5	11.2		10.9	.3	.2	11.1	.1
	17.5–18	200	14.1	14.1	13.4	.7	.5	13.9	.2
7	18–18.5	235.5	15.3	14.1	14.0	1.3	.6	14.6	.7
	18.5–19	183.5	13.5		13.1	.4	.4	13.5	0
9	19–19.5	125.5	11.3		11.0	.3	.3	11.3	0
	19.5–20	57.5	7.6		8.0	−.4	.2	8.2	−.6
11	20–20.5	31.5	5.6		5.4	.2	0	5.4	.2
	20.5–21	8	2.8		3.3	−.5	−.1	3.2	−.4
13	21–21.5	3.5	1.9		2.0	−.1	−.1	1.9	0
	21.5–22	2	1.4	1.6	1.6	−.2	−.1	1.5	−.1
15	22–22.5	2.5	1.6	1.4	1.1	.5	0	1.1	.5
		0	0		.4	−.4	0	.4	−.4
		0	0		0	0	0	0	0
	(Sum)	(1050)							

*** "Back add" is final smooth = sum of both 3RSSH columns. "Final rough" = Root MINUS Back add.**

Note: Half counts arise from dividing borderline fathers, putting one-half in each bin.

B) LETTER VALUES for ROUGH--"zero" line at each end included

#17

M8	0				
H4h	.1	−.2	.3		
E2h	.2	−.4			

	.4h	
f	.5h	−.6h
	one	xxx
F	1.0	

The results are pictured in exhibits **2** (the smooth) and **3** (the rough). Notice the smoothness and near symmetry of exhibit 2. Note the way in which fences, eighths, hinges, and median are shown in exhibit 3. The horizontal marks nearest the rough are from panel B of exhibit 1; those further away are from a set of reference values for any root rough, namely:

$$M = 0$$
$$H = \pm .34$$
$$E = \pm .58$$
$$f = \pm 1.36$$

Experience suggests that the actual behavior of such a rough will be somewhat more compressed than these standard values––but not as much more compressed as in this example, which we can thus suspect of being selected to give an unusually good fit.

symmetry

We clearly make it easier to describe a pattern of counts if that pattern is symmetrical. If a simple re-expression will do this for us, it is almost certain to be worthwhile to use it. In practice, a rather substantial fraction of all sets of measurements will need re-expression.

Chemical concentrations that are neither too large nor too restricted in range are a good example, whether the concentrations are in rocks, water, blood, grain, or tissue. It should become almost automatic to try logs on any such concentration.

exhibit **1** of chapter 17 (continued)

P) PROBLEMS

1a) Make a stem-and-leaf for the final rough and check the letter-value display given in panel B.

1b) Rerough with 3RSSH one more time. Are the residuals improved? Why/why not?

1b2) Do a letter-value display for the rough of 1b and compare with panel 3. Comment

1c) Check the first 3RSSH of panel A.

1d) Check the second 3RSSH of panel A.

1e) Do a modified (section 7E) letter-value display. Comment.

S) SOURCE

K. Pearson and A. Lee 1903. On the laws of inheritance in man. I. Inheritance of physical characters. *BMTA 2:* 357–462. Text table at page 367

concentrations of gold

Exhibit 4 deals with 1536 measurements of gold concentrations — expressed as pennyweight (dwt) values (per ton) — in assays of samples from the City Deep mine, Central Witwatersrand, South Africa, as reported by Koch and Link.

Exhibits 5 and 6 show the smooth and the rough, respectively. We can see essentially nothing about symmetry, since the smooth we have does not even

exhibit 2 of chapter 17: forearm lengths

The smooth (smoothed roots of fathers per half-inch for the Pearson and Lee data; based on exhibit 1)

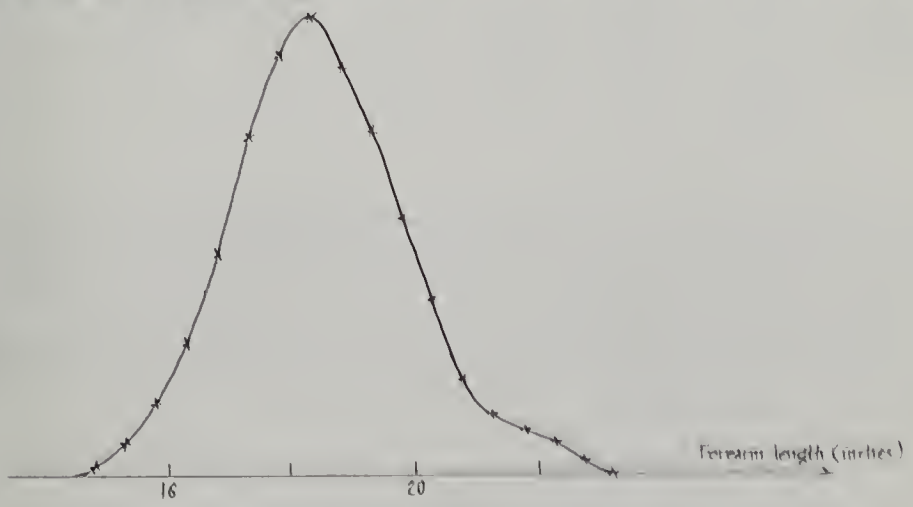


exhibit 3 of chapter 17: forearm lengths

The rough—in a standard format (for the smooth of exhibit 2)—(includes ONE zero line at each end)

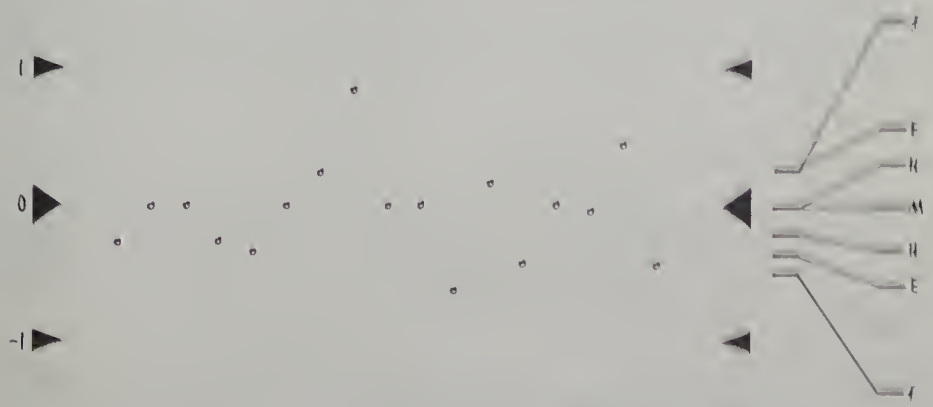


exhibit 4 of chapter 17: gold assays

Smoothed roots and their rough for 1536 gold values (in dwt) for the City Deep mine

A) DATA, ROOTS, SMOOTHS and ROUGHS

Bin #	Interval	Count of samples	Root of count	3RSSH	First rough	3RSSH	Add back	Final rough
	(0 to 5)	(910)						
1	5 to 10	208	14.4	14.4	0	0	14.4	0
2	10 to 20	198	14.1	13.4	.7	.2	13.6	.5
3	20 to 40	124	11.1	10.8	.3	.3	11.1	0
4	40 to 80	52	7.2	7.8	−.6	.2	8.0	−.8
5	80 to 160	34	5.8	5.4	.4	0	5.4	.4
6	160 to 320	8	2.8	3.2	−.4	0	3.2	−.4
7	320 to 6400	2	1.4	1.4	0	0	1.4	0
8			0	.4	−.4	0	.4	−.4
(Sum)		(1536)	0	0	0	0	0	0

Notes

1. We cannot add even one line at the low side, analogous to the two added at the high side, because we do not know how many of the 910 samples assaying between 0 and 5 actually assayed between 2.5 and 5.0. (The data was given in intervals of length 5 for raw assays.)
2. Using intervals corresponding to doubling of the raw values, conveniently called octaves, is often convenient when raw measurements are given and log measurements are to be analyzed.

B) LETTER VALUES for FINAL ROUGH

#8					.8	
M4h	0					
H2h	.2	−.4	.6	f	1.1	−1.3
E1h	.4h	−.6	1.0h		xxx	xxx

P) PROBLEMS

4a) S. B. Flacker and H. A. Brischle, 1944. "Measuring the local distribution of *Ribes*," *Ecology* 28: 288–303; (Figure 3 on page 29) give counts of 2.5 acre blocks (in Clearwater National Forest), showing different numbers of *Ribes* plants as follows:

Plants:	0	1	2 or 3	4 to 7	8 to 15
# Blocks:	4	3	7	4	8

Plants:	16 to 31	32 to 63	64 to 127	128 to 255	256 up
# Blocks:	8	16	6	4	4

Analyze as above.

S) SOURCE

George S. Koch, Jr. and Richard F. Link, 1970. *Statistical Analysis of Geological Data*, John Wiley and Sons, New York. Table 6.5 at page 216.

reach the peak. Nevertheless the rough (i) looks just reasonably irregular and (ii) has its letter values spread out about as we might expect--somewhat more pulled in than the standard letter values, but not nearly as much less as in exhibit 3. So far so good.

exhibit 5 of chapter 17: gold assays

The smooth (of root counts)

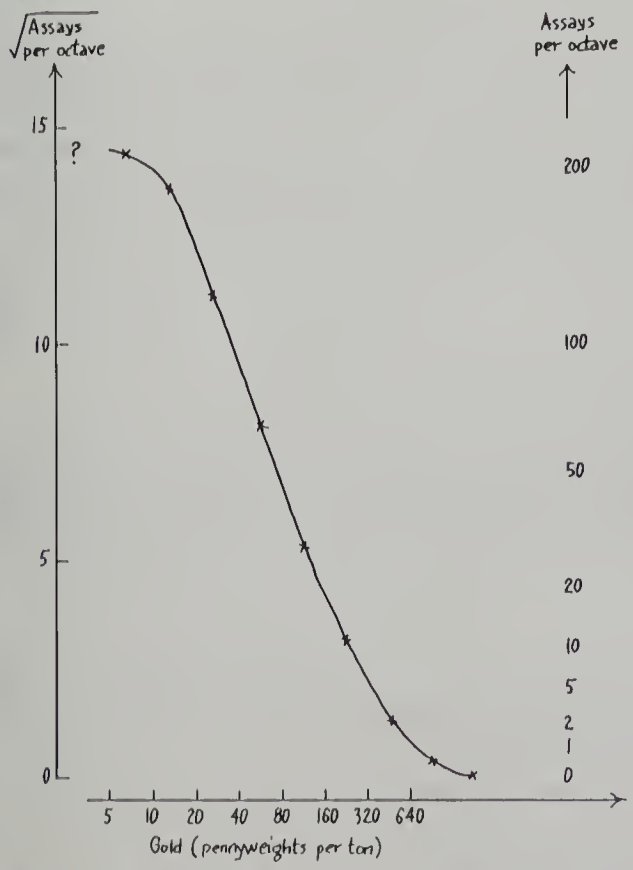
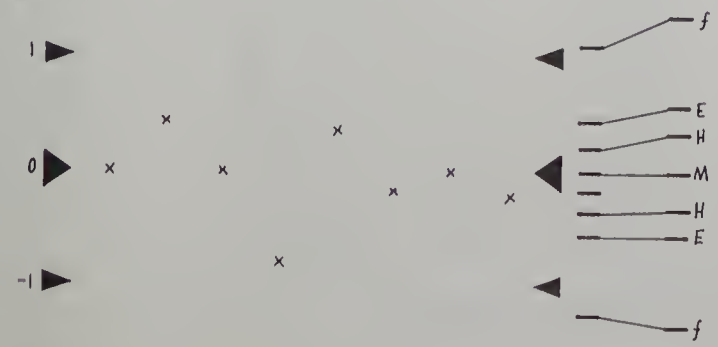


exhibit 6 of chapter 17: gold assays

The rough (for the smooth of exhibit 5)



review questions

How do strings of counts arise? Why do we say "bin"? What is the general principle for dealing with counts? What example did we start with? What did we do with it? What did we see? What is our standard format for roughs associated with smoothed roots? How should we use our reference values for fEHMHEf? Is symmetry useful in a smooth? Why/why not? What do we expect to do with chemical concentrations? What example was next? Did we treat it differently? Why/why not? What did we see?

17B. Counts of basic counts

One case where the bins seem made for us arises when the observations are themselves counts--number of books per home, number of children per family, number of flashes of light in an eighth of a minute. Since we now have two kinds of counts, it pays us to be careful, calling the observations "basic counts" so that we are dealing with "counts of basic counts".

Sometimes we can go right ahead taking each basic count as a bin, and treating each bin as of equal width. (In a later section of this chapter we will learn an alternative approach.)

corn borers in field corn

Exhibit 7 shows the calculation for the numbers of corn borers (pupae and 5th instar larvae) in 3205 plants of field corn, as counted by McGuire, Brindley, and Bancroft. We have to be a little careful at the low end, because there CANNOT be LESS than zero borers in a plant. There MIGHT have

exhibit 7 of chapter 17: field corn

Smoothed roots and their rough for 3205 counts of borers--pupae and 5th instar larvae only--in plants of field corn

A) DATA, ROOTS, SMOOTHS and ROUGHS

	Root of counts			Final	
Counts	raw	smoothed	twiced	rough	(letters values)
588	24.2	24.2	25.5	−1.3	
807	28.4	26.4	27.7	.7	
741	27.2	25.9	27.0	.2	(−1.7h f)
479	21.9	22.3	22.9	−1.0	(−1.0 E)
328	18.1	17.6	17.6	.5	(−.4h H)
159	12.6	12.9	12.6	0	(.0h M)
67	8.2	8.4	8.1	.1	(.4 H)
22	4.7	5.0	4.7	0	(.7 E)
5	2.2	3.0	2.8	−.6	(.7 f)
7	2.6	2.1	2.2	.4	(out: xxx)
2	1.4	1.2	1.4	.0	
0	0	.4	.4	−.4	
(3205)				(0)	

been fathers with forearm lengths between 14.5 and 15 inches, so we could and should adjoin a 0 below our first count in exhibit 1. We cannot and should not adjoin 0 at −1 in exhibit 7.

Plotting the rough is left for the reader to do. This time the rough is about the size it ought to be--perhaps even a shade larger. (If the low residuals at basic counts of 0 and 3 were strays, we could sense a trend, but we have no reason to believe they are strays.) Exhibit **8** shows the smooth. This time we have lost not all, but most, of one side of a possibly symmetric smooth.

exhibit **7** of chapter 17 (continued)

P) PROBLEMS

7a) Try applying 3RSSH to the final rough of panel A, and adding back again.

7a2) What are the letter values now? How do they compare with (i) the reference values, (ii) the letter values of panel A?

7a3) Plot the corresponding rough.

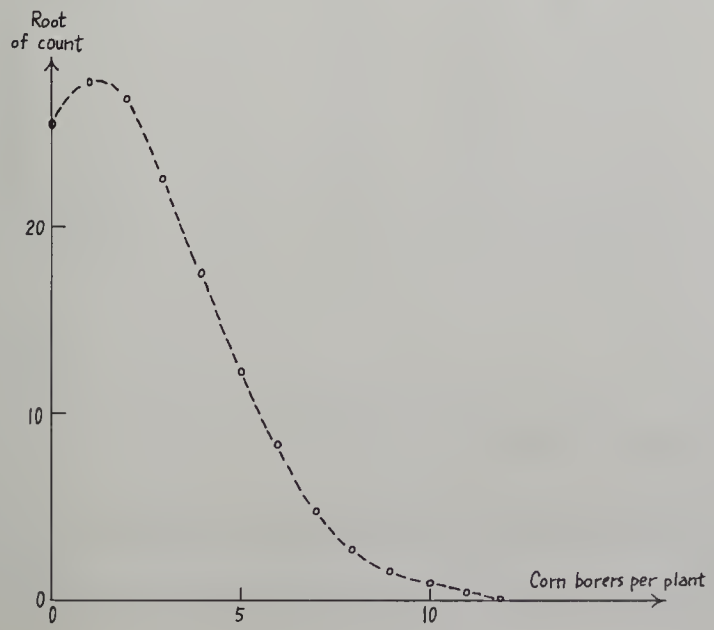
7b) McGuire *et al.* also give the results of counting total borers per plant. The bin counts, beginning with bin zero, were: 355, 600, 781, 567, 411, 2105, 135, 42, 17, 11, 11. Copy the analysis for this data.

S) SOURCE

J. U. McGuire, T. A. Brindley, and T. A. Bancroft (1957). The distribution of European corn borer larvae *Pyrausta nubilalis* (Hbn.), in field corn. *Biometrics 13:* 65–78. Distribution 2 on page 75.

exhibit **8** of chapter 17: field corn

The smooth for exhibit 7



Sometimes, when we are dealing with counts of basic counts, it is not satisfactory to use equal widths for the bins. This usually happens because the pattern, if we did this, would be very long-tailed and hard to use. To avoid this, we often need to go over to roots of basic counts or logs of basic counts. When we go over to roots, we usually need to use unequal widths (in roots) for our bins, something we have not yet come to. (We will give an example shortly.)

octaves for logs

When we go to started logs of basic counts, however, we can have equal intervals by working in octaves. Labelling the bins 1, 2 to 3, 4 to 7, 8 to 15, 16 to 31, and so on, corresponds to making cuts at 0.5, 1.5, 3.5, 7.5, 15.5, 31.5, and so on. For "counts plus 0.5" the cuts are now at 1, 2, 4, 8, 16, 32, and so on, so the bins for "log of (count plus 0.5)" are of equal widths. Doing this is convenient, and often works well.

exhibit **9** of chapter 17: valley birds

Smoothed roots and their rough for numbers of breeding pairs of birds by octave, Quaker Run Valley, New York

A) DATA, ROOTS, SMOOTHS and ROUGHS

Bin #	Number of breeding pairs	Count of species	Root of count			Final rough	Letter values
			raw	3RSSH	twice		
1	0.5 to 1.5	2	1.4	1.4	1.4	0	
2	1.5 to 3.5	5	2.2	2.2	2.3	−.1	
3	3.5 to 7.5	9	3.0	2.8	3.0	0	(out: xxx)
4	7.5 to 15.5	10	3.2	3.0	3.2	0	(−1.0 f)
5	15.5 to 31.5	8	2.8	3.0	3.2	−.4	(−.4 E)
6	31.5 to 63.5	13	3.6	3.0	3.1	.5	(−.3 H)
7	63.5 to 127.5	6	2.4	3.0	3.0	−.6	(0 M)
8	127.5 to 255.5	9	3.0	3.0	3.0	0	(.1h H)
9	255.5 to 511.5	11	3.3	2.9	2.8	.5	(.5 E)
10	511.5 to 1023.5	4	2.0	2.5	2.2	−.2	(.8h f)
11	1023.5 to 2047.5	3	1.7	1.5	1.4	.3	(out: xxx)
12	2047.5 to 4095.5	0	0	.4	.4	−.4	
	(sum)	(80)	0	0	0	0	

P) PROBLEMS

9a) Try 3RSSH'ing the final rough. What do you conclude?

S) SOURCE

F. W. Preston (1948). "The commonness, and rarity, of species," *Ecology* 29: 254–283. (Table 1–A on page 258.)

A. A. Saunders (1936). Ecology of the birds of Quaker Run Valley, Alleghany State Park, New York. *New York State Museum Handbook* 16.

breeding pairs of birds

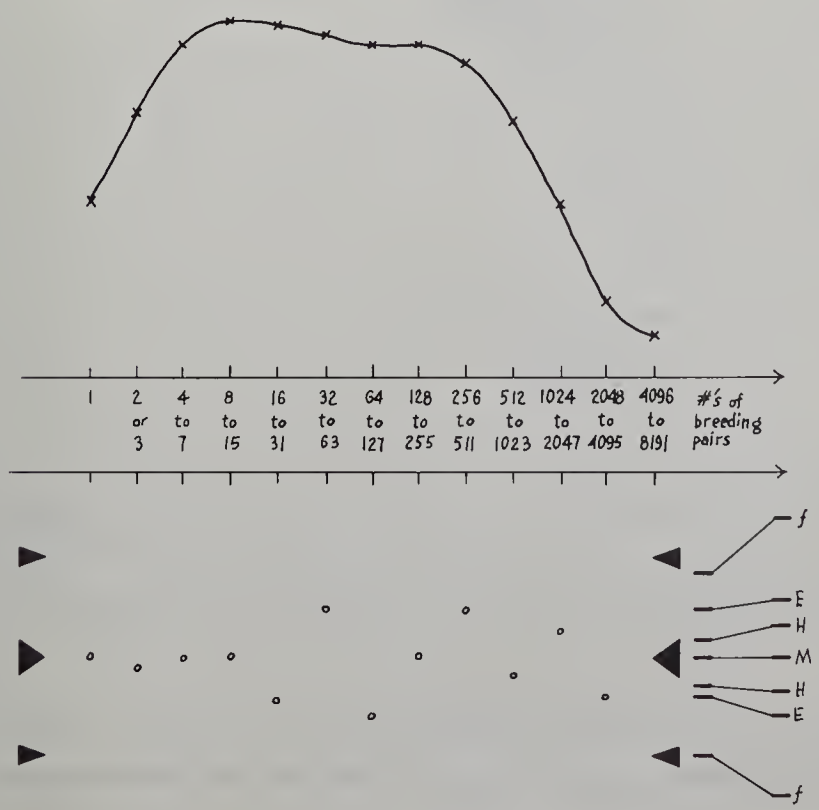
Exhibit **9** shows the arithmetic, where the basic counts are the numbers of breeding pairs of each species of bird in Quaker Run Valley, New York, as reported by Saunders via Preston. We have not tried to add zero lines for fractional numbers of species.

With these small numbers, exhibit **10** easily accommodates both the smooth and the rough. Again, we have a reasonably symmetric smooth--and a well-behaved rough.

Consider carefully the mess we would have had if we had stayed with raw basic counts. We see that 2 species had 1 pair each, and a total of 16 species had up to 7 pairs each, almost 2 species per basic count. Between 512 and 1023 pairs there were 4 species, and between 1024 and 2047 there were 3. No fixed interval for raw numbers of pairs makes any sense. Either it is too short to help us above 511, or so long as to pack almost all the pairs into the first interval. Without octaves, our situation would have been unhappy.

exhibit **10** of chapter 17: valley birds

The smooth and the rough (of roots of counts of species by numbers of breeding pairs, from exhibit 9)



more problems

Exhibit **11** contains data and wording for a variety of problems.

review questions

Can we treat binned counts just like binned measurements? Always? What example was next? What did we see? And after that?

exhibit **11** of chapter 17: problems

Data and further problems on smoothed roots and their roughs

A) COUNTS of YEAST CELLS in a HAEMOCYTOMETER

Basic Count	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**
# counted	103	202	268	143	32	17	10	0

B) COUNTS of PLANTS of *Salicornia europea*

in 625cm² squares	
0*	23334455556789
1	1122344778
2	002336888999
3	01146799
4*	00235789
5	000011222345
6	011234578889
7	00334567889
8*	228
9	116
10	04
11	07
12*	7
13	23

in 2500cm² squares	
0*	3445555567889
1	12334566789
2	11122356
3	00355689
4*	7
5	8
6	169
7	
8*	0
9	
1**	01, 02, 04, 29, 35, 39
·	52, 71, 73
2**	25, 32, 35

P) PROBLEMS

11a) Can we use octaves for the data of panel A? Why/why not? Do the best you can.

11b/c) Same questions and instructions for the left/right data set of panel B.

S) SOURCES

For A: Student, 1907. "On the error of counting with a haemocytometer," *Biometrika* **5**: 351–360. Example (ii) on page 356.

For B: E. Ashby, 1935, "The quantitative analysis of vegetation," *Annals of Botany* **47**: 779–803. Table III (page 190) and Table IV.

17C. Fitting to smoothed roots

Our smooths make nice pretty pictures, so much so that we want to go further and try to fit some simple shape to them--or to a re-expression of them. The simplest formula that gives a smooth symmetrical peak will be one that, in the Pearson and Lee data, for example, has a simple straight-line dependence of

(some expression of root count)

ON

(forearm length MINUS peak forearm length)^2

(The square accounts for both the peaking and the symmetry, as we shall see.)

We shall write "shift" for the difference between where we are and the peak. Thus, for the present,

B = peak PLUS shift,

shift = B MINUS peak,

fit = constant PLUS (tilt)(shift)^2

In any given set of data, then, there are three things for us to do:

- find the peak.
- choose the re-expression,
- fit the line.

finding the peak

Using the values near the peak to find where the peak lies is often very difficult. In particular, small irregularities can throw us far off. How does our eye judge where the peak is, GIVEN that the smooth looks symmetrical? Mainly by moving down from the peak, choosing two points at the same height, and splitting the difference. For simplicity (especially when we come to arithmetic), we always start from a plotted point. Exhibit **12** shows how this can be done graphically for the Pearson and Lee data (exhibits 1 to 3) on forearm lengths. Notice a slight tendency for the peak count to be lower when we work at a higher level. Exhibit **13** shows how the same thing can be done arithmetically for the same data.

The left side of panel B shows the resulting picture of smoothed roots against squared shift. Note that we have used one character, ×, on one side of the peak, and another, ○, on the other. This helps us to see whether one side is systematically behaving differently from the other (a little of this shows here)--or whether the behavior on one side, or on both, is only irregular. We see a reasonably well-defined curve in this exhibit, but nothing like a straight line. Clearly, a different expression of the root smooth is needed.

choosing an expression

We know how to set about choosing an expression (see chapter 6)--beginning with a few chosen points. Looking at exhibit 13, we first select

(0, 15), (10, 7.5), (20, 3.9). Looked at hard, the three immediately suggest taking logs of the root smooth--since 0, 10, 20 are equally spaced, while 15, 7.5, 3.9 change by nearly a constant factor. (The numerical values of the logs are at the right of exhibit 13.) The righthand side of panel B shows a plot of log smoothed roots against squared shift. This picture clearly calls for a straight line. We have a reasonable re-expression and are ready to press on. (But notice, toward the right of the picture, that all the ○'s are higher and all the ×'s are lower. This speaks of a tendency to lack of symmetry about our chosen peak.)

making the fit

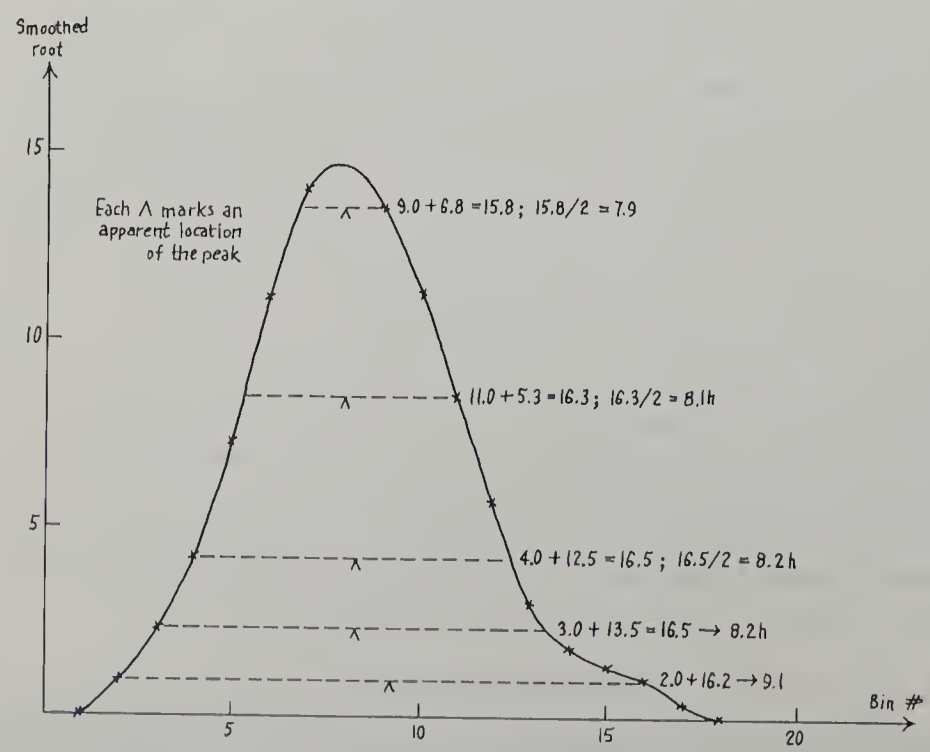
Exhibit **14** shows the same picture, with a straight line drawn in. Since this line goes through (0, 1.17), (50, −.24), its equation is

$$\log(\text{smoothed roots}) = 1.17 - 0.0282\ (\text{squared difference}).$$

The result of using this is set out in exhibit **15** with the residuals shown in panel B. While this plot shows moderate irregularity, it feels a little trendy--and its

exhibit **12** of chapter 17: forearm length

Locating the peak through midpoints (shown "∧") of secants



letter values are broader than the reference. Thus we wonder whether the fit is good enough. Another way to ask this question is to smooth the residuals--arithmetic in panel A and display in panel C. The smoothed residuals are large enough, as the last picture shows--going out to the hinges of the unsmoothed residuals--to make us wonder about a better fit.

review questions

What do we want our fits to look like? What will we call "shift"? What dependence on shift will be simplest, among those that might look right? What will depend simply on shift^2? How do we find a peak? Would this work for an unsymmetrical distribution. With what example did we start? With what results? What did we use three selected points for?

exhibit **13** of chapter 17: forearm lengths

Finding the peak for the Pearson and Lee data of exhibit 1

A) ROOT SMOOTH, its MATCHING ACROSS, and SQUARED DIFFERENCE

Bin #	Smoothed roots	Matched step	Sum of steps	Step minus 8.25	Square of same	log smoothed roots
1	.2	17.0	18.0	−7.25	52.6	−.70
	1.0	16.2	18.2	−6.25	39.1	.00
3	2.5	13.5	16.5	−5.25	27.6	.40
	4.3	12.5	16.5	−4.25	18.1	.63
5	7.3	11.3	16.3	−3.25	10.6	.86
	11.2	10.1	16.1	−2.25	5.1	1.05
7	13.8	--		−1.25	1.6	1.14
	14.4	--		−.25	.1	1.16
9	13.5	6.9	15.9	.75	.6	1.13
	11.3	6.1	16.1	1.75	3.1	1.05
11	8.2	5.2	16.2	2.75	7.6	.91
	5.4	4.4	16.4	3.75	14.1	.73
13	3.2	3.4	16.4	4.75	22.6	.50
	1.9	2.6	16.6	5.75	33.1	.28
15	1.5	2.4	17.4	6.75	45.6	.18
	1.1	2.1	18.1	7.75	60.1	.04
17	.4	1.5	18.5	8.75	76.6	−.40
	0	median peak is	16.5 8.25			

Example: 7.3 comes at 5, and again between 11 and 12, where interpolation suggests it comes at 11.4. Next 5 PLUS 11.4 = 16.4, given in the "sum of steps" column. *Note.* The increase of the "sum-of-steps" value toward the tail indicates a tendency to lack of symmetry about 8.25, the positive tail appearing longer.

→

exhibit **13** of chapter 17 (continued)

B) PLOTS AGAINST SQUARED SHIFT

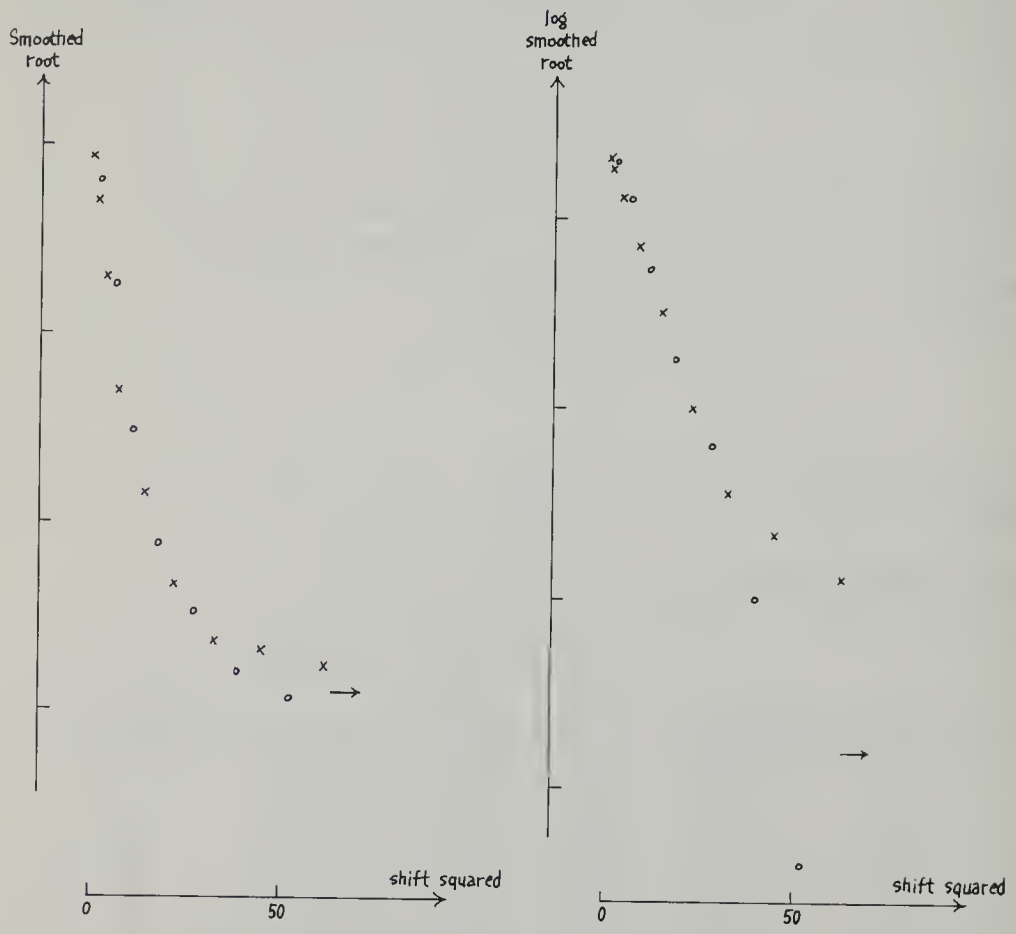


exhibit **14** of chapter 17: forearm lengths

The final plot, with a straight line inserted

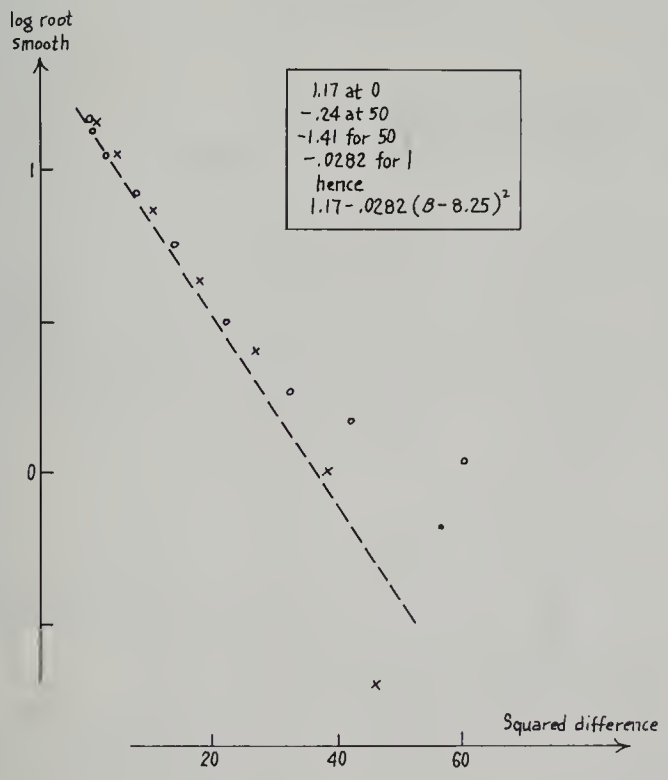


exhibit **15** of chapter 17: forearm lengths

Calculation of the fit corresponding to the straight line of exhibit 14 (first-fit)

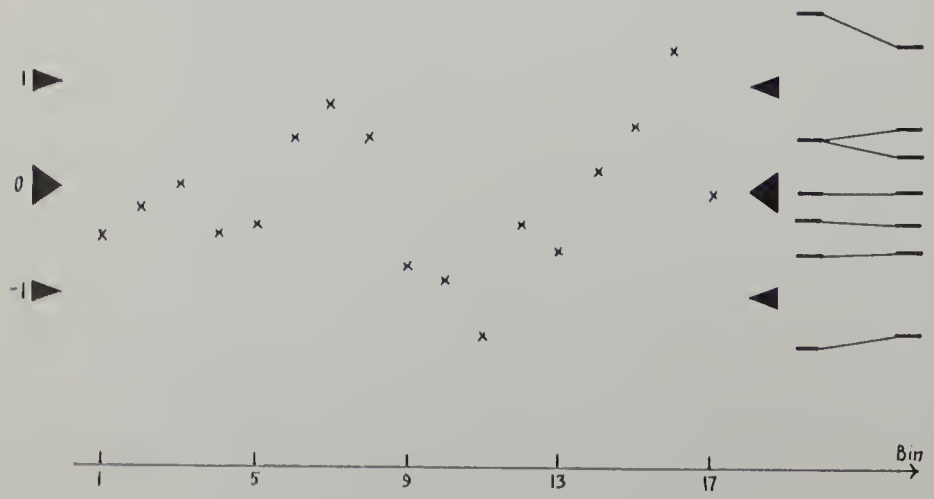
A) ARITHMETIC––peak at 8.25, fit as $1.17 - .0282\,(\text{difference})^2$

Interval	Bin #	Fitted values: log root	Fitted values: root	observed root	The rough*	Residual	3RSSH	(letter values**)
					0			
15–15.5	1	−.31	.5	0	.2	−.5	−.2	
		.07	1.2	1.0	0	−.2	−.2	
	3	.39	2.5	2.5	0	0	−.2	
		.66	4.6	4.1	−.2	−.5	0	
	5	.87	7.4	7.0	−.3	−.4	.4	(out: none)
		1.03	10.7	11.2	0	.5	.5	(−1.5 f)
18–18.5	7	1.12h	13.3	14.1	.3	.8	.5	(−.6 E)
		1.17	14.8	15.3	.9	.5	.2	(−.3 H)
	9	1.15h	14.3	13.5	0	−.8	−.5	(0 M)
		1.08h	12.2	11.3	0	−.9	−.8	(.5 H)
	11	.95h	9.0	7.6	−.6	−.7	−.8	(.5 E)
		.77h	6.0	5.6	.2	−.4	−.6	(1.7 f)
	13	.53h	3.4	2.8	−.4	−.6	−.3	(out: none)
		.24	1.7	1.9	0	.2	.2	
22–22.5	15	−.11	.8	1.4	−.1	.6	.5	
		−.52h	.3	1.6	.5	1.3	.6	
	17	−.99	.1	0	−.4	−.1	.6	

Examples: $1.12h = 1.17 - (.0282)(1.25)^2$

* from exhibit 1
** for unsmoothed residuals

B) The RESIDUALS PICTURED––as is



17D. Corn borers, wheat prices, and Student's simulations

Let us turn back next to exhibit 7, with its counts of corn borers per plant. The root smooth comes to a maximum and turns over, going down for just one point. Zero borers gives 24.2, which interpolates to about 2.7 borers on the other side of the smooth. So we may as well start with a trial peak at 1.3 borers. Exhibit **16** has the arithmetic. The left side of panel B shows log smoothed roots against shift-squared, where we see that our line is at best a first approximation.

Going back to panel A, we calculate the residuals from the line, and then plot them again--at the right side of panel B. We now see that we can fit a reasonably good line to the first six points--marked "o"--if we forget the other five--marked "×". Doing this makes some sense, because the first six involve the larger counts and the more stable logs.

We have now fitted, writing B for borers per plant:

$$1.45 - .031(\text{shift})^2 = 1.45 - .031(B - 1.3)^2$$

PLUS

$$-.005 + .002(B - 1.3)^2$$

to our log smoothed roots, a total of

$$1.435 - .029(B - 1.3)^2.$$

With the new fit, the residuals are not beautiful, as exhibit **17** pictures. Two things are notable:

- ◇ the residuals for the two points closest to the peak are high--one is outside.
- ◇ the residuals as a whole are too positive.

exhibit **15** of chapter 17 (continued)

C) The SMOOTHED RESIDUALS PICTURED

Both of these are natural consequences of the peak being too narrow for our smoothing technology. Fitting to "log raw roots" may, in this case, be better than fitting to "log smoothed roots".

If we had wished, we could have tried an additional fit linear in B, rather than linear in $(B - 1.3)^2$. Adding such a fit would shift the fitted peak.

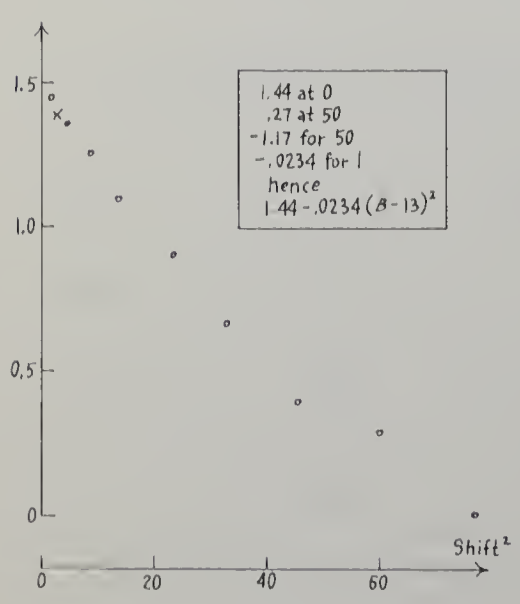
exhibit **16** of chapter 17: field corn

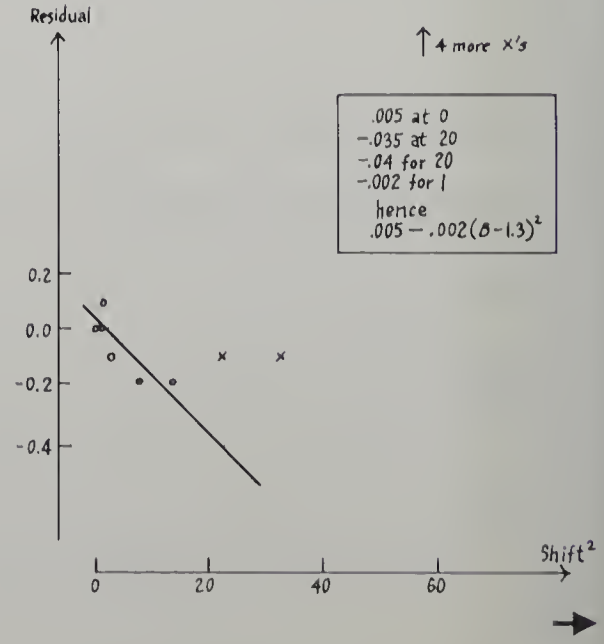
A trial fit to the corn borer data (smooth from exhibit 15, peak at 1.3 from text)

A) FITS and RESIDUALS

Borers per plant	Smoothed root	log smoothed root	(shift)^2	1.45 −.031 × (shift)^2	Resid. log root	.005 −.002 (shift)^2	Improved fit: log root	Improved fit: root	Improved fit: Residual	(Letter values)
0	25.5	1.41	1.7	1.40	.01	−.00	1.40	25.1	.9	
	27.7	1.44	.1	1.44	.00	−.01	1.45	28.2	.2	
2	27.0	1.43	.5	1.43	.00	−.00	1.43	26.9	.3	(out: one)
	22.9	1.36	2.9	1.37	−.01	−.00	1.37	23.4	−1.5	(1.7 f)
4	17.6	1.25	7.3	1.27	−.02	−.01	1.26	18.2	−.1	(.7 E)
	12.6	1.10	13.7	1.12	.02	−.02	1.10	12.6	0	(.5h H)
6	8.1	.91	22.1	.92	.01	−.04	.88	7.8	.6	(.2 M)
	4.7	.67	32.5	.68	−.01	−.06	.62	4.2	.5	(−.2 H)
8	2.8	.45	44.9	.39	.06	−.08	.31	2.0	.2	(-.9 E)
	2.2	.34	59.3	.05	.29	−.11	.06	.9	1.3	(1.3h f)
10	1.4	.15	75.7	−.33	.48	−.15	−.18	.7	.7	(out: xxx)
	.4	−.40	94.1	−.76	.36	−.18	−.94	.1	.3	
12	(0)	(L)								

B) PLOTS





Student's simulations

In 1908 Student--a chemist, later brewmaster for Guinness--published two papers which did much to found modern statistics. In both he simulated the behavior of certain quantities one might like to calculate from batches of numbers. He did this by preparing 750 batches of four numbers (or 750 batches of 8) and calculating the values of the interesting quantity for each batch. He used the heights and the left middle finger measurements of 3000 criminals, writing these on 3000 pieces of cardboard, shuffling well and drawing them out "at random". (For the batches of eight, each criminal appeared in two batches.) Some of his results are shown in exhibit **18**. Since two of his collections of counts are for bins of unequal widths, their analysis will have to wait till the next section. The third can now be treated in just the same way we have treated other sets of counts, bin after bin.

exhibit **16** of chapter 17 (continued)

P) PROBLEMS

16a) Carry one extra decimal through panel A, beginning with the log smoothed root. Is the increased accuracy worth the effort?

16b) (Class exercise). Try a variety of different fits of the same form, calculating and plotting residuals for each.

16c) Plot the final residuals against B, rather than $(B - 1.3)^2$, fit a line, add to fit. Discuss the effect.

16d) Plot the unsmoothed log roots against $(B - 1.3)^2$. Complete the fit. Discuss the results.

exhibit **17** of chapter 17: field corn

Residuals from the new fit

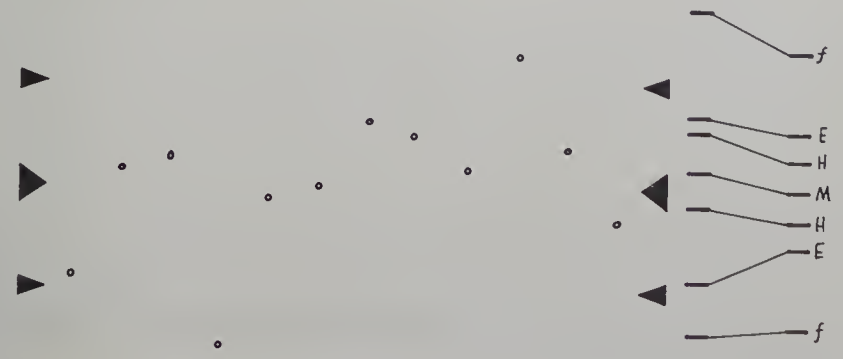


exhibit **18** of chapter 17: simulations

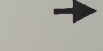
Results of two of Student's simulations

A) The DATA

Bin	Values of correlation coefficient	Counts of these
1	(to −.98)	(1)
2	−.97 to −.93	0
3	−.92 to −.88	0
4	−.87 to −.83	1
5	−.82 to −.78	7
6	−.77 to −.73	10
7	−.72 to −.68	10
8	−.67 to −.63	15
9	−.62 to −.58	14
10	−.57 to −.53	15
11	−.52 to −.48	18
12	−.47 to −.43	24
13	−.42 to −.38	18
14	−.37 to −.33	27
15	−.32 to −.28	36
16	−.27 to −.23	33
17	−.22 to −.18	43
18	−.17 to −.13	45
19	−.12 to −.08	26h
20	−.07 to −.03	26h
21	−.02 to +.02	34
22	+.03 to +.07	42h
23	+.08 to +.12	27h
24	+.13 to +.17	34
25	+.18 to +.22	23h
26	+.23 to +.27	36h
27	+.28 to +.32	33h
28	+.33 to +.37	28h
29	+.38 to +.42	19
30	+.43 to +.47	24
31	+.48 to +.52	22
32	+.53 to +.57	15
33	+.58 to +.62	13h
34	+.63 to +.67	9
35	+.68 to +.72	2h
36	+.73 to +.77	7
37	+.78 to +.82	5
38	+.83 to +.87	3
39	+.88 to +.92	–
40	+.93 to +.97	–
41	(.98 up)	–

Values of "Student's t"	Counts of t for heights	Counts of t for fingers
to −3.05	9	4
−3.05 to −2.05	14h	15h
−2.05 to −1.55	11h	18
−1.55 to −1.05	33	33h
−1.05 to −.75	43h	44
−.75 to −.45	70h	75
−.45 to −.15	119h	122
−.15 to .15	151h	138
.15 to .45	122	120h
.45 to .75	67h	71
.75 to 1.05	49	46h
1.05 to 1.55	26h	36
1.55 to 2.05	16	11
2.05 to 3.05	10	9
3.05 on	6	6

Note. 750 batches of eight for the correlation coefficient; two sets of 750 batches of four for what is now called "Student's t".



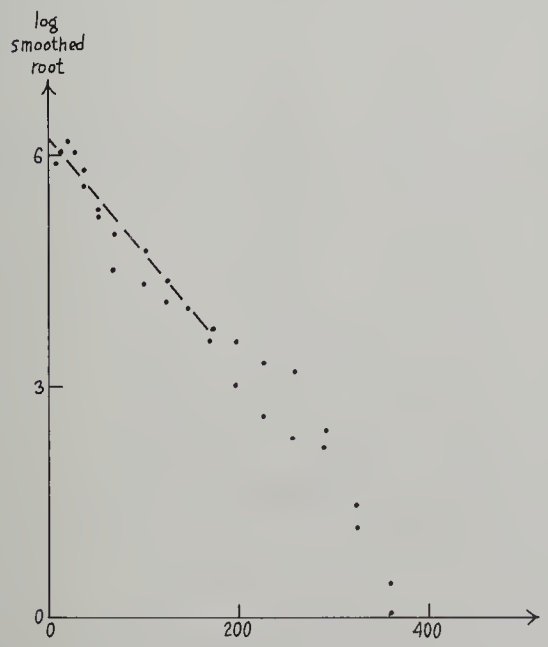
The result of taking the peak at 0.0 is shown in exhibit **19**. Clearly, we do NOT need to take logs in this case. We find

$$6.3 - .0158(\text{shift in bins})^2$$

for our fit, and we find the residuals shown in exhibit 19. These look quite reasonable, being of the right size and suitably irregular. (When they are smoothed they show rather neat, small wiggles, suggesting no trace of remaining structure.)

exhibit **18** of chapter 17 (continued)

B) PLOT--for the correlation coefficient



P) PROBLEMS

18a) Find roots, smoothed roots and their roughs for the correlation coefficient counts, using bins given. Compare results with exhibit 24.

18b) Do the same with half as many bins, each twice as wide.

S) SOURCES

Counts of "*t*": Student, 1908. "The probable error of a mean." *Biometrika* 6: 1–25.
Counts of correlation coefficients: Student, 1908. "Probable error of a correlation coefficient," *Biometrika* 6: 302–310.
See alternatively: pp. 1–10 and pp. 11–34 of *"Student's" Collected papers* edited by E. S. Pearson and John Wishart, n.d. Biometrika Office, Cambridge University press.

wheat price changes

In 1953, Maurice Kendall published a study of price series from which we can take counts of weekly changes in wheat prices from 1883 to 1934. Exhibit **20** has the arithmetic. The first rough has values of 3.4 and 2.9 for the two center bins--in words, the peak is too sharp for our smoother. This is easily dealt with by splitting the smooth between these two. Taking the peak at zero, we see that log smoothed roots do not give a straight line against B^2, but that reciprocal smoothed roots may. Fitting a line by eye gives fitted roots which leave reasonable residuals. Exhibit **21** shows both root rough and residuals. In each case, both plain and modified letter values are shown--modified next to plot, and plain between these and the reference values. The fit is not wonderful, but not unsatisfactory.

exhibit **19** of chapter 17: simulations

The residuals, alone above and with their smooth below (numbers based on exhibit 18).

exhibit **20** of chapter 17: price changes

Changes in wheat prices from one week to the next, 1883–1934. Bins are 2 cents a bushel wide. Bin numbers are all odd and are centers of changes. (Thus −13 covers −14 to −12, etc.) Smooth 3RSSH-twice. Peak assumed at zero.

A) ARITHMETIC

Bin #*	Count in bin	Root of count	Smoothed root	Their rough	Split smoothed roots	Their rough
L	(2)		.0	.0	.0	.0
−17	h	0.7	.8	−.1	.8	−.1
−15	5	2.2	2.0	.2	2.0	.2
−13	8h	2.9	2.6	.3	2.6	.3
−11	7h	2.7	3.3	−.6	3.3	−.6
−9	24	4.9	4.5	.4	4.5	.4
−7	47h	6.9	6.9	.0	6.9	.0
−5	114h	10.7	10.7	.0	10.7	.0
−3	263	16.2	17.0	−.8	17.2	−1.0
−1	722	26.9	23.5	3.4	26.4	.5
1	708h	26.6	23.7	2.9	25.9	.7
3	284h	16.9	17.1	−.2	16.2	.7
5	100	10.0	10.5	−.5	9.3	.7
7	50	7.1	6.5	.6	6.5	.6
9	17	4.1	4.1	.0	4.1	.0
11	10	3.2	3.0	.2	3.0	.2
13	5	2.2	2.2	.0	2.2	.0
15	4h	2.1	2.1	.0	2.1	.0
17	4h	2.1	1.9	.2	1.9	.2
19	1h	1.2	1.5	−.3	1.5	−.3
21	3h	1.9	0.5	1.4	0.5	1.4
H	(2)		.2	−.2	.2	−.2

*** Midvalue as a change in cents per bushel––changes on a division value assigned half to each bin.**

B) LETTER VALUES for ROUGH––plain and modified

#22

M11h	.1		
H 6	.5	−.0	.5
E 3h	.7	−.2h	.9h

*#19 = 22 − (1/2)(6)

*M10	.2		
*H 5h	.5h	−.h	.6
*E 3	.7	−.3	1.0

	.7h	
f	1.2h	−.7h
	one	one
F	2.0	−1.5
	xxx	xxx

	.9	
f	1.4h	−.9h
	xxx	one
F		−1.8h
		xxx

outside both = −1.0, outside one = 1.4

exhibit **20** of chapter 17 (continued)

C) ARITHMETIC CONTINUED

squared shift	log smoothed roots	1000/ smoothed roots	37 + 2.256 × squared shift	Fitted root	Root residual
361			851.5	1.2	−1.2
289	−.10	1250	689.0	1.5	−.8
225	.30	500	544.6	1.8	.4
169	.41	385	418.3	2.4	.5
121	.51	303	310.0	3.2	−.5
81	.65	222	219.8	4.5	.4
49	.84	145	147.6	6.8	.1
25	1.03	93	93.4	10.7	.0
9	1.24	58	57.3	17.5	−1.3
1	1.42	38	39.3	25.5	1.4
1	1.41	39	39.3	25.5	1.1
9	1.21	62	57.3	17.5	−.6
25	.97	108	93.4	10.7	−.7
49	.81	154	147.6	6.8	.3
81	.61	244	219.8	4.5	−.4
121	.48	333	310.0	3.2	.0
169	.34	455	418.3	2.4	−.2
225	.32	476	544.6	1.8	.3
289	.28	526	689.0	1.5	.6
361	.18	667	851.5	1.2	.0
441	−.30	2000	1032.0	1.0	.9
529	−.70	5000	1230.4	0.8	−.8

D) LETTER VALUES for RESIDUALS

#22

M11h	0		
H 6	.4	−.6h	1.0h
E 3h	.7h	−.8	1.5h

	1.6	
f	1.9	−2.1
	xxx	xxx

*#18h = 22 − (1/2)(3)

*M 9h	0		
*H 5	.4h	−.6h	1.2
*E 3	.9	−.8	1.7

	1.6h	
f	2.1	−2.3
	xxx	xxx

P) PROBLEMS:

20a) Kendall also gave data on changes of investment share prices as follows: (21): 13, 15, 25, 46, 66, 81; 91; 44, 32, 14, 15, 8; (14) where these counts apply to intervals centered as follows:

(Low); −11.5 to −1.5; 0.5; 2.5 to 10.5; (High).
Make an analysis parallel to the three panels above.

S) SOURCE

M. G. Kendall, 1953."The analysis of economic time-series—Part I: Prices." *J. Roy. Stat. Soc.* **A116** 11-25.

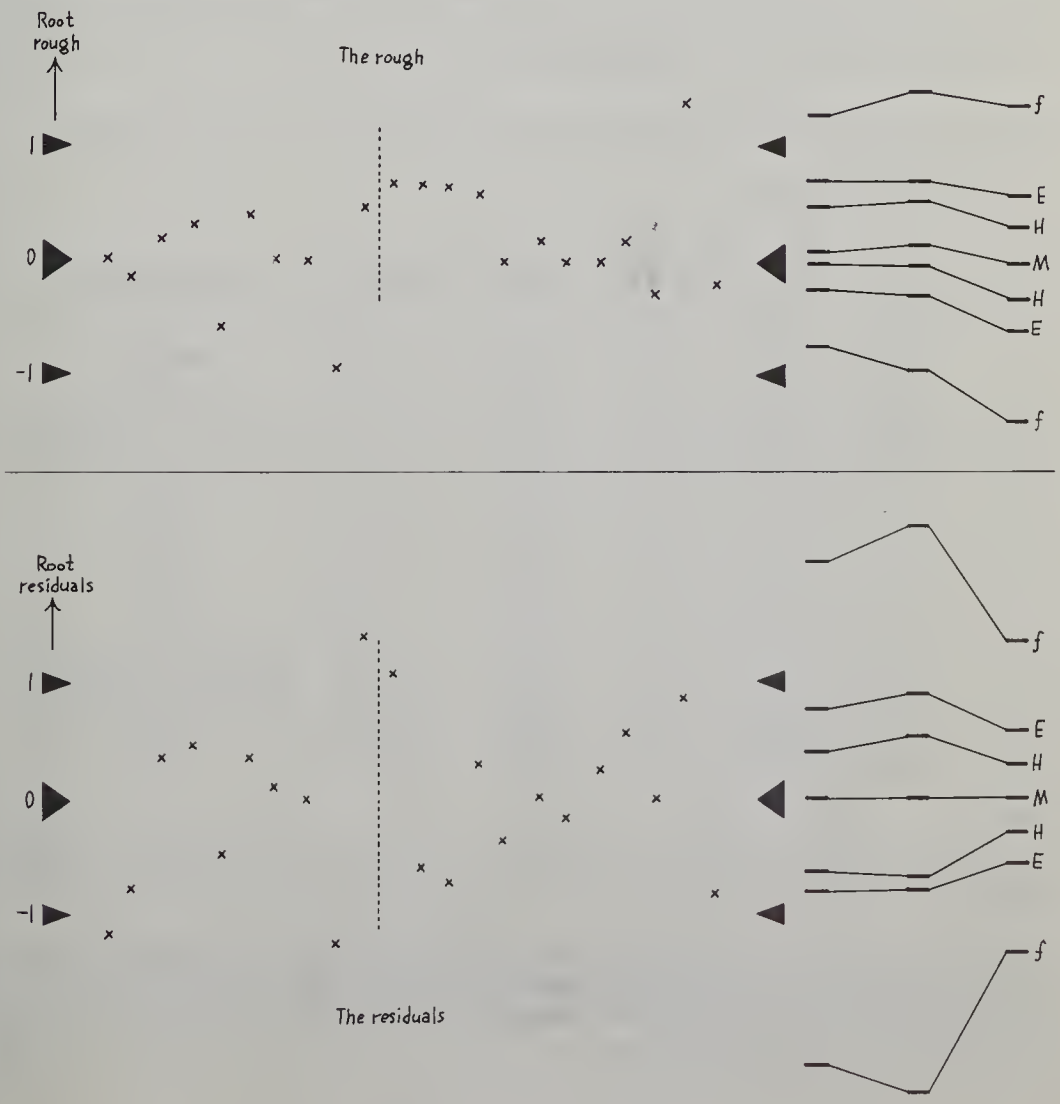
(Reprinted in P. Cootner, ed. 1964, 1969. *The Random Character of Stock Market Prices*, MIT Press.)

review questions

What example came first? How did we guess a peak? Why did we have to guess? What did we do next? With what result? Whose data did we turn to next? What were these counts based upon? What result did we reach? What example came next? With what result? How did rough and residuals compare? What would you suspect from this?

exhibit **21** of chapter 17: price changes

Root rough and root residuals for exhibit 20



17E. Bins of unequal width

When the widths of our bins differ, we must work with

$$\sqrt{\frac{\text{counts}}{\text{bin width}}},$$

instead of just

$$\sqrt{\text{counts}}.$$

To see that this is so, make up an equal-width bin example where the counts vary between 90 and 110, and experiment by combining two or three small bins into one large one. You will find that dividing counts by bin width is essential.

Since the bins are not of equal width, their centers are not equally spaced.

exhibit **22** of chapter 17: simulation

Root smooth and root rough for Student's simulation of *t* using heights

A) BIN ARITHMETIC

Bin #	Bin dimensions				3RSSH	Getting back
	Start	End	Width	Center		
1	Low	−3.05	---	---		
	−3.05	−2.05	1.0	−2.55	−2.55	hold
3	−2.05	−1.55	.5	−1.80	−1.86	go 6/54 toward #4
	−1.55	−1.05	.5	−1.30	1.32	go 2/40 toward #5
5	−1.05	−.75	.3	−.90	−.92	go 2/32 toward #6
	−.75	−.45	.3	−.60	−.60	hold
7	−.45	−.15	.3	−.30	−.30	hold
	−.15	.15	.3	.00	.00	hold
9	.15	.45	.3	.30	.30	hold
	.45	.75	.3	.60	.60	hold
11	.75	1.05	.3	.90	.92	go 2/32 toward #10
	1.05	1.55	.5	1.30	1.32	go 2/40 toward #11
13	1.55	2.05	.5	1.80	1.86	go 6/54 toward #12
	2.05	3.05	1.0	2.55	2.55	hold
15	3.05	High	---	---		

Notes.

1. Bins with one open side (#1, #15) have no useful definitions of width or center.
2. After hanning, bin #3 ends up at −1.86, #4 at −1.32. To bring #3 after hanning, back to −1.80, where it started, requires moving .06, while going all the way to #4 would move 0.54. Hence moving .06/.54 = 1/9 of the way to #4 should do the trick. And so on.

→

This leaves us concerned about the use of hanning. We could:

◇drop hanning and use only 3RSS twice.

◇whenever we hann, hann the sequence of bin centers as well as the sequence of partly smoothed roots, and interpolate back.

We will leave the first of these almost entirely to problems.

Student's *t*

Let us try the second approach for another of Student's simulations, as already shown in exhibit 18. Exhibit **22** has the arithmetic. Note that since we smooth

$$\sqrt{\frac{\text{count}}{\text{bin width}}},$$

we must multiply the final rough by the square root of the bin width in order to find a rough in terms of root counts. Exhibit **23** shows the rough, which looks suitably irregular and of about the right spread.

exhibit **22** of chapter 17 (continued)

B) SECOND SMOOTHING--first smoothing in panel C of exhibit 24.

Bin #	First rough	3RSSH	Moved back*	Final smooth	Final rough	Square root width	Rough** (in roots)
2	0	−.3	(−.3)	3.5	.3	1.00	.3
3	−.9	−.3	−.3	5.4	−.6	.71	−.4
4	−.3	−.3	−.3	8.1	0	.71	0
5	0	−.3	−.3	11.7	.3	.55	.2
6	−.3	0	(0)	15.6	−.5	.55	−.3
7	1.1	.8	(.8)	19.7	.3	.55	.2
8	2.3	1.2	(1.2)	21.4	1.1	.55	.6
9	1.3	1.1	(1.1)	20.0	.2	.55	.1
10	−.8	.6	(.6)	16.4	−1.4	.55	−.8
11	.5	.1	.1	12.4	.4	.55	.2
12	−1.2	0	0	8.5	1.2	.71	−.8
13	0	0	0	5.7	0	.71	0
14	0	0	0	3.2	0	1.00	0

* No move as large as ±0.05 is called for.
** Rough (in roots) = (final rough) TIMES (root of bin width)

Note that the final smooth and final rough above are in root of (counts/width), NOT yet in root of counts.

→

exhibit **22** of chapter 17 (continued)

C) LETTER VALUES for ROOT ROUGH

#13	(three zeros)			
M 7		.1		
H 4	.2		−.2	.4
E 2h	.2h		−.6h	.9h
		.6		
f	0.9h		−.10h	
	xxx		xxx	

P) PROBLEMS

22a) Repeat for Student's data based on finger measurements (righthand column of exhibit 18).

22b) H. Chavez, M. Contreras G., and T. P. E. Hernandez D. (1968, "On the coast of Tamaulipas." *International Turtle and Tortoise Society Journal*, vol. 2, no. 5, pp. 16 to 19 and 27 to 34.) gave the following counts for the duration (in days) of incubation for 1664 eggs of the ridley turtle.

days:	**50**	**51**	**52**	**53**	**54**	**55**	**56**	**57**	**58**	**59**	**60**	**61**	/ /	**64**	**65**
# eggs:	77	122	10	321	725	180	162	21	14	6	9	14		1	1

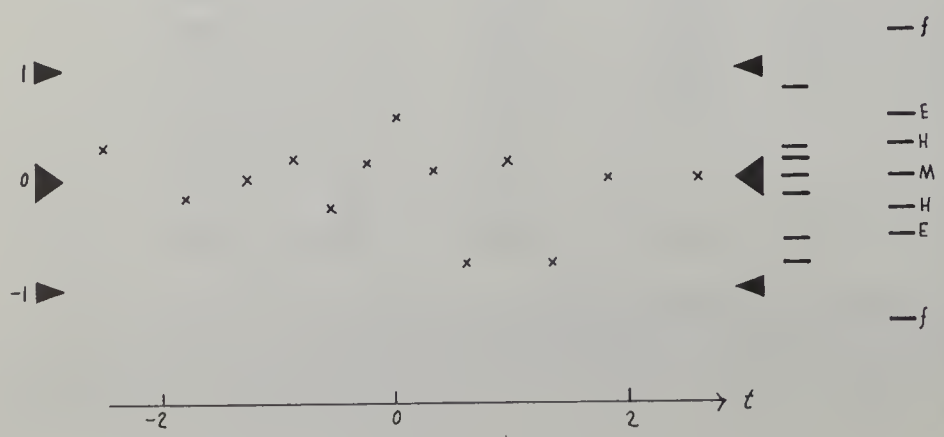
Carry out a careful analysis using $\sqrt{\text{days}}$. Comment.

22c) Carry out a careful analysis using logs. Comment.

22d) Do the same using $\sqrt{-49 + \text{time in days}}$. Comment.

exhibit **23** of chapter 17: simulations

The rough of the fit of exhibit 22?



fitting the smooth

Exhibit **24** carries on this analysis, doing the usual peak-finding procedure. The left side of panel D shows the plot of smoothed raw counts against shift-squared--very curved. The center of the same panel shows log smoothed roots against shift-squared--plenty curved. So we go on to reciprocals, as in the right side of the same panel. At last our picture is straight, so we proceed to calculate first a fitted reciprocal smooth, then a fitted root smooth, then residuals on a "root of count over width" basis, then residuals on a root-of-count basis. Exhibit **25** plots the latter residuals, which are reasonably irregular and of about the right width. (They tend to be slightly positive, and to tilt a little, suggesting that a modified fit might be a shade better.)

exhibit **24** of chapter 17: simulation

Fitting the smooth of exhibit 22 (peak found to be at 8.00)

A) The ARITHMETIC

Bin #	Final smooth	Match. bin #	Half sum	Bin center	(shift)^2	100/ smooth root	Fit = 4.8 + 4.32 × $(B - 8)^2$	Fit to smooth	Root of Resid.	Root of bin width	Resid. (in roots)
	3.5	13.8	7.9	−2.55	6.50	28.6	32.9	3.0	.8	1.0	.8
3	5.4	13.1	8.1	−1.8	3.24	18.5	18.8	5.3	.5	.71	.4
	8.1	12.1	8.0	−1.3	1.69	12.3	12.1	8.3	−.2	.71	−.1
5	11.7	11.1	8.0	−.9	.81	8.5	8.3	12.0	0	.55	0
	15.8	10.1	8.0	−.6	.36	6.3	6.4	15.6	−.3	.55	−.2
7	19.7	9.1	8.0	−.3	.09	5.1	5.2	19.2	.8	.55	.4
	21.4			0	0	4.7	4.8	20.8	1.7	.55	.9
9	20.0	7.2	8.1	.3	.09	5.0	5.2	19.2	1.0	.55	.6
	16.4	6.2	8.0	.6	.36	6.1	6.4	15.6	−.6	.55	−.3
11	12.1	5.1	8.0	.9	.81	8.3	8.3	12.0	.8	.55	.4
	8.5	4.1	8.0	1.3	1.69	11.8	12.1	8.3	1.0	.71	.7
13	5.7	3.1	8.0	1.8	3.24	17.5	18.8	5.3	.4	.71	.3
	3.2			2.55	6.50	31.2	32.9	3.0	.2	1.00	.2

B) LETTER VALUES for the RESIDUALS

#13 one zero

M	7	.2		
H	4	.4	−.2	.6
E	2h	.7	−.3h	1.0h

	.9	
f	1.3	−1.1
	xxx	xxx

→

exhibit **24** of chapter 17 (continued)

C) ROOTING and FIRST SMOOTHING

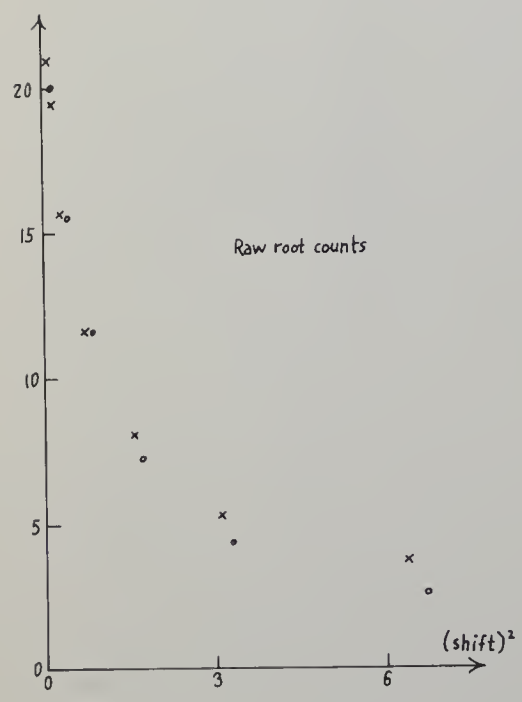
Bin #	Bin width	Bin count	Ratio	Root	3RSSH	Moved back*	First rough
	1.0	14h	14h	3.8	3.8	(3.8)	0
3	.5	11h	23	4.8	5.4	5.7	−.9
	.5	33	66	8.1	8.2	8.4	−.3
5	.3	43h	145	12.0	11.8	12.0	0
	.3	70h	235	15.3	15.6	(15.6)	−.3
7	.3	119h	398	22.0	18.9	(18.9)	1.1
	.3	151h	505	22.5	20.2	(20.2)	2.3
9	.3	122	407	20.2	18.9	(18.9)	1.3
	.3	67h	225	15.8	15.8	(15.8)	−.8
11	.3	49	163	12.8	12.0	12.3	.5
	.5	26h	53	7.3	8.2	8.5	−1.2
13	.5	16	32	5.7	5.4	5.7	0
	1.0	10	10	3.2	3.2	(3.2)	0

* Unchanged values in ( ).

5.6 = 5.3 + (1/9)(8.2 − 5.3),

8.4 = 8.2 + (1/15)(11.8 − 8.2), etc

D) three plots



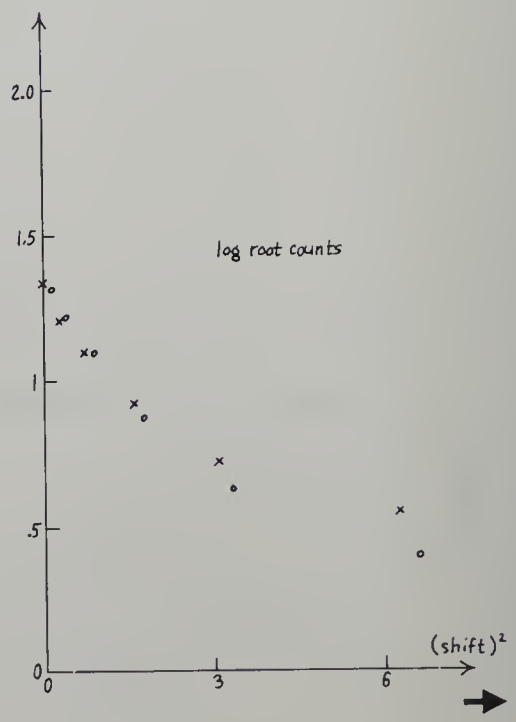
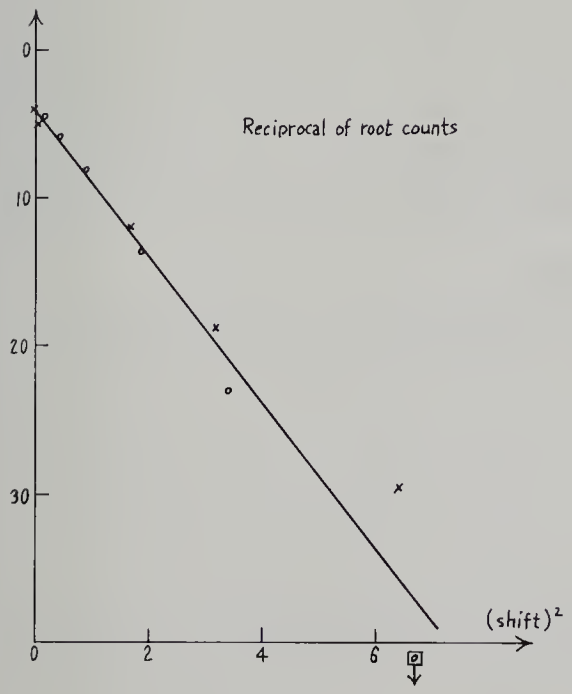


exhibit **24** of chapter 17 (continued)

(panel D continued)



P) PROBLEMS

24a) Try .03 as the peak location, and carry through to a plot of residuals. Compare with exhibit 44.

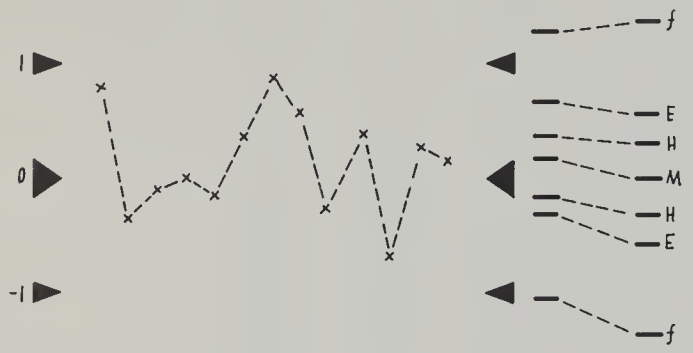
24b) Do the same for −.03 as the peak location.

24c) Fit a straight line to log smoothed roots instead, and plot residuals. Comment.

24d) Do panel A for Student's data on fingers (from panel A of exhibit 18).

exhibit **25** of chapter 17: simulations

The root residuals (plotted against bin number)--(numbers in exhibit 24)



The form of the fit that we have, which is much better than we could have found using either raw roots or log roots, is

$$\frac{1}{\sqrt{\text{count/width}}} = 4.8 + 4.32\ (\text{shift})^2,$$

a form different from most of those we have so far seen.

review questions

What do we work with when the widths of our bins differ? Why? What concerns us about our smoothing? What are our choices? What could we do? What did we do? To what example did we turn? What changes in our calculations are needed? Can we use the final rough as it stands? Why/why not? How did our rough look? What about fitting? Can we use the residuals as they stand? What form did we fit? Had we met it before?

17F. Double roots

Many sets of counts of basic counts deserve expression by some sort of square root. Since bin widths are important, we need to use such an expression not only for the integer counts that we observe, but also for the half-integer counts that mark the bin ends. Convenience and ease of understanding is particularly worthwhile when provided for the half-integer bin ends.

It is fortunate, accordingly, that the most frequently useful expression is the so-called double-root

$$\sqrt{2 + 4\,(\text{count})},$$

for which the endpoints will reduce to

$$\sqrt{2 + 4\,(\text{integer} + 0.5)} = \sqrt{4 + 4\,(\text{integer})} = 2\sqrt{\text{integer} + 1}.$$

Thus, the bin for basic count = 0, a basic count which would be expressed as $\sqrt{2}$, is thought of as stretching from $\sqrt{0} = 0$ to $\sqrt{4} = 2$. (If we had not started our roots, the left end of the bin for basic count = 0 would have been at the square root of a negative number--something hard to plot.)

table

Exhibit **26** gives all the most useful numbers for double roots up to a basic count of 25. For each basic count, there are given, in order:

- ◇the value of the double root.
- ◇the value of the hanned double root.
- ◇the fraction of move-back required.
- ◇the bin width (one endpoint minus the other).
- ◇the root of the bin width (convenient if root count is given).
- ◇the start of each bin (its end is given as the start of the next).
- ◇the center of each bin.

The third item is quite helpful. Double-roots force us to unequal bins and thus to move back. Having the move-back fractions already calculated saves us much arithmetic.

When we come to peaking and fitting our double roots, it is optional--and seems usually desirable--to treat the bins as if they were placed at the hanned double roots instead of at bin centers. (This shifts the "zero" bin by .41, the "one" bin by .04, the "two" bin by .02, the next four by .01, and the rest by .00. Except for the "zero" bin, all changes are either quite small or very small, and the "zero" bin change often helps.

polonium scintillations

Exhibit **27** gives the arithmetic for counts by Rutherford and Geiger of the number of scintillations (flashes of light) in intervals of time of fixed length (1/8 minute). Each scintillation indicated the arrival of one α-particle from the radioactive decay of a single atom of polonium in their sample. This body of data, and its interpretation by Rutherford, Geiger, and (in an appendix) Bateman, are classics in the study of radioactive decay and the establishment of its random character. The upper panel of exhibit **28** shows the rough, which (i) is moderately irregular, (ii) is not excessively wide, (iii) shows the effects of the failure of our smooth to follow as closely as we might wish the rather narrow peak (in root ratios) presented by the data.

exhibit **26** of chapter 17: reference table

Useful numbers for double roots up to a basic count of 25

Basic count	Double root		Move back	Width		Bin start	Bin center
	raw	hanned		raw	root		
0	1.41	1.41	hold	2.000	1.414	0.00	1.00
1	2.45	2.37	up 1/9.4	.828	.910	2.00	2.41
2	3.16	3.13	up 1/17.8	.636	.798	2.83	3.15
3	3.74	3.72	up 1/26	.536	.732	3.46	3.73
4	4.24	4.23	up 1/34	.472	.687	4.00	4.24
5	4.69	4.68	up 1/42	.427	.653	4.47	4.69
6	5.10	5.09	up 1/50	.393	.627	4.90	5.10
7	5.48	5.47	up 1/58	.365	.604	5.29	5.47
8	5.83	5.83	up 1/66	.343	.586	5.66	5.83
9	6.16	6.16	up 1/74	.325	.570	6.00	6.16
10	6.48	6.48	up 1/82	.309	.556	6.32	6.48
11	6.78	6.78	up 1/90	.295	.543	6.63	6.78
12	7.07	7.07	up 1/98	.283	.532	6.93	7.07
13	7.35	7.35	up 1/106	.272	.522	7.21	7.35
14	7.62	7.62	up 1/114	.263	.513	7.48	7.61
15	7.87	7.87	up 1/122	.254	.504	7.75	7.87
16	8.12	8.12	up 1/130	.246	.496	8.00	8.12
17	8.37	8.36	up 1/138	.239	.489	8.25	8.37
18	8.60	8.60	up 1/146	.233	.482	8.49	8.60
19	8.83	8.83	up 1/154	.226	.475	8.72	8.83
20	9.06	9.06	up 1/162	.221	.470	8.94	9.05
21	9.27	9.27	up 1/170	.216	.464	9.16	9.27
22	9.49	9.49	up 1/178	.211	.459	9.38	9.49
23	9.70	9.70	up 1/186	.206	.454	9.59	9.69
24	9.90	9.90	up 1/194	.202	.449	9.80	9.90
25	10.10	10.10	up 1/202	.198	.445	10.00	10.10

Note. For basic counts of 3 or more, "move back" = 1/(2 + 8 (basic count)).

P) PROBLEMS

26a) show the calculations for basic count = 1, keeping enough decimals to check the values given.

26b) Do the same for basic count = 4.

26c) And for basic count = 17.

exhibit **27** of chapter 17: polonium scintillations

Rutherford and Geiger's data on the radioactive decay of polonium (events are scintillations caused by alpha-particles)

A) COUNTS, ROOTS, ROOT RATIOS, and FIRST SMOOTH

Events per 1/8 minute	Counts for bin	Root count	Root ratio*	3RSSH	Move back	First rough
0	57	7.6	5.3	5.3		0
	203	14.2	15.6	15.3	16.2	.6
2	383	19.6	24.6	24.0	24.3	.3
	525	22.9	31.3	29.6	29.7	1.6
4	532	23.1	33.6	31.2	31.2	2.4
	408	20.2	30.9	29.8	29.7	1.2
6	273	16.5	26.4	25.8	25.7	.7
	139	11.8	19.5	19.2	19.1	.4
8	45	6.7	11.5	12.9	12.8	−1.3
	27	5.2	9.1	8.8		.3
10	10	3.2	5.7	6.0		−.3
	4	2.0	3.7	3.8		−.1
12	0	0	0	2.4		2.4
	1	1.0	1.9	1.9		0
14	1	1.0	2.0	1.4		.6
	0	0	0	.5		−.5
16	0	0	0	0		
(total)	(2607)					

* Root ratio $= \sqrt{\frac{\text{count}}{\text{width}}} = \frac{\text{Root count}}{\sqrt{\text{width}}}$ where "$\sqrt{\text{width}}$" comes from exhibit 26.

B) SECOND SMOOTH––starting with 3RSSH of first rough

Bin #	3RSSH	Moved back	Added back	Rough	Rough (in roots)	Adjusted smooth*
0	0	T	5.3	0	0	5.3
1	.1	h	16.3	−.7	−.7	16.3
2	.4	e	24.7	−.1	−.1	24.7
3	.8		30.5	.8	.6	31.1
4	1.1	S	32.3	1.3	.9	33.5
5	1.1	a	30.8	.1	.1	31.4
6	.6	m	26.5	−.1	−.1	26.5
7	.4	e	19.5	0	0	19.5
8	.2		13.0	−1.5	−.9	3.0
9	0		8.8	.3	.2	8.8
10	−.1		5.9	−.2	−.1	5.9
11	−.1		3.7	0	0	3.7
12	0		2.4	−2.4	1.3	2.4
13	0		1.9	0	0	1.9
14	0		1.4	.6	.3	1.4
15	0		0.5	−.5	−.3	.5
16	0		0	0	0	0.0

* Adjusted by judgment (a) at the peak, (b) in the far right tail––principally here as a basis for fitting.

→

Before going on to fitting, done in panel C of exhibit 27, we have adjusted the smooth in two places, as noted at the foot of panel B.

The lower panel of exhibit **28** shows the residuals, which are neither good nor horrible. In order to fit the smoothed (and adjusted) root ratios moderately far away from the peak, we had to move the peak of the fit about one bin number lower than the peak of the data, thus creating the large hill in the plot of residuals. Clearly, the residuals are wider than we would like, and two points are outside the reference fences (though not the self fences).

exhibit **27** of chapter 17 (continued)

C) FIT––using $1.52 - .12(\text{hanned} - 4.0)^2$ for log (root) ratio

Bin #	Hanned dbl. rt.	Shift	Shift^2	log*	↓	↓	Fitted log	Fitted ratio	Residual ratio	Residual root
0	1.41	−2.59	6.7	.72	P		.72	5.2	.1	.1
1	2.37	−1.63	2.7	1.21	L		1.20	15.0	−.2	−.2
2	3.13	−.87	.8	1.39	O		1.42	26.3	−1.7	−1.4
3	3.72	−.28	.1	1.49	T		1.51	32.4	−1.1	−.8
4	4.23	.23	.1	1.53			1.51	32.4	1.2	.8
5	4.68	.68	.5	1.50			1.46	28.8	2.1	1.4
6	5.09	1.09	1.2	1.42	U		1.38	24.0	2.4	1.5
7	5.47	1.47	2.2	1.29	S		1.26	18.2	1.3	.8
8	5.83	1.83	3.3	1.11	E		1.12	13.2	−1.7	−1.0
9	6.16	2.16	4.7	.94	D		.96	9.1	0	0
10	6.48	2.48	6.2	.77			.78	6.0	−.3	−.2
11	6.78	2.78	7.7	.51	H		.60	4.0	−.3	−.2
12	7.07	3.07	9.4	.38	E		.39	2.4	−2.4	−1.3
13	7.35	3.35	11.2	.28	R		.18	1.5	.4	−.2
14	7.62	3.62	13.1	.15	E		−.05	.9	1.1	.6
15	7.87	3.87	15.0	.30			−.28	.5	−.5	−.2
16	8.12	4.12	17.0	L			−.52	.3	−.3	−.1

* log of adjusted fit from panel B

P) PROBLEMS

27a) Make the plot used in panel C and check––or improve on––the choice of fit.

27b) Find a similar set of data and repeat all these calculations.

27c) Replot exhibit 28 against hanned location, rather than against bin number.

S) SOURCE

E. Rutherford and H. Geiger, 1910. "The probability variations in the distribution of α-particles." (*The London, Edinburgh and Dublin) Philosophical Magazine and Journal of Science* **20**: 698–704. Table on page 701.)

The double-root

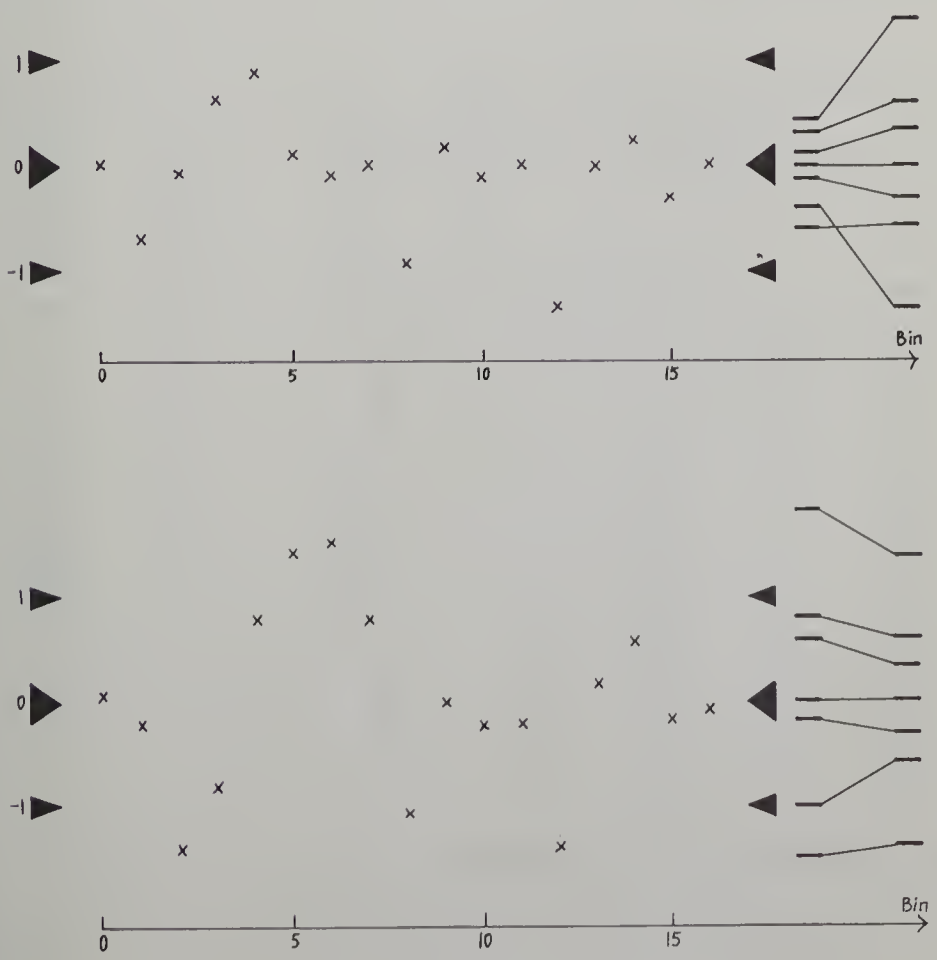
$$\text{location} = \sqrt{2 + 4\,\text{count}},$$

$$\text{width} = \sqrt{4 + 4 \cdot \text{count}} - \sqrt{4 \cdot \text{count}},$$

has allowed us to do (a) a reasonably good job of smoothing the Rutherford and Geiger data and (b) a job of fitting it that may be bearably good. (There seem to be further systematic things going on in the residuals, but it is hard to put words to them.)

exhibit 28 of chapter 17: polonium scintillations

The rough and the residuals from exhibit 27



review questions

What is a double-root? Why might we want one? What are the bin boundaries for basic counts 1, 2, and 3? Of what quantities did we give a table? How are they used? Where do we usually treat double-root values as located? Is this unusual? To what example did we then turn? What were the results? What did we do before fitting? How well did we do with our fit? What do we think of double-roots?

17G. Cautionary examples

It is human, it would seem, to jump from "thus-and-so is a satisfactory fit" to "thus-and-so is the (only) proper fit". The best way to inhibit such natural--but dangerous--leaping at conclusions seems to be to look at some examples where it cannot be right to jump. The simplest such are examples where TWO DIFFERENT FITS are reasonably SATISFACTORY. (It cannot be right to jump to the conclusion that BOTH are the ONLY proper fit!) Let us do a pair of examples, choosing cases where we were marginally satisfied with previous fits. At the same time, we can show how we can polish up our fits by finding and using "desired peaks".

exhibit 29 of chapter 17: forearm lengths

Calculations for root-reciprocals of actual root counts

A) FAST FIT taking PEAK at 7--shift^2 = $(B-7)^2$

B = Bin #	Recip-root of root count*	shift^2	↓	.255 + .0127× shift^2	Fitted root count	Root resid
1	1.000	36	M	.712	2.0	−1.0
	.632	25	A	.572	3.1	−.6
3	.494	16		.458 —	4.8	−.7
	.378	9	K	.369	7.3	−.3
5	.299	4	E	.306	10.7	.5
	.266	1		.268	13.9	.2
7	.256	0		.255	15.4	−.1
	.272	1	A	.268	13.9	−.4
9	.297	4		.306	10.7	.6
	.363	9		.369	7.3	.3
11	.423	16	P	.458	4.8	.8
	.598	25	L	.572	3.1	−.3
13	.725	36		.712	2.0	−.1
	.845	49	O	.877	1.3	+.1
15	.791	64	T	1.068	0.9	+.7

* This is $1/\sqrt{\text{root count}}$ later called just "recip-root".

Note that .255 + .0127(shift)^2 comes from plot called for after the "shift^2 column".

→

forearm lengths again

We fitted the counts of fathers' forearm lengths moderately well with

$$\log \sqrt{\text{count}} = 1.17 - .0282(\text{B} - 8.25)^2.$$

Let us try a gently differing expression, replacing "log" by "1/root". Exhibit **29** has the computations, starting by assuming "peak" = 7 as a reasonable eye-catching and eye-chosen start. The plotting for a fit is left to the reader. The residuals from

$$\text{recip-root of } \sqrt{\text{count}} = .255 + .0127(\text{B} - 7)^2$$

then seem--see the end of panel A of exhibit 29--to be quite satisfactory. Indeed, they compare favorably with those of exhibit 15, where we fitted to log root instead of recip-root. But we might do still better!

exhibit **29** of chapter 17 (continued)

B) DESIRED SHIFT and PEAK, then REFIT with PEAK = 7.15--shift^2 = $(B = 7.15)^2$.

Desired		↓				
shift*	peak	↓	new shift^2	.255 + .0127 shift^2	Fitted root count	Root resid
7.66	8.66	M	37.8	.735	1.8	−.8
5.45	7.45	A	26.5	.592	2.9	−.4
4.34	7.34		17.2	.473	4.5	−.4
3.11	7.11	K	9.9	.381	6.9	.1
1.86	6.86	E	4.6	.313	10.2	1.0
.93	6.93		1.3	.272	13.5	.6
.28	(6.72)		.0	.255	15.4	−.1
1.16	6.84	A	.7	.264	14.3	−1.8
1.82	7.18		3.4	.298	11.3	0
2.92	7.08		8.1	.358	7.8	−.2
3.64	7.36	P	14.8	.443	5.1	.5
5.20	6.80	L	23.5	.554	3.3	−.5
6.08	6.92		34.2	.689	2.1	−.2
6.82	7.18	O	46.9	.851	1.4	0
6.50	8.50	T	61.6	1.037	.9	.7

* recip-root = .255 + .0127 $(B$ − desired shift$)^2$, so that desired shift = $\sqrt{\text{recip-root} - .255}/.0127$.

Note that 7.15 comes from plot called for after the "desired peak" column.

P) PROBLEMS

29a) Calculate and plot the desired peaks for the fit of exhibit 23 to this same data.

29a2) How do you suggest continuing? Try what you suggest.

peak polishing (rest of section is optional)

To see if we can do better, we try a convenient polishing technique--asking, for each bin, which value of "peak" in

$$.255 + .0127(B - \text{peak})^2$$

will reproduce the observed count. As the equations at the foot of panel B (of exhibit 29) make clear, it is relatively easy to calculate the "desired shift" and the corresponding "desired peak". This is done in panel B, plotting of the results being left to the reader. When we plot--or when we look hard at panel B--we see that all but the first and last points are in relatively good agreement with one another.

The median of the twelve well-behaved values of desired peak is 7.15, so the remainder of panel B calculates the fit for "peak = 7.15", with results slightly better than for "peak = 7".

The real merit of this desired-peak analysis is not the change from 7.00 to 7.15, but rather the fact that a constant peak does very well, even when we try to look very hard, leaving the peak free to vary.

The upper panel of exhibit **30** shows the final residuals, which are quite well behaved--one low point being an exception. These are to be compared with panel B of exhibit 15, where we were left wondering about a better fit. Perhaps we now have one.

polonium scintillations again

We can now apply the desired peak technique to the polonium-scintillations example, where (exhibit 27) we earlier found

$$\log \sqrt{\frac{\text{count}}{\text{width}}} = 1.52 - .12(\text{L} - \text{peak})^2$$

to be a reasonable fit.

Let us see what a careful look can dredge up. Starting from the original log root ratios, and these two fitted constants, we find the arithmetic at the left of exhibit **31**. Exhibit **32** shows the desired peaks--and a fit by a broken straight line. The rest of exhibit 31 carries this fit through to root residuals, which are plotted in the lower part of exhibit 30. The two-straight-line fit, which corresponds to using one fit on the lower and a different one on the upper end, is clearly quite good. When we compare with the lower panel of exhibit 28, where we called the residuals "neither good or horrible," we see that we have again improved the fit considerably.

comment

Clearly, calculating desired peaks offers us a useful way to look more closely at our fits, sometimes going along with exactly the sort of simple fit we

Residuals from two alternative fits (see exhibits 29 and 31 for numbers)

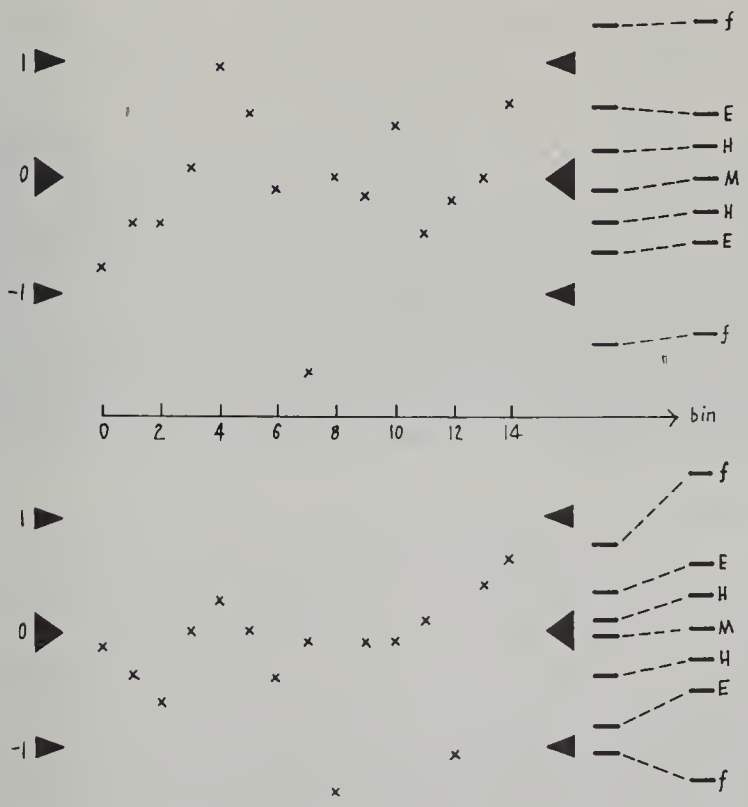


exhibit 31 of chapter 17: polonium scintillations

The desired peak calculation for another example

A) The ARITHMETIC--using $1.52 - .125\,(\text{hanned} - 4.0)^2$

B= Bin #	L= Bin loc	log root ratio	Desired shift	Desired peak	↓ TURN TO EXHIBIT 32 ↓	Fitted peak 3.77 + .11L	Fitted peak 4.63 − .092L	1.52 − .125x (L − fit)²	Fitted root ratio	Root ratio resid	Root resid
0	1.41	.727	2.52	3.93		3.92		.732	5.4	−.1	−.1
	2.45	1.195	1.61	4.06		4.04		1.204	16.0	−.4	−.4
	3.16	1.390	1.02	4.18		4.12		1.405	25.4	−.8	−.6
	3.74	1.496	.44	4.18		4.18		1.496	31.3	0	0
4	4.24	1.526	imag	4.24		4.24	4.24	1.520	33.1	.5	.3
	4.69	1.490	.49	4.20		4.29	4.20	1.490	30.9	0	0
6	5.10	1.421	.89	4.21			4.16	1.410	25.7	−.7	−.4
	5.48	1.290	1.36	4.12			4.13	1.292	19.6	−.1	−.1
8	5.83	1.059	1.92	3.91			4.09	1.142	13.9	−2.4	−1.4
	6.16	.960	2.12	4.04			4.06	.969	9.3	−.2	−.1
10	6.48	.755	2.47	4.01			4.03	.770	5.9	−.2	−.1
	6.78	.566	2.76	4.02			4.01	.561	3.6	.1	.1
12	7.07	--					3.98	.326	2.1	−2.1	−1.1
	7.35	--					3.95	.075	1.2	.7	.4
14	7.62	--					3.93	−.182	.7	1.3	−.7

started with, sometimes calling for a more complex fit. We can be happy to have such a tool. The main issue of this section, however, is broader than a specific technique.

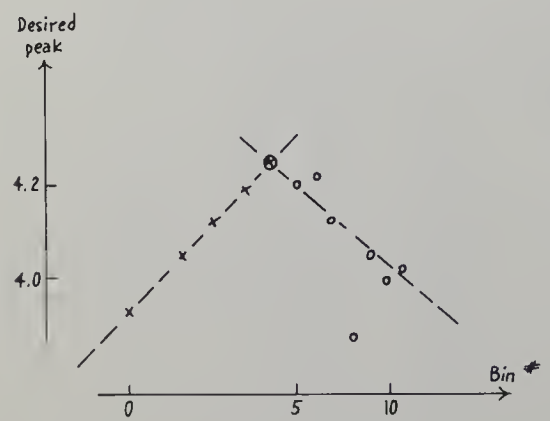
We have used desired peaks to show that finding a reasonable fit does not mean that we are unlikely to do better with some other fit. The changes we have considered are quite appreciable: A log expression of root count was replaced by a recip-root expression; a single log root ratio fit was replaced by a pair of log root ratio fits involving quite different constants. The moral is clear: **If such large changes need not greatly distort the fit, smaller changes need not distort the fit appreciably. Even when we see a very good fit--something we know has to be a very useful summary of the data--we dare not assume that what is fitted is of exactly the right form; we dare not believe we have found a natural law.**

review questions

What two purposes does this section have? How did we try to meet them? What example did we start with? What did we fit? How well did it do? Compared to what? What is a "desired shift"? A "desired peak"? Can we use them to improve a fit? How? What happened in the example? Where did we turn next? What did we do? What happened? Can we use part of two fits? Why/why not? How good was our result? By itself? Compared to an earlier fit? What is the moral toward which this section points? How does it point?

exhibit **32** of chapter 17: polonium scintillations

Analysis of the desired peaks from exhibit 31 and display of the resulting residuals



17H. How far have we come?

This chapter has focused on the analysis of sequences of counts in successive bins and a technique of fitting designed to work for peaked sequences.

The fitting technique first notes that symmetrically peaked curves have to fall away from their peak by an amount determined by the SQUARE of the shift away from the location of the peak, and then seeks a re-expression of the curve's height so that this dependence on the squared shift is linear.

As the many examples have shown, these two techniques together work surprisingly well, especially as we have seen by using a display of the roughs that is particularly fitted for examining roughs arising from counts.

We are now ready:

- to start in on a sequence of bin counts by taking (square) roots and smoothing the results carefully.
- to plot the rough and comparing its fEHMHEf with some standard values.
- to locate the peak by splitting the difference between points of equal height.
- to plot (varied expressions of) the smooth against the squared shift (the square of location MINUS peak) and find (hopefully) an expression for the smooth for which a straight line is an adequate fit.
- to deal with bins of unequal width by replacing $\sqrt{\text{count}}$ by $\sqrt{\text{count/width}}$, whether the inequality of widths was inevitable or the result of our own choice.
- to deal with basic counts in terms of $\sqrt{2 + 4\ (\text{basic count})}$.
- to use an analysis in terms of "desired peaks" to improve a first fit.

We now understand that:

- bin boundaries can be described in terms of amounts, balances, or basic counts.
- bin widths are sometimes fixed (e.g., numbers of children) and sometimes at our choice (e.g., octaves for larger counts).
- that "thus and so is a satisfactory fit" DOES NOT mean that "thus and so" is "the proper fit", or "the best fit", or "a natural law".

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Product-ratio plots--going beyond bins

18

One problem that arises in a variety of fields is dealing with very stretched-tailed patterns of basic counts--or amounts. The sizes of different cities and towns in a region--the occurrences of different words in a play, novel, or text--even the numbers of species in each genus of some kind of plant or animal--these are but three examples. Two goals are appropriate:

- ◇ we want to be able to picture such distributions so that we can look at and compare them.
- ◇ we would like to find some way to describe most of their behavior in some very simple way.

(Of course, once we have done this, we will want to look at the "rest" of the behavior, also.) By now, these kinds of goals should be quite familiar.

Techniques we have already learned do moderately well in producing usable pictures. Octave bins for basic counts tame many long-tailed distributions, but they miss detail that is often meaningful. This is not surprising. We should be able to learn rather more from a count of 1647 when we preserve it in detail than when we say only that it is between 1024 and 2047.

In this chapter, then, we look at procedures that focus on individual counts, rather than bins--at least for the largest counts--striving first for pictures we can compare and then for simple descriptions and the consequent residuals.

These techniques can also be used for other very stretch-tailed distributions, like those of the areas of lakes and ponds (in a country or a continent), that do not involve counts. We shall leave such examples to the reader.

review questions

What problem arises? What goals are reasonable? Are they familiar? How well will what we have already learned do? Will we use bins in this chapter?

18A. Sizes and counts

We mentioned sizes of lakes and ponds--clearly not basic counts--as a possible example. Even if such sizes are not basic counts, they often share, because they are given to limited precision, a difficulty that is very common with basic counts: tied basic counts or tied sizes. The classical approach to this

kind of data, put forward by George Kingsley Zipf, spoke of "rank" and "size" and proceeded to their product, which

"Zipf's law"

said was constant. The question of "which rank" was never clearly addressed, perhaps because of emphasis on large sizes of basic counts. When many basic counts of "1" occur, as is so often the case, their "rank" requires careful definition. (In the Pushkin example to which we will shortly turn, words of basic count 1 occupy ranks 2400 to 4783. What rank should we use for "size 1"?)

We will use the "completed rank", which we will often abbreviate as

c'rank,

the largest rank assigned to that basic count (or size), the number of individuals attaining at least that basic count, or size.

Our next exhibits will look at three examples where the basic counts are:

- the number of those papers, abstracted in *Science Abstracts*, A, for 1961 and classified as on Electrical Properties of Solids, that appeared in a given journal.
- the number of those papers, abstracted in *Science Abstracts*, A, for 1961 and classified as on Atomic and Molecular Physics, that appeared in a given journal.
- the number of appearances of a given Russian word (grammatically defined) in Pushkin's *A Captain's Daughter.*

The long-tailedness of all three is shown by the following information about extreme occurrences:

	"Most"	"1's"	"Total"	"#"
Elec. Prop. of Solids:	255	46	1342	118
Atomic and Molec. Phys:	372	39	1339	107
The Captain's Daughter:	1160	2384	29345	4783

Thus, for *The Captain's Daughter*, one word occurred 1160 times, while 2384 words appeared only once and there were a total of 29,345 occurrences of 4,783 different words.

Exhibit **1** has the numbers; exhibit **2** shows the three root-residual plots for the fits indicated in panel B of exhibit 1, which give root-count-per-octave --or its logarithm--as linear in bin number. We see nothing distinctive, nothing alerting. Yet at least one of these three examples has a peculiarity well worth noticing.

The five highest basic counts in each example are:

Elec. Prop. of Solids:	255	130	128	75	69
Atomic and Molec. Phys:	372	79	61	53	50
The Captain's Daughter:	1160	777	724	582	479

exhibit **1** of chapter 18: three sets of occurrences

Three long-tailed counts by octave bins

A) DATA by OCTAVE

Bin # B	Basic counts	Count	Root smooth	Root fit	Root resid
	1	46	6.8	5.6	1.2
1	2 to 3	23	4.8	4.9	−.1
	4 to 7	15	3.9	4.3	−.4
3	8 to 15	16	3.6	3.6	.4
	16 to 31	11	3.0	3.0	.3
5	32 to 63	2	1.9	2.3	−.9
	64 to 127	2	1.4	1.7	−.3
7	128 to 255	3	1.0	1.0	.7
	256 to 511	0	.4	.4	−.4
9	512 to 1023				
	1024 to 2047				

Bin # B	Basic counts	Count	Root smooth	Root fit	Root resid
	1	39	6.2	5.6	.6
1	2 to 3	20	4.5	4.9	−.4
	4 to 7	13	3.6	4.2	−.6
3	8 to 15	13	3.6	3.5	.1
	16 to 31	13	3.4	2.8	.8
5	32 to 63	7	2.4	2.1	.5
	64 to 127	1	1.4	1.4	−.4
7	128 to 255	0	.5	.7	−.7
	256 to 511	1	0	0	1.0
9	512 to 1023				
	1024 to 2047				

Bin # B	Basic counts	Count	Root smooth	Root fit	Root resid
	1	2384	48.8	50.1	−1.3
1	2 to 3	1280	35.8	34.7	1.1
	4 to 7	580	24.1	24.0	.1
3	8 to 15	280	16.7	16.6	.1
	16 to 31	133	11.5	11.5	0
5	32 to 63	70	8.4	7.9	.5
	64 to 127	31	5.6	5.5	.1
7	128 to 255	11	3.3	3.8	−.5
	256 to 511	10	3.2	2.6	.6
9	512 to 1023	3	1.7	1.8	−.1
	1024 to 2047	1	1.3	1.3	−.3
		(0)	(0)	.9	(−.9)

→

exhibit **1** of chapter 18 (continued)

B) The FITS

Bins are (started) octaves; #0 contains basic count 1, #1 contains basic counts 2 and 3.

Electrical Prop. of Solids: $\sqrt{(\text{counts/bin})} = 5.6 - .65B$

Atomic and Molecular Physics: $\sqrt{(\text{counts/bin})} = 5.6 - .7B$

The Captain's Daughter: $\log_{10} \sqrt{(\text{counts/bin})} = 1.70 - .16B$

C) The LETTER VALUES

(Elec. Prop. of Solids)

#	(no zeros)		
M5	−.1		
H3	.4	−.4	.8
E2	.7	−.4	1.1

	1.2	
f	1.6	−1.6
	xxx	xxx

(Atomic and Molec. Phys)

#9	(one zero)		
M5	.1	.4	
H3	.6	−.4	1.0
E2	.8	−.6	1.4

	1.5	
f	2.1	−1.9
	xxx	xxx

(*The Captain's Daughter*)

#11	(one zero)		
M6	.1		
H3h	.3	−.4	.7
E2	.6	−.9	1.5

	1.0h	
f	1.3h	−1.4h
	one	xxx

P) PROBLEMS

1a) In view of the steady––and slight––tendency to positive residuals for *The Captain's Daughter* (how shown?) try $\log_{10} \sqrt{\text{counts/bin}} = .01 + 1.70 - 16B$, finding residuals and letter values. Is it an improvement?

S) SOURCES

(for Elec. Prop. of Solids, and Atomic and Molec. Phys.): Tables 20 and 22, respectively, of Stella Keenan and Pauline Atherton, 1964, *The Journal Literature of Physics* (AIP/DRP PA1(1964)), 156 pp., American Institute of Physics, New York.

(for "The Captain's Daughter"): G. Herdan, 1966. *The Advanced Theory of Langauge as Choice and Chance* (*Kommunikation und Kybernetik in Einzeldarstellungen. Band* 4), Springer, New York. Table 32 at page 97.

His source: Harry H. Josselson, 1953. *The Russian Word Count: (and Frequency Analysis of Grammatical Categories of Standard Literary Russian)*, Wayne University Press, Detroit.

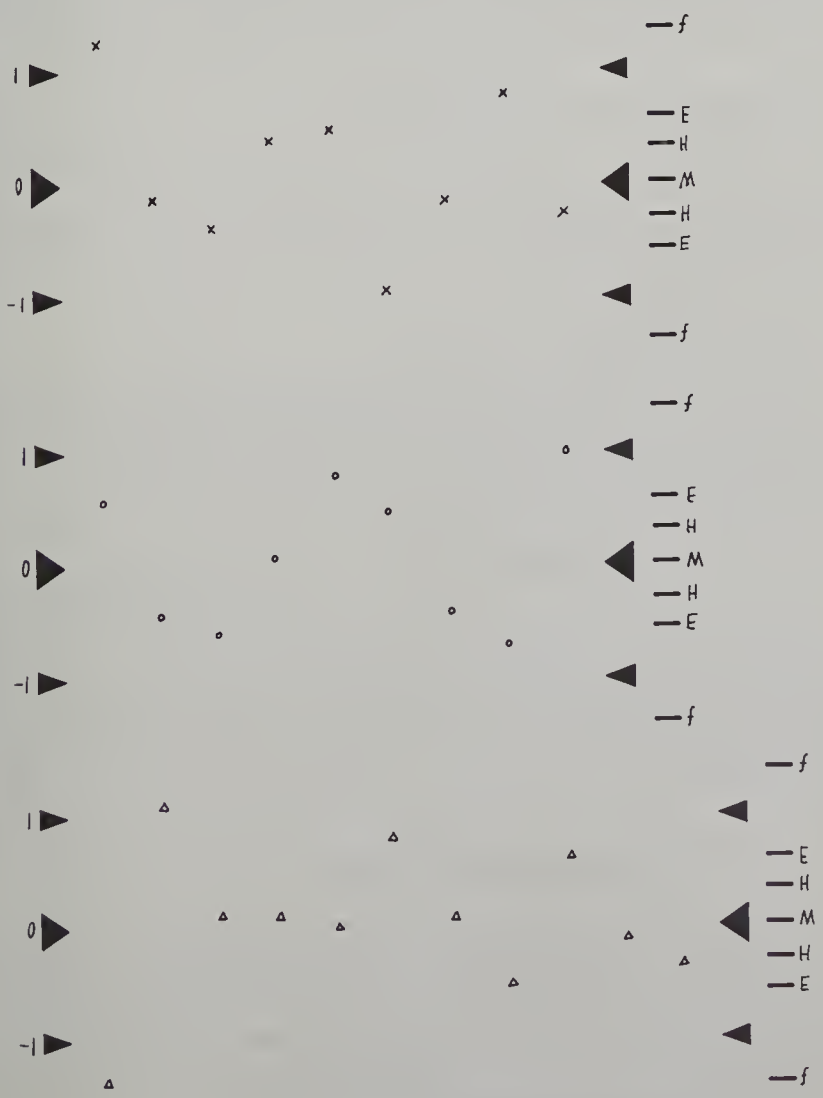
Clearly, the highest basic count for Atomic and Molecular Physics is much more detached from the other high basic counts than is the case for either of the other two examples. We ought to have some way to trap such indications. How?

We can get a better idea if we look at the roots of the five high basic counts, together with their differences, where we find, as roots:

Elec. Prop. of Solids (root counts):	16.0	11.4	11.3	8.7	8.3
Atomic and Molec. Phys. (root counts):	19.3	8.9	7.8	7.3	7.1
The Captain's Daughter (root counts):	34.1	27.9	26.9	24.1	21.9

exhibit **2** of chapter 18: three sets of occurrences

The residuals from fits to octave bins for the three examples of exhibit 1



and, as differences of these roots:

Elec. Prop. of Solids (diffs):	4.6	0.1	2.6	0.4
Atomic and Molec. Phys. (diffs):	10.4	1.1	0.5	0.2
The Captain's Daughter (diffs):	6.2	1.0	2.8	2.2

In each case the highest and the second highest are quite well separated--so well separated that asking about "the highest" is very close to asking about a specific name. If we were asking about a name, we would expect to begin by taking the square root of its count.

The total number of species--here journal names or Russian words--involved is another count. Again we would naturally take a square root.

a possible approach

At one extreme many individuals (many journals, many words) appear only once; at the other extreme, one individual (one journal, one word) appears very many times. There is a vague symmetry between two counts:

- ◇ the number of appearances = the basic count.
- ◇ the number of individuals appearing at least that many times = the completed rank (from above).

Perhaps we should find an analysis that

- ◇ combines these pairs of counts.
- ◇ treats them symmetrically.
- ◇ involves the square root of the largest number of appearances (= the root of the largest basic count) at one extreme and the square root of the total number of individuals (= the root of the largest rank) at the other.

review questions

Why do we need another way to look at long-tailed distributions of counts? What examples did we start with? What did the roughs show? What did the largest individual values show? How could we let them show this effectively? What two counts available for each basic count have a vague symmetry? What is a c'rank?

18B. Product-ratio analysis

What are the easy ways to combine x and y, treating them symmetrically? One is to form $x + y$ and $x - y$, which do not sound promising here. Another, when x and y are both never zero, is to form xy and x/y--or, better than x/y itself, if both x and y are always positive (as here), xy and $\log(x/y)$. This is a little more promising start. When $x = 1$, we want to involve $\sqrt{y}$. When $y = 1$, we want to involve $\sqrt{x}$. Clearly, $\sqrt{xy}$ will do just this for us.

So we ought to look at what we get when we look at

$$\sqrt{(\text{basic count})\ (\text{completed rank above})}$$

in terms of

$$\log((\text{basic count})/(\text{completed rank}))$$

--at root of PRODUCT against log of RATIO. This seems even more plausible when we recall that Zipf stressed the PRODUCT. (We are not as optimistic as he was. We hope that the product will be manageable, but hardly that it will prove to be constant.)

Exhibit **3** illustrates the operations of such a product–ratio analysis on a slightly smaller example. This exhibit combines a detailed computation using all detail--in the left four columns--and computation based on two sets of half-octaves--to be described below. Note that the "root of prod" column, whose values are the square roots of the products of

- ◇ each basic count (basic), and
- ◇ the total number of cases (here authors) with at least that basic count (c'rank),

BEGINS with $\sqrt{46}$, where 46 is the largest basic count, and ENDS with $\sqrt{721}$, where 721 is the total number of authors (as we planned).

half-octave calculation

It is enough to take (basic, c'rank) pairs at or near the half-octave values --e.g., 1, (1h), 2, 3, 4, 6, 8, 11, 16, 22, . . . , of both basic and c'rank, stopping each once basic = c'rank has been crossed. In practice, we would make the calculations in the lines of exhibit 3 only for the two sets of half-octaves, thus saving eight of the lines started in exhibit 3--and saving many more lines in larger examples.

display

Exhibit **4** plots "root of PRODUCT" against log of RATIO. The straightness of the plot is notable. Giving the two ends corresponding to 721 authors with at most 46 contributions must tell most of the story. (Here the root of PRODUCT has been very manageable, and yet far from constant.)

Some related problems are set in exhibit **5.**

review questions

How did we combine "basic" and "c'rank"? Why choose these combinations? What is a product–ratio analysis? Do we have to look at every basic count? How can we use half-octave steps? Are half-octaves sacred? What example did we try? How did the product–ratio plot look?

exhibit **3** of chapter 18: econometric activity

Numbers of authors contributing various basic counts of papers, 1933–1952––counts combine meetings of the Econometric Society and the pages of its journal *Econometrica*.

A) The NUMBERS––detailed and half-octave calculations

	Detailed calculation				values for half-octaves only*			
					√PROD		log of RATIO	
Contr = basic count	Count of authors	Ind. c'rank	√ of† PROD	log of‡ RATIO	basic	c'rank	basic	c'rank
46	(1)	1	6.8	1.66		6.8		1.66
37	(1)	2	8.6	1.27		8.6		1.27
30	(1)	3	9.5	1.00		9.5		1.00
28	(2)	5	11.8	.75				
24	(1)	6	12.0	.60		12.0		.60
23	(1)	7	12.7	.52				
18	(1)	8	12.0	.35		12.0		.35
17	(2)	10	13.0	.23				
16	(1)	11	13.3	.16	13.3		.16	
14	(2)	13	13.5	.03				
13	(3)	16	14.4	−.09		14.4		−.09
12	(2)	18	14.7	−.18				
11	(4)	22	15.6	−.30	15.6		−.30	
9	(1)	23	14.4	−.41				
8	(11)	34	16.5	−.63	16.5		−.63	
7	(6)	40	16.7	−.76				
6	(23)	63	19.4	−1.02	19.4		−1.02	
5	(14)	77	19.6	−1.19				
4	(40)	117	21.6	−1.47	21.6		−1.47	
3	(61)	178	23.1	−1.77	23.1		−1.77	
2	(107)	285	23.9	−2.15	23.9		−2.15	
1	(436)	721	26.9	−2.86	26.9		−2.86	

*** Half-octave sequences of PROD and RATIO combined.**
† Examples: 6.8 = $\sqrt{(48)(1)}$, 8.6 = $\sqrt{(37)(2)}$, and so on to 26.9 = $\sqrt{(1)(721)}$.
‡ Examples: 1.66 = log 46/1, 1.27 = log 37/2, and so on to −2.86 = log 1/721.

P) PROBLEMS

3a) Fit a straight line to (1.66, 6.8) and (−2.86, 26.9) and use to give fitted roots. Find and plot residuals. Comment.

3b) Interpolate values for basic = 1h and c'rank = 1h, and smooth the resulting basic and c'rank sequences. Compare with raw, and comment.

S) SOURCE

D. H. Leavens, 1953. (Communication) *Econometrica* 21: 630–632.

exhibit **4** of chapter 18: econometric activity

Plot of root of PRODUCT against log of RATIO (numbers in exhibit 3)

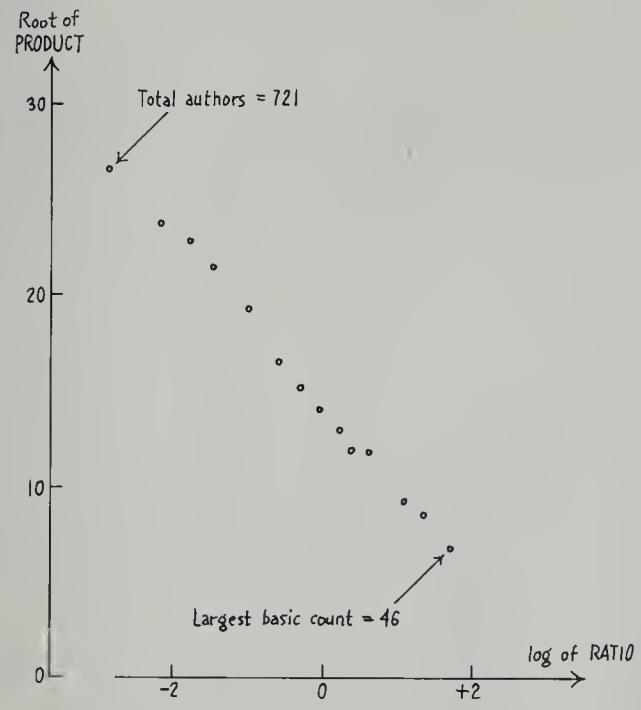


exhibit **5** of chapter 18: three participations.

Half-octave summaries of three other sets of participation data

A) HIGH-BASIC-COUNT ENDS

	# of contributions per individual = basic		
c'rank	Math†	Physics†	Chemistry†
1	70	--	346
2	42	48	114
3	39	37	109
4	35	34	107
6	27	27	84
8	21	25	78
11	20(10)*	24	68
16	13(15)	18	57
22		16	54(21)
32			47
44			41

→

18C. Forcing the unusual to be noticed

Exhibit **6** gives the data, in half-octave form, for a total of nine different subfields of physics, including the two that appeared in exhibit 1. Exhibit **7** shows the plot of "root of PRODUCT" vs. "log of RATIO" for the example "Atomic and Molecular Physics" that attracted our attention earlier. We see a reasonably peaky plot, with one point--that for the most popular journal--far away from a continuation of the others. We seem to have had a "whopper" thrust upon us, without having to look for it specifically. (This is of course what we want our displays to do.)

exhibit **5** of chapter 18 (continued)

B) LOW-BASIC-COUNT ENDS

Basic count	# with as many or more contributions = c'rank		
	Math	Physics	Chemistry
1	278	1325	6891
2	145	541	2900
3	102	337	1841
4	78	210	1348
6	55	127	877
8	36	80	633
11	23	48	419
16	12	22	250
22		14	149
32			68
44			41

*** Entries in parenthesis (after columns) are the actual c'ranks used--see Note (1).**

†For mathematics: Members of the Chicago Section of the American Mathematical Society, 1897–1992.

†For physics: Up to 1900 from Auerbach's *Geschichtstafeln der Physik*.

†For chemistry: Authors with last names beginning with A or B, in *Chemical Abstracts*, 1907–1916.

Notes

1. Since our definitions are in terms of basic counts, half-octave values of ranks may fall in the middle of a basic count, and thus be unusable. In such cases, we have used the nearest usable value LESS than the half-octave value.

2. Note that columns in panel B do NOT correspond directly to those in panel A.

P) PROBLEMS

5a/b/c) Make a plot of the raw root-of-product and log-of-ratio values for Math/Physics/Chemistry.

5d/e/f) (helped by (5a/b/c)). Make a plot of the smoothed root-of-product and log-of-ratio values for Math/Physics/Chemistry.

exhibit **6** of chapter 18: physics papers

Half-octave summaries for nine subfields of physics--all papers in each abstracted in *Science Abstracts*, A, for 1961

A) HIGH-BASIC-COUNT END--c'rank and corresponding basic counts; data for ranks 5, 7, 9, and 10 included for curiosity (see previous exhibit for use of ( )'s).

c'rank	Astro-phys.	Fluids + Gases	Elec. + Mag.	Nuclear Phys.	Elem. Part.	Atomic + Molec.	Solid State	ElProp Solids	MagProp Solids
1	99	116	152	391	174	372	219	255	168
2	77	83	135	368	160	79	208	130	165
3	71	60	132	231	157	61	196	128	122
4	55	57	119	184	123	53	160	95	57
6	43	36	2	127	73	41	152	36	53
8	34	31(7)	75	89	29	38	110	27	45
11	24	25	64(9)	54	18	27	84	24	38
16	16	17(15)	43(14)	40(15)	12	22(15)	49	19(15)	18
22	8(21)	15(21)	31(21)	30(21)		16	38	14(20)	12
32			22	17			20(31)		
(Total)	(906)	(1206)	(2342)	(2902)	(1238)	(1339)	(3059)	(1342)	(1245)
5	49	52	83	155	110	50	155	69	53
7	40	31	81	102	31	40	117	33	45
9	31	30	74	75	27	35	100	26	40
10	27	29	64	59	21	31	94	24	35

B) HIGH-C'RANK END--basic count and corresponding c'ranks.

1	115	154	182	141	97	107	184	118	87
2	73	104	124	81	60	68	125	72	62
3	51	80	102	78	48	57	102	58	49
4	39	68	81	67	42	46	84	49	38
6	27	50	69	66	31	36	68	40	31
8	21	41	58	52	24	33	55	34	28
11	17	31	48	45	17	27	42	29	21
16	16	17	35	34	12	22	37	18	16
22	13	11	32	30		15	30	12	15
32			20	20			26		

S) SOURCE

Stella Keenan and Pauline Atherton 1964. *The Journal Literature of Physics* (AIP/DRP PAI (1964)). American Institute of Physics, New York.

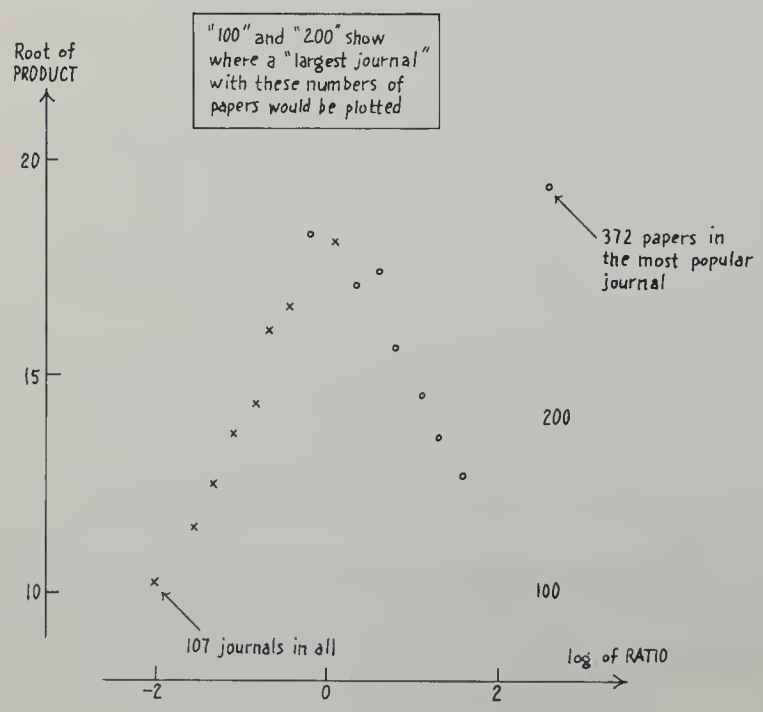
Specifically: Table 8 on pages 12–13 for Astrophysics, Table 12 on pages 18–20 for Physics of Fluids and Gases, Table 16 on pages 24–26 for Electricity and Magnetism (also Plasma), Table 17 on pages 27–29 for Nuclear Physics, Table 18 on pages 29–30 for Elementary Particles, Table 20 on pages 32–33 for Atomic and Molecular Physics, Table 21 on pages 34–36 for Solid State Physics, Table 22 on pages 37–38 for Electrical Properties of Solids, Table 24 on pages 40–41 for Magnetic Properties of Solids.

The reasonable conclusion is that this journal--the *Journal of Chemical Physics*--plays a special role for this subfield. One natural approach is to set this journal--with its 372 papers on Atomic and Molecular Physics--aside, and analyze the others. Doing this will reduce every rank by one, changing both the "root of the PRODUCT" and "log of the RATIO" values for each cut. (It will also move the high-basic-count ends away from being half-octaves, but we agreed to do such things freely.) Thus the low-basic end will involve (1, 106), (2, 67), . . . , (22, 14), while the high-basic end will involve (79, 1), (61, 2), (53, 3), (41, 5), (38, 7), (27, 10), (22, 14) and (16, 21). Exhibit **8** shows the corresponding plot, which is now quite symmetrical, with a peak close to "log of ratio" = 0.0. Clearly, once *J. Chem. Phys.* is removed, the other journals show coherent, though two-segment, behavior. So our best description to date of the distribution of those papers on Atomic and Molecular Physics abstracted in 1961 thus consists of two parts:

- ◇ one special journal, *J. Chem. Phys.* with 372 papers.
- ◇ a mass of other journals--106 of which had at least one of the 967 remaining papers--whose "$\sqrt{\text{product}} - \log \text{ratio}$" plot is well described by the two straight lines joining $(-\log 106, \sqrt{106})$ to $(0, \sqrt{335})$ to $(\log 79, \sqrt{79})$.

exhibit **7** of chapter 18: physics papers

Product–ratio plot for papers in atomic and molecular physics



Interestingly enough, when we examine in similar detail the data for Electrical Properties of Solids, which we early took as a reference standard of nice behavior, we find not one very special journal but five somewhat special journals. (See exhibit **9** for this, and other problems.)

exhibit **8** of chapter 18: physics papers

The plot for Atomic and Molecular Physics after removal of the *Journal of Chemical Physics*

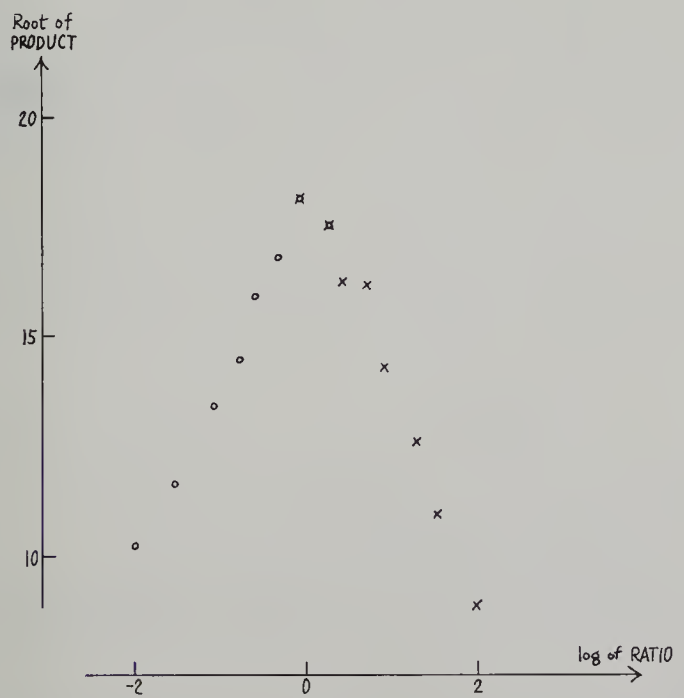


exhibit **9** of chapter 18: problems

Some problems involving product–ratio plots

9a/b/c/d) Make plots like exhibit 7 for "Astrophys"/"Fluids and Gases"/"Elec. + Mag."/"Nuclear Physics", and identify any unusual values.

9a2/b2/c2/d2) Omit the unusual journals, if any, found in (9a/b/c/d), and make a plot for the others. Comment.

9e/f/g/h) As in (9a/b/c/d), for "Elem. Part. "/"Solid State"/"Elect. Solids"/"Mag. Prop. Solids".

9e2/f2/g2/h2) As in (9a2/b2/c2/d2), for the corresponding category.

review questions

What example attracted attention earlier? How did its product–ratio plot behave? What was the diagnosis? What did we do then? With what outcome? How would you describe the position we reached?

18D. Comparisons between collections

Turning back to exhibit 5, we would clearly like to compare these three cases to see to what extent they behave similarly and to what extent differently. Direct comparison of the kind of plots we are so far making will not help us much. With 6891 contributions in chemistry and only 278 in mathematics, we can certainly not expect either the largest basic count--or the number of occurrences with basic count one--to be anywhere near the same.

Clearly, we need to make an allowance for the size of the body of data we are looking at. This we can, and will, do in different ways. All will be such that, for example, doubling the count for every basic count, thus doubling each c'rank, will not affect the plot. We like this property, because it means that "getting twice as much data, exactly similar" has no effect on our plot.

The easy way to do this is to choose a number a and a basic count b and to divide every c'rank by a times the c'rank for b (for that set of data).

By replacing c'rank by c'rank/ac_b, we replace PRODUCT, the root of the product of "basic" and "c'rank", by

$$\frac{\text{PRODUCT}}{\sqrt{ac_b}}$$

and RATIO, the log of the ratio of basic to c'rank, by

$$\text{RATIO} + \log(ac_b).$$

Thus we have shrunk the vertical coordinate of our plot by a constant factor and shifted its horizontal coordinate by an additive constant. The shape of our plot is thus left unchanged.

If we take $a = b$, we will get a point at $(\log a^2, 1.000)$ for any set of data. If we do the same for several sets of data, their (raw) plots will have to cross (or touch) at this point.

If we take $a = b = 1$, we will divide every c'rank by the c'rank for basic count 1, which is the total number of cases, thus using

$$\text{completed fraction} = \frac{\text{completed rank}}{\text{total \# of cases}}$$

in place of the c'rank. This is sometimes nice, but doesn't always do what we might like, as we shall soon see.

Exhibit **10** shows the result of taking $a = b = 6$ for the three data sets of exhibit 4. Clearly we can do a very good job of making comparisons here. For basic counts above 6, the three plots are rather similar, with Physics falling fastest and Chemistry showing appreciable curvature. For basic counts below 6,

all three follow straight lines moderately well, though the slopes of the lines are (a) quite different from one another and (b) appreciably, moderately, or very different from the slopes for the same data for basic counts greater than 6.

Exhibit **11** sets some problems.

review questions

Do we need to allow for differences in size of data collections? Why/why not? What were a and b? How were they used? What happened when $a = 6$? What examples did we try? What did they show?

exhibit **10** of chapter 18: three participations

Three collections matched at basic count 6 ($a = b = 6$)

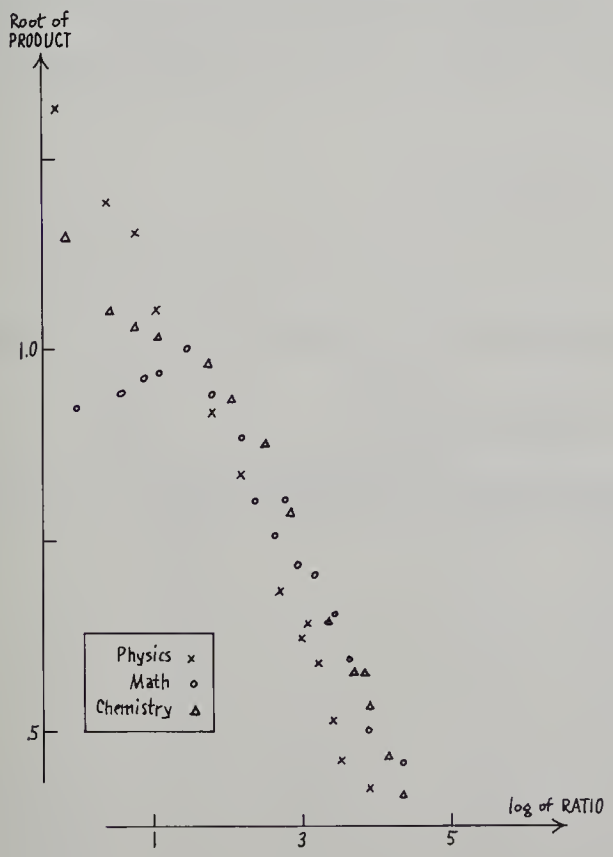


exhibit **11** of chapter 18: problems

Some problems on matched plots, suitable for teamwork

11a) (can be divided among 4; 3 calculations and 1 plot). Take $a = b = 10$ and make the three plots like those in exhibit 10. Discuss and comment.

11b) (can be divided among 4; 3 calculations and 1 plot). Do the same for $a = b = 3$.

18E. Looking at the smallest basic count

Now that we are used to looking at our plots with more and more care, we notice that the point for the smallest basic count, so far basic count one, is not always on the line through the neighboring points. It is easy to experiment with the results of changing the total count, which is, of course, the c'rank for the lowest basic count, and seeing where the point goes. This amounts to deleting (or adding) some "species" with the smallest basic count.

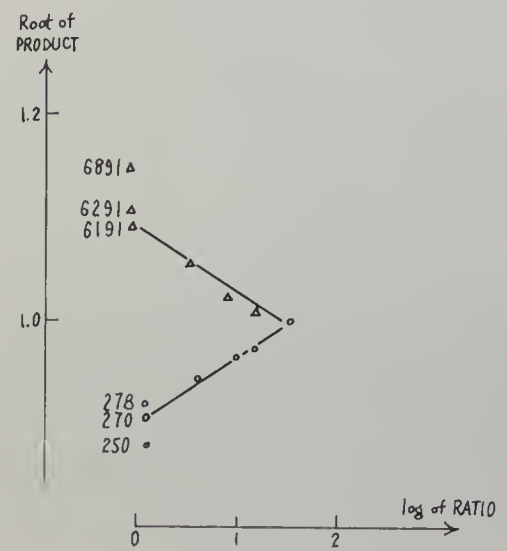
Exhibit **12** shows a part of exhibit 10 with such alternatives plotted. For chemistry, we might find 6251, corresponding to 3351 (=6251 − 2900) authors of one contribution, a better fit to the line than the actual 6891, corresponding to 3991 (=6891 − 2900) such authors. Since $\sqrt{3351} = 58$ and $\sqrt{3991} = 63$, we are likely to find the difference worth appreciable comment (5 is a considerable difference for two square roots of actual counts). For mathematics, on the other hand, 270 giving 125, instead of 278 giving 133, corresponds to $\sqrt{270} = 16.4$ in place of $\sqrt{278} = 16.7$, which does not deserve comment.

review questions

What did we start to do in this section? Was it easy? Was it effective?

exhibit **12** of chapter 18: three participations

Part of exhibit 10 with points for alternative c'ranks for basic count 1



18F. When zeros are counted

There is another situation in which product–ratio plots are helpful, at least for comparative purposes. If we have a binning of counts where zeros can occur and tails are quite stretched out, as for instance in these instances for which data is given in exhibit **13**:

- ◇ numbers of fleas on each of 209 live rats,
- ◇ numbers of isopods found under each of 122 boards,
- ◇ numbers of bacterial clumps seen in each of 400 microscope fields of milk, or
- ◇ numbers of lice on the heads of each of 1083 Hindu male prisoners,

we can start our basic count, so that the log of the ratio gives no trouble. This seems most likely to be helpful when there are more "zeros" than any other single basic count in the original data. (Thus the milk-film example may be marginal.) Exhibit **14** shows the standard (but unsmoothed) product–ratio plots for the four examples. Clearly, lice are behaving differently, while fleas, isopods, and clumps show related patterns.

exhibit **13** of chapter 18: ecological counts

Some data where started counts give helpful product–ratio plots

Rats, with ≥ fleas		Boards, with ≥ isopods		Fields, with ≥ clumps		Heads, with ≥ headlice	
Basic	c'rank	Basic	c'rank	Basic	c'rank	Basic	c'rank
0	209	0	122	0	400	0	1083
1	146	1	94	1	344	1	461
2	114	2	66	2	240	2	355
3	84	3	52	3	160	3	305
5	58	5	33	5	56	5	243
7	39	7	20	7	20	7	209
10	26	10	11	8	11	10	168
15	14	11	8	9	6	15	130
20	8	13	6	10	3	21	96
26	6	14	5			14	5
46	4	15	3			40	44
48	3	17	2			47	33
61	2	10	7			58	22
83	1					74	16
						11	11
						129	8
						149	6
						239	4
						170	3
						303	2
						385	1

→

exhibit **13** of chapter 18 (continued)

S) SOURCES

For rat fleas on 209 live rats in Mobile, Alabama: L. C. Cole, 1946, "A theory for analyzing contagiously distributed populations." *Ecology* 27: 329–341. Table VI on page 338.

For *Trachelipus Rathkei* (an isopod) under 122 boards: L. C. Cole, 1946. "A theory for analyzing contagiously distributed populations." *Ecology* 27: 329–341. Table V at page 337. His source: L. C. Cole, Ecological Monographs 16.

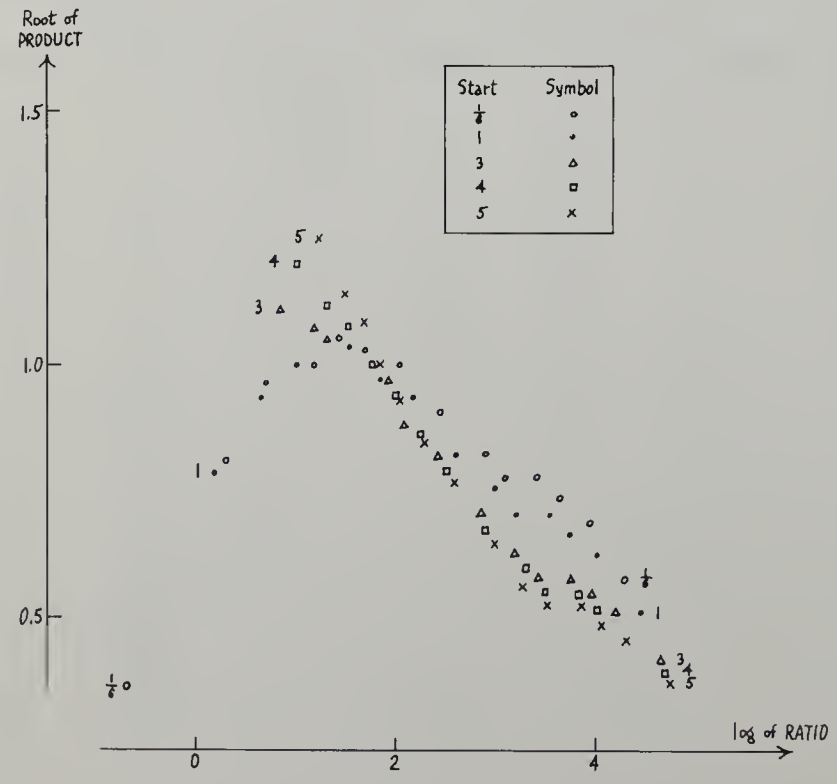
For bacterial clumps in each of 400 microscope fields in a milk farm: M. E. Morgan, P. MacLeod, E. O. Anderson, and C. I. Bliss, 1951. "A sequential procedure for grading milk by microscopic counts." *Storrs Agricultural Experiment Station Bulletin* 276. Used by: C. I. Bliss, 1953. "Fitting the negative binomial distribution to biological data." *Biometrics* 9: 176–196. Table 2 on page 186.

For lice of all stages on the heads of 1083 Hindu male prisoners in Cannamore, South India in 1937–39: C. I. Bliss, 1953. "Fitting the negative binomial distribution to biological data." *Biometrics* 9: 176–196. Table 8 on page 194. His source: F. J. Anscombe, 1950."Sampling theory of the negative binomial and logarithmic series of distributions." (Full data at page 11 of the source for exhibit 19, below).

Original source: P. A. Buxton 1940. "Studies on populations of headlice. III." Material from South India *Parasitology* 32: 296.

exhibit **14** of chapter 18: ecological counts

Fleas on rats with five different starts added to all basic counts (matched with $a = b = 3$)



We can use any start that seems helpful, 1/6, 1/4, 1/2, 1, 2, 3, 4, 5, or what have you. Exhibit 14 shows what five possibilities do with the rat data from exhibit 13. (For later convenience we used $a = b = 3$.) All of them give reasonably manageable (= reasonably easily describable) results, but a start of 4 seems to give the simplest result.

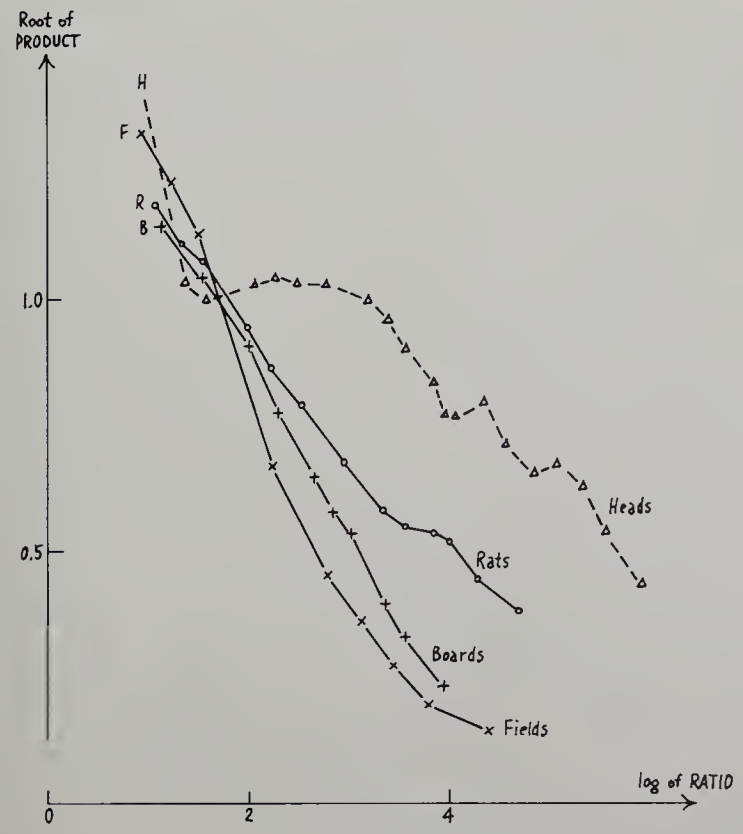
Exhibit **15** shows the plots for start 4 (and $a = b = 3$) for the four sets of data in exhibit 13. We see that:

- ◇ most of each plot is crudely straight.
- ◇ the points for basic count = 1 tend to look discrepant.
- ◇ the tilt of the "heads" plot is appreciably less than the others there.

The effectiveness, in exhibit 14, of a substantial start in straightening out our plot makes us wonder whether some of our earlier plots, where a start was NOT forced by counts of zero, might also be straightened out. Inquiry into this is left to the reader.

exhibit **15** of chapter 18: ecological counts

All four sets of counts, started by 4 and matched at basic count 3



review questions

What is a started basic count? Why might we want to use it? To what examples did we apply it? With what results? Where else might we have tried it?

18G. Under the microscope

Three of the plots in exhibit 15 are somewhat similar in slope. A rough fit gives

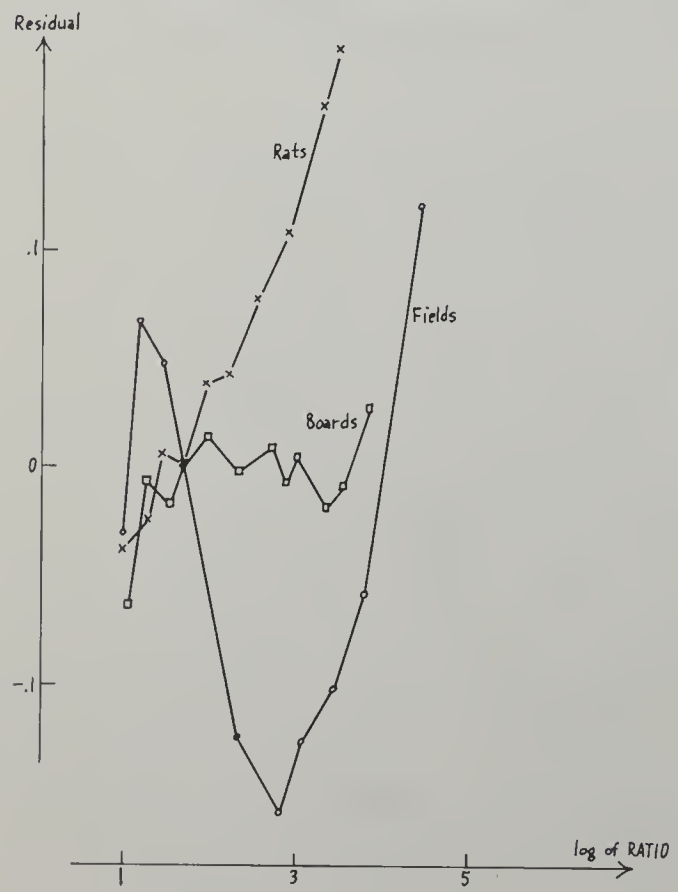
$$\text{PROD} = 1.000 - 0.36\,(\text{RATIO} - 1.69) = 1.608 - 0.36L,$$

where PROD is root of product and RATIO is log of ratio using a start of 4 and $a = b = 4$. Given a fit, it is by now almost automatic for us to take residuals, with the result shown in exhibit **16.** We can now distinguish the behavior of the three data sets quite sharply.

The smallest basic count, here zero, is acting up again, so exhibit **17** sets some problems.

exhibit **16** of chapter 18: ecological counts

Residuals (of root of product with start = 4, $a = b = 3$) from a general straight-line fit (formula in text)



While something could be gained by fitting a line to the "heads" data (see problem (17d)), we do not need a microscope to see how far the zero-basic-count point is out of line for this data. As exhibit **18** shows, a reduction in the zero-basic-count count of from 300 to 400 would be required to bring this point in line. These correspond to differences in roots of

$$\sqrt{622} - \sqrt{322} = 7.0,$$

$$\sqrt{622} - \sqrt{222} = 10.0,$$

exhibit **17** of chapter 18: ecological counts

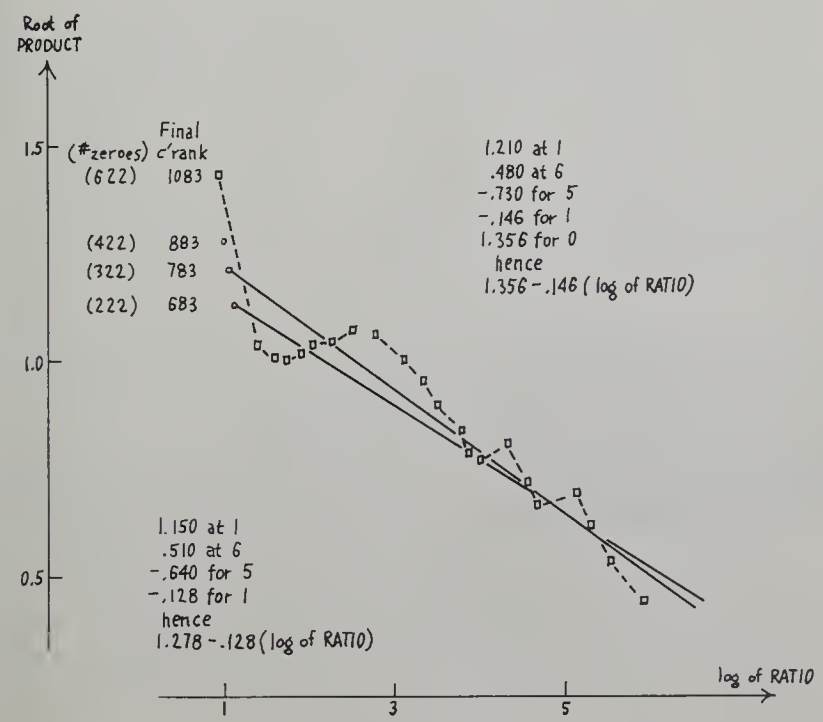
Problems on "using the microscope" (started counts)

17a/b/c) Find what c'rank--and hence what count for basic count = 0--would bring the basic-count-zero point into line on exhibit 16 for the rats/boards-/fields data. Note the difference in roots, $\sqrt{\text{assigned \# of zeros}} - \sqrt{\text{given \# of zeros}}$ for your case, and comment.

17d) Fit a line to the corresponding product–ratio plot for "heads", and plot the residuals (as in exhibit 16).

exhibit **18** of chapter 18: ecological counts

Headlice per head plot (started by 4 and matched at 3) with two possible straight lines and some other c'ranks for basic count one



and can only be considered large. A reasonable interpretation would be that the criminals involved were divided into two groups: one, providing about a third of the heads, being nearly free from lice, while the remainder had louse infestations that behaved simply in our product–ratio plot.

Exhibit **19** poses some closing problems.

review questions

What standard process lets us "use the microscope"? What did we try it on? How did it work?

exhibit **19** of chapter 18: data and problems

Some problems involving data on species per genus assembled by Williams

A) INSECTS

MacGillivray		Beare + D		Stainton		South + E		Edwards		Butler + C	
c'rank	basic	c'rank	basic	c'rank	basic	c'rank	basic	c'rank	basic	c'rank	basic
1	200	1	155	1	40	1	43	1	56	1	41
2	70	2	77	2	26	2	18	2	26	2	33
3	60	3	64	3	23	4	17	3	18	4	21
4	31	4	53	4	18	6	13	4	15	5	20
8	30	6	40	6	15	7	11	6	12	7	15
12	27	9	37	7	13	10	10	7	11	8	13
16	21	11	33	10	11	15	8	11	8	11	9
20	17	16	25	16	8	22	6	15	6	13	8
43	11	22	21	27	6	42	4	17	4	25	8
59	8	35	16	52	4	67	3	19	3	25	4
70	6	70	11	77	3	118	2	28	2	30	3
107	4	96	8	117	2	357	1	47	1	40	2
129	3	134	6	249	1					71	1
171	2	211	4								
352	1	286	3								
		412	2								
		803	1								

MacGillivray: *Coccid*ae (scale insects) of the world (p. 22 of source)
Beare + D: *Coleoptera* of Britain, 1904 (p. 24 of source)
Stainton: *Macrolepidoptera* (excl. Butterflies) of Britain, 1857 (p. 24 of source)
South + E: *Macrolepidoptera* (excl. Butterflies) of Britain, 1939 (p. 24 of source)
Edwards: *Cicadina* of Britain, 1894 (p. 28 of source)
Butler + C: *Cicadina* of Britain, 1942 (p. 28 of source)

→

exhibit **19** of chapter 18 (continued)

B) BIRDS and PLANTS

Witherby ++		Bentham + H		Babington		Willis	
c'rank	basic	c'rank	basic	c'rank	basic	c'rank	basic
1	15	1	47	1	97	1	525
5	13	2	21	2	74	2	500
6	12	3	17	3	48	3	325
11	11	4	16	5	31	4	320
17	8	7	15	6	29	6	250
22	6	8	14	8	21	8	190
43	4	12	12	10	15	12	180
60	3	16	11	16	13	16	150
92	2	22	10	26	11	21	125
199	1	32	8	38	8	31	105
		45	6	60	6	42	65
		88	4	117	4	63	60
		139	3	162	3	88	45
		223	2	255	2	123	30
		479	1	510	1	151	22
						195	16
						264	11
						350	8
						400	6
						488	4
						565	3
						691	2
						1000	1

Witherby ++: Birds, including subspecies, of Britain, 1941 (p. 29 of source)
Bentham + H: Flowering plants of Britain, 1906 (p. 30 of source)
Babington: Flowering plants of Britain, 1922 (p. 30 of source)
Willis: Plants and ferns of the world (1000 genera, p. 30 of source)

P) PROBLEMS

19a/b/c/d/e/f) Make an unstarted product–ratio plot with $a = b = 1$ for the MacGillivray/Beare + D/Stainton/South + E/Edwards/Butler + C data.

19g/h/i/j) Do the same for the Witherby ++/Bentham + H/Babington/Willis data.

19a2/b2/c2/d2/e2/f2) Choose a start between 2 and 6, and superpose the corresponding plot on that made in problem (19a/b/c/d/e/f). Compare; conjecture what start would give a reasonably straight line, if any.

19g2/h2/i2/j2) Do the same on the result of (19g/h/i/j).

19a3/b3/c3/d3/e3/f3) Use the start conjectured in problem (19a2/b2/c2/d2/e2/f2), making the plot, and commenting on its straightness.

19g3/h3/i3/j3) Do the same for 19g2/h2/i2/j2.

19cd4) Compare the results of (19c3) and (19d3). Comment. Make any additional plot needed.

18H. How far have we come?

This chapter has treated long-tailed distributions, mainly of counts, with attention to the individual largest values. It has centered on the relation of root-of-PRODUCT to log-of-RATIO, of

$$\sqrt{(\text{basic count or size})(\text{completed rank})}$$

to

$$\log (\text{basic count or size})/(\text{completed rank}),$$

which often proves to be simply describable and almost always plots usefully.

We are now ready:

◇to make product–ratio plots, using either all (basic, c'rank) pairs or only those corresponding (perhaps roughly) to half-octave sequences of "basic" and "c'rank".

◇to use $a = b =$ (a convenient integer) to bring several such plots through a common point (by shrinking or magnifying vertically and sliding horizontally).

◇to start our basic counts, either when necessary (zeros counted) or convenient (simplifying our plot's appearance).

◇to form simple residuals and thus put the "shape" of stretched-tail distributions under a microscope.

commentary

There may be readers who will react rather negatively to the plots of this chapter, on the ground that they "do not know what the axes mean", an argument with some--but only very little--face plausibility. It is, the writer

exhibit **19** of chapter 18 (continued)

19ef4) Same for (19c3) and (19f3).

19hi4) Same for (19h) and (19i).

19k) (teaser) Look up the source and plot some of the data there on number of generations in families.

S) SOURCE

C. B. Williams 1944 "Some applications of the logarithmic series and the index of diversity to ecological problems." *J. Ecology* 32: 1–44. (See this source for references to original sources.)

believes, fair to argue that very meaningful analyses have taken place in this chapter--meaningful in that:

- ◇ the bulk of the behavior of several stretched-tail distributions has been concentrated into a few numbers.
- ◇ the rest has been exhibited in the form of simple residuals.

In our analysis of fleas on rats, for instance, we could summarize in terms of

- ◇ a start of 4,
- ◇ a largest basic count of 84 out of 209 rats,

and present the rest of the behavior in terms of residuals (compare exhibit 16) from a line through the points corresponding to the (basic, c'rank) pairs (209, 1) and (1, 84).

This has taken us a long way forward. We can compare two such distributions quite effectively, and can detect many of their idiosyncrasies. **We can do this without requiring "a feeling" or "an intuitive understanding" of what the coordinates in our plot mean.**

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9★	627
10★	629
11	631
12	632
13★	633
14★	635
15	636
19D	
16★	638
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18	640
19★	643
20	644
19H	

Shapes of distribution 19

Suppose we have a batch of values, and that we want to talk about how they are "distributed." Chapter 17 took a rather flexible approach involving a fair amount of mechanism: bins, counts in bins, bin densities = counts/width, square roots of bin densities, smoothing, and then, in the special case of near symmetry, simple fitting to our usual re-expressions. Chapter 18 took a rather different approach to a special case--to the case where there are many instances of very small values, but a small number of very large ones. In the present chapter, we will take a third approach, based on comparing the observed distribution with a reference standard, so that the residuals can tell us how what is before us deviates from that standard. Again we deal with only a special case, but this time one which is much more common, one involving batches of data whose distribution is not too different, perhaps when suitably re-expressed, from the chosen reference standard.

To speak of a distribution is to speak of where the values are. We are now familiar (see chapter 15) with:

- re-expressing fractions.
- looking at several different cuts.

If we know the fraction of values "to the left"--or "to the right"--of each and every possible cut, we know where all the values are. So we can tackle the distribution problem by working with the parallel fractions "above" or "below" each cut. Working with these parallel fractions will help us look at shapes of distribution and show us how to exhibit actual behavior--actual shape of distribution--by comparing observed counted fractions with a standard reference shape, thus generating pictures of the deviation of the actual from the standard.

As we ought to expect, exact matching of the standard almost never happens.

We will also, in this chapter, go back to chapter 2, one of our very early chapters, and build upon the use of letter values as one way to conveniently summarize how the values of our batch are spread out. As well as halving outwards in terms of letter values, we will, alternatively, also work inwards by half-octaves.

review questions

What approaches have which previous chapters taken toward looking at shapes of distribution? What approach will we take in the present chapter?

How will what we are to do be related to chapter 15's counted fractions? What will we need as a basis of comparison? What will we compare with it? How often do we expect very close agreement with our reference standard? What tool will we take from which very early chapter? What related approach will we also follow?

19A. Looking at shapes of distribution

We want simple expressions related to distributions--expressions that come near enough for rough descriptions of the residuals. To ask for raw fractions or ss-fractions to behave simply in terms of cut position would be too much. What we need is a tailing off of the values at either end, a situation where extreme fractions change much more slowly than middle fractions. What we need is tail-stretching.

Perhaps the re-expressions of fractions we have been using will help us. Either froots or flogs--presumably based on ss-fractions--might behave simply enough for us to be able to use at least frequently:

- a simple fit of froots or flogs to cut position,
- AND a description of the residuals

as an adequate description of distributions of values.

Chevrolet prices

Exhibit **1** tries this out, for the batch of seventeen Chevrolet prices of exhibit 3 of chapter 1. The first results are plotted in exhibit **2**, where we see a fairly steady rise. Showing the results for all possible cuts, however, gives us both isolated points, wherever one or more values fall at that cut, and horizontal line segments (which together cover all other values). From the point of view of a first understanding of what we are doing, showing both points and horizontal lines seems good. Once we get the general picture and want to look harder, however, we gain (both in visibility of result and in ease of plotting) by showing only the points--either isolated or, as in exhibit **3**, lightly connected.

Taking the eye-fitted line of exhibit 3 back to panel A of exhibit 1, we find the first residuals, shown in exhibit **4**. The additional fitting that is now strongly indicated is left to a problem, as is the repetition of the whole process using froots instead of flogs. Those who do these problems will find a much closer fit by a straight line when froots are used. How are we to interpret this?

shapes

We have an intuitive idea of what we mean by the shape of a distribution of values--more precisely, perhaps an intuitive idea of what does not change a shape. To take a toy example with only five values, the sets 1, 4, 9, 16, 25, and 10, 40, 90, 160, 250, and 101, 104, 109, 116, 125 all have the same shape, one

exhibit **1** of chapter 19: Chevrolet prices

Calculations for displaying re-expressed fractions against cut values, including first residuals

A) INITIAL CALCULATIONS

Cut	Less than PLUS 1/2 equal	Same +1/6*	ss-fraction	flog		first fit†	first residual	
149	0	s	.0096	−2.32	T			T
150	h	hs	.038	−1.61	U	−1.48	−.13	U
151	1	1s	.067	−1.31	R			R
249	1	1s	.067	−1.31	N			N
250	1h	1hs	.096	−1.12	T	−1.32	+.20	T
251–687	2	2s	.125	−.97	O			O
688	2h	2hs	.154	−.85		−.60	−.25	
689–694	3	3s	.183	−.75	E			E
695	3h	3hs	.211	−.66	X	−.58	−.08	X
696–794	4	4s	.240	−.58	H			H
795	5	5s	.298	−.43	I	−.42	−.01	I
796–894	6	6s	.356	−.30	B			B
895	7h	7hs	.442	−.12	I	−.25	+.13	I
896–1098	9	9s	.529	.06	T			T
1099	9h	9hs	.558	.12	S	.08	+.04	4
1100–1165	10	10s	.587	.17	2			
1166	10h	10hs	.615	.23	A	.19	+.04	
1167–1332	11	11s	.644	.30	N			
1333	11h	11hs	.673	.36	D	.47	−.11	
1334–1498	12	12s	.702	.43				
1499	12h	12hs	.731	.50	3	.74	−.24	
1500–1692	13	13s	.760	.58				
1693	13h	13hs	.789	.66		1.06	−.40	
1694–1698	14	14s	.817	.75				
1699	14h	14hs	.846	.85		1.07	−.22	
1700–1774	15	15s	.875	.98				
1775	15h	15hs	.904	1.13		1.20	−.08	
1776–1894	16	16s	.933	1.31				
1895	16h	16hs	.982	1.61		1.40	+.21	
1896–	17	17s	.9904	2.32				

*Here "s" = "and 1/6" and the divisor is 17ss = 17 1/3.
† See exhibit 3.

P) PROBLEMS

1a) Use the additional fit found in exhibit 14 to find second residuals. Plot and comment.

1b) Repeat the calculations using froots instead of flogs. Plot, compare, comment.

1c) Find a batch of 15 to 25 values of interest to you and analyze it as above.

exhibit **2** of chapter 19: Chevrolet prices

Complete plot (all possible cuts) of flog against cut values (from exhibit 11)

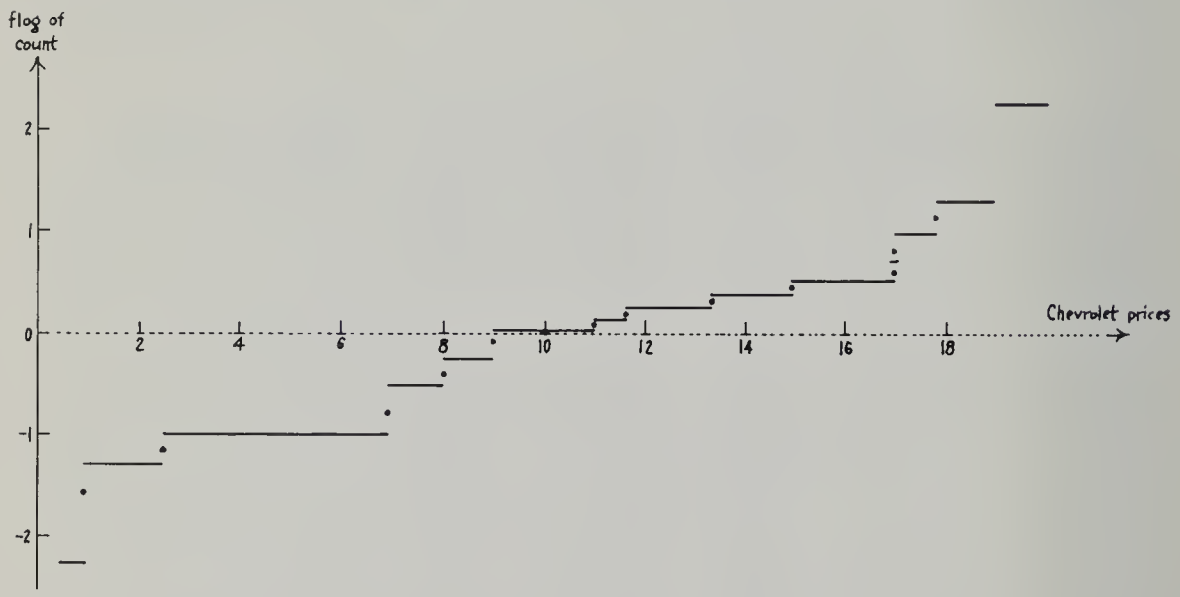
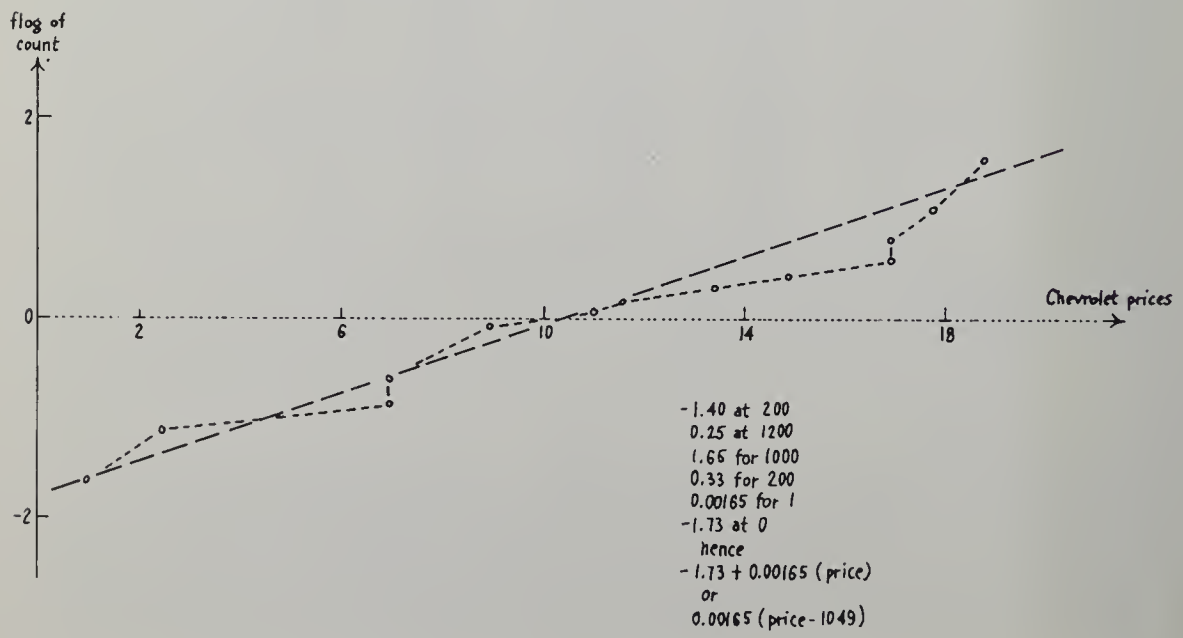


exhibit **3** of chapter 19: Chevrolet prices

Working plot (points only, lightly connected) and eye-fitted line of flog against cut value for 17 Chevrolet prices (from exhibit 11)



that is tighter on the left than on the right. We insist that the sorts of changes we have just made:

- ◇multiplying each value by the same constant (in our example, by 10), or
- ◇adding the same constant (in our example, 100) to each value,

do not change shape. If this is so, then no combination of such changes will affect shape either. (Why?) **Shape** is essentially that which is NOT affected by TRIVIAL re-expression.

If a particular re-expression of "fraction above" is a "linear function of cut position"--is (constant) PLUS ((another constant) times (cut position))--then the shape of our distribution is the same as when the constants are zero and one, as when

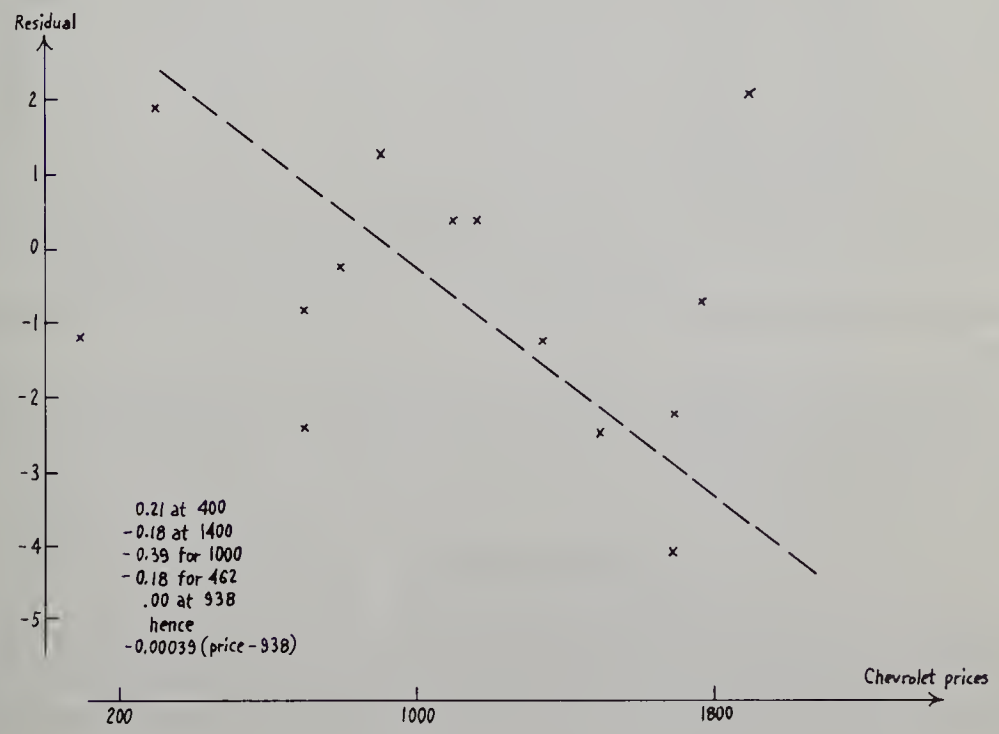
the same re-expression of fraction above = cut position,

which we can take as defining the standard distribution corresponding to that particular re-expression.

Recall now the result of problem (1b).

exhibit **4** of chapter 19: Chevrolet prices

Residuals from the eye-fitted line of exhibit 13 and a supplementary line



We can now say: The shape of the distribution of the seventeen Chevrolet prices is closer to the shape corresponding to froots than it is to the shape corresponding to flogs. In general, we can compare the shape of any batch with either or both of these standard shapes.

binned data

We have carried through the analysis of the 17 Chevrolet prices in the fullest possible detail. We already noticed that it was surely enough to work with only the values that actually occurred. In most cases we do not want to go into even that much detail, and we frequently could not, even if we wished to, since many sorts of data are binned before we can get our hands on them.

One sort of example where unbinned data is not available--and would be unmanageable if it were--is data on individual income tax returns. Exhibit **5** deals with the 59,085,182 individual income-tax returns for 1958 reported by the Internal Revenue Service, specifically at their binning by adjusted gross income. As exhibit **6** shows:

- ◇ for incomes up to $10,000, the flogs are roughly linear in the square root of the cut.
- ◇ for incomes above $15,000, the flog is roughly linear in the log of the cut.
- ◇ both "roughly linear" behaviors involve a detectable curvature (clear enough if residuals are found and plotted).

The words we have just used give a qualitative description of the distribution of adjusted gross income as reported by those filing income-tax returns. This can be made quantitative by specifying fitted lines--and could be made still more precise by fitting quadratics to the segments (in place of lines).

The purpose of this example is NOT intended to show a new and wonderful representation for the shape of the adjusted gross income distribution--this we heartily disclaim. (Accordingly, we do not expect this particular representation to work particularly well for other data.) Rather it is to show:

- ◇ **that binned data can be dealt with by the same approach as individual data.**
- ◇ **that even fairly complex shapes of distribution can be seized upon with reasonable approximation when we use the simplest of methods.**

These are its true purposes.

review questions

What does "how distributed" mean? Have we expressed an answer before? Where? How? Can we usefully describe patterns of distribution with (counted) fractions? Why/why not? If we can, are the fractions likely to need

exhibit **5** of chapter 19: adjusted gross income

Analysis of a binning of U.S. individual income tax returns for 1958 according to their adjusted gross income

A) DATA and CALCULATIONS

Upper edge of bin	Count	Information for cuts						
		Cum ↓ (*)	Cum ↑ (*)	Cum ↓	Cum ↑	flog	$\sqrt{\text{cut}}$	log cut
$600	3,950,030	3,950,030		6.73%		−1.32	.78	
$1,000	3,060,247	7,010,277		11.9%		−1.00	1.00	
$1,500	4,120,276	11,130,553		19.0%		−.72	1.22	
$2,000	3,570,536	14,701,089		25.0%		−.55	1.41	
$2,500	3,689,218	18,390,307		31.3%		−.39	1.58	
$3,000	3,723,909	22,114,216		37.7%		−.23	1.73	
$3,500	3,742,848	25,857,064		44.0%		−.12	1.87	
$4,000	3,729,578	29,586,642	29,114,282	50.4%	49.6%	+.01	2.00	3.60
$4,500	3,745,242		25,369,040		43.2%	.14	2.12	3.65
$5,000	3,639,977		21,729,063		37.0%	.27	2.24	3.70
$6,000	6,375,555		15,353,508		26.2%	.82	2.45	3.78
$7,000	4,676,947		10,676,561		18.2%	.75	2.65	3.84
$8,000	3,226,844		7,449,717		12.7%	.96	2.83	3.90
$9,000	2,171,701		5,278,016		8.99%	1.16	3.00	3.95
$10,000	1,452,594		3,825,422		6.52%	1.33	3.16	4.00
$15,000	2,488,095		1,337,327		2.28%	1.88	3.87	4.18
$20,000	588,262		749,065		1.28%	2.17		4.30
$25,000	264,732		484,333		.825%	2.39		4.40
$50,000	369,939		114,394		.195%	3.12		4.70
$100,000	91,715		22,679		.0386%	3.93		5.00
$150,000	14,080		8,599		.0146%	4.42†		
$200,000	3,863		4,736		.0081%	4.72		
$500,000	3,956		780		.00133%	5.62		
$1,000,000	536		244		.00042%	6.20		
	244							

(Total = 59,085,182 which includes 384,258 with no gross income that have been excluded. Thus working total = 58,700,924.)

*** Bothering with the added 1/6 would obviously be foolish.**

† The last four values in this column have to be found from flog = $1.1513 \log_{10}(f/(1-f))$.

P) PROBLEMS

5a) Find a set of 8 to 12 binned counts of interest to you and analyze it along the lines just illustrated.

5b) Do the same for a set of 13 to 17 binned counts.

5c) Do the same for a set of 18 or more binned counts.

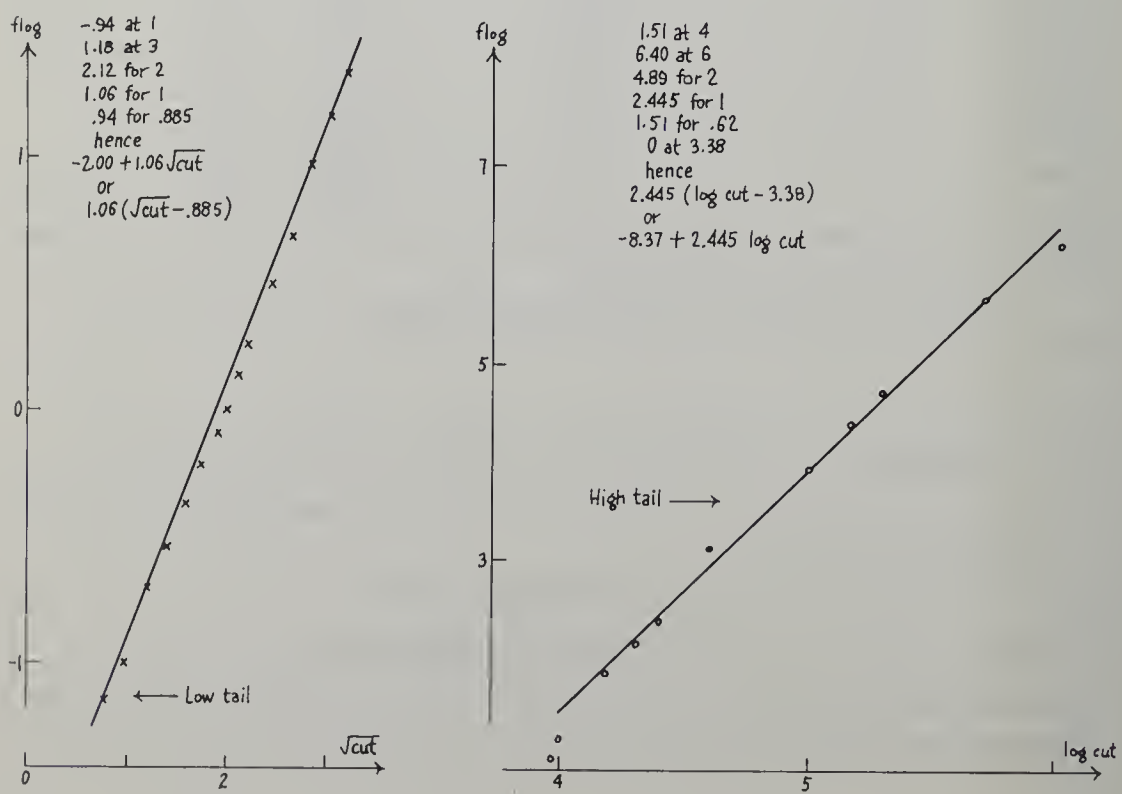
S) SOURCE

U.S. Internal Revenue Service, 1958.

re-expression? Why/why not? If we can't, what should we do instead? What did we take as an example? Did it need re-expression? Why/why not? And when we displayed the results? Using how many styles of display? Why/why not? Did there seem to be a reasonable fit? What did we do next? Why? What was the overall result? What do we want to mean by "shape of distribution"? What ought not change such a shape? What is a standard distribution corresponding to a re-expression? How can we compare the shape of distribution of a batch with such a standard? What do we now learn from the previous data? Do we need individual values or can we use counts in bins? What example did we try next? Could we have worked with individual values? How did things come out? Could we fit the left tail reasonably? How? Could we fit the right tail reasonably? How? Need the re-expression that works well in one tail also work well in the other? Why/why not? Did we think the representation we found for our data ought to be used for other similar data? Why/why not?

exhibit **6** of chapter 19: adjusted gross income

The two tails plotted--and eye fitted--separately (59,085,182 individual income-tax returns in 1958, from exhibit 15)



19B. The Gaussian reference

Although the shape of distribution associated with the froot re-expression of fractions matched the shape of our seventeen Chevrolet prices quite well, this is not a very common occurrence. Accordingly, this shape would not be a good choice as a single standard shape.

The shape of distribution associated with flog re-expression of fractions is much more often a close match to data. Indeed, so far as data goes, this shape of distribution is one that might quite reasonably be taken as the single standard. Both for specific reasons--mainly more closely associated with mathematical theory than with data--and, perhaps, because of accidents, this was not done.

The reference standard for shapes of distribution has long been the shape associated with the name of Gauss, who combined mathematical genius with great experience with the highest-quality data of his day--that of surveying and astronomy. Later writers have made the mistake of thinking that the Gaussian (sometimes misleadingly called normal) distribution was a physical law to which data must adhere--rather than a reference standard against which its discrepancies are to be made plain. It has taken a long time to recover from such unwarranted simplification, but much progress has been made.

Today we can use the Gaussian shape of distribution in a variety of ways to our profit. We can:

- ◇use it freely as a reference standard, as a standard against which to assess the actual behavior of real data--doing this by finding and looking at deviations.
- ◇use it, cautiously but frequently, as a crude approximation to the actual behavior, both of data itself and of quantities derived from data.

In using the Gaussian shape as such an approximation, we owe it to ourselves to keep in mind that real data will differ from the Gaussian shape in a variety of ways, so that treating the Gaussian case is only the beginning.

Mathematically, it would be much neater if both:

- ◇we knew EXACTLY how values are distributed.
- ◇this exact knowledge could be expressed SIMPLY.

In the real world we can never expect either of these to hold. **We are indeed lucky if:**

- ◇we know APPROXIMATELY how values are distributed.
- ◇this approximation is itself NOT TOO FAR from being simple.

To accept these facts is never an easy task, never something rapidly learned. **It is, however, an inevitable necessity.**

Exhibit **7** sets the Gaussian shape out as a re-expression of fractions. We have chosen, in order to tie ourselves to something more easily assessed, to scale this re-expression so that 25% and 75% correspond to ±1 (rather than

exhibit 7 of chapter 19: reference table

The Gaussian shape as a re-expression of fractions with 25% and 75% re-expressed as ±1 (take sign of answer from head of column giving %)

+	.0%	.2%	.4%	.6%	.8%	1.0%	−
50%	.000	.007	.015	.022	.030	.037	49%
51	.037	.045	.052	.059	.067	.074	48
52	.074	.082	.089	.097	.104	.112	47
53	.112	.119	.126	.134	.141	.149	46
54	.149	.156	.164	.171	.179	.186	45
55%	.186	.203	.201	.209	.216	.224	44%
56	.224	.231	.239	.246	.254	.262	43
57	.262	.269	.277	.284	.292	.299	42
58	.299	.307	.314	.322	.330	.337	41
59	.337	.345	.351	.360	.368	.376	40
60%	.376	.383	.391	.397	.406	.414	39%
61	.414	.422	.430	.437	.445	.453	38
62	.453	.461	.469	.476	.484	.492	37
63	.492	.500	.508	.514	.523	.532	36
64	.532	.539	.547	.555	.563	.571	35
65%	.571	.579	.587	.595	.603	.612	34%
66	.612	.620	.628	.636	.644	.652	33
67	.652	.660	.669	.677	.685	.693	32
68	.693	.702	.710	.718	.727	.735	31
69	.735	.744	.752	.760	.769	.777	30
70%	.777	.786	.795	.803	.812	.820	29%
71	.820	.829	.838	.846	.855	.864	28
72	.864	.872	.882	.890	.900	.909	27
73	.909	.918	.927	.936	.945	.954	26
74	.954	.962	.972	.981	.991	1.000	25

+	.0% .5%	.1% .6%	.2% .7%	.3% .8%	.4% .9%	.5% 1.0%	−
94	2.305	2.318	2.330	2.343	2.356	2.369	5.5
94.5	2.311	2.382	2.396	2.410	2.424	2.438	5
95%	2.438	2.493	2.468	2.483	2.498	2.514	4.5%
95.5	2.514	2.529	2.545	2.562	2.579	2.596	4
96	2.595	2.613	2.630	2.649	2.667	2.686	3.5
96.5	2.686	2.705	2.726	2.746	2.767	2.782	3
97	2.782	2.766	2.833	2.857	2.881	2.906	2.5
97.5%	2.906	2.932	2.958	2.986	3.015	3.045	2
98	3.045	3.076	3.109	3.143	3.179	3.217	1.5
98.5	3.217	3.258	3.301	3.346	3.396	3.449	1
99	3.449	3.507	3.571	3.643	3.724	3.819	.5
99.5	3.819	3.932	4.074	4.276	4.582	X	0
	1.0% .5%	.9% .4%	.8% .3%	.7% .2%	.6% .1%	.5% .0%	

+	.00%	.02%	.04%	.06%	.08%	.10%	−
98.0%	3.045	3.051	3.057	3.063	3.070	3.076	1.9%
98.1	3.076	3.082	3.089	3.096	3.102	3.109	.8
98.2	3.109	3.116	3.123	3.129	3.136	3.143	.7
98.3	3.143	3.150	3.157	3.165	3.172	3.179	.6
98.4	3.179	3.187	3.194	3.202	3.210	3.217	.5
98.5%	3.217	3.225	3.233	3.241	3.249	3.258	1.4%
98.6	3.258	3.266	3.274	3.283	3.292	3.301	.3
98.7	3.301	3.309	3.318	3.328	3.337	3.346	.2

75%	1.000	1.009	1.019	1.028	1.036	1.047	**24%**
76	1.047	1.056	1.066	1.076	1.085	1.095	**23**
77	1.095	1.105	1.115	1.125	1.135	1.145	**22**
78	1.145	1.154	1.165	1.175	1.185	1.195	**21**
79	1.195	1.209	1.216	1.226	1.237	1.248	**20**
80%	1.248	1.258	1.269	1.279	1.291	1.302	**19%**
81	1.302	1.313	1.324	1.335	1.346	1.351	**18**
82	1.357	1.368	1.380	1.391	1.402	1.414	**17**
83	1.414	1.426	1.438	1.450	1.462	1.474	**16**
84	1.474	1.487	1.499	1.511	1.524	1.537	**15**
85%	1.537	1.549	1.562	1.575	1.588	1.602	**14%**
86	1.602	1.615	1.629	1.642	1.656	1.670	**13**
87	1.670	1.684	1.698	1.713	1.727	1.742	**12**
88	1.742	1.757	1.772	1.787	1.804	1.818	**11**
89	1.818	1.834	1.850	1.867	1.883	1.900	**10**
90%	1.900	1.917	1.934	1.951	1.970	1.988	**9%**
91	1.988	2.006	2.025	2.044	2.063	2.083	**8**
92	2.083	2.103	2.124	2.144	2.166	2.184	**7**
93	2.188	2.210	2.233	2.257	2.280	2.305	**6**
94	2.305	2.330	2.356	2.382	2.410	2.438	**5**
	1.0%	**.8%**	**.6%**	**.4%**	**.2%**	**.0%**	

+							–
	.0%	**.1%**	**.2%**	**.3%**	**.4%**	**.5%**	
	.5%	**.6%**	**.7%**	**.8%**	**9%**	**1.0%**	
90%	1.000	1.909	1.917	1.926	1.943	1.943	**9.5%**
90.5	1.943	1.951	1.960	1.970	1.978	1.988	**9**
91	1.988	1.997	2.006	2.016	2.025	2.034	**8.5**
91.5	2.034	2.044	2.054	2.063	2.073	2.083	**8**
92	2.083	2.093	2.103	2.113	2.124	2.134	**7.5**
92.5%	2.134	2.144	2.155	2.166	2.171	2.188	**7%**
93	2.188	2.199	2.210	2.222	2.233	2.244	**6.5**
93.5	2.244	2.257	2.269	2.280	2.293	2.305	**6**
	1.0%	**.9%**	**.8%**	**.7%**	**.6%**	**.5%**	
	.5%	**.4%**	**.3%**	**.2%**	**.1%**	**.0%**	

98.8	3.346	3.356	3.366	3.376	3.386	3.396	**.1**
98.9	3.396	3.406	3.417	3.427	3.438	3.449	**1.0**
99.0%	3.449	3.460	3.472	3.483	3.495	3.507	**.9%**
99.1	3.507	3.520	3.532	3.545	3.558	3.571	**.8**
99.2	3.571	3.585	3.599	3.613	3.628	3.643	**.7**
99.3	3.643	3.659	3.674	3.690	3.707	3.724	**.6**
99.4	3.724	3.742	3.760	3.779	3.799	3.819	**.5**
99.5%	3.819	3.840	3.861	3.884	3.908	3.932	**.4%**
99.6	3.932	3.958	3.984	4.013	4.042	4.074	**.3**
99.7	4.074	4.107	4.143	4.181	4.222	4.276	**.2**
99.8	4.276	4.316	4.370	4.431	4.501	4.582	**.1**
99.9	4.582	4.679	4.802	4.971	5.249	X	**0.0%**
	.10%	**.08%**	**.06%**	**.04%**	**.02%**	**.00%**	

+							–
	.000%	**.001%**	**.002%**	**.003%**	**.004%**	**.005%**	
	.005%	**.006%**	**.007%**	**.008%**	**.009%**	**.010%**	
99.50%	3.819	3.829	3.840	3.851	3.861	3.872	**.45%**
99.55	3.872	3.884	3.895	3.908	3.919	3.932	**.4**
99.60	3.932	3.945	3.958	3.971	3.984	3.998	**.35**
99.65	3.998	4.013	4.027	4.042	4.058	4.074	**.3**
99.70	4.074	4.090	4.107	4.125	4.143	4.162	**.25**
99.75	4.162	4.181	4.201	4.222	4.244	4.267	**.2%**
99.80	4.276	4.291	4.316	4.341	4.370	4.400	**.15**
99.85	4.400	4.431	4.465	4.501	4.539	4.582	**.1**
99.90	4.582	4.628	4.629	4.736	4.802	4.878	**.05**
99.95	4.878	4.971	5.065	5.249	5.513	X	**0.0%**
	.010%	**.009%**	**.008%**	**.007%**	**.006%**	**.005%**	
	.005%	**.004%**	**.003%**	**.002%**	**.001%**	**.000%**	

matching $f - (1 - f)$ near 50%, for example). Exhibit **8** sets out some problems to be treated using this re-expression. (Chapter 20, below, will relate this shape to certain of the fits of chapter 14.)

review questions

What is a Gaussian distribution? What is it useful for? What is it dangerous for? Have there been any superstitions about it? What would be mathematically neat? Is the real world neat? What task does all this set us? How do we make a Gaussian distribution available for use?

19C. Using letter values to look at shapes of distribution

We have become accustomed to summarizing batches in terms of letter values--of a median, two hinges, two eighths, and so on. For most purposes "a hinge is a hinge" and "an eighth is an eighth". For our present purposes we find it convenient to look somewhat closer.

exhibit **8** of chapter 19: data and problems

Some problems using the Gaussian re-expression of fractions and the Gaussian standard with which distribution shapes are often compared

A) AGES at MARRIAGE of 235,252 spinsters

Age range	Isserlis data
15–19	17546
20–24	118542
25–29	70411
30–34	20241
35–39	5873
40–44	1706
45–49	636
50–54	171
55–59	64
60–64	28
65–69	23
70 over	11

P) PROBLEMS

8a1) Convert the Isserlis data first to cumulative fractions, and reexpress these fractions on the Gaussian scale.

8a2) Plot these reexpressed fractions in as many ways as are necessary to (i) make a good fit and (ii) reveal the characteristics of the residuals.

S) SOURCE

L. Isserlis 1917. "On the representation of statistical data." *Biometrika* 11: 418–425. Table VI at page 423.

Were we to work with batches of 7, 8, 9, 10, 11, or 12 observations, the ss-fractions corresponding to median (M), hinges (H), eighths (E) and so on (D, C, B, . . .) are as indicated in exhibit **9**. The ss-fraction for M is always 0.500--right on the nose. The ss-fractions for H's, which we might like to have = .250, run

.295, .260, .286, .258, .279, .257.

exhibit **9** of chapter 19: example

The ss-fractions corresponding to MHE . . . , for batches of 7 to 12 (using both *h* = 1/2 and s = 1/6, so that ss = 1/3)

A) BATCHES of 7 or 8

depth	ss-count	ss-fraction
# 7		
M 4	3hs	.500
H 2h	2s	.295 (vs .250)
E 1h	1s	.159 (vs .125)
D 1	hs	.091 (vs .062)
	(7ss)	

depth	ss-count	ss-fraction
# 8		
M 4h	4s	.500
H 2h	2s	.260 (vs .250)
E 1h	1s	.140 (vs .125)
D 1	hs	.080 (vs .062)
	(8ss)	

B) BATCHES of 9 or 10

depth	ss-count	ss-fraction
# 9		
M 5	4hs	.500
H 3	2hs	.286 (vs .250)
E 2	1hs	.179 (vs .125)
D 1h	1s	.125 (vs .062)
C 1	hs	.072 (vs .031)
	(9ss)	

depth	ss-count	ss-fraction
# 10		
M 5h	5s	.500
H 3	2hs	.258 (vs .250)
E 2	1hs	.161 (vs .125)
D 1h	1s	.133 (vs .062)
C 1	hs	.065 (vs .031)
	(10ss)	

C) BATCHES of 11 or 12

depth	ss-count	ss-fraction
# 11		
M 6	5hs	.500
H 3h	3s	.279 (vs .250)
E 2	1hs	.147 (vs .125)
D 1h	1s	.103 (vs .062)
C 1	hs	.059 (vs .031)
	(11ss)	

depth	ss-count	ss-fraction
# 12		
M 6h	6s	.500
H 3h	3s	.257 (vs .250)
E 2	1hs	.135 (vs .125)
D 1h	1s	.095 (vs .062)
C 1	hs	.054 (vs .031)
	(12ss)	

→

Those for E's (which we might like to be .125) run

.159, .140, .179, .161, .147, .135.

The pattern is clear. As the batch size creeps up, these ss-fractions run down toward their nominal values, and then jump away again (every 2 in count for H, every 4 for E, every 8 for D, etc.). On the whole, though, they get steadily closer to the target values.

When we want to be extra (perhaps sometimes unduly) careful, then, we have a choice:

- ◇ work with H, E, ... as they are, making allowance for what ss-fraction each corresponds to.
- ◇ go back and replace our H-, E-, etc., values by values at other depths (for E for a batch of 7, for example, we want the ss-count to be (.125)(7.33) = 0.91, so that we need to work at a depth about .34/.68 nearer the upper extreme, namely 1.24 (instead of at 1h = 1.50).

Taking the latter choice is a lot of repeated work, since we have to go 24% of the way from depth 1 to depth 2 every time we deal with a batch of 7. Going the former route is a lot of work, but it is work that can be over and done with, since we can, for each batch size, first ask what ss-fraction corresponds to the H, E, D, or whatever (for that batch size) and then re-express the resulting ss-fraction once and for all.

Exhibit 12 will, when we reach it, show the result for batches of sizes 1 to 100, while exhibit 13 will give approximate formulas to be used for batches of more than 100. Before we turn to these exhibits, however, we shall look at how the results are to be used, taking the $\sqrt{\text{height}}$ of 219 volcanoes--given in stem-and-leaf form in exhibit 11 of chapter 3--as our example. Exhibit **10** has the calculations, first for H's, E's, D's, and C's separately (panels A to D), then for all as part of a letter-value-display (panel E), and finally all this with mid's included (panel F). The final step, shown in panel G, is to list smoothed values

exhibit **9** of chapter 19 (continued)

P) PROBLEMS

9a) Provide similar information for batches of 13 or 14.

9b) Provide similar information for batches of 15 or 16.

9c) Provide similar information for batches of 17 or 18.

9d) Provide similar information for batches of 19 or 20.

9abcd2) Compare results for batches of 13 to 20 with what you expected from 7 to 12. Discuss.

of mids and "25% pseudospreads". Each 25% pseudospread would be 2.00 for a distribution (of an infinite number of values) with exactly Gaussian shape, and 25% points at ±1.00. Equally, each would be the 25% spread of the distribution if the batch followed a Gaussian distribution exactly (to do which it would have to include infinitely many values). We know that it almost never

exhibit **10** of chapter 19: volcano heights

Using M, H, E, ... to look at shape of distribution for 219 volcano $\sqrt{\text{height}}$'s

A) The HINGES––at depth 55h

hinges:	61 and	98	
H-spread:	37		
divisor:	0.995		(1.00 would do)
ratio:	37.2		

B) The EIGHTHS––at depth 28

eighths:	49 and	110	
E-spread:	61		
divisor:	1.705		(1.71 would do)
ratio:	35.8		

C) The D's––at depth 14h

D's:	40 and	118	
D-spread:	78		
divisor:	2.254		(2.27 would nearly do)
ratio:	34.6		

D) The C's––at depth 7h

C's:	30 and	132	
C-spread:	102		
divisor:	2.738		(2.76 would probably do)
ratio:	37.3		

E) the EXTENDED LETTER-VALUE DISPLAY

# 219			spr		pseudo-spread
M 110	80				
H 55h	61	98	37	(0.995)	37.2
E 28	49	110	61	(1.705)	35.8
D 14h	40	118	78	(2.254)	34.7
C 7h	30	132	102	(2.738)	37.4
B 4	25	140	115	(3.165)	36.3
A 2h	23h	140	116h	(3.482)	33.4
Z 1h	18	140	122	(3.806)	32.1
Y 1	14	140	126	(3.95)	31.9

→

will follow a Gaussian distribution exactly. (In particular, no human ever worked with a batch with infinitely many values.) So we call such values

pseudospreads

to remind us that they will tell us about a particular spread--here the 25% spreads--when a very rare thing happens. They are useful to us for all that,

exhibit **10** of chapter 19 (continued).

F) The Same with MIDS INCLUDED

#219		Mid		Spr		25% Pseudo-Spread
M	110		80			
H	55h	61	(79h)	98	37	(.995) 37.2
E	28	49	(79)	110	61	(1.705) 35.8
D	14h	40	(79)	118	78	(2.244) 37.6
C	7h	30	(81)	132	102	(2.738) 37.3
B	4	25	(82h)	140	115	(3.165) 36.3
A	2h	23h	(82)	140	116h	(3.482) 33.4
Z	1h	18	(79)	140	122	(3.806) 32.1
Y	1	14	(77)	140	126	(3.95) 31.9

G) The MIDS and 25%-PSEUDOSPREADS after SMOOTHING--by 3RSS

letter	mid	25%	div^2
M	79h	(35.8)	0
H	79h	35.8	.99
E	79h	35.8	2.91
D	79h	35.8	5.08
C	81	35.8	7.5
B	82	35.8	10.0
A	82	33.4	12.1
Z	79	32.1	14.5
Y	77	31.9	15.6

P) PROBLEMS

10a) Do the same for the heights (not $\sqrt{\text{heights}}$) of the 219 volcanoes. Discuss the result.

10b) Do the same for the log heights (not $\sqrt{\text{heights}}$) of the 219 volcanoes. Discuss the result.

10c) Do the same for the 82 areas of the counties of Mississippi (see exhibit 4 of chapter 1). Discuss the result.

10d) Find a batch of 80 to 150 values that interest you and do the same.

10e) Find a batch of 150 to 300 values that interest you and do the same.

S) SOURCE

Exhibit 11 of chapter 3.

because by their CHANGES they tell us how the shape of distribution of the batch compares with the Gaussian shape.

Exhibit **11** plots the (smoothed) mids and 25% pseudospreads, using the square of the divisors as the horizontal coordinate. We see the mids waving first one way and then the other, leaving us with no particular impression of skewness. The 25% pseudospreads start at about 37 and eventually decrease to 32, suggesting a possible (slight) tendency toward squeezed-in tails.

Exhibit **12**, as advertised, contains the divisors for batches of up to 100; exhibit **13** contains approximate formulas for use with larger batches.

Exhibit **14** deals with a smaller example--one calling for exhibit 12 rather than for exhibit 13. The resulting mids and 25% pseudospreads are plotted in exhibit **15**. Both show a trend. The trend in mids corresponds to skewness with high values stretched farther out than low values, the fitted line being least satisfactory at the median. The 25% pseudospreads have a trend corresponding to stretched tails; the fitted line fits fairly well (as might other shapes of curve).

exhibit **11** of chapter 19: volcano heights

Plots of 25% pseudospread and mid against (Gaussian) ratio for $\sqrt{\text{heights}}$ of 219 volcanoes (from exhibit 10)

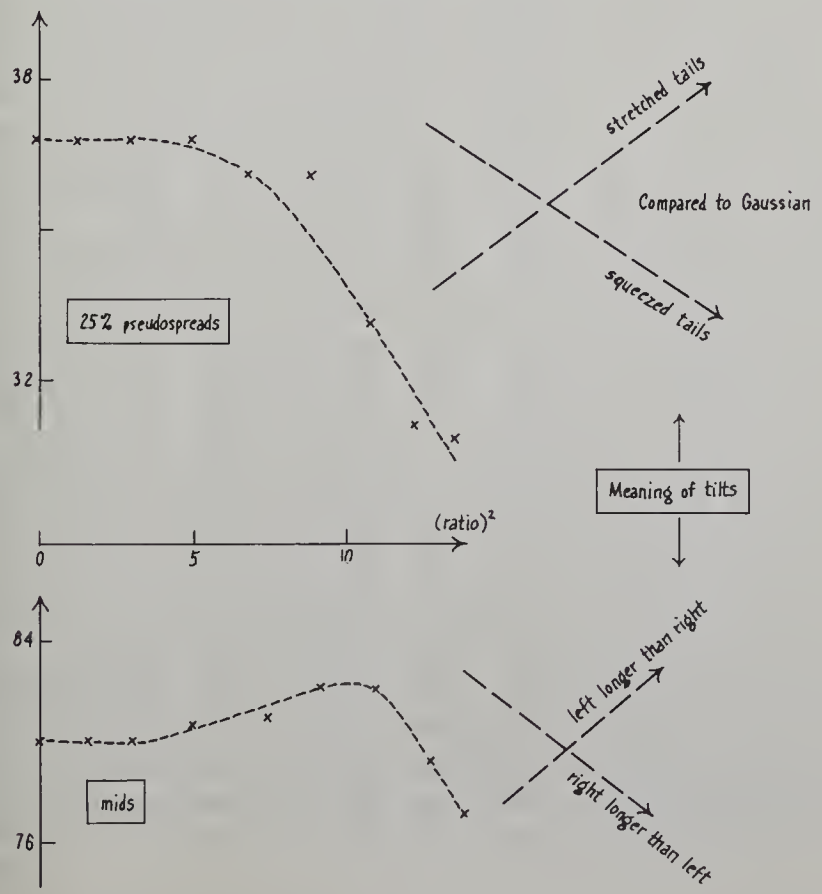


exhibit **12** of chapter 19: reference table

Factors to get pseudospreads from HED. . . spreads for batches of ≤100

Batch size	H	E	D	C	B	A	Batch size	H	E	D	C	B	A	Z
3	.57	1.25	X	X	X	X	**45**	.97	1.63	2.19	2.65	2.89	3.23	
4	.91	1.51	X	X	X	X	**46**	.99	1.65	2.21	2.67	2.90	3.24	
5	.72	1.15	1.70	X	X	X	**47**	.97	1.67	2.22	2.68	2.91	3.25	
6	.94	1.33	1.86	X	X	X	**48**	.99	1.69	2.24	2.69	2.93	3.26	
7	.80	1.48	1.98	X	X	X	**49**	.97	1.63	2.14	2.53	2.94	3.28	
8	.95	1.60	2.08	X	X	X	**50**	.99	1.65	2.16	2.54	2.95	3.29	
9	.84	1.36	1.70	2.17	X	X	**51**	.97	1.67	2.17	2.56	2.96	3.30	X
10	.96	1.47	1.79	2.25	X	X	**52**	.99	1.69	2.19	2.51	2.98	3.31	X
11	.87	1.55	1.87	2.32	X	X	**53**	.97	1.64	2.20	2.58	2.99	3.32	X
12	.97	1.63	1.95	2.38	X	X	**54**	.99	1.66	2.21	2.60	3.00	3.33	X
13	.89	1.46	2.01	2.44	X	X	**55**	.97	1.67	2.23	2.61	3.01	3.34	X
14	.97	1.53	2.07	2.49	X	X	**56**	.99	1.69	2.29	2.62	3.02	3.35	X
15	.90	1.59	2.12	2.54	X	X	**57**	.97	1.64	2.16	2.63	3.03	3.36	X
16	.98	1.65	2.17	2.58	X	X	**58**	.99	1.66	2.17	2.67	3.04	3.32	X
17	.91	1.51	1.93	2.22	2.62	X	**59**	.97	1.67	2.18	2.60	3.05	3.38	X
18	.98	1.56	1.98	2.26	2.66	X	**60**	.99	1.69	2.20	2.67	3.06	3.39	X
19	.92	1.61	2.02	2.30	2.69	X	**61**	.97	1.65	2.21	2.68	3.07	3.40	X
20	.98	1.66	2.06	2.34	2.73	X	**62**	.99	1.66	2.22	2.69	3.08	3.41	X
21	.93	1.55	2.10	2.37	2.76	X	**63**	.98	1.68	2.23	2.70	3.09	3.42	X
22	.98	1.59	2.14	2.41	2.79	X	**64**	.99	1.69	2.25	2.71	3.10	3.43	X
23	.93	1.63	2.17	2.44	2.82	X	**65**	.98	1.65	2.17	2.58	2.89	3.11	3.44
24	.99	1.67	2.20	2.47	2.85	X	**66**	.99	1.66	2.19	2.59	2.90	3.12	3.45
25	.94	1.57	2.03	2.50	2.87	X	**67**	.98	1.68	2.20	2.60	2.91	3.13	3.45
26	.99	1.61	2.06	2.52	2.90	X	**68**	.99	1.69	2.21	2.61	2.92	3.14	3.46
27	.94	1.64	2.09	2.55	2.92	X	**69**	.98	1.65	2.22	2.62	2.93	3.15	3.47
28	.99	1.67	2.12	2.57	2.94	X	**70**	.99	1.67	2.23	2.63	2.94	3.16	3.48
29	.95	1.59	2.15	2.60	2.96	X	**71**	.98	1.68	2.24	2.64	2.95	3.16	3.48
30	.99	1.62	2.17	2.62	2.99	X	**72**	.99	1.69	2.25	2.65	2.96	3.18	3.49
31	.95	1.65	2.20	2.64	3.01	X	**73**	.98	1.66	2.19	2.66	2.97	3.18	3.50
32	.99	1.68	2.22	2.66	3.02	X	**74**	.99	1.67	2.19	2.67	2.97	3.19	3.51
33	.95	1.60	2.08	2.44	2.68	3.04	**75**	.98	1.68	2.20	2.65	2.98	3.20	3.52
34	.99	1.63	2.11	2.46	2.79	3.05	**76**	.99	1.69	2.21	2.69	2.99	3.21	3.53
35	.96	1.65	2.13	2.48	2.72	3.08	**77**	.98	1.66	2.22	2.70	3.00	3.21	3.53
36	.99	1.68	2.15	2.50	2.74	3.10	**78**	1.00	1.67	2.23	2.70	3.01	3.22	3.54
37	.96	1.61	2.17	2.52	2.75	3.11	**79**	.98	1.68	2.24	2.71	3.01	3.23	3.54
38	.99	1.64	2.19	2.54	2.78	3.13	**80**	1.00	1.69	2.25	2.72	3.02	3.24	3.55
39	.96	1.66	2.21	2.55	2.79	3.14	**81**	.96	1.66	2.19	2.61	3.03	3.24	3.56
40	.99	1.68	2.23	2.57	2.81	3.16	**82**	1.00	1.67	2.20	2.62	3.04	3.25	3.56
41	.96	1.62	2.12	2.59	2.83	3.17	**83**	.98	1.68	2.21	2.63	3.05	3.26	3.57
42	.99	1.64	2.14	2.61	2.84	3.17	**84**	1.00	1.69	2.22	2.64	3.06	3.27	3.58
43	.95	1.66	2.15	2.62	2.56	3.20	**85**	.98	1.66	2.23	2.65	3.06	3.27	3.59
44	.99	1.68	2.17	2.64	2.87	3.21	**86**	1.00	1.67	2.24	2.65	3.07	3.28	3.60

→

exhibit **12** of chapter 19 (continued)

Batch size	H	E	D	C	B	A	Z
87	.98	1.68	2.25	2.66	3.07	3.28	3.60
88	1.00	1.69	2.25	2.67	3.08	3.29	3.60
89	.98	1.66	2.20	2.67	3.09	3.30	3.61
90	1.00	1.67	2.21	2.68	3.09	3.30	3.61
91	.98	1.68	2.22	2.69	3.10	3.31	3.62
92	1.00	1.69	2.22	2.70	3.11	3.32	3.63
93	.98	1.67	2.23	2.71	3.11	3.32	3.63
94	1.00	1.68	2.24	2.71	3.12	3.33	3.64
95	.98	1.69	2.25	2.72	3.13	3.33	3.64
96	1.00	1.70	2.25	2.73	3.13	3.34	3.65
97	.98	1.67	2.21	2.64	2.98	3.35	3.66
98	1.00	1.68	2.21	2.64	2.98	3.35	3.66
99	.98	1.69	2.72	2.65	2.99	3.36	3.67
100	1.00	1.70	2.23	2.66	3.00	3.37	3.67

exhibit **13** of chapter 19: reference table

Factors to get 25% pseudospreads from HED. . . spreads for batches of size ≥100

A) The APPROXIMATE FACTORS--divide into indicated spread to get pseudo-H-spread

for H-spreads: $1.00 - \frac{2}{n}\left(1 - .86\left(\frac{n}{2} \text{ Mod } 1\right)\right)$

for E-spreads: $1.71 - \frac{3}{n}\left(1 - .80\left(\frac{n}{4} \text{ Mod } 1\right)\right)$

for D-spreads: $2.27 - \frac{5}{n}\left(1 - .77\left(\frac{n}{8} \text{ Mod } 1\right)\right)$

for C-spreads: $2.76 - \frac{10}{n}\left(1 - .76\left(\frac{n}{16} \text{ Mod } 1\right)\right)$

for B-spreads: $3.19 - \frac{15}{n}\left(1 - .76\left(\frac{n}{32} \text{ Mod } 1\right)\right)$

for A-spreads: $3.58 - \frac{30}{n}\left(1 - .75\left(\frac{n}{64} \text{ Mod } 1\right)\right)$

for Z-spreads: $3.94 - \frac{55}{n}\left(1 - .75\left(\frac{n}{128} \text{ Mod } 1\right)\right)$

for Y-spreads: $4.28 - \frac{110}{n}\left(1 - .75\left(\frac{n}{256} \text{ Mod } 1\right)\right)$

for X-spreads: $4.59 - \frac{200}{n}\left(1 - .75\left(\frac{n}{512} \text{ Mod } 1\right)\right)$

for W-spreads: $4.89 - \frac{380}{n}\left(1 - .75\left(\frac{n}{1024} \text{ Mod } 1\right)\right)$

→

exhibit **13** of chapter 19 (continued)

B) EXAMPLE of USE––for $n = 137$

	Halvings	Fractions*	1 − (const)(fraction)	Resulting factor
H	$n/2 = 68.5$	.5	$(1 - \quad) = .57$	$1.00 - \frac{2}{137}(.57) = 0.99$
E	$n/4 = 34.25$	.25	$(1 - \quad) = .80$	$1.71 - \frac{3}{137}(.80) = 1.69$
D	$n/8 = 17.125$	.125	$(1 - \quad) = .90$	$2.27 - \frac{5}{137}(.90) = 2.24$
C	$n/16 = 8.5625$	.56	$(1 - \quad) = .57$	$2.76 - \frac{10}{137}(.57) = 2.72$
B	$n/32 = 4.28125$	.28	$(1 - \quad) = .79$	$3.19 - \frac{15}{137}(.79) = 3.10$
A	$n/64 = 2.1406$	.14	$(1 - \quad) = .90$	$3.58 - \frac{30}{137}(.90) = 3.38$
Z	$n/128 = 1.0703$	.07	$(1 - \quad) = .95$	$3.94 - \frac{55}{137}(.95) = 3.56$
Y	$n/256 = .5352$	.54	$(1 - \quad) = .60$	$4.28 - \frac{110}{137}(.60) = 3.80$

*** These are values of "*n*/divisor Mod 1" for "*n*/divisor" shown in first column.**

C) IMPORTANT NOTE

The capital M in "x Mod 1" MEANS "x MINUS the next LOWER integer"; thus:

Explicit form	Fraction
4.001 Mod 1 = .001	.001
4.000 Mod 1 = 1.000	1.000
3.999 Mod 1 = .999	.999

and so on.

D) ARITHMETIC SAVER––making successive division easy

$$
\begin{aligned}
137 &= 68 \cdot 2 + 1 \\
&= 34 \cdot 4 + 1 \\
&= 17 \cdot 8 + 1 \\
&= 8 \cdot 16 + 9 \\
&= 4 \cdot 32 + 9 \\
&= 2 \cdot 64 + 9 \\
&= 1 \cdot 128 + 9 \\
&= 0 \cdot 128 + 137
\end{aligned}
\qquad
\begin{aligned}
219 &= 109 \cdot 2 + 1 \\
&= 54 \cdot 4 + 3 \\
&= 27 \cdot 8 + 3 \\
&= 13 \cdot 16 + 11 \\
&= 6 \cdot 32 + 27 \\
&= 3 \cdot 64 + 27 \\
&= 1 \cdot 128 + 91 \\
&= 0 \cdot 256 + 219
\end{aligned}
$$

→

exhibit **13** of chapter 19 (continued)

Examples of mental calculation:

$$27 \cdot 8 = 26 \cdot 8 + 1 \cdot 8$$
$$= 13 \cdot 16 + 8$$
$$3 + 8 = 11$$

$$13 \cdot 16 = 12 \cdot 16 + 1 \cdot 16$$
$$= 6 \cdot 32 + 16$$
$$16 + 11 = 27$$

P) PROBLEMS

13a) Find the factors for H, E, D, C, B, and A values, for $n = 137$. (Show your work.)

13b) Same for $n = 257$.

13c) Same for $n = 173$.

13d) Same for $n = 1079$.

exhibit **14** of chapter 19: county areas

Mids and 25% pseudospreads for the logs of the areas of the 83 counties of Michigan

A) The FULL LETTER-VALUE DISPLAY

	mid			spr		25% pseudo-spread	div²	
M42		1.76					0.0	S
H21h	1.73h	(1.81h)	1.89h	.16	(.98)	.16	1.0	E
E11	1.69	(1.83h)	1.98	.29	(1.68)	.17	2.8	E
D6	1.66	(1.86h)	2.07	.41	(2.21)	.19	4.9	
C3h	1.59	(1.84)	2.10	.51	(2.63)	.19	6.9	
B2	1.54	(1.87)	2.20	.66	(3.05)	.22	9.3	
A1h	1.52h	(1.88)	2.23	.70h	(3.26)	.22	10.6	
Z1	1.51	(1.88h)	2.26	.75	(3.57)	.21	12.7	

↓ ↓ EXHIBIT 15

P) PROBLEMS

14a) Imitate panel A for the areas (not log areas) of the 83 counties.

14a2) Plot the results of (14a1). →

exhibit **14** of chapter 19 (continued)

14b) Same for the root areas.

14b2) Plot the results of (14b1).

14ab3) Compare the three analyses of the 83-county batch (panel A, (14a2), (14b2)). Discuss the comparison. What re-expression would be most symmetrical?

14c) Find a batch of 80 to 125 values that interest you and imitate panel A.

14c2) Plot the results of (14c).

14d) Find a batch of 150 to 250 values that interest you and imitate panel A.

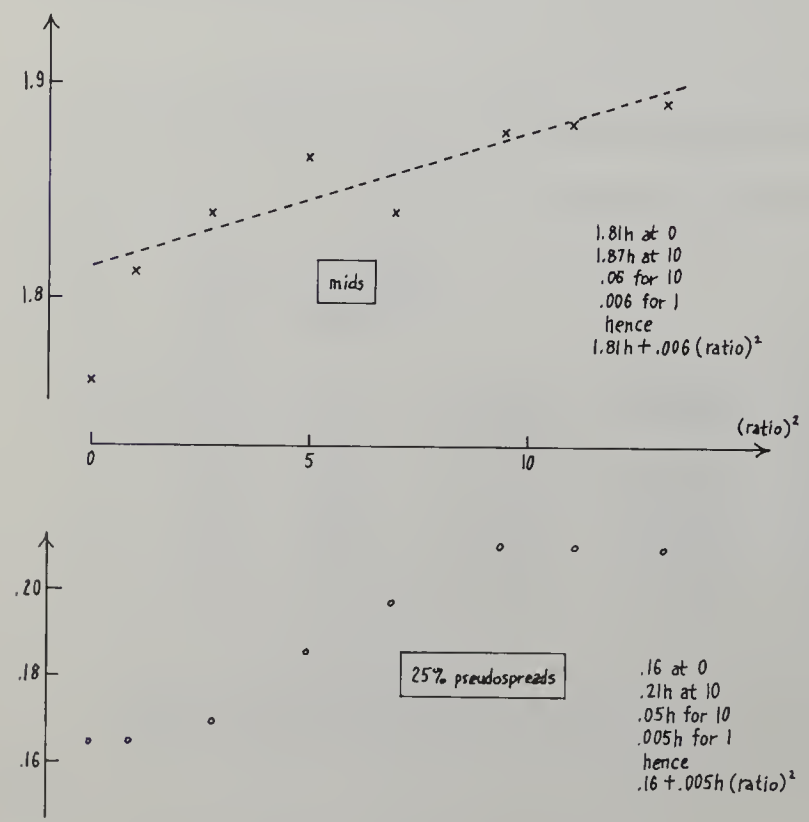
14d2) Plot the results of (14d).

S) SOURCE

Exhibit 5 of chapter 3

exhibit **15** of chapter 19: Michigan counties

Plots of mids and 25% pseudospreads for the logs of the land areas of the 83 counties of Michigan (from exhibit 14)



The relations

$$\text{mid} = 1.81h + .006\,(\text{div})^2$$

and

$$25\%\ \text{pseudospread} = 0.16 + 0.005h\,(\text{div})^2$$

thus describe the shape, spread, and location of these 83 points with moderate precision.

review questions

Can we assign ss-fractions to letter values? What happens if we try? In how many ways could we be extra careful? What are they? Which do we choose? Why? How hard is it to package the results? What example did we try the result on first? What did the results look like? What is a mid? What is a pseudospread? What do we plot mids against? Pseudospreads? How do these plots behave in our examples? What do we put in a table for use in finding 25% pseudospreads? What do we do for larger batches? What example did we do next? What happened? What formulas summarized our plot? Did we do a further example? What happened?

19D. Pushback technique (optional section)

Now that we have seen how hard we can look at shapes using mids (found directly, without a standard) and 25% spreads (found using a Gaussian standard), we would like to look at the individual observations--or individual letter values--as effectively as possible. A variety of simple pushback techniques let us do this easily in a way where we find ourselves doing other useful things.

We shall start from the letter-values, and the (Gaussian) 25% pseudospreads found from them. We let Q be

$$\text{one-half the median of the 25\% pseudospreads.}$$

For the example of exhibit 14--the log areas of the 83 counties of Michigan--for instance, the median is 0.19, so that $Q = 0.10$. We can now proceed as in exhibit **16**, multiplying Q by the appropriate value of "ratio", and subtracting the result from each of the letter values. Exhibit **17** shows the smoothed differences plotted against ratios. Two sections are reasonably well fitted by (segments of) straight lines, which, when we add back the amounts

$$Q\,(\text{ratio}) = .1\,(\text{ratio})$$

that we had subtracted before plotting exhibit 17, become

$$\text{(for ratios below } -1.5) \qquad 1.86 + .1\,(\text{ratio})$$

$$\text{(for ratios above } +1.0) \qquad 1.741 + 0.048\,(\text{ratio}).$$

(We could fit the middle--rather more roughly--with a third straight segment if we wished.)

If we want to describe the shape of distribution of these 83 logs of areas of counties of Michigan, we can now say, AMONG OTHER THINGS:

- ◇ each tail behaves rather like the Gaussian standard in shape.
- ◇ the righthand (positive) tail is longer than the lefthand (negative) one.

By dealing with the two tails separately, we are able, for this example, to say simpler things than we said in the last section, where we worked with location

exhibit **16** of chapter 19: Michigan counties

Pushback analysis (letter-value version) of the logs of the areas of the 83 counties of Michigan ($Q = .10$)

A) CALCULATIONS

Depth	Letter	Obs'd	Ratio	$Q\times$ Ratio	Diff*	3R		S		↓ NOW LOOK AT EXHIBIT 17	Fits	Resid
1	Z	1.51	−3.57	−.36	1.87		1.86				1.503	+.007
1h	A	1.52h	−3.26	−.33	1.85h	1.85h					1.534	−.009
2	B	1.54	−3.05	−.31	1.85	1.85		1.85			1.555	−.015
3h	C	1.59	−2.63	−.26	1.85	1.85		1.86	1.85h		1.597	−.007
6	D	1.66	−2.21	−.22	1.88	1.86		1.85	1.85h		1.639	+.021
11	E	1.69	−1.68	−.17	1.86	1.86		1.86	1.85		1.692	−.002
21h	H	1.73h	−.98	−.10	1.83h	1.83h					(1.762)	(−.027)
42	M	1.76	.00	.00	1.76	1.79h		1.79h			(1.741)	(+.019)
21h	H	1.89h	.98	.10	1.79h	1.79h		1.79h			1.886	+.009
11	E	1.98	1.68	.17	1.81	1.81					1.990	−.010
6	D	2.07	2.21	.22	1.85	1.84					2.068	+.002
3h	C	2.10	2.63	.26	1.84	1.85					2.130	−.030
2	B	2.20	3.05	.31	1.89	1.89					2.192	+.008
1h	A	2.23	3.26	.33	1.90	1.90					2.223	+.007
1	Z	2.26	3.57	.36	1.90	1.90					2.269	−.009

Notes

1. "Ratio" contains signed values of "factors" from exhibit 12.
2. Area enclosed in [ ] could be omitted and the next column filled in by symmetry. (It is convenient to fill it in for use in plotting.)

* **Diff = "obsd" MINUS "Q × Ratio"**

P) PROBLEMS

16a) Make a similar calculation and plot for the root volcano height data of exhibit 10. How many straight lines seem needed? Discuss and explain.

S) SOURCE

Exhibit 14 above.

and spread--with mids and 25% pseudospreads. In other examples, the opposite can and will happen--the location and spread pictures will be simpler. We need to have both approaches.

a half-octave version

As a way of describing the general shape of a batch, using its letter values (as we have just done) is quite reasonable and effective. If we want to focus our attention on individual values, we can usually do somewhat better.

In exhibit 16, the depths involved were 1, 1h, 2, 3h, 6, 11, ... Clearly, 1, 2, 6, and 11 tell us about individual values, while 1h wastefully tells us about a mixture of depths 1 and 2. Moreover, and perhaps confusingly, 3h tells us about a mixture of depths 3 and 4. With the same amount of arithmetic we could deal with 1, 2, 3, 4, 6, 11, ..., instead, finding only results each of which tells us about an individual point. If our concern is with individual values, this is a better thing to do.

Exhibit **18** offers values of "ratio" for such a "half-octave" choice of depths and batch sizes up to 100. Exhibit **19** applies these to the logs of areas of counties of Michigan. Exhibit **20** illuminates this application.

exhibit **17** of chapter 19: Michigan counties

Plot of adjusted letter values against (Gaussian) ratios (pushback analysis, from exhibit 16)

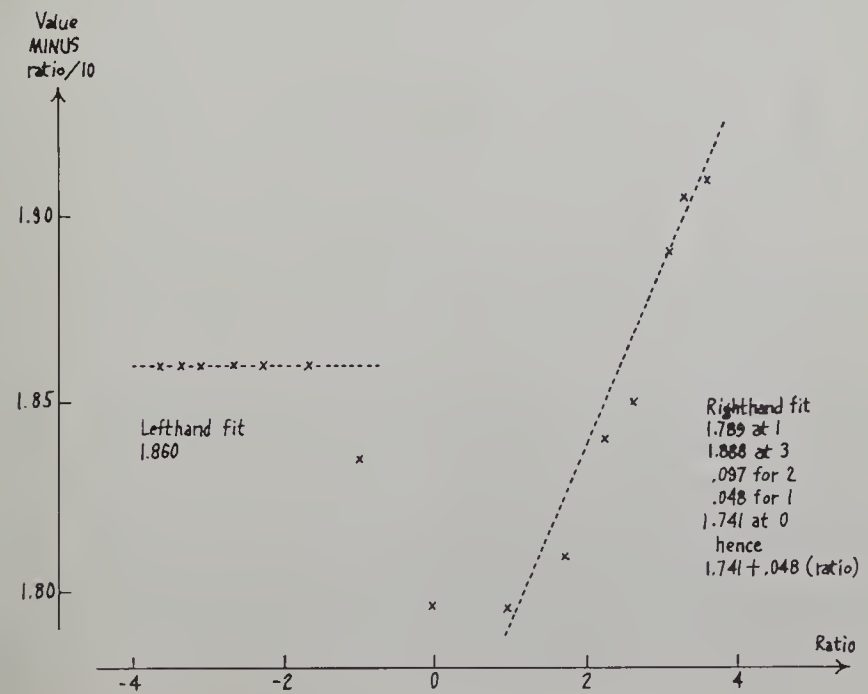


exhibit **18** of chapter 19: reference table

Values of standard ratios for points at half-octave depths (standard is the Gaussian shape)

Batch count	Ratio for point at depth 1	2	3	4	6	8	11	16	22
1	.00	X	X	X	X	X	X	X	X
2	.84	X	X	X	X	X	X	X	X
3	1.25	.00	X	X	X	X	X	X	X
4	1.51	.44	X	X	X	X	X	X	X
5	1.71	.72	.00	X	X	X	X	X	X
6	1.86	.94	.30	X	X	X	X	X	X
7	1.98	1.11	.52	.00	X	X	X	X	X
8	2.08	1.25	.69	.22	X	X	X	X	X
9	2.17	1.37	.84	.40	X	X	X	X	X
10	2.25	1.47	.96	.55	X	X	X	X	X
11	2.32	1.56	1.07	.68	.00	X	X	X	X
12	2.38	1.63	1.16	.79	.15	X	X	X	X
13	2.44	1.71	1.25	.89	.28	X	X	X	X
14	2.49	1.77	1.33	.98	.39	X	X	X	X
15	2.54	1.83	1.39	1.05	.49	.00	X	X	X
16	2.58	1.88	1.45	1.12	.58	.11	X	X	X
17	2.62	1.93	1.51	1.19	.66	.22	X	X	X
18	2.66	1.98	1.57	1.25	.73	.31	X	X	X
19	2.70	2.02	1.62	1.30	.81	.39	X	X	X
20	2.73	2.06	1.66	1.36	.87	.46	X	X	X
21	2.76	2.10	1.71	1.40	.93	.53	.00	X	X
22	2.79	2.19	1.75	1.45	.98	.60	.08	X	X
23	2.82	2.17	1.79	1.49	1.03	.66	.16	X	X
24	2.85	2.20	1.82	1.53	1.08	.71	.23	X	X
25	2.87	2.24	1.86	1.57	1.13	.77	.30	X	X
26	2.90	2.27	1.89	1.61	1.17	.82	.36	X	X
27	2.92	2.29	1.92	1.64	1.21	.86	.41	X	X
28	2.94	2.32	1.95	1.67	1.25	.91	.47	X	X
29	2.97	2.35	1.98	1.71	1.28	.94	.52	X	X
30	2.99	2.37	2.01	1.74	1.32	.99	.56	X	X
31	3.01	2.39	2.03	1.76	1.35	1.02	.61	.00	X
32	3.03	2.42	2.06	1.79	1.38	1.06	.65	.06	X
33	3.05	2.44	2.08	1.82	1.41	1.10	.69	.11	X
34	3.06	2.46	2.11	1.84	1.44	1.13	.73	.16	X
35	3.08	2.48	2.13	1.87	1.47	1.16	.77	.21	X
36	3.10	2.50	2.15	1.89	1.50	1.19	.80	.26	X
37	3.11	2.52	2.17	1.92	1.53	1.22	.84	.30	X
38	3.13	2.54	2.19	1.94	1.55	1.25	.87	.34	X
39	3.15	2.56	2.21	1.96	1.57	1.27	.90	.39	X
40	3.16	2.51	2.23	1.98	1.60	1.30	.93	.42	X
41	3.17	2.58	2.25	2.00	1.62	1.33	.96	.46	X
42	3.19	2.61	2.27	2.02	1.64	1.35	.99	.49	X

→

exhibit **18** of chapter 19(continued)

Batch count	Ratio for point at depth										
	1	**2**	**3**	**4**	**6**	**8**	**11**	**16**	**22**	**32**	**44**
43	3.20	2.62	2.29	2.04	1.66	1.37	1.02	.53	.00		
44	3.22	2.64	2.30	2.06	1.69	1.40	1.04	.56	.04		
45	3.23	2.65	2.32	2.07	1.71	1.42	1.07	.59	.08		
46	3.24	2.67	2.34	2.09	1.73	1.49	1.09	.62	.12		
47	3.25	2.68	2.35	2.11	1.74	1.46	1.12	.65	.16		
48	3.27	2.70	2.37	2.13	1.76	1.48	1.14	.68	.19		
49	3.28	2.71	2.38	2.14	1.78	1.40	1.16	.70	.23		
50	3.29	2.72	2.40	2.16	1.79	1.52	1.19	.73	.26		
51	3.30	2.74	2.41	2.17	1.82	1.54	1.21	.73	.29	X	X
52	3.31	2.75	2.43	2.19	1.83	1.56	1.23	.76	.32	X	X
53	3.32	2.76	2.44	2.20	1.85	1.58	1.25	.80	.35	X	X
54	3.33	2.77	2.45	2.22	1.86	1.59	1.27	.83	.38	X	X
55	3.34	2.79	2.47	2.23	1.88	1.61	1.29	.85	.41	X	X
56	3.35	2.80	2.48	2.24	1.90	1.63	1.31	.87	.43	X	X
57	3.36	2.81	2.49	2.26	1.91	1.64	1.32	.89	.46	X	X
58	3.37	2.82	2.50	2.27	1.92	1.66	1.34	.91	.49	X	X
59	3.38	2.83	2.51	2.28	1.94	1.68	1.36	.94	.51	X	X
60	3.39	2.84	2.53	2.30	1.95	1.69	1.38	.96	.53	X	X
61	3.40	2.85	2.54	2.31	1.97	1.71	1.39	.97	.56	X	X
62	3.41	2.86	2.55	2.32	1.98	1.72	1.41	.99	.58	X	X
63	3.42	2.87	2.56	2.33	1.99	1.73	1.42	1.01	.60	.00	X
64	3.43	2.88	2.57	2.34	2.01	1.75	1.44	1.03	.62	.03	X
65	3.44	2.89	2.58	2.35	2.02	1.76	1.45	1.05	.65	.06	X
66	3.45	2.90	2.59	2.37	2.03	1.78	1.47	1.07	.67	.08	X
67	3.45	2.91	2.60	2.38	2.04	1.79	1.48	1.08	.69	.11	X
68	3.46	2.92	2.61	2.39	2.05	1.80	1.50	1.10	.71	.14	X
69	3.47	2.93	2.62	2.40	2.07	1.81	1.51	1.12	.72	.16	X
70	3.48	2.94	2.63	2.41	2.08	1.83	1.53	1.13	.74	.19	X
71	3.49	2.95	2.64	2.42	2.09	1.84	1.54	1.15	.76	.21	X
72	3.49	2.96	2.65	2.43	2.10	1.85	1.55	1.16	.78	.23	X
73	3.50	2.97	2.66	2.44	2.11	1.86	1.57	1.18	.80	.25	X
74	3.51	2.97	2.67	2.45	2.12	1.87	1.58	1.19	.91	.28	X
75	3.52	2.98	2.68	2.46	2.13	1.89	1.59	1.21	.83	.30	X
76	3.52	2.99	2.69	2.47	2.14	1.90	1.60	1.22	.85	.32	X
77	3.53	3.00	2.70	2.48	2.15	1.91	1.62	1.23	.86	.34	X
78	3.54	3.01	2.71	2.49	2.16	1.92	1.63	1.25	.88	.36	X
79	3.54	3.01	2.71	2.50	2.17	1.93	1.64	1.26	.89	.38	X
80	3.55	3.02	2.72	2.50	2.18	1.94	1.65	1.27	.91	.40	X
81	3.56	3.03	2.73	2.51	2.19	1.95	1.66	1.29	.92	.42	X
82	3.57	3.04	2.74	2.52	2.20	1.96	1.67	1.30	.94	.43	X
83	3.57	3.05	2.75	2.53	2.21	1.97	1.68	1.31	.95	.45	X

→

exhibit **18** of chapter 19 (continued)

Batch count	Ratio for point at depth 1	2	3	4	6	8	11	16	22	32	44
84	3.58	3.05	2.75	2.54	2.22	1.98	1.69	1.32	.97	.47	X
85	3.58	3.06	2.76	2.55	2.23	1.99	1.71	1.34	.98	.49	X
86	3.59	3.07	2.77	2.55	2.24	2.00	1.72	1.35	1.00	.50	X
87	3.60	3.07	2.78	2.56	2.25	2.01	1.73	1.36	1.01	.52	.00
88	3.60	3.08	2.78	2.57	2.25	2.02	1.74	1.37	1.02	.54	.02
89	3.61	3.09	2.79	2.58	2.26	2.03	1.74	1.38	1.04	.55	.04
90	3.62	3.09	2.80	2.59	2.27	2.04	1.76	1.39	1.05	.57	.06
91	3.62	3.10	2.81	2.59	2.28	2.04	1.77	1.41	1.06	.58	.08
92	3.63	3.11	2.81	2.60	2.29	2.05	1.78	1.42	1.07	.60	.10
93	3.63	3.11	2.82	2.61	2.30	2.06	1.79	1.43	1.08	.61	.12
94	3.64	3.12	2.83	2.62	2.30	2.07	1.79	1.44	1.10	.63	.14
95	3.64	3.13	2.83	2.62	2.31	2.08	1.80	1.45	1.11	.64	.16
96	3.65	3.13	2.84	2.63	2.32	2.09	1.81	1.46	1.12	.66	.17
97	3.65	3.14	2.85	2.64	2.33	2.10	1.82	1.47	1.13	.67	.19
98	3.66	3.15	2.85	2.64	2.34	2.10	1.83	1.48	1.14	.68	.21
99	3.67	3.15	2.86	2.65	2.34	2.11	1.84	1.49	1.15	.70	.23
100	3.67	3.16	2.87	2.66	2.35	2.12	1.85	1.50	1.16	.71	.24

B) For BATCHES of MORE than 100 and DEPTH 1

Take "standard ratio" = $2.97\sqrt{-0.45 + \log_{10}(\text{batch size})}$

C) For BATCHES of MORE than 100 and DEPTHS >1

Use––in (A) or (B), as appropriate––the standard ratio for DEPTH = 1 and the following altered batch sizes:

For depth	Batch size used for depth 1
2	$\frac{1}{5}$(2(batch size) − 1)
3	$\frac{1}{4}$(batch size − 1)
4	$\frac{1}{11}$(2(batch size) − 3)
6	$\frac{1}{17}$(2(batch size) − 5)
8	$\frac{1}{23}$(2(batch size) − 7)
11	$\frac{1}{16}$(batch size − 5)
16	$\frac{1}{47}$(2(batch size) − 15)
22	$\frac{1}{65}$(2(batch size) − 21)

Example and check: For batch size = 100, (B) above gives 3.70 for depth 1 (using (A) directly gives 3.67). For depth 2, we use batch size = $\frac{1}{5}$(199) = 39.8 whence (A) gives 3.16 (using (A) directly gives 3.16). For depth 3, we use batch size = $\frac{1}{4}$(99) = 24.75, whence (A) gives 2.87 (using (A) directly gives 2.87). For depth 4, we use batch size = $\frac{1}{11}$(197) = 17.91 which gives 2.66 (using (A) directly gives 2.66). For depth 6, we use $\frac{1}{17}$(195) = 11.47, whence (A) gives 2.35 (using (A) directly gives 2.35). For depth 8, we use $\frac{1}{23}$(193) = 8.39, whence (A) gives 2.12 (using (A) directly gives 2.12). For depth 11, $\frac{1}{16}$(95) = 5.938 gives 1.85 (direct: 1.85). For depth 16, $\frac{1}{47}$(185) = 3.94 gives 1.49 (direct: 1.50). For depth 22, $\frac{1}{65}$(179) = 2.75 gives 1.15 (direct: 1.16).

review questions

What is our aim in this section? Where do we start from? What do we calculate? How do we proceed? How do the descriptions found here compare with those of the last section? Which is always better? Why/why not? If we want to concentrate more on individual values in the batch, how can we do this? Will it take much more effort? Why/why not? What did we put in a table to make calculation easy? What did we do next? Could we do pushbacks for every individual value in the batch? Would it be worthwhile?

exhibit **19** of chapter 19: county areas

Pushback analysis (123468... version) of the logs of the areas of the counties of Michigan (Q = 10)

A) CALCULATIONS

Depth or Pt	Obs'd	Ratio	$Q \times$ Ratio	Diff	Add'l fit	2nd resid	3R	S	Rough
1	1.51	−3.57	−.36	1.87				1.85	+.02
2	1.54	−3.05	−.30	1.84			1.85		−.01
3	1.57	−2.75	−.28	1.85			1.85		.00
4	1.61	−2.53	−.25	1.86			1.86		.00
6	1.66	−2.21	−.22	1.88			1.88	1.88	.00
8	1.68	−1.97	−.19	1.88			1.88	1.88	.00
11	1.69	−1.68	−.17	1.86			1.86		.00
H	1.73h	−.98	−.10	1.83h				1.84	.00
M	1.76	0	0	1.76	.00	1.76	1.76		.00
H	1.89h	.98	.10	1.79h	.04	1.75h	1.75h		.00
11	1.98	1.68	.17	1.81	.07	1.74	1.74		.00
8	2.01	1.97	.19	1.81	.08	1.73	1.74		−.02
6	2.07	2.21	.22	1.85	.09	1.76	1.73		.03
4	2.08	2.53	.25	1.83	.10	1.73	1.73		.00
3	2.12	2.75	.28	1.83	.11	1.72	1.73		−.01
2	2.20	3.05	.30	1.90	.12	1.78	1.76		−.02
1	2.26	3.57	.36	1.90	.14	1.76	1.76		.00

Notes: "Diff" contains unsmoothed "pushed-back" values, which are plotted in exhibit 20.

P) PROBLEMS

19a) Repeat for the raw areas.

19a2) Plot the result.

19b) Repeat for the root areas.

19b2) Plot the result.

S) SOURCE

Exhibit 5 of chapter 3.

19H. How far have we come?

This chapter has focussed on comparing counted fractions with a standard shape of distribution, doing this in various terms:

- ◇ all individual values.
- ◇ some binning of the values.
- ◇ letter values for the batch.
- ◇ half-octave-depth values for the batch.

As a reference standard from which to measure the deviations of the real world, we have introduced the Gaussian distribution, and we have been careful to avoid thinking of real data as in any sense "exactly Gaussian".

We are now ready to:

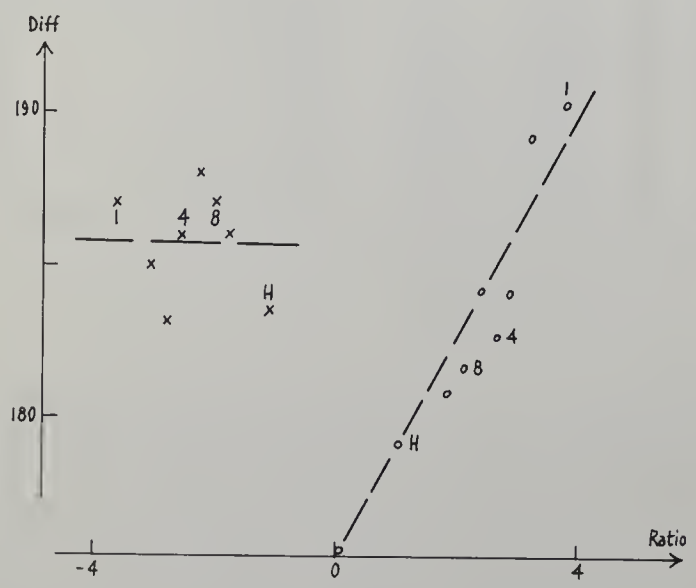
- ◇ approach the description of distributions by first finding flogs (or froots) for each of several cut locations and then describing the relationship by the combination of a fit (flog to location) and residuals (of flogs).
- ◇ do this using two or more different fits in different sections.

Moreover we are also ready to use letter values--or half-octave-depth values--to study shapes of distribution in a somewhat different way:

- ◇ by converting letter values into mids and 25% pseudospreads--by plot-

exhibit **20** of chapter 19: county areas

Plot of pushed back values (from "diff" column of exhibit 19)



ting the result against--and then fitting to--the square of the divisor used to find the corresponding pseudospread.

◇by alternatively using simple pushback on either letter values or half-octave-depth values.

And we have begun to understand the proper nature and relevance of the Gaussian distribution (sometimes misleadingly called the "normal" distribution), in particular understanding that:

◇the Gaussian is a reference standard, *not* an example of what in fact is-- that it is something against which to compare the actual behavior of real data. (This can be by residuals--including pushed-back values--or by the behavior of mids and 25% pseudospreads.)

◇we often can, and even should, use it as an initial approximation to actual behavior.

◇in dealing with distributions of the real world, we are very lucky if

◇we know APPROXIMATELY how values are distributed.

◇this approximation is itself not TOO far from being simple.

review questions 647

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Mathematical distributions 20

Suppose we have a binning involving many, many appearances--say a binning of U.S. taxpayers according to reported income (as in exhibit 5 of chapter 19). We could think of subdividing our bins more and more finely--if the incomes were in thousand-dollar bins, we could surely go to hundred-dollar bins, and probably much further, especially if we could have more individuals to bin.

It is natural for a mathematician, once started on the idea of narrower bins and more individuals, to push on until the bins are infinitely narrow and the individuals are infinitely many. The result is a mathematical distribution, a concept that has:

- ◇many uses in the theories of statistics and data analysis.
- ◇real convenience as a background against which to interpret and understand actual binnings.

It will clearly be convenient to try to represent the distributions we work with by simple formulas--but we must remember that this is only a hope--indeed that it is more reasonable to hope for only reasonable approximation by simple formulas. (Even this is asking a lot.)

This chapter uses the idea of mathematical distribution to tie together what we have been doing in the last two chapters. In particular, it looks more closely into what a binning suggests about a corresponding mathematical distribution, and into how the various mathematical distributions suggested by the last two chapters compare with one another.

review questions

In principle, how small bins could we use? In practice? What is a mathematical density? A mathematical distribution? How does a mathematician's mind get to them? What are mathematical distributions good for? Can we expect real distributions to be given by simple formulas? Exactly or approximately? What is our aim in this chapter? What are two important parts of what we will do?

20A. Binnings vs. distributions

continuity and discreteness

If we are going to think about arbitrarily narrow bins, we have to think of observed values, each of which is given to arbitrarily many decimals. To make arbitrarily narrow bins useful for any real data--assuming we could get as much of it as was asked for--each of these decimals would have to have entries not all the same. (Numbers, all of which were like 1063.2130000..., 2039.1170000..., and 1557.6420000..., might possibly use bins of width 0.001, but bins of width 0.0001 would tell us nothing more.) Many mathematical distributions assume that the "observations" are given to arbitrarily many decimals; an observation matching the given value to that number of decimals will come along if we wait long enough.

To the man with data, such unrealism hardly looks helpful, but to the worker with theory such an assumption turns out to make many things simpler. Since all the man with data can hope for is an approximation, and since an approximation to a few decimal places is often more than he deserves, he soon learns not to worry very often about this particular sort of unrealism. (But he may have to worry about theoretical consequences that others have drawn from it.)

Other kinds of mathematical distributions assume that only isolated values are possible. These values are usually, as a practical convenience, taken to be integers (which may, of course, have to be reexpressed).

fractions, densities, cumulatives

When we think of dealing with more and more--and then many more--"observations", we soon start to find counts unhandy. Instead, it becomes natural to think about fractions of the form

$$\frac{\text{count}}{\text{total count}} = \text{fraction}.$$

(Let us not worry about "+1/6" here--we need not, since all counts are getting larger and larger) and, once our bins get very narrow, we tend to work with the results of dividing by bin width

$$\frac{\text{fraction}}{\text{bin width}} = \text{(average) density},$$

in close analogy to what we did when we dealt with counts for bins of unequal width.

One reasonable way to describe a mathematical distribution is to say how the

(mathematical) density

depends on the value concerned. This will usually be a density for an "infinitely narrow" bin, meaning a bin so narrow that making it narrower will not change

the density--to the precision we are concerned with. (Hidden in this definition is a more detailed assumption about the smoothness of distribution of the "observations" that we need to know is there, but which we can lump in with the other idealizations as part of "all is approximation anyway".) Panel A of exhibit **1** shows a few values of the mathematical density of a particular distribution.

Another way to describe a mathematical distribution is in terms of a

$$\text{cumulative fraction} = \frac{\left(\begin{matrix}\text{count of all}\\ \text{values} < \text{cut}\end{matrix}\right) \text{ plus } \left(\begin{matrix}\text{half of count of}\\ \text{those equal to the cut}\end{matrix}\right)}{\text{total count}}$$

defined for any way of cutting the values into lower (to the left of the cut) and higher (to the right of the cut). We can do this by a formula--or at least for all practical purposes, by a numerical table--from either of which we can find the cumulative fraction, given the cut. Panel B (of exhibit 1) shows a few lines of such a table for the same distribution. Panel C gives a rather longer list of values from such a table for the same distribution. We can do equally well by a formula--or table--from which we can find the cut, given the cumulative fraction.

exhibit **1** of chapter 20: sample description

A few values of the mathematical density, the cumulative distribution, and the representing function (% point function) for the standard distribution used in chapter 19 (with hinges at ±1)

A) A few VALUES of the MATHEMATICAL DENSITY

cut = Value of what is distributed	Density*
−5	.0009
−4	.0071
−3	.035
−2	.108
−1	.214
0	.269
1	.214
2	.108
3	.035
4	.0071
5	.0009

B) A few VALUES of the CUMULATIVE DISTRIBUTION

cut = Value of what is distributed	Fraction below = Cumulative distribution
−5	.0004
−4	.0035
−3	.0215
−2	.0887
−1	.2500
0	.5000
1	.7500
2	.9113
3	.9785
4	.9965
5	.9996

*** Mathematical density.**

→

Note that, just as panel B leads from −1 to 0.25, so panel C leads from 0.25 back to −1. (And that, as panel C leads from 0.75 to +1, so panel B leads from +1 back to 0.75.)

Such formulas, if they exist, are said to give the "cumulative distribution" (or the "cumulative") when they give cumulative fraction in terms of cut, or to give "the representing function" or "the percent-point function" when they give the cut in terms of the cumulative fraction. (A good numerical table can, of course, be used either way, giving whichever we want in terms of whichever we have. Such a table thus gives both cumulative and representing function--to an indicated approximation.) We are, by now, familiar with tables of representing functions for a few distributions--disguised as "re-expressions of fractions" in the case of froots and flogs, or as "ratios" in the Gaussian case.

exhibit 1 of chapter 20(continued)

C) Selected VALUES of the REPRESENTING FUNCTION (or PERCENT POINT FUNCTION)

Fraction below = Cumulative fraction	Cumulative percent	cut = Value of what is distributed
.001	(0.1%)	−4.58
.01	(1%)	−3.45
.05	(5%)	−2.44
.10	(10%)	−1.90
.15	(15%)	−1.54
.20	(20%)	−1.25
.25	(25%)	−1.00
.30	(30%)	−.78
.40	(40%)	−.38
.50	(50%)	.00
.60	(60%)	.38
.70	(70%)	.78
.75	(75%)	1.00
.80	(80%)	1.25
.85	(85%)	1.54
.90	(90%)	1.90
.95	(95%)	2.44
.99	(99%)	3.45
.999	(99.9%)	4.58

P) PROBLEMS

1a/b/c/d) Below what value do about 9%/45%/98%/99.9% of the values lie?

1e/f/g/h/i) At what value does the density amount to .1/.2/.01/.03/.001?

1j) How do panels A and C interrelate?

A few kinds of mathematical distributions--two, three, or a half-dozen, depending on your definition of simple--have simple formulas for all three of density, cumulative, and representing function. Rather more have simple formulas for one or two of the three. Others have not very simple formulas--some of these are quite important--or even quite complex formulas for one or two or three of the triad: representing function, density, or cumulative. This will not bother us here because--at least in analyzing data--we will always do one of two things:

- ◇rely on some quite simple formula.
- ◇rely on a table (which gets rid of any need for a formula).

review questions

Do most real distributions fit simple formulas? Could arbitrarily narrow bins be useful? When? In theory? In practice? Is a "continuity" assumption realistic? Why/why not? Does lack of realism here really matter in practice? Why/why not? What is a (mathematical) density? How is it related to any actual batch of numbers (pick the closest one)? To any mathematical distribution (pick the closest one)? What is a cumulative fraction? How can we give a cumulative fraction for all or many cuts? What is a cumulative distribution? A representing function? A percent-point function? How many of these are related? Which, how, and how closely? What are the three ways to specify a mathematical distribution? Do many mathematical distributions have simple formulas for all three ways? About how many? Dare we expect to stick to mathematical distributions with at least one simple formula? Why/why not? What do we do here (in this book)?

20B. Densities for distributions vs. densities for binnings

We learned, in chapter 17, to fit many binnings. We routinely fitted

$$\text{some simple expression of } \sqrt{\frac{\text{count}}{\text{bin width}}}$$

with a simple function of bin location--usually one constant plus another constant times (bin location − peak)^2. (When the bins were of equal width, we just took all their widths to be 1.0 and talked about an expression of $\sqrt{\text{count}}$.)

The first change we need to make is to replace "count" by fraction, and start with

$$\text{some simple expression of } \sqrt{\frac{\text{fraction}}{\text{bin width}}},$$

which is, of course,

$$\text{some simple expression of } \sqrt{\text{density}}.$$

correction for curvature

The next thing we have to worry about is the difference between

average density

for some not too narrow bin width, which is the only thing we can calculate from data, and

(mathematical) density

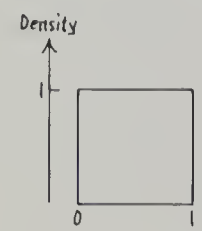
mathematically defined, which is what appears in the mathematical distribution. Fortunately, a simple approximate correction is good enough for essentially every case that comes up in practice. So let us learn what it is.

The simplest kinds of mathematical distributions are shown in exhibit 2;

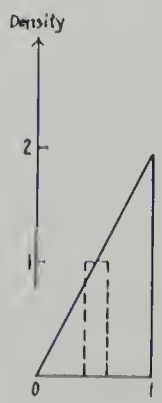
exhibit 2 of chapter 20: various densities

Densities for some very simple mathematical distributions

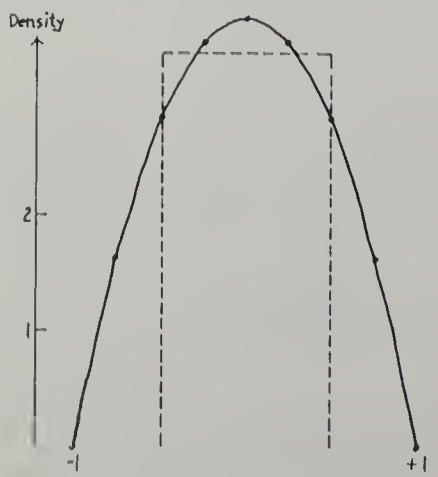
A) Rectangular = constant density



B) Triangular = linear density



C) Quadratic density (enlarged scale)

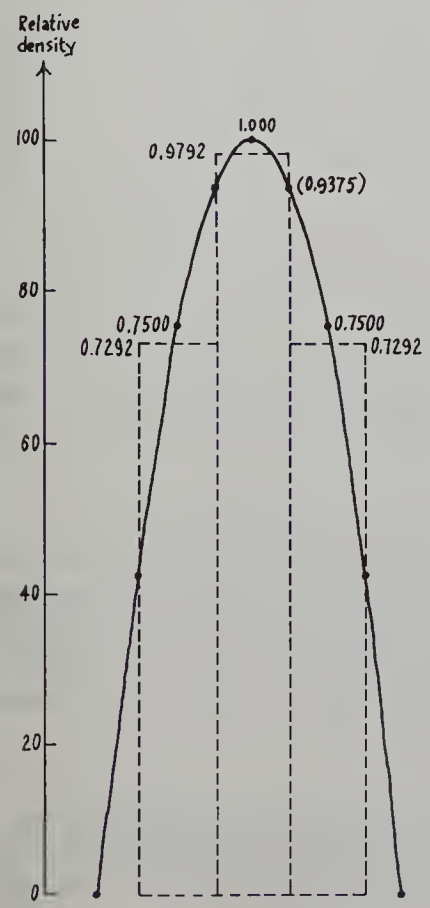


constant density, linear density, quadratic density. The dashed rectangles on the last two show what average densities we would get for particular bins. We see that, for the linear density, the average density is the same as the mathematical density at the center of the bin. Apparently we will need no correction when the density is linear--presumably also when the average density is linear.

For the quadratic density, and the rather wide, centrally placed bin shown, the average density is appreciably below the mathematical density at the center of the bin. We do need an adjustment. Exhibit **3** shows the same situation in more detail, using a convenient vertical scale. For three adjacent bins of equal size, the average densities are 0.7292, 0.9792, and 0.7292. If these were linear--and how clearly they are not!--the skip mean involving the two outer

exhibit **3** of chapter 20: quadratic density

Some average and mathematical densities for a particular example



values, here .7292, would equal the central value, here .9792. Thus the difference

average density MINUS skip mean of average densities,

here equal to

$$.9792 - .7292 = .25,$$

is a natural measure of bend.

The correction needed here is

$$1.0000 - .9792 = .0208,$$

which is exactly 1/12 of .25. So it is reasonable to take, as a plausible correction,

$$\frac{\text{average density MINUS skip mean of average densities}}{12},$$

which means that we use, as the adjusted average density,

$$\text{average density PLUS } \frac{\left(\begin{array}{c}\text{average density MINUS skip} \\ \text{mean of average density}\end{array}\right)}{12}.$$

Note that this adjustment is only an approximation. If $\sqrt{\text{adjusted count}}$ differs from $\sqrt{\text{count}}$ by more than a few units, we should not trust the approximation too far--though the adjusted value will usually be much better than the unadjusted one. (Better approximations are more trouble than we care to face in this volume.)

Exhibit **4** shows how this works for an age-of-death example, one for deaths by coronary disease. The main feature of this exhibit is the small size of the adjustments to $\sqrt{\text{count}}$, only two of which are larger than ± 1.1. (The adjustment of -2.6 applies at ages 95–99, where expert judgment tells us that the accuracy of ages is very dubious.) The adjustments are small, even though there are a total of nearly half a million carefully tabulated deaths.

review questions

What is the aim of this section? How did we begin? What is an average density? A (mathematical) density? What are some very simple mathematical distributions? What mathematical distribution was a simple leading case? What formula did this suggest? How firmly do we believe in the corresponding answers? What example did we try? How big were the adjustments? What happened when we used wider bins?

20C. Tables and pictures comparing two sets of shapes of distribution

In chapter 17 we introduced seven choices of expression for $\sqrt{\text{count/width}}$ that might behave simply when plotted against $(B - \text{peak})^2$ or $(L - \text{peak})^2$.

exhibit **4** of chapter 20: coronary deaths

Age binning and adjustment for 406,499 deaths assigned to cause 420.1--heart disease specified as involving coronary arteries--in the U.S. during 1965

A) DATA and CALCULATIONS

Bin*	Count	$\sqrt{\text{count}}$	>†	Diff	Diff/12	Adj count	Adj $\sqrt{\text{count}}$	Adj in $\sqrt{\text{count}}$
0–4	32	(5.7)	5	27	2.2	34	5.8	(+.2)
5–9	10	(3.2)	22	−12	−1.0	9	3.0	(−.2)
10–14	12	(3.5)	27h	−15h	−1.3	11	3.3	(−:2)
15–19	45	(6.7)	68	−23	−1.9	43	6.6	(−.1)
20–24	124	(11.1)	193h	−69h	−5.8	118	10.9	(−.2)
25–29	342	(18.5)	637	−295	−24.6	317	17.8	(−.7)
30–34	1150	(33.9)	1940	−790	−65.8	1084	32.9	(−1.0)
35–39	3538	(59.5)	4900	−1362	−113.5	3424	58.5	(−1.0)
40–44	8650	(93.0)	9666	−1016	−84.7	8565	92.5	(−.5)
45–49	15794	(125.7)	17414	−1620	−135.0	15659	125.1	(−.6)
50–54	26178	(161.8)	26252h	−75h	−6.2	26172	161.8	(0)
55–59	36711	(191.6)	36955	−244	−20.3	36691	191.5	(−.1)
60–64	47732	(218.5)	47441	291	24.2	47756	218.5	(0)
65–69	58171	(241.2)	56200	1971	164.2	58335	241.5	(+.3)
70–74	64668	(254.3)	59577	5091	424.2	65092	255.1	(+.8)
75–79	60983	(246.9)	55021	5962	496.8	61480	248.0	(+1.1)
80–84	45374	(213.0)	43137h	2236h	186.4	45560	213.4	(+.4)
85–89	25292	(159.0)	27336h	−2044h	−170.4	25122	158.5	(−.5)
90–94	9299	(96.4)	13654h	−4355h	−363.0	8936	94.5	(−.9)
95–99	2017	(44.9)	4787h	−2770h	−230.9	1786	42.3	(−2.6)
100–‡	276	(16.6)	1008h	−732h	−61.0	215	14.7	(−1.9)

(Age not stated: 101)

*** Age in years.**
† Split mean of count (not of ($\sqrt{\text{count}}$)).
‡ All taken as 100–104.

P) PROBLEMS

4a) Combine the counts in 15-year bins (0–14, 15–29, etc.) and make a similar calculation. Are the adjustments larger?

4a2) Divide the adjusted root counts thus obtained by $\sqrt{3}$, and compare the results with those for the 5-year intervals at the center of every fifteen-year interval. How do they compare?

4b/4b2) Do the same for 15-year bins (5–19, 20–34, etc.).

4c/4c2) Do the same for 15-year bins (10–24, 25–39, etc.).

4d) Why did we choose to divide by $\sqrt{3}$ in (4a2)?

S) SOURCE

Vital Statistics of the United States, 1965, Volume II-Mortality, Part A. **U.S. Dept. Health, Education and Welfare, 1967. Pages 1–132 and 1–133.**

After the discussion of the last section, we are likely to be interested in the seven shapes of distribution whose $\sqrt{(\text{mathematical})\ \text{density}}$, when correspondingly expressed, is a linear function of $(B - \text{peak})^2$.

In chapter 19 we introduced three shapes of distribution--two deriving from convenient re-expressions of fractions and the third being the reference standard (the Gaussian). We may reasonably want to compare these shapes, both with each other and with the seven just mentioned.

It is now time to say that the Gaussian shape is exactly the shape for which $\log\sqrt{\text{density}}$ is linear in $(x - \text{peak})^2$--the reference standard of chapter 19 is exactly the middle shape of the seven of chapter 17. Thus, seven and three give nine, not ten.

Exhibit **5** compares the expressions of fractions corresponding to all nine distribution shapes, each adjusted to hinges = +1 for easy comparison. We know that this table can also be used as a skeleton table of representing functions. With 75% forced to give 1.000, we should not be surprised that the expressions of 70% are a lot alike (all but one between .789 and .764 (one at .726), as are those at 80% (1.221 to 1.279 and 1.376) and 60% (.388 to .362 and .325). Even at 90% things do not vary tremendously (1.727 to 2.141 and 3.077). At 99% things are more different (2.45 to 5.94 and 31.82). And so more and more, the farther we go into the tails.

Exhibit **6** gives the corresponding table of cumulatives, again matched to have hinges at ±1.0. Again, behavior between −1.5 and +1.5 is reasonably steady.

Exhibit **7** gives the corresponding relative densities. For larger deviations, the influence of the increasingly stretched tails of the distributions to the right of exhibit 7--and the increasingly squeezed-in tails of the distributions to the left of the same exhibit--is clearly evident. (The distributions are quite different--but smoothly and reasonably so.) Exhibit **8** pictures the spacings of the letter values. For the nine distributions, we see that the tails stretch more and more in the order r_4 (very squeezed), r_6, r_{10}, Gauss, t_7, t_3, t_1 (very very stretched), with:

- ◇ the shape corresponding to froots close to, but more squeezed than, r_4.
- ◇ the shape corresponding to flogs close to t_7, extreme tails being less stretched.
- ◇ the t_1 shape very much more tail-stretched than the others.

Those who use flogs or froots clearly straddle the Gaussian neatly (as would one who used t_7 and r_4--thus taking either $(\sqrt{\text{density}^2})$ or $1/(\sqrt{\text{density}^{1/2}})$ as linear in $(\text{Shift})^2$).

identification

Some of the readers who reach this, the last section of the main body of this book, will have previously been introduced to classical statistics. They may recognize such terms as "Student's t", "Pearson's r", or "product-moment correlation". These paragraphs are for them--others are encouraged to skip.

exhibit **5** of chapter 20: tabular comparison

Nine expressions of fractions, adjusted to "hinges = ±1".

A) TABLE

	Value of appropriate multiple of									
+	"froot"	r_4	r_6	r_{10}	"Gauss"	"flog"	t_7	t_3	t_1	−
50%	.000	.000	.000	.000	.000	.000		.000	.000	**50%**
60%	.388	.386	.382	.380	.376	.369	.370	.362	.325	**40%**
70%	.789	.787	.783	.781	.777	.771	.772	.764	.726	**30%**
80%	1.221	1.226	1.234	1.235	1.248	1.261	1.260	1.279	1.376	**20%**
90%	1.727	1.752	1.802	1.842	1.90	2.000	1.990	2.141	3.077	**10%**
92%	1.85	1.88	1.998	2.00	2.38	2.22	2.21	2.43	3.89	**8%**
95%	2.05	2.10	2.21	2.30	2.44	2.68	2.68	3.08	6.31	**5%**
98%	2.32	2.40	2.605	2.77	3.04	3.54	3.54	4.55	15.89	**2%**
99%	2.45	2.54	2.81	3.05	3.45	4.18	4.22	5.94	31.82	**1%**
99.2%	2.48	2.58	2.86	3.12	3.57	4.39	4.44	6.45	39.7	**.8%**
99.5%	2.53	2.64	2.97	3.28	3.82	4.82	4.92	7.64	63.7	**.5%**
99.8%	2.61	2.73	3.13	3.52	4.27	5.65	5.92	10.59	159.1	**.2%**
99.9%	2.64	2.77	3.22	3.68	4.58	6.29	6.73	13.35	318.3	**.1%**
99.92%	2.65	2.78	3.24	3.72	4.68	6.49	7.00	14.4	398.	**.08%**
99.95%	2.68	2.80	3.29	3.81	4.88	6.92	7.60	16.9	636.	**.05%**
99.98%	2.69	2.83	3.36	3.96	5.25	7.75	8.85	23.0	1591.	**.02%**
99.99%	2.70	2.85	3.40	4.05	5.51	8.38	9.95	29.0	3180.	**.01%**
(100%)	(2.73)	(2.88)	(3.56)	(4.34)						

Notes

"froot"	marks the shape of distribution associated with froots
r_4	means $(\sqrt{\text{density}})^2$ is linear in $(x - \text{peak})^2$
r_6	means $(\sqrt{\text{density}}$ is linear in $(x - \text{peak})^2$
r_{10}	means $(\sqrt{\text{density}})^{1/2}$ is linear in $(x - \text{peak})^2$
"Gauss"	marks the Gaussian shape, for which log density is linear in $(x - \text{peak})^2$
"flog"	marks the shape of distribution associated with flogs
t_7	means $1/(\sqrt{\text{density}})^{1/2}$ is linear in $(x - \text{peak})^2$
t_3	means $1/\sqrt{\text{density}}$ is linear in $(x - \text{peak})^2$
t_1	means $1/(\sqrt{\text{density}})^2$ is linear in $(x - \text{peak})^2$

P) PROBLEMS

5a) Plot all eight representing functions (on the same plot) against the one associated with flogs. Discuss.

5b/c/d/e) Use these % values the way the letter values were used in section 19C to find (Gaussian) 25% pseudospreads and mids for the 1st/2nd/3rd/4th column. Plot against (Gaussian representing function)^2.

5f/g/h/i) Do the same for the 6th/7th/8th/9th column.

exhibit **6** of chapter 20: tabular comparison

Nine cumulative distributions, each adjusted to "hinges = ±1" for easy comparison

A) OUT to ±5

	Cumulative for appropriate multiple of								
Cut	"froot"	r_4	r_6	r_{10}	"Gauss"	"flog"	t_7	t_3	t_1
−5	X	X	X	X	.0004	.0041	.0046	.0157	.0526
−4.5	X	X	X	X	.0012	.0071	.0075	.0206	.0696
−4	X	X	X	.0001	.0035	.0122	.0124	.0274	.0780
−3.5	X	X	.0000	.0022	.0091	.0209	.0208	.0378	.0886
−3	X	X	.0043	.0113	.0215	.0357	.0352	.0528	.1024
−2.5	.0033	.0124	.0259	.0349	.0459	.0683	.0593	.0759	.1711
−2	.0571	.0678	.0734	.0803	.0887	.1000	.0990	.1118	.1476
−1.5	.1422	.1446	.1490	.1620	.1558	.1614	.1608	.1672	.1872
−1	.2500	.2500	.2500	.2500	.2500	.2500	.2500	.2500	.2500
−.5	.3716	.3711	.3700	.3691	.3680	.3660	.3663	.3638	.3524
0	.5000	.5000	.5000	.5000	.5000	.5000	.5000	.5000	.5000
1	.7500	.7500	.7500	.7500	.7500	.7500	.7500	.7500	.7500
2	.9429	.9372	.9266	.9197	.9113	.9000	.9010	.8882	.8524
3	X	X	.9957	.9887	.9785	.9643	.9648	.9472	.8976
4	X	X	X	.9999	.9965	.9878	.9876	.9725	.9304
5	X	X	X	X	.9995	.9959	.9954	.9843	.9474

(For column labels, see note on exhibit 5.)

B) FARTHER OUT

	Cumulative for appropriate multiple of				
Cut	Gauss	"flog"	t_7	t_3	t_1
−1000				.0005	.00032
−300				.0007	.00106
−100				.0011	.00317
−30		.00000	.00000	.0016	.0106
−10		.00002	.00010	.00232	.0317
−10		.00002	.00010	.00232	.0303
−9		.00005	.00018	.00314	.03522
−8		.00015	.00037	.00439	.03968
−7	.00000	.00046	.00080	.00637	.04517
−6	.00003	.00137	.00186	.00972	.05287
−5	.00037	.00410	.00464	.01574	.06283

(For column labels, see note on exhibit 5.)

P) PROBLEMS

6a) For what cut are most cumulative distributions close to 37%? To 63%?

6b) Estimate, by eye and mind, the nine cumulative fractions corresponding to a cut at −1.8.

6c) The same, as nearly as may be, for a cut near −2.8.

6d) Or near −3.8.

exhibit **7** of chapter 20: tabular comparison

Nine mathematical densities, each for a distribution adjusted to "hinges = ±1" for easy comparison

A) The TABLE

	Density for appropriate multiple of								
Cut	"froot"	r_4	r_6	r_{10}	"Gauss"	"flog"	t_7	t_3	t_1
0	.2588	.2605	.2636	.2651	.2689	.2747	.2737	.2811	.3183
±.5	.2523	.2526	.2532	.2537	.2540	.2549	.2549	.2556	.2546
±1.	.2320	.2291	.2235	.2196	.2142	.2050	.2071	.1969	.1591
±1.5	.1962	.1898	.1782	.1706	.1613	.1487	.1499	.1358	.0979
±2.	.1404	.1346	.1233	.1163	.1084	.0989	.0992	.0887	.0836
±2.5	.0398	.0641	.0675	.0669	.0650	.0622	.0617	.0571	.0439
±3.	X	X	.0220	.0300	.0348	.0378	.0369	.0370	.0318
±3.5			.0003	.0089	.0166	.0225	.0217	.0245	.0340
±4.			X	.0011	.0071	.0132	.0127	.0166	.0187
±4.5					.0027	.0077	.0074	.0115	.0150
±5.					.0009	.0045	.0044	.0081	.0122
±5.5					.0003	.0028	.0027	.0059	.0102
±6.					.00010	.0015	.0016	.0044	.0086
±6.5					.0000	.0009	.0010	.0033	.0074
±7.					.0000	.0005	.0006		
±8.						.0002	.0003	.0016	.0049
±9						.0001	.0001	.0010	.0039
±10						.0000	.0001	.0007	.0032
±11							.0000	.0005	.0026
±12								.0003	.0022

P) PROBLEMS

7a) What is the mathematical density for t_3 at −2.5? at −4.5? at +4.5?

7b) Same for r_4 at each of the three cuts in turn.

7c/d/e/f/g) Make up some problems of your own.

The seven distributions of chapter 17, where simple re-expressions of the square root of the density are linear in the square of the shift from the peak, all become conventional mathematical distributions, once we replace average density (as used in chapter 17) by mathematical density. Some will like to know that the correspondence is:

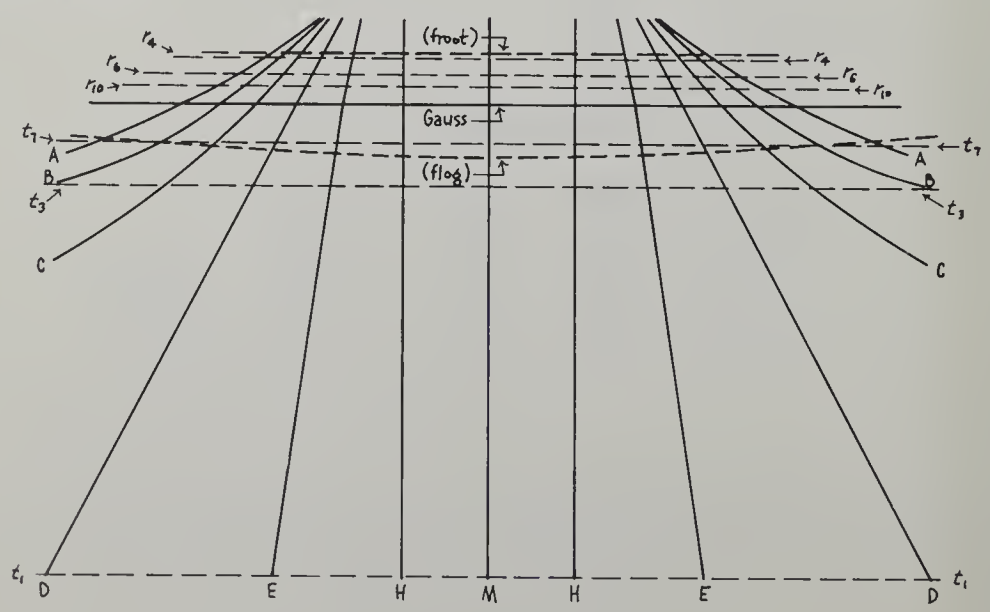
$(\sqrt{\text{density}})^2$	distribution of (product-moment) r on 4 df.
$(\sqrt{\text{density}})$	distribution of r on 6 df.
$(\sqrt{\text{density}})^{1/2}$	distribution of r on 10 df.
$\log \sqrt{\text{density}}$	Gaussian distribution.
$(1/\sqrt{\text{density}})^{1/2}$	distribution of Student's t on 7 df.
$(1/\sqrt{\text{density}})$	distribution of Student's t on 3 df.
$(1/\sqrt{\text{density}})^2$	distribution of Student's t on 1 df (Cauchy).

further comment

The distributions of r (of t) are those usually called by those names and apply to the product-moment correlation coefficient or to Student's t, when the

exhibit **8** of chapter 20: mathematical distributions

Behavior of the letter values for the nine shapes of distribution (Gaussian shape solid, shapes corresponding to froots and flogs heavier dashes, *r* and *t* shapes lighter dashes)



underlying values follow a Gaussian distribution EXACTLY and the population correlation (or difference) is IDENTICALLY zero. (When the underlying distribution, as always, is nonGaussian, the actual null distributions (those for zero population correlation or zero population difference) for observed r and observed t will be somewhat different--usually NOT catastrophically different but, particularly for r, often appreciably different.

review questions

What shapes of distribution from chapter 17 may interest us? What was tried out as behaving nicely in that chapter? What modification is now considered? What shapes of distribution did we introduce in chapter 19? Were they given in terms of observed distributions or mathematical ones? How many were there in chapter 17? In chapter 15? How many all together? Explain your answer! What have we given tables for in this section?

21A. Our relationship to the computer 663
21B. What has been omitted? 664
21C. How should the past chapters look different? 665
21D. What have we been introduced to? 666

Postscript 21

What ought to be said to those who have come through twenty chapters of diverse techniques? Where do they stand? In what ways should earlier parts be looked at differently in the light of later parts? What sorts of techniques have not been begun? Why? Brief answers to such questions are due all those who have worked their way to this point.

21A. Our relationship to the computer

Today the world is going through a process of computerization that is culturally rapid but humanly slow. The writer learned to calculate with a hand-crank desk calculator, and looked upon electric motor drive as a great advance. (Much better facilities than those that then cost rather more than half as much as a new car are now available for less than twenty dollars, and are being sold to housewives as a shopping aid.) A decade later, the writer helped a little with John von Neumann's computer project, which set the pattern for modern computing systems--large, medium, or small. After about a second decade, the first large array of statistical programs for computer use (Dixon's BIOMED, later BIMED, later BMD) began to appear. The two decades that followed have seen great diversification of approaches, techniques, and packages. Nothing in statistical computing a decade old looks at all up to date--and we would find it hard to say where we are to be a decade hence.

This book focusses on paper and pen(cil)--graph paper and tracing paper when you can get it, backs of envelopes if necessary--multicolor pen if you can get it, routine ballpoint or pencil if you cannot. Some would say that this is a step backward, but there is a simple reason why they cannot be right: Much of what we have learned to do to data can be done by hand--with or without the aid of a hand-held calculator--LONG BEFORE one can find a computing system--to say nothing of getting the data entered. It will be a long time until this fails to be so. Even when every household has access to a computing system, it is unlikely that "just what we would like to work with" will be easily enough available. Now--and in the directly foreseeable future--there will be a place for hand calculation.

This is not to undersell the virtues of good computing systems, large or small. Given only a little more data than one is willing to tackle by hand, they become necessary. Long before that they become a great convenience. Many of the things we have learned to do in this book are relatively novel. It will be a

while before most computing systems are set up to furnish the facilities we all would like. Installation is never trivial; appreciable effort will be needed at every place where such facilities are made available.

But there is hope. Paul Velleman, with support from the National Science Foundation, has been putting together facilities adapted to the use of such techniques in research, a number of versions of SNAP-IEDA (stemming from Michael Godfrey's initial efforts) are in use here and there, and a variety of facilities have been provided locally to help with the teaching of this book's preliminary edition. Perhaps there will come to be supplementary accounts, complete with programs, meshing computer systems with the text of this book.

Actually, the attitudes that underlie the writing of this book, and many of the techniques used in it owe much to modern computers. Today's understanding of how well elementary techniques work owes much to both mathematical analysis and experimentation by computer. You couldn't see the computer in the 661 pages that have gone before, but its shadow lay heavily on many of them.

21B. What has been omitted?

If we were to ask a diverse group of well-informed users which statistical techniques were most used, we would get diverse answers. Most, I am sure, would mention regression and analysis of variance, though in special areas of application these words might be drowned out by contingency-table analysis, life tables, bioassay, or even item and factor analysis. How is it that we have seen only the barest of introductions to any of these?

We have fitted lines, but almost everyone would feel that "regression" is multiple regression. We have made row-PLUS-column fits--and their extensions to somewhat more complex models and to three factors--but most would agree that "real analysis of variance" begins when you must choose between error terms, something we have not done. Why have we not gone more deeply into the main highways of today's data analysis?

A number of reasons can be given--all plausible, and some, perhaps, correct. These include:

- ◇ these main-highway methods have conventionally been thought of as confirmatory rather than exploratory (even though their most important uses are often in exploration).
- ◇ these main-highway methods tend to involve more computing than anyone is prepared to face by hand.
- ◇ understanding what is really going on in some main-highway methods is far from trivial.
- ◇ first things should come first.

Any one or two of these would be enough reason for not covering these methods here.

When the work that led to this book began--in preparation for the teaching of a course in the spring of 1968--only a few of the techniques we have worked through were available. Many of them have been called forth first by the course and then by the book. Over these same years, we have learned a lot about the main-highway techniques, enough so that it may soon be time to start to develop an account of these techniques either as exploration, or as confirmation, or as both.

A quarter century ago, W. G. Cochran said that "Regression is the worst-taught part of statistics". The continuing truth of his statement has not kept (multiple) regression from being used perhaps more than any other technique. It has had many outstanding successes--and it has helped to mislead many. It will continue to be a main-highway technique--one you will and should learn if you become an analyzer of data--very few can do without it. Just do not expect it to be very easy to learn both just what it is about and just what cautions are needed in its use--another reason why we could not crowd it in here. (But see Mosteller and Tukey, 1977.)

21C. How should the past chapters look different?

A course of reading, like a course of instruction, has a beginning, a middle, and an end. It may well loop back and forth, but often only at the price of confusion.

We started with elementary techniques, including plotting scatters of points, and it was not till chapters 7 and 8 that we learned about smoothing and middling. Were someone to write a "second course in exploratory data analysis," one that, like a second course in high-school French, went over the same ground just a little more deeply, it would draw on chapters 7 and 8 when it started on scatter plots, for, once you know about smoothing, middling, and even delineation (chapter 9), it is both very natural and very useful to equip each and every scatter diagram either with one or more automatically-produced middles or with judgment-based smooths of same.

It was an early lesson that we gain a lot by plotting the points. It was a later lesson, not emphasized as hard as it should be, perhaps, that you often do not learn enough until you smooth or middle the points. Plotting the points is often not enough.

Both the scatter and its smooth or middle can help us. The scatter is still our best way to recognize the utterly unanticipated--the smooth or middle is a better way to recognize the more or less anticipated. (And the rough may be a way to recognize what neither scatter nor smooth shows clearly.) Having gone through our techniques once, we can move toward a more balanced attitude toward them, and toward wider uses for many.

In a more detailed way, the fitting techniques of chapters 10 and 11 can be used to organize and simplify the delineation of point scatters. Our results could be essentially the same, but the calculations would be neater, and one computer program, if available, would do for both purposes.

These are but examples. Every serious reader should look for additional ones, taking a personal responsibility for further organizing and usefully digesting the various techniques we have learned to use in isolation.

21D. What have we been introduced to?

When we try to structure what we have done, we find overlapping relationships, as the following list suggests (chapter numbers in parentheses):

Batches (1, 2, 3, 4, 15)

Lines and trends (5, 6, 7, 8)

Smooths and roughs (7, 8, 9, 16, 17)

Scatters (8, 9, 17)

Two-way analyses (10, 11, 12, 13)

Counted data (scattered throughout, 17, 18, 19)

It will help most readers to review, at least generally, what has been said under each of these broad topics as a whole. Some will then find it useful to consider how each one overlaps with or interrelates to each of the others.

References

F. Mosteller and J. W. Tukey, 1977, Data Analysis and regression: A second course in statistics. Addison-Wesley, Reading, Mass.

The Holy Bible (King James Version). Revelation, Chapter 13, Verse 18.

Glossary

(entries beginning with punctuation marks and digits will be found on page 676)

adjacent value: value nearest to, but still within, an inner fence.

adjacent point: point on the adjacent polygon.

adjacent polygon: the polygon exactly surrounding all points not outside.

adjusted density: in chapter 16, observed density adjusted for observed curvature.

all: abbreviation for value of common term; for something included (by addition) in all values of the fit.

amount: value that can never be less than zero and is not a count.

average density: (fraction of counts)/(bin width)

balances: variates that can be either + or − and have no obvious bounds.

basic count: a number of occurrences, as distinguished from the numbers of bins/cases/individuals/species for which a particular number of occurrences are observed.

batch: a set of values of similar meaning, obtained however. (By no means necessarily anything like a sample.)

bin: a cell or interval along a scale, into which individual values do or don't fall.

blurring: representing smoothed values as the centers of vertical bars (of carefully chosen length).

box-and-whisker plots: a box from hinge to hinge, barred at the median, with "whiskers" to (a) the extremes (b) the innermost identified values (c) the adjacent values.

break table: a no-interpolation table working through the "breaks"—the arguments at which the functional value changes. (Also called "critical table".)

broken median: horizontal bars at the median of each piece (as defined by letter cuts) and the portions of the letter cuts needed to connect them.

clarity: ratio of a measure of size of fit (usually highest value MINUS lowest value) to a measure of size of residual (usually a letter spread). (E.g. E-clarity, H-clarity.)

coded residual: display of the rough value of a residual as one of few (often ◎, ○, ∘, ·, ×, ×, ⨻, or ⧺).

col: frequently used to mean a value depending upon column alone.

col-extra: an additional quantity depending upon column.

column effect: see effect.

common term: something included, by addition, in all values of a fit.

comparison values: values of (row)(col)/(all) for a PLUS fit.

completed fraction: ratio of completed rank to total number of cases.

completed rank: *See* c'rank.

condensed fit: result of coalescing fits and medianing residuals when fitted "row" [or "col"] values come too close to one another.

condensed residuals: see condensed fit.

coordinate: a function of position, often defined by its level traces.

count: integer (positive or zero) obtained by counting.

counted fraction: a fraction obtained by counting both a total and the number that "do", and taking the ratio.

counting in: assigning an extreme value rank 1, the next value 2, and so on.

c'rank: the largest rank assigned to a given basic count (or size).

cross median: that point, often not among the given points, each of whose two coordinates is the median of the corresponding coordinates of the given points.

cumulative: a formula (or table) giving cumulative fractions in terms of cuts.

cutting value: one of a sequence of values of x chosen to divide a batch of (x, y) pairs into slices.

D: a letter value at depth $\frac{1}{2}(1 + e)$, where e is the integer part of the depth of an E; crudely, a sixteenth.

delineation: a set of smoothed letter traces, crossed by the letter cuts used to begin their definition; a display of traces and cuts (often each D, E, H, M, H, E, D).

density: see average density, mathematical density.

dependency ratio: (actual MINUS median)/(most positive MINUS median), unless this is negative, when denominator becomes (median MINUS most negative).

dependency trace: a trace of dependency ratio (usually smoothed) against slice position.

depth: lesser of rank up and rank down.

diagnostic plot: After a PLUS analysis, a plot of residuals against comparison values. (The middle half, in terms of comparison values, is often replaced by its cross median.)

double flog: See fflog.

double lines: indication that the things they separate are to be added together.

double root: $\sqrt{2 + 4\ (\textit{observed count})}$ (and sometimes $\sqrt{1 + 4(\textit{fitted count})}$).

D-spread: high D-value MINUS low D-value.

D-trace: a smooth curve indicating the behavior of D-values of slices, usually passing through points obtained by smoothing D-to-E or E-to-D differences and subtracting or adding the result to the appropriate E-trace.

E: a letter value at depth $\frac{1}{2}(1 + h)$, where h is the integer part of the depth of a hinge; crudely an eighth, sometimes called one.

E-box: closed figure made up of E-traces and E-cuts. (*See* letter traces *and* letter cuts.)

eff: abbreviation for "effect".

effect: allocation of part of several observed values to a circumstance (more generally to a combination of circumstances).

end-value rule: using $y_1 = \textit{median}(x_1,\ y_2,\ 3y_2 - 2y_3)$ as the smoothed end value (x = data, y = smooth).

E-spread: high E-value MINUS low E-value.

E-trace: a smooth curve indicating the behavior of E-values of slices, usually passing through points obtained by smoothing E-to-H or H-to-E differences and subtracting or adding the result to the appropriate hinge trace.

E-value: *See* E.

extreme: a highest or lowest value of a batch. (As of depth 1, often labelled "1".)

factor: a circumstance described in terms of specified versions.

far out: a value beyond the outer fences.

fcent: a folded percent, given by $p - (100 - p) = 2p - 100$.

fence: inner fences are one step outboard of the hinges; outer fences are two steps outboard.

fence polygon: in a plane, the inner polygon defined by all possible fence lines.

fenced letter display: a fairly complete letter display, including fence values and identification of outside points.

fflog: a doubly folded logarithm; a double flog, a sum and difference of four terms. (*See* definition of flog and Section 15E.)

fit: an incomplete description of the data, usually selected from a class of possible descriptions, sometimes by a specified procedure.

flog: a folded logarithm, specifically

$$\frac{1}{2}\log_e (\textit{fraction that ``do''}) - \frac{1}{2}\log_e (\textit{fraction that ``don't''}),$$

usually best taken as

$$\log_e \sqrt{\textit{number that ``do''} + \frac{1}{6}} - \log_e \sqrt{\textit{number that ``don't''} + \frac{1}{6}}.$$

folded root: see froot.

folded log: see flog.

four steps (of median polish): a frequently adequate stage of analysis reached by removing row and column medians (alternately) twice each. (Also called "two cycles of median polish".)

froot: a folded square root, specifically

$$\sqrt{2\,(\textit{fraction that ``do''})} - \sqrt{2\,(\textit{fraction that ``don't''})}.$$

Gaussian reference: the Gaussian shape of distribution regarded as something from which we assess deviations (almost certainly not zero) rather than as something that might really happen.

Gaussian shape: all distributions obtained from

$$\frac{1}{\sqrt{2\pi}} e^{-x^2/2}\, dx$$

by linear change of scale and location.

grade: any variate whose values are ordered, but which we do not care to consider numerical.

h: following a digit, means "and a half".

H: as a letter value, a hinge, a value whose depth is $\frac{1}{2}(1 + m)$, where m is the integer part of the depth of the median; crudely, a quartile.

H: in smoothing, hanning (which *see*).

half-octave values: values proceeding by two steps to the octave--as 1, –, 2, 3, 4, 6, 8, 11, 16, . . .

hanning: a smoothing process alternatively describable as the result of (i) two repetitions of running means of 2; (ii) taking the mean of each given value and its skip mean; (iii) a moving average with weights 1/4, 1/2, 1/4.

hinge: letter value at depth $\frac{1}{2}(1 + m)$, where m is the integer part of the depth of the median; crudely, a quartile.

hinge trace: a smooth curve indicating the behavior of hinges of slices, usually passing through points obtained by smoothing H-to-M or M-to-H differences and subtracting or adding the result to a median trace.

H-spread: high hinge MINUS low hinge.

H-value: *See* H.

integer part: if i is an integer, and f a fraction with $0 < f < 1$, the integer part of $i + f$ is i.

interpolated rank: the value at interpolated rank $i + f$ (i an integer, f a fraction) is $(1 - f)x_i + fx_{i+1}$, where x_i has rank i and x_{i+1} has rank $i + 1$.

ladder of expressions: the increasing sequence of powers (first negative powers, then the log, then positive powers).

leaf: in a stem-and-leaf display, the righthand part of one of the values displayed.

letter cut: a vertical cut defined by a slight outward displacement of a letter value.

letter-value difference: difference between adjacent letter values (e.g., H-to-M or M-to-H).

letter display: *See* letter-value display.

letter trace: a trace which tries to divide the points roughly like a letter value, doing this not only globally but somewhat locally.

letter value: one of the values tagged . . . , B, C, D, E, H, M, H, E, D, C, B, A, Z, . . . , where the depth of each letter is almost half-way from the depth of the letter value next nearer the median to 1 (the depth of an extreme). (Extremes and the like are sometimes included.)

letter-value display: a somewhat formalized display of the letter values. (It may include: fence positions, number outside, spreads, mids, identifications, and/or pseudospreads.)

level curve: curve along which some function, usually a possible coordinate, takes a given value (elsewhere also called *level line*).

level line: line along which some function takes a given value.

level trace: a level line or level curve.

log: a logarithm (probably to base 10).

$\log_e$: a natural or Napierian logarithm (to base 2.71828 . . . , $\log_e u = 2.30259 \log_{10} u$).

$\log_{10}$: a common logarithm, a log to base 10.

M: as a letter value, the median.

mathematical density: limiting value of average density (as bin width $\to 0$ and bin count $\to \infty$).

median: value of depth $\frac{1}{2}(1 + n)$, where there are n values in the batch; the middle value when all values are arranged in nondecreasing--or nonincreasing--order.

median polish: a process of (finding, adding elsewhere, and subtracting here) medians alternately. (In a two-way table, alternately those of rows and those of columns.)

median removal: *See* median polish.

median trace: a middle trace obtained by using medians (as most middle traces are).

mid: indicates the mean of two letter values (e.g. midhinge, mideighth, midD, midextreme).

middle trace: trace leaving roughly half of all points above and below itself, both semilocally and globally--in practice, usually the result of smoothing the cross medians of a set of pieces.

mixed leaves: leaves in a single stem-and-leaf display involving different numbers of leaf digits.

modified (hinges, medians, etc.): result of weighting exact zeroes $\frac{1}{2}$.

negative reciprocal: the result of dividing a chosen negative constant by the given number.

octave: values (perhaps counts) between A and 2A (for some A), or intervals of such form.

outside point: one outside the inner fence polygon.

outside value: a value outside the inner fences (and, if "far out" is also mentioned, inside the outer fences).

peak: inferred highest point of fit (will usually not fall at a bin center).

piece: those data points between two adjacent letter cuts.

plurality: difference between a fraction and its complement, $f - (1 - f) = 2f - 1$.

PLUS fit: a row-PLUS-column fit (or one of its simplest generalizations).

PLUS-one fit: a fit that can be put in either--and hence both--of the forms:

all PLUS row PLUS col PLUS (row)(col)/constant

constant PLUS (row*)(col*)

PLUS-one plot: a two-way plot appropriate to a PLUS-one fit (one set of parallel lines and one set of lines with a common intersection when raw response is represented).

PLUS-one plot of residuals: coded residuals placed where intersections in a PLUS-one plot would fall.

PRODUCT: in product–ratio analysis, basic count times c'rank (from above).

product–ratio analysis: calculation and use of product–ratio plots; study of the relation of root of PRODUCT to log of RATIO.

product–ratio plots: plots of $\sqrt{(\textit{basic count})\ (\textit{final rank from above})}$ against $\log((\textit{basic count})/(\textit{final rank from above}))$, or of the result of changing "final rank" to "final fraction".

(25%) pseudospread: the spread between 25% points of a standard distribution shape (usually Gaussian) when the spread between p% points is that observed.

R: in "3R" means "3 repeated as many times as necessary to settle down."

range: high extreme MINUS low extreme.

rank: ordinal number of value when values are made nondecreasing (rank up) or nonincreasing (rank down).

RATIO: in product–ratio analysis, basic count divided by c'rank (from above).

reciprocal: the result of dividing 1 by the given number (and hence, loosely, the result of dividing any chose constant, instead, by the given number).

reduced delineation: an E-box in solid lines, M traces/cuts dashed within box, H-traces/cuts dashed outside it.

re-expression: expressing the same information by different numbers, as when y's are replaced by log y's.

re-matching: re-pairing of given y's with given x's, usually to make the re-matched relationship as strong (negative or positive) as possible.

reroughing: smoothing a rough and adding this smooth to the original smooth.

residual: result of subtracting fit from data; thus, what is not accounted for by incomplete description (fit).

response: variate, or value of variate, thought of, at least potentially and temporarily or schematically, as influenced by (or associated with) certain factors or circumstances.

root: square root.

root rough: the rough of the square roots of the given counts.

root smooth: the smooth of the square roots of the given counts.

rough: data MINUS smooth.

row: often means something depending upon row alone.

row effect: *See* effect.

row-extra: an additional quantity depending upon row alone (often appears multiplied by constant and col).

row-PLUS-column fit: a PLUS fit, a fit of the form "all PLUS row PLUS col" (for generalizations, see Sections 12A and 13D).

row-TIMES-column fit: a TIMES fit, a fit of the form "(row*) TIMES (col*)" (for generalizations, see 12A and 13D).

running median of 3: for a sequence, at each i the median of x_{i-1}, x_i, and x_{i+1}.

S: in smoothing, treating peaks/valleys of length 2 by splitting the sequence between the 2 and applying first the end-value rule to each piece and then 3R to the result.

schematic plot: a schematic representation of a batch, including a hinge box, barred at the median with dashed whiskers extending to the adjacent values terminated by (dashed) bars, and with all outside values identified.

schematic (x, y) plot: a schematic representation of a batch, including an E-box, an adjacent polygon, and identified outside points.

sequence: an ordered set of (x, y) data (where x may have been suppressed), in which x proceeds by steps of equal size.

shape of distribution: that which is common to the distributions of all trivial re-expressions of some quantity.

shift: displacement of value considered from the peak.

skip mean: in smoothing, $(x_{j-1} + x_{j+1})/2$ placed at j.

slice: those (x, y) pairs whose x-values lie between chosen cutting values.

sliced coordinate: one used to determine letter cuts, and thus stress and various traces.

smooth: a sequence of smoother appearing values, calculated from the given values; the process of such calculation.

smoother: a process dividing a given sequence into smooth and rough (which sum, term by term, to the given sequence).

split count: a count of "those less" PLUS one-half of "those equal".

squeezed stem-and-leaf display: a display with 5 stems (usually labelled "*", "T", "F", "S", and "·") for each stem number.

SS: S applied once, and then applied once again.

ss: stands for "started split".

ss-count: the value of "$(\text{count below}) + \frac{1}{2}(\text{count equal}) + \frac{1}{6}$".

ss-fraction: the value of "$\left[(\textit{count below}) + \frac{1}{2}(\textit{count equal}) + \frac{1}{6}\right]$ divided by $\left(\textit{total count} + \frac{1}{3}\right)$".

standard cutting points: for coding residuals, 1 or 2 H-spreads outboard of hinges. (Distinguish from fences. Nonstandard cutting points often used.)

start: a constant added before taking a log or root (or etc.). (Here 1/6 is popular. Elsewhere 1/10, 1/4, or 1 may appear.)

started count: for product–ratio analysis, the basic count increased by 1; (in general a count increased by a constant).

stem: a line in a stem-and-leaf display.

stem-and-leaf display: a generalized two-digit display, in which the lefthand portion of the values displayed is given by a stem value, while the righthand portion makes up a leaf (leaves follow stem juxtaposed if uniformally one digit, follow separated by commas if some or all involve 2 or more digits).

stem label: the stem value, followed (if there are 2 or more stems per value) by a character from "∗", "T", "F", "S" and "·", the whole preceding a vertical line.

stem-value: that part of the label of a stem which consists of the common left-hand portion of the values recorded in that line.

step: 1.5 times the H-spread.

stretched stem-and-leaf: a stem-and-leaf display, with 2 stems per value, these being distinguished with "∗" and "·".

three-way analysis: analysis of one response according to three kinds of circumstances and their combinations.

tilt: an (x, y) data display is tilted if subtracting some bx from y, with $b \neq 0$, is needed to make the display appear "untilted".

TIMES fit: a row-TIMES-column fit (or one of its simplest generalizations).

traced coordinate: one whose slice medians define the broken median, etc.

trimean: $\frac{1}{4}(\textit{low hinge}) + \frac{1}{2}(\textit{median}) + \frac{1}{4}(\textit{high hinge})$.

trivial re-expression: using $a + bx$ for some fixed a and b, instead of x.

twicing: reroughing with the same smoother used to make the initial smooth and rough.

two-way analysis: analysis of one response according to two kinds of circumstances.

two-way plot: a representation of a fit by two families of lines––one family for each kind of circumstance, one line for each circumstance––such that the y-coordinate of each intersection gives the corresponding fitted value.

two-way plot of residuals: coded residuals, each plotted where the corresponding intersection of the two-way plot would fall.

two-way schematic plot: *See* schematic (x, y) plot.

untilted: such that no subtraction of bx from y makes the (x, y) display seem more nearly horizontal.

untilted analysis: in smoothing, schematic plotting, or delineation, an analysis of $(x, y - bx)$ values for a suitable b, whose results are then restored to (x, y) form.

version: a version may be qualitative (when it is likely to be called a "category" or "class") or quantitative (when it is likely to be called a "level"). (Ordered versions may also be called "levels".)

wandering schematic plot: a description of an (x, y) batch involving hinge and median traces and the adjacent polygon (points outside that polygon will ordinarily be plotted and identified).

",": in smoothing, used to separate components related by reroughing or twicing.

#: a count, often a total count or batch size.

***:** may indicate a place that can be filled.

***:** when followed by a letter-value term or symbol (as in *E, *-fences, *H, *-hinge, *-letter-value, *M, etc.), indicates that exact zeroes have each been given a weight of 1/2.

1: in letter values, an extreme.

3: in smoothing, running medians of 3.

5-number summary: values of extremes, hinges, and median, values of 1HMH1.

7-number summary: values of 1EHMHE1.

9-number summary: values of 1DEHMHED1.

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exhibit **2** of chapter 15: reference table

Pluralities, folded roots, folded logarithms--alternative expressions for counted fractions (take sign of answer from head of column giving %)

A) MAIN TABLE

+	Plur.	froot	flog	−	+	Plur.	froot	flog	−
50%	use	.00	use	**50%**	**85%**	.70	.76	.87	**15%**
51	→	.02	←	**49**	**86**	.72	.78	.91	**14**
52		.04		**48**	**87**	.74	.81	.95	**13**
53		.06		**47**	**88**	.76	.84	1.00	**12**
54		.08		**46**	**89**	.78	.87	1.05	**11**
55%	use	.10	use	**45%**	**90.0%**	.80	.89	1.10	**10.0%**
56	→	.12	←	**44**	**90.5**	.81	.91	1.13	**9.5***
57		.14		**43**	**91**	.82	.92	1.16	**9**
58		.16		**42**	**91.5**	.83	.94	1.19	**8.5**
59		.18		**41**	**92**	.84	.96	1.22	**8**
60%	use	.20	use	**40%**	**92.5%**	.85	.97	1.26	**7.5%**
61	→	.22	←	**39**	**93**	.86	.99	1.29	**7 ***
62		.24		**38**	**93.5**	.87	1.01	1.33	**6.5**
63	.26	.26	.27	**37**	**94**	.88	1.02	1.37	**6**
64	.28	.28	.29	**36**	**94.5**	.89	1.04	1.42	**5.5**
65%	.30	.30	.31	**35%**	**95.0%**	.90	1.06	1.47	**5.0%**
66	.32	.32	.33	**34**	**95.5**	.91	1.08	1.53	**4.5**
67	.34	.35	.35	**33**	**96**	.92	1.10	1.59	**4**
68	.36	.37	.38	**32**	**96.5**	.93	1.12	1.65	**3.5**
69	.38	.39	.40	**31**	**97**	.94	1.15	1.74	**3**
70%	.40	.41	.42	**30%**	**97.2%**	.94	1.16	1.77	**2.8%**
71	.42	.43	.45	**29**	**97.4**	.95	1.17	1.81	**2.6**
72	.44	.45	.47	**28**	**97.6**	.95	1.18	1.85	**2.4**
73	.46	.47	.50	**27**	**97.8**	.96	1.19	1.90	**2.2**
74	.48	.50	.52	**26**	**98.0**	.96	1.20	1.95	**2.0**
75%	.50	.52	.55	**25%**	**98.2%**	.96	1.21	2.00	**1.8%**
76	.52	.54	.58	**24**	**98.4**	.97	1.22	2.06	**1.6**
77	.54	.56	.60	**23**	**98.6**	.97	1.24	2.13	**1.4**
78	.56	.59	.63	**22**	**98.8**	.98	1.25	2.21	**1.2**
79	.58	.61	.66	**21**	**99.0**	.98	1.27	2.30	**1.0**
80%	.60	.63	.69	**20%**	**99.2%**	.99	1.28	2.41	**0.8%**
81	.62	.66	.72	**19**	**99.4**	.99	1.30	2.55	**0.6**
82	.64	.68	.76	**18**	**99.6**	.99	1.32	2.76	**0.4**
83	.66	.71	.79	**17**	**99.8**	1.00	1.35	3.11	**0.2**
84	.68	.73	.83	**16**	**100.0%**	1.00	1.41	*	**0.0**

For more detail for flogs of fractions beyond 1% or 99%, see page 500.

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Values of 199, (b) for every
10th count integers ≤ 20

A) TAB ntegers up to 20

	0						7	8	9
00	−.9(						.98	1.05	1.11
10	1.1(						1.42	1.45	1.48
20	1.5(						1.65	1.67	1.69
30	1.7(						1.81	1.82	1.83
40	1.8						1.93	1.94	1.95
50	1.9(						2.02	2.03	2.04
60	2.0						2.10	2.11	2.12
70	2.1						2.17	2.18	2.19
80	2.1						2.23	2.24	2.25
90	2.2						2.29	2.29	2.30
100	2.3(						2.34	2.34	2.35
110	2.3						2.38	2.39	2.39
120	2.3						2.42	2.43	2.43
130	2.4						2.46	2.46	2.47
140	2.4						2.50	2.50	2.50
150	2.51						2.53	2.53	2.53
160	2.5						2.56	2.56	2.57
170	2.57						2.59	2.59	2.59
180	2.6(						2.62	2.61	2.62
190	2.6						2.64	2.64	2.65

B) TABLE for COUNTS from 200 to 3090--grab nearest; don't interpolate

	00	10	20	30	40	50	60	70	80	90
200	2.65	2.67	2.70	2.72	2.74	2.76	2.78	2.80	2.82	2.84
300	2.85	2.87	2.88	2.90	2.91	2.93	2.94	2.96	2.97	2.98
400	3.00	3.01	3.02	3.03	3.04	3.05	3.07	3.08	3.09	3.10
500	3.11	3.12	3.13	3.14	3.15	3.16	3.16	3.17	3.18	3.19
600	3.20	3.21	3.21	3.22	3.23	3.24	3.25	3.25	3.26	3.27
700	3.28	3.28	3.29	3.30	3.30	3.31	3.32	3.32	3.33	3.34
800	3.34	3.35	3.35	3.36	3.37	3.37	3.38	3.38	3.39	3.40
900	3.40	3.41	3.41	3.42	3.42	3.43	3.43	3.44	3.44	3.45
1000	3.45	3.46	3.46	3.47	3.47	3.48	3.48	3.49	3.49	3.50
1100	3.50	3.51	3.51	3.52	3.52	3.52	3.53	3.53	3.54	3.54
1200	3.545	3.549	3.553	3.557	3.562	3.566	3.569	3.573	3.577	3.581
1300	3.585	3.589	3.593	3.597	3.600	3.604	3.608	3.611	3.615	3.619
1400	3.622	3.626	3.629	3.633	3.636	3.640	3.643	3.647	3.650	3.653
1500	3.657	3.660	3.663	3.667	3.670	3.673	3.676	3.679	3.683	3.686
1600	3.689	3.692	3.695	3.698	3.701	3.704	3.707	3.710	3.713	3.716
1700	3.719	3.722	3.725	3.728	3.731	3.734	3.737	3.739	3.742	3.745
1800	3.748	3.751	3.753	3.756	3.759	3.762	3.764	3.767	3.770	3.772
1900	3.775	3.777	3.780	3.783	3.785	3.788	3.790	3.793	3.795	3.798

For continuation in text, for counts above 1900, see pages 504–505.

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Values of $\sqrt{\text{count} + \frac{1}{6}}$ (a) for integer counts from 0 to 199, (b) for every 10th count from 200 to 3090, (c) for half-integer counts < 20

A) TABLE for COUNTS up to 199--see panel C for half-integers < 20

	0	1	2	3	4	5	6	7	8	9
00	.41	1.08	1.47	1.78	2.04	2.27	2.48	2.68	2.86	3.03
10	3.19	3.34	3.49	3.63	3.76	3.89	4.02	4.14	4.26	4.38
20	4.49	4.60	4.71	4.81	4.92	5.02	5.12	5.21	5.31	5.40
30	5.49	5.58	5.67	5.76	5.85	5.93	6.01	6.10	6.18	6.26
40	6.34	6.42	6.49	6.57	6.65	6.72	6.79	6.87	6.94	7.01
50	7.08	7.15	7.22	7.29	7.36	7.43	7.49	7.56	7.63	7.69
60	7.76	7.82	7.88	7.95	8.01	8.07	8.13	8.20	8.26	8.32
70	8.38	8.44	8.50	8.55	8.61	8.67	8.73	8.78	8.84	8.90
80	8.95	9.01	9.06	9.12	9.17	9.23	9.28	9.34	9.39	9.44
90	9.50	9.55	9.60	9.65	9.70	9.76	9.81	9.86	9.91	9.96
100	10.01	10.06	10.11	10.16	10.21	10.26	10.30	10.35	10.40	10.45
110	10.50	10.54	10.59	10.64	10.68	10.73	10.78	10.82	10.87	10.92
120	10.96	11.01	11.05	11.10	11.14	11.19	11.23	11.28	11.32	11.37
130	11.41	11.45	11.50	11.54	11.58	11.63	11.67	11.71	11.75	11.80
140	11.84	11.88	11.92	11.97	12.01	12.05	12.09	12.13	12.17	12.21
150	12.25	12.29	12.34	12.38	12.42	12.46	12.50	12.54	12.58	12.62
160	12.66	12.70	12.73	12.77	12.81	12.85	12.89	12.93	12.97	13.01
170	13.04	13.08	13.12	13.16	13.20	13.24	13.27	13.31	13.35	13.39
180	13.42	13.46	13.50	13.53	13.57	13.61	13.64	13.68	13.72	13.75
190	13.79	13.83	13.86	13.90	13.93	13.97	14.01	14.04	14.08	14.11

B) TABLE for COUNTS from 200 to 3090--grab nearest; don't interpolate

	00	10	20	30	40	50	60	70	80	90
200	14.15	14.50	14.84	15.17	15.50	15.82	16.13	16.44	16.74	17.03
300	17.33	17.61	17.89	18.17	18.44	18.71	18.98	19.24	19.50	19.75
400	20.00	20.25	20.50	20.74	20.98	21.22	21.45	21.68	21.91	22.14
500	22.36	22.59	22.81	23.03	23.24	23.46	23.67	23.88	24.09	24.29
600	24.50	24.70	24.90	25.10	25.30	25.50	25.69	25.89	26.08	26.27
700	26.46	26.65	26.84	27.02	27.21	27.39	27.57	27.75	27.93	28.11
800	28.29	28.46	28.64	28.81	28.99	29.16	29.33	29.50	29.67	29.84
900	30.00	30.17	30.33	30.50	30.66	30.82	30.99	31.15	31.31	31.47
1000	31.63	31.78	31.94	32.10	32.25	32.41	32.56	32.71	32.87	33.02
1100	33.17	33.32	33.47	33.62	33.77	33.91	34.06	34.21	34.35	34.50
1200	34.64	34.79	34.93	35.07	35.22	35.36	35.50	35.64	35.78	35.92
1300	36.06	36.20	36.33	36.47	36.61	36.74	36.88	37.02	37.15	37.28
1400	37.42	37.55	37.69	37.82	37.95	38.08	38.21	38.34	38.47	38.60
1500	38.73	38.86	38.99	39.12	39.24	39.37	39.50	39.63	39.75	39.88
1600	40.00	40.13	40.25	40.38	40.50	40.62	40.75	40.87	40.99	41.11
1700	41.23	41.35	41.47	41.60	41.72	41.83	41.95	42.07	42.19	42.31
1800	42.43	42.55	42.66	42.78	42.90	43.01	43.13	43.25	43.36	43.48
1900	43.59	43.71	43.82	43.93	44.05	44.16	44.27	44.39	44.50	44.61

For continuation in text, for counts above 1900, see page 507.

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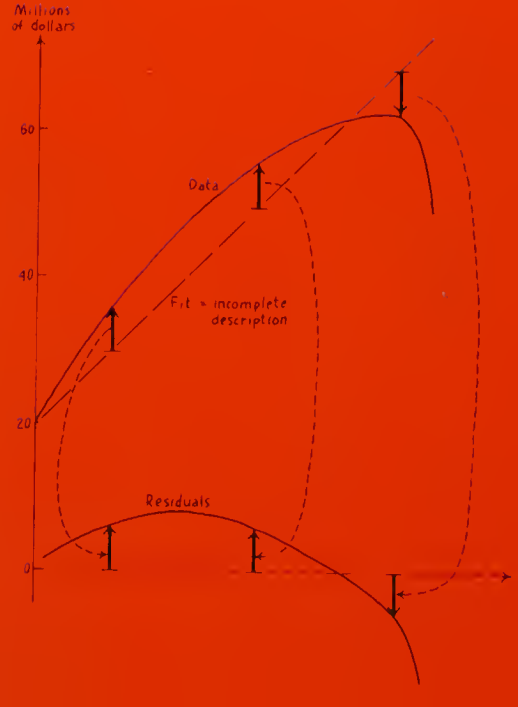
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